

ADVANTEST®

ADVANTEST CORPORATION

*T5500 Series
Tester Utility
Program
Reference Manual*

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*Applicable OS
ASX/U-51*

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- Description of the T5591 test system in this manual applies to the T5591R unless otherwise notified.

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Preface

Purpose of This Document

This manual explains useful tools (utility programs) to use testers.

Related Manuals

ATL-51 Test Plan Program Reference Manual(No. : 8431498)

ATL-51 Pattern Program Reference Manual(No. : 8256292)

ATL-51 Pattern Program Reference Manual (T5376, T5375, T5371, T5336 and T5311)(No. : 8289058)

ATL-51 Socket Program Reference Manual(No. : 8256293)

ATL-51 Scramble Program Reference Manual(No. : 8256294)

T5500 Series Tester Controller Commands Reference Manual(No. : 8256297)

ATLworks Tool User's Manual(No. : 8311174)

ASX/U-51 Hybrid Controller User's Manual(No. : 8409598)

ASX/U-51 Hybrid Controller Reference Manual(No. : 8409600)

Error Code Reference Manual(No. : 8141091)

XATL/U-51 Cross Compiler User's Manual(No. : 8256237)

ADVANsite Test System Software Installation/System Start User's Manual VOL1 Basic OS Installation(No. : 8227728)

ADVANsite Test System Software Installation/System Start User's Manual VOL2 Tester Environment Software Installation(No. : 8227729)

T5500 Series Control Box/Trigger Panel User's Manual(No. : 8256298)

T5300 Series Flash Memory Parallel Test Function Description(No. : 8239605)

T5500 Series Memory Test Systems Calibration Program Reference Manual(No. : 8256296)

T5500 Series Memory Test Systems High-Precision Calibration Program Reference Manual(No. : 8311137)

T5500 Series Memory Test System High-Precision Calibration2 Program Reference Manual(No. : 8350393)

Manual Performance Board PBTYP E Specifications(No. : 8292985)

T5300/T5500+M3700 Series GPIB Handler Control Utility Program Specifications(No. : 8269079)

ADVANSite System Introduction(No. : 8227727)

1. Pin Monitor (/PM)

The pin monitor utility displays the settings of specified tester units on the VKT or printer. This statement is used when the system is in one of the following states:

- stopped in "MANUAL HOLD" mode.
- looping in "MANUAL RECYCLE HOLD" mode.
- stopped at the debugger break point

1. 1 Syntax

```
/PM {Global switch} Unit name↵  
/PM {Global switch} Symbol name↵
```

Enter the above command from the VKT while the controller is operating. The conditions set for the specified units and the units defined by symbol names are displayed on the VKT.

1. 1. 1 Global Switch

The following switches can be specified globally:

/L

Outputs to printer.

/W

Sets 132-column mode for the pin monitor screen by fixed-format or the monitor screen at Pn specification.

/S

Outputs to the output unit selected by S command.

/C

Displays the conditions for the symbol-defined pin.

/M

Displays the pin conditions using a symbol name.

/R

Displays a pin condition and then displays the conditions for the unit connected to the displayed pin.

/I

Displays the VSn settings, as well as the voltage setting for each PPS being used.

/A

Displays all Cycle Palette data.

/B

Displays the driver level settings for all Child settings of each pin.

1. 1. 2 Unit Name and Symbol Name

Multiple unit names or symbol names can be simultaneously specified.

If the unit name or symbol name is omitted, the condition of each unit is displayed in fixed format.

The /PM command accepts the following unit names:

Pn

Displays the connection state for Pn.

PDn

Uses a mnemonic display to show the connection state for PDn.

PCn

Uses a mnemonic display to show the connection state for PCn.

Tip

To specify pin numbers (the suffix n of Pn, PCn and PDn) in the pin monitor utility program, use the numbers described in the test plan program. Pin number is specified in both ascending and descending orders.

VS_n

Displays the setting of VS_n.

PPS_n

Displays the set value of PPS_n.

On the T5581 and T5581H, displays the setting when the 16VPPS option is installed.

DC

Displays the setting of DC.

ISVM

Displays the setting of ISVM.

VSIM

Displays the setting of VSIM.

IN_n

Displays the set value of IN_n or INH_n.

On the T5376, the settings of IN_n and INH_n are displayed.

OUT_n

Displays the set value of OUT_n.

DCLP

Displays the set value of DCLP.

VTn

Displays the set value of VTn.

ILn

Displays the set value of ILn.

PCON

Displays the set value of PCON.

VCON

Displays the set value of VCON.

LCON

Displays the set value of LCON.

RATE <m>

Displays the set value of TG RATE.

<m> indicates a timing set number. When omitted, it is 1. The default value is 1 (applicable for all TG-related units given below).

ACLKn <m>

Displays the set value of TG ACLKn.

BCLKn <m>

Displays the set value of TG BCLKn.

CCLKn <m>

Displays the set value of TG CCLKn.

STRBn <m>

Displays the set value of TG STRBn

WSTRBn <m>

Displays the set value of the TG window comparison timing WSTRBn.

DRELn <m>

Displays the set value of TG DRELn.

DRETn <m>

Displays the set value of TG DRETn.

TGCLKn <m>

Displays the set value of TG TGCLKn.

PCLKn

Displays the set value of TG PCLKn.

REF

Displays the name of the reference-on unit.

TSn

Displays the set values of TG timing sets from n to n+3.

LIMIT

Displays the setting value of LIMIT DC, VS_n, or JUDGE.

MPAT

The specified pattern file name and SCRAM file name are displayed.

SETTLING

Displays the set values of SETTLING TIME.

A symbol name is a pin name which is defined in a socket program. Only pin names are valid.

1. 1. 3 Pressing the Return Key Only

Once a pin monitor command has been specified, the same command can be executed repeatedly by pressing the return key (CR key) only.

If the unit or symbol name is omitted, however, this function is invalid.

1. 2 Function

1. 2. 1 Pn

Displays the connection state for Pn.

BANK

Bank number (This is not displayed on the T5377, T5376, T5375, T5371, T5336 or T5311.)

Displayed messages and their meanings are as follows.

"A"	The same setting has been made for each bank.
"1"	Bank1
"2"	Bank2
"3"	Bank3 (2WAY or 4WAY on the T5592 or T5591)
"4"	Bank4 (2WAY or 4WAY on the T5592 or T5591)

When "!" is displayed to the left of a number, A, B, C and DRE clocks stored in banks do not work. (ACLK is not used on the T5593 or T5592.)

DC

Connection of the DC test unit.

TMDC

Specification of connecting DC measurement unit to terminal channel (This is displayed on the T5593)

GND

Connection of the GND relay. (This is not displayed on the T5593 and T5592.)

IN

VI unit number connected.

On the T5377, T5376, T5375, T5371, T5336 and T5311, INHn is displayed as Hn. (n indicates the INH number.)

HSC

Specification of a high-speed clock driver. (This is displayed only on the T5593.)

ACLK

BCLK

CCLK

Clock number used. (ACLK is not displayed on the T5593 and T5592.)

PCLK

PCLK specification (This is not displayed on the T5593, T5592, T5377, T5376, T5375, T5371, T5336 or T5311.)

MODE

Waveform mode.

LEVEL

Driver initial level specified by the PIN statement.

The current level of the hardware unit driver is shown in parentheses.

(This is not displayed on the T5591.)

RINV

Specification of real-time invert mode.

DMX

Specification of data multiplex mode (This is displayed only on the T5336 and T5311.)

SDM

Specification of data swap mode (This is not displayed on the T5593 and T5592.)

PSM

Specification of pattern swap mode (This is not displayed on the T5336 and T5311.)

SMC

Specification of swap mode control (This is not displayed on the T5336 and T5311.)

RDSM

Specification of real-time data swap mode.

UTL

Connection of the UTL relay (This is not displayed on the T5593, T5592, T5377, T5376, T5375, T5371, T5336 or T5311.)

PPAT

Specification of the DBM as a pattern to shape the driver waveform (This is not displayed on the T5377, T5376, T5375, T5371, T5336 or T5311.)

DRDLY

Number of delay cycles for the driver pattern (This is not displayed on the T5336 and T5311.)

DRPA

Specification of dummy cycle driver pattern for port A (This is displayed only on the T5593.)

DRPB

Specification of dummy cycle driver pattern for port B (This is displayed only on the T5593.)

SCFIX

Specification of the same cycle fix (This is displayed only on the T5336 and T5311.)

RINHA

Specification of real-time inhibit ACLK (This is displayed only on the T5336 and T5311.)

RINHB

Specification of real-time inhibit BCLK (This is displayed only on the T5336 and T5311.)

RINHC

Specification of real-time inhibit CCLK (This is displayed only on the T5336 and T5311.)

IOC

Specification of the common I/O pin.

IOCP

Specification of the mode in which to use a comparator channel as an I/O common channel (This is displayed only on the T5593.)

DRE

DRE signal used.

Displayed messages and their meanings are as follows:

"1"	DRE1
"2"	DRE2
"C12"	C12 (This is displayed only on the T5593 and T5592.)
"C13"	C13 (This is displayed only on the T5593 and T5592.)
"C14"	C14 (This is displayed only on the T5593 and T5592.)
"C15"	C15 (This is displayed only on the T5593 and T5592.)
"P"	(This is not displayed on the T5593, T5592, T5377, T5376, T5375, T5371, T5336 and T5311.)

PDRE

Specification of PDRE (This is displayed only on the T5593 and T5592.)

Displayed messages and their meanings are as follows:

"P"	PDRE
-----	------

DRENRZ

Specification of driver-enabled area takes place in NRZ or DNRZ mode.

Displayed messages and their meanings are as follows:

"-"	DRERZ
"1"	DRENRZ
"D"	DREDNRZ (This is displayed only on the T5593, T5592, T5377, T5376 and T5375.)

DRON

Initial state of the common I/O pin is the driver.

DREDLY

Number of delay cycles for the DRE pattern (This is not displayed on the T5336 and T5311.)

DREPA

Specification of dummy cycle DRE pattern for DREA. (This is displayed only on the T5593.)

DREPB

Specification of dummy cycle DRE pattern for DREB (This is displayed only on the T5593.)

DRECLK

DRE clock number used.

DBMFI

Specification of the mode for disabling the INHDBM signal (This is displayed only on the T5593.)

OUT

VO unit number connected.

TERM

Connection of the terminator using the OUT instruction.

Displayed messages and their meanings are as follows:

"1"	Connects the TERM relay. (Excluding the T5593, T5592 and T5591) Connects the comparator-side TERM relay and driver-side TERM relay. (On the T5593, T5592 and T5591)
"C"	Connects the comparator side TERM relay only. (On the T5593, T5592 and T5591)
"D"	Connects the driver side TERM relay only. (On the T5592 and T5591)

VT

Terminator level number used

IL

Programmable load number used (This is not displayed on the T5593 and T5592.)

ILALLn is displayed as "An". (n indicates the ILALL number.)

On the T5592**STRB**

Strobe number used.

CPE

Control pattern number used in strobe number.

HZ

Specification of Hi-Z used in strobe number.

On the T5377, T5376, T5375, T5371, T5336 and T5311**STRB**

Strobe number used.

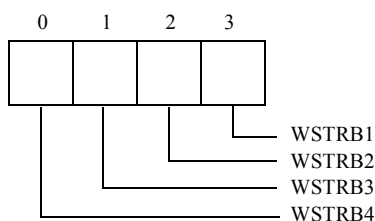
Displayed messages and their meanings are as follows:

"1"	STRB1
"2"	STRB2
"3"	STRB3
"4"	STRB4
"A"	STRB1, STRB2
"B"	STRB1, STRB4
"C"	STRB3, STRB2
"D"	STRB3, STRB4

WSTRBn is displayed as "Wn". (n indicates the WSTRB number.)

When MWS mode is specified, however, it is displayed as "Mn". The n is displayed in hexadecimal notation and corresponds to each bit as shown below.

WSTRB4 is not displayed on the T5377, T5376 and T5375.



CPE

Control pattern number used in strobe number.

Displayed messages and their meanings are as follows:

"1"	CPE1
"2"	CPE2
"3"	CPE3
"4"	CPE4
"A"	CPE1, CPE2
"B"	CPE1, CPE4
"C"	CPE3, CPE2
"D"	CPE3, CPE4

HZ2

Specification of Hi-Z used in strobe number (This is not displayed on the T5336 and T5311.)

On test systems other than the T5593, T5592, T5377, T5376, T5375, T5371, T5336 and T5311

STRB1

Odd-numbered phase strobe number used.

CPE

Control pattern number used in odd-numbered phase strobe.

HZ

Specification of Hi-Z used in odd-numbered phase strobe.

STRB2

Even-numbered phase strobe number used.

CPE

Control pattern number used in even-numbered phase strobe.

HZ

Specification of Hi-Z used in even-numbered phase strobe.

WSTRBn is displayed as "Wn". (n indicates the WSTRB number.)

On the T5593

STRB1

Odd-numbered phase strobe number used.

CPE

Control pattern number used in odd-numbered phase strobe.

HZ

Specification of Hi-Z or real time Hi-Z used in odd-numbered phase strobe.

Displayed messages and their meanings are as follows:

"1"	HZ
"A"	RHZA
"B"	RHZB

STRB2

Even-numbered phase strobe number used.

CPE

Control pattern number used in even-numbered phase strobe.

HZ

Specification of Hi-Z or real time Hi-Z used in odd-numbered phase strobe.

Displayed messages and their meanings are as follows:

"1"	HZ
"A"	RHZA
"B"	RHZB

MATCH

Specification of the match mode (This is not displayed on the T5593 and T5592.)

PCPE

Specification of PCPE, CCPE, ACPE (This is not displayed on the T5377, T5376, T5375, T5371, T5336 or T5311.)

Displayed messages and their meanings are as follows:

"P"	PCPE
"C"	CCPE (This is not displayed on the T5593 and T5592.)
"A"	ACPE (This is not displayed on the T5593 and T5592.)

EINV

Specification of inversion mode for the expected data (This is not displayed on the T5336 and T5311.)

ARID

Specification of the ARIRAM enable mode (This is not displayed on the T5593.)

LOAD

Connection of the LOAD relay (This is not displayed on the T5593 and T5592.)

DCLP

Specification of the dynamic clamp (This is not displayed on the T5593.)

EXP

Specification of the expected pattern output (Port A or Port B), FIXL or FIXH (This is not displayed on the T5336 and T5311.)

EXPDLY

Number of delay cycle for the expected patter. (This is not displayed on the T5336 and T5311.)

CPEDLY

Number of delay cycle for the CPE pattern (This is not displayed on the T5336 and T5311.)

ESM

Specification of the swap mode of the expected data (This is not displayed on the T5593, T5592, T5377, T5376, T5375, T5371, T5336, and T5311.)

BMUX

Specification of main bank and sub-bank (This is not displayed on the T5592, T5377, T5376, T5375, T5371, T5336 and T5311.)

Displayed messages and their meanings are as follows:

"1"	BMAIN
"2"	BSUB
"3"	BMUX

MSM

Specification of the mode used for the SUBPG option in 2 WAY interleave mode. (This is displayed only on the T5593.)

DSS

Specification of data (This is not displayed on the T5377, T5376, T5375, T5371, T5336 and T5311.)

CYP

Cycle palette number (This is displayed only on the T5593, T5592, T5377, T5376, T5375 and T5371.)

Displayed messages and their meanings on the T5593 are as follows:

"An"	Cycle palette A
"Bn"	Cycle palette B

(n represents a cycle palette number)

Display example for the T5591

/PM P1-3.J

```

B DG I A B C P M LRSPSRUPD IDDDD D O T V I S CH S CH MPEALDEECE BD
A CN N C C C C O EIDSM DTPR ORRRR R U E T L T PZ T PZ ACIROCXXPS MS
N D L L L L D VNMMCSLAD CEEOE E T R R E R E TPNIALPPPEM US
K K K K K E EV M TL NND C M B B CEVDDP DD X
L Y R L L 1 2 H LL
Z Y K YY

33 1 -- 1 - 1 1 - FIXL 0-----0 ----- - 1 - - - 1 1- - -- -----B00- 31 <FL,-,-,FL>
129 1- - -- --- A
2 -- 1 - 2 1 - FIXL 0-----0 ----- - 1 - - - - -- 2 2- -----B00- 31 <FL,-,-,FL>
130 2- --- A
3 -- 1 - 1 1 - FIXL 0-----0 ----- - 1 - - - 1 1- - -- -----B00- 31 <FL,-,-,FL>
129 1- - -- --- A
4 -- 1 - 2 1 - FIXL 0-----0 ----- - 1 - - - - -- 2 2- -----B00- 31 <FL,-,-,FL>
130 2- --- A

RESET DRIVER : EXECUTED
/RSTDRV : DISABLE

```

Tip

- On the T5593 and T5592, pin data information is displayed in two lines.
In this case, DRE, DRECLK, STRB, CPE, HZ, PCPE, EINV, ARID and EXP on T5592 are displayed as follows:
1st line: Displays the settings for odd-numbered DRECLK1 through DRECLK128 and STRB1 through STRB127.
2nd line: Displays the settings for odd-numbered DRECLK129 through DRECLK256 and STRB129 through STRB255.
In this case, DRE, DRECLK, STRB1, CPE, HZ, STRB2, CPE, HZ, PCPE, EINV and EXP on the 5593 are displayed as follows:
1st line: displays the settings of DRECLK1 through DRECLK128 and STRB1 through STRB128.
2nd line: displays the settings of DRECLK129 through DRECLK256 and STRB129 through STRB256.
- For the I/O pin in the system where STRB129 through STRB256 are available, the pin condition is always displayed in two lines. The STRB1, CPE, HZ, STRB2, CPE, HZ, PCPE, EINV, ARID and EXP displays have the following meanings.
1st line: The STRB1-128 setting is displayed.
2nd line: The STRB129 through STRB256 setting is displayed.

Example

```

. . . . A . . . . I S CH S CH MPEALDEECE . . . .
C . . . . L T PZ T PZ ACIROCXXPS
L R E R E TPNIALPPEM
K B B CEVDDP DD
1 2 H LL
YY
33 . . . . !1 . . . . - 1 1- 2 2- -----B--- . . . .
129 1- 130 2- --- A ← setting of STRB129-256

```

- When both *ACLK* and *STRB129-256* are set on the *T5591*, "!" is displayed to the left of the disabled clock number or strobe number. In the example above, *ACLK* is disabled.
- On the *T5586* or *T5585* used in 1 WAY, when both the odd-numbered phase strobe numbers of the *STRB129-STRB256* group and the even-numbered phase strobe numbers of the *STRB1-STRB128* group or *WSTRB1- WSTRB64* are set simultaneously, "!" is displayed to the left of each strobe number.
- When *HSCMAIN* is specified on the *T5593*, pin data information is displayed always in two lines. The first line displays *HSCMAIN*, and the second displays *HSCSUB*.

1. 2. 2 PDn

Uses a mnemonic display to show the connection state for PDn.

RESET DRIVER

Execution state of the RESET DRIVER statement immediately before the MEAS/START MPAT statement is executed, or of driver output level initialization when the RESET DRIVER ENABLE/DISABLE setting program (/RSTDRV) specifies ENABLE.

/RSTDRV

The RESET DRIVER ENABLE/DISABLE setting program (/RSTDRV) setting

◆ Example of a Mnemonic Display of Pins 1 to 3 (The OPEN Pin is Displayed as "OPEN")

When the same TG data is set in all banks.

Display example for the T5591

```
/PM PD1-3.┘

PD1  = IN1 ACLK1  FIXL  LINIT DRDLY0 DSS1 <FL>
PD2  = IN1 ACLK1  FIXL  LINIT DRDLY0 DSS1 <FL>
PD3  = IN1 ACLK1  FIXL  LINIT DRDLY0 DSS1 <FL>

RESET DRIVER : EXECUTED
/RSTDRV      : DISABLE
```

When different TG data is set for each bank.

On the T5593, T5592, T5591, T5586, T5585, T5581, T5581H, T5581P and T5571P, data for each bank is displayed.

Display example for the T5591

```
/PM PD1-3.┘

PD1  (BANK1) = IN1 ACLK1  FIXL  LINIT DRDLY0 BMUX DSS1 <FL>
      (BANK2) = IN1 ACLK2  FIXL  LINIT DRDLY0 BMUX DSS1 <FL>
PD2  (BANK1) = IN1 ACLK1  FIXL  LINIT DRDLY0 BMUX DSS1 <FL>
      (BANK2) = IN1 ACLK2  FIXL  LINIT DRDLY0 BMUX DSS1 <FL>
PD3  (BANK1) = IN1 ACLK1  FIXL  LINIT DRDLY0 BMUX DSS1 <FL>
      (BANK2) = IN1 ACLK2  FIXL  LINIT DRDLY0 BMUX DSS1 <FL>

RESET DRIVER : EXECUTED
/RSTDRV      : DISABLE
```



Tip

When "!" is displayed to the left of banks, the A, B, C and DRE clocks stored in them do not work.
When HSCMAIN is specified on the T5593, pin data information is displayed always in two lines.
The first line displays HSCMAIN, and the second displays HSCSUB.

1. 2. 3 PCn

Uses a mnemonic display to show the connection state for PCn.

◆ Display Example for the T5591

```
/PM PC33.↓  
  
PC33  = OUT1 [STRB1 CPE1] [STRB2 CPE2] EXPB EXPDLY0 CPEDLY0 DSS1 <FL,-,-,FL>
```

 Tip

On the T5593, T5592, T5591, T5586, T5585, T5581, T5581H, T5581P and T5571P, the PCn statement settings are displayed as pin data D.

The example shown below is for when PC33=OUT1, STRB1, <D0> are specified.

```
PC33=OUT1 [STRB1 CPE1] EXPB EXPDLY0 CPEDLY0 DSS1 <FL, -, -, D0>
```

1. 2. 4 VS_n

Displays the setting of VS_n.

◆ Display Examples

```
/PM VS1.↓  
  
VS1   =  1.000V R16V M4UA 400.0MA -400.0MA T39.00US
```

1. 2. 5 PPS_n

Displays the set value of PPS_n.

◆ Display Examples

Displays the voltage setting of the specified PPS for each active station.

```
/PM PPS33-40.↓  
  
STN1 PPS33 = 1.000V   PPS34 = 1.100V   PPS35 = 1.200V   PPS36 = 1.300V  
      PPS37 = 1.400V   PPS38 = 1.500V   PPS39 = 1.600V   PPS40 = 1.700V  
  
STN2 PPS33 = 1.700V   PPS34 = 1.600V   PPS35 = 1.500V   PPS36 = 1.300V  
      PPS37 = 1.300V   PPS38 = 1.200V   PPS39 = 1.100V   PPS40 = 1.000V
```

1. 2. 6 DC

Displays the setting of DC.

◆ Display Examples

```
/PM DC.␣
DC      =  OPEN
```

1. 2. 7 ISVM

Displays the setting of ISVM.

◆ Display Examples

```
/PM ISVM.␣
ISVM    =  8.000UA R8UA M16V
```

1. 2. 8 VSIM

Displays the setting of VSIM.

◆ Display Examples

```
/PM VSIM.␣
VSIM    =  10V  R16V  M8UA
```

1. 2. 9 INn

Displays the set value of INn or INHn.

◆ Display Examples

```
/PM IN1.␣
IN1     =  3.100V  1.100V
```

On the T5376, the driver output levels of INn and INHn are displayed as shown below.

```
/PM IN1.␣
IN1     =  3.100V  1.100V      INH1    =  10.00V  5.000V
```

1. 2.10 OUTn

Displays the set value of OUTn.

◆ Display Examples

```
/PM OUT1.↓
```

```
OUT1 = 2.200V 1.200V
```

1. 2.11 DCLP

Displays the set value of DCLP.

◆ Display Examples

```
/PM DCLP.↓
```

```
DCLP = 2.000V 0.000V
```

1. 2.12 VTn

Displays the set value of VTn.

◆ Display Examples

```
/PM VT1.↓
```

```
VT1 = -3.000V
```

1. 2.13 ILn

Displays the set value of ILn.

◆ Display Examples

```
/PM IL1.↓
```

```
IL1 = -10.00MA 10.00MA
```

1. 2.14 PCON

Displays the set value of PCON.

◆ Display Examples

```
/PM PCON.␣
```

```
PCON    = #FOF
```

1. 2.15 VCON

Displays the set value of VCON.

◆ Display Examples

```
/PM VCON.␣
```

```
VCON    = #FOF
```

1. 2.16 LCON

Displays the set value of LCON.

◆ Display Examples

```
/PM LCON.␣
```

```
LCON = 7
```

1. 2.17 RATE <m>

Displays the set value of TG RATE <m>.

<m> indicates a timing set number. When omitted, it is 1. The default value is 1 (applicable for all TG-related units given below).

◆ Display Examples

```
/PM RATE <1>.␣
```

```
RATE    = <1> 1.000000000MS
```


1. 2.18 ACLKn <m>

Displays the set value of TG ACLKn <m>.

◆ Display Examples

```
/PM ACLK1-3 <1>┐  
  
ACLK1 = <1> 1.100000000US  
ACLK2 = <1> OPEN  
ACLK3 = <1> OPEN
```

1. 2.19 BCLKn <m>

Displays the set value of TG BCLKn <m>.

◆ Display Examples

```
/PM BCLK1-3 <1>┐  
  
BCLK1 = <1> 1.100000000US  
BCLK2 = <1> OPEN  
BCLK3 = <1> OPEN
```

1. 2.20 CCLKn <m>

Displays the set value of TG CCLKn <m>.

◆ Display Examples

```
/PM CCLK1-3 <1>┐  
  
CCLK1 = <1> 1.100000000US  
CCLK2 = <1> OPEN  
CCLK3 = <1> OPEN
```

1. 2.21 STRBn <m>

Displays the set value of TG STRBn <m>.

◆ Display Examples

If an offset is specified using the SET OFFSET statement, the value after incorporating the offset value is displayed in parentheses.

```
/PM STRB1-2 <1>┘
```

```
STRB1 = <1> 50.000000000US ( 40.000000000US )
```

```
STRB2 = <1> 60.000000000US ( 50.000000000US )
```

↑
Value specified by STRB

↑
Value after operating the offset value

1. 2.22 WSTRBn <m>

Displays the setting value of the TG window comparison timing WSTRBn <m>.

◆ Display Examples

```
/PM WSTRB1 <1>┘
```

```
WSTRB1 = <1> 10.000000000US , 50.000000000US
```

1. 2.23 DRELn <m>

Displays the set value of TG DRELn <m>.

◆ Display Examples

```
/PM DREL1 <1>┘
```

```
DREL1 = <1> 10.000000NS
```

1. 2.24 DRETn <m>

Displays the set value of TG DRETn <m>.

◆ Display Examples

```
/PM DRET1 <1>┘
```

```
DRET1 = <1> 910.000000NS
```

1. 2.25 TGCLKn <m>

Displays the set value of TG TGCLKn <m>.

◆ Display Examples

```
/PM TGCLK1 <1>┘
```

```
TGCLK1 = <1> 10.000000NS
```

1. 2.26 PCLKn

Displays the set value of TG PCLKn.

◆ Display Examples

```
/PM PCLK1┘
```

```
PCLK1 = 11.000000NS
```

1. 2.27 REF

Displays the name of the reference-on unit.

◆ Display Examples

```
/PM REF┘
```

```
VS1 ... ON      IN1 ... ON      VT1 ... ON      IL1 ... ON
VS2 ... ON      IN2 ... ON      VT2 ... ON      IL2 ... ON
```

1. 2.28 TSn

Displays the set values of TG timing sets from n to n+3.

◆ Display Examples

/PM TS1.┘									
	TS1		TS2		TS3		TS4		
RATE	1.000MS		1.000MS		1.000MS		1.000MS		
ACLK1	100.0US		110.0US		120.0US		130.0US		
ACLK2	200.0US		210.0US		220.0US		230.0US		
ACLK3	300.0US		310.0US		320.0US		330.0US		
ACLK4	400.0US		410.0US		420.0US		430.0US		
B,CCLK1	100.0US	100.0US	110.0US	110.0US	120.0US	120.0US	130.0US	130.0US	
B,CCLK2	200.0US	200.0US	210.0US	210.0US	220.0US	220.0US	230.0US	230.0US	
B,CCLK3	300.0US	300.0US	310.0US	310.0US	320.0US	320.0US	330.0US	330.0US	
B,CCLK4	400.0US	400.0US	410.0US	410.0US	420.0US	420.0US	430.0US	430.0US	
B,CCLK5	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	
B,CCLK6	600.0US	0.000S	610.0US	0.000S	620.0US	0.000S	630.0US	0.000S	
B,CCLK7	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	
B,CCLK8	0.000S	800.0US	0.000S	810.0US	0.000S	820.0US	0.000S	830.0US	
DRE1	10.00US	910.0US	10.00US	910.0US	10.00US	910.0US	10.00US	910.0US	
STRB1	30.00US		40.00US		50.00US		60.00US		
STRB2	30.00US		40.00US		50.00US		60.00US		



Tip

When the /PM TG unit is specified, time data is displayed up to a 1/1000 ps digit.

Time data (ps) on the T5592 is displayed with an accuracy of seven decimal places.

Time data (ps) on the T5377, T5376 and T5375 is displayed with an accuracy of five decimal places.

Time data (ps) on the T5593 is displayed with an accuracy of eight decimal places.

1. 2.29 LIMIT

Displays the setting value of LIMIT DC, VS_n, or JUDGE.

◆ Display Examples

```
/PM LIMIT DC.↓
```

```
LIMIT DC = 50.00UA 40.00UA
```

```
/PM LIMIT VS1.↓
```

```
LIMIT VS1 = 50.00UA 40.00UA
```

```
/PM LIMIT JUDGE.↓
```

```
LIMIT JUDGE = 50.00UA 40.00UA
```

1. 2.30 MPAT

The specified pattern file name and SCRAM file name are displayed.

◆ Display Examples

```
/PM MPAT.↓
```

```
MPA UPMM0001  
[MAIN] SCR UPMC0001  
[SUB] SCR UPMC0001
```

1. 2.31 SETTLING

Displays the set values of SETTLING TIME.

◆ Display Examples

```
/PM SETTLING.↓
```

```
SETTLING TIME = 3.000MS 10.00MS
```

1. 2.32 Symbol Name

The pin conditions of the symbol-defined pins are displayed along with the symbol names.

◆ Display Examples

When ABCD1 through 3 are defined as pin symbol names in the socket program, these pins conditions are displayed as follows.

/PM ABCD1-34

	B	DG	I	A	B	C	P	M	LRSPSRUPD	IDDDD	D	O	T	V	I	S	CH	S	CH	MPEALDEECE	BD
	A	CN	N	C	C	C	C	O	EIDSMDTPR	ORRRR	R	U	E	T	L	T	PZ	T	PZ	ACIROCXXPS	MS
	N	D		L	L	L	L	D	VNMMCSLAD	CEBOE	E	T	R			R	E	R	E	TPNIALPPME	US
	K			K	K	K	K	E	EV	M	TL	NND	C	M		B		B	CEVDDP	DD	X
									L		Y	R	L	L		1		2	H	LL	YY
											Z	Y	K								
ABCD1	A	--	1	-	-	-	-	FIXL	0-----0	-----	-	-	-	-	-	--	-	-	-----	-1	<FL,-,-,FL>
ABCD2	A	--	1	-	-	-	-	FIXL	0-----0	-----	-	-	-	-	-	--	-	-	-----	-1	<FL,-,-,FL>
ABCD3	A	--	1	-	-	-	-	FIXL	0-----0	-----	-	-	-	-	-	--	-	-	-----	-1	<FL,-,-,FL>

```
RESET DRIVER : EXECUTED
/RSTDRV      : DISABLE
```

1. 2.33 Global Switch /C

The pin conditions of the symbol-defined pins are displayed along with the pin numbers. This switch can be used for symbol names only.

◆ Display Examples

The conditions for pins 1 through 3 that are defined with symbol names in the socket program are displayed along with the pin numbers.

/PM/C ABCD1-3

	B	DG	I	A	B	C	P	M	LRSPSRUPD	IDDDD	D	O	T	V	I	S	CH	S	CH	MPEALDEECE	BD		
	A	CN	N		C	C	C	O	EIDSM	TPR	ORRRR	R	U	E	T	L	T	PZ	T	PZ	ACIROCXXPS	MS	
	N	D		L	L	L	L	D	VNMMCSLAD	CEBOE	E	T	R				R	E	R	E	TPNIALPPPM	US	
	K			K	K	K	K	E	EV	M	TL	NND	C	M			B		B	CEVDDP	DD	X	
									L		Y	R	L	L			1		2	H	LL	YY	
												Z	Y	K									
1																							
2		A	--	1	-	-	-	-	FIXL	0	-----	0	-----	-	-	-	-	-	--	-	--	-----	-1 <FL,-,-,FL>
2		A	--	1	-	-	-	-	FIXL	0	-----	0	-----	-	-	-	-	-	--	-	--	-----	-1 <FL,-,-,FL>
3		A	--	1	-	-	-	-	FIXL	0	-----	0	-----	-	-	-	-	-	--	-	--	-----	-1 <FL,-,-,FL>

```
RESET DRIVER : EXECUTED
/RSTDRV      : DISABLE
```

1. 2.34 Global Switch /M

Displays the pin conditions using a symbol name.

This switch can be used for the Pn, PDn and PCn unit only.

◆ Display Examples

The conditions for the pins that are defined with symbols in the socket program are displayed along with the symbol names.

/PM/M P1-3,↓

```

      B DG I A B C P M      LRSFSRUPD IDDDD D O T V I S CH      S CH MPEALDEECE BD
      A CN N C C C C O      EIDSMDFPR ORRRR R U E T L T PZ      T PZ ACIROCXXPS MS
      N D L L L L D      VNMMCSLAD CEEOE E T R R E R E TPNIALPPEM US
      K K K K K E      EV M TL NND C M B B CEVDDP DD X
                        L Y R L L 1 2 H LL
                        Z Y K YY

ABCD1 A -- 1 - - - - FIXL 0-----0 ----- - - - - - - - - -1 <FL,-,-,FL>
ABCD2 A -- 1 - - - - FIXL 0-----0 ----- - - - - - - - - -1 <FL,-,-,FL>
ABCD3 A -- 1 - - - - FIXL 0-----0 ----- - - - - - - - - -1 <FL,-,-,FL>

```

```

RESET DRIVER : EXECUTED
/RSTDRV       : DISABLE

```

1. 2.35 Global Switch /R

Displays a pin condition and then displays the conditions of the unit connected to the displayed pin. This switch can be used for the Pn, PDn and PCn unit only.

◆ Display Examples

After displaying pin conditions, the states of the IN, OUT, VT, IL and REF units connected to pins are displayed.

Data on the TGs connected to pins is also displayed on the T5593, T5592, T5591, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5377, T5376, T5375 and T5371.

When PDn is specified, OUT, VT, IL and STRB are not displayed.

However, if a pin is specified as DVT on the T5593, VT of driver side is displayed.

When PCn is specified, IN, ACLK, BCLK, CCLK, DREL and DRET are not displayed.

Only reference-on units are displayed for REFERENCE ON.

When the same TG data is set in all banks:

When the 1WAY or 2WAY interleave mode is used on the T5591:

/PM/R P33.1

B	DG	I	A	B	C	P	M	LRSPSRUPD	IDDDD	D	O	T	V	I	S	CH	S	CH	MPEALDEECE	BD
A	CN	N	C	C	C	C	O	EIDSDTPT	ORRRR	R	U	E	T	L	T	PZ	T	PZ	ACIROCXXP	MS
N	D		L	L	L	L	D	VNMMC	SLAD	C	E	T	R		R	E	R	E	T	NIALPP
K			K	K	K	K	E	EV	M	TL	N	N	D		C	M	B	B	CEVDDP	DD
								L		Y	R	L	L				1	2	H	LL
										Z	Y	K								YY

```

33      A -- 1  1  1  1 - RZX    0-----0 -----  - - - - -  - - - - -  -1 <FL,-,-,FL>

```

IN = 3.100V 100.0MV OUT = 0.000V 0.000V VT(CP) = 0.000V VT(DR) = 0.000V
REFERENCE ON =

[illegible][illegible]

```
RESET DRIVER : EXECUTED
/RSTDRV      : DISABLE
```

When the 4WAY interleave mode is used on the T5591:

STRB2 and STRB130 are displayed as follows:

[illegible]

When the 1WAY or 2WAY interleave mode is used on the T5591:

When setting WSTRB, WSTRB1 and WSTRB65 (instead of STRB1, STRB2, STRB129 and STRB130) are displayed as shown below.

[illegible]

When the 1WAY interleave mode is used on the T5586, T5585, T5581, T5581H, T5581P or T5571P:

```

B D G I A B C P M L R S P S R U P D I D D D D D O T V I S C H S C H M P E A L D E E C E B D
A C N N C C C C C O E I D S M D T P R O R R R R R U E T L T P Z T P Z A C I R O C X X P S M S
N D L L L L L D V N M M C S L A D C E E O E E T R R E T P N I A L P P E M U S
K K K K K E E V M T L N N D C M B B C E V D D P D D X
L Y R L L 1 2 H L L
Z Y K Y Y

```

33 A -- 1 1 1 1 - XOR 1(1)-----0 ----- - - - - - - - - - - - - - - - - -1 <FL,-,-,FL>

IN = 3.100V 100.0MV OUT = 0.000V 0.000V VT = 0.000V IL = 0.000A 0.000A

REFERENCE ON =

	TS1	TS2	TS3	TS4	TS5	TS6	TS7	TS8
RATE	100.0US	200.0US	300.0US	400.0US	0.000S	0.000S	0.000S	0.000S
ACLK	11.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
BCLK	22.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
CCLK	33.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
DREL	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
DRET	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
STRB1	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
STRB2	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S

	TS9	TS10	TS11	TS12	TS13	TS14	TS15	TS16
RATE	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
ACLK	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
BCLK	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
CCLK	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
DREL	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
DRET	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
STRB1	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
STRB2	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S

RESET DRIVER : EXECUTED

/RSTDRV : DISABLE

When the 2WAY interleave mode is used on the T5586, T5585, T5581, T5581H, T5581P or T5571P:

STRB2 is displayed as follows:

[illegible]

When setting WSTRB on the T5586, T5585, T5581, T5581H, T5581P and T5571P
STRB1 and STRB are not displayed but WSTRB1 is as shown below.

WSTRB1	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S

On the T5371

DG	I	A	B	C	M	L	RSPSR	D	IDDD	D	D	O	T	V	I	S	CHMEALDE	E	C	C
CN	N	C	C	C	O	E	IDSMD	R	ORRR	R	R	U	E	T	L	T	PZAIROCX	X	P	Y
D		L	L	L	D	V	NMCS	D	CEEO	E	E	T	R			R	E2TNIALP	P	E	P
		K	K	K	E		E	V	M	L	NN	D	C	M		B	CVDDP	D	D	
						L				Y	R	L	L			H		L	L	
										Z	Y	K						Y	Y	
33	--	1	1	1	1	RZX	0(0)-----	0	----	-	-	1	-	-	-	2	1-----	0	0	1 <FL,-,-,FL>
IN = 3.100V 100.0MV OUT = 0.000V 0.000V VT = 0.000V IL = 0.000A 0.000A																				
REFERENCE ON =																				
		TS1	TS2	TS3	TS4	TS5	TS6	TS7	TS8											
RATE		100.0US	200.0US	300.0US	400.0US	0.000S	0.000S	0.000S	0.000S											
ACLK		11.00US	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S											
		0.00US	0.00US	0.00US	0.00US	0.00US	0.00US	0.00US	0.00US										←	*1
BCLK		22.00US	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S											
		0.00US	0.00US	0.00US	0.00US	0.00US	0.00US	0.00US	0.00US										←	*2
CCLK		33.00US	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S											
		0.00US	0.00US	0.00US	0.00US	0.00US	0.00US	0.00US	0.00US										←	*3
DREL		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S											
DRET		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S											
STRB2		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S										←	*4
		TS9	TS10	TS11	TS12	TS13	TS14	TS15	TS16											
RATE		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S											
ACLK		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S											
		0.00US	0.00US	0.00US	0.00US	0.00US	0.00US	0.00US	0.00US										←	*1
BCLK		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S											
		0.00US	0.00US	0.00US	0.00US	0.00US	0.00US	0.00US	0.00US										←	*2
CCLK		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S											
		0.00US	0.00US	0.00US	0.00US	0.00US	0.00US	0.00US	0.00US										←	*3
DREL		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S											
DRET		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S											
STRB2		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S										←	*4
RESET DRIVER : NOT EXECUTED																				
/RSTDRV : DISABLE																				

*1 When SDM is specified, ACLK data is displayed.

*2 When SDM is specified, BCLK data is displayed.

*3 When SDM is specified, CCLK data is displayed.

*4 The specified STRB number and data are displayed.

When different TG data is set for each bank.

On the T5593, T5592, T5591, T5586, T5585, T5581, T5581H, T5581P and T5571P, data for all banks are displayed.

On the T5581

```

B DG I A B C P M L RSPSRUPD IDDDD D O T V I S CH S CH MPEALDEECE BD
A CN N C C C C O E IDSMDTPR ORRRR R U E T L T PZ T PZ ACIROCXXPS MS
N D L L L L D V NMMCSLAD CEEOE E T R R E R E TPNIALPPEM US
K K K K K E E V M TL NND C M B B CEVDDP DD X
L Y R L L 1 2 H LL
Z Y K YY

33 1 -- 1 1 1 1 - RZX 0(0)-----0 ----- - - - - - - - - - - 31 <FL,-,-,FL>
2 -- 1 2 1 1 - RZX 0(0)-----0 ----- - - - - - - - - - - 31 <FL,-,-,FL>

IN = 3.100V 100.0MV OUT = 0.000V 0.000V VT = 0.000V IL = 0.000A 0.000A
REFERENCE ON =

BANK = 1
      TS1      TS2      TS3      TS4      TS5      TS6      TS7      TS8
RATE 100.0US 200.0US 300.0US 400.0US 0.000S 0.000S 0.000S 0.000S
ACLK 11.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
BCLK 22.000US 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
CCLK 33.000US 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
DREL 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
DRET 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
STRB1 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
STRB2 -----
      TS9      TS10     TS11     TS12     TS13     TS14     TS15     TS16
RATE 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
ACLK 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
BCLK 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
CCLK 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
DREL 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
DRET 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
STRB1 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
STRB2 -----

BANK = 2
      TS1      TS2      TS3      TS4      TS5      TS6      TS7      TS8
RATE 100.0US 200.0US 300.0US 400.0US 0.000S 0.000S 0.000S 0.000S
ACLK 15.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
BCLK 22.000US 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
CCLK 33.000US 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
DREL 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
DRET 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
STRB1 -----
STRB2 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
      TS9      TS10     TS11     TS12     TS13     TS14     TS15     TS16
RATE 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
ACLK 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
BCLK 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
CCLK 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
DREL 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
DRET 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S
STRB1 -----
STRB2 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S 0.000S

RESET DRIVER : EXECUTED
/RSTDRV      : DISABLE

```

On the T5592 and T5591

BANK1 through BANK2 are displayed in the 1-WAY mode and BANK1 through BANK4 are displayed in the 2-WAY and 4-WAY modes.

1. 2.36 Global Switch /I

Displays the VS_n settings, as well as the voltage setting for each PPS being used.
This switch can be used for the VS_n unit only.

◆ Display Examples

Displays the VS1 setting and then the voltage setting of the PPS assigned to VS1 for each DUT on the active station.

/PM/I VS1.↓					
VS1 = 1.000V R16V M4UA 400.0mA -400.0mA T39.00US					
DUT1	PPS33	1.000V	DUT9	PPS33	1.700V
DUT2	PPS34	1.100V	DUT10	PPS34	1.600V
DUT3	PPS35	1.200V	DUT11	PPS35	1.500V
DUT4	PPS36	1.300V	DUT12	PPS36	1.400V
DUT5	PPS37	1.400V	DUT13	PPS37	1.300V
DUT6	PPS38	1.500V	DUT14	PPS38	1.200V
DUT7	PPS39	1.600V	DUT15	PPS39	1.100V
DUT8	PPS40	1.700V	DUT16	PPS40	1.700V
↑		Voltage value of PPS _n	↑		Voltage value of PPS _n
DUT at the first station			DUT at the second station		

Tip

- For VS with a parallel drive, the minimum PPS number is displayed.
- Station 2 is displayed only in 2-STN parallel test.

1. 2.37 Global Switch /A

Displays all Cycle Palette data.
This switch can be used for the P_n, PD_n and PC_n unit only.

◆ Display Examples

- When Pn is specified.

```
/PM/A P33.↓
```

```

DG I A B C M L RSPSR D IDDD D D O T V I S CHMEALDE E C C
CN N C C C O E IDSMD R ORRR R R U E T L T PZAIROCX X P Y
D L L L D V NMMCS D CEE E E T R R E2TNIALP P E P
K K K E E V M L NN D C M B CVDDP D D
L Y R L L H L L
Z Y K Y Y

```

```

33 -- 1 1 1 1 RZX 0(0)----- 0 ---- - - 1 - - - 2 1----- 0 0 1 <D0,-,SD0>
2 <D1,-,SD1>
3 <D2,-,SD2>
4 <D3,-,SD3>
5 <D4,-,SD4>
6 <D5,-,SD5>
7 <D6,-,SD6>
8 <D7,-,SD7>
9 <D8,-,SD8>
10 <D9,-,SD9>
11 <D10,-,SD10>
12 <D11,-,SD11>
13 <D12,-,SD12>
14 <D13,-,SD13>
15 <D14,-,SD14>
16 <D15,-,SD15>

```

```

RESET DRIVER : EXECUTED
/RSTDRV      : DISABLE

```

- When PDn is specified.

```
/PM/A PD1.↓
```

```
PD1 = IN1 ACLK1 FIXL LINIT DRDLY0
```

```

CYP1 <C1> CYP2 <C2>
CYP3 <C3> CYP4 <C4>
CYP5 <C5> CYP6 <C6>
CYP7 <C7> CYP8 <C8>
CYP9 <C9> CYP10 <C10>
CYP11 <C11> CYP12 <C12>
CYP13 <C13> CYP14 <C14>
CYP15 <C15> CYP16 <C16>

```

```

RESET DRIVER : EXECUTED
/RSTDRV      : DISABLE

```

1. 2.38 Global Switch /B

Displays the driver level settings for each pin of all Child settings.

This switch can be used for Pn and PDn unit only.

◆ Display Examples

When standard driver pins (which are used for parallel tests on the T5377, T5376 and T5375) are specified, the following data is displayed.

/PM/B PD1↓

PD1 = IN1 FIXL LINIT(0) DRDLY0 DRE1 DREDLY0 CYP1 <FL>

STN1

A1	5.000V	1.000V	B1	5.000V	1.000V	C1	5.000V	1.000V	D1	5.000V	1.000V
A2	5.000V	1.000V	B2	5.000V	1.000V	C2	5.000V	1.000V	D2	5.000V	1.000V
A3	5.000V	1.000V	B3	5.000V	1.000V	C3	5.000V	1.000V	D3	5.000V	1.000V
A4	5.000V	1.000V	B4	5.000V	1.000V	C4	5.000V	1.000V	D4	5.000V	1.000V
A5	5.000V	1.000V	B5	5.000V	1.000V	C5	5.000V	1.000V	D5	5.000V	1.000V
A6	5.000V	1.000V	B6	5.000V	1.000V	C6	5.000V	1.000V	D6	5.000V	1.000V
A7	5.000V	1.000V	B7	5.000V	1.000V	C7	5.000V	1.000V	D7	5.000V	1.000V
A8	5.000V	1.000V	B8	5.000V	1.000V	C8	5.000V	1.000V	D8	5.000V	1.000V

E1	5.000V	1.000V	F1	5.000V	1.000V	G1	5.000V	1.000V	H1	5.000V	1.000V
E2	5.000V	1.000V	F2	5.000V	1.000V	G2	5.000V	1.000V	H2	5.000V	1.000V
E3	5.000V	1.000V	F3	5.000V	1.000V	G3	5.000V	1.000V	H3	5.000V	1.000V
E4	5.000V	1.000V	F4	5.000V	1.000V	G4	5.000V	1.000V	H4	5.000V	1.000V
E5	5.000V	1.000V	F5	5.000V	1.000V	G5	5.000V	1.000V	H5	5.000V	1.000V
E6	5.000V	1.000V	F6	5.000V	1.000V	G6	5.000V	1.000V	H6	5.000V	1.000V
E7	5.000V	1.000V	F7	5.000V	1.000V	G7	5.000V	1.000V	H7	5.000V	1.000V
E8	5.000V	1.000V	F8	5.000V	1.000V	G8	5.000V	1.000V	H8	5.000V	1.000V

STN2

A1	5.000V	1.000V	B1	5.000V	1.000V	C1	5.000V	1.000V	D1	5.000V	1.000V
A2	5.000V	1.000V	B2	5.000V	1.000V	C2	5.000V	1.000V	D2	5.000V	1.000V
A3	5.000V	1.000V	B3	5.000V	1.000V	C3	5.000V	1.000V	D3	5.000V	1.000V
A4	5.000V	1.000V	B4	5.000V	1.000V	C4	5.000V	1.000V	D4	5.000V	1.000V
A5	5.000V	1.000V	B5	5.000V	1.000V	C5	5.000V	1.000V	D5	5.000V	1.000V
A6	5.000V	1.000V	B6	5.000V	1.000V	C6	5.000V	1.000V	D6	5.000V	1.000V
A7	5.000V	1.000V	B7	5.000V	1.000V	C7	5.000V	1.000V	D7	5.000V	1.000V
A8	5.000V	1.000V	B8	5.000V	1.000V	C8	5.000V	1.000V	D8	5.000V	1.000V

E1	5.000V	1.000V	F1	5.000V	1.000V	G1	5.000V	1.000V	H1	5.000V	1.000V
E2	5.000V	1.000V	F2	5.000V	1.000V	G2	5.000V	1.000V	H2	5.000V	1.000V
E3	5.000V	1.000V	F3	5.000V	1.000V	G3	5.000V	1.000V	H3	5.000V	1.000V
E4	5.000V	1.000V	F4	5.000V	1.000V	G4	5.000V	1.000V	H4	5.000V	1.000V
E5	5.000V	1.000V	F5	5.000V	1.000V	G5	5.000V	1.000V	H5	5.000V	1.000V
E6	5.000V	1.000V	F6	5.000V	1.000V	G6	5.000V	1.000V	H6	5.000V	1.000V
E7	5.000V	1.000V	F7	5.000V	1.000V	G7	5.000V	1.000V	H7	5.000V	1.000V
E8	5.000V	1.000V	F8	5.000V	1.000V	G8	5.000V	1.000V	H8	5.000V	1.000V

RESET DRIVER : NOT EXECUTED
/RSTDRV : DISABLE

Tip

- The IN or INH (driver level) data set for the pin number "n," which is specified by Pn or PDn, is displayed.
- When no pins are specified, data for the pin for which the driver level has been set is displayed.
- For the LV (Level) driver pin of the T5377 or T5375, low level data is not displayed.

- When INH is selected for the HV(High Voltage) level driver pin of the T5376, low-level data is not displayed.
- For parallel tests using two test stations, data set for both Station 1 and Station 2 is displayed separately.

1. 2.39 When the Unit Name and Symbol Name are Omitted

The settings of each unit are displayed in the fixed format.

◆ Display Examples

Input /PM↓. The condition of the tester is displayed in fixed format as shown below.

This is explained using a display example for the T5581.

/PM↓

*** DRIVER LEVEL ***

IN1	=	3.100V	100.0MV
IN2	=	3.200V	200.0MV
IN3	=	3.300V	300.0MV
IN4	=	3.400V	400.0MV
IN5	=	3.500V	500.0MV
IN6	=	3.600V	600.0MV
IN7	=	3.700V	700.0MV
IN8	=	3.800V	800.0MV
IN9	=	3.900V	900.0MV
IN10	=	4.000V	1.000V
IN11	=	4.100V	1.100V
IN12	=	4.200V	1.200V
IN13	=	4.300V	1.300V
IN14	=	4.400V	1.400V
IN15	=	4.500V	1.500V
IN16	=	4.600V	1.600V

On the T5377, T5375, T5371, T5336 and T5311, if INH is selected for the driver, this is displayed with "INH." On the T5376, the driver output levels of INn and INHn are displayed as shown below.

/PM

*** DRIVER LEVEL ***

IN1	=	5.000V	1.000V	INH1	=	10.00V	5.000V
IN2	=	5.000V	1.000V	INH2	=	10.00V	5.000V
IN3	=	5.000V	1.000V	INH3	=	10.00V	5.000V
IN4	=	5.000V	1.000V	INH4	=	10.00V	5.000V
IN5	=	5.000V	1.000V	INH5	=	10.00V	5.000V
IN6	=	5.000V	1.000V	INH6	=	10.00V	5.000V
IN7	=	5.000V	1.000V	INH7	=	10.00V	5.000V
IN8	=	5.000V	1.000V	INH8	=	10.00V	5.000V
IN9	=	5.000V	1.000V	INH9	=	10.00V	5.000V
IN10	=	5.000V	1.000V	INH10	=	10.00V	5.000V
IN11	=	5.000V	1.000V	INH11	=	10.00V	5.000V
IN12	=	5.000V	1.000V	INH12	=	10.00V	5.000V
IN13	=	5.000V	1.000V	INH13	=	10.00V	5.000V
IN14	=	5.000V	1.000V	INH14	=	10.00V	5.000V
IN15	=	5.000V	1.000V	INH15	=	10.00V	5.000V
IN16	=	5.000V	1.000V	INH16	=	10.00V	5.000V

*** COMPARATOR LEVEL ***

OUT1	=	2.100V	1.100V	VT1	=	-3.000V	IL1	=	-10.00MA	10.00MA
OUT2	=	2.200V	1.200V	VT2	=	-3.000V	IL2	=	-11.00MA	11.00MA
OUT3	=	2.300V	1.300V	VT3	=	-2.500V	IL3	=	-12.00MA	12.00MA
OUT4	=	2.400V	1.400V	VT4	=	-2.000V	IL4	=	-13.00MA	13.00MA
OUT5	=	2.500V	1.500V	VT5	=	-1.500V	IL5	=	-14.00MA	14.00MA
OUT6	=	2.600V	1.600V	VT6	=	-1.000V	IL6	=	-15.00MA	15.00MA
OUT7	=	2.700V	1.700V	VT7	=	-500.0MV	IL7	=	-16.00MA	16.00MA
OUT8	=	2.800V	1.800V	VT8	=	0.000V	IL8	=	-18.00MA	17.00MA
OUT9	=	2.900V	1.900V	VT9	=	500.0MV	IL9	=	-18.00MA	18.00MA
OUT10	=	3.000V	2.000V	VT10	=	1.000V	IL10	=	-19.00MA	19.00MA
OUT11	=	3.100V	2.100V	VT11	=	1.500V	IL11	=	-20.00MA	20.00MA
OUT12	=	3.200V	2.200V	VT12	=	2.000V	IL12	=	-21.00MA	21.00MA
OUT13	=	3.300V	2.300V	VT13	=	2.500V	IL13	=	-22.00MA	22.00MA
OUT14	=	3.400V	2.400V	VT14	=	3.000V	IL14	=	-23.00MA	23.00MA
OUT15	=	3.500V	2.500V	VT15	=	3.500V	IL15	=	-24.00MA	24.00MA
OUT16	=	3.600V	2.600V	VT16	=	4.000V	IL16	=	-25.00MA	25.00MA

*** DYNAMIC CLAMP ***

DCLP = 8.480V -3.488V

On the T5593, T5592 and T5591, IL is not displayed.

```
*** VS UNIT ***

VS1 = 10.00V R10V M4UA 0.000A 0.000A
VS2 = -10.00V R10V M40UA 460.8MA -460.8MA
VS9 = 2.000V R8V M20UA 102.4MA -102.4MA
VS10 = -2.000V R8V M400UA 102.4MA -102.4MA
VS11 = 10.00V R16V M400UA 102.4MA -102.4MA
VS12 = 10.00V R16V M4MA 102.4MA -102.4MA

*** DC MEASUREMENT UNIT ***

VSIM = 1.000V R2V M8MA 11.20MA -11.20MA

*** PATH CON ***

PCON = 0

*** LOAD ***

LCON = 0

*** VS OPEN ***

VCON = 0

*** REFERENCE ***

VS1 ... ON      IN2 ... ON      VT3 ... ON      IL4 ... ON
DC ... ON
DCLP... ON
  ↑           ↑           ↑           ↑
Column of displaying VS, DC and DCLP  Column of displaying IN  Column of displaying VT  Column of displaying IL
----- STRIKE ANY KEY TO CONTINUE -----
```

When the display up to REFERENCE is ended, the VKT display pauses. Continue the display by pressing any key. If the VKT is selected as the output device using the S command, "---STRIKE ANY KEY TO CONTINUE---" is not displayed.

	TS1		TS2		TS3		TS4	
RATE	100.00US		200.00US		300.00US		400.00US	
ACLK1	11.00US		12.00US		13.00US		14.00US	
ACLK2	21.00US		22.00US		23.00US		24.00US	
ACLK3	31.00US		32.00US		33.00US		34.00US	
ACLK4	41.00US		42.00US		43.00US		44.00US	
B,CCLK1	12.00US	13.00US	14.00US	15.00US	16.00US	17.00US	18.00US	19.00US
B,CCLK2	22.00US	23.00US	24.00US	25.00US	26.00US	27.00US	28.00US	29.00US
B,CCLK3	32.00US	33.00US	34.00US	35.00US	36.00US	37.00US	38.00US	39.00US
B,CCLK4	42.00US	43.00US	44.00US	45.00US	46.00US	47.00US	48.00US	49.00US
B,CCLK5	52.00US	53.00US	54.00US	55.00US	56.00US	57.00US	58.00US	59.00US
B,CCLK6	62.00US	63.00US	64.00US	65.00US	66.00US	67.00US	68.00US	69.00US
B,CCLK7	0.000S	73.00US	0.000S	75.00US	0.000S	77.00US	0.000S	79.00US
B,CCLK8	0.000S	83.00US	0.000S	85.00US	0.000S	87.00US	0.000S	89.00US
DRE1	5.000US		4.000US		3.000US		2.000US	
DRE2	6.000US		7.000US		8.000US		9.000US	
STRB1	40.00US		45.00US		50.00US		55.00US	
STRB2	50.00US		55.00US		60.00US		65.00US	
STRB3	60.00US		65.00US		70.00US		75.00US	
STRB4	70.00US		75.00US		80.00US		85.00US	
WSTRB3	30.00US	50.00US	31.00US	51.00US	32.00US	52.00US	33.00US	53.00US
----- STRIKE ANY KEY TO CONTINUE -----								

On the T5593, T5592, T5591, T5586, T5585, T5581, T5581H, T5581P, and T5571P up to the maximum number of ACLK, BCLK, CCLK, DRE and STRB/WSTRB used are displayed.

BCLK and CCLK are displayed up to the bigger maximum number of either BCLK or CCLK.

On the system with PCLK, PCLK 1 to 4 are displayed following STRB/WSTRB.

On the T5336 and T5311, all of the parameters ACLK, BCLK, DRE and STRB/WSTRB that can be used are displayed.

On the T5377, T5376, T5375, T5371, all the ACLK, BCLK, CCLK and DRE numbers that are currently being used are displayed and all available STRB and WSTRB numbers are displayed.

The value after incorporation of the offset is displayed here.

/PM can display TG timing sets 1 through 4.

To display other timing sets, enter /PM TS_n.

2. *Data Set (/unit name)*

The data set utility is used to change the setting states for one or more tester units during test program execution.

```
/Unit name=Set value↓
```

The data set utility directly changes the contents of the specified unit register. Test program itself is not changed.

This utility is normally used to temporarily change unit settings when the tester stops in the MANUAL or HOLD mode or when it loops in the MANUAL, RECYCLE, or HOLD modes.

Limitation

- This utility changes data by executing a program compiled by XATL-51. Therefore, this utility cannot be used in the system where XATL-51 is not installed.
- This utility cannot be used when the directory has been changed by U command mode 7 (LIST&CHANGE CUSTOMER NAME).
(Error message "\$icdin.sys: No such file or directory" is displayed.)

The data scan command accepts the following unit names:

VS_n

Changes the setup value of VS_n

VSIM

Changes the setup value of VSIM.

ISVM

Changes the setup value of ISVM.

IN_n

Changes the setup value of IN_n.

INH_n

Changes the setup value of INH_n (HV driver). Cannot be specified on the T5500 series.

OUT_n

Changes the setup value of OUT_n.

IL_n

Changes the setup value of IL_n. Cannot be specified on the T5592 and T5591.

VT_n

Changes the setup value of VT_n.

DCLP

Changes the setup value of DCLP.

RATE

Changes the setup value of TG RATE.

Timing set must be specified as <m> to the right of the equal sign (=). The default value for <m> is 1. More than one timing set cannot be specified. The default value is 1 (applicable for all TG-related units given below).

ACLKn

Changes the setup value of TG ACLKn. Cannot be specified on the T5592.

BCLKn

Changes the setup value of TG BCLKn.

CCLKn

Changes the setup value of TG CCLKn.

STRBn

Changes the setup value of TG STRBn. Even numbers cannot be specified on the T5592.

WSTRBn

Changes the setup value of TG WSTRBn. Cannot be specified on the T5592.

DRELn

Changes the setup value of TG DRELn.

DRETn

Changes the setup value of TG DRETn.

TGCLKn

Changes the setup value of TG TGCLKn.

Pn

Changes the setup of driver pin n and comparator pin n.

PDn

Changes the setup of driver pin n.

PCn

Changes the setup of comparator pin n.

RON**ROF****SRON****SROF****LIMIT**

Changes the setup of LIMIT.

LIMITDC

Changes the setup of LIMIT DC.

LIMITVS_n

Changes the setup of LIMIT VS_n.

LIMITJUDGE

Changes the setup of LIMIT JUDGE.

SETTLING TIME

Changes the setup of SETTLING TIME.

TGTS

Changes the base value of the timing set at time of execution.

PC

Specifies the start address to execute the pattern program.

PCLK_n

Changes the setup value of PCLK_n. Can be specified on the T5591.

◆ **Example of Setting 5V for VS1**

```
/VS1 = 5V.↓
```

◆ **Example of Setting 5US for BCLK3 Timing Set 2**

```
/BCLK3 = <2>5US.↓
```

◆ **Example of Setting 5V for VS1 and VS2**

```
/VS1 = 5V & VS2 = 5V.↓
```

When unit names are delimited by & in data set as shown in example above, several units can be specified.

◆ **Example of Setting OPEN for Pins 1, 3 to 6, 9 and 15**

```
/P1, 3-6, 9, 15 = OPEN.↓
```

If the same condition is specified for several pins, these pins can be specified in one-data set command by using one or more commas (,) as shown above.

If these pins have consecutive pin numbers, the first pin is specified as n1 and the last pin is specified as n2, and these two pin numbers are connected with a "-."

Start and stop pin numbers may be in either ascending or descending order.

◆ Example of Changing Data Selector When Pn, PDn or PCn is Used

```
/PD1=IN1, RZO, BCLK1, CCLK1, <X0>.┘
```

— Tip —

When the specification of data selector is omitted, "FL" is set.

```
/PD1=IN1, RZO, BCLK1, CCLK1.┘
```

3. *Data scan (/DS)*

The data scan utility enables the setting of a tester unit to be changed in real time from the VKT keyboard.

```
/DS unit name=minimum value, maximum value, step value↵
```

Enter the above command from the VKT while the controller is operating.

The /DS command sets the initial data scan conditions, that is, the allowable range of set values for the specified unit.

After the specifications are made, each of the following inputs causes the operation listed with it:

↑

Increases the set value of the specified unit by the specified step value.
(This operation is performed by pressing the [SHIFT] and [6] keys.)

←

Decreases the set value of the specified unit by the specified step value.
(This operation is performed by pressing the [SHIFT] and [-] keys.)

?

Displays the initial data scan conditions.

The minimum and maximum values indicate the lower and upper limits between which the set value can be changed. The set value cannot exceed the lower or upper limit. If the set value is already smaller than the lower limit or greater than the upper limit, the set value is not changed.

The unit of data specified under initial data scan conditions must conform to that for the specified device.

For example, the unit for VS must be V and the unit for TG must be S.

The data scan command cannot specify more than one device name.

The data scan command accepts the following unit names:

VS_n

Specifies VS_n.

VSIM

Specifies VSIM.

ISVM

Specifies ISVM.

HIN_n

Specifies the High level of IN_n.

LIN_n

Specifies the Low level of IN_n.

HINHn

Specifies the High level of INHn. Cannot be specified on the T5500 series.

LINHn

Specifies the Low level of INHn. Cannot be specified on the T5500 series.

HOUTn

Specifies the High level of OUTn.

LOUTn

Specifies the Low level of OUTn.

VTn

Specifies VTn.

RATE

Specifies TG RATE.

Timing set must be specified to the right of the equal sign (=), as <m>. The default value for <m> is 1. The default value is 1 (applicable for all TG-related units given below).

ACLKn

Specifies TG ACLKn. These parameters cannot be specified on the T5593 and T5592.

BCLKn

Specifies TG BCLKn.

CCLKn

Specifies TG CCLKn.

STRBn

Specifies TG STRBn. Even numbers cannot be specified on the T5592.

WSTRBLn

Specifies the TG window comparison timing (leading). These parameters cannot be specified on the T5593 and T5592.

WSTRBTn

Specifies the TG window comparison timing (trailing). These parameters cannot be specified on the T5593 and T5592.

DRELn

Specifies TG DRELn.

DRETn

Specifies TG DRETn.

TGCLKn

Specifies TG TGCLKn.

HILn

Specifies the High level of ILn. These parameters cannot be specified on the T5593, T5592 and T5591.

LILn

Specifies the Low level of ILn. These parameters cannot be specified on the T5593, T5592 and T5591.

PCLKn

Specifies PCLKn. Can be specified on the T5591.

The data scan utility is usually used according to the following procedure:

The applications of the data scan utility are effective when waveforms are monitored using an oscilloscope.

Data Scan Procedure

- 1) Generate a loop during execution of the appropriate DC parametric or functional test item by using the control box. That is, turn on the ENABLE, MANUAL, RECYCLE and HOLD switches in the control box.
- 2) Enter **/DS** and specify unit to scan.

Enter "↑" using the [SHIFT] and [6] keys or "←" using the [SHIFT] key and the [-] key to increase or decrease the set value of the specified unit.
- 3) To execute the data scan utility for another unit, enter a **/DS** command again.

◆ Example of Data Scan

/DS HIN1=2V, 4V, 0.1V Sets the data scan conditions for the high level of IN1.

DSCAN CONDITION IS ... HIN1=2V, 4V, 0.1V ? Entering this statement displays the conditions.

/PM IN1↓ Current value of IN1

IN1=3.100V, 100.0MV

HIN1= 3.200V	}	Increases by 0.1 V by entering ↑ each time.
HIN1= 3.300V		
HIN1= 3.400V		
HIN1= 3.300V	}	Decreases by 0.1 V by entering ← each time.
HIN1= 3.200V		
HIN1= 3.100V		
HIN1= 3.000V		

/PM IN1↓ Current value of IN1

IN1=3.000V, 100.0MV

/PM RATE ACLK1↓

RATE=<1> 100 .0US

ACLK1=<1> 1.100US

Current value of RATE and ACLK1

/DS ACLK1=900NS, 1.3US, 100NS↓ Sets the data scan conditions for ACLK1.

ACLK1= 1.200US (100.0US)	}	Increases by ↑ Do not exceed the maximum value.
ACLK1= 1.300US (100.0US)		
ACLK1= 1.300US (100.0US)		
ACLK1= 1.300US (100.0US)		
ACLK1= 1.200US (100.0US)	}	Decreases by ←. Do not exceed the minimum value.
ACLK1= 1.100US (100.0US)		
ACLK1= 1.000US (100.0US)		
ACLK1= 900.0NS (100.0US)		
ACLK1= 900.0NS (100.0US)		
ACLK1= 900.0NS (100.0US)		

/ACLK1=99US↓ ↑ Value of RATE

/DS ACLK1=90US, 110US, 500NS↓ Changes the ACLK1 data scan conditions.

ACLK1= 99.50US (100.0US)	}	↑
ACLK1= 100.0US (100.0US)		
ACLK1= 100.5US (100.0US)		
ACLK1= 100.0US (100.0US)	}	←
ACLK1= 99.50US (100.0US)		
ACLK1= 99.00US (100.0US)		

 Tip

- When VSIM and ISVM are specified as a scan unit, the applied voltage and current ranges are displayed for each.

```
/DS VSIM = 4.5V, 5.5V, 0.1V┘
VSIM = 5.500V R8V
VSIM = 5.400V R8
```

└──────────┘ Applied voltage range

- If the set maximum and minimum values exceed the VSIM applied voltage range or the ISVM applied current range, the scanning range is set to conform to the tester hardware applied voltage/current ranges.

For example, assume that the following setting is made in the test plan program: SVM = -10UA, R80UA, M8, 1V, -4U At this time, if the operator scans in the following range. /DS ISVM = -100UA, -4UA, 2UA┘ the setting value -100 UA is invalid since the hardware range is set to ± 80 UA and scanning is only possible up to -80 UA.

- If the minimum value and the maximum value do not satisfy the following condition, message "DATA ERROR" is output. Re-enter values.

Minimum value < Maximum value

Example

```
/DS VS1=5V 4V -0.1V┘
DATA ERROR
```

- If VSn is specified as a scan unit during the bump test, the unit cannot be scanned.
 - Be sure to specify the increment to the value over the minimum resolution of each unit.
-

◆ Example of Data Scan in Parallel Test (8 DUTs Per Station)

```

/DS HOUT1=0V, 5V, 0.1V.◀ Sets the data scan conditions for the high level of OUT1.

DSCAN CONDITION IS ...◀ HOUT1=0V, 5V, 0.1V? Entering this statement displays the
                        condition.

/PM OUT1.◀ Current value of OUT1

OUT1      =2.600V, 0V

DATA-SCAN      <STN#=1>
                DUT1   DUT2   DUT3   DUT4   DUT5   DUT6   DUT7   DUT8
HOUT1= 2.700V  PASS   PASS   PASS   FAIL   PASS   PASS   PASS   PASS
HOUT1= 2.800V  PASS   PASS   PASS   FAIL   PASS   PASS   PASS   PASS
HOUT1= 2.900V  PASS   FAIL   FAIL   FAIL   PASS   PASS   PASS   PASS
HOUT1= 3.000V  FAIL   FAIL   FAIL   FAIL   FAIL   PASS   PASS   FAIL
HOUT1= 3.100V  FAIL   FAIL   FAIL   FAIL   FAIL   FAIL   FAIL   FAIL
HOUT1= 3.200V  FAIL   FAIL   FAIL   FAIL   FAIL   FAIL   FAIL   FAIL
HOUT1= 3.100V  FAIL   FAIL   FAIL   FAIL   FAIL   FAIL   FAIL   FAIL
HOUT1= 3.000V  FAIL   FAIL   FAIL   FAIL   FAIL   PASS   PASS   FAIL
HOUT1= 2.900V  PASS   FAIL   FAIL   FAIL   PASS   PASS   PASS   PASS
HOUT1= 2.800V  PASS   PASS   PASS   FAIL   PASS   PASS   PASS   PASS
HOUT1= 3.700V  PASS   PASS   PASS   FAIL   PASS   PASS   PASS   PASS
HOUT1= 3.600V  PASS   PASS   PASS   FAIL   PASS   PASS   PASS   PASS
HOUT1= 2.500V  PASS   PASS   PASS   PASS   PASS   PASS   PASS   PASS
HOUT1= 2.400V  PASS   PASS   PASS   PASS   PASS   PASS   PASS   PASS
HOUT1= 2.300V  PASS   PASS   PASS   PASS   PASS   PASS   PASS   PASS

/PM OUT1.◀ Current value of OUT1

```

4. *Debugger (/DEBUG)*

The debugger utility debugs user programs and has functions which enable the following:

- A breakpoint can be set on any line to interrupt program execution.
- The contents of variables used by the user program can be referenced or updated.
- The source program can be referenced.
- Statements can be deleted, modified, or added on any line. Any statement which can be used in a utility data set can be modified or added.
- The set values of tester units can be directly referenced or changed.

4. 1 Debugger Activation

The debugger can be activated by entering any of the three commands below from the VKT.

```
/DEBUG file name␣      ..... (a)
/DEBUG␣                ..... (b)
CTRL/D                 ..... (c)
```

CTRL/D in (c) indicated by indicates that the [CTRL] key and [D] key are pressed at the same time. When the debugger is activated, the debugger indicates % and waits for a command to be entered by the operator.

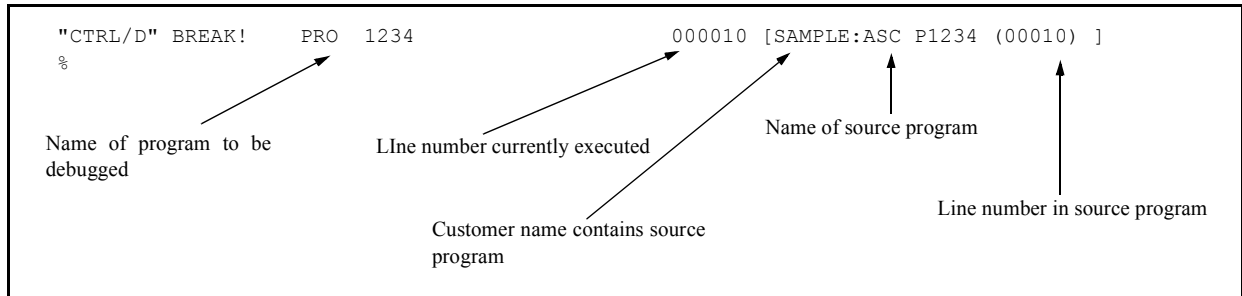
When the command indicated by (a) is entered, the program of the specified file name is called to be debugged.

When the command indicated by (b) or (c) is entered, the program being executed or the last executed program with the PRO attribute is called to be debugged.

```
/DEBUG PRO 1234
%
```

_____ The debugger waits for an operator command to be entered from the VKT.

When CTRL/D is entered, the debugger outputs the following message and waits for a command to be entered:



4. 2 Referencing or Changing the Name of the Program Being Debugged

Enter \$N to reference or change the name of the program being debugged.

%\$N PRO 1234	000010	[SAMPLE:ASC P1234 (00010)]	The program name is changed from PRO1234 to SUB ABC.
? SUB ABC.↵			
%\$N SUB ABC	000000	[SAMPLE:ASC ABC (00000)]	
? ↵			
%			

Program is not changed with only ↵ entered.

If a non-existent file is specified with a \$N command, files to be debugged are cleared.

The following message is output.

```
FILE DOES NOT EXIST : Filename
```

Tip

\$ indicates that the [ESC] key is to be pressed. This is applicable throughout this manual.

4. 3 Referring the Source Program

\$F

Displays the entire source program.

n/

Displays the line n only.

n</

Displays the source program from line n to the last line.

n1<n2/

Displays the lines n1 to n2.

./

Redisplays the line being displayed.

.± n/

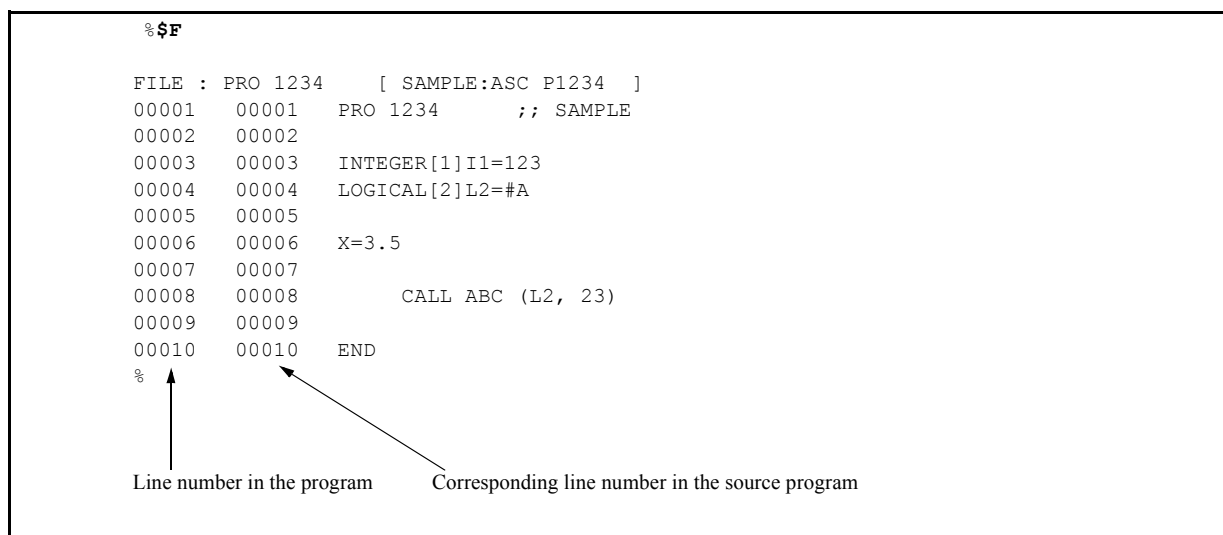
Displays the line n lines prior to or following the line being displayed.

<LF>

Displays the line immediately following the line being displayed.

↑

Displays the line immediately prior to the line being displayed.



In addition to line number, the debugger enable labels used by the program to be specified.

Tip

- To set the system to wait for the debugger command during source program reference, press any alphanumeric key. If the CTRL/C key is pressed, the program will stop. The space key can not be used.
- If 100 < n/, only the hundredth line is displayed.
- 0/ is the same as ./.

- $\uparrow$ (*up arrow*) stands for *[SHIFT]+[6] keys*.
-

4. 4 Breakpoints

When a breakpoint is set for a program, the program is interrupted there, and the debugger is activated.

n\$B

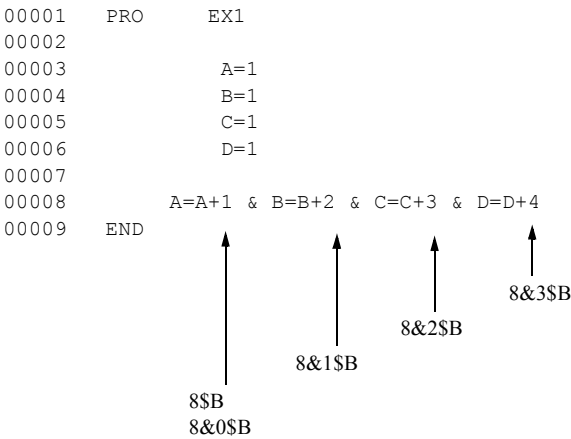
Sets a breakpoint at line n.

n&m\$B

Sets a breakpoint at position m in line number n.

Specifies a line where multiple statements are written as a breakpoint

Example



If variable name! line number\$B is entered, the specified variable name is automatically displayed when the program is interrupted at the breakpoint:

```
%6$B
%I1!8$B    Set breakpoints on lines 6 and 8.
%
```

If \$B is entered, the debugger displays breakpoints settings.

```
%$B
01B  PRO  1234      6 [ P1234 - 6 ]
02B  PRO  1234      8 [ P1234 - 8 ]    VN=I1
%
```

↑
Breakpoint number. When deleting the
breakpoint, this number is used.

↑
Indicates that variable number 1 is specified
at this breakpoint.

When referring to breakpoints in the source program, the * mark before the point shows the last point executed.

```

%$F
FILE : PRO 1234      [ SAMPLE:ASC P1234 ]
00001 00001  PRO 1234      ;; SAMPLE
00002 00002
00003 00003  INTEGER[1 ]I1=123
00004 00004  LOGICAL[2 ]L2=#A
00005 00005
00006 00006 01B X=3.5
00007 00007
00008 00008 02B      CALL ABC ( L2 , 23 )
00009 00009
00010 00010  END
%
```

Indicates that a breakpoint is set on line 6.

Indicates that a breakpoint is set on line 8.

Set the program in READY state (ready to start).

```

%$R
BREAK! 01  PRO 1234      6 [ P1234 - 6 ]
%$P
BREAK! 02  PRO 1234      8 [ P1234 - 8 ]
%
```

Breakpoint number. When the program is executed, the program is interrupted on line 6 and the debugger is activated.

Resumes execution of the program interrupted at the breakpoint.

The program is interrupted on line 8. The content of variable I1 is automatically displayed.

I1=123

Entering \$P resumes execution of the interrupted program. The program is interrupted again when the next breakpoint is detected.

Entering n\$P resumes execution of the program and ignores the next n breakpoints.

If a variable name is specified for one of the breakpoints above, the line number and the contents of the variable are displayed.

```

BREAK! 01  PRO 1234      6 [ P1234 - 6 ]
%1$P
PRO 1234  8 [ P1234 - 8 ] I1 = 123
```

Resumes program ignoring the following breakpoint one time after being interrupted on line 6.

Program is not interrupted but display is performed as variable number is specified.

Space

Displays CONTINUE and resumes execution of the interrupted program. The program is interrupted again when the next breakpoint is detected.

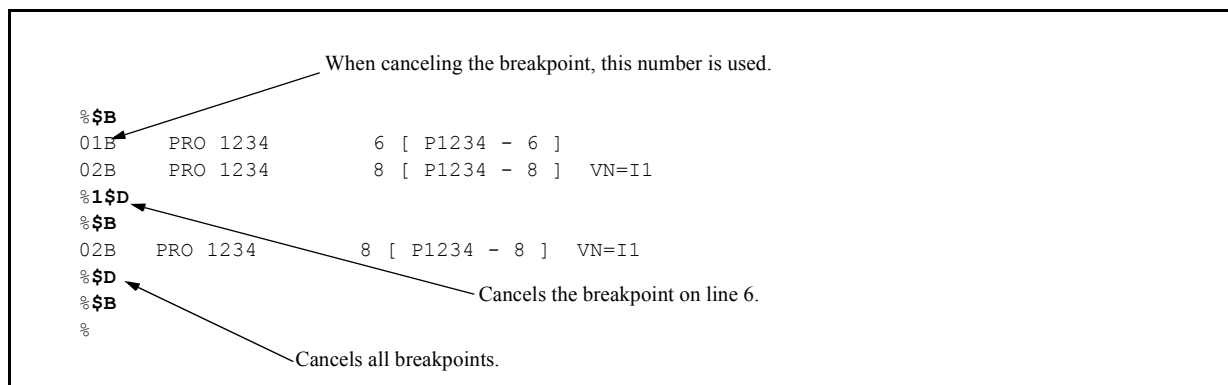
Operates similarly to \$P.

\$D

Cancels all breakpoints set currently.

n\$D

Cancels the nth breakpoint. Cancels the nth breakpoint. Note that this n is breakpoint number, not a line number.



Limitation

No breakpoint can be set between a GOSUB statement with an argument and the ARGUMENT statement.

If a breakpoint is set between these statements, the following message is displayed and the breakpoint is ignored: CALL POINT ABORTED

'E\$B

Sets an error break.

'E\$D

Cancels the error break

The \$D command can also be used to cancel an error break.



Tip

An error break can also be reset using the u command, one of the controller commands.

However, if the error break was reset using the u command instead of 'E\$D or \$D, and then the debugger is started, the error break setting is restored.

```

"CTRL/D" BREAK! PRO 345          000004          [ SAMPLE:ASC P345 (00004) ]
%$F

FILE : PRO 345          [ SAMPLE:ASC P345 ]
00001 00001 PRO 345
00002 00002
00003 00003 X=0
00004 00004 Y=X+5V ← The program generates an error on line 4.
00005 00005
00006 00006 END
%$R

RUN-TIME ERROR !      CODE=3156      (000079EB)
PRO      - 345
TEST     - 0
LIME1    - 4      [ P345-1-4 ]
LIME2 (&) - 0
0000EE00 IM.CODE=5
01500000 00000000 0000C859 00000015
00000004 00940000 00007A5F 000079E4

"CTRL/D" BREAK! PRO 345 000004 [ SAMPLE:ASC P345 (00004) ]
%'E$B ← Sets an error break.
%$B
ERROR BREAK ← Indicates that an error break has been set
%$R

ERROR BREAK! CODE : 3156 PRO 345 000004 [SAMPLE:ASC P345 (00004) ]
%'E$D ← Resets
%$B      an error break
%$R
      ↑
      If an error occurs, this message is displayed and the debugger is activated.

RUN-TIME ERROR      CODE=3156      ( 000079EB )
PRO      - 345
TEST     - 0
LIME1    - 4      [P345-1-4]
LIME2 (&) - 0
0000EE00 IM.CODE=5
01500000 00000000 0000C859 00000015
00000004 00940000 00007A5F 000079E4

```

4. 5 Referring or Updating the Variable Contents

\$L

Displays the contents of all variables used by the program.

The screenshot shows the output of the `%$L` command. It lists variables with their names, types, sizes, and contents. Annotations point to specific parts of the output:

- Variable name:** Points to the variable name 'X'.
- Variable type:** Points to the variable type '*R'.
- Variable size:** Points to the variable size '[3]'.
- Variable contents:** Points to the variable contents '0.3500E+01'.

An asterisk ("*") indicates that this variable number is non-specific type (that is, a GENERAL declaration is made or no type declaration is made for the variable.)

Variable type. When a variable is a non-specific type, the type of the variable is determined by the loaded data.

Variable type	Meaning	Variable size
I	INTEGER	Word number
L	LOGICAL	Word number
R	REAL	Word number
C	CHARACTER	Word number
S	STRING	Word number
DS	DSTRING	Word number

When A is indicated at the end of the contents of a variable, it indicates that the variable is an argument.

The size of any string variable is indicated by using the `$X` or `1$X` command.

If the value passed from the main program is a constant in the subroutines, `CONSTANT` is displayed as follows:

The screenshot shows the output of the `%$L` command for variables ARG1, ARG2, and AC. ARG1 and ARG2 are constants, and AC is an integer.

```

%$L
ARG1      CONSTANT      2
ARG2      CONSTANT      8
AC        *I [3]        6
  
```

The data display mode can be changed by using `$M`.

1\$X

Displays the relative addresses of all variables.

The execution-time address of a variable is its relative address plus its offset address which is determined during program execution.

1\$X									
I1	700		I	[1]	#8088	#0000	#02BC		
L2	701		L	[2]	#808D	#0000	#02BD		
X	703			[3]	#8082	#0000	#02BF		
1\$X									
I1	0	\$	I	[1]	#8088	#0000	#0000		
L2	1	\$	L	[2]	#808D	#0000	#0001		
X	3	\$		[3]	#8082	#0000	#0003		
%									

Variable address

A dollar sign (\$) indicates that the variable address is a relative address.

A blank indicates that the variable is a non-specific type.

Indicates an internal code of a variable without conversion.

Variable name (space)

Displays and changes the contents of the specified variable.

Variable name

Displays the execution-time address of the specified variable.

Variable name\$

Displays the relative address of the specified address. In this case, the \$ key instead of the ESC key is pressed.

The variable contents displayed depend on the last mode specified with \$M.

1\$X	*R	[1]	0.3500E+01	#2F
1\$X	*L	[2]	0000002F	12
1\$X	*L	[3]	12	
%				

Unchanged

Changed to decimal 12.

Changed to hexadecimal 2F.

The type of a non-specific type of variable is automatically switched according to the loaded data.

The return key is usually entered as a terminator. <LF> and ↑ can also be used.

Enter <LF> to display the contents of the next variable.

Enter ↑ to display the contents of the previous variable.

%I1	I [1]	123<LF>
L2	L [2]	000000A<LF>
X	*I [3]	12↑
L2	L [6]	0000000A↑
I1	I [1]	123↓
%		

The variables are displayed in the same order as in the program.

4. 6 Adding, Modifying, or Deleting Statements

Line number\$I

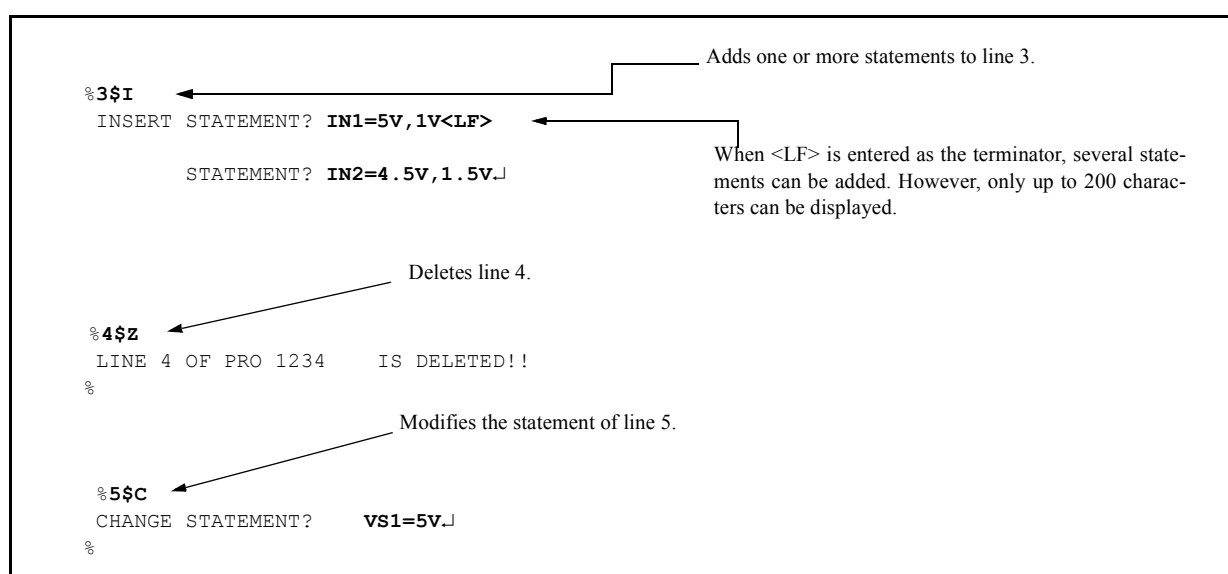
Adds one or more statements to a specified line.

Line number\$C

Replaces a statement on a specified line with another statement.

Line number\$Z

Deletes a statement from a specified line.



A location at which a statement is added, modified, or deleted is called a call point.

(A breakpoint is one type of callpoint.)

Statement addition, modification, or deletion does not have any effect on the source program.

That is, the source program does not contain any information on the changes that occurred. To confirm what statements have been added, modified, or deleted, the following command can be used:

\$O

Displays what statements have been added, modified, or deleted.

```

%$O
*****
001      PRO 1234                      3      .... INSERT [ SYS CPTABL01 ]
IN1= 5V, 1V
IN2= 4.5V, 1.5V
*****
002      PRO 1234                      4      .... DELETE [ SYS CPTABL02 ]
*****
003      PRO 1234                      5      .... CHANGE [ SYS CPTABL03 ]
VS1= 5V
%

```

Callpoint number.
When resetting the callpoint, this number is used.

Added or changed statement

Line number

Distinguish of addition and change, deletion

File name that has changed data in

```

%$F
FILE : PRO 1234      [ SAMPLE:ASC P1234 ]
00001  00001      PRO 1234      ;; SAMPLE
00002  00002
00003  00003  01I  IN1=5V, 1V
                     IN2=4.5V, 1.5V
... ORIGINAL ...>
00004  00004  02Z  LOGICAL [1] L2=#A
00005  00005  03I  VS1=5V
... ORIGINAL ...>
00006  00006      X=3.5
00007  00007
00008  00008      CALL ABC (L2, 23)
00009  00009
00010  00010      END
%

```

Indicates that line 5 has been modified to VS= 5V.

Indicates that line 4 has been deleted.

Indicates that statements IN1 and IN2 have been added to line 3.

n\$K

Cancels the nth callpoint.

\$K

Cancels all callpoints. (No break point is canceled.)

```

%2$K ← Cancels the second call point
%$O
*****
001  PRO 1234          3    .... INSERT  [ SYS CPTABL01 ]
IN1=5V, 1V
IN2=4.5V, 1.5V
*****
003  PRO 1234          5    .... CHANGE  [ SYS CPTABL03 ]
VS1=5V
%

%$K ← Cancels all call points.
%$O
%
```

All statements can be deleted. However, only a limited number of statements can be added or modified.

Statements which can be specified for addition or modification include:

- VSn
- VSIM
- ISVM
- INn
- OUTn
- VTn
- RATE
- ACLKn
- BCLKn
- CCLKn
- STRBn
- DRELn
- DRETn
- TGCLKn
- Pn
- RON
- ROF

These 17 statements are tester-related statements.

They can be written in the same format as in a source program, but they cannot use any variable.

The TG-related statements cannot specify more than one timing set. These conventions are the same as those applying to the data set utility.

In addition to the statements above, the following statements can be specified for addition or modification:

- CALL file name (argument)
- STOP "Character string" , ? Characterstring?, /
- PAUSE "Character string" , ? Characterstring?, /
- GOTO LINE . ± n
- SEND EXT (address) data
- WRITE "Character string", ? Characterstring?, /

These statements also cannot use any variable.

The WRITE statement cannot specify any output unit.

4. 7 Referencing or Changing the Set Values of Tester Units

Unit name.↵

Displays the set value of the specified unit.

These commands accept all specifications which can be made using the utility pin monitor.

Unit name=specified value.↵

Directly changes the set value of the specified unit to the specified value.

This command has the same function as the data set utility command.

n\$N

n is a nesting level number.

```
%$N    SUB   ABC           000005    [SAMPLE:ASC ABC (00005)]  
?↵  
%0$N  
%$N    PRO   1234          000008    [SAMPLE:ASC P1234 (00008)]  
?↵  
%
```

Indicates PRO 1234 by specifying nesting level number 0.

```
%%$s  
LEVEL-00 ** PRO 1234      00008 & 00  
LEVEL-01   SUB ABC       00005 & 00  
LEVEL-02   LIB DEBUG     01117 & 00  
LEVEL-03   LIB STACKTYP  00037 & 00  
%
```

Indicates that the program to be debugged was changed to PRO 1234.

4. 9 Displaying Information on Programs in the User Area

\$Q

Displays the programs which are currently contained in the user area.

The screenshot shows the output of the \$Q command. It is divided into two main sections. The top section, titled '*** USER AREA INFORMATION ***', provides summary statistics: TOTAL (198828), LEFT (161294), USED (37534), *V-AREA (1690), and *U-AREA (35844), each with its corresponding hexadecimal range. The bottom section lists five programs: PRO 1234, SUB ABC, LIB DEBUG, SYS DEBUG.BF, and LIB CDIRTYPE. To the right of these program names is a detailed list of memory addresses and directory names. Annotations with arrows point to specific parts of the output: a '%' symbol indicates a protected file; an arrow points to the 'Byte number' column; another points to the 'Word number' column; a third points to a 'Memory address'; and a fourth points to a 'Directory name'.

```

%$Q
*** USER AREA INFORMATION ***
TOTAL      198828      (**99414)      [#00E7A9-#026BFF]
LEFT       161294      (**80647)      [#00EAF6-#0225FD]
USED       37534       (**18767)
*V-AREA    1690        (***845)      [#00E7A9-#00EAF6]
*U-AREA    35844       (**17922)      [#0225FD-#026BFF]

1 PRO 1234      184 #00026BA3 #000000B8 #00000044 ADI$DP1
2 SUB ABC      178 #00026B4A #000000B2 #00000044 ADI$DP1
3 LIB DEBUG    31282 #00022E31 #00007A32 #00000840 DP0
4 SYS DEBUG.BF 12 #00022E2B #0000000C #00000000 DP0
5 LIB CDIRTYPE 4188 #000225FD #0000105C #000001F6 DP0
%
  
```

Annotations in the diagram:

- % Indicates that the file in memory is protected.
- Byte number
- Word number
- Memory address
- Directory name

TOTAL

Size of the entire user area

LEFT

Total size of free areas in the user area

USED

Total size of area currently used in the user area

V-AREA

Total size of variable areas

U-AREA

Total size of program areas

LIB DEBUG

Debugger main program

SYS DEBUG.BF

Debugger work file

LIB CDIRTYPE

Debugger utility that processes the SQ command

4.10 Specifying the Output Unit to be Used by the Debugger

\$V

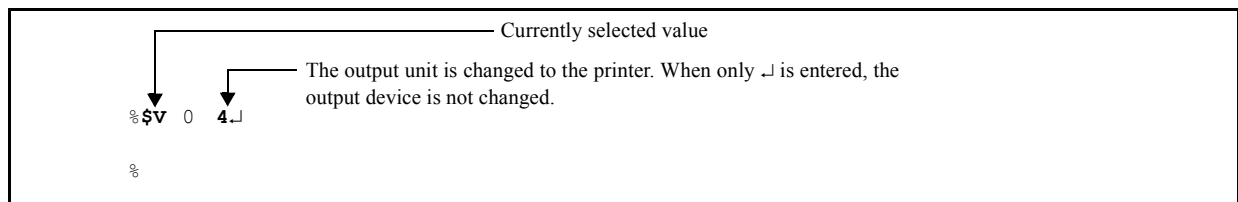
Specifies the output device using the numbers 0, 1, 2, 3, and 4.

0,2,3 are to VKT and 4 is to LPT.

1 selects the output device specified by key command S.

The output unit specified by this command is used for the following commands:

- \$F
- n</
- n1<n2/
- \$L
- \$X
- \$U



4.11 Tracing the Program

\$T
Sets trace mode in which the program being executed is traced.

Trace mode

- 0 Does not trace the program.
- 1 Traces the program.
- 2 Traces the program, pausing on each line.

BREAK! 01 PRO 1234 2 [P1234 - 2]

%\$T 0 1J ← Set trace mode to the debugger.

%\$P

PRO -1234 3 [P1234-1-3]

PRO -1234 4 [P1234-1-4] Indicates the execution state of the program.

PRO -1234 5 [P1234-1-5]

PRO -1234 6 [P1234-1-6]

PRO -1234 7 [P1234-1-7]

PRO -1234 8 [P1234-1-8]

SUBR -ABC PRO -1234 2 [ABC-1-2]

SUBR -ABC PRO -1234 3 [ABC-1-3]

SUBR -ABC PRO -1234 4 [ABC-1-4]

BREAK! 02 SUB ABC 4 [ABC - 4]

%

Indicates that the subroutine ABC is called by PRO 1234.

$n_1 < n_2$ \$W
Traces the program from line n_1 to n_2 .

Variable name! $n_1 < n_2$ \$W
Traces the program from line n_1 to line n_2 and displays the contents of the specified variable.

\$W
Displays the execution state of the program within the specified trace ranges. Up to four trace ranges can be specified.

\$Y
Cancels all trace specifications.

n\$Y

Cancels the nth trace specification.

```
%3<5$W
%$N PRO 1234 000011 [ T3340:ASC P1234 ( 00011 ) ]? SUB ABC
%ARG1! 3<4$W
%A! 4<4$W
%$W
1T FROM PRO 1234 3
   TO PRO 1234 5
2T ARG1
   FROM SUB ABC 3
   TO SUB ABC 4
3T A
   FROM SUB ABC 4
   TO SUB ABC 4
%$R

/PRO 1234
PRO - 1234 3 ( P1234-1-3 )
PRO - 1234 4 ( P1234-1-4 )
PRO - 1234 5 ( P1234-1-5 )
SUB ABC 3 ARG1=#0000000A
SUB ABC 4 ARG1=#0000000A A=33
ARG1+ARG2= 33
```

When a variable is specified, the source program name is not displayed.

\$J

Displays the program lines which are executed before the debugger is activated (LINE HISTORY function). (LINE HISTORY)

The LINE HISTORY function is effective when the debugger sets a breakpoint or when 1\$J is specified.

```
%%$J
EXECUTED LINE HISTORY
-10 5
- 9 6
- 8 7
- 7 8
- 6 2
- 5 3
- 4 4
- 3 5
- 2 6
- 1 9
0 10
```

← Number of the line immediately prior to the line being executed

← Number of the line being executed

The LINE HISTORY function can display up to 10 lines. Because no file name is displayed, when a CALL statement is being executed, this function must be used carefully.

n\$J

Enables or disables the LINE HISTORY function. n=0 DISABLE n=1 ENABLE

n=0 DISABLE

n=1 ENABLE

4.12 Mathematical Calculation

, = the answer

The calculation to be performed is entered where "" appears; the answer appears when "=" is pressed.

```
% ,15+3-4*2+10/5= 12.↵
%
```

\$H

Enables the answer for the entered expression to be changed from the current notational system to another.

The notational systems which can be specified are octal, decimal and hexadecimal. The decimal notational system is the default.

```
%$H      0      8.↵      Selects octal notation.
% ,65= $101 ↵
%$H      8      16.↵     Selects hexadecimal notation.
% ,65= #41 ↵
%
```

Tip

If a notational system other than octal or hexadecimal is specified, the decimal notational system is used.

When # or \$ is entered, the answer can be converted into the appropriate radix without using the \$H command.

↑ ↑ ↑ ↑ ↑ ↑
 % ,65 = 65 #41 \$101 'A & 32 1 ↵
 Output by radix of \$H \$ indicates the value in octal notation. The answer and the remaining of data divided by 2 is displayed when & is entered.
 # indicates the value in hexadecimal notation. "'": Following character is displayed as character.

Limitation

Neither parentheses nor functions can be used in these mathematical expressions.

If real numbers are used in an expression, and octal or hexadecimal notation is selected, the answer is displayed as E-type data.

```
% ,10.0+5.5= 15.50000 # 0.1550000E+02 $0.1550000E+02 ↵  
%  
%$H 0 16↵  
% ,10.5+5= #0.1550000E+02 ↵
```


4.13 Canceling All Debug Modes

\$A

Cancels all debug modes including breakpoint, callpoint and trace modes.

%**\$A**

DEBUG MODES ARE INITIALIZED!

CLEARED BREAK POINT

CLEARED CALL POINT

CLEARED TRACE TABLE

\$ H 0

\$ M 0

\$ T 0

\$ V 0

%

4.14 Returning from the Debugger

\$P

Resumes execution of the program interrupted by a breakpoint.

n\$P

Has the same function as \$P, but ignores the next n breakpoints.

\$R

Sets the program to a start wait (READY) state.

n\$R

Effective when the program specified by \$N has the PRO attribute or is interrupted. This command starts or resumes execution of the program from line n.

4.15 Other Commands

\$U

Displays the names of the labels used by the program and the corresponding line numbers.

n↵

Displays the contents of tester address n.

n₁<# n₂ (# n₃)↵

Displays the contents of tester memory from address n₁ to address n₂ in increments of n₃.

n (# n₂) /

Displays the contents of tester memory from address n₁ to the last address in increments of n₂.

/↵

Enables the specified utility to be directly used during debugger execution.

The utility command name is entered where appears.

4.16 Debugger Commands

**Tip**

\$ indicates the ESC key.

\$A

Cancels all debug modes including breakpoint, callpoint and trace modes.

\$B

Displays information on the currently set breakpoints.

n\$B

Sets a breakpoint on line n.

'E\$B

Sets an error break.

n\$C

Replaces the statement of line n with the specified statement. (Callpoint)

\$D

Cancels all breakpoints set currently.

n\$D

Cancels the nth breakpoint.

'E\$D

Cancels the error break

\$F

Displays the source program in a list format.

\$H

Specifies the notational system for answers in the mathematical calculation mode.

n\$I

Inserts the specified statement in line n. (Callpoint)

\$J

Displays LINE HISTORY information.

0\$J

Disables the LINE HISTORY function.

1\$J

Enables the LINE HISTORY function.

\$K

Cancels all callpoints.

n\$K

Cancels the nth callpoints.

\$L

Displays the names and contents of all variables.

\$M

Sets the display mode in which variable contents are to be displayed. (0 to 7)

\$N

Specifies the name of the program to be debugged.

n\$N

Indicates that the program of the specified nesting level n is to be debugged.

\$O

Displays information on the currently set callpoints.

\$P

Resumes execution of a program interrupted at a breakpoint.

n\$P

Resumes execution of the program interrupted at a breakpoint and ignores the next n breakpoints.

\$Q

Displays the usage state of the user area.

\$R

Returns from the debugger and sets the program to a start wait state.

n\$R

Starts a program to be debugged from the nth line when it has the PRO attribute.

\$S

Displays the contents of the stack.

\$T

Sets the debugger to trace mode.

\$U

Displays the names of the labels used by the program and the corresponding line numbers.

\$V

Specifies the output device to be used by the debugger.

\$W

Displays the currently set trace ranges.

n₁<n₂\$W

Sets a trace range.

\$X

Displays the names and execution-time addresses of all variables used by the program.

1\$X

Displays the names and relative addresses of all variables used by the program.

\$Y

Cancels all trace ranges.

n\$Y

Cancels the nth trace range.

n\$Z

Deletes the statement of line n. (Callpoint)

n/

Displays line n of the source program.

n</

Displays the source program from line n to the last line.

n₁< n₂/

Displays the source program from line n₁ to line n₂.

xxx_

Displays the contents of the variable xxx.

xxx:

Displays the execution-time address of the variable xxx.

xxx\$

Displays the relative address of the variable xxx.

 **Tip**

\$ indicates the [\$] key, not the [ESC] key, in this case.

yyy↵

Displays the set value of the tester device yyy.

yyy=data↵

Directly sets the specified data for the tester device yyy.

#n↵

Displays the contents of tester address n.

n₁< # n₂ (# n₃)↵

Displays the contents of tester memory from address n₁ to address n₂ in increments of n₃.

n₁(# n₂)/

Displays the contents of tester memory from address n₁ to the last address in increments of n₂.

/.....↵

Enables use of the utility command without changes.

5. *Two-Dimensional Shmoo Plot (SHM2)*

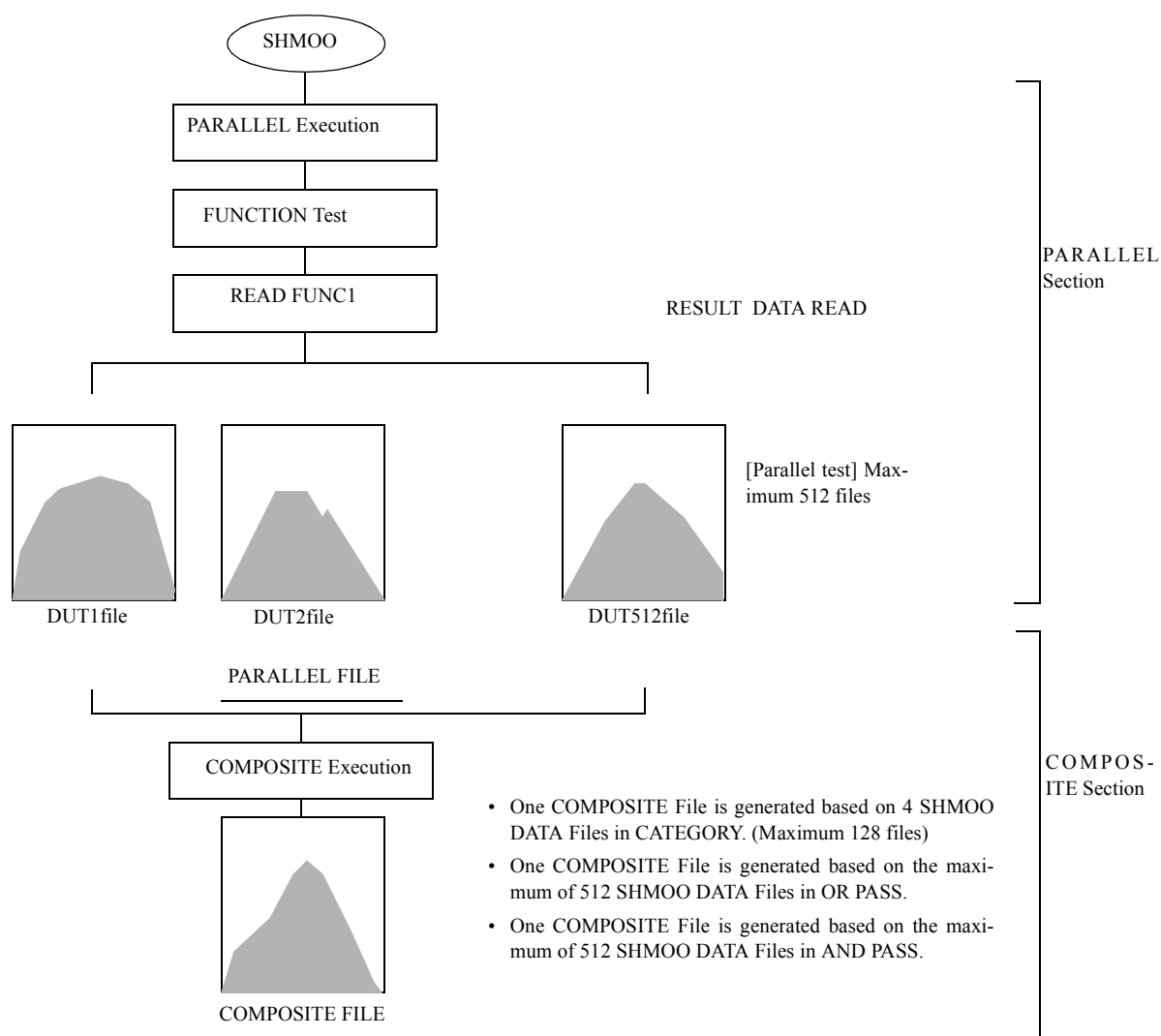
This section describes the two-dimensional shmoo plot (SHM2).

5. 1 SHM2 Overview

- The two-dimensional shmoo plot utility outputs two-dimensional shmoo plots, changing the X axis and Y axis parameters.
The user can set the X and Y axes, specify shmoo operation mode, and specify or change comments easily using the VKT in conversational mode.
Prepared shmoo conditions can be saved into a file. Main functions are as follows:
 - X, Y and T shmoo conditions can be changed in the interactive mode.
This feature reduces the time required to prepare shmoo conditions and allows quicker testing of memory devices.
 - All the current/voltage and timing settings saved before shmoo execution can be restored after execution.
 - A number of conditions can be specified for the X and Y axes, facilitating the preparation of more effective reports.
 - The skip function is available during shmoo operation.
 - The MODIFY function is available for the X axis.
 - Test pattern names and start PC can be changed in the utility.
 - Prepared shmoo conditions can be saved to a disk as a file for future use and restored by specifying the file name.
 - Calibration can be performed when HIN or LIN is specified as the X, Y, or T parameter.
 - Separate tracking conditions can be specified.
 - The original data of the parameters set by the test plan program are indicated by arrows on the X and Y axes.
 - ROF can be executed within the shmoo. It is used when a device is replaced.
- Prepare the report of V-I characteristics by executing the DC parametric test for the untested device and plotting the result.
- Output the shmoo plot, increasing the size of the X and Y axes using logarithmic scale to compact the wide range of operating test.
- There are two ways to use the shmoo program:
 - A shmoo for a test item is independently plotted from the keyboard instead of writing it in the test plan program. Normally, execute MANUAL HOLD in the any MEAS MPAT statement to use the TEST item shmoo.

- Shmoo is plotted continuously using the condition file created by the operation above, by specifying the CALL statement (used to execute Shmoo) in the main program in advance.
- Shmoo is plotted simultaneously for the devices currently being tested (CDUT), and the results are saved to the hard drive as a parallel file.
- Under the following conditions, multiple parallel files and composite files are combined and saved to the disk as a composite file.
CATEGORY(Maximum 4 file)
OR PASS(Maximum 512 file)
AND PASS(Maximum 512 file)
- Output the existing shmoo data files (parallel file and composite file) to the VKT or LPT.

- PARALLEL SHMOO and COMPOSITE SHMOO Outline



- Execution Time comparison of PARALLEL SHMOO and FUNCTION SHMOO
 The following table shows comparisons of execution times for the shmoo (FUNCTION SHMOO) when parallel-testing devices, and for parallel shmoo, creating a number of parallel files equal to the number of parallel-tested devices.
 The parallel shmoo is more effective when parallel-testing large numbers of devices and devoting substantial time to pattern execution.

Measurement conditions

On the T5581 test system, the function shmoo and the parallel shmoo are executed in the test program which calls these shmoos between the START TIMER statement and the READ TIMER statement to execute the measurement on the TP4.

Shmoo screen size is 50×20 .

The inhibit mode (selected by SELECT MPAT INHIBIT) is used for pattern operation.

Execution time

(Actual measurement values)

Parallel test condition	Function shmoo	Parallel shmoo
8 DUTs simultaneous measurement pattern step 4 K	31.0 sec.	10.8 sec.
8 pattern step 512 K	448.0 sec.	63.3 sec.
64 DUTs simultaneous measurement pattern step 4 K	354.4 sec.	61.2 sec.
64 DUTs simultaneous measurement pattern step 512 K	3690.2 sec.	110.2 sec.

(Comparison rate of execution time)

Parallel test condition	Function shmoo	Parallel shmoo
8 DUTs simultaneous measurement pattern step 4 K	1	0.35
8 pattern step 512 K	1	0.14
64 DUTs simultaneous measurement pattern step 4 K	1	0.17
64 DUTs simultaneous measurement pattern step 512 K	1	0.03

- Bump test is executed by changing the BURST BLOCK data of the test plan program and the pattern program as a parameter.
- The test is executed with consideration for any output pin (max. 4 pins) of multiple-bit configuration devices, and sends its results in combined form to the VKT or LPT.
- In executing the shmoo (except DC shmoo), the PASS/FAIL of the range to be specified is output according to the display mode.
 - When SHM2 is called from the test plan program, the number of fails for each DUT in the specified range, as well as a total number of fails for each DUT, can be obtained.
 - When SHM2 is called from the test plan program, a range and a display mode for shmoo can be specified using an argument.

- When SHM2 is called from the test plan program, the device number to be tested in the functional shmoo and the DC shmoo or the device number to be displayed in the parallel shmoo can be specified using an argument.

5. 2 Using SHM2

Set the test mode to MANUAL and HOLD mode, repeat an MEAS statement that executes an appropriate FUNCTIONAL test, and activate SHM2.

```
/SHM2.
```

The following menu is displayed.

```
*** 2-D SHMOO FIRST MENU ***
      . CHANGE *1
      . START
      . PRINT
      . HEADER
      . WRITE
      . REFERENCE OFF

** NOTE **

PF1:SHMOO EXIT
```

*1 _ indicates a cursor.

Select the item you want to execute using [↑] and [↓] key, and press the return key.

Exit the shmoo program using the [PF1] key.

CHANGE

Specifies or updates the shmoo conditions of the X and/or Y axes.

START

Begins shmoo operation according to the specified shmoo conditions.

PRINT

Displays headers and shmoo conditions in a list on the VKT screen.

HEADER

Enters comments such as the device name (DEVICE), shmoo number (SHMOO) and test condition (COND).

Change lot name, lot name subscript and sample No.

WRITE

Stores shmoo conditions in a data file.

REFERENCE OFF

Performs REFERENCE OFF in the time sequence defined by the test plan program. (Execution of the SROF statement)

◆ Setting of Lot Name, Lot Name Subscript, Sample No. and Comments

When HEADER is selected, lot name, sample number, lot name subscript and comments can be entered.

When the following display appears, enter the required information in the order displayed.

Up to 24 characters can be entered for the lot name. Pressing the space key specifies the lot number of the main program.

Up to 12 characters can be input for DEVICE, up to 10 characters for SHMOO#, and up to 40 characters COND.

LOT	:	-----	_	
		↑		Current lot name
SUBLOT	:	-----	_	
		↑		Current lot name subscript
SAMPLE	:	-----	_	
		↑		Current sample No.
DEVICE	:	-----	_	
SHMOO#	:	-----	_	
COND	:	-----	_	
		↑		Current comment
			↑	VKT cursor

If the current comments need not be changed, press the return key.

When all items have been entered, the screen returns to the FIRST MENU.

A DEVICE comment can also be entered using /HEADER.

If an incorrect character is entered, the character can be corrected by pressing the DELETE key.

This correction function is effective until the return key is pressed.

◆ Setting of Shmoo Condition

Select CHANGE to specify conditions for the X and Y axes.

```
*** CHANGE MENU ***  
  . X-AXIS,Y-AXIS,TRACKING  
  . OPTION-AXIS  
  . FUNCTION  
  . DC  
  . MODE  
  . BUMP  
  . 4BIT CATEGORY SHMOO  
  . SPEC. MODE  
  
** NOTE **  
  PF1:FIRST MENU
```

At this time, if the X-AXIS, Y-AXIS, and TRACKING items are selected from this menu, the X-axis, Y-axis and TRACKING conditions are displayed in the following format. Position the cursor at the item to be set or changed by pressing [↑], [↓], [←], and [→] keys and enter data.

If the current comments need not be changed, press the return key.



Tip

Use the [PF1] key to return to the FIRST MENU.

```

X-UNIT  =                Y-UNIT  =
X-START = 0                Y-START = 0
X-MODIFY = 0
X-DELT  = 0                Y-DELT  = 0
X-STEP  = 50              Y-STEP  = 20
X-STOP  = 0                Y-STOP  = 0
X-NAME  =                Y-NAME  =
XDISP-A = 1,000          YDISP-A = 1,000
XDISP-B = 0                YDISP-B = 0
XCALI   =                YCALI   =
XSPEC-U = 0                YSPEC-U = 0
XSPEC-L = 0                YSPEC-L = 0

TRACK NO.  X/Y  T-UNIT  T-ADD2    TCAL
1
2
3
4
5
6
7

0
0
0
0
0
0
0

0
0
0
0
0
0
0

** NOTE **
PF1 : CHANGE MENU          PF3 : FIRST MENU
PF2 : START MENU           PF4 : UNIT NAME LIST DISPLAY

```



Tip

Use the [PF1] key to return to the CHANGE MENU.

Use the [PF2] key to return to the START MENU.

Use the [PF3] key to return to the FIRST MENU.

Use the [PF4] key to display the following programmable unit names:

***** UNIT NAME LIST *****

ACLK1-128,BCLK1-128,CCLK1-128 : input clock timing edge
 PCLK1-4 : PLL clock timing edge
 RATE : test period clock
 DREL1-256 : leading edge timing of the driver enable clock
 DRET1-256 : trailing edge timing of the driver enable clock
 STRB1-256 : output strobe for edge comparison
 WSTRBL1-128 : leading edge timing for window comparison
 WSTRBT1-128 : trailing edge timing for window comparison
 HIN1-16 : high level voltage for the driver
 LIN1-16 : low level voltage for the driver
 HOUT1-16 : high level voltage for the comparison
 LOU1-16 : low level voltage for the comparison
 VT1-16 : voltage for the 50 ohm dut terminator line
 VS1-12 : voltage for the power source pins
 HINH1-16 : high level voltage for the HV driver
 LINH1-16 : low level voltage for the HV driver
 HIL1-16 : programmable load current on high level output
 LIL1-16 : programmable load current on low level output
 HDCLP : high level dynamic clamp
 LDCLP : low level dynamic clamp
 TIMER : time for REG MPAT TIMER

*** NOTE ***

STRIKE ANY KEY TO CONTINUE

***** UNIT NAME LIST *****

VSIM : voltage for VSIM
 ISVM : current for ISVM
 HVSIM : voltage for HVSIM
 IMDC : measure current mode with VSIM
 VMDC : measure voltage mode with ISVM
 HIMDC : measure current mode with HVSIM
 IMVS1-12 : measure current mode with VS
 BVS1-12 : voltage for the power source pins on BURST
 BVT1-16 : voltage for the 50 ohm dut terminator line on BURST
 BHIN1-16 : high level voltage for the driver on BURST
 BLIN1-16 : low level voltage for the driver on BURST
 BHOUT1-16 : high level voltage for the comparator on BURST
 BLOUT1-16 : low level voltage for the comparator on BURST
 BTIME : BURST wait time
 OVSI-12 : voltage for the power source pins on OBM
 OTIME : wait time on OBM

*** NOTE ***

STRIKE ANY KEY TO RETURN

The input formats of these conditions are shown below.

◆ Setting of Conditions for X and Y Axes

X-UNIT, Y-UNIT

Select one of the following as a unit name for the X and Y axes of the shmoo.

There are units that can not be specified depending on test system.

➔ For details, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498 \)](#).

- (The TIMER executes REG MPAT TIMER.)

VS1 to 12

VT1 to 16 ^{*1}

HIN1 to 16 LIN1 to 16

HOUT1 to 16 LOUT1 to 16

HIL1 to 16 LIL1 to 16

HDCLP LDCLP

HINH1 to 16

LINH1 to 16

ISVM

VSIM

HVSIM

- Burst unit ^{*2}

BVS1 to 12(m) ^{*3}

BVT1 to 16(m) ^{*1}

BHIN1 to 16(m) BLIN1 to 16(m)

BHOUT1 to 16(m) BLOUT1 to 16(m)

BTIME(m)

OVS1 to 12(n) ^{*3}

OTIME(n)

- RATE <1 to 16>

ACLK1 to 128 <1 to 16> BCLK1 to 128<1 to 16> CCLK1 to 128 <1 to 16>

DREL1 to 256 <1 to 16> DRET1 to 256<1 to 16>

STRB1 to 256 <1 to 16>

WSTRBL1 to 128 <1 to 16> WSTRBT1 to 128 <1 to 16>

PCLK1 to 4 (When PCLK is specified, the START PCLK statement is executed in the shmoo.

On completion of the execution, the previous PCLK operating status is restored.)

**1 When VT is specified on the T5593, T5592 and T5591, the same value is set for the comparator and driver. BVT cannot be specified when VT is used to set data for both driver and comparator in the BURST BLOCK module.*

*2 (m) and (n) of the burst unit specify line number or block number displayed on BURST BLOCK output. The following shows the example of block number.

- "m" indicates the line number and specifies in 1 to 1024 of the integer value.
- "n" indicates the block number and specifies in 1 to 16 of the integer value.

Example

BURST BLOCK OUTPUT in the TEST PLAN PROGRAM : BURST

*** BURST BLOCK DISPLAY ***				DATE	
BURST					
BURST SET NO.	LINE NO.	UNIT	VALUE		
1	1	VS1	3.000V	→	BVS1 (1)
	2	VS2	5.000V	→	BVS2 (2)
2	3	IN1	5.000V,1.000V	→	BHIN1 (3) , BLIN1 (3)
3	4	OUT1	3.000V,1.000V	→	BHOUT1 (4) , BLOUT1 (4)

BURST BLOCK OUTPUT in the PATTERN PROGRAM : OBM

*** OBM DISPLAY ***			DATE		
BLOCK NO.		UNIT	VALUE		
1		VS1	3.000V	→	OVS1 (1)
		VS2	5.000V	→	OVS2 (1)
2		VS9	5.000V	→	OVS9 (2)

*3 On the T5593, T5377, T5376 and T5375, the BVS and OVS ranges are determined by the VS statement settings, as shown in the following table.

Type/Model	VS statement setting		BVS and OVS ranges
	Applied voltage value	Range (R)	
T5377 T5376 T5375	0 V to 5 V	5V	0 V to 5 V
	0 V to 16 V	16V	0 V to 16 V
	-5 V to -2 mV	16V	-5 V to -2 mV
	0 V to 20 V	20V	0 V to 20 V
	-20 V to -2 mV	20V	-20 V to -2 mV
T5593	0 V to 5 V	16V	0 V to 5 V
	-5 V to -2 mV	16V	-5 V to -2 mV

Range (R) settings of 5 V and 16 V are used for the system standard PPS. 20V is the range of 20VPPS.

XSTART, YSTART

Specify the initial values for the X and Y axes.

XDELT, YDELT

Specify the step values for the X and Y axes.

XSTEP, YSTEP

Specify the number of test steps the X and Y axes.

Maximum numbers of steps are 100 for X and 9999 for Y.

XSTOP, YSTOP

Specify the final values for the X and Y axes.

XNAME, YNAME

Used as comments on X, Y axes. (Up to 14 characters)

YDISP-A, YDISP-B

Specify the coefficient and offset value used to display the Y-axis. Assuming, for example, that the Y value is "y", the value of $A \times y +$ is displayed.

A represents a real number and B represents a real number expressed in the same units as the YSTART value.

A=1.000 and B=0 in the initial state.

XDISP-A, XDISP-B

Have the same functions as YDISP-A and YDISP-B.

XCALI, YCAL

Specify the number of calibration increments for the X and Y axes. The number of calibration increments must be specified as a positive integer. This number is effective only when HIN or LIN is selected in the UNIT item.

To perform calibration, MODE must be selected from the CHANGE MENU screen, and ENABLE(=1) must be selected in response to message "CAL ENABLE(1) OR DISABLE(0)=".

When an ordinary shmoo is plotted, the calibration facility need not be specified. However, this facility must be specified to plot a shmoo when driver levels (HINn, LINn) are selected in the X, Y and T parameters, or when fluctuation of the rise and fall times cannot be ignored in evaluating shmoos at sites such as high voltage stations.

 Limitation

- The operation of the shmoo with calibration takes approx. two minutes per point. (Some variation may occur depending on the number of pins used.)
- The shmoo with calibration function cannot be used for DC shmoo.
- The calibration setting is disabled on the T5593, T5592 and T5591. (The menu is not displayed, either.)

XSPEC-U, YSPEC-U

Specify the upper limit values for the X and Y axes.

➔ For details, refer to the description for selecting items of "X-AXIS, Y-AXIS, TRACKING" at [5. 10. 2 "Setting of Specification Value" on page 5-91](#).

XSPEC-U, YSPEC-U

Specify the lower limit values for the X and Y axes.

➔ For details, refer to the description for selecting items of " X-AXIS, Y-AXIS, TRACKING " at [5. 10. 2 "Setting of Specification Value" on page 5-91](#).

XMODIFY

Usually set at 0. When set to 1, the initial value of the X axis is the result of the following expression:

Actual initial value of the X axis = Y-axis value + XSTART value.

That is, the initial value of the X axis can be changed according to the Y-axis value.

This method is used to convert [Figure 5-1](#) into [Figure 5-2](#) which is easier to read when a dynamic RAM device, for example, is to be evaluated. In [Figure 5-1](#), /CAS DELAY values are specified on the Y axis, and STROBE for access timing from /CAS is specified on the X axis.

Figure 5-1 Example of Shmoo without XMODIFY

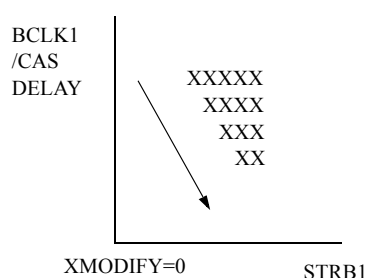
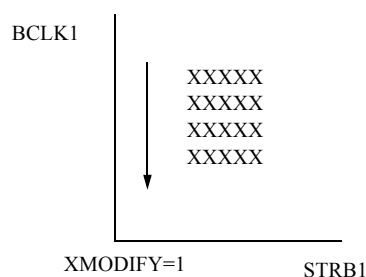
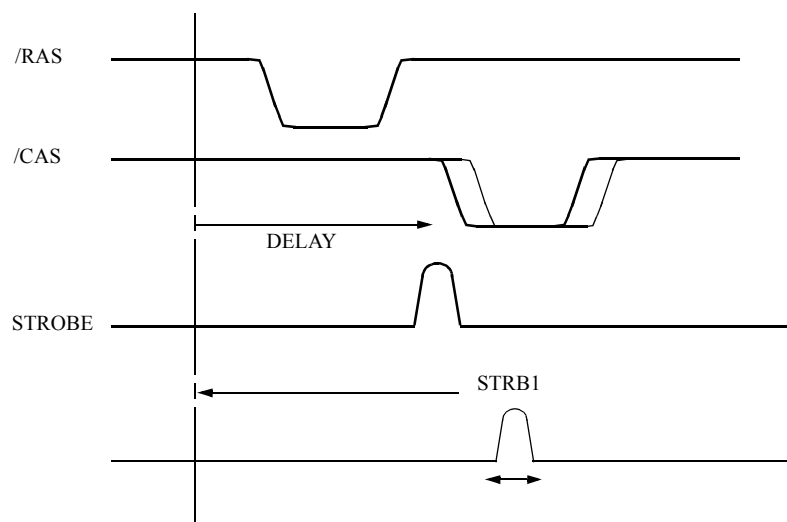


Figure 5-2 Example of Shmoo with XMODIFY



Data in [Figure 5-1](#) must be read diagonally. Data in [Figure 5-2](#) must be read vertically.

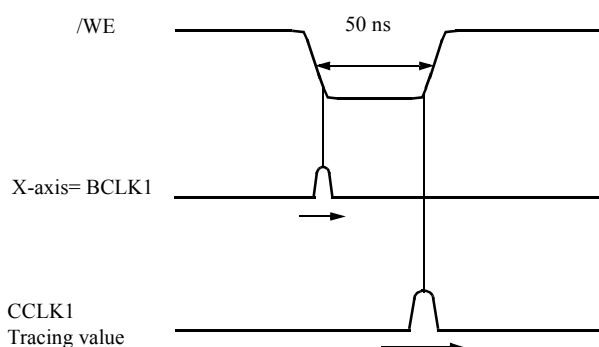
[Figure 5-3](#) shows example of DELAY changed with /CAS along the Y axis. In this case, the initial values of STROBE of the X axis are displayed diagonally as shown in [Figure 5-1](#).

Figure 5-3 Memory Signal and XMODIFY

◆ Setting of TRACKING Conditions

Unit names which must be modified and tracked according to the X and Y axes, and axis values to be increased can be specified. A maximum of seven tracking values for the X and Y axes can be specified.

The /WE signal in the memory device is operated in the /RZO waveform mode by using BCLK1 and CCLK1, as shown in the figure below.



The /WE signal in the memory device is operated in the M1 waveform by using BCLK1 and CCLK1, as shown in the figure below.

The /WE waveform pulse width is constant, and DELAY is changed on the Y axis. When BCLK1 is specified in the Y axis, the TPAR1 set as the tracking value is set as shown below to maintain the /WE waveform pulse width at 50 ns.

TRACK NO.	X/Y	T-UNIT	T-ADD2	TCAL
1	Y	CCLK1	50.00NS	0

X, Y

Specifies the axis to be tracked. Specify using the X and Y symbols.

T-UNIT

Specifies the name of the unit to be tracked.

T-ADD2

Sets the value to be added to the current value of the axis to be tracked.

The unit of the value must match that of the value of the axis to be tracked.

TCAL

Specifies the number of calibration increments for the axis to be tracked as a positive integer.

When calibration is specified for the axis to be tracked, the smaller of the tracking value and the number of calibration increments specified for the axis to be tracked is given higher priority.

Effective only when HIN or LIN is specified in the TCALI tracking value.

Tip

To cancel specification of a tracking unit set once, enter the space and the return key when specifying T-UNIT.

◆ Example of Setting X-AXIS, Y-AXIS, TRACKING

```

X-UNIT=STRB1      Y-UNIT=HIN1
XSTART=0.000S     Y-START=6.000V
XMODIFY=0
XDELT=2.000NS     YDELT=-500.0MV
XSTEP=50          YSTEP=15
XSTOP=98.00NS     YSTOP=1.000V
XNAME=STROBE      YNAME=HIM LEVEL
XDISP-A=1.000     YDISP-A=1.000
XDISP-B=0.000S    YDISP-B=0.000V
XCALI=0           YCALI=0
XSPEC-U=0         YSPEC-U=0
XSPEC-L=0         YSPEC-L=0

```

TRACK No.	X/Y	T-UNIT	T-ADD2	TCAL
1	Y	HIN1	1.300V	5
2			0	0
3			0	0
4			0	0
5			0	0
6			0	0
7			0	0

◆ Setting of Separate Tracking Conditions

Independent tracking is used to set conditions independent of the X- and Y-axis setting values. A uniform interval is set for normal tracking; independent tracking can set at different intervals.

A maximum of 100 separate tracking conditions can be specified for the X and Y axes by selecting OPTION AXIS from the CHANGE MENU screen.

The screen format is as follow:

ITEM	Y/X	UNIT	START	DELT
1			0	0
2			0	0
3			0	0
4			0	0
5			0	0
6			0	0
7			0	0
8			0	0
9			0	0
10			0	0
11			0	0
12			0	0
13			0	0
14			0	0
15			0	0
16			0	0
17			0	0
18			0	0
19			0	0
20			0	0
** NOTE **				
LF : NEXT PAGE		PF1 : CHANGE MENU		PF3 : FIRST MENU

- Feeds the page with [LF] key.
 - Use the [PF1] key to return to the CHANGE MENU.
 - Use the [PF3] key to return to the FIRST MENU.
- Move the cursor with [↑], [↓], [→], and [←] keys and enter the following data.

X, Y

Specifies the axis to be set.

UNIT

Specify the unit name. It can be specified by the same method as that used for the Y and X axes.

START

Specify the initial value.

DELT

Specify the step value.

X, Y

Specifies the axis to be set.

UNIT

Specify the unit name. It can be specified by the same method as that used for the Y and X axes.

START

Specify the initial value.

DELT

Specify the step value.

- To delete a data item set once, position the cursor at the item and enter the space and the return key.
- To change a data item set once, position the cursor at the item and enter data.
- X/Y, UNIT, START and DELT must be entered in this order.

An example of a separate tracking condition setting is as follows.

ITEM	X/Y	UNIT	START	DELT
1	Y	HIN1	5.000V	-100.0mV
2				
.				
.				
.				

◆ Setting of FUNCTION Pattern

Select FUNCTION from the CHANGE MENU.

MAIN PAT(0), CHANGE PAT(1) = __

Select MAIN PAT(0) to give higher priority to the user program pattern (MPAT).

Select CHANGE PAT(1) to specify a pattern different from the user program pattern. When CHANGE PAT(1) is selected, the following message is displayed:

FUNCTION = __

In response to this message, enter a FUNCTION name as a character string consisting of 32 characters or less.

MAIN PC(0), CHANGE PC(1) = __

Select MAIN PC(0) when giving priority to the user program "PC".

Select CHANGE PC (1) when specifying a different PC from the user program. The following message is displayed.

START PC = ____



Tip

When CHANGE PAT(1) is selected, the FUNCTION pattern (MPAT) is transferred to the PG. The pattern is not deleted from the PG even after shmoo execution or SHM2 termination.

◆ Specifying Pattern by DC Shmoo

*** DC MEASURE CONDITION ***

•WAIT TIME

*** NOTE ***

PF1:FIRST MENU

WAIT TIME

WAIT TIME FROM START MPAT TO START DC (OR VS) WHEN DYNAMIC DC SHMOO EXECUTE

WAIT TIME = ____

When executing DYNAMIC DC, input the waiting time for START DC from START MPAT.

◆ Specification of Shmoo Operation Mode

When MODE is selected from the CHANGE MENU screen, the messages below are displayed. Enter data sequentially.

PSKIP = ____

PAGE = ____

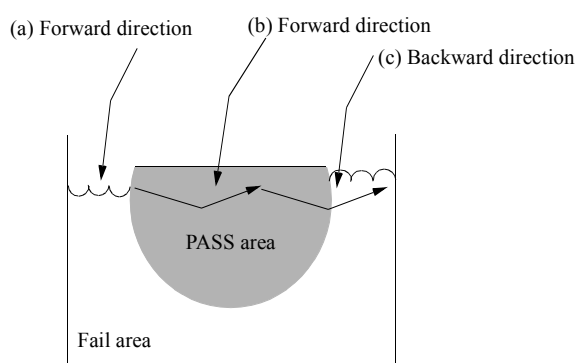
CALL ENABLE (1) OR DISABLE (0) = ____

RON/ROF MODE DISABLE (0) OR ENABLE (1) = ____

◆ PSKIP

When shmoo is executed, the operation of the unit specified for the X axis differs, depending on the specified PSKIP value, as follows:

If a shmoo plot indicates a pass area and a fail area as shown below, the test was performed as follows:



(a) Forward direction

The test was performed according to the XDELTA step value in the forward direction until PASS was detected.

(b) Forward direction

When PASS was detected, the test was performed, skipping according to the PSKIP value. When PSKIP=1, the test was performed according to the XDELTA step value. When PSKIP=2, the test was performed every two XDELTA step values. Either test was performed until FAIL was detected.

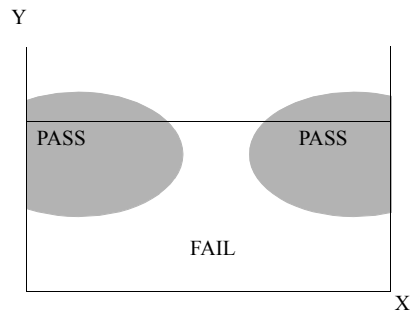
(c) Backward direction

When FAIL was detected, the test was performed according to the XDELTA step value in the backward direction.

Operations (a), (b) and (c) were repeated until the last X-axis value was detected.

If the X-axis value was exceeded when a skip operation was performed in the (b) forward direction, the test was performed from the last X-axis value in the backward direction.

When PSKIP is set to 1, the test is performed for all X-axis points. When an appropriate value for PSKIP is specified, the shmoo plot below can be tested at both ends of each pass area.



PSKIP is a timesaving function for units with a large memory and for use of a pattern such as a GALLOPING pattern test because it needs much time to run.

PSKIP = 1

All-point test (initial value)

PSKIP = 0

Tests at both ends of each pass area.

(PSKIP determined to be more than 100.)

PSKIP = n

Tests every n points,

that is, at the interval of n-1 points in each pass area. ($2 \leq n \leq 100$)

◆ PAGE

PAGE = 1

Clears the VKT screen or feeds one page in the line printer when a shmoo plot is output.

PAGE = 0

Does not perform the operation described for "PAGE=1."

◆ CAL ENABLE (1) OR DISABLE (0)

In normal shmoo operation, DISABLE (0) is specified. The details are described later.

The procedure to output a shmoo plot is briefly explained here.

When CAL ENABLE (1) is selected, the number of calibration increments specified for the X, Y and T parameters is effective, and calibration is performed.

When CAL DISABLE(0) is selected, calibration is not performed even if the number of calibration increments is specified.

When CAL ENABLE (1) is selected, the following is displayed:

```
PAGE = 0                      ↵
CAL ENABLE (1) OR DISABLE (0) = 0 1↵
VAR NAME    =
            AA↵
FILE TAG    =
            BB↵
CAL STRING =
DRREF = 2.4V, .....↵
```

For "VAR NAME =", input the string variable name where the calibration conditions are stored when CALB is specified in the user program.

Enter as "FILE TAG =", the acronym (26 characters or less) of the name of the CAL attribute file used when calibration is performed in the shmoo.

For "CAL STRING =", input the calibration condition to be added or changed. If VAR NAME and FILE TAG are not specified, they must be specified.

➔ Refer to [5. 6 "SHM2 Operation Using the Calibration Facility" on page 5-44](#) for the shmoo with the calibration facility.

◆ RON/ROF MODE DISABLE (0) OR ENABLE (1)

Specifies the mode turning reference on/off when a pattern is started in FUNCTION SHMOO.

A default is specified at DISABLE (0), and no reference-off is executed.

This function provides a shmoo plot including power-on for each point.

◆ BUMP Mode Setting and BURST BLOCK Output

BURST

The data which is transferred to the tester unit is provided to the BURST BLOCK area of the test plan program in advance. When the pattern generation is stopped using the pause instruction in the pattern program, the data defined are transferred to the tester unit in sequence, and the DUT is tested under these conditions.

OBM

The data which is transferred to the tester unit is provided to the BURST BLOCK area of the pattern program in advance. The data are transferred to each unit using the OUT instruction, and the DUT test is executed.

BUMP test is performed by changing the parameters of two BURST BLOCK data described above in the shmoo.

```

*** BUMP MENU ***
- BUMP SHMOO
• BURST BLOCK
• OBM

WORK AREA

*** NOTE***
PF1:CHANGE MENU PF3:FIRST MENU

```

Tip

- Use the **[PF1]** key to return to the **CHANGE MENU**.
- Use the **[PF3]** key to return to the **FIRST MENU**.

◆ When the BUMP SHMOO is Selected

When BUMP SHMOO is selected, the following message is displayed.

```
BUMP SHMOO DISABLE/BURST/OBM(0/1/2) : 0 ____
```

The execution mode of FUNCTION/PARALLEL SHMOO in the START MENU is changed.
The execution mode has the following three types:

- When inputting 0, the normal shmoo operation is performed.
- When 1 is entered, the BURST BLOCK setting data is changed in the test plan program, and the shmoo operation is executed.
- When 2 is entered, the BURST BLOCK setting data is changed in the pattern program, and the shmoo operation is executed.

◆ When the BURST BLOCK is Selected

When BURST BLOCK is selected, the following message is displayed.

```
BURST BLOCK NAME : BB1 ____
```

Input the BURST BLOCK variable name defined in the test plan program.

Output the contents of the specified BURST BLOCK variable name.

When only inputting a return, the input waiting state terminates and changes to the waiting state for selecting BUMP MENU.

Limitation

- BURST BLOCK NAME is valid only when it is 32 characters or less.

- The size of the BURST BLOCK handled in the shmoo is 4 K (4096) half word. A unit setting for the BURST BLOCK greater than 4 K half word is invalid.

*** BURST BLOCK DISPLAY ***				DATE:1998/11/26 00:00:00	
BURST BLOCK NAME: BB1					
BURST SET NO.	LINE NO.	UNIT	VALUE		
1	1	VS1	5.000V		
	2	IN1	3.000V.	2.000V	
	3	OUT1	2.000V.	*	
2	4	WAIT TIME	10.00MS		
3	5	VS2	200.0MV		
	6	IN2	1.000V.	-2.000V	
	7	OUT2	9.000V.	100.0MV	

*** NOTE ***					
PF1 : BUMP MENU		LF : CONTINUE			
PF2 : CHANGE MENU		P : PRINT OUT			

Tip

- Use the [PF1] key to return to the BUMP MENU.
- Use the [PF2] key to return to the CHANGE MENU.
- Use the [LF] key to feed the page
- Use the [P] key to print the display BURST BLOCK data out.
- When the BURST unit for shmoo does not exist in the specified BURST BLOCK variables, the following message is displayed on the screen. " Available unit does not exist on OBM or BURST for BUMP SHMOO "
- A unit displayed by * can not execute.
- This screen is prepared for setting a unit parameter of each axis (X/Y tracking and independent tack- ing), referring to the line number and unit name indicated on the above screen.

◆ Example of BURST BLOCK Output

Test plan program

```
PRO    BR
:
LOGI[2] BSTBUF(1024)=0
BURST BLOCK BEGIN(BSTBUF)
    BURST SET
        VS1=3V
        VS2=5V
    BURST RESET
    IN1=5V,1V
    OUT1=3V,1V
BURST BLOCK END
:
SEND MPAT PAT1
MEAS MPAT(4,BSTBUF)
:
END
```

Output result

```
*** BURST BLOCK DISPLAY ***    DATE:-----
BURST BLOCK NAME:--
BURST SET NO.  LINE NO.  UNIT   VALUE
    1           1      VS1    3.000V
           2      VS2    5.000V
    2           3      IN1    5.000V,1.000V
    3           4      OUT1   3.000V,1.000V
```


◆ When OBM is Selected

When selecting OBM (OUT BUFFER MEMORY), the reference of OBM (OUT BUFFER MEMORY) can be displayed.

*** OBM(OUT BUFFER MEMORY) DISPLAY ***			DATE:1998/11/26 00:00:00
BLOCK NO.	UNIT	VALUE	
1	VS1	5.000V	
2	WAIT TIME	10.00MS	
3	VS2	200.0MV	
*** NOTE ***			
PF1 : BUMP MENU		LF : CONTINUE	
PF2 : CHANGE MENU		P : PRINT OUT	

 Tip

- Use the [PF1] key to return to the BUMP MENU.
- Use the [PF2] key to return to the CHANGE MENU.
- Use the [LF] key to feed the page.
- Use the [P] key to print the display BURST BLOCK data out.
- When the BURST unit for shmoo does not exist in the specified BURST BLOCK variables, the following message is displayed on the screen. " Available unit does not exist on OBM or BURST for BUMP SHMOO "
- This screen is prepared for setting a unit parameter of each axis (X/Y tracking, and independent tracking), referring to the block number and unit name indicated on the above screen.

◆ Example of OBM Output

Pattern program

```

MPAT  OPAT1
REGISTER
.
.
BURST BLOCK BEGIN(OB0)
      VS2=5V
BURST BLOCK END
BURST BLOCK BEGIN(OB1)
      VS1=3V
BURST BLOCK END
START  #0
OUT OB0
.
OUT OB1
.
.
END

```

Output result

```

*** OBM DISPLAY ***      DATE:-----

BLOCK NO.      UNIT      VALUE
   1           VS2      5.000V
   2           VS1      3.000V

```

◆ 4BIT CATEGORY SHMOO

Pass/fail distribution is output to the categorization table with considering for any output pins (maximum 4 pins) of the multiple-bit configuration devices.

When selecting 4 BIT CATEGORY SHMOO in the CHANGE MENU, the following menu is displayed.

```

**** 4 BIT CATEGORY SHMOO ****

.4 BIT CATEGORY DISABLE/ENABLE(0/1):  0      —

. BIT 0                      : 33
. BIT 1                      : 34
. BIT 2                      :
. BIT 3                      : 36

*** NOTE***
PF1:CHANGE MENU PF3:FIRST MENU

```

 Tip

- Use the [$\uparrow$] and [$\downarrow$] keys or the return key to move the cursor.
- Use the [PF1] key to return to the CHANGE MENU.
- Use the [PF3] key to return to the FIRST MENU.

◆ 4 BIT CATEGORY DISABLE/ENABLE(0/1)

The execution mode of FUNCTION/PARALLEL SHMOO in the START MENU has changed. There are two types of execution modes, which are as follows:

- When inputting 1, the execution mode is set to 4 BIT CATEGORY to perform the shmoo operation.
- When inputting 0, the normal shmoo operation is performed.

 Tip

- When inputting the return key only, the current setting value is valid.
- When the setting value is 0, return to the "CHANGE MENU."

◆ BIT0 to BIT3

An object pin number is defined.

Specify the file sequential number with an integer from 1 to 512.

Release the pin setting with the [C] key.

Omitted BITn's are output as " 0 " (fail) in the categorization at the shmoo operation.

◆ 4 BIT CATEGORIZATION

Define the object pin numbers in 4 BIT CATEGORY as BIT0, BIT1, BIT2 and BIT3 in the specified order.

The test results for each pin are overlapped and categorized into 1 to F.

Categorization	Object pin number			
	BIT 3	BIT 2	BIT 1	BIT 0
Space or period	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
A	1	0	1	0
B	1	0	1	1
C	1	1	0	0
D	1	1	0	1
E	1	1	1	0
F	1	1	1	1

Object pin number is the one defined to BIT 0 to BIT 3.

0 means that test results for each pin are fail.

1 means that test results for each pin are pass.

◆ Example of Execution

- Pin condition of 4 BIT CATEGORY SHMOO

```

4 BIT CATEGORY DISABLE/ENABLE (0/1) : 1
BIT 0                               : 33
BIT 1                               : 34
BIT 2                               : 35
BIT 3                               : 36

```

- The test results for each pin (PASS/FAIL:1/0)

BIT3 36	BIT2 35	BIT1 34	BIT0 33		Category data
1	1	1	1	→	"F"
1	1	1	0	→	"E"
1	1	0	1	→	"D"
1	1	0	0	→	"C"
0	0	0	0	→	" "
0	0	0	1	→	"1"
0	0	1	0	→	"2"
0	0	1	1	→	"3"
.	.	.	.		
.	.	.	.		
.	.	.	.		

- Plot data which is output by above conditions or the test results.

FEDC 123...

Test results output

◆ Mode Setting for Output of Specification Value

If SPEC.MODE is selected in the CHANGE MENU, the following message is displayed.

Enter data.

SPEC.ENABLE (1) OR DISABLE (0) : —

Selecting " 1 " The following message is displayed.

Enter data.

DISPLAY DEFAULT (0) , REVERSE (1) , REPLACE (2) OR MARKER (3) : —

Mode setting above is related to the specification value output.

➔ For details, refer to the setting of [5. 10. 1 "Mode Setting for Output of Specification Value" on page 5-87.](#)

◆ Displaying of Shmoo Condition

When all required data is entered, return to the FIRST MENU screen and select PRINT. The message shown below is displayed.

To output data to the VKT, select "VKT." To output data to both the VKT and the LPT, select "LPT & VKT." When either of the above is selected, the shmoo conditions specified up to that time are displayed.

- VKT
- LPT & VKT

◆ Execution of Shmoo

When START is selected from the FIRST MENU screen, the following menu is displayed:

```

**** START MENU ****
  •FUNCTION SHMOO      •VKT
  •STATIC DC SHMOO    •LPT AND VKT
  •DYNAMIC DC SHMOO   •S-COMMAND
  •COPY-OUT
  •HEADER
  •SELECT DUT NO.
  •PARALLEL SHMOO
  •COMPOSITE SHMOO

*** NOTE ***
    PF1:FIRST MENU

```

Tip

- Use the [PF1] key to return to the FIRST MENU.
- If S-COMMAND is selected, the shmoo plot is output to a file specified by the key command [S] key.

◆ COPY-OUT

To see the Shmoo plot without starting the PG, select COPY-OUT, and select the following message which will be executed.

```
FUNCTION/STATIC DC/DYNAMIC DC : 0/1/2 =
```

Select "VKT", "LPT" AND VKT", or "S-COMMAND." The values of the X and Y axes can be checked while spaces are present in the shmoo program contents.

The shmoo program has a 5000-character buffer. After shmoo is executed, the result is stored in this buffer as symbols. Therefore, the result can be displayed any number of times by using the COPY-OUT function.

This buffer is cleared under the following conditions:

- When data for X and Y axes in the CHANGE menu are changed.
- When /SHM2 is started.
- When a shmoo program is executed from the START MENU screen.

◆ HEADER

Header information output when a shmoo is executed can be controlled.

When a shmoo program is started by entering /SHM2↵, header information can be output.

To suppress header information, select HEADER. The following message is displayed:

NO HEADER!!

As a result, no header information is output.

To cancel the suppression of header information, select HEADER again. The following message is displayed:

HEADER ADDITION!!

Header information is also output.

◆ SELECT DUT NO

Specify DUT NO. of the shmoo to be plotted.

When SELECT DUT No. is selected, the message below is displayed. Specify a DUT number.

DUT# = —

Up to two DUT NO.'s can be specified with values of 1 to 512.

A DUT NO. can be specified using up to two integers between 1 and 512. To plot shmoos of all DUTs set in the station, enter ALL↵, or A↵.

To cancel the current specification, enter CLEAR↵ or C↵.

◆ Entry Example

DUT# = 4 12 5, 13 ↵

Currently set value

Newly set value

 Tip

As the shmoo plot DUT NO.'s use numbers of DDUT, the DUT NO starts from 1 even when a measurement is executed at the 2nd station using the program which has been compiled with the socket program for the one station parallel test.

◆ FUNCTION SHMOO

When an output device is specified after SHMOO is selected, the result is displayed each time a point is executed.

An example is shown in the log file described later. If the result is incorrect, modify according to the previously described procedure, and execute again.

If they exceed the setting range, a question mark (?) is displayed for the range.

To interrupt shmoo program execution, press the [B] key. The shmoo program is interrupted after the current point is displayed and the message below is output.

BREAK! EXECUTION ABORTED!!

 Tip

When the shmoo program is interrupted, only the part of the shmoo program which has been executed is stored in the 5000-character buffer.

When DISABLE is selected in the RON/ROF MODE specifying operating mode: SRON is performed as an operation within the shmoo program prior to a double loop. Then, when the double loop ends or is interrupted, SROF is performed.

➔ For details, refer to ["RON/ROF MODE DISABLE \(0\) OR ENABLE \(1\)" on page 5-23](#).

Shmoo program conditions for the X and Y axes and for T are saved for all level timing values, and after the double DO loop ends, they are recovered. Therefore, the main program conditions specified when shmoo was called remain unchanged. Only the pattern specified by FUNCTION is changed, and it remains in the PG.

When the axis unit is VSn, the clamp current values are processed so that the values specified in the test plan program are valid.

There is no restriction on the step value for the Y axis. Overflow characters are not stored in the 5000-character buffer, nor is data written to the buffer or indicated by the COPY-OUT function.

To plot such a large shmoo, output it directly to the VKT or LPT when SHMOO is selected.

An asterisk (*) indicates PASS, and an equal sign (=) indicates a point skipped by using the PSKIP function.

A blank () or a dot (.) indicates FAIL.

Dots (.) are used to make the shmoo plot easy to read.

When 4 BIT CATEGORY is set to ENABLE(1) and PSKIP value is more than two and all results of tested each pin which is specified in 4 BIT CATEGORY SHMOO are all pass, SKIP operation is executed.

If a DUT number that does not exist is selected and the shmoo program is executed, the message below is displayed.

```
DUT# : n NOT EXIST !
STRIKE ANY KEY!
```

◆ STATIC DC SHMOO, DYNAMIC DC SHMOO

FUNCTION SHMOO display PASS/FAIL information at each X or Y-axis. When the source unit is specified at the shmoo in the DC parametric test, a measured current or voltage is displayed with an asterisk (*).

The STATIC DC SHMOO does not execute START MPAT but the DC shmoo.

The DYNAMIC DC SHMOO executes START MPAT, executing the DC SHMOO.

When DYNAMIC DC SHMOO is selected, the following message is displayed.

```
START MPAT(3) / START MPAT(3)* : 0/1 = 0
```

When START MPAT(3) is executed, input 0. When START MPAT(3)* is executed, input 1.

The following gives notes on the DC shmoo.

- The TRACKING parameter and its unit must be correctly specified.
- Specification of units such as VSIM, ISVM and HVSIM VS must not be duplicated.
(For example, VSIM is specified at X-axis and HVSIM is specified at TRACKING for the same X-axis.)
- The test unit must be specified at either X or Y axis.

To interrupt DC shmoo program execution, press the [B] key.

The following message is displayed, and the menu command is waited.

```
BREAK! EXECUTION ABORTED
```

➔ For details of the DC shmoo parameter input, refer to [5. 3 "DC SHMOO Use Method" on page 5-37](#).

◆ Filing Shmoo Condition as Data

After the shmoo plot is output, the shmoo conditions can be stored in a disk file for future use or included in the test plan program as a CALL statement.

When WRITE is selected from the FIRST MENU screen, the following message is displayed.

```
NEW DATA FILE-NAME = ?      _
```

In response to the message, enter a data file name consisting of up to 32 characters. Note that if the current customer file has the same name as a data file, it is deleted unconditionally.

The shmoo program conditions are stored in data file AQ2.SM in /SHM2. Therefore, the previously used shmoo program conditions can be reused after /SHM2 ends. However, these conditions are not stored if the CTRL and C keys are pressed simultaneously or the RESET switch is turned on during execution of /SHM2.

If the current customer file contains no data file, the data file read function searches for a customer DP0. This operation must be performed carefully when the DP0 has the same name as a file.

The file name S3P01 is entered in response to the message below.

```
NEW DATA FILE-NAME = ?  S2P01.↓
```

When the shmoo is to be started, the following is entered: **/SHM2 S2P01,1.↓**

As a result, the shmoo condition is S2P01. Then the system starts measurement without entering the conversational mode.

When the data file name /SHM2.↓ is entered, the same operation as that specified by **/SHM2 AQ2 . SM.↓** is performed.

◆ Execution of the Reference-off

When REFERENCE OFF is selected from the FIRST MENU screen, REFERENCE is turned off according to the TIME statement defined by the test plan program.

(REFERENCE OFF is used to re-execute the shmoo after replacing devices.) DC SHMOO Use Method

5. 3 DC SHMOO Use Method

In the shmoo related to the general function test, though the PASS/FAIL information in each point of X and Y axes is displayed, for the shmoo of DC parametric test, the measured value of current or voltage is displayed with an asterisk (*) when the source unit is set.

When the shmoo in the DC parametric test is plotted, each X axis and Y axis has a source unit and a measuring unit.

The source unit

There are units that can not be specified depending on test system.

➔ For details, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498 \)](#).

VS1 to 12, VT1 to 16^{*1}, HIN1 to 16,

LIN1 to 16, LIL1 to 16, HIL1 to 16, HOUT1 to 16, LOUT1 to 16, HDCLP,

LDCLP

TIMER, ISVM, VSIM and HVSIM

HINH1 to 16, LINH1 to 16

RATE <1 to 16>

ACLK1 to 128 <1 to 16>, BCLK1 to 128 <1 to 16>, CCLK1 to 128 <1 to 16>

DREL1 to 256<1 to 16>, DRET1 to 256<1 to 16>

STRB1 to 256 <1 to 16>, WSTRBL1 to 128 <1 to 16>, WSTRBT1 to 128 <1 to 16>

PCLK1 to 4^{*2}

**1 When VT is specified on the T5593, T5592 and T5591, the same value is set for the comparator and driver.*

**2 When PCLK is specified, the START PCLK statement is executed in the shmoo. On completion of the execution, the previous PCLK operating status is restored.*

The measuring unit

IMVS1 to 12 (On the T5581 and T5581H, IMVS1, 2, 9 to 12)

IMDC, MDC (Pin specification)

HIMDC, HIMDC (Pin specification) (Can only be used on the T5581 and T5581H)

VMDC, VMDC (Pin specification)

In IMDC, HIMDC and VMDC, pin specification is possible. In this case, the shmoo plot is created for each specified pin. The pin specification method is the same as the pin statement.

➔ For details, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498 \)](#).

◆ Example of Specification

IMDC (PD1-2, 5, 7)

HIMDC (PC1, 10)

VMDC (P33)

However, the pin type changed as shown below cannot be specified.

IMDC(PD1-2, PC33)

Tip

In the MEAS DC statement where the DC is not connected to the pin beforehand, when the DC SHMOO is performed by starting /SHM2, use the specification of the measuring pin.

Example: When the following messages are described in the main program

MEAS DC (Pin specification)

And, when the DC SHMOO is performed by starting /SHM2 in the MEAS DC statement, after connecting the DC to the pin beforehand do not use the specification of the measuring pin.

Example: When the following messages are described in the main program

Pin specification = DC

MEAS DC

In the DC SHMOO, for XSTART, YSTART, XDELTA, YDELTA, XSTEP, YSTEP, XSTOP and YSTOP on X and Y axes, input from the CHANGE menu.

When the DC SHMOO is performed, the following setting is also possible.

- When executing DYNAMIC DC SHMOO, input waiting time of START DC from START MPAT.

For this condition, select the DC from the CHANGE menu to input.

AUTO specification of measurement unit

When the measurement unit is set, " AUTO " can be specified as XSTART, YSTART, XDELTA and YDELTA.

- When "AUTO" is specified as START value:

$$XSTART = \text{Minimum value of the measurement value}$$

$$XSTART = \text{Maximum value of the measurement value}$$
- When "AUTO" is specified as DELTA value:

$$XDELTA = \frac{(\text{Max. value of the measurement value} - \text{Min. value of the measurement value})}{XSTEP}$$

$$YDELTA = \left(\frac{(\text{Max. value of the measurement value} - \text{Min. value of the measurement value})}{YSTEP} \right)$$

Hysteresis plot

The XSTEP and YSTEP hysteresis of the source unit can be specified. Input Hn (n: The number of steps).

(Example) XSTEP=H50.↓

Specification of number of steps

For the DC SHMOO, up to 100 steps can be set for both XSTEP and YSTEP.

The number of times measurement is performed

Measurement is performed the number of times specified in the MEAS DC and START DC statements, or the MEAS VS and START VS statements before the shmoo execution. The result is the average value of the measured data.

```

PRO SAMPLE

CALL SHM2 (".....") (1)
:
MEAS VS1 ! 6          (2) Executes /SHM2
CALL SHM2 (".....") (3)
:
MEAS DC ! 7           (4) Executes /SHM2
CALL SHM2 (".....") (5)
:
MEAS VS12 ! 9         (6) Executes /SHM2
CALL SHM2 (".....") (7)

END

```

- For (1)
Measurement is performed once when IMDC, HIMDC, VMDC, or IMVS1-12 is selected for the X or Y axis.
- For (2) and (3)
Measurement is performed six times when IMVS1 is selected for the X or Y axis. Measurement is performed once when IMVS2-12 is selected.
- For (4) and (5)
Measurement is performed seven times when IMDC, HIMDC, or VMDC is selected for the X or Y axis.
- For (6) and (7)
Measurement is performed nine times when IMVS12 is selected for the X or Y axis. Measurement is performed six times when IMVS1 is selected and is performed once when IMVS2-11 is selected.

5. 4 How to Use Log Scale Shmoo

" DELT " is used on a logarithmic scale based in the input elements.
It can be set in the " Ldelt " units or in the " LAUTO " mode.

◆ " Ldelt " Unit Setup

Select either " X-AXIS, Y-AXIS, TRACKING " or " OPTION-AXIS " on the CHANGE menu, and input data.

- Entering XDELTA, YDELTA in ["Setting of Conditions for X and Y Axes" on page 5-11](#)

XSTART = 1.000 MS

XDELTA = L2.000 NS

XSTEP = 50

If these data are input, data A on axis X has the following value:

$$A = 1.000\text{MS} + (2.000)^{n-1} \text{ NS } (1 \leq n \leq 50)$$

- Entering T-ADD in ["Setting of TRACKING Conditions" on page 5-15](#)

T-ADD2 = L5.000NS

If this data is input, tracking data T associated with axis X of (a) has the following value:

$$T = A + (5.000)^{n-1} \text{ NS } (1 \leq n \leq 50)$$

- Entering DELT in ["Setting of Separate Tracking Conditions" on page 5-17](#)

START = 1.000 US

DELTA = L3.000 US

If these data are input, separate tracking data D has the following value: $D = 1.000\text{US} + (3.000)^{n-1} \text{ US } (1 \leq n \leq 50)$

◆ " LAUTO " Setup

Select the " X-AXIS, Y-AXIS, TRACKING " option on the CHANGE menu, and input data.

If " LAUTO " is set for the DELTA value, the DELTA value is determined automatically as the natural logarithm according to the already input START and STOP values.

◆ Input Example of X-AXIS and Y-AXIS

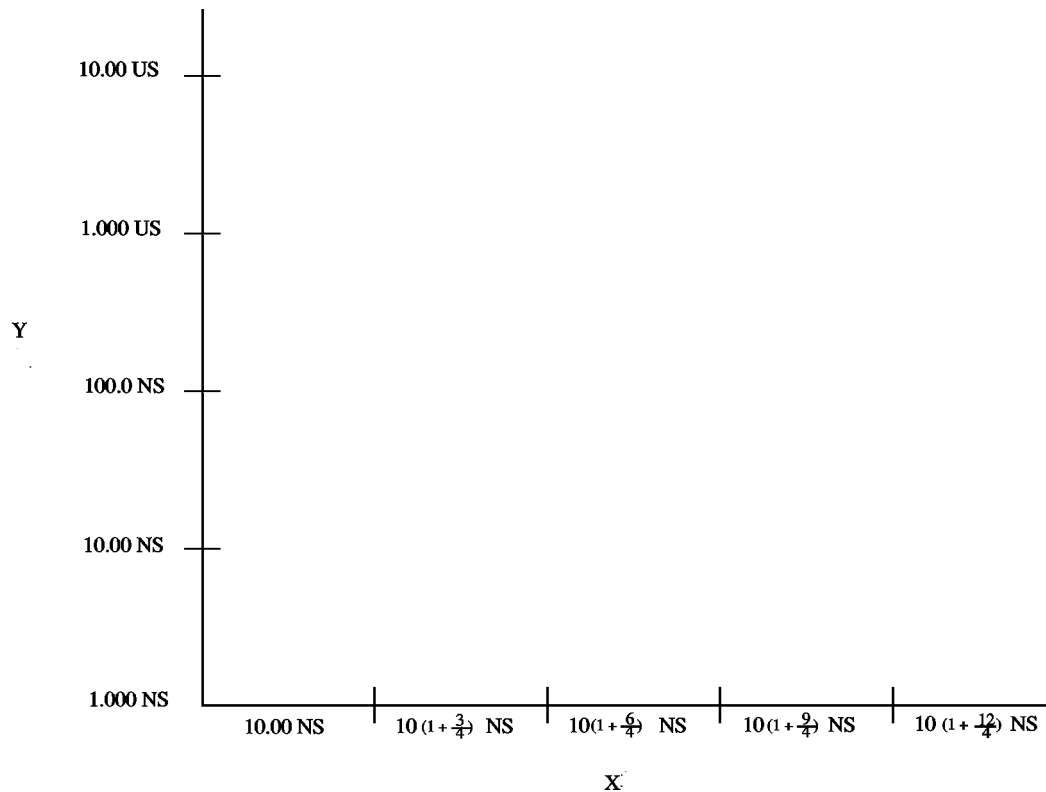
XSTART=10.00 NS YSTART=1.000 NS

XDELTA=LAUTO YDELTA=LAUTO

XSTEP=5 YSTEP=5

XSTOP=10.00 US YSTOP=10.00 US

If these data are input, the set value of each step on axes X and Y are as follows:



Limitation

- Specify the delt value for the Ldelt unit as follows: $0 < \text{delt} < 1000$.
- "LAUTO" cannot be specified for separate tracking.
- The following must be satisfied for "LAUTO" setup: START value > 0 , STOP value > 0 and START value $\neq$ STOP value.

5. 5 SHM2 Autoscaling

Generally, the three conditions (the initial value, the step value and the number of steps) must be specified for each parameter. The autoscaling facility automatically calculates the three conditions based on the value of the parameter in the test program.

◆ Using the Utility Program

The START, DELT and STEP values are determined by selecting X-AXIS, Y-AXIS and TRACKING from the CHANGE MENU screen and entering a unit name.

The unit name to be set must be the same as the current unit name.

◆ Scaling Range

A scaling range is set in the 0 and ∞ directions from the original pointer.

Auto scaling setting is shown in [Table 5-1](#) and [Table 5-2](#). When the autoscaling range exceeds the unit setting range, the autoscaling range is 50 steps, 20 steps from the unit limit value in the 0 or ∞ direction.

Table 5-1 X-axis Autoscaling Setting

Unit name	DELT value	Number of steps in the 0 direction	Number of steps in the ∞ direction
RATE, ACLK, BCLK, CCLK, DREL, DRET, STRB, WSTRBL, WSTRBT	2NS	24	26
PCLK	250PS	24	26
HIN, HOUT, LIN, LOUT, VT, HINH, LINH, BHIN, BHOUT, BLIN, BLOUT, BVT	40MV	24	26
HIL, LIL	100UA	24	26
HDCLP, LDCLP	32MV	24	26
VS, VSIM, HVSIM, BVS, OVS	40MV	9	11
ISVM	5% of set value	9	11
IMDC, HIMDC, IMVS, VMDC	AUTO	9	11
TIMER	2% of set value	24	26

* The default values for the BTIME and OTIME setting are listed below:

START value = 3 ms

DELT value = 10 μ s

STOP value = 3.49 ms

Table 5-2 Y-axis Autoscaling Setting

Unit name	DELT value	Number of steps in the 0 direction	Number of steps in the ∞ direction
RATE, ACLK, BCLK, CCLK, DREL, DRET, STRB, WSTRBL, WSTRBT	2NS	9	11
PCLK	250PS	9	11
HIN, HOUT, LIN, LOUT, VT, HINH, LINH, BHIN, BHOUT, BLIN, BLOUT, BVT	100MV	9	11
HIL, LIL	200UA	9	11
HDCLP, LDCLP	64MV	9	11
VS, VSIM, HVSIM, BVS, OVS	100MV	9	11
ISVM	5% of set value	9	11
IMDC, HIMDC IMVS, VMDC	AUTO	9	11
TIMER	5% of set value	9	11

* The default values for the BTIME and OTIME setting are listed below:

START value = 3.19 ms

DELT value = -10 μ s

STOP value = 3 ms

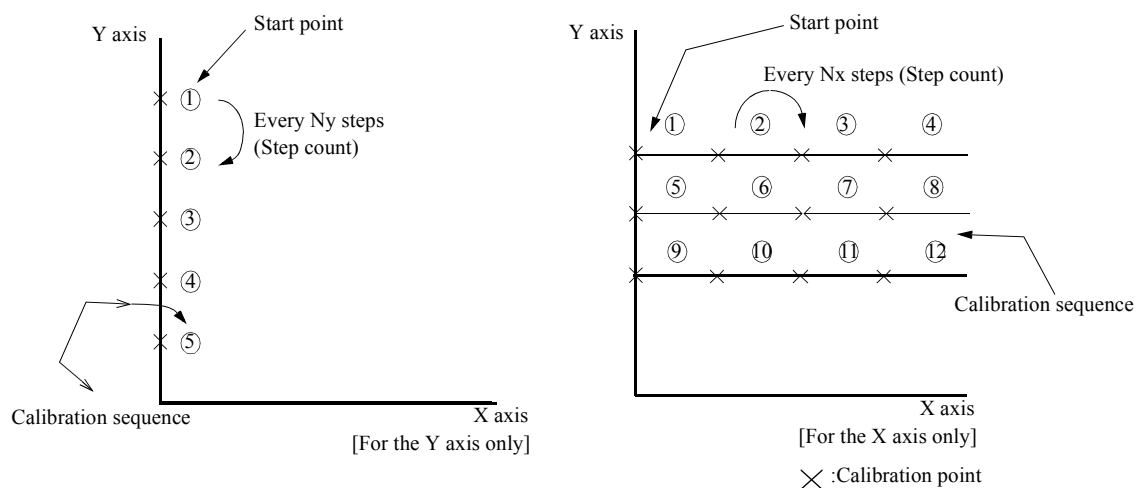
5. 6 SHM2 Operation Using the Calibration Facility

The shmoo with the calibration facility can be used when the rising and falling edges are changed in such the EPROM driver station.

Tip

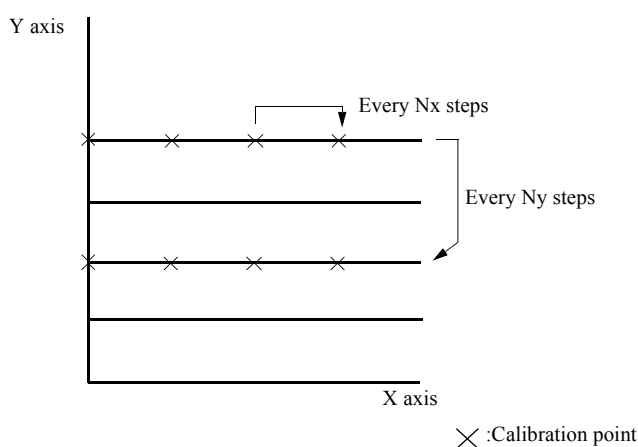
The operation of the shmoo with calibration takes approximately two minutes per point. (This depends on the number of pins used.)

• Calibration points

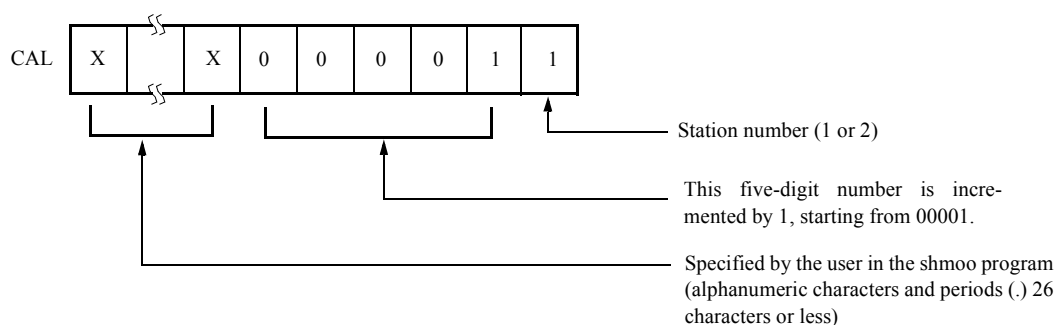


The calibration facility set for each axis parameter operates separately.

Therefore, the following shmoo with calibration cannot be plotted by specifying calibration for both the X and Y axes



- File names with CAL attributes



The previously set CAL attribute names are not deleted.

Tip

When a shmoo with calibration is run, many files with CAL attributes are created in the disk according to the number of points specified for each axis. Be very careful when executing calibration for file control (for maintenance, deletion and name of the file).

- Specifying the string variable name of the calibration facility
CALB in the test plan program must always be specified by a STRING variable name as follows:

```
STRING JJ (1000)
JJ = "DRREF=2.4V, ..."
CALL CALB ( "64K", JJ) }
```

Correct

If a character string is specified directly as shown below, it is not transferred to the shmoo character string.

```
CALL CALB ( "64K", "DRREF = 2.4V, ....." )
```

Mistake

- An additional character string specified by a shmoo program can be used by adding it to the STRING variable name in the test plan program. If the specifications are the same, the most recent specification is effective.

5. 7 SHM2 Runtime Error

- If an execution-time error occurs during shmoo program execution, an error code, an object file name, an object file line number, a customer name, a source file name and a source file line number are displayed.

An error message is then displayed, and the menu is displayed again.

```
ERROR CODE: 3316      SUB SPL2      00559      [ZETSU:  AP401      (00559)]
               ↑           ↑           ↑           ↑           ↑
            Error  Object file  L i n e  Customer  Source  L i n e
            Codes   name        number   name      file name number

-----ERR3-----

3316  MPAT file has still not been transferred to the test vector generator.

                        Error message
```

- When the MPAT file is specified to execute the shmoo program, this file is searched for in the current customer file.

If there is no MPAT file in the customer file, it is searched for in the DP0.

If there is no MPAT file in the DP0, an error message is displayed.

Tip

After displaying an error code, it takes some time until the corresponding error message is displayed.

5. 8 Examples of Using SHM2

- Set values

```
DEVICE: DRAM256K                      ! DATE:      1998/08/14 19:13:42
SHMOO#:                               ! DATA FILE =AQ2.SM
COND:
DUT#= 1
-----Y-AXIS-----X-AXIS-----
UNIT  =HOUT1                          ! STRB1          !
START =3.800V                         ! 50.00NS        !
MODIFY=                              ! 0              !
DELT  =-200.0MV                      ! 1.000NS        !
STOP  =1.000V                        ! 99.00NS        !
STEP  =15                            ! 50             !
NAME  =VOH                           ! ACCESS         !
DISP-A=1                             ! 1              !
DISP-B=0.000V                       ! 0.000S         !
CALL  =                              !                !
SPEC-U=0                             ! 0              !
SPEC-L=0                             ! 0              !

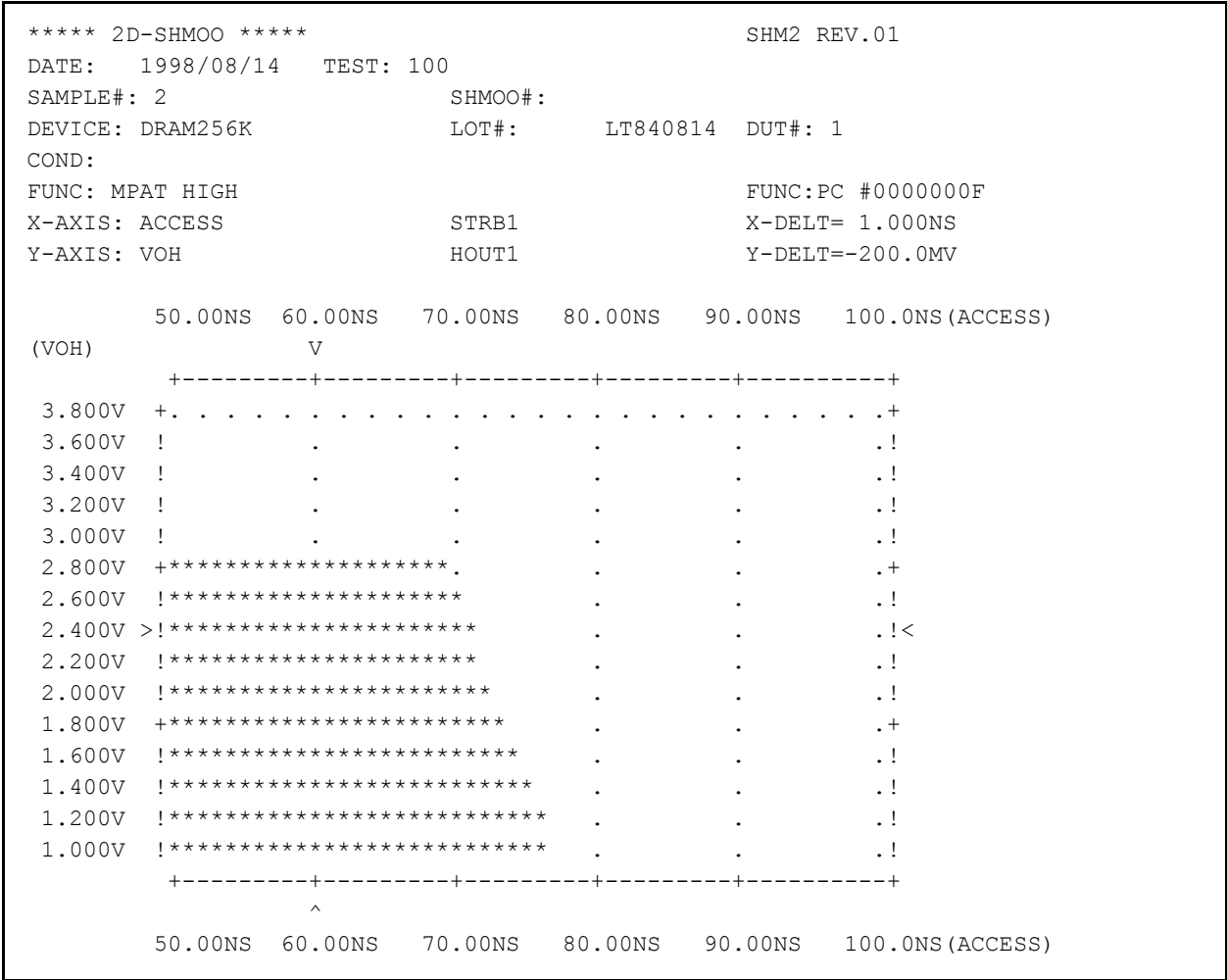
PSKIP = 1                            PAGE = 1          CALI DISABLE

SPEC. DISABLE                        DISPLAY DEFAULT
RON/ROF ENABLE                       BUMP SHMOO DISABLE      4BIT CATEGORY DISABLE

MPAT (CHANGE PAT)                    FUNCTION = HIGH

PC (CHANGE PC)                       START PC = #0000000F
```

• Execution result



In the figure above, "." or space indicates FAIL.
* indicates PASS.

5. 9 Using the Parallel and Composite Shmoo

5. 9. 1 Start

When MANUAL HOLD or MANUAL RECYCLE HOLD are done using a MEAS statement including the arbitrary function test, start SHM2.

```
/SHM2.

**** 2-D SHMOO FIRST MENU ****

      . CHANGE
      _ START *1
      . PRINT
      . HEADER
      . WRITE
      . REFERENCE OFF

**** NOTE ****
      PF1: SHMOO EXIT
```

*1 _ indicates a cursor.



Tip

Exit the shmoo program with the [PF1] key.
Selecting START allows the next menu to appear.

```
**** START MENU ****

      . FUNCTION SHMOO
      . STATIC DC SHMOO
      . DYNAMIC DC SHMOO
      . COPY OUT
      . HEADER
      . SELECT DUT NO.
      _ PARALLEL SHMOO *1
      _ COMPOSITE SHMOO

      . VKT
      . LPT AND VKT
      . S-COMMAND

**** NOTE ****
      PF1: FIRST MENU
```

*1 _ indicates a cursor.

Tip

Use the [PF1] key to return to the FIRST MENU.

- PARALLEL SHMOO

The system waits for a selection after PARALLEL SHMOO MENU is displayed.

- COMPOSITE SHMOO

The system waits for a selection after COMPOSITE SHMOO MENU is displayed.

5. 9. 2 PARALLEL SHMOO

```
**** PARALLEL SHMOO ****
```

```

— EXECUTE PARALLEL          . VKT
. EXECUTE PARALLEL & COMPOSITE . LPT AND VKT
. CHANGE                    . S-COMMAND
. COPY OUT

```

```
**** NOTE ****
```

```
PF1 : START MENU
```

When PARALLEL SHMOO is selected, the above menu is displayed. Select the item you want to execute using [↑] and [↓] keys, and press the return key.

Use the [PF1] key to return to the START MENU.

- EXECUTE PARALLEL

Start to operate parallel shmoo under the condition for shmoo data file name specified in the CHANGE menu.

- EXECUTE PARALLEL & COMPOSITE

Start to operate parallel shmoo under the condition for shmoo data file name specified in the CHANGE menu, and start to execute composite shmoo with created parallel file.

- CHANGE

Use CHANGE to set and change the condition for the shmoo data file name.

- COPY OUT

Specify the parallel file name and start copy.

◆ CHANGE Selection

The CHANGE menu operation depends on parallel test conditions as follows:

A

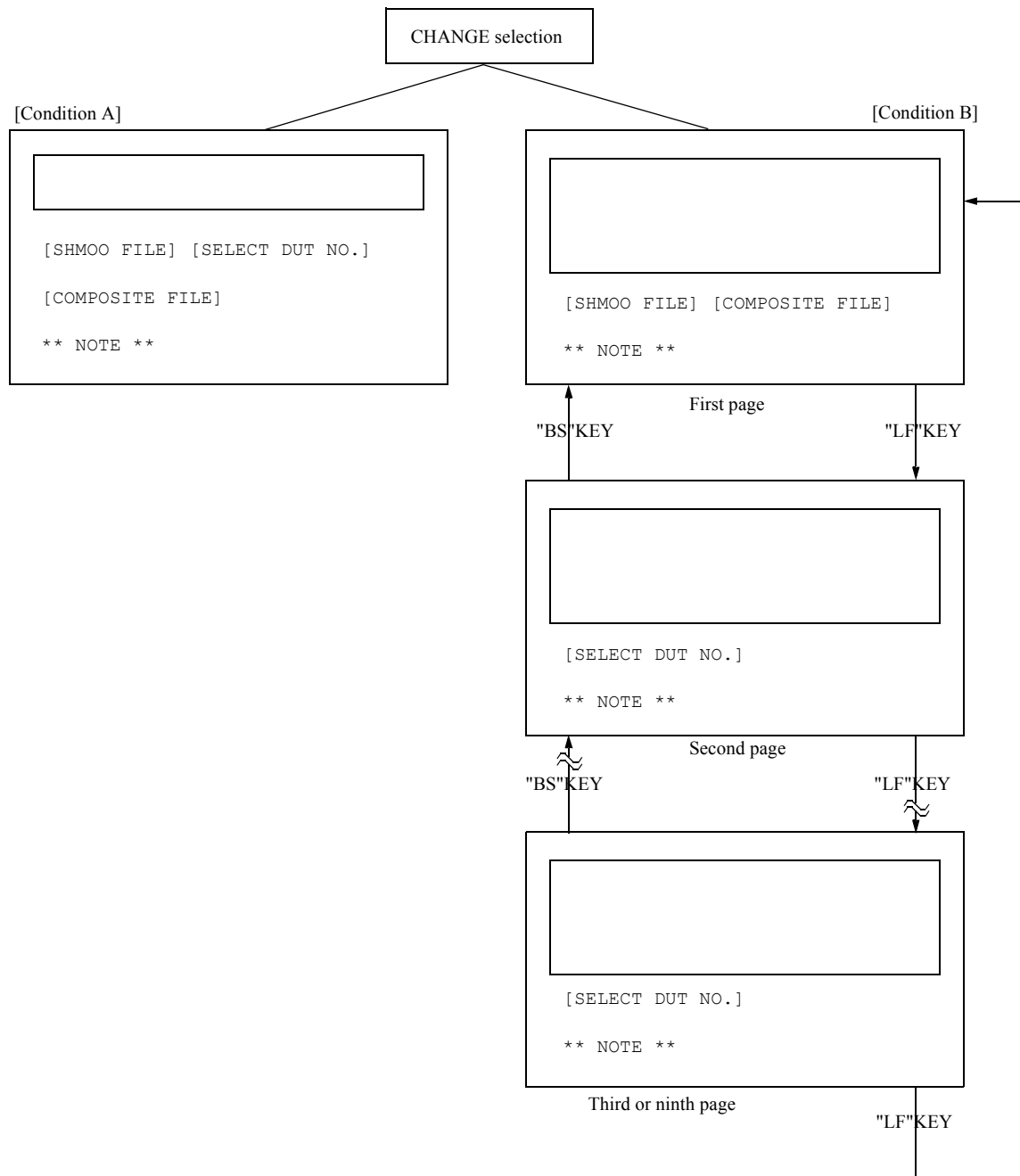
Parallel test of 16 or less DUTs per one station.

B

Parallel test of 32 or more DUTs per station.

For parallel test of 64 or fewer DUTs per test station, three pages of data entry screen are available.

For 128 or more DUTs, nine pages are available and COMPOSITE (1 to 128) data entry is allowed.



[Condition A]

CURRENT DUT ON TESTING																																				
STN1 =	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16																				
	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32																				
[SHMOO FILE] [SELECT DUT NO.]																																				
FIX(0) OR SAVE(1) =																COMPOSITE(1) =																				
FILE TAG NAME =																COMPOSITE(2) =																				
FILE SEQ NO. =																COMPOSITE(3) =																				
[COMPOSITE FILE]																COMPOSITE(4) =																				
																COMPOSITE(5) =																				
FIX(0) OR SAVE(1) =																COMPOSITE(6) =																				
FILE TAG NAME =																COMPOSITE(7) =																				
FILE SEQ NO. =																COMPOSITE(8) =																				
**** NOTE ****																																				
PF1 : PARALLEL SHMOO MENU																PF3 : FIRST MENU																				
PF2 : START MENU																PF4 : SKIP MODE																				

Select the item you want to set or change using [$\uparrow$], [$\downarrow$], [$\rightarrow$], and [$\leftarrow$] keys, and enter data.

 Tip

- Use the [**PF1**] key to return to the *PARALLEL SHMOO MENU*.
- Use the [**PF2**] key to return to the *START MENU*.
- Use the [**PF3**] key to return to the *FIRST MENU*.
- Use the [**PF4**] key to move the cursor to the beginning of input parts of [*CURRENT DUT ON TESTING*], [*SHMOO FILE*], [*COMPOSITE FILE*] and [*SELECT DUT NO.*].

[Condition B]

First page

CURRENT DUT ON TESTING																																				
STN1 =	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16																				
	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32																				
STN2 =	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48																				
	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64																				

[SHMOO FILE]	[COMPOSITE FILE]
FIX(0) OR SAVE(1) =	FIX(0) OR SAVE(1) =
FILE TAG NAME =	FILE TAG NAME =
FILE SEQ NO. =	FILE SEQ NO. =

**** NOTE ****

PF1 : PARALLEL MENU	PF3 : FIRST MENU	LF : NEXT PAGE
PF2 : START MENU	PF4 : SKIP MODE	

Select the item you want to set or change using [↑], [↓], [→], and [←] keys, and enter data.

Tip

- Use the [PF1] key to return to the PARALLEL SHMOO MENU.
- Use the [PF2] key to return to the START MENU.
- Use the [PF3] key to return to the FIRST MENU.
- Use the [PF4] key to move the cursor to the beginning of input parts of [CURRENT DUT ON TESTING], [SHMOO FILE] and [COMPOSITE FILE].
- Use the [LF] key to return to the second page.

Second page

CURRENT DUT ON TESTING																
STN1 =	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
STN2 =	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64

[SELECT DUT NO.]

COMPOSITE (1) =	COMPOSITE (9) =
COMPOSITE (2) =	COMPOSITE (10) =
COMPOSITE (3) =	COMPOSITE (11) =
COMPOSITE (4) =	COMPOSITE (12) =
COMPOSITE (5) =	COMPOSITE (13) =
COMPOSITE (6) =	COMPOSITE (14) =
COMPOSITE (7) =	COMPOSITE (15) =
COMPOSITE (8) =	COMPOSITE (16) =

**** NOTE ****

PF1 : PARALLEL MENU	PF3 : FIRST MENU	LF : NEXT PAGE
PF2 : START MENU	BS : PREV PAGE	

Select the item you want to set or change using [↑], [↓], [→], and [←] keys, and enter data.

Tip

- Use the [PF1] key to return to the PARALLEL SHOMOO MENU.
- Use the [PF2] key to return to the START MENU.
- Use the [PF3] key to return to the FIRST MENU.
- Use the [LF] key to proceed to the third page.
- Return to the first page with the [BS] key.

Third page

CURRENT DUT ON TESTING																																			
STN1 =	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16																			
	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32																			
STN2 =	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48																			
	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64																			

[SELECT DUT NO.]

COMPOSITE (17) =

COMPOSITE (18) =

COMPOSITE (19) =

COMPOSITE (20) =

COMPOSITE (21) =

COMPOSITE (22) =

COMPOSITE (23) =

COMPOSITE (24) =

**** NOTE ****

PF1 : PARALLEL MENU

PF2 : START MENU

COMPOSITE (25) =

COMPOSITE (26) =

COMPOSITE (27) =

COMPOSITE (28) =

COMPOSITE (29) =

COMPOSITE (30) =

COMPOSITE (31) =

COMPOSITE (32) =

PF3 : FIRST MENU

BS : PREV PAGE

LF : NEXT PAGE

Select the item you want to set or change using [↑], [↓], [→], and [←] keys, and enter data.

Tip

- Use the [PF1] key to return to the PARALLEL SHMOO MENU.
- Use the [PF2] key to return to the START MENU.
- Use the [PF3] key to return to the FIRST MENU.
- Use the [LF] key to return to the first page.
For parallel test of 128 or more DUTs per one test station, the fourth page is displayed when the "LF" key is pressed.
- Use the the [BS] key to return to the second page.

◆ Selecting OBJECT DUT

The DUT number (CDUT) tested currently is displayed, and the DUT number for parallel shmoo is displayed in reverse.

The DUT for parallel shmoo is selected and canceled.

[Condition A]

CURRENT DUT ON TESTING																
STN1 =	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
STN2 =	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32

[Condition B] For parallel test of 64 or fewer DUTs using one test station

CURRENT DUT ON TESTING																
STN1 =	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
STN2 =	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64

[Condition B] For parallel test of 128 or more DUTs per one test station

CURRENT DUT ON TESTING																	
P : PREV DUT																N : NEXT DUT	
STN1 =	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	
	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	
	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	
	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	
	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	
	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	
	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	

- Using the [↑], [↓], [→], and [←] keys or space key,
- move the cursor to the number to be selected or canceled, and enter the CR (carriage return).
- To recover cancel, move the cursor to the number to be recovered and enter the CR (carriage return) key.
- To select all keys, enter the [A] key.
- When the condition is B for parallel test of 128 or more DUTs per one test station, the DUT selection screen is displayed for every 128 DUTs.
Press the [N] key to proceed to the next group, or the [P] key to return to the previous group.
The [N] and [P] keys can be used when the cursor is inside the DUT selection area.

◆ File Mode

The following two modes are used to handle files.

This mode is enabled when EXECUTE PARALLEL and EXECUTE PARALLEL & COMPOSITE are executed.

FIX Mode (when 0 is input)

This mode does not execute automatic increment of file sequence No.

Only the file specified currently is saved as a result.

SAVE Mode (when 1 is input)

This mode executes automatic increment of file sequence No.

Only the file specified currently is saved as a result.

— Tip

• Fix mode

When there is the same file as the one in FIX mode, rewrite the content of the file without displaying the message.

• SAVE mode

When there is the same file as the one in SAVE mode, update the file sequence No. and create the file.

◆ Inputting the File Tag Name

Input the file tag name with a period, comma, \$ and fewer than three alphanumeric characters.

◆ Inputting the File Sequential No.

Input the file sequential number using an integer from 1 to 999.

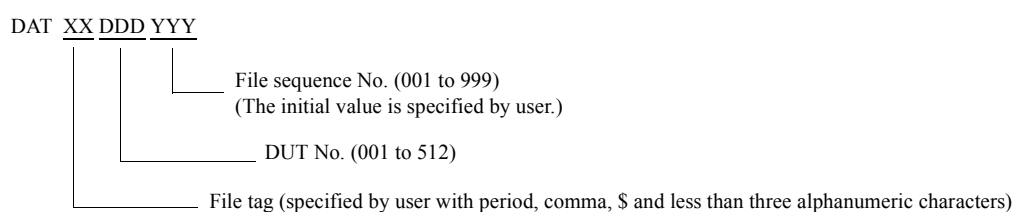
— Tip

When EXECUTE PARALLEL and EXECUTE PARALLEL & COMPOSITE are executed in the SAVE mode, the sequence No. is incremented automatically.

◆ Parallel File Name

Determination of the file name with file tag, DUT No. and sequence No. are as follows:

File name structure



◆ Inputting the DUT Definition Part (SELECT DUT NO.)

Define the DUT to execute a composite shmoo (category).

Input the DUT No. that is displayed in reverse in the CURRENT DUT ON TESTING.

So, up to four DUT Nos. can be specified every blocks.

The parallel file for the entered DUT No. is subject to composite shmoo. When EXECUTE PARALLEL & COMPOSITE is executed, composite shmoo is executed to the block defined by DUT.

The method of specifying is as follows.

Select the item you want to set or change using $\uparrow$, $\downarrow$, $\rightarrow$, and $\leftarrow$ keys, and enter data.

```
[SELECT DUT NO. ]
COMPOSITE (1) = 1,2,3,4 *1
COMPOSITE (2) =
COMPOSITE (3) = 5 to 8 *2
COMPOSITE (4) = 9,11 to 13 *3
COMPOSITE (5) =
COMPOSITE (6) = 14 to 16,10 *4
COMPOSITE (7) =
COMPOSITE (8) = 32 to 29 *5
```

(1) to (8) are the block No. of composite file.

*1 applies to the parallel files of DUT No. 1, 2, 3 and 4.

**2 applies to the parallel files of DUT No. 5, 6, 7 and 8.*

**3 applies to the parallel files of DUT No. 9, 11, 12 and 13.*

**4 applies to the parallel files of DUT No. 14, 15, 16 and 10.*

**5 applies to the parallel files of DUT No. 32, 31, 30 and 29.*

DUT No.'s are categorized in order.

➔ See "Selecting EXECUTE CATEGORY" on page 5-77 for categorization and categorized composite.

◆ Composite File Name File Name Structure

The composite file name contains the file tag, sequence No. and block No. The structure of the file name is as follows.

File name structure

DAT XX DDD YYY

File sequence No. (001 to 999)
(The initial value is specified by user.)

DUT No. (001 to 128)

File tag (specified by user with period, comma, \$ and less than three alphanumeric characters)

◆ Selecting EXECUTE PARALLEL

Selecting EXECUTE PARALLEL executes parallel shmoo.

While the result of tested SELECT DUT No. in the START menu is output to the VKT or printer, it is filed according to the file condition specified in the CHANGE menu.

Tip

- When there are more than two values of PSKIP and all results of tested same points (X- and Y-axis) of DUT for parallel shmoo are pass, SKIP is done.
- When 4 BIT CATEGORY is set to ENABLE (1), the PSKIP value is more than two, and all results of tested each pin specified in 4 BIT CATEGORY SHMOO at the same points (X- and Y-axis) of DUT for parallel shmoo are pass, SKIP is executed.
- Stop parallel shmoo using the [B] key.

• Example of Execution

[Condition A]

CURRENT DUT ON TESTING															
STN1 =	1	2	3	4	5	6	7	8							
STN2 =	9	10	11	12	13	14	15	16							
[SHMOO FILE]															
FIX(0) OR SAVE(1)= 0															
FILE TAG NAME = AB															
FILE SEQ NO. = 001															

[Condition B]

CURRENT DUT ON TESTING																
STN1 =	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
STN2 =	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64
[SHMOO FILE]																
FIX(0) OR SAVE(1)= 0																
FILE TAG NAME = AB																
FILE SEQ NO. = 001																

Therefore, the following parallel files are created.

AB001001	AB009001
AB002001	AB010001
AB003001	AB013001
AB004001	AB014001
AB007001	AB015001
AB008001	AB016001

◆ Selecting EXECUTE PARALLEL & COMPOSITE

Selecting EXECUTE PARALLEL & COMPOSITE executes parallel shmoo.

Composite shmoo is executed every block with file conditions specified by the created parallel file and the CHANGE menu.

The result of composite shmoo is filed and is output to the VKT or printer.

Tip

- Use the **[B]** key to stop parallel shmoo and composite shmoo.
- Composite shmoo is executed for the block defined by DUT.
- Category composite is executed.

- Example of Execution

[Condition A]

CURRENT DUT ON TESTING															
STN1 =	1	2	3	4	5	6	7	8							
STN2 =					9	10	11	12	13	14	15	16			

[SHMOO FILE]	[COMPOSITE DUT NO.]
FIX(0) OR SAVE(1)= 0	COMPOSITE(1) = 1-4
FILE TAG NAME = AB	COMPOSITE(2) =
FILE SEQ NO. = 002	COMPOSITE(3) = 7,8
	COMPOSITE(4) = 13-16
[COMPOSITE FILE]	COMPOSITE(5) =
FIX(0) OR SAVE(1)= 0	COMPOSITE(6) =
FILE TAG NAME = CP	COMPOSITE(7) =
FILE SEQ NO. = 001	COMPOSITE(8) =

[Condition B]
First page

CURRENT DUT ON TESTING

STN1 =	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
STN2 =	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64

[SHMOO FILE]

FIX(0) OR SAVE(1)= 0

FILE TAG NAME = AB

FILE SEQ NO. = 002

[COMPOSITE FILE]

FIX(0) OR SAVE(1)= 0

FILE TAG NAME = CP

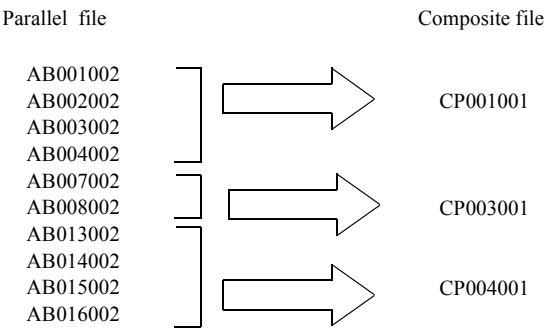
FILE SEQ NO. = 001

Second page

[SELECT DUT NO.]

COMPOSITE(1) = 1-4	COMPOSITE(9) =
COMPOSITE(2) =	COMPOSITE(10) =
COMPOSITE(3) = 7,8	COMPOSITE(11) =
COMPOSITE(4) = 13-16	COMPOSITE(12) =
COMPOSITE(5) =	COMPOSITE(13) =
COMPOSITE(6) =	COMPOSITE(14) =
COMPOSITE(7) =	COMPOSITE(15) =
COMPOSITE(8) =	COMPOSITE(16) =

Therefore, the following parallel and composite files are created.
(The same result is obtained independent of Condition A or B.)



◆ Selecting COPY OUT

The parallel file on the disk is output to the VKT or printer.

When COPY OUT is selected, the following message is displayed. Input sequential data.

FILE TAG NAME = ____ (a)

FILE DUT NO. = ____ (b)

FILE SEQ NAME = ____ (c)

(a) Inputting the Parallel File

Specifies the tag name of the parallel file. Specify the file tag name a with period, comma, \$ and less than three alphanumeric characters.

(b) Inputting the DUT No.

Specify the DUT No. of the parallel file. Specify the file sequential number with an integer from 1 to 512. The method of specifying is as follows.

- FILE DUT NO. = 2-5
Output the files of DUT No.s 2, 3, 4 and 5.
- FILE DUT NO. = 1<
Output the present files of DUT No.s 1 to 512.
- FILE DUT NO. = 10>
Output the present files of DUT No.s 1 to 10.

(c) Inputting the Sequence No.

Specify the sequence number of the parallel file. Specify the file sequential number with an integer from 1 to 999.

The method of specifying is as follows.

- FILE SEQ NO. = 1-3
Output the file of sequence No.'s 1, 2 and 3.
- FILE SEQ NO. = 1<
Output the present file of sequence No.'s 1 to 999.
- FILE SEQ NO. = 5>
Output the present file of sequence No.'s 1 to 5.

5. 9. 3 COMPOSITE SHMOO

```

**** COMPOSITE SHMOO ****
. EXECUTE      CATEGORY          . VKT
. EXECUTE      OR PASS           . LPT AND VKT
. EXECUTE      AND PASS          . S-COMMAND
. MODE (SEQUENTIAL)
— CHANGE
. COPY OUT

**** NOTE ****
      PF1 : START MENU

```

When COMPOSITE SHMOO is selected, the above menu appears. Select the item you want to execute using [↑] and [↓] keys, and press the return key.

Use the [PF1] key to return to the START MENU.

- EXECUTE CATEGORY
Executes composite category under the conditions for shmoo data file name specified in the CHANGE menu (selected in MODE).
- EXECUTE OR PASS
Executes OR PASS composite under the conditions for shmoo data file name specified in the CHANGE menu (selected in MODE).
- EXECUTE AND PASS
Executes AND PASS composite under the condition for shmoo data file name specified in the CHANGE menu (selected in MODE).
- MODE
Changes the composite execution mode.
- CHANGE
Sets and changes the condition for shmoo data file name.
- COPY OUT
Specify the composite file name and start copy.

◆ CHANGE Selection

COMPOSITE SHMOO uses SEQUENTIAL and RANDOM as composite execution modes. In this section, the condition for executing them are set and changed.

The following message appears when selecting CHANGE.

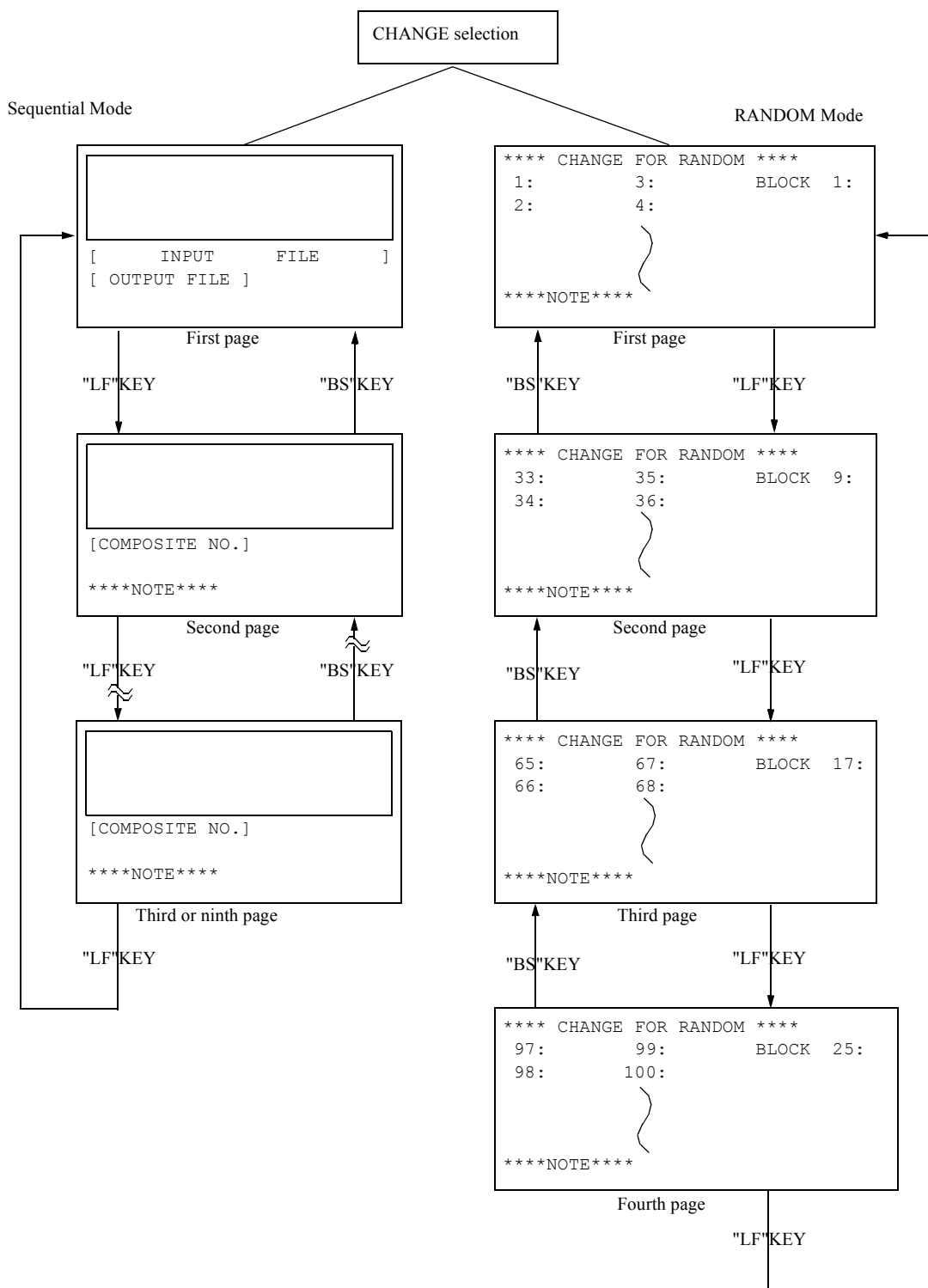
Input 0 or 1.

SEQUENTIAL MODE / RANDOM MODE (0/1)? = 0

When 0 is input, the SEQUENTIAL CHANGE menu is displayed.

When 1 is input, the RANDOM CHANGE menu is displayed.

The CHANGE screen configuration is as follows:



◆ Sequential Mode

The sequential mode executes composite shmoo with several shmoo data files (parallel and composite files) having the same tag name and sequence No. on the disk as the file name.

First page

CURRENT SHMOO FILE															
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64
65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96
97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112
113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128

[INPUT FILE]	[OUTPUT FILE]
FILE TAG NAME =	FILE TAG NAME =
FILE SEQ NO. =	FILE SEQ NO. =

**** NOTE ****

PF1 : PARALLEL MENU	PF3 : FIRST MENU	LF : NEXT PAGE
PF2 : START MENU	PF4 : SKIP MODE	

Select the item you want to set or change using [↑], [↓], [→], and [←] keys, and enter data.

Tip

- Use the [PF1] key to return to the PARALLEL SHMOO MENU.
- Use the [PF2] key to return to the START MENU.
- Use the [PF3] key to return to the FIRST MENU.
- Use the [PF4] key to move the cursor to the beginning of input parts of [CURRENT SHMOO FILE], [INPUT FILE] and [OUTPUT FILE].
- Use the [LF] key to return to the second page.

Second page

CURRENT SHMOO FILE															
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64
65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96
97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112
113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128

[COMPOSITE NO.]

COMPOSITE (1) =	COMPOSITE (9) =
COMPOSITE (2) =	COMPOSITE (10) =
COMPOSITE (3) =	COMPOSITE (11) =
COMPOSITE (4) =	COMPOSITE (12) =
COMPOSITE (5) =	COMPOSITE (13) =
COMPOSITE (6) =	COMPOSITE (14) =
COMPOSITE (7) =	COMPOSITE (15) =
COMPOSITE (8) =	COMPOSITE (16) =

**** NOTE ****

PF1 : PARALLEL MENU	PF3 : FIRST MENU	LF : NEXT PAGE
PF2 : START MENU	BS : PREV PAGE	

Select the item you want to set or change using [↑], [↓], [→], and [←] keys, and enter data.

Tip

- Use the [PF1] key to return to the COMPOSITE SHMOO MENU.
- Use the [PF2] key to return to the START MENU.
- Use the [PF3] key to return to the FIRST MENU.
- Use the [LF] key to proceed to the third page.
- Use the [BS] key to return to the first page.

Third page

CURRENT SHMOO FILE															
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64
65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96
97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112
113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128

[COMPOSITE NO.]

COMPOSITE (17) =	COMPOSITE (25) =
COMPOSITE (18) =	COMPOSITE (26) =
COMPOSITE (19) =	COMPOSITE (27) =
COMPOSITE (20) =	COMPOSITE (28) =
COMPOSITE (21) =	COMPOSITE (29) =
COMPOSITE (22) =	COMPOSITE (30) =
COMPOSITE (23) =	COMPOSITE (31) =
COMPOSITE (24) =	COMPOSITE (32) =

**** NOTE ****

PF1 : PARALLEL MENU	PF3 : FIRST MENU	LF : NEXT PAGE
PF2 : START MENU	BS : PREV PAGE	

Select the item you want to set or change using [↑], [↓], [→], and [←] keys, and enter data.

Tip

- Use the [PF1] key to return to the COMPOSITE SHMOO MENU.
- Use the [PF2] key to return to the START MENU.
- Use the [PF3] key to return to the FIRST MENU.
- Use the [LF] key to return to the first page.
If there are more than 129 shmoo data files, the fourth page of input screen is displayed.
- Use the [BS] key to return to the second page.

Selecting the Object File

The DUT No. or block No. of shmoo data file (on the disk) that contains the tag name and sequence number specified in [INPUT FILE] is displayed,

and the number for composite shmoo is displayed in reverse. In this section, the composite object file shmoo is selected and canceled.

CURRENT SHMOO FILE															
P : PREV FILE								N : NEXT FILE							
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64
65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96
97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112
113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128

- Using the [↑], [↓], [→], and [←] keys or space key,
- move the cursor to the number to be selected or canceled, and enter a carriage return.
- To select all keys, enter the [A] key.



Tip

Inputting the File Tag Name

Inputting the File Tag Name

Input the file tag name with a period, comma, \$ and less than three alphanumeric characters.

Inputting the File Sequential No.

Input the file sequential number with an integer from 1 to 999.

If there are 129 or more shmoo data files, P : PREV FILE and N : NEXT FILE are displayed, groups of 128 data files can be displayed at a time.

Press the [N] key to proceed to the next group,

or the [P] key to return to the previous group.

The [N] and [P] keys can be used when the cursor is inside the DUT number selection area.

Inputting the DUT Definition Part (COMPOSITE NO.)

Define the DUT to execute composite shmoo (category).

Input the DUT No. or the block number that is displayed in reverse in the CURRENT SHMOO FILE.

So, up to four DUT numbers or block numbers can be specified every blocks.

The parallel file for entered DUT No. or the composite file for the block No. is subject to composite shmoo. When EXECUTE CATEGORY is executed, composite shmoo is executed to the block defined by DUT.

The method of specifying is as follows.

Select the item you want to set or change using [↑], [↓], [→], and [←] keys, and enter data.

[COMPOSITE NO.]

COMPOSITE(1) = 1,2,3,4 *1	COMPOSITE(9) =
COMPOSITE(2) =	COMPOSITE(10) =
COMPOSITE(3) = 5 to 8 *2	COMPOSITE(11) =
COMPOSITE(4) = 9,11 to 13 *3	COMPOSITE(12) =
COMPOSITE(5) =	COMPOSITE(13) =
COMPOSITE(6) = 14 to 16,10 *4	COMPOSITE(14) =
COMPOSITE(7) =	COMPOSITE(15) =
COMPOSITE(8) = 32 to 29 *5	COMPOSITE(16) =

**1 applies to the files of DUT No.'s or block No. 1, 2, 3 and 4.*

**2 applies to the files of DUT No.'s or block No.'s 5, 6, 7 and 8.*

**3 applies to the files of DUT No.'s or block No.'s 9, 11, 12 and 13.*

**4 applies to the files of DUT No.'s or block No.'s 14, 15, 16 and 10.*

**5 applies to the files of DUT No.'s or block No.'s 32, 31, 30 and 29.*

DUT No. are categorized in order.

➔ See ["Selecting EXECUTE CATEGORY" on page 5-77](#) for categorization and categorized composite.

Determining the Composite Object File

- For Category Composite

The composite object file is determined by tag name and sequence number in the INPUT FILE and the DUT number specified in the DUT definition part.

- For OR PASS, AND PASS Composite

The composite object file is determined by tag name and sequence No. in the INPUT FILE and the DUT No. that is displayed in reverse in the CURRENT SHMOO FILE

Composite file Created

- For Category Composite

The composite file name is determined by tag name and sequence number in the OUTPUT FILE and the block number specified in the DUT definition part. Category composite is executed per block, so up to 128 composite files are created.

- For OR PASS, AND PASS Composite

The composite file with block No.1 is created by tag name and sequence No. in the OUTPUT FILE.

◆ RANDOM Mode

The RANDOM mode executes composite shmoo using several shmoo data files (parallel and composite files) whose consist of different tag name and sequence No. on the disk as a file name.

First page

```

**** CHANGE FOR RANDOM ****

[ INPUT FILE ]                [ OUTPUT FILE ]

1:          3:                BLOCK 1 :
2:          4:                BLOCK 2 :
5:          7:                BLOCK 3 :
6:          8:                BLOCK 4 :
9:          11:               BLOCK 5 :
10:         12:               BLOCK 6 :
13:         15:               BLOCK 7 :
14:         16:               BLOCK 8 :
17:         19:
18:         20:
21:         23:
22:         24:
25:         27:
26:         28:
29:         31:
30:         32:

**** NOTE ****
      PF1 : COMPOSITE SHMOO MENU    PF3 : FIRST MENU      LF : NEXT PAGE
      PF2 : START MENU              PF4 : SKIP MODE

```

Select the item you want to set or change using [↑], [↓], [→], and [←] keys, and enter data.

Tip

- Use the [PF1] key to return to the COMPOSITE SHMOO MENU.
- Use the [PF2] key to return to the START MENU.
- Use the [PF3] key to return to the FIRST MENU.
- Use the [PF4] key move the cursor to the beginning of input parts or [INPUT FILE] left, [INPUT FILE] right and [OUTPUT FILE].
- Use the [LF] key to return to the second page.

Second page

```

**** CHANGE FOR RANDOM ****

[ INPUT FILE ]                [ OUTPUT FILE ]

33:          35:              BLOCK 9 :
34:          36:
37:          39:              BLOCK 10:
38:          40:
41:          43:              BLOCK 11:
42:          44:
45:          47:              BLOCK 12:
46:          48:
49:          51:              BLOCK 13:
50:          52:
53:          55:              BLOCK 14:
54:          56:
57:          59:              BLOCK 15:
58:          60:
61:          63:              BLOCK 16:
62:          64:

**** NOTE ****
      PF1 : COMPOSITE SHMOO MENU    PF3 : FIRST MENU      LF : NEXT PAGE
      PF2 : START MENU              PF4 : SKIP MODE       BS : PREV PAGE

```

Select the item you want to set or change using [↑], [↓], [→], and [←] keys, and enter data.

 Tip

- Use the [PF1] key to return to the COMPOSITE SHMOO MENU.
- Use the [PF2] key to return to the START MENU.
- Use the [PF3] key to return to the FIRST MENU.
- Use the [PF4] key move the cursor to the beginning of input parts or [INPUT FILE] left, [INPUT FILE] right and [OUTPUT FILE].
- Use the [LF] key to go to the next page.
- Use the [BS] key to return to the previous page.

Third page

```

**** CHANGE FOR RANDOM ****

[ INPUT FILE ]                [ OUTPUT FILE ]

65:          67:              BLOCK 17:
66:          68:
69:          71:              BLOCK 18:
70:          72:
73:          75:              BLOCK 19:
74:          76:
77:          79:              BLOCK 20:
78:          80:
81:          83:              BLOCK 21:
82:          84:
85:          87:              BLOCK 22:
86:          88:
89:          91:              BLOCK 23:
90:          92:
93:          95:              BLOCK 24:
94:          96:

**** NOTE ****
      PF1 : COMPOSITE SHMOO MENU    PF3 : FIRST MENU      LF : NEXT PAGE
      PF2 : START MENU              PF4 : SKIP MODE       BS : PREV PAGE

```

Select the item you want to set or change using [↑], [↓], [→], and [←] keys, and enter data.

 Tip

- Use the [PF1] key to return to the COMPOSITE SHMOO MENU.
- Use the [PF2] key to return to the START MENU.
- Use the [PF3] key to return to the FIRST MENU.
- Use the [PF4] key move the cursor to the beginning of input parts or [INPUT FILE] left, [INPUT FILE] right and [OUTPUT FILE].
- Use the [LF] key to go to the next page.
- Use the [BS] key to return to the previous page.

Fourth page

```

**** CHANGE FOR RANDOM ****

[ INPUT FILE ]                [ OUTPUT FILE ]

97:          99:              BLOCK 25:
98:          100:             BLOCK 26:
101:         103:             BLOCK 27:
102:         104:             BLOCK 28:
105:         107:             BLOCK 29:
106:         108:             BLOCK 30:
109:         111:             BLOCK 31:
110:         112:             BLOCK 32:
113:         115:
114:         116:
117:         119:
118:         120:
121:         123:
122:         124:
125:         127:
126:         128:

**** NOTE ****
      PF1 : COMPOSITE SHMOO MENU    PF3 : FIRST MENU      LF : NEXT PAGE
      PF2 : START MENU              PF4 : SKIP MODE       BS : PREV PAGE

```

Select the item you want to set or change using [↑], [↓], [→], and [←] keys, and enter data.

 Tip

- Use the [PF1] key to return to the COMPOSITE SHMOO MENU.
- Use the [PF2] key to return to the START MENU.
- Use the [PF3] key to return to the FIRST MENU.
- Use the [PF4] key move the cursor to the beginning of input parts or [INPUT FILE] left, [INPUT FILE] right and [OUTPUT FILE].
- Use the [LF] key to return to the first page.
- Use the [BS] key to return to the previous page.

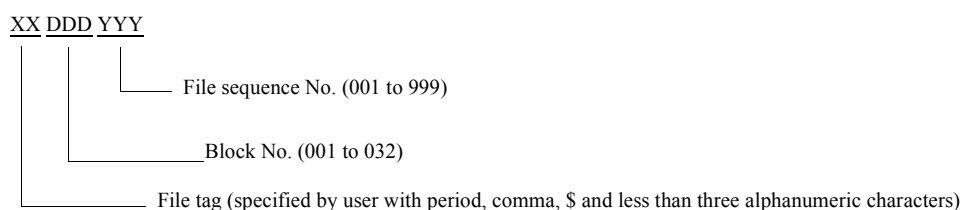
Inputting INPUT FILE

Input the parallel and composite file names which are on the disk.

Inputting OUTPUT FILE

Input the composite file name in the following format.

File name structure



Determining the Composite Object File

- For Category Composite
Divide the [INPUT] into 32 blocks (each four block), correspond the specified file to the block No. of the [OUTPUT FILE], and determine the composite object file.
- For OR PASS, AND PASS Composite
All files specified in the INPUT FILE is to be the composite object file.

Composite file Created

- For Category Composite
The composite file is created by file name specified in the [OUTPUT FILE]. Because category composite is executed per block, up to 32 composite files are created.
- For OR PASS, AND PASS Composite
The composite file is created by the first file name specified in the [OUTPUT FILE].

The relationship between [INPUT FILE] and [OUTPUT FILE] in block No. is as follows.

[INPUT FILE]	[OUTPUT FILE]	
1: 3: 2: 4:	BLOCK 1 :	Block No. 1
5: 7: 6: 8:	BLOCK 2 :	Block No. 2
9: 11: 10: 12:	BLOCK 3 :	Block No. 3
.	.	.
.	.	.
.	.	.
.	.	.
.	.	.
.	.	.
117: 119: 118: 120:	BLOCK30 :	Block No. 30
121: 123: 122: 124:	BLOCK 31 :	Block No. 31
125: 127: 126: 128:	BLOCK 32 :	Block No. 32



Tip

For the composite object file, specify the file whose X-axis step value is the same as that of the Y-axis step. The first file and the file whose X-axis or Y-axis step value is different are automatically excluded from the composite file.

◆ Selecting EXECUTE CATEGORY

Using the composite object file specified in the CHANGE menu, categorize up to four files per block for overlap. File the result on the disk with the specified composite file name and output it to the VKT or printer.



Tip

Use the **[B]** key to stop composite shmoo.

◆ Categorization

Define composite object files to the following FILE1, FILE2, FILE3 and FILE4 in specified order, overlap the results of test (PASS/FAIL) for points (X-axis, Y-axis), and categorize them into 1 to F. Categorization

Table 5-3 Categorization

Categorization	Composite file			
	FILE 4	FILE 3	FILE 2	FILE 1
Space or period	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
A	1	0	1	0
B	1	0	1	1
C	1	1	0	0
D	1	1	0	1
E	1	1	1	0
F	1	1	1	1

Object file is parallel or composite file.

1 indicates PASS. PASS data contains "*", "=", "1 to 9" and "A to F."

0 indicates FAIL. FAIL data contains "space" and "?."

◆ File Definition

- For sequential mode

First page

CURRENT SHMOO FILE

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64
65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96
97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112
113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128

[SHMOO FILE]

[COMPOSITE FILE]

FILE TAG NAME = AA

FILE TAG NAME = CC

FILE SEQ No. = 001

FILE SEQ NO. = 001

Second page

[COMPOSITE NO.]

COMPOSITE(1) = 1-4 *1

COMPOSITE(2) = 5,10-8 *2

COMPOSITE(3) =

COMPOSITE(4) =

COMPOSITE(5) =

COMPOSITE(6) =

COMPOSITE(7) =

COMPOSITE(8) =

COMPOSITE(9) =

COMPOSITE(10) =

COMPOSITE(11) =

COMPOSITE(12) =

COMPOSITE(13) =

COMPOSITE(14) =

COMPOSITE(15) =

COMPOSITE(16) =

If files are set as above, they are defined as follows.

*1 FILE1 AA001001
FILE2 AA002001
FILE3 AA003001
FILE4 AA004001

*2 FILE1 AA005001
FILE2 AA010001
FILE3 AA009001
FILE4 AA008001

- For random mode

[INPUT FILE]		[OUTPUT FILE]	
*1	1: AA001002	3: AA003002	BLOCK 1 : CC001001
	2: AA002002	4: AA004002	
*2	5: AA005002	7: AA009002	BLOCK 2 : BB001001
	6: AA010002	8: AA008003	
	}		}
	125:	127:	BLOCK 32:
	126:	128:	

If files are set as above, they are defined as follows.

```
*1 FILE1 AA001002
   FILE2 AA002002
   FILE3 AA003002
   FILE4 AA004002

*2 FILE1 AA005002
   FILE2 AA010002
   FILE3 AA009002
   FILE4 AA008003
```

Example of Execution

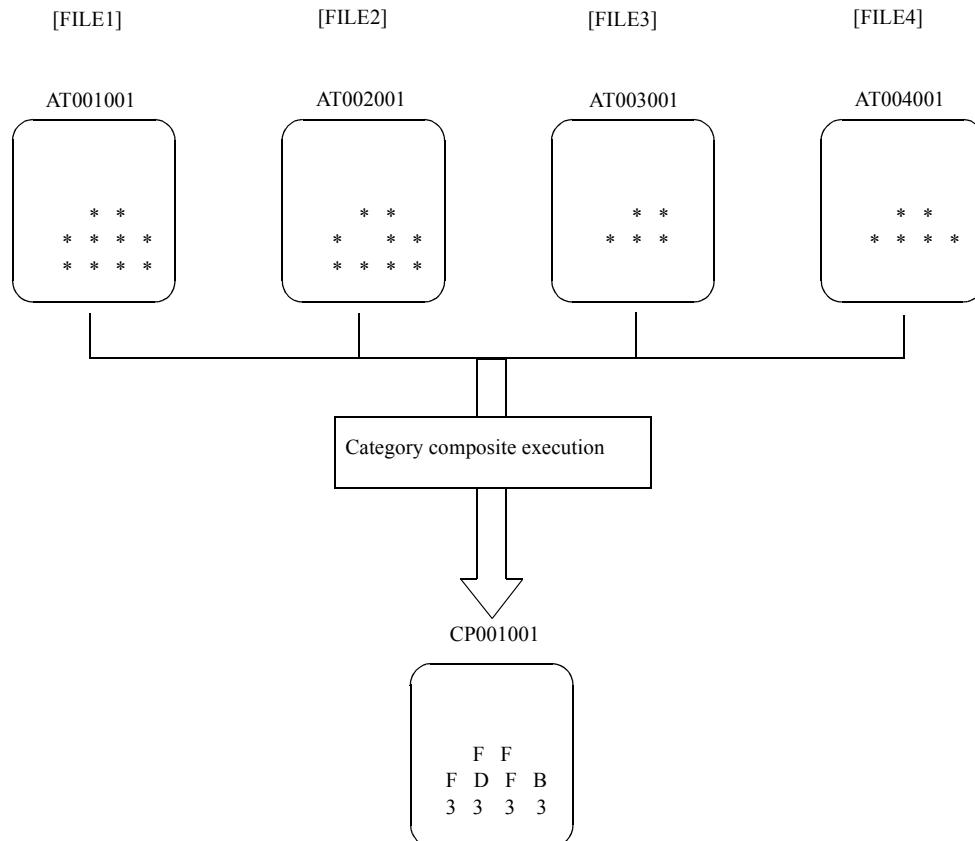
- File conditions in the SEQUENCE mode

[INPUT FILE]	[COMPOSITE NO.]
FILE TAG NAME = AT	COMPOSITE(1)=1, 2, 3, 4
FILE SEQ NO. = 001	
[OUTPUT FILE]	
FILE TAG NAME = CP	
FILE SEQ NO. = 001	

- File conditions in the RANDOM mode

[INPUT FILE]	[OUTPUT FILE]
1: AT001001 3: AT003001	BLOCK 1 : CP001001
2: AT002001 4: AT004001	

When the file condition in the SEQUENTIAL mode is the same as that in the RANDOM mode and category composite is executed, the following composite files are created and output to the VKT or the printer.



◆ Selecting EXECUTE OR PASS

Take the logical sum (OR) of the result of test for all same points (X- and Y-axis) within the composite file specified by CHANGE menu.

File the result on the disk with specified composite file name and output it to the VKT or printer.

If only one PASS is found in OR, the result is PASS.

If the result is PASS, it is *. If it is FAIL, it is space or period.

Tip

Use the **[B]** key to stop composite shmoo.

PASS data contains "*", "=", "1 to 9" and "A to F."

FAIL data contains "space" and "?."

◆ Selecting EXECUTE AND PASS

Take the logical product (AND) of the result of test for all of the same points (X- and Y-axis) with in the composite file specified by CHANGE menu.

File the result on the disk with specified composite file name and output it to the VKT or printer.

If all files are PASS is found in AND, the result is PASS.

If the result is PASS, it is *. If it is FAIL, it is a space or period.

Tip

Use the **[B]** key to stop composite shmoo.

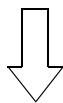
PASS data contains "*", "=", "1 to 9" and "A to F."

FAIL data contains "space" and "?."

◆ Selecting MODE

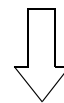
When MODE is selected, the composite execution mode (SEQUENTIAL and RANDOM) is changed.

_MODE (SEQUENTIAL)



.MODE (RANDOM)

_MODE (RANDOM)



.MODE (SEQUENTIAL)

Example of Execution in SEQUENTIAL Mode

First page

CURRENT SHMOO FILE															
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64
65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96
97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112
113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128

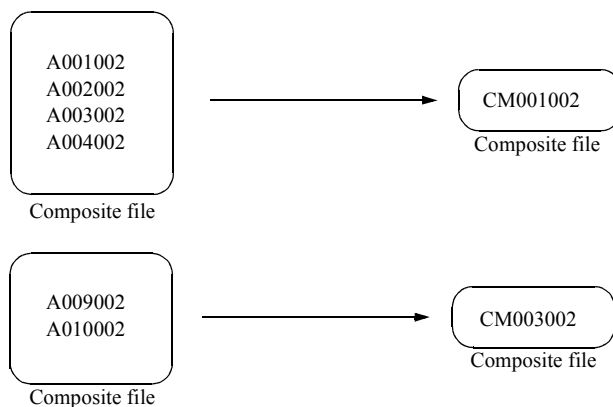
[SHMOO FILE]	[COMPOSITE FILE]
FILE TAG NAME = A	FILE TAG NAME = CM
FILE SEQ NO. = 002	FILE SEQ NO. = 002

Second page

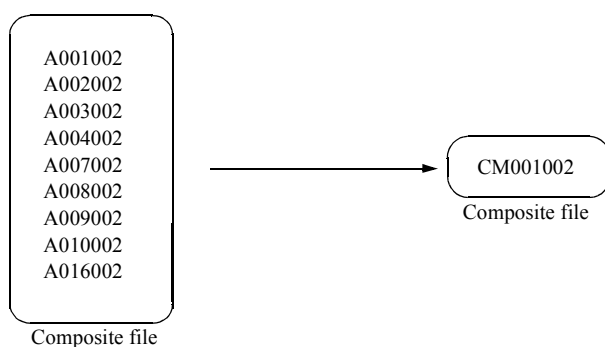
[COMPOSITE NO.]	
COMPOSITE(1) = 1,2,3,4	COMPOSITE(9) =
COMPOSITE(2) =	COMPOSITE(10) =
COMPOSITE(3) = 9,10	COMPOSITE(11) =
COMPOSITE(4) =	COMPOSITE(12) =
COMPOSITE(5) =	COMPOSITE(13) =
COMPOSITE(6) =	COMPOSITE(14) =
COMPOSITE(7) =	COMPOSITE(15) =
COMPOSITE(8) =	COMPOSITE(16) =

When composite shmoo is executed according to the above set, the following composite files are created:

- For Category Composite



- For OR (AND) PASS Composite

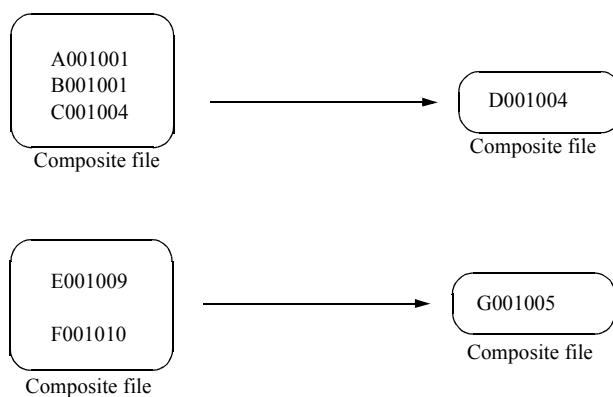


Example of Execution in RANDOM Mode

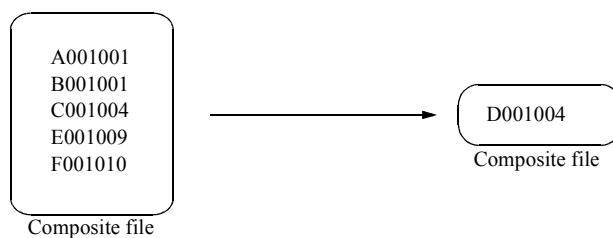
[INPUT FILE]		[OUTPUT FILE]	
1: A001001	3: C001004	BLOCK 1 : D001004	
2: B001001	4:		
5: E001009	7:	BLOCK 2 : G001005	
6: F001010	8:		
:		:	
:		:	
:		:	
125:	127:	BLOCK 32:	
126:	128:		

When composite shmoo is executed according to the above set, the following composite files are created.

- For Category Composite



- For OR (AND) PASS Composite



◆ Selecting COPY OUT

The composite file on the disk is output to the VKT or printer.

When COPY OUT is selected, the following message is displayed. Input sequential data.

FILE TAG NAME = ____ (a)

FILE BLOCK NO. = ____ (b)

FILE SEQ NAME = ____ (c)

(a) Inputting the Composite File

Specifies the tag name of a composite file. Specify the file tag name with period, comma, \$ and less than three alphanumeric characters.

(b) Inputting the Block No.

Specify the block No. of the composite file.1 Specify the file sequential number with integer from 1 to 128. The method of specifying is as follows.

- FILE BLOCK NO. = 2-5
Output the files of block No.'s 2, 3, 4 and 5.

- FILE BLOCK NO. = 1<
Output the present files of block No.'s 1 to 128.
- FILE BLOCK NO. = 6>
Output the present files of block No.'s 1 to 6.

(c) Inputting the Sequence No.

Specify the sequence number of the composite file. Specify the file sequential number using integers from 1 to 999.

The method of specifying is as follows.

- FILE SEQ NO. = 1-3
Output the file of sequence No.'s 1, 2 and 3.
- FILE SEQ NO. = 1<
Output the present file of sequence No.'s 1 to 999.
- FILE SEQ NO. = 5>
Output the present file of sequence No.'s 1 to 5.

5.10 Using of Specification Value Output

In executing the shmoo (except DC shmoo), the PASS/FAIL of the range to be specified is output according to the display mode.

For outputting the specification value, the following three settings are required.

- ENABLE/DISABLE for output of specification value
- Display mode for output of specification value
- Specification value

Each setting is explained in accordance with the operation procedure.

5.10. 1 Mode Setting for Output of Specification Value

Activate the SHM2 while it is stopped using MANUAL HOLD or MANUAL RECYCLE HOLD including the optional function tests.

```
/SHM2.␣

*** 2-D SHMOO FIRST MENU ***
      ____ CHANGE
        . START
        . PRINT
        . HEADER
        . WRITE
        . REFERENCE OFF

** NOTE **
      PF1:SHMOO EXIT
```

Select CHANGE to display CHANGE MENU.

```
*** CHANGE MENU ***

•X-AXIS, Y-AXIS and TRACKING
•OPTION-AXIS
•FUNCTION
•DC
•MODE
•BUMP
•4BIT CATEGORY SHMOO
  ____SPEC. MODE

** NOTE **
      PF1: FIRST MENU
```

Select SPEC. MODE The following message is displayed.

```
SPEC. ENABLE(1) OR DISABLE(0) : __
```

ENABLE/DISABLE of the specification value output when the shmoo (exclude DC shmoo) is executed is specified.

- When 1 is input, the specification value will be output according to the display mode.
- When 0 is input, the specification value is not output. A usual shmoo is executed.

Default is DISABLE(0).

It was specified that ENABLE(1) would be selected with X-AXIS, Y-AXIS and TRACKING of CHANGE MENU and outputs the specification value according to XSPEC-U, XSPEC-L, YSPEC-U and YSPEC-L.

For COMPOSITE SHMOO, the value of XSPEC-U, XSPEC-L, YSPEC-U, and YSPEC-L of the file specified for the first turn is continued to the file specified since the second turn.

When the file specified for the first iteration is SPEC. DISABLE(0), the specification value will not be output to the file specified after the second turn.

Selecting "1" The following message is displayed.

```
DISPLAY DEFAULT(0), REVERSE(1), REPLACE(2) OR MARKER(3) : __
```

Specify a display mode of spec value output. Input 0 to 3. Input 0 to 3.

The details of each display mode are shown in the following table.

Display mode

- DISPLAY DEFAULT(0)

	Other than 4BIT CATEGORY SHMOO COMPOSITE CATEGORY SHMOO	4BIT CATEGORY SHMOO COMPOSITE CATEGORY SHMOO
VKT	White/black reverse	White/black reverse
LPT	Character replacement	Marker display
DISK ^{*1}	Character replacement	Marker display

- DISPLAY REVERSE(1) ^{*2}

	Other than 4BIT CATEGORY SHMOO COMPOSITE CATEGORY SHMOO	4BIT CATEGORY SHMOO COMPOSITE CATEGORY SHMOO
VKT	White/black reverse	White/black reverse
LPT	White/black reverse, however, not always White/black reverse in dependence on the printers	White/black reverse, however, not always White/black reverse in dependence on the printers
DISK ^{*1}	White/black reverse	White/black reverse

- DISPLAY REPLACE(2) ^{*3}

	Other than 4BIT CATEGORY SHMOO COMPOSITE CATEGORY SHMOO	4BIT CATEGORY SHMOO COMPOSITE CATEGORY SHMOO
VKT	Character replacement	Not execute ^{*4}
LPT	Character replacement	Not execute ^{*4}
DISK ^{*1}	Character replacement	Not execute ^{*4}

- MARKER(3)

	Other than 4BIT CATEGORY SHMOO COMPOSITE CATEGORY SHMOO	4BIT CATEGORY SHMOO COMPOSITE CATEGORY SHMOO
VKT	Marker display	Marker display
LPT	Marker display	Marker display
DISK ^{*1}	Marker display	Marker display

^{*1} Disk file specified using the [S] key of the key command.

Only when SHM2 is called from a main program with the call statement, this making to the disk file is effective.

^{*2} When selecting the DISPLAY REVERSE (1), the following warning message is displayed.

Warning! Can't print reversed characters on the printers that do not
support the function.

Strike any key.

**3 When selecting the DISPLAY REVERSE (2), the following warning message is displayed.*

Warning! Can't execute 4bit category shmoo or composite category shmoo.

Strike any key.

**4 When the shmoo is executed, the following message is displayed and the shmoo cannot be executed.*

Error! Can't execute 4bit category shmoo or composite category shmoo
with the display mode 2.

Strike any key.

- Example of white/black reverse.

```

*****
*****
** *****
** *****
** *****
*****
*****
*****

```

- Example of character replacement.

```

*****
*****
** $$$$###
** $$$$###
** $$$$###
*****
*****
*****

```

The character display within the range of the specification value is replaced as follows:

PASS: "*" → "\$"

FAIL: "space", "." → "#"

- Example of marker display.

```

      ++++++
      |      |
      | ***** |
      | ***** |
      | ***** |
+    | ***** |    +
+    | ***** |    +
+    | ***** |    +
      | ***** |
      | ***** |
      | ***** |
      ++++++

```

Specification value range is displayed in the marker in the outside frame of the shmoo plot.

5.10. 2 Setting of Specification Value

```

*** CHANGE MENU ***

__ X-AXIS, Y-AXIS, TRACKING
•OPTION-AXIS
•FUNCTION
•DC
•MODE
•BUMP
•4BIT CATEGORY SHMOO
•SPEC. MODE

** NOTE **
    PF1 : FIRST MENU

```

When the item of X-AXIS, Y-AXIS, TRACKING selected, the contents of X- and Y-axis settings and tracking setting are displayed

```

X-UNIT = STRB1          Y-UNIT = HOUT1
XSTART = 22.00NS        YSTART = 2.000V
XMODIFY= 0
XDELT  = 2.000NS        YDELT  = -100.0MV
XSTEP  = 50             YSTEP  = 20
XSTOP  = 120.0NS        YSTOP  = 100.0MV
XNAME  =                YNAME  =
XDISP-A= 1              YDISP-A= 1
XDISP-B= 0.000S         YDISP-B= 0.000V
XCALI  =                YCALI  =
XSPEC-U= 82.00NS        YSPEC-U= 1.000V
XSPEC-L= 42.00NS        YSPEC-L= 110.0MV

TRACK NO. X/Y T-UNIT T-ADD2 TCAL
1          0          0
2          0          0
3          0          0
4          0          0
5          0          0
6          0          0
7          0          0

** NOTE **
    PF1 : CHANGE MENU      PF3 : FIRST MENU
    PF2 : START MENU       PF4 : UNIT NAME LIST DISPLAY

```

Enter the spec value.

Satisfy the following conditions.

```
XSPEC-U ≥ XSPEC-L
YSPEC-U ≥ YSPEC-L
```

When the value which does not satisfy these conditions is entered, the following error occurs.

```
Error! Illegal upper or lower spec. value.
      Strike any key.
```

```
X-UNIT = STRB1          Y-UNIT = HOUT1
XSTART = 22.00NS        YSTART = 2.000V
XMODIFY= 0
XDELT  = 2.000NS        YDELT  = -100.0MV
XSTEP  = 50              YSTEP  = 20
XSTOP  = 120.0NS        YSTOP  = 100.0MV
XNAME  =                 YNAME  =
XDISP-A= 1              YDISP-A= 1
XDISP-B= 0.000S         YDISP-B= 0.000V
XCALI  =                 YCALI  =
XSPEC-U= 0              YSPEC-U= 0
XSPEC-L= 0              YSPEC-L= 0
```

TRACK NO.	X/Y	T-UNIT	T-ADD2	TCAL
1				0
2				0
3				0
4				0
5				0
6				0
7				0

** NOTE **

PF1 : CHANGE MENU
PF2 : START MENU

PF3 : FIRST MENU
PF4 : UNIT NAME LIST DISPLAY

When specifying " 0 " as the initial state, each value is displayed as follows:

The initial state becomes 0.

XSPEC-U is 0

Value of one with large XSTART or XSTOP.

XSPEC-L is 0

Value of one with small XSTART or XSTOP.

YSPEC-U is 0

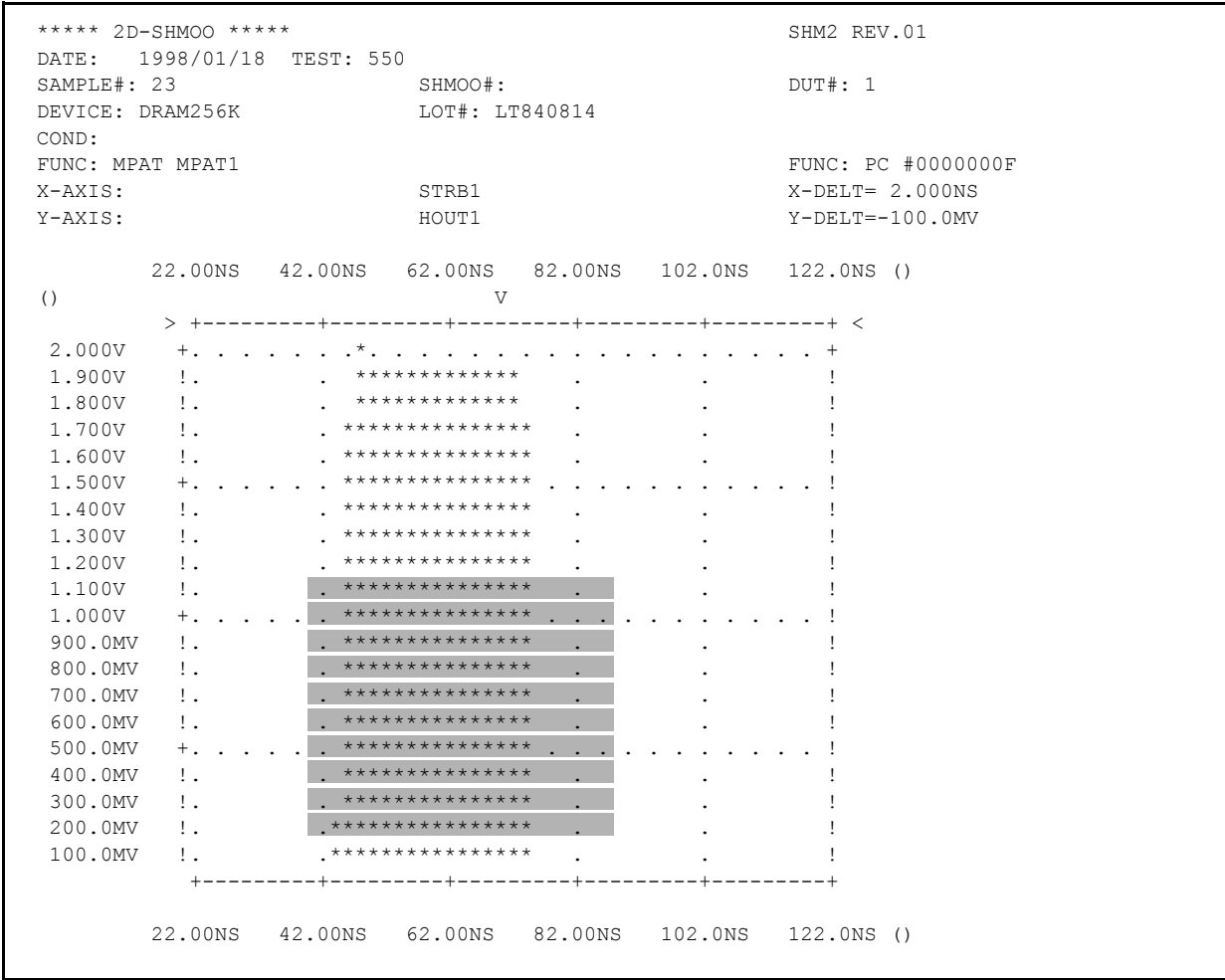
Value of one with large YSTART or YSTOP.

YSPEC-L is 0

Value of one with small YSTART or YSTOP.

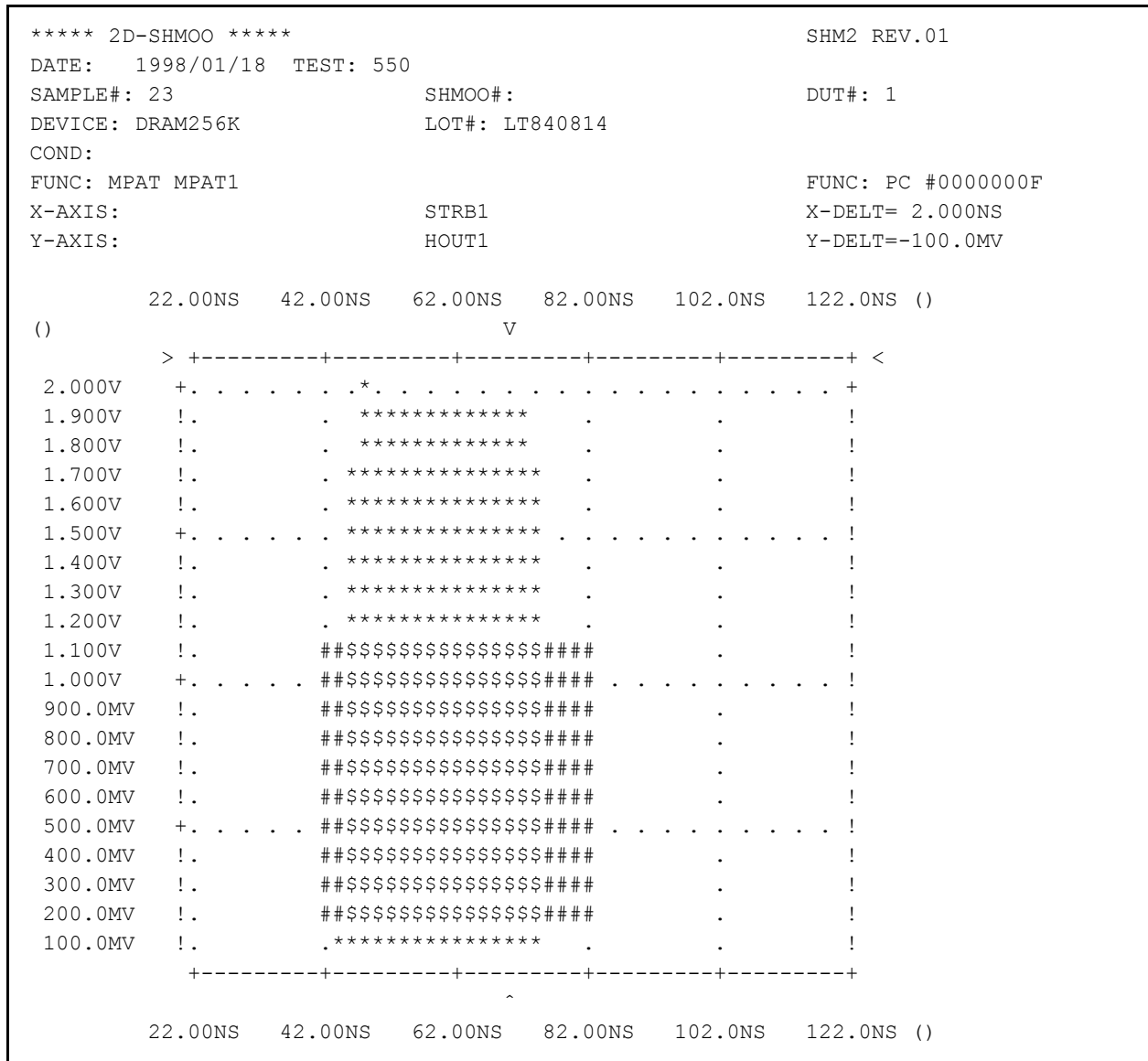
5.10. 3 Example of Execution

Specify SPEC.
ENABLE and DISPLAY DEFAULT for the display mode to execute the FUNCTION SHMOO.
• Example of VKT output



5. 10 Using of Specification Value Output

• Example of LPT output



5.11 Call from Test Plan Program

◆ Overview

To execute a shmoo plot built into the main program, call the shmoo by using the CALL statement.
In this case, the output unit can be determined using the [S] key.

```
PRO XXX

TEST XX
  Level setting
  Timing setting
  CALL SHM2("S2P01,1,FTBL/F,IERR/E")
      :
```

◆ Syntax

```
CALL SHM2("flnm,tmd,[dut/D,dmd/M,xsu-xsl/X,ysu-ysl/Y,fcnt/F,tfcnt/T,err/E]")
```

Write " at the top and at the end of an argument.
Arguments in square brackets are unordered.

◆ Argument

- For the input

flnm

Specifies a shmoo condition file. Write flnm at the top of arguments flnm must be specified.
tmd must be specified.

- When the specified file does not exist, the following message is displayed.

```
FILE DOES NOT EXIST !
      : DAT XXXX
          ↑
      Specified file name
```

- When the specified file is not recognized as a shmoo condition file, the following message is displayed.

```
ILLEGAL PARAMETER FILE!
```

```
:  DAT  XXXX      PRCODE= XXXX
```

```
      ↑
```

```
File name
```

```
      ↑
```

```
First word of the file (Four digits hexadecimal number)
```

When this message is displayed, specify a correct file name and re-execute the program.

tmd

Specifies a test mode by a integer type constant from 1 to 9. Write tmd at the second of arguments.

tmd must be specified.

- | | |
|----|--|
| 1: | Executes FUNCTION SHMOO in START MPAT(3) |
| 2: | Executes STATIC DC SHMOO |
| 3: | Executes START MPAT(3) of DYNAMIC DC SHMOO |
| 4: | Executes START MPAT(3)* of DYNAMIC DC SHMOO |
| 5: | Executes EXECUTE PARALLEL in START MPAT(3) |
| 6: | Executes EXECUTE PARALLEL & COMPOSITE in START MPAT(3) |
| 7: | Executes FUNCTION SHMOO in START MPAT(5) |
| 8: | Executes EXECUTE PARALLEL in START MPAT(5) |
| 9: | Executes EXECUTE PARALLEL & COMPOSITE in START MPAT(5) |

However, execute FUNCTION SHMOO, EXECUTE PARALLEL or EXECUTE PARALLEL & COMPOSITE in START MPAT (4) if a test mode selected from 1, or 5 to 9 is specified, and BURST is specified for BUMP Mode in a shmoo condition file.

When BURST is not specified, the menu is output, interactive mode takes effect.

dut/D

Specifies DUT No. to be tested in the functional shmoo and the DC shmoo or DUT No. to be displayed in the parallel shmoo.

If dut/D is specified, the DUT number set in a shmoo condition file is ignored and shmoo is executed using the DUT number defined by dut/D.

When tmd=1, 2, 3, 4 is specified, xsu-xsl/X can be specified.

dut/D can be omitted.

dut can be specified using an integer-type constant of 1 to 512, variable or "*."

"If "*" is specified as dut, shmoo is executed for all devices set in the station.

If variable is specified, it must be of 1-word or 2-word of integer type.

Up to two dut/Ds can be specified. If two dut/Ds are specified, they must be specified individually. If more than three dut/Ds are specified, the third and later dut/Ds are ignored.

Example

```
CALL SHM2 (" ....., 1/D, 2/D, .....,")
```

dmd/M

Specified a display mode of spec value output.

When tmd=1, 5, 6 is specified, xsu-xsl/X can be specified.

dmd/M can be omitted.

dmd can be specified by integer type constant of 0 to 3 or variable.

If variable is specified, it must be 1-word or 2-word of integer type.

xsu-xsl/X

Specifies upper limit value (xsu) and lower limit value (xsl) of X axis spec value.

When tmd=1, 5, 6 is specified, xsu-xsl/X can be specified.

Both xsu and xsl must be specified.

If xsu-xsl/X is specified, the X axis spec value set in a shmoo condition file is ignored and shmoo is executed using the X axis spec value defined by xsu-xsl/X.

xsu-xsl/X can be omitted.

xsu or xsl is specified in a real number constant or variable with the same unit as that of the value specified by the X axis unit in the shmoo condition file A constant can coexist with a variable. The variable must be defined using a 3-word, real number type.

Example

```
CALL SHM2 (" ....., 5.0V-1.0V/X, .....,")
REAL[3] XSL=1.0V
CALL SHM2 (" ....., 5.0V-XSL/X, .....,")
```

 Tip

If xsu-xsl/X is specified when the spec mode of the shmoo condition file is DISABLE, the spec mode is considered to be ENABLE and the spec display mode is considered to be DEFAULT and shmoo is executed.

ysu-ysl/Y

Specifies upper limit value (ysu) and lower limit value (ysl) of Y axis spec value.

Specification method is the same as that of xsu-xsl/X.

- For the output

fcnt/F

Specifies the variable which returns the number of fails within the spec value range for each DUT.

When tmd=1, 5, 6 is specified, xsu-xsl/X can be specified.

fcnt/F can be omitted.

fcnt must be defined using a 1-word or 2-word, integer-type array variable.

The number of factors in the array should be more than the number of devices to be tested in parallel.

(The number of factors $\geq$ The number of devices to be tested in parallel)

- Relations between data and destination variable

Variable (1) $\leftarrow$ Fail count of DUT1

Variable (2) $\leftarrow$ Fail count of DUT2

:

Variable (128) $\leftarrow$ Fail count of DUT128

-1 is returned to the variable (n) for DUTn which is not an object of shmoo.

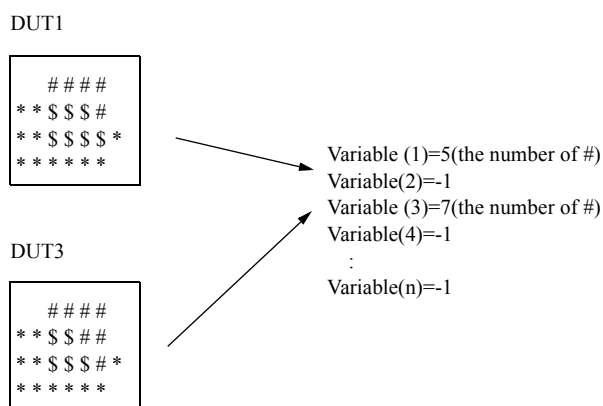
When 1-word type variable is specified, 32767 is returned if the number of fails exceeds 32767.

When a 2-word type variable is specified, 2147483647 is returned if the number of fails exceeds 2147483647.

- Count method of number of fails

(Example 1)

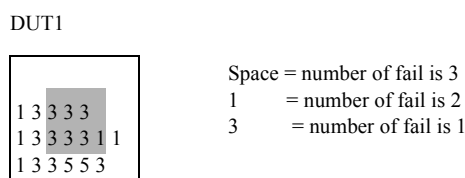
Functional shmoo and Parallel shmoo When DUT1 and DUT3 are to be tested.



(Example 2)

Functional shmoo and Parallel shmoo When DUT1 is to be tested.

When 4BIT CATEGORY is ENABLE (BIT0, BIT1 and BIT2 are defined.)



Variable(1) = space (number of fail is 3) $\times$ 5+1 (number of fail is 2) $\times$ 1+3 (number of fail is 1) $\times$ 6
= 23

Variable(2)= -1
 Variable(3)= -1
 Variable(4)= -1
 :
 Variable(n)= -1

tfcnt/T

Specifies the variable which returns a total number of fails within the spec value range for the devices to be tested.

When tmd=1, 5, 6 is specified, xsu-xsl/X can be specified.

tfcnt/T can be omitted.

tfcnt must be defined by a 1-word or 2-word, integer type variable.

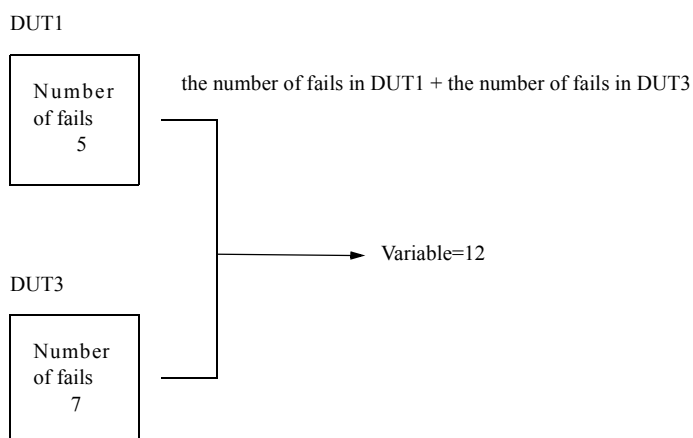
When 1-word type variable is specified, 32767 is returned if the total number of fails is over 32767.

When a 2-word type variable is specified, 2147483647 is returned if the total number of fails is over 2147483647.

- Count method of total number of fails

(Example 1)

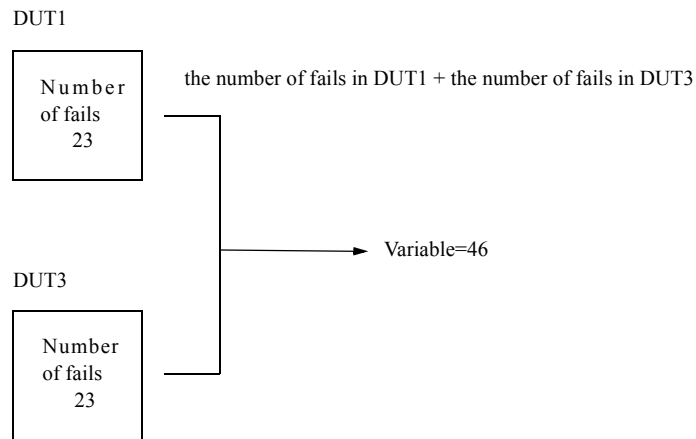
Functional shmoo and Parallel shmoo when DUT1 and DUT3 are to be tested.



(Example 2)

Functional shmoo and Parallel shmoo when DUT1 and DUT3 are to be tested.

When 4BIT CATEGORY is ENABLE (BIT0, BIT1 and BIT2 are defined.)



(Example 3)

Parallel & Composite shmoo

If the DUT numbers of composite shmoo are the following:

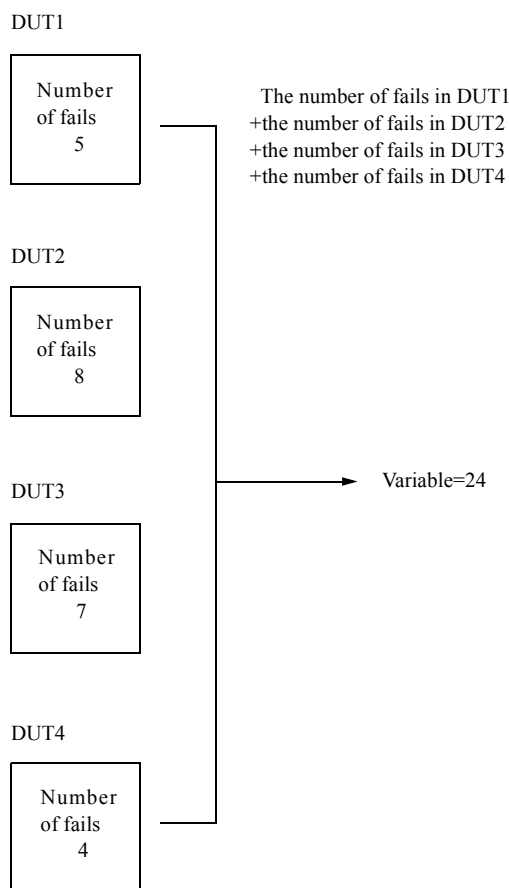
COMPOSIT(1)=1, 2

COMPOSIT(2)=3, 4

COMPOSIT(3) = No specification

:

COMPOSIT(32) = No specification



 Tip

When the DUT number of composite shmoo is overlapped as follows, the variable is equal to "the number of fails in DUT1 + the number of fails in DUT2 + the number of fails in DUT3."

COMPOSIT(1)=1, 2

COMPOSIT(2)=1, 3

err/E

Specifies a variable which returns an error code when an error has occurred.

err/E must be specified if the third and after arguments are specified

If only err/E is specified in square brackets, it is ignored.

err must be defined by 1-word or 2-word of integer type variable.

When an error occurs on the third or later argument specification, shmoo is not executed and an error code is returned to err.

➔ For details, refer to [5. 11. 1 "Error Code Table" on page 5-102.](#)

When err/E is not specified and other arguments are described correctly, the following message is displayed. In this case, select the next processing.

```
Error! code=0004 in SHM2
Please select next processing.
      CONTINUE(0) OR STOP(1) : __
```

If CONTINUE(0) is selected, shmoo is continued according to the specified arguments.

5.11. 1 Error Code Table

#0001

"/" is not specified in the third and later arguments.

#0002

A letter other than A to Z is specified just after / in the arguments.

#0003

No data is specified before / in the argument.

#0004

/E is not specified for the third argument or later.

#0005

The variable name of the argument begins with a non-alphanumeric character.

#0006

The test plan program from which the shmoo is called does not exist.

#0007

The variable specified in the argument is not defined in the test plan program from which the shmoo is called.

#0008

The type of the specified variable in the argument is not permitted.

#0009

Value other than variables, numerals and * are specified for the dut.

#000A

A value other than 1 to 512 is specified for a dut.

#000B

A value other than 0 to 3 is specified as dmd.

#000C

The number of fcnt array elements specified by the argument is not enough.

#000D

/F, /T, /X, /Y and /M cannot be specified when tmd=2, 3, 4 is specified.

#000E

The unit name of the X and Y axes must be specified for the condition file.

#000F

The unit of xsu, xsl, ysu or ysl specified by the argument is different from that of X and Y axes in the condition file.

#0010

The specified value of xsu-xsl or ysu-ysl does not satisfy "upper limit $\geq$ lower limit."

#0011

The value of xsu-xsl or ysu-ysl is not written in the format of " upper limit - lower limit ."

#0012

/F and /T cannot be specified when no spec value is specified.

6. *Three-Dimensional Shmoo Plot (SHM3)*

This section describes the Three-Dimensional Shmoo Plot (SHM3).

6. 1 SHM3 Overview

- In addition to the two-dimensional shmoo plot function, the three-dimensional shmoo plot function is available with the Z axis added to output a three-dimensional shmoo plot by changing three parameters. Like the two-dimensional shmoo function, the three-dimensional shmoo function can set parameters in the interactive mode, execute them, and store the shmoo conditions in files. Main functions are as follows:
 - X, Y, Z and T shmoo conditions can be changed in the interactive mode. This reduces the time required to prepare shmoo conditions and allows quicker testing of memory devices.
 - All the voltage/current and timing settings saved before shmoo execution can be restored after execution.
 - A number of conditions can be specified for the X, Y and Z axes, resulting in the preparation of more effective reports.
 - The MODIFY function is available for the X and Z axis.
 - Test pattern names and start PC can be changed in the utility.
 - Prepared shmoo conditions can be saved in a disk as a file for future use. and restored by specifying the file name.
 - When HIN or LIN is specified in the parameter (Y, X, Z, T), calibration can be performed by specification.
 - Independent tracking can be set.
 - The original data of the parameters set by the test plan program are indicated by arrows on the X and Y axes.
 - ROF can be executed within the shmoo. It is used when a device is replaced.
- There are two ways to use the shmoo program:
 - A shmoo for a test item is independently plotted from the keyboard instead of writing it in the test plan program. Normally, execute MANUAL HOLD in the any MEAS MPAT statement to use shmoo in the TEST item.
 - CALL statements to call shmoo are incorporated into the test plan program, and shmoos are plotted continuously by the operation above.

6. 2 Using SHM3

6. 2. 1 Conversational Mode

- 1) To plot a three-dimensional shmoo in the interactive mode, run and half a test program in MANUAL and HOLD mode at a MEAS statement including a functional test, and start SHM3.

```
/SHM3.↓
```

The following menu is displayed.

```
*** 3-D SHMOO FIRST MENU ***
  _ CHANGE *1
    . START
    . PRINT
    . HEADER
    . WRITE
    . REFERENCE OFF

** NOTE **
PF1:SHMOO EXIT
```

**1 _ indicates a cursor.*

Tip

Exit the shmoo program with [PF1] key.

- 2) Select the item you want to execute using the [↑] and [↓] keys, and press the return key.

CHANGE

Sets and changes shmoo conditions (X, Y and Z axes).

START

Starts shmoo operation according to the shmoo conditions.

PRINT

Displays headers and shmoo conditions in a list on the VKT screen.

HEADER

Enters comments such as the device name (DEVICE), shmoo number (SHMOO) and test condition (COND).

WRITE

Stores shmoo conditions in a data file.

REFERENCE OFF

Performs REFERENCE OFF in the time sequence defined by the test plan program. (Execution of the SROF statement)

- Setting of comments.

When HEADER is selected, comments can be entered.

DEVICE : -----*1 _*2

SHMOO#: -----*1 _*2

COND: -----*1 _*2

*1 *Current comment*

*2 *VKT cursor*

When the above display appears, enter the required information in the order displayed.

A maximum of 12 characters can be entered for a comment for DEVICE, 10 for SHMOO and 40 for COND.

If the current comments need not be changed, press the return key.

After all comments are entered, the screen returns to the initial menu.

A DEVICE comment can also be entered using /HEADER.

If an incorrect character is entered, the character can be corrected by pressing the [DELETE] key.

This correction function is effective until the return key is pressed.

- Setting of shmoo condition.

Select CHANGE to specify conditions for the X, Y and Z axes.

The following menu is displayed.

```
***CHANGE MENU***
. X-AXIS, Y-AXIS, Z-AXIS, TRACKING
. OPTION-AXIS
. FUNCTION
. MODE

**NOTE**
PF1:FIRST MENU
```

When the X-AXIS, Y-AXIS, Z-AXIS, TRACKING items are selected, the contents of X-, Y-, X-axis settings and tracking setting are displayed.

Select the item you want to set or change using the [↑], [↓], [→], and [←] keys, and enter data.

If the current comments need not be changed, press the return key.



Tip

Use the [PF1] key to return to the FIRST MENU.

The input formats of these conditions are shown below.

```
X-UNIT =          Y-UNIT =          Z-UNIT =
XSTART = 0         YSTART = 0         ZSTART = 0
XMODIFY= 0         ZMODIFY= 0
XDELT  = 0         YDELT  = 0         ZDELT  = 0
XSTEP  = 50        YSTEP  = 20        ZSTEP  = 10
XNAME  =          YNAME  =          ZNAME  =
XDISP-A= 1,000     YDISP-A= 1,000     ZDISP-A= 1,000
XDISP-B= 0         YDISP-B= 0         ZDISP-B= 0
XCALI  =          YCALI  =          ZCALI  =

TRACK NO. X/Y/Z T-UNIT T-ADD2   TCAL
  1                0            0
  2                0            0
  3                0            0
  4                0            0
  5                0            0
  6                0            0
  7                0            0

** NOTE **
PF1 : CHANGE MENU      PF3 : FIRST MENU
PF2 : START MENU      PF4 : UNIT NAME LIST DISPLAY
```

 Tip

Use the [PF1] key to return to the CHANGE MENU.

Use the [PF2] key to return to the START MENU.

Use the [PF3] key to return to the FIRST MENU.

Pressing the [PF4] key displays the following programmable unit names:

```

***** UNIT NAME LIST *****

ACLK1-128, BCLK1-128, CCLK1-128 : input clock timing edge
PCLK1-4      : PLL clock timing edge
RATE        : test period clock
DREL1-256    : leading edge timing of the driver enable clock
DRET1-256    : trailing edge timing of the driver enable clock
STRB1-256    : output strobe for edge comparison
WSTRBL1-128 : leading edge timing for window comparison
WSTRBT1-128 : trailing edge timing for window comparison
HIN1-16      : high level voltage for the driver
LIN1-16      : low level voltage for the driver
HOUT1-16     : high level voltage for the comparator
LOUT1-16     : low level voltage for the comparator
VT1-16       : voltage for the 50 ohm dut terminator line
HINH1-16     : high level voltage for the HV driver
LINH1-16     : low level voltage for the HV driver
VS1-12       : voltage for the power source pins
HIL1-16      : programmable load current on high level output
LIL1-16      : programmable load current on low level output
HDCLP        : high level dynamic clamp
LDCLP        : low level dynamic clamp

*** NOTE ***
STRIKE ANY KEY TO RETURN

```

The input formats of these conditions are shown below.

- Setting of conditions for X, Y and Z axes

X-UNIT, Y-UNIT, Z-UNIT

Depending on a given test system, there are units that cannot be specified.

➔ For details, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498\)](#).

Select one of the followings as a unit name for the X, Y, Z axes of the shmoo.

VS1 to 12

VT1 to VT 16 when VT is specified on the T5593, T5592, and T5591, the same value is set for the comparator and driver.

HIN1 to 16 LIN1 to 16

HOUT1 to 16 LOUT1 to 16

HINH1 to 16 LINH1 to 16

HIL1 to 16 LIN1 to 16

HDCLP LDCLP

RATE <1 to 16>

ACLK1 to 128 <1 to 16> BCLK1 to 128 <1 to 16> CCLK1 to 128 <1 to 16>

DREL1 to 256 <1 to 16> DRET1 to 256 <1 to 16>

STRB1 to 256 <1 to 16> WSTRBL1 to 128 <1 to 16> WSTRBT1 to 128 <1 to 16>

PCLK1 to 4 (When PCLK is specified, the START PCLK statement is executed in the shmoo. On completion of the execution, the previous PCLK operating status is restored.)

XSTART, YSTART, ZSTART

Specify the initial values for the X, Y and Z axes.

XDELTA, YDELTA, ZDELTA

Specify the step values for the X, Y and Z axes.

XSTEP, YSTEP, ZSTEP

Specify the number of test steps for the X, Y and Z axes.

The maximum numbers of steps are 100 for X, 9999 for Y and 36 for Z.

XNAME, YNAME, ZNAME

Used as comments on X, Y and Z axes. (Up to 14 characters)

YDISP-A, YDISP-B

Specify the coefficient and offset value used to display the Y-axis. Assuming, for example, that the Y value is "y", the value of $A \times y + B$ is displayed.

A represents a real number and B represents a real number expressed in the same units as the YSTART value.

A = 1.000 and B = 0 in the initial status.

XDISP-A, XDISP-B

Have the same functions as YDISP-A and YDISP-B.

ZDISP-A, ZDISP-B

Have the same functions as YDISP-A and YDISP-B.

XCALI, YCALI, ZCALI

Specify the number of calibration increments for the X, Y and Z axes. The number of calibration increments must be specified as a positive integer. This number is effective only when HIN or LIN is selected in the UNIT item.

To perform calibration, MODE must be selected from the CHANGE MENU screen, and ENABLE(=1) must be selected in response to message "CAL ENABLE(1) OR DISABLE(0)=".

When an ordinary shmoo is plotted, the calibration facility need not be specified. However, this facility must be specified to plot a shmoo in the following cases: The driver levels (HINn, LINn) are selected in the X, Y and T parameters. Fluctuation of the rise and fall times cannot be ignored in evaluating shmoos at sites such as the EPROM driver station.

Limitation

The operation of the shmoo with calibration takes approximately two minutes per point. (Time depends on the number of pins used.)

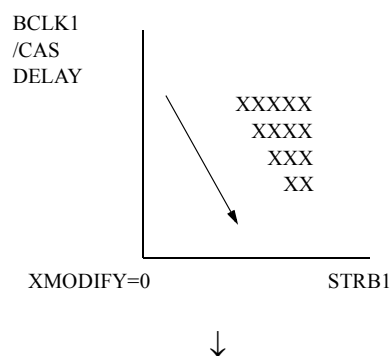
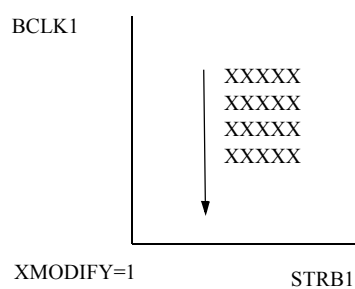
The calibration setting is disabled on the T5593, T5592 and T5591. (The menu is not displayed, either.)

XMODIFY

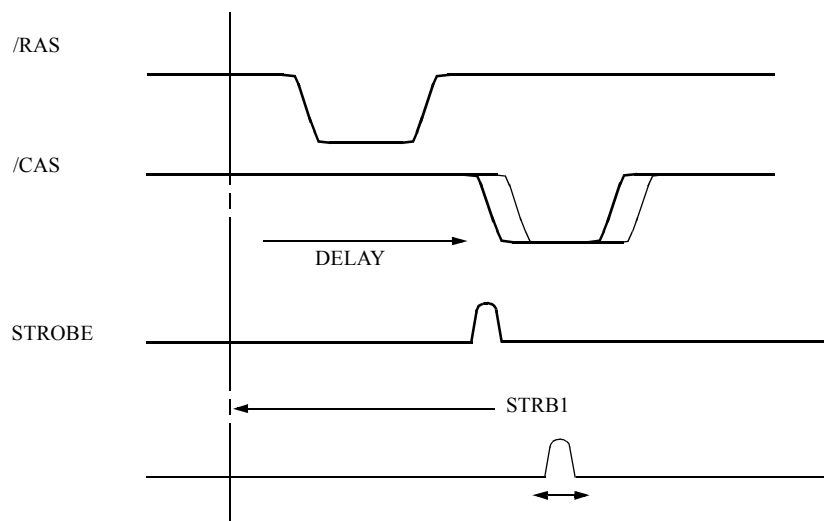
Usually set at 0. When set at 1, the initial value of the X axis is the result of the following expression:

Actual initial value of the X axis = Y-axis value + XSTART value.

That is, the initial value of the X axis can be changed according to the Y-axis value. This method is used to convert [Figure 6-1](#) into [Figure 6-2](#) which is easier to read when a dynamic RAM device, for example, is to be evaluated. In [Figure 5-1](#), /CAS DELAY values are specified on the Y axis, and STROBE for access timing from /CAS is specified on the X axis.

Figure 6-1 Plot without MODIFY**Figure 6-2 Plot with MODIFY**

Data in [Figure 6-1](#) must be read diagonally. Data in [Figure 6-2](#) must be read vertically. [Figure 6-3](#) shows an example of DELAY changed with /CAS along the Y axis. In this case, unless changed each time, the initial values of STROBE of the X axis are displayed diagonally as shown in [Figure 6-1](#).

Figure 6-3 Memory Signal and XMODIFY**ZMODIFY**

Usually set at 0. When set at 1, the initial value of the Z axis is the result of the following expression:

Actual initial value of the X axis = Y-axis value + ZSTART value.

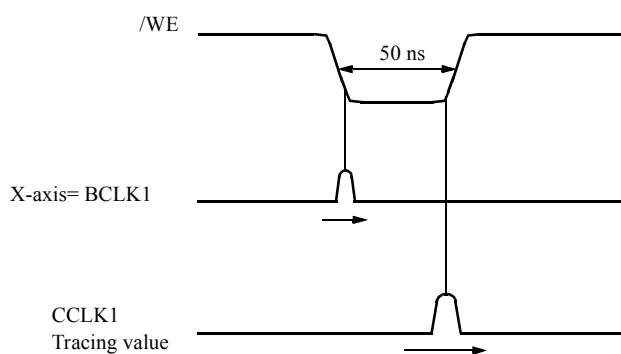
➔ For details, refer to ["XMODIFY" on page 6-7](#).

- **Setting of TRACKING conditions**

Unit names which must be modified and tracked according to the X, Y and Z axes and axis values to be increased can be specified.

A maximum of seven tracking values for the X, Y and Z axes can be specified.

As shown in the figure below, /WE signal of a memory device is operated as a /RZO waveform using BCLK1 and CCLK1.



The /WE signal in the memory device is operated in the M1 waveform by using BCLK1 and CCLK1, as shown in the figure below.

The /WE waveform pulse width is constant, and DELAY is changed on the Y axis. When BCLK1 is specified in the Y axis, the TPAR1 set as the tracking value is set as shown below to maintain the /WE waveform pulse width at 50 ns.

TRACK NO.	X/Y/Z	T-UNIT	T-ADD2	TCAL
1	Y	CCLK1	50.00NS	0

X/Y/Z

Specifies the axis to be tracked. Uses X, Y, Z to specify.

T-UNIT

Specifies the name of the unit to be tracked.

T-ADD2

Sets the value to be added to the current value of the axis to be tracked.

The unit of the value must match that of the value of the axis to be tracked.

TCAL

Specifies the number of calibration increments for the axis to be tracked as a positive integer.

When calibration is specified for the axis to be tracked, the smaller of the tracking value and the number of calibration increments specified for the axis to be tracked is given higher priority.

Effective only when HIN or LIN is specified in the TCALI tracking value.



Limitation

To cancel specification of a tracking unit set once, enter the space and the return key when specifying T-UNIT.

Tracking numbers must be specified serially.

When TPAR1, TPAR1 and TPAR3 are specified, they are all determined to be effective. But if TPAR1, TPAR2 and TPAR4 are specified and TPAR3 is omitted, TPAR4 is ignored and only TPAR1 and TPAR2 are determined to be effective.

- Setting separate tracking conditions

Independent tracking is used to set conditions independent from the X- and Y-axis setting values. Normal tracking sets the same interval; independent tracking can set different intervals.

A maximum of 100 separate tracking conditions can be specified for the X, Y, and Z axes by selecting OPTION-AXIS from the CHANGE MENU screen.

The screen format is as follows:

ITEM	Y/X/Z	UNIT	START	DELT
1			0	0
2			0	0
3			0	0
4			0	0
5			0	0
6			0	0
7			0	0
8			0	0
9			0	0
10			0	0
11			0	0
12			0	0
13			0	0
14			0	0
15			0	0
16			0	0
17			0	0
18			0	0
19			0	0
20			0	0
** NOTE **				
LF: NEXT PAGE PF1: CHANGE MENU PF3: FIRST MENU				

Use the [LF] key to feed the page.

Use the [PF1] key to return to the CHANGE MENU.

Use the [PF3] key to return to the FIRST MENU.

Move the cursor using [↑], [↓], [→], and [←] keys, and enter the following data.

X/Y/Z

Specify the axis to be set.

UNIT

Specify the unit name. It can be specified by the same method as that used for the Y and X axes.

START

Specify the initial value.

DELT

Specify the step value.

To delete a data item set once, position the cursor at the item and enter the space and the return key.

To change a data item set once, position the cursor at the item and enter data.

X/Y/Z, UNIT, START and DELT must be entered in this order.

An example of a separate tracking condition setting is as follows.

ITEM	Y/X/Z	UNIT	START	DELT
1	Y	HIN1	5.000V	-100.0mV
2				
.				
.				
.				

- Setting of FUNCTION pattern

Select FUNCTION from the CHANGE MENU.

```
MAIN PAT(0), CHANGE PAT(1) = _
```

Select MAIN PAT(0) to give higher priority to the user program pattern (MPAT).

Select CHANGE PAT(1) to specify a pattern different from the user program pattern.
When CHANGE PAT(1) is selected, the following message is displayed:

```
FUNCTION = _
```

In response to this message, enter a FUNCTION name with a character string consisting of 32 characters or less.



Tip

When CHANGE PAT(1) is selected, the FUNCTION pattern (MPAT) is transferred to the PG. The pattern is not deleted from the PG even after shmoo execution or /SHM3 exit.

- Specification of shmoo operation mode

When MODE is selected from the CHANGE MENU screen, the messages below are displayed. Enter data sequentially.

PAGE =

—

CAL ENABLE (1) OR DISABLE (0) =

—

PAGE

PAGE = 1: clears the VKT screen and feeds one page for the line printer when a shmoo plot is output.

PAGE = 0: does not perform the operation above.

CAL ENABLE (1) OR DISABLE (0)

In normal shmoo operation, DISABLE (0) is specified.

Details are described later. The procedure to output a shmoo is briefly explained here.

When CAL ENABLE(1) is selected, the number of calibration increments specified for the X, Y, Z and T parameters is effective, and calibration is performed.

When CAL DISABLE(0) is selected, calibration is not performed even if the number of calibration increments is specified.

When CAL ENABLE (1) is selected, the following is displayed:

```
PAGE = 0                ↵
CAL ENABLE (1) OR DISABLE (0) = 0  1↵
VAR NAME  =
          AA↵
FILE TAG  =
          BB↵
CAL STRING =
DRREF = 2.4V,  ..... ↵
```

For "VAR NAME =", input the string variable name where the calibration conditions are stored when CALB is specified in the user program.

Enter as "FILE TAG =", the acronym (26 characters or less) of the name of the CAL attribute file used when calibration is performed in the shmoo.

For "CAL STRING =", input the calibration condition to be added or changed.

Be sure to specify "VAR NAME" and "FILE TAG" unless they have been specified previously.

➔ See [6. 4 "SHM3 Operation Using the Calibration Facility" on page 6-22](#) for shmoo with the calibration facility.

- 3) When all required data is entered, return to the FIRST MENU screen, and select PRINT. The message below is displayed.

To output data to the VKT, select VKT. To output data to both the VKT and the LPT, select LPT & VKT. When either of the above is selected, the shmoo conditions specified up to that time are displayed.

```
. VKT
. LPT & VKT
```

- 4) When START is selected from the FIRST MENU screen to output a shmoo plot, the following menu is displayed:

```
****START MENU****
. SHMOO                . VKT
. COPY-OUT             . LPT AND VKT
. HEADER               . S-COMMAND
. SELECT DUT NO.

***NOTE***
PF1:FIRST MENU
```

If S-COMMAND is selected, the shmoo plot is output to a file specified by the key command [S].



Tip

Use the [PF1] key to return to the FIRST MENU.

COPY-OUT

To check the finished shmoo plot composition without starting the pattern generator, select COPY-OUT and then select either VKT or LPT AND VKT. This displays the X-, Y- and Z-axis values with no shmoo data.

The shmoo program has a 5000-character buffer. After the shmoo is executed, the result is stored in this buffer as symbols. Therefore, the result can be displayed any number of times by using the COPY-OUT function.

This buffer is cleared under the following conditions:

- When /SHM3 is started.
- When SHMOO is selected from the START MENU screen, and a shmoo program is executed.

If the conditions of the X, Y and Z axes are changed after shmoo program execution, the symbols in the shmoo plot are not displayed in plot order, but an outline of the X, Y and Z axes can be seen.

HEADER

Header information output when a shmoo is executed can be controlled.

When a shmoo program is started by entering /SHM3↵, header information can be output.

To suppress header information, select HEADER. The following message is displayed:

NO HEADER!!

As a result, no header information is output.

To cancel the suppression of header information, select HEADER again. The following message is displayed:

HEADER ADDITION!!

Header information is also output.

SELECT DUT NO.

Specify DUT NO. of the shmoo to be plotted.

When SELECT DUT No. is selected, the message below is displayed. Specify DUT number.

DUT# =

—

DUT NO. can be specified up to 2 using integers from 1 to 512.

To plot shmoos for all DUTs set in the station, enter ALL↵, or A↵.

To cancel the current specification, enter CLEAR↵ or C↵.

Example

DUT# =4 12 5, 13<LF>

↑

↑

Newly set value

Currently set value

 Tip

As the shmoo plot DUT NOs use DUT numbers, DUT NO starts from 1 even when a measurement is executed at the 2nd station using the program which has been compiled with the socket program for single-device parallel testing.

SHMOO

When an output device is specified after SHMOO is selected, a line of the Y axis is executed and displayed.

An example is shown in the log file described later.

If the result is incorrect, modify according to the previously described procedure, and execute again.

Notes on the three-dimensional shmoo appear below.

- Are the TRACKING parameter specification and unit correct?
- Do the set values of the strobes exceed the set range (four times the cycle time) when these values are changed?

If they exceed it, a question mark (?) is displayed for the range.

- When VIH (HIN) and VIL (LIN) are specified, $VIH < VIL$?

If so, a question mark (?) is displayed for the range.

To interrupt shmoo program execution, press the [B] key. The shmoo program then enters the menu command wait state.

BREAK! EXECUTION ABORTED!!

 Tip

When the shmoo program is interrupted, only the part of the shmoo program which has been executed is stored in the 5000-character buffer.

SRON is performed as an operation within the shmoo program prior to a triple loop. Then, when the triple loop ends or is interrupted, SROF is performed.

Shmoo program conditions for the X, Y and Z axes and for T are saved for all level timing values, and after the triple DO loop ends, they are recovered. Therefore, the main program conditions specified when shmoo was called remain unchanged. Only the pattern specified by FUNCTION is changed, and it remains in the pattern generator.

When the axis unit is VSn, the clamp current values are processed so that the values specified in the test plan program are effective.

There is no restriction on the step value for the Y axis. Overflow characters are not stored in the 5000-character buffer, and no data is written to the buffer or indicated by the COPYOUT function. To plot such a large shmoo, output it directly to the VKT or LPT when SHMOO is selected.

PASS display uses a symbol of the level at which the third dimension loop (Z axis loop) passes first.

A blank () or a dot (.) indicates FAIL.

A dot (.) is used as a separator to make a shmoo plot easy to read.

If a DUT number that does not exist is selected and the shmoo program is executed, the message below is displayed.

```
DUT# : n NOT EXIST!
STRIKE ANY KEY!
```

- 5) After the shmoo plot is output, the shmoo program conditions can be stored in a disk file for future use in the test plan program as a CALL statement.

When WRITE is selected from the FIRST MENU screen, the following message is displayed.

```
NEW DATA FILE-NAME = ?
```

In response to the message, enter a data file name consisting of up to 32 characters.

Note that if the current customer file has the same name as a data file, it is deleted unconditionally.

The shmoo program conditions are stored in data file AQ3.SM in /SHM3. Therefore, the previously used shmoo program conditions can be reused after /SHM3 ends. However, these conditions are not stored if the CTRL and C keys are pressed simultaneously or the RESET switch is turned on during execution of /SHM3.

If the current customer file contains no data file, the data file read function searches for a customer DP0. This operation must be performed carefully when the DP0 has the same name as a file.

The file name S3P01 is entered in response to the message below.

```
NEW DATA FILE-NAME = ? S3P01.␣
```

When the shmoo is to be started, creating a data file as name S3P01, for example, the shmoo condition becomes S3P01 by entering **/SHM3 S3P01,1**. Then the system starts measurement without entering the conversational mode.

When the data file name /SHM3 is entered, the same operation as that specified by **/SHM3 AQ3.SM** is performed.

- 6) When REFERENCE OFF is selected from the FIRST MENU screen, REFERENCE is turned off according to the TIME statement defined by the test plan program.

(REFERENCE OFF is used to re-execute the shmoo after replacing devices.) DC SHMOO Use Method

6. 2. 2 Call from Test Plan Program

- To execute a shmoo plot built into the main program, call the three-dimensional shmoo by using the CALL statement.

```
PRO   xxx
TEST  xx
      Level setting
      Timing setting
CALL SHM3( S3P01, n )
      :
      :
```

Always specify "n". If it is omitted, the shmoo menu is displayed and shmoo plotting is delayed.

Executes FUNCTION SHMOO in START MPAT(3) if n=1

Executes FUNCTION SHMOO in START MPAT(5) if n=7

- In this case, the output unit can be determined using the [S] key.

Tip

- If a different name is specified as the shmoo condition file, the following message is displayed:

```
FILE DOES NOT EXIST !
      : DAT   xxx
      :       ↑
      :       Different name
```

- *When the specified file is not recognized as a shmoo condition file, the following message is displayed.*

ILLEGAL PARAMETER FILE!

: DAT xxxx PRCODE = xxxx

↑

↑

File name

First word of the file (Four digits hexadecimal
number)

When this message is displayed, specify a correct file name and re-execute the shmoo program.

6. 3 SHM3 Autoscaling

Generally, three conditions (the initial value, the step value and the number of steps) must be specified for each parameter. The autoscaling facility automatically calculates the three conditions based on the value of the parameter in the test program.

Method of Use

The START, DELT and STEP values are determined by selecting X-AXIS, Y-AXIS, Z-AXIS and TRACKING from the CHANGE MENU screen and entering a unit name.

The unit name to be set must be the same as the current unit name.

Scaling range

A scaling range is set in the 0 and ∞ directions from the original pointer.

Auto scaling setting is shown in [Table 6-1](#), [Table 6-2](#) and [Table 6-3](#). When the autoscaling range exceeds the unit setting range, the autoscaling range is the 50 steps, 20 steps or 10 steps from the unit limit value in the 0 or ∞ direction.

Table 6-1 X-axis Autoscaling Setting

Unit name	DELT value	Number of steps in the 0 direction	Number of steps in the ∞ direction
RATE, ACLK, BCLK, CCLK, DREL, DRET, STRB, WSTRBL, WSTRBT	2 NS	24	26
PCLK	250 PS	24	26
VS, HIN, HOUT, LIN, LOUT, VT, HINH, LINH	40 MV	24	26
HIL, LIL	100 UA	24	26
HDCLP, LDCLP	32 MV	24	26

Table 6-2 Y-axis Autoscaling Setting

Unit name	DELT value	Number of steps in the 0 direction	Number of steps in the ∞ direction
RATE, ACLK, BCLK, CCLK, DREL, DRET, STRB, WSTRBL, WSTRBT	2 NS	9	11
PCLK	250 PS	9	11
VS, HIN, HOUT, LIN, LOUT, VT, HINH, LINH	100 MV	9	11

Unit name	DELT value	Number of steps in the 0 direction	Number of steps in the ∞ direction
HIL, LIL	200 UA	9	11
HDCLP, LDCLP	64 MV	9	11

Table 6-3 Z-axis Autoscaling Setting

Unit name	DELT value	Number of steps in the 0 direction	Number of steps in the ∞ direction
RATE, ACLK, BCLK, CCLK, DREL, DRET, STRB, WSTRBL, WSTRBT	2 NS	4	6
PCLK	250 PS	4	6
VS, HIN, HOUT, LIN, LOUT, VT, HINH, LINH	200 MV	4	6
HIL, LIL	400 UA	4	6
HDCLP, LDCLP	128 MV	4	6

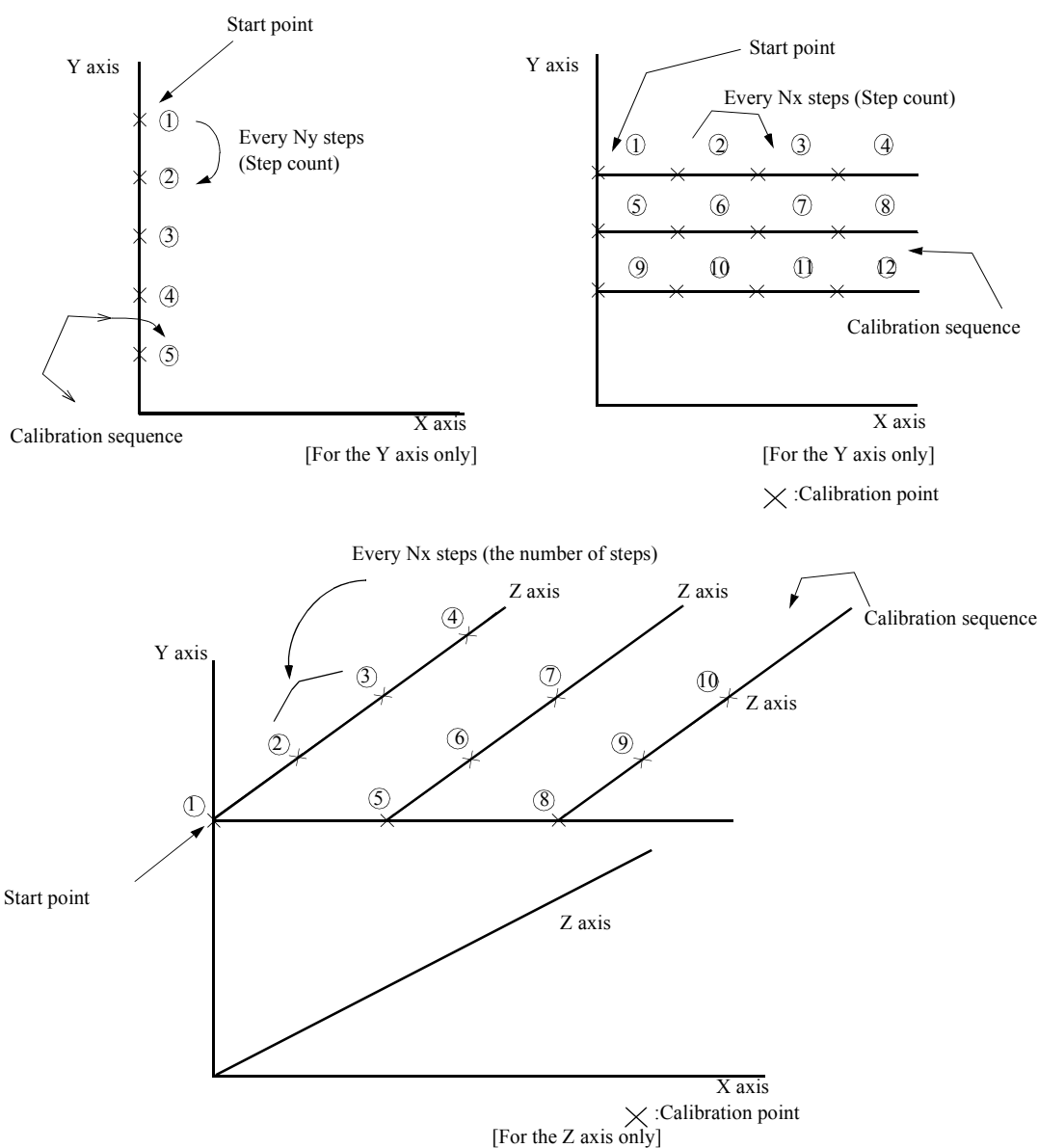
6. 4 SHM3 Operation Using the Calibration Facility

Shmoo with calibration can be used when the rising and falling edges are changed as with the EPROM driver station.

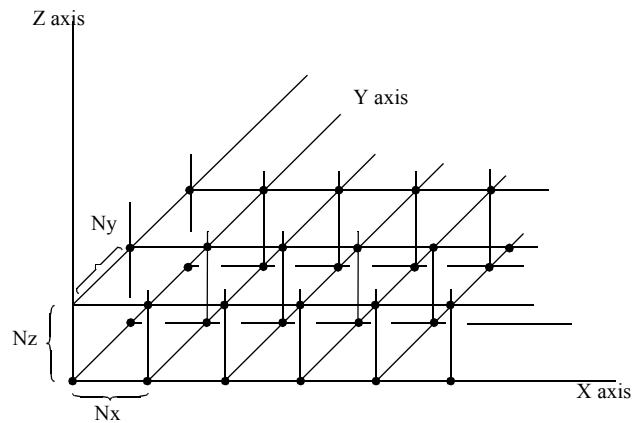
Tip

*The operation of the shmoo with calibration takes approximately two minutes per point.
(Time depends on the number of pins used.)*

- Calibration points



Three-dimensional calibration image when calibration steps of the X, Y and Z axes are N_x , N_y , and N_z , respectively.



N_x , N_y , N_z

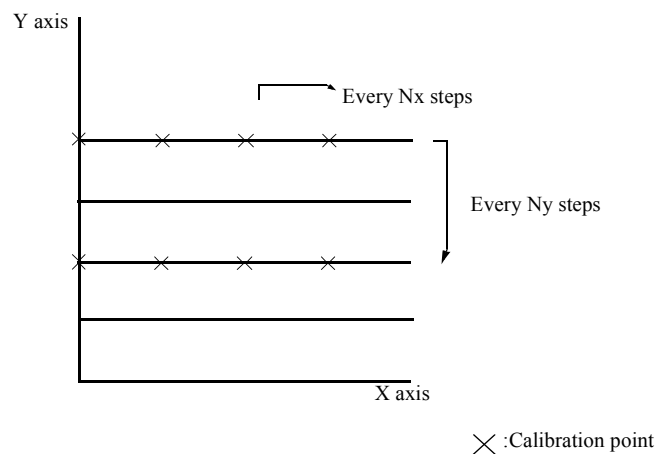
Calibration steps of the X, Y and Z axes

. (Point)

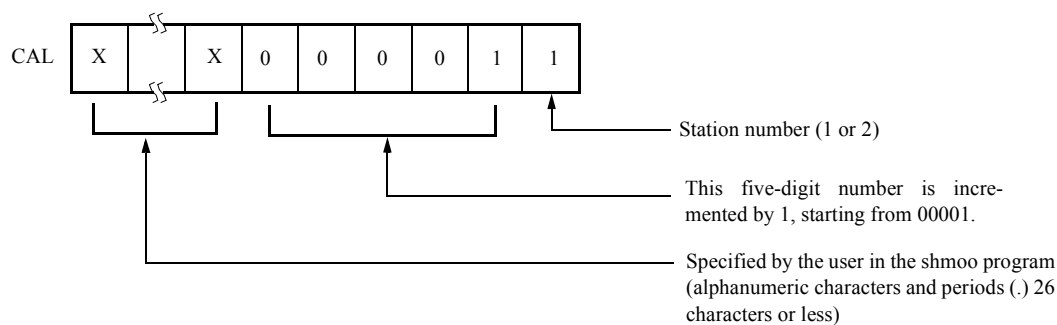
Calibration point

A calibration is performed unconditionally at the start point of each axis.

Calibration facility set for each axis parameter operates separately. Therefore, the following shmoo with calibration cannot be plotted by specifying calibration for both the X and Y axes.



- File names with CAL attributes



The previously set CAL attribute names are not deleted.

Tip

When a shmoo with calibration is run, many files with CAL attributes are created on the disk according to the number of points specified for each axis. Be very careful when executing calibration for file control (for maintenance, deletion and name of the file).

- Specifying the string variable name of the calibration facility
CALB in the test plan program must always be specified by a STRING variable name as follows:

```
STRING JJ (1000)
JJ = "DRREF=2.4V, ..."
CALL CALB ( "64K", JJ) } Correct
```

If a character string is specified directly as shown below, it is not transferred to the shmoo character string.

```
CALL CALB ( "64K", "DRREF = 2.4V, ....." ) Mistake
```

- An additional character string specified by a shmoo program can be used by adding it to the STRING variable name in the test plan program. If the specifications are the same, the most recent specification is effective.

6. 5 SHM3 Runtime Error

- If an execution-time error occurs during shmoo program execution, an error code, an object file name, an object file line number, a customer name, a source file name and a source file line number are displayed. An error message is then displayed, and the menu is displayed again.

```

ERROR CODE: 3316      SUB SPL3      00559      [ZETSU:      AP401      (00559)]
              ↑          ↑          ↑          ↑          ↑          ↑
            Error Object file Line Customer Source Line
            Codes  name      number name   file name number

-----ERR3-----

3316      MPAT file has still not been transferred to the test vector generator.

                        Error message

```

- When the MPAT file is specified to execute the shmoo program, the file is searched for in the current customer file.
If there is no MPAT file in the customer file, it is searched for in the DP0.
If there is no MPAT file in the DP0, an error message is displayed.

Tip

After an error code and other information are displayed, an error message may not be displayed immediately.

6. 6 Example of Using SHM3

Set values

```

DEVICE   : DRIVER WAVE FORM           ! DATE:      1998/08/14 20:14:32
SHMOO# :                               ! DATA FILE =AQ3.SM
COND:
DUT#= 1
-----Y-AXIS-----X-AXIS-----Z-AXIS-----
UNIT  =HOUT1                !  STRB1                !  HIN1
START =3.000V                !  0.000S                !  1.300V
MODIFY=                      !  0                      !  0
DELT  =-200.0MV              !  2.000NS                !  200.0MV
STEP  =15                    !  50                     !  10
NAME  =HOUT                  !  ACCESS                 !  HIN LEVEL
DISP-A=1.000                 !  1                      !  1.000
DISP-B=0.000V                !  0.000S                !  0.000V
CALI   =                     !                          !  0
PAGE   =1                    ! CALI DISABLE
MPAT (MAIN PAT)
```


7. *Parallel Test Monitor Program (/MPDISP)*

This section describes Parallel Test Monitor Program (/MPDISP)

7. 1 MPDISP Overview

The parallel test monitor displays the test results and yields of the MUT sorted by the contactor when more than one data item is tested in parallel using T5500 SERIES, handlers and probers.

This function enables the operation of handlers and probers and the progress of each device test to be checked.

 Tip

/MPDISP is compatible with a maximum of 64 DUTs for one station, or a maximum of 128 DUTs for two stations.

7. 2 MPDISP Operation

- 1) Start the simultaneous monitor program.

```
/MPDISP_
```

- 2) Selecting execution

```
ENABLE (1) /DISABLE (0) = 0
```

↑
↑
 Currently set data Cursor

To perform the parallel test monitor, enter 1.

When 0 is entered, the parallel test monitor is not executed.

- 3) When 1 is entered in step 2, the following prompting message is displayed:

```
MONITORING CYCLE (0-999) = 2
```

↑
↑
 Currently set data Cursor

Specify the number of DUTs in a range between 0 to 999 as the number of displayed results. (0 and 1 have the same effect.)

```

DISPLAY DUT ENABLE (1) /DISABLE (0)
DUT  1-16 = 1   0
DUT 17-32 = 1   0
DUT 33-48 = 0   1
DUT 49-64 = 1   1
DUT 65-80 = 0   0
DUT 81-96 = 0   0
DUT 97-112 = 0  1
DUT113-128 = 0  -

```

↑
↑
 Currently set data Cursor

To display the measurement result of a DUT on the monitor, enter "1" at the DUT No.

When 0 is entered, no measurement result is displayed.

To proceed to the next DUT No., press the return key.

When "1" is entered for each DUT No., test results are displayed for the DUTs by the number of socket program parallel tests.

The above operation is all specification for parallel test monitor program conditions.

7. 3 MPDISP CRT Display Format

When ENABLE (1) is selected in 7. 2 "MPDISP Operation" on page 7-2 2 and the test plan program is executed, the following data is output:

MP-DISPLAY			
TOTAL= 12004		YIELD= 98.64 (%)	
ST1/BUSY		ST2/BUSY	
X-ADD= 12	Y-ADD= 36	X-ADD= 26	Y-ADD= 48
CHIP COUNTER= 32		CHIP COUNTER= 32	
YIELD=97.44 (%) WAFER#= 3767		YIELD=97.02 (%) WAFER#= 3764	

DUT#	PASS/FAIL	CATE	YIELD (%)
1	PASS	3	96.84
2	FAIL	21	95.88
3	PASS	5	99.45
4	----		98.44
5	*3	2	96.82
6	*2	1	97.22
7	----		95.00
8	PASS		99.00
9	PASS	5	95.45
10	PASS		98.88
11	FAIL		99.76
12	PASS	46	98.96
13	PASS		97.33
14	PASS		95.28
15	FAIL	27	98.32
16	PASS	32	96.50

- TOTAL
- Number of MUTs measured so far
- STn/
- Station number and handler or prober status
- WAFER#
- Number of wafers
- X-ADD
- X coordinate of the chip to be measured
- Y-ADD
- Y coordinate of the chip to be measured
- CHIP COUNTER
- Number of chips measured so far in a wafer
- YIELD
- Yield
- CATE
- Specified category number

*n

Number of failures in the same category

Indicates that these DUTs are measured.

Tip

Display (b) is in the unit of 16 DUTs independent of the station No. Therefore, DUTs displayed in (b) do not correspond to station number at the same columns in (a).

8. Header Command

This section describes Header Command.

8. 1 /HSET, /HD

The header command can output a simple message to a line printer.

```
/HSET
/HD
```

1) Enter the following command in the controller mode.

```
/HSET.␣
HEADER = DEV NO. 1234 ␣
      ↑
      The current header contents are output.
```

To perform no modification or to terminate HSET, press the return key only.

When the header contents are to be changed to new contents, set as follows:

```
HEADER = DEV NO. 1234 12147 NO.1.␣
HEADER = 1247 NO.1.␣
      ↑
      Newly changed
```

A maximum of 50 characters (only one line) can be entered.

2) Specify an output device by pressing the [S] key.

3) When /HD.␣ is entered, the character string input by HSET in 2 is output to the output device specified in 1.

Tip

If /HD is executed without executing /HSET, the following error message is displayed:

```
ERROR! KEY-IN      "/HSET"
```

8. 2 /LN

Comments on the lot number output in a shmoo plot can be entered by using the following command:

`/LN`

A maximum of 20 characters can be entered as a lot number.
Operation is the same as that of /HSET.

8. 3 /SN

Sample number output in a shmoo plot can be entered by using the following command:

/SN

A sample number of up to eight digits can be input.

Operation is the same as that of /HSET.

9. *Enlarged Character Header*

This section describes the enlarge character header.

9. 1 Enlarged Character Header Overview

The three types of enlarged header commands shown in [Table 9-1](#) can output a simple message to a line printer in enlarged characters.

Table 9-1 ENLARGED HEADER COMMAND

Program name	Enlarged character size	Start up
HD1	7 characters (vertical), 5 characters (horizontal)	Conversational mode or program call
HD2	12 characters (vertical), 12 characters (horizontal)	Conversational mode or program call
TITLE	12 characters (vertical), 12 characters (horizontal) (For the date output, 7 × 5 characters)	Conversational mode

◆ Character Limitation for Enlarged Character Header

[Special characters]

/: Line feed

/*: Page break

//: / output

[Maximum number of characters to be output]

HD1

- For outputting to the VKT
11 characters/line, 2 lines/screen
- For outputting to the LPT
11" × 15": for this size, 18 characters/line, 6 lines/screen
11" × 11": for this size, 12 characters/line, 6 lines/screen

HD2

- For outputting to the VKT
5 characters/line, 1 lines/screen
- For outputting to the LPT
11" × 15": for this size, 9 characters/line, 4 lines/screen

11" × 11": for this size, 6 characters/line, 4 lines/screen

[Output character type]

Numbers

0 1 2 3 4 5 6 7 8 9

Alphabetic character

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz

Other characters

Space ! " # \$ % ') (* + , - . / : ; < = > ? @ [\ ^

9. 2 /HD1, /HD2

The types of characters that can be output are shown in ["Character Limitation for Enlarged Character Header" on page 9-1](#).

9. 2. 1 Conversational Mode-I

- 1) Enter the following command in the controller mode.

```
/HSET.␣
HEADER =   ABC   ␣
           ↑
       The current header contents are output.
```

To perform no modification or to terminate HSET, press the return key only.

When the header contents are to be changed to new contents, set as follows:

```
HEADER = ABC      /*64K-/DRAM.␣
HEADER = /*64K-/DRAM.␣
           ↑
       Newly changed
```

A maximum of 50 characters (only one line) can be entered.

- 2) Specify an output device by pressing the [S] key.
- 3) Start the enlarged character header program.

```
/HD1.␣ (or /HD2.␣)
```

- 4) Character strings set with HSET in [1](#) are output to the output device specified in [2](#).

Tip

If /HD1 (or /HD2) is executed without executing /HSET, the following error message is displayed:

```
ERROR!  KEY-IN  "/HSET"
```

9. 2. 2 Conversational Mode-II

Enter the following command with a direct specification of output characters.

```
/HD1 Characters to be output.␣ (or /HD2 Character string to be output.␣)
```

The character string specified is output in enlarged characters to the output device selected by the [S] key.

Output example is shown below.

◆ Example of Output When /HD1 /*64K-/DRAM.␣ is Entered

```

  66      4   K   K
    6      44  K   K
  6       4 4   K  K
6666     4 4   KK   -----
  6   6   44444  K  K
  6   6      4   K   K
    666     4   K   K

DDD      RRRR    AAA  M   M
D  D    R   R  A   A  MM  MM
D   D   R   R  A   A  M  M  M
D   D   RRRR   AAAAA M  M  M
D   D   R  R   A   A  M   M
D   D   R   R  A   A  M   M
DDD     R   R  A   A  M   M
```

◆ Example of Output When /HD2 /*64K-/DRAM. are Entered

6666666666	44	KK	KK	
666666666666	4444	KK	KK	
6666	664444	KK	KK	
6666	4444	KK	KK	
6666	4444	KK	KK	
666666666666	4444	KKKKKKK		-----
66666666666666	4444	KKKKKKK		-----
666666	66444444444444	KK	KK	
666666	66444444444444	KK	KK	
666666	6644	KK	KK	
66666666666666	44	KK	KK	
666666666666	44	KK	KK	
DDDDDDDDDD	RRRRRRRRRRR	AAAAAA	MM	MM
DDDDDDDDDDDD	RRRRRRRRRRRR	AAAAAAA	MMM	MMM
DDDD	DDRRRR	AA	AA	MMMMMMMM
DDDD	DDRRRR	AA	AA	MMMMMM
DDDD	DDRRRR	AA	AA	MMMMMM
DDDD	DDRRRRRRRRRR	AAAAAAAAAAAA	MM	MM
DDDD	DDRRRRRRRRRR	AAAAAAAAAAAA	MM	MM
DDDD	DDRRRR	AA	AA	MM
DDDD	DDRRRR	AA	AA	MM
DDDD	DDRRRR	AA	AA	MM
DDDDDDDDDDDD	RRRR	AA	AA	MM
DDDDDDDDDD	RRRR	AA	AA	MM

9. 2. 3 CALL Statement Format-I

Characters can be specified as shown below by including the following statement in the program:

CALL HD1 ("Character string to be output")

CALL HD2 ("Character string to be output")

Example

CALL HD2 ("/*64-/DRAM")

9. 2. 4 CALL Statement Format-II

Characters can be specified by including the following string variable in the program:

```
STRING  xx(n) =  "Character string to be output"
```

Number of characters

Variable name

```
CALL  HD1 (xx)
```

OR

```
CALL  HD2 (xx)
```

9. 3 /TITLE

The TITLE utility program is used to output headers with enlarged character strings, date and time.

9. 3. 1 Operation

- 1) Specifies the comment character string.

```
/TITLE Character string_
```

- 2) TITLE command outputs a character string to the output device specified with the [S] key in the following sequence:

- 1) A line feed is made.

- 2) Up to 7 characters can be output on an LPT using the 11" × 15" size of paper.

The character string specified with the TITLE statement is output in the HD2 type enlarged characters.

- 3) The current date and time are output in the HD1 type enlarged characters.

◆ **Example of Output When /TITLE ABC↵ is Entered**

AAAAAA		BBBBBBBBBB		CCCCCCCCC	
AAAAAAA		BBBBBBBBBB		CCCCCCCCC	
AA	AA	BB	BB	CC	CC
AA	AA	BB	BB	CC	
AA	AA	BB	BB	CC	
AAAAAAAAA		BBBBBBBBBB		CC	
AAAAAAAAA		BBBBBBBBBB		CC	
AA	AA	BB	BB	CC	
AA	AA	BB	BB	CC	
AA	AA	BB	BB	CC	CC
AA	AA	BBBBBBBBBB		CCCCCCCCC	
AA	AA	BBBBBBBBBB		CCCCCCCCC	

1	999	999	888	00	77777	1	999		
11	9 9	9 9	8 8	/	0 0	7	/	11	9 9
1	9 9	9 9	8 8	/	0 0	7	/	1	9 9
1	9999	9999	888	/	0 0	7	/	1	9999
1	9	9	8 8	/	0 0	7	/	1	9
1	9	9	8 8	/	0 0	7	/	1	9
111	99	99	888	00	7	111	99		
	1	4	222	4	4	1			
	11	44	2 2	44	44	11			
	1	4 4	:	2	4 4	:	4 4	1	
	1	4 4	:	2	4 4	:	4 4	1	
	1	444444	2	44444	44444	1			
	1	4	:	2	4	:	4	1	
	111	4	:	22222	4	:	4	111	

10. Pattern Tracer (/MTRACE)

This section describes Pattern Tracer (/MTRACE).

10. 1 Pattern Tracer Overview

The pattern tracer is a tool used to debug the microinstruction program for the algorithmic pattern generator (ALPG).

The pattern tracer follows the registers at each step and checks the operation of each program step. To start the pattern tracer, enter the following command in the controller mode.

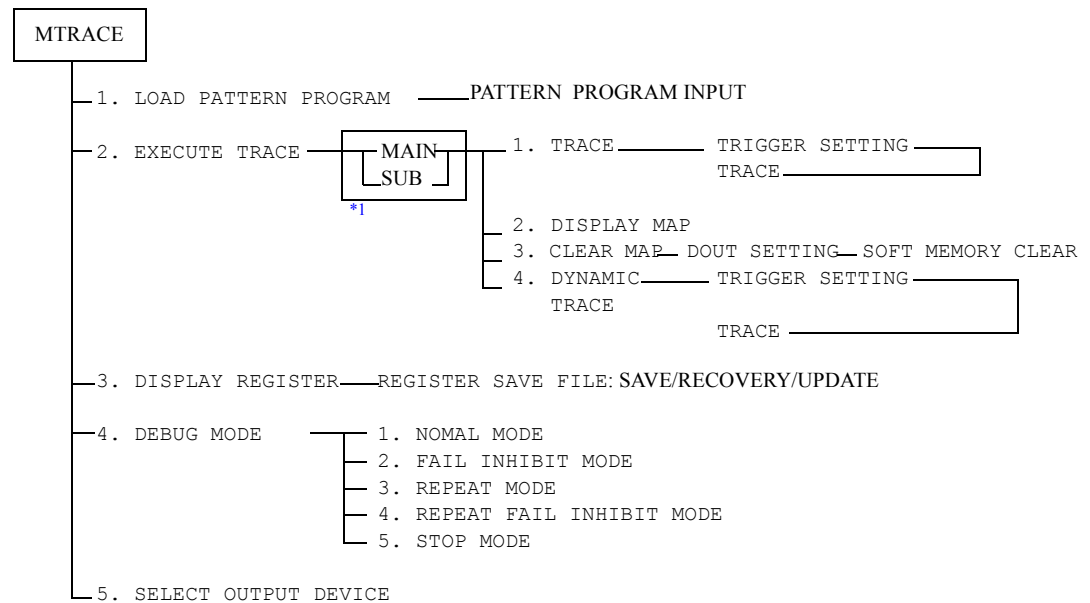
```
/MTRACE Pattern program name␣*1
```

**1 Pattern program name can be omitted.*

The pattern tracer has the following main functions.

- /MTRACE traces registers, displays maps and performs logical checks by comparing the read data with the pattern data stored in /MTRACE, in read cycles.
- Transfers a new pattern program to the ALPG.
- Checks and modifies the contents of the registers in the pattern program.
- Reduces the register data to 16 word or 64 word memory for visual pattern checking.

- MTRACE configuration



**1 This function can be used on the T5593, T5591, T5586, T5585, T5581, T5581H and T5581P (which use 1-WAY mode), the T5593 (which uses 2-WAY mode with SUBPG option), and on the T5591 (which use 1 WAY mode).*

10. 2 Functions of Pattern Tracer

When this utility is initiated, the following screen is displayed showing a wait state for menu selection.

```
*** T5500/T5300 SERIES MPAT TRACE MENU (REV. 1.00) ***
  1. LOAD PATTERN PROGRAM
  2. EXECUTE TRACE
  3. DISPLAY REGISTER
  4. DEBUG MODE
  5. SELECT OUTPUT DEVICE

- SELECT NO. = (CR: EXIT)
```

To terminate this utility, press the return key.

If a pattern program was already transferred before MTRACE was started, the pattern program is transferred again using the SEND MPAT statement in MTRACE.

As a result, the retransferred pattern program register values are set for each register when MTRACE is completed.

10. 2. 1 LOAD PATTERN PROGRAM

```
CURRENT MPAT NAME .... SAMPLE
NEW MPAT OR NO-LOAD(CR) = PATTERN PROGRAM NAME␣
```

Enter any name of a pattern program.

10. 2. 2 EXECUTE TRACE

```
SELECT DISPLAY PG (1=MAIN PG    2=SUB PG) = __
Displayed on the T5593, T5591, T5586, T5585, T5581, T5581H and T5581P (which use 1-
WAY mode), the T5593 (which uses 2-WAY mode with SUBPG option), and on the T5591
(which use 1 WAY mode).
```

Selects PG to be traced.

If the return key only is pressed and no selections have been made, the system assumes that the MAIN PG has been selected.

```

*** 2. EXECUTE TRACE MENU ***
TRACE
2. DISPLAY MAP
CLEAR MAP
4. DYNAMIC TRACE
[WORK]
- SELECT NO. = (CR: EXIT)

```

Select a content to be displayed from the above menu items.

To terminate this utility, press the return key only.

Each item in the menu is described below.

➔ Referred to [10. 3 "Details of Trace Function" on page 10-13](#) for the details of use.

- TRACE

Trace the pattern program.

- DISPLAY MAP

Display the condition in the memory after trace execution for 4×4 matrix only.

- CLEAR MAP

Specify the data bit displayed in CLEAR MAP, DYNAMIC trace and delete the data displayed in DISPLAY MAP.

- DYNAMIC TRACER

Display the trace result of the pattern program for 4×4 , 8×8 matrix only immediately.

10. 2. 3 Display Register

```
SELECT DISPLAY PG (1=MAIN PG 2=SUB PG) = __
```

Select MAIN or SUB PG for REGISTER data.

If the return key only is pressed and no selections have been made, the system assumes that the MAIN REGISTER has been selected.

Limitation

This function can be used on the T5593, T5591, T5586, T5585, T5581, T5581H and T5581P (which use 1-WAY mode), the T5593 (which uses 2-WAY mode with SUBPG option), and on the T5591 (which use 1 WAY mode).

```
SELECT PC = __
```

Input the PC number of the program.

If the return key only is input, the registers currently set are displayed.

Entering the PC number of the program executes the following statement and displays the register which the pattern was resent.

```
REG MPAT PC=.....
SEND MPAT File name to be traced
```

Screen display for the T5500 series

```
*** 3. DISPLAY REGISTER *** MAIN PG ←Inverse display

1:TIMER  = 00100F   19:CPE1  = 000000 [ 0 ]  37:D1    = 000000
2:TREFRE = 00100F   20:CPE2  = 000002 [ R ]  38:D2    = 000000
3:TPAUSE = 00100F   21:CPE3  = 000002 [ R ]  39:D3B   = XXXXXX *1
4:DBMA   = 000000   22:CPE4  = 000002 [ R ]  40:D4B   = XXXXXX *1
5:ISP    = 000000   23:DRE1  = 000003 [ R ]  41:TPH   = 000000000
6:BAR    = 000000   24:DRE2  = 000003 [ W ]  42:DCMR  = 000000000
7:CFLG   = 000000   25:RINV  = 000002 [ W ]  43:RCD   = 000000
8:OUTBLK = 000000   26:LMAX  = 00000F      44:CCD   = 000000
9:RLMAX  = 000000   27:HMAX  = 00000F      45:DCD   = 000000
10:IDX1   = 0000000  28:XMAX  = 00000F      46:XOS   = 000000
11:IDX2   = 0000000  29:YMAX  = 00000F      47:YOS   = 000000
12:IDX3   = 0000FF   30:ZMAX  = 0000FF      48:XT    = 000000
13:IDX4   = 0000EE   31:XMASK = 000FFD      49:YT    = 000000
14:IDX5   = 000000   32:YMASK = 000FFD      50:TPH2  = 000003FFF
15:IDX6   = 000000   33:XH    = 000000      51:DCMR2 = 000000000
16:IDX7   = 000000   34:YH    = 000000      52:D5    = 000000000
17:IDX8   = 000000   35:ZH    = 000000      53:D6    = 000000000
*:PC      = 000000   36:NH    = 000000

- SELECT NO. = ____ ( CR:DATA FILE RECOVERY/SAVE/UPDATE , EXIT )
```

*1 Since D3B and D4B cannot be read, XXXXXX is displayed. However, data can be changed.

Display screen for the T5592

*** 3. DISPLAY REGISTER ***					
1:TIMER	= 00100F	19:CPE1	= 000000 [0]	37:D1	= 000000
2:TREFRE	= 00100F	20:CPE2	= 000002 [R]	38:D2	= 000000
3:TPAUSE	= 00100F	21:CPE3	= 000002 [R]	39:D3B	= XXXXXX *1
4:DBMA	= 000000	22:CPE4	= 000002 [R]	40:D4B	= XXXXXX *1
5:ISP	= 000000	23:DRE1	= 000003 [R]	41:TPH	= 000000000
6:BAR	= 000000	24:DRE2	= 000003 [W]	42:DCMR	= 000000000
7:CFLG	= 000000	25:RINV	= 000002 [W]	43:RCD	= 000000
8:OUTBLK	= 000000	26:LMAX	= 00000F	44:CCD	= 000000
9:RLMAX	= 000000	27:HMAX	= 00000F	45:DCD	= 000000
10:IDX1	= 0000000	28:XMAX	= 00000F	46:XOS	= 000000
11:IDX2	= 0000000	29:YMAX	= 00000F	47:YOS	= 000000
12:IDX3	= 0000FF	30:ZMASK	= 0000FF	48:XT	= 000000
13:IDX4	= 0000FE	31:XMASK	= 000FFD	49:YT	= 000000
14:IDX5	= 000000	32:YMASK	= 000FFD	50:TPH2	= 000003FFF
15:IDX6	= 000000	33:XH	= 000000	51:DCMR2	= 000000000
16:IDX7	= 000000	34:YH	= 000000	52:D5	= 000000000
17:IDX8	= 000000	35:ZH	= 000000	53:D6	= 000000000
**:PC	= 000000	36:NH	= 000000	54:DBMLPB	= 000000
				55:DBMLPE	= 000000
- SELECT NO. = ____ (CR:DATA FILE RECOVERY/SAVE/UPDATE , EXIT)					

*1 Since D3B and D4B cannot be read, XXXXXX is displayed. However, data can be changed.

Display screen for the T5336 and T5331

```

*** 3. DISPLAY REGISTER ***

1:TIMER = 00100F    20:CPE1 = 000000 [ 0 ]    39:D3B = XXXXXX *1
2:TREFRE = 00100F    21:CPE2 = 000002 [ R ]    40:D4B = XXXXXX *1
3:TPAUSE = 00100F    21:CPE3 = 000002 [ R ]    39:D3B = XXXXXX
4:DBMA = 000000      22:CPE4 = 000002 [ R ]    40:D4B = XXXXXX
5:ISP = 000000       23:DRE1 = 000003 [ W ]    41:TPH = 000000000
6:BAR = 000000       24:DRE2 = 000003 [ W ]    42:DCMR = 000000000
7:CFLG = 000000      25:RINV = 000002 [ R ]    43:RCD = 000000
8:OUTBLK = 000000    26:LMAX = 00000F          44:CCD = 000000
9:RLMAX = 000000     27:HMAX = 00000F          45:DSD = 000000
10:IDX1 = 0000000     28:XMAX = 00000F          46:XOS = 000000
11:IDX2 = 0000000     29:YMAX = 00000F          47:YOS = 000000
12:IDX3 = 000000FE    30:ZMAX = 0000FF          48:XT = 000000
13:IDX4 = 000000FE    31:XMASK = 000FFD          49:YT = 000000
14:IDX5 = 00000000    32:YMASK = 000FFD          50:TPH2 = 000003FFF
15:IDX6 = 00000000    33:XH = 000000          51:DCMR2 = 000000000
16:IDX7 = 00000000    34:YH = 000000          52:D5 = 000000000
17:IDX8 = 00000000    35:ZH = 000000          53:D6 = 000000000
18:IDX8 = 0000000     37:D1 = 000000
**:PC = 000000       38:D2 = 000000

- SELECT NO. =      ( CR:DATA FILE RECOVERY/SAVE/UPDATE , EXIT )

```

**1 Since D3B and D4B cannot be read, XXXXXX is displayed. However, data can be changed.*

Display screen for the T5371

*** 3. DISPLAY REGISTER ***					
1:TIMER	= 00100F	20:CPE1	= 000000 [0]	39:D3B	= XXXXXX
2:TREFRE	= 00100F	20:CPE2	= 000002 [R]	38:D3B	= XXXXXX *1
3:TPAUSE	= 00100F	21:CPE3	= 000002 [R]	39:D4B	= XXXXXX *1
4:TRAS	= 000000	23:CPE4	= 000002 [R]	42:DCMR	= 000000
5:ISP	= 000000	24:DRE1	= 000003 [W]	43:RCD	= 000000
6:BAR	= 000000	25:DRE2	= 000003 [W]	44:CCD	= 000000
7:CFLG	= 000000	26:RINV	= 000002 [R]	45:DSD	= 000000
8:RC	= 000000	27:LMAX	= 00000F	46:XOS	= 000000
9:OUTBLK	= 000000	28:HMAX	= 00000F	47:YOS	= 000000
10:RLMAX	= 000000	29:XMAX	= 00000F	48:XT	= 000000
11:IDX1	= 0000000	30:YMAX	= 00000F	49:YT	= 000000
12:IDX2	= 0000000	31:XMASK	= 000FFD	50:TPH2	= 000000
13:IDX3	= 00000FF	32:YMASK	= 000FFD	51:DCMR2	= 000000
14:IDX4	= 00000FE	33:XH	= 000000	52:D5	= 000000
15:IDX5	= 0000000	34:YH	= 000000	53:D6	= 000000
16:IDX6	= 0000000	35:ZH	= 000000	54:MR1	= 000000
17:IDX7	= 0000000	36:NH	= 000000	55:ZD	= 000000
**:PC	= 000000	36:D1	= 000000	54:ZD	= 000000
- SELECT NO. = (CR:DATA FILE RECOVERY/SAVE/UPDATE , EXIT)					

*1 Since D3B and D4B cannot be read, XXXXXX is displayed. However, data can be changed.

Display screen for the T5376 and T5375

*** 3. DISPLAY REGISTER ***			
1:TIMER = 00100F	19:IDX10 = 00000000	37:YMAX = 000000	55:XT = 000000
2:TREFRE= 00100F	20:IDX11 = 00000000	38:XMASK = 000000	56:YT = 000000
3:TPAUSE= 00100F	21:IDX12 = 00000000	39:YMASK = 000000	57:TPH2 = 00000000
4:TRAS = 000000	22:IDX13 = 00000000	40:XH = 000000	58:DCMR2 = 00000000
5:ISP = 000000	23:IDX14 = 00000000	41:YH = 000000	59:D5 = 00000000
6:BAR = 000000	24:IDX15 = 00000000	42:ZH = 000000	60:D6 = 00000000
7:CFLG = 000000	25:IDX16 = 00000000	43:NH = 000000	61:MR1 = 000000
8:OUTBLK= 000000	**PC = 000000	44:D1 = 000000	62:ZD = 000000
9:RLMAX = 000000	27:CPE1 = 000000[0]	45:D2 = 000000	63:ND = 000000
10:IDX1 = 00000000	28:CPE2 = 000000[0]	46:D3B = XXXXXX *1	64:BD = 000000
11:IDX2 = 00000000	29:CPE3 = 000000[0]	47:D4B = XXXXXX *1	65:BH = 000000
12:IDX3 = 00000000	30:CPE4 = 000000[0]	48:TPH = 000000000	66:FPSEL = 000000[----]
13:IDX4 = 00000000	31:DRE1 = 000000[0]	49:DCMR = 000000000	
14:IDX5 = 000000FE	32:DRE2 = 000000[0]	50:RCD = 000000	
15:IDX6 = 000000FE	33:RINV = 000000[0]	51:CCD = 000000	
16:IDX7 = 00000000	34:LMAX = 000000	52:DSD = 000000	
17:IDX8 = 00000000	35:HMAX = 000000	53:XOS = 000000	
18:IDX9 = 00000000	36:XMAX = 000000	54:YOS = 000000	
- SELECT NO. = (CR:DATA FILE RECOVERY/SAVE/UPDATE , EXIT)			

*1 Since D3B and D4B cannot be read, XXXXXX is displayed. However, data can be changed.

Display screen for the T5593

*** 3. DISPLAY REGISTER ***			
1:TIMER = 00100F	20:IDX12 = 00000000	39:YMASK = 000FFD	58:DCMR2 = 00000000
2:TREFRE= 00100F	21:IDX13 = 00000000	40:XH = 000000	59:D5 = 00000000
3:TPAUSE= 00100F	22:IDX14 = 00000000	41:YH = 000000	60:D6 = 00000000
4:DBMA = 000000	23:IDX15 = 00000000	42:ZH = 000000	61:DBMLPB= 000000
5:ISP = 000000	24:IDX16 = 00000000	43:NH = 000000	62:DBMLPE= 3FFFFFF
6:CFLG = 000000	**PC = 000000	44:D1 = 000000	63:ZD = 000000
7:OUTBLK= 000000	26:CPE1 = 000000[0]	45:D2 = 000000	64:ND = 000000
8:RLMAX = 0000FF	27:CPE2 = 000002[R]	46:D3B = XXXXXX *1	65:ND2 = 000000
9:IDX1 = 00000000	28:CPE3 = 000002[R]	47:D4B = XXXXXX *1	66:ND3 = 000000
10:IDX2 = 00000000	29:CPE4 = 000002[R]	48:TPH = 000000000	67:ND4 = 000000
11:IDX3 = 00000000	30:DRE1 = 000003[W]	49:DCMR = 000000000	68:ND5 = 000000
12:IDX4 = 00000000	31:DRE2 = 000003[W]	50:RCD = 000000	69:ND6 = 000000
13:IDX5 = 00000000	32:RINV = 000002[R]	51:CCD = 000000	70:ND7 = 000000
14:IDX6 = 00000000	33:LMAX = 000FFF	52:DSD = 000000	71:BD = 000000
15:IDX7 = 00000000	34:HMAX = 000FFF	53:XOS = 000000	72:BD2 = 000000
16:IDX8 = 00000000	35:XMAX = 000FFF	54:YOS = 000000	73:BD3 = 000000
17:IDX9 = 00000000	36:YMAX = 000FFF	55:XT = 000000	74:BD4 = 000000
18:IDX10 = 00000000	37:ZMAX = 0000FF	56:YT = 000000	75:BD5 = 000000
19:IDX11 = 00000000	38:XMASK = 000FFD	57:TPH2 = 000000000	76:BD6 = 000000
77:BD7 = 000000			
78:BH = 000000			
79:D1B = 000000			
80:D1C = 000000			
81:D1D = 000000			
82:D1E = 000000			
83:D1F = 000000			
84:D1G = 000000			
85:D1H = 000000			
86:D2B = 000000			
87:D2C = 000000			
88:D2D = 000000			
89:RHZA = 000000[0]			
90:RHZB = 000000[0]			
91:ASCNT = 000000[0]			
92:ASINI = 000000[0]			
93:FPSEL = 000000[----]			
- SELECT NO. = (CR:DATA FILE RECOVERY/SAVE/UPDATE , EXIT)			

*1 Since D3B and D4B cannot be read, XXXXXX is displayed. However, data can be changed.

Display screen for the T5377

*** 3. DISPLAY REGISTER ***				
1:TIMER = 00100F	20:IDX11 = 00000000	39:YMASK = 000000	58:DCMR2 = 000000000	77:BD6 = 000000
2:TREFRE= 00100F	21:IDX12 = 00000000	40:XH = 000000	59:D5 = 000000000	78:BD7 = 000000
3:TPAUSE= 00100F	22:IDX13 = 00000000	41:YH = 000000	60:D6 = 000000000	79:D1B = 000000
4:TRAS = 000000	23:IDX14 = 00000000	42:ZH = 000000	61:MR1 = 000000	80:D1C = 000000
5:ISP = 000000	24:IDX15 = 00000000	43:NH = 000000	62:ZD = 000000	81:D1D = 000000
6:BAR = 000000	25:IDX16 = 00000000	44:D1 = 000000	63:ND = 000000	82:D1E = 000000
7:CFLG = 000000	**PC = 000000	45:D2 = 000000	64:BD = 000000	83:D1F = 000000
8:OUTBLK= 000000	27:CPE1 = 000000[0]	46:D3B = XXXXXX *1	65:BH = 000000	84:D1G = 000000
9:RLMAX = 000000	28:CPE2 = 000000[0]	47:D4B = XXXXXX *1	66:FPSEL = 000000[----]	85:D1H = 000000
10:IDX1 = 00000000	29:CPE3 = 000000[0]	48:TPH = 000000000	67:ND2 = 000000	86:D2B = 000000
11:IDX2 = 00000000	30:CPE4 = 000000[0]	49:DCMR = 000000000	68:ND3 = 000000	87:D2C = 000000
12:IDX3 = 00000000	31:DRE1 = 000000[0]	50:RCD = 000000	69:ND4 = 000000	88:D2D = 000000
13:IDX4 = 00000000	32:DRE2 = 000000[0]	51:CCD = 000000	70:ND5 = 000000	89:TPH1B = 000000000
14:IDX5 = 000000FE	33:RINV = 000000[0]	52:DSD = 000000	71:ND6 = 000000	90:TPH1C = 000000000
15:IDX6 = 000000FE	34:LMAX = 000000	53:XOS = 000000	72:ND7 = 000000	91:TPH1D = 000000000
16:IDX7 = 00000000	35:HMAX = 000000	54:YOS = 000000	73:BD2 = 000000	92:TPH2B = 000000000
17:IDX8 = 00000000	36:XMAX = 000000	55:XT = 000000	74:BD3 = 000000	93:TPH2C = 000000000
18:IDX9 = 00000000	37:YMAX = 000000	56:YT = 000000	75:BD4 = 000000	94:TPH2D = 000000000
19:IDX10 = 00000000	38:XMASK = 000000	57:TPH2 = 000000000	76:BD5 = 000000	

- SELECT NO. = (CR:DATA FILE RECOVERY/SAVE/UPDATE , EXIT)				

*1 Since D3B and D4B cannot be read, XXXXXX is displayed. However, data can be changed.

Tip

If AUTOSCAN COUNTER is enabled by using the AUTOSCAN utility, the ASCNT register value is stored in the RHZA and RHZB registers every time the MEAS MPAT or START MPAT statement is executed. The ASCNT register value cannot be stored by MEAS MPAT(1) or START MPAT(1).

Table 10-1 Values of Registers CPE1, CPE2, CPE3, CPE4, DRE1, DRE2 and RINV, and Their Displays

Register value	Display	Register value	Display	Register value	Display	Register value	Display
#02	R	#0A	C4	#13	C11	#1B	C19
#03	W	#0B	C5	#14	C12	#1C	C20
#04	M1	#0C	C6	#15	C13	#1D	C21
#05	M2	#0D	C7	#16	C14	#1E	C22
#06	C0	#0E	PREF	#17	C15	#1F	C23
#07	C1	#10	C8	#18	C16		
#08	C2	#11	C9	#19	C17		
#09	C3	#12	C10	#1A	C18		

Table 10-2 Values of a Register FPSEL and Their Displays

Register value	Display	Register value	Display
#0	---	#7	FP6
#1	FP0	#8	FP7
#2	FP1	#9	FP8
#4	FP3	#A	FP9
#5	FP4	#B	FP10
#6	FP5		

Specifying a register number moves the cursor to the specified register position so that the register data can be changed.

If the input data is prefixed with a symbol "#", the system recognizes the input data as hexadecimal. Also, the register can be specified the register name instead of the register number.

Entering the return key only, data file of register data can be created and SAVE, RECOVERY and UPDATE operations can be made for the file.

```
- SELECT NO. = __ ( 1=RECOVERY 2=SAVE 3=UPDATE CR: EXIT )
```

To terminate the processing, press the only return key again without making any selections.

The following message appears when selecting RECOVERY.

The contents of former data file restores by specifying a file name

```
- FILE NAME = __ ( CR:PREVIOUS )
```

The following message appears when selecting SAVE.

The contents of registers currently displayed are saved in a file by specifying a file name.

```
- FILE NAME = ( CR:PREVIOUS )
```

When selecting UPDATE, the restored data file is updated.

10. 2. 4 DEBUG MODE

```
*** 4. DEBUG MODE MENU ***  
  
NORMAL MODE  
2. FAIL INHIBIT MODE  
3. REPEAT MODE  
4. REPEAT FAIL INHIBIT MODE  
5. STOP MODE  
  
- SELECT NO. =      (CR:EXIT)
```

```
KEY-IN "E" TO BREAK !!
```

DEBUG MODE is used for observing ALPG output by an oscilloscope.

Entering E key terminates output and returns to the menu selection screen.

Entering the only return key in the menu selection screen terminates this process.

10. 2. 5 SELECT OUTPUT DEVICE

```
OUTPUT CRT(0), CRT & LPT(1) = __
```

Select an output device.

0 = CRT

1 = CRT & LPT

10. 3 Details of Trace Function

10. 3. 1 TRACE

SELECT DATA

PG OUT (1), SCRAM OUT (2), PRESCRAM OUT (3), PRESCRAM & SCRAM OUT (4)

SELECT NO. (1-4) = ____

Select output data to be traced.

1 = Address data output from the pattern generator

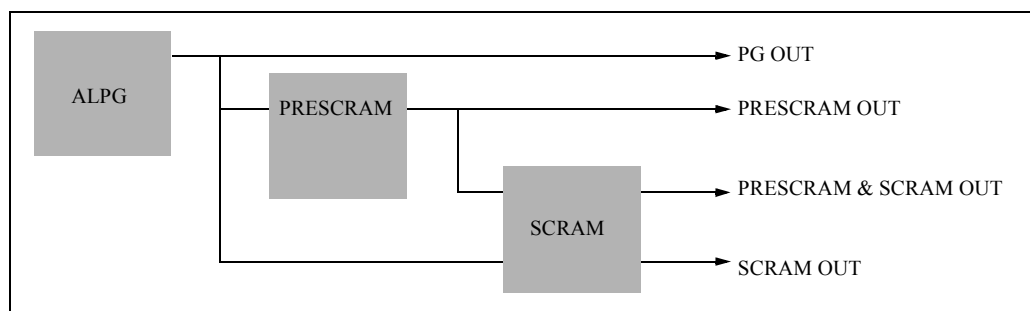
2 = Address data output through the address descrambler

3 = Address data output through the address prescrambler

4 = Address data output through the prescrambler and the descrambler

If the return key only is entered, it is regarded as PG OUT.

Refer to the following flow chart for the output of the address data.



If 3 or 4 is selected, the address data is output through the pre-scrambler whether or not the SELECT PRESCRAM ADDRESS statement is specified.

In the T5593, the X address is output by using the XALD instruction. In the T5377, the X and Y addresses are output by using the XALD instruction and YALD (YASL) instruction.

If 2 or 4 is selected and then SEPARATE is specified in the SELECT SCRAM MODE statement, the address data output through the scrambler to the DUT is traced.

On the T5336 and T5311, the prescrambler (PRESCRAM) is not installed.

FLAG SENSE INSTRUCTION WILL BE EXECUTED AND INTERNAL FLAG 0 OR 1 ? ____

Prohibits or permits an interruption of flag sense instructions during pattern trace.

Cannot be selected for the T5593.

0=Prohibits an interruption

1=Permits an interruption

Display screen for the T5500 series, T5336 and T5311

```

*1
  2.1 TRACE (MPAT NAME:MTRM0002 [MAIN]
1. TRIGGER  PC  X    Y  TS  R  W  M1  M2  C  STEP
           000 00000 00000 01          00 0000000000
2. COUNT    0    3.DISPLAY STEP:      0    4.CONTINUE

STEP  PC  X    Y  D    D    MUT          0...4...8...C..F
*2

COMMAND = *3
B:BREAK  E:EXIT  G:GO  T:TRIGGER  F:FORWARD  P:PAUSE  SPACE:CONTINUE

```

Display screen for the T5592 and T5371

```

*1
  2.1 TRACE (MPAT NAME:MTRM0002 [MAIN]
1. TRIGGER  PC  X    Y  TS  R  W  M1  M2  C  STEP
           000 00000 00000 01          00 0000000000
2. COUNT    0    3.DISPLAY STEP:      0    4.CONTINUE

STEP  PC  X    Y  D    D    MUT          CYP          0...4...8...C..F
*2

COMMAND = *3
B:BREAK  E:EXIT  G:GO  T:TRIGGER  F:FORWARD  P:PAUSE  SPACE:CONTINUE

```

Display screen for the T5377, T5376 and T5375

```

*1
  2.1 TRACE (MPAT NAME:MTRM0002 )
1. TRIGGER  PC  X    Y  TS  R  W  M1  M2  C  STEP
           000 00000 00000 01          00 0000000000
2. COUNT    0    3.DISPLAY STEP:      0    4.CONTINUE

STEP  PC  X    Y  D.X.FDA D2.X.FDB MUT          CYP          0...4...8...C..F
*2

COMMAND = *3
B:BREAK  E:EXIT  G:GO  T:TRIGGER  F:FORWARD  P:PAUSE  SPACE:CONTINUE

```

Display screen for the T5593

```

*1
+-----+ 2. 1 TRACE (MPAT NAME:MTRM0002) +-----+
| 1. TRIGGER PC      X      Y TS  R W M1 M2  C  STEP |
| 000 00000 00000 01 0000000000 |
| 2. COUNT:         0 3. DISPLAY STEP:         0 4. CONTINUE |
+-----+
| STEP  PC  X      Y      D      D      MUT      CYA CYB      0...4...8...C...F |
| 0 000 00000 00000 000000 000000 R W M1 M2 T01 C00000000000000000000000000000 01 01 CP1 CP2 CP3 CP4 DR1 DR2 ,..... > ? |
+-----+
| COMMAND = *3 |
| B: BREAK E: EXIT G: GO T: TRIGGER F: FORWARD P: PAUSE SPACE: CONTINUE |

```

*1 Display area of trigger, step count, etc.

➔ For details, refer to "Display Area of Trigger, Step Count, Etc." on page 10-15.

*2 Display area of trace data

➔ For details, refer to "Display Area of Trace Data" on page 10-17.

*3 Command display section

➔ For details, refer to "Command Display Section" on page 10-18.

◆ Display Area of Trigger, Step Count, Etc.

Trigger can be specified by entering T at ^{*3}, then the following message appears on the upper right of the screen.

```
*SELECT NO. __
```

1 Enter any one of 1 to 4.

Entering the return key only, terminates trigger setting mode and returns to a state waiting for command input.

- 1 = Trigger Specification

```

Trigger item  ──────────> PC  X      Y  TS  R  W  M1  M2  C  STEP
Trigger data  ──────────> 001 00001 00001 01          01 0000000000
                    <a>  <b>  <c>  <d> <e> <f> <g> <h> <i> <j>

```

A trigger can be set at the desired point by specifying <a> to <j>.

Multiple specifications mean their logical and (AND).

The following procedure shows an example of trigger setting.

To break at the point of X=#1F.AND.Y=#3FF.

1) Move the cursor to the display position of X using [←] and [→] keys.

2) Place the cursor to X, then hit the return key to select X.

When X is selected, the system enters a wait state until a break address is input.

3) Enter the break address.

Leading # means hexadecimal entry.

(Terminator is the return key.)

4) Move the cursor to the display position of Y using [←] and [→] keys.

Specify a break address of Y using a procedure similar to that in 2 and 3.

5) To release a selection mode of trigger items, hit [.] key in the step 3 above.

6) When trigger data entry is completed, the trigger item is highlighted.

7) 1 to To terminate trigger setting mode after repeating steps of 3, hit [PF1] key.

- Setting ranges of trigger data

<a>

PC #0 to #3FF (#0 to #1FF for T5336 or T5311)
(#0 to #7FF for T5593, T5377, T5376 or T5375)

X #0 to #FFFF (#0 to #3FFFF for T5593, T5377, T5376 or T5375)

<c>

Y #0 to #FFFF (#0 to #3FFFF for T5593, T5377, T5376 or T5375)

<d>

TS #1 to #10

<i>

C #0 to #F (#0 to #B for T5336 or T5311)
(#0 to #17 for T5593, T5377, T5376 or T5375)

<j>

STEP #0 to #7FFFFFFF



Limitation

Trigger item STEP cannot be specified together with any other items at a time.
If trigger data is set, pattern trace is executed from the beginning.

- 2 = COUNT

Specifies a cycle to break.

When COUNT3 for example, a break occurs at the third cycle of the specified trigger condition and the pattern trace starts.

- 3 = DISPLAY STEP

Specifies the number of steps to be displayed after a break point.

The number of steps after a break can be specified.

- 4 = CONTINUE

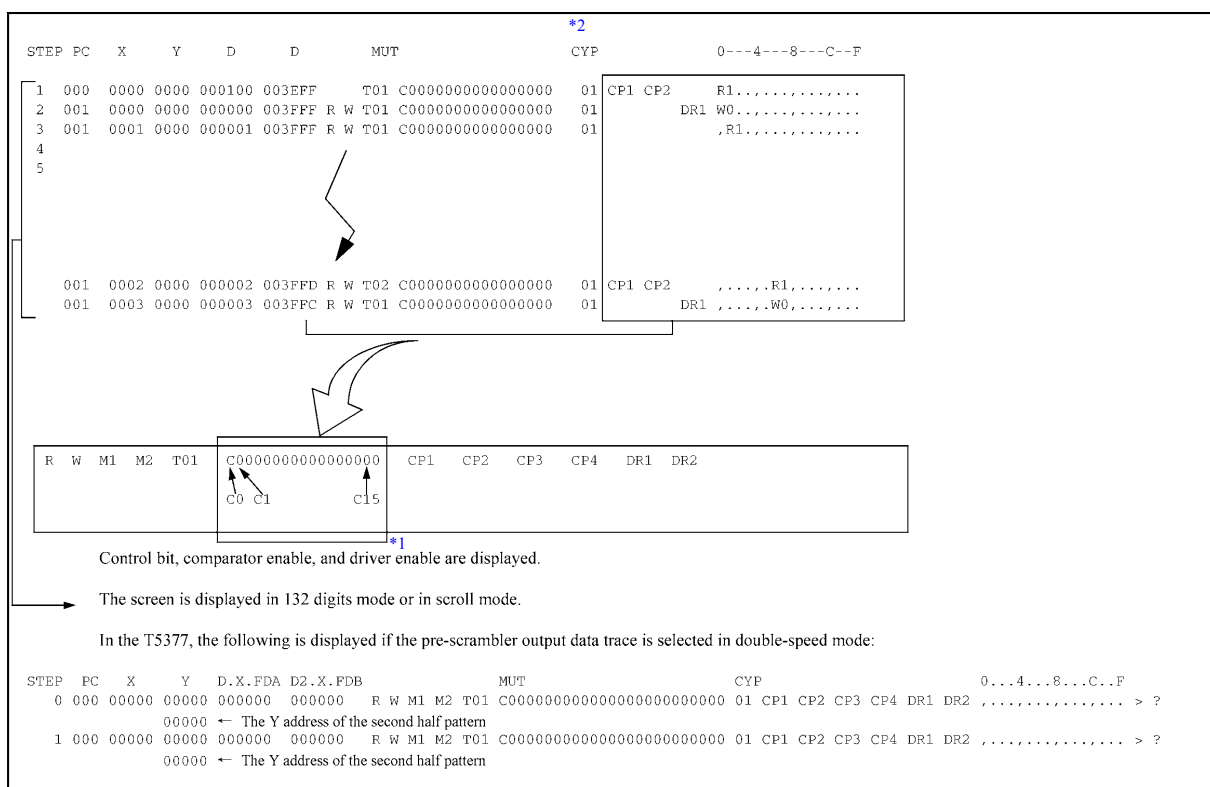
1 Function to automatically repeat pattern trace in the conditions set by 1 to 3.

Enter 4 at *SELECT NO. _ to select this mode. CONTINUE is highlighted.

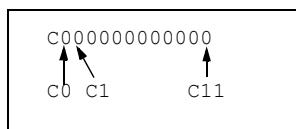
Another input of 4 at *SELECT NO._ clears the mode.

◆ Display Area of Trace Data

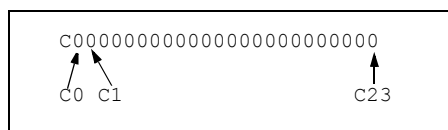
Screen display



*1 On the T5336 and T5331, the trace data is displayed as shown below.



On the T5593, T5377, T5376 and T5375, the trace data is displayed as shown below.



*2 Cycle palette data is displayed only on the T5593, T5592, T5377, T5376, T5375 and T5371.
The example shown below is for the T5593.

CYA

Cycle palette A

CYB

Cycle palette B

◆ Command Display Section

COMMAND =
B: BREAK E: EXIT G: GO T: TRIGGER F: FORWARD P: PAUSE SPACE: CONTINUE

Enter a character of [B], [E], [G], [T], [F], [P] and space. Each has the following functions.

Command name	Operation
B	Interrupts pattern trace.
E	Stops pattern trace.
G	Starts pattern trace.
T	Performs trigger specification.
F	1-step trace. (Pause on each step)
P	Temporarily halts pattern trace and waits for a command input.
Space	Resumes pattern trace temporarily halted.

Tip

- Specifications in the *SELECT PDS* statements, *SELECT PDS DATA* statements or *DATA DEFINE* in the pattern program are displayed for "D" (data bit) at tracing.

On the T5500 series

SELECT PDS DATA	Display of pattern tracer
D/D	D/D
D/CBM2	D/CBM2
D/FDB	D/FDB
CBM1/D	CBM1/D
CBM1/CBM2	CBM1/CBM2
CBM1/FDB	CBM1/FDB
FDA/D	FDA/D
FDA/CBM2	FDA/CBM2
FDA/FDB	FDA/FDB

On the T5371, T5336 and T5311

SELECT PDS DATA	Display of pattern tracer
D.XOR.FDA/D2	D.X.FDA/D2
D.XOR.FDA/FDB	D.X.FDA/FDB
D/D2	D/D2
FDA/D2	FDA/D2
D/FDB	D/FDB
FDA/FDB	FDA/FDB

On the T5377, T5376 and T5375

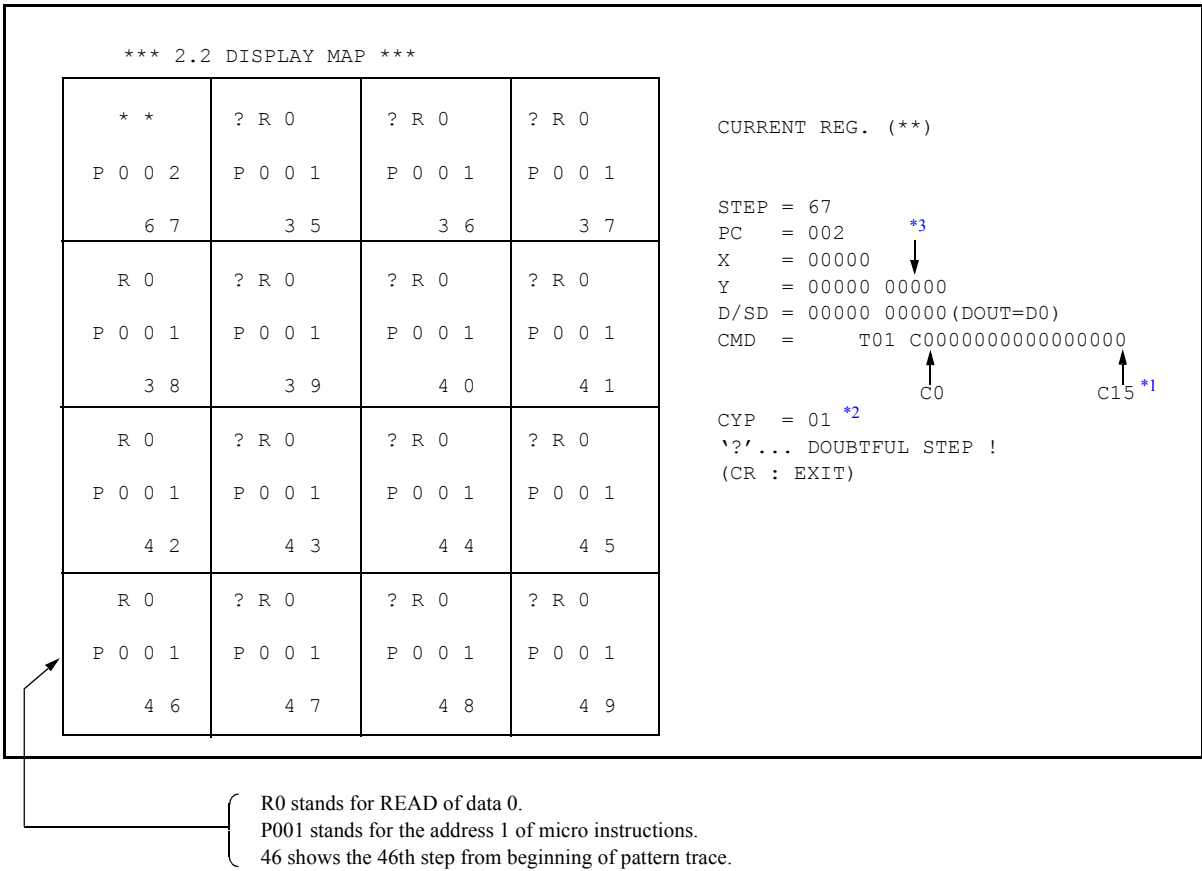
SELECT PDS DATA	Display of pattern tracer
D.XOR.FDA/D2	D.X.FDA/D2
D.XOR.FDA/FDB	D.X.FDA/FDB
D/D2	D/D2
FDA/D2	FDA/D2
D/FDB	D/FDB
FDA/FDB	FDA/FDB
D.XOR.FDA/D2.XOR.FDB	D.X.FDA/D2.X.FDB
D/D2.XOR.FDB	D/D2.X.FDB
FDA/D2.XOR.FDB	FDA/D2.X.FDB

- On the T5500 series, when a trigger is set at the trace, the step may not be displayed correctly in the following cases: (Display of STPS may shift by 3 to 9 steps.)
At 1WAY
When the trigger is set between two steps just before STPS/STFL and STPS/STFL in the pattern program.
At 2WAY
When the trigger is set between four steps just before STPS/STFL and STPS/STFL in the pattern program.
At 4WAY
When the trigger is set between eight steps just before STPS/STFL and STPS/STFL in the pattern program.

10. 3. 2 DISPLAY MAP

Map display is performed in a reduced form of 4 × 4 matrix by execution of 10. 3. 1 "TRACE" on page 10-13.

Screen display



*1 On the T5336 and T5311, C0 through C11 are displayed.

On the T5593, T5377, T5376 and T5375, C0 through C23 are displayed.

*2 On the T5593, CYPA and CYPB are displayed.

*3 In the T5377, if the pre-scrambler output data trace is selected and TRACE is executed in double-speed mode, the Y address of the second half pattern is also displayed.

The final value data of the result of [10. 3. 1 "TRACE" on page 10-13](#) is displayed.

To terminate displaying, press the return key.

10. 3. 3 CLEAR MAP

```
DOUT(D0-17 OR SD0-17) = *1
CLEAR WITH 16-SOFT MEMORY ? YES(1) OR NO(0) = *2
```

*1 Specify data bit. Specify D0 to D17 or SD0 to SD17.

*2 Specify whether to delete the data used for DISPLAY MAP.

Does not delete.

Deletes.

10. 3. 4 DYNAMIC TRACER

The following message appears after selecting the DYNAMIC TRACE.

```
SELECT
  4 × 4  MATRIX(1)
  8 × 8  MATRIX(2) = __
```

To select the matrix size.

1 = 4 × 4 matrix size

2 = 8 × 8 matrix size

```
SELECT DATA
PG OUT(1), SCRAM OUT(2), PRESCRAM OUT(3), PRESCRAM & SCRAM OUT(4)
SELECT NO. (1-4) = __ (a)

FLAG SENSE INSTRUCTION WILL BE EXECUTED AND INTERNAL FLAG 0 OR 1 ? __ (b)
```

Select output data to be traced. The menu shown here is for the T5500 series.

On the T5336 and T5311, the following menu is displayed.

```

SELECT DATA
  PG      OUT (1)
  SCRAM   OUT (2)  _   (a)

FLAG SENSE INSTRUCTION WILL BE EXECUTED AND INTERNAL FLAG 0 OR 1 ? _ (b)

```

➔ Refer to [10. 3. 1 "TRACE" on page 10-13](#) for inputting the data of (a) and (b).

The dynamic trace of each size is explained here.

The register values such as XMAX, YMAX and IDX1 must be set according to the matrix size in REGISTER (3) mode in advance.

◆ 4 × 4 Dynamic Trace

*1

2.4 DYNAMIC TRACE (4*4)

```

1. TRIGGER PC  X    Y    TS  R  W  M1  M2  STEP
           000 00000 00000 00              00000000000
2. COUNT  0    3. DISPLAY STEP  000 4. CONTINUE

```

	X0	X1	X2	X3
Y0	W0 PC01 1234	W0 PC01 1235		
Y1				
Y2				
Y3				

Current position
Inverse display

*2

```

STEP PC  0---4---8---C---F
1234 001 W0.....

```

*3

COMMAND = _ G:GO F:FORWARD R:REVERSE C:CLR
I:INIT B:BREAK E:EXIT P:PAUSE T:TRIGGER

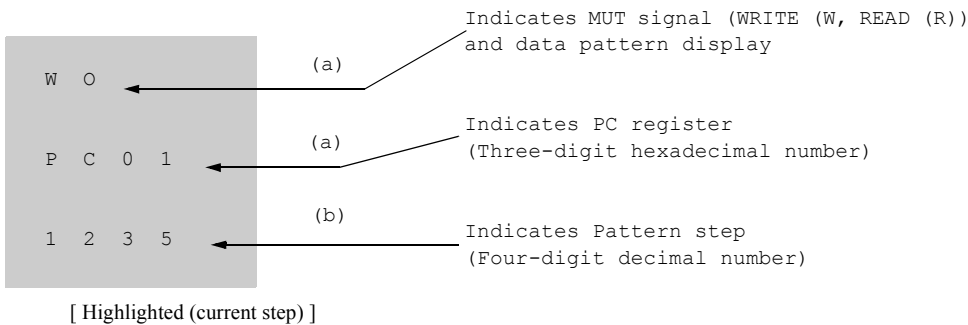
*4

CRT screen

_ indicates a cursor.

- *1 Specify TRIGGER
 - ➔ For details, refer to [10. 3. 1 "TRACE" on page 10-13.](#)
- *2 4 × 4 matrix display section
 - ➔ For details, refer to ["4 × 4 Matrix Display Section: Data Description" on page 10-23.](#)
- *3 Register trace display section
 - ➔ For details, refer to ["Register Trace Display Section" on page 10-24.](#)
- *4 Command display section
 - ➔ For details, refer to ["Command Display Section" on page 10-24.](#)

◆ 4 × 4 Matrix Display Section: Data Description



(a) MUT signal and data pattern display

MUT control signals are as follows:

[T][R][W][M1][M2][TSn]

$$\left[\begin{array}{c} [C1] [C1] \dots [C15] \\ RAm \end{array} \right]$$

Here, signals [R] and [W] and pattern data are displayed as data. The Dout number specified in MAP-CLEAR is used.

- [R]
Cycle written in the R instruction.
Compares the data with the data in Software Memory.
- [W]
Cycle written in the W instruction
- Pattern data
Data used in [R] and [W]

(b) Indicates PC register

The address to start the pattern program, that is played in a hexadecimal three-digit number.

(c) Pattern step display

Displays the step after tracing starts. Displayed as a decimal ten-digit number. (1 to 999999999)
The maximum pattern step is 999999999. When it is exceeded, **** is displayed.

Enter [G], [F], [R], [P], [I], [E], [B], [C], [T] and space key.

Command name	Operation
G	Performs trace step. (Continuous step operation)
F	Operates one step each time the [F] key is pressed.
R	Reverse step operation. Performs one reverse trace step each time the [R] key is pressed. (Up to 16 steps)
P	Temporarily stops the current trace step operation, and the input wait status is entered.
I	Stops and initializes the pattern generator (ALPG). The input wait status of a new start is entered.
B	Interrupts trace step.
E	Ends DYNAMIC TRACE mode.
C	Deletes data in the matrix currently being displayed.
T	Performs trigger specification.
Space	Resumes pattern trace temporarily halted.

When the cursor is displayed in the command display section, a command can be entered. The current position and display sections of matrix and register trace are displayed inversely. The procedure is shown below.

- 1) Execute the trace step by pressing the [G] key.
- 2) Pause at any steps by pressing the [P] key.
- 3) Backtrack steps one by one by pressing the [R] key.
Up to 16 steps can be backtracked.
- 4) Advance from the point to which backtracking was performed to the step at which the pause occurred by pressing the [F] or [G] key to check upon each register.
- 5) When the display becomes difficult to see as the steps advance or the pattern progress is checked, pause by pressing the [P] key once, and clear the currently displayed data by pressing the [C] key.

If a pass, fail, or interrupt occurs or the TIMER or OUT operation is performed when the PG stops, they are displayed in the command display section.

If an error is detected in a MUT command R step by comparing the software memory ^{*1}, the cell upper part of the matrix display section (WO is displayed in the example) is inversed, and "?" is displayed in the register trace display section.

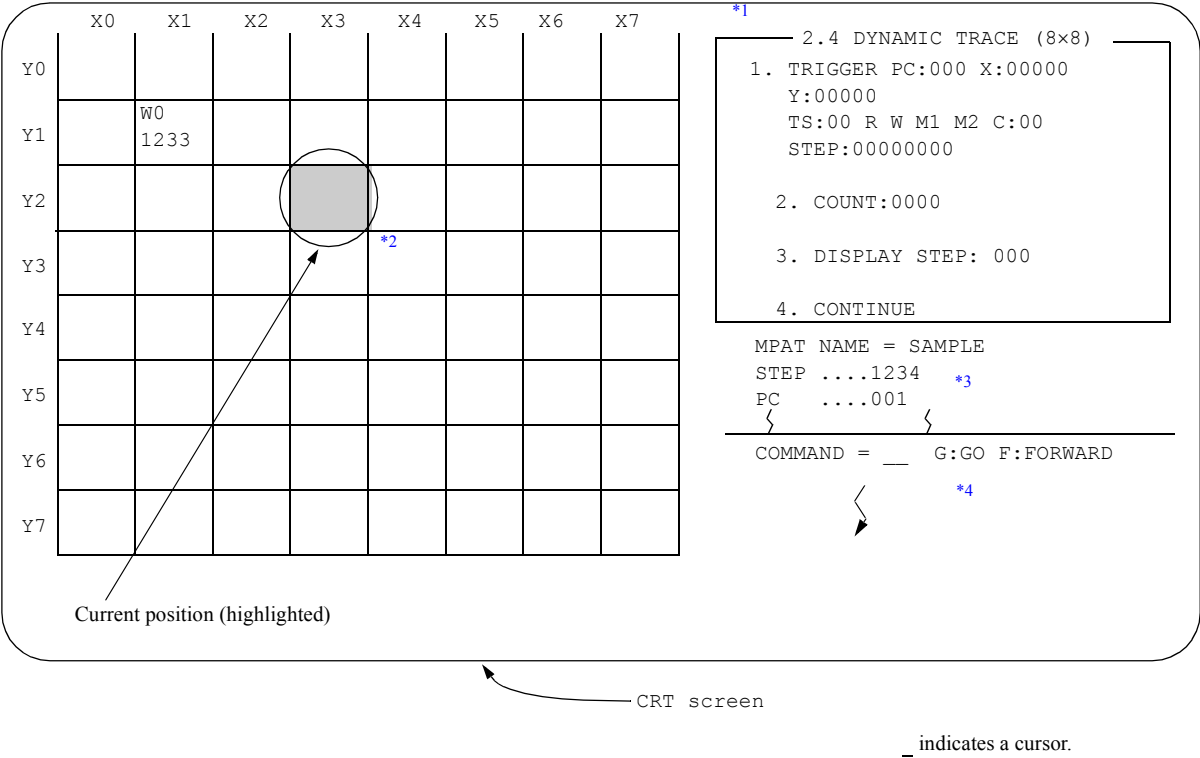
When the X or Y address exceeds 4×4 , > is displayed on the right side of the register trace display section.

When continuous advancing is performed from the step backtracked to in (d) and (e) above by pressing the G key, the [P] key (pause) is ineffective until the step to be paused at is reached. Therefore, steps are traced one by one by pressing the [F] key, instead of the [P] key.

If the [P] key is pressed when backtracking is not performed, the data in the register display section is sorted so that the current step is at the bottom row. When backtracking is performed by pressing the [R] key, the backtracked position moves upwards in step units each time the key is pressed.

**1 Software memory*
Stores the pattern data for 16 steps of the latest WRITE cycle, X and Y addresses, and other items.
Software memory is compared during the read cycle.
The W operation is performed prior to the R operation in a cycle in which both R and W are specified.
The software is used in comparing the ALPG output. The X and Y addresses contain the values before they are stored in memory even though the scramble memory is used.

◆ 8 × 8 Dynamic Trace



- *1 Specify TRIGGER*
→ For details, refer to [10. 3. 1 "TRACE" on page 10-13.](#)
- *2 8 × 8 matrix display section*
→ For details, refer to ["8 × 8 Matrix Display Section" on page 10-27.](#)
- *3 MUT Command display section*
→ For details, refer to ["MUT Command Display Section" on page 10-27.](#)

*4 Command display section

➔ For details, refer to ["Command Display Section" on page 10-28.](#)

◆ 8 × 8 Matrix Display Section

Data to be displayed is the same as in the 4 × 4 matrix display data. However, data on the microinstruction is not displayed.

The current position is highlighted. (The current step is displayed in MUT command display section.)

Steps are displayed in decimal four-digit numbers. When the step number is greater than 9999, the number minus 10000 is displayed. (The original number is displayed in the MUT command display section.)

◆ MUT Command Display Section

Only the step during trace step ([G] key) is displayed. Other MUT command data is also displayed during pause ([P] key), advance ([F] key) and reverse ([R] key).

Screen display

```
MPAT NAME ... MARCH[MAIN] *1

STEP .... 1234 *2
PC ..... 001 *3
X ..... 00003 *4
Y ..... 00002 *4
D/SD .... 00000 00000 *5
DOUT .... 000 *6
CMD ..... W
T01 C00000000000000000 *7
CYP ..... 01 *8
```

*1 Current MPAT file name and selected PG

Current MPAT file name loaded in PG and selected PG

*2 Pattern step

Displayed up to 10 digits in decimal number.

*3 Indicates PC register

Displayed up to 3 digits in hexadecimal number.

*4 X and X address

X and Y addresses are displayed in hexadecimal five-digit numbers.

In the T5377, if the pre-scrambler output data trace is selected and TRACE is executed in double-speed mode, the Y address of the second half pattern is also displayed.

*5 Pattern data

Pattern data is displayed.

*6 DOUT number

The DOUT number specified by MAP-CLEAR is displayed as a decimal two-digit number.

***7 MUT signal**

Displays each MUT control signal. Displays the contents of R, W, M1 and M2. On the T5500 series, C0 to C15 are displayed as sixteen-digit binary numbers; on the T5336 and T5311, C0 to C11 are displayed as twelve-digit binary numbers; on the T5593, T5377, T5376 and T5375, C0 to C23 are displayed as twenty-four-digit binary numbers. Comparator enable or driver enable information is also displayed.

***8 Cycle palette display**

Displayed up to 2 digits in hexadecimal number.

Displayed on the T5593, T5592, T5377, T5376, T5375 and T5371 systems.

On the T5593, CYPA and CYPB are displayed as follows:

CYPA/B.	01	01
	↑	↑
	CYPA	CYPB

◆ Command Display Section

```
COMMAND = __ G:GO F:FORWARD
R:REVERSE I:INIT B:BREAK E:EXIT
P:PAUSE C:CLEAR T:TRIGGER
```

The command above has the same meaning as the command of the 4×4 dynamic trace.

The use is the same as that of the 4×4 dynamic tracer command.

If an error is detected in an MUT command R step by comparing memory with the software memory, the upper part of the matrix display cell (WO is displayed in the example.) is inverted.

If the X or Y address exceeds 8×8 , it is not displayed in the matrix display section. Set register values such as XMAX, YMAX and IDX1 according to the matrix size, as described previously.

When 8×8 MATRIX (2) is first started by pressing the [G] key, only the number of steps is displayed in the MUT command display section described in.

Once the [P] or [F] key is pressed, the current register data is displayed. The data except that in the STEP column remains unchanged until the next [P], [F], or [R] key is pressed.

If a PASS, FAIL, or interrupt occurs, the data is displayed above the COMMAND=.

If the TIMER or OUT operation is performed when the PG stops, a message is displayed under C : CLEAR DISPLAY.

11. Program Call (/MLOAD)

This utility is used to call the specified program into the user area. It can also apply protection to the called program at this time.

```
/MLOAD file name.┘ *1  
/MLOAD +P file name.┘ *2
```

**1 Calls the specified program into the user area.*

**2 Calls the specified program into the user area and applies protection to the called program.*

If the user area overflows during program execution, all programs stored in the user area are deleted, but protected programs are not deleted.

However, protected programs are deleted unconditionally when the user area is cleared by the "U" command or when they exit from the controller.

- Global switch

/L

Outputs a called program to the printer.

/D

If a specified program is already present in the user area and is protected, it cannot be called. /MLOAD/D can cancel the protection and call the program.

- File name

More than one file name can be specified at the same time. However, the number of characters including the command cannot exceed 100.

When the file specifier is omitted, it is determined to be PRO.

When the second or subsequent file specifier is omitted, it is determined to be the same as the preceding specifier.

12. User Area Display (/MLIST)

MLIST displays the service status of the user area under the control of the controller.

```
/MLIST.↓ *1
/MLIST/L.↓ *2
```

*1 Displays the service status on the VKT.

*2 Outputs the service status to the printer.

/MLIST ↓				
*** USER AREA INFORMATION ***				
TOTAL	6526454 (3263227)		[#0CFBDB-#3EC6D6]	
LEFT	6508088 (3254044)		[#0D08AF-#3EAFCB]	
USED	18366 (**9183)			
*V-AREA	6568 (**3284)		[#0CFBDB-#0D08AF]	
*U-AREA	11798 (**5899)		[#3EAFCB-#3EC6D6]	
1.PRO	123	2800	#3EC15E	#000AF0 #0030 ADI\$DP1
2.LIB	TCLI	4370	#3EB8D5	#001112 #0150 DP0
3.LIB	MLIST	272	#3EB84D	#000110 #0042 DP0
4.LIB	CDIRTYPE	4356	#3EAFCB	#001104 #020E DP0

Number of bytes Word Memory address

TOTAL

Size of the entire user area

LEFT

Total size of free areas in the user area

USED

Size of currently used user area

V-AREA

Variable area size

U-AREA

Program area size

LIB TCLI

Utility which processes key commands

LIB MLIST

Utility which processes the MLIST command

LIB CDIRTYPE

Utility which displays service status of user area

13. *BUFF* Program Specification (/BUFF)

When program select of the control box is 0, PRO BUFF is selected automatically. It can be replaced with a program name.

```
/BUFF.␣  
STATION NUMBER = ____ *1  
PREVIOUS FILE PRO   xxx  
SPECIFY FILE PRO    ____ *2
```

**1 Specify a station number. Programs can be specified for each station independently.*

**2 Specify a program name.*

Press the Return key if no change is needed.

14. Execution Program Specification (/PRO)

PRO is used to specify a program name from the VKT and execute the program without operating program select of the control box.

/PRO	program name↓	*1
/PRO/1	program name↓	*2
/PRO/2	program name↓	*3

- *1 The specified program is executed at the current station.
(The current station refers to the station used by the preceding program.)
- *2 The specified program is executed at station 1.
- *3 The specified program is executed at station 2.

15. Histogram Data Reduction (HIDAR)

This chapter describes the Histogram Data Reduction (HIDAR) program.

15. 1 HIDAR Overview

HIDAR (Histogram DATA Reduction) is a utility program that is used to create a histogram.

This section describes histogram data reduction (HIDAR) functions, operating procedures, and relevant statements.

When the display instruction is executed, the integrated value can be output as a histogram.

The HIDAR function has the following configuration:

- Control key commands

.J

Sets conditions of HIDAR.

.H

Specifies output of a histogram and so forth.

- Statements

MEAS DC statement

MEAS VS statement

JUDGE statement

15. 1. 1 Function

- HIDAR can independently collect distribution information for each station.
- The conditions and information set for HIDAR are maintained after the tester power is turned off.
The sequence of data can be released by using key command C.
- Distribution information is not integrated during the PASS/FAIL judgment test (MEAS/C DC).
- Resolution for histogram generation has two kinds; auto resolution and fixed resolution.

Auto resolution

Resolution for histogram generation is determined by measurement data and measurement range.

Fixed Resolution

Resolution for histogram generation is determined by maximum and minimum values of measurement data and number of divisions.

- Histogram ranking

Auto resolution

Histogram ranking is estimated (and set automatically) on the basis of the first set of measurement data and the number of cells.

Fixed Resolution

Histogram ranking is made according to maximum and minimum values of measurement data and number of divisions.

- Determination of necessary resolution for histogram generation

Auto resolution

Determined according to the measuring mode and measuring range.

➔ For more information on resolution, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498\)](#).

If the resolution cannot be set by a VS, VSIM, or ISVM statement, it is determined as follows:

- When integers are as measured values
Resolution: 1
- When the unit of measurement data is a unit of time
Resolution: 1 ns (1×10^{-9} s)
- When the unit of measurement data is a real-number type other than time (s)
The resolution is determined by the following formula with the mantissa as 1.00.

$$(\text{Measurement data}) \times \frac{1}{10000}$$

ExampleThe resolution is calculated in the following way when the unit of measurement data is 20.5 R (ohm)

$$20.5R \times \frac{1}{10000} = 20.5 \times 10^{-3}R$$

The above formula gives: resolution = $1.00 \times 10^{-3}R$.

- When operations are done in the storage destination of the measured data in the MEAS statement and when the operation result is given in a unit different from that of the measured data

The settings of the resolution are determined with one of the operations described previously depending on the unit of the data after operation.

Example

$$\text{MEAS DC : } x = f(x)$$

↑
↑

Storage destination variable name Arithmetic expression

Fixed Resolution

Resolution for histogram generation is determined by maximum and minimum values of measurement data and number of divisions.

Resolution (r) is taken by the following formula in the condition that the maximum value of measurement data is (a), the minimum value of measurement data is (b), and the number of divisions is (n):

$$r = \frac{a - b}{n - 1}$$

(Example 1)

The resolution is calculated in the following way when the maximum value of measurement data is 10 V, the minimum value of measurement data is 0 V, and the number of divisions is 10:

$$\frac{10 \text{ V} - 0 \text{ V}}{10 - 1} = 1.111 \text{ V}$$

$$\text{Resolution} = 1.111 \text{ V}$$

When integers are used as measured values, the calculated resolution is rounded down to an integer.

In a histogram, the maximum value is used as the highest cell value and the minimum value is used as the lowest cell value. However, when the measured values are assigned to integer variable, the maximum value is occasionally different from the highest cell value because the measured values are always rounded down to integers.

(Example 2)

When the maximum and minimum values are 7 and 0, respectively, and the number of divisions is 5, the resolution is calculated as shown below:

$$\frac{7 - 0}{5 - 1} = 1.75$$

The resolution equals 1 when figures below the decimal point are discarded.

The histogram is determined as follows:

(HISTOGRAM)					
NO. OF CELLS : 5			RESOLUTION : 1		
CELL	%	HISTOGRAM		COUNT	CUM%
	0.000 !			0	0.000
Least significant					
cell →	0	0.000 !		0	0.000
	1	20.00 ! *		1	20.00
	2	20.00 ! *		1	40.00
	3	20.00 ! *		1	60.00
Most significant					
cell →	4	20.00 ! *		1	80.00
		0.000 !		0	80.00
20.00% / PLOT CHARACTER			* : PASS # : FAIL		

- The maximum total number of tests is 2147483648. (A 31-bit counter is used.)
- Distribution data can be totaled within a specified range.
Abnormal values can be excluded from distribution data by specifying the maximum and minimum value of measured data.

15. 1. 2 Output Information

Output data is mainly divided into the following three parts: The HEADER part includes the following data:

- HEADER part
 - Device name
 - Lot number
 - DUT interface name
 - Operator name
 - Comments
 - Satellite number
 - Station number
 - Date and time
- CLASSIFICATION part
 - .J command comment and statement class name
 - Program name from which data is collected
 - Total number of tests
 - Number of passes, low fails, and high fails (upper fails)
 - Upper and lower limits of the pass area
 - Number of $\pm$ overflow measured data items
 - Mean value
 - Maximum and minimum measured data

Standard deviation

3 sigma (standard deviation $\times$ 3)

Variance

Maximum and minimum values of collected data

- HISTOGRAM part

Number of histogram divisions (NO. OF CELLS)

Resolution

Histogram

Percentage per character

15. 1. 3 Designation Method

HIDAR can be specified by using the .J command or a statement. Distribution data can be totaled independently of each other.

Therefore, when the .J command specification and statement specification overlap, distribution data is totaled according to the conditions set by each of them, and the results of both the .J command and statement specifications are output when the .H command is issued.

The .J command specification differs from statement specification as follows:

- The .J command can specify a station number, but a statement cannot specify it.
A statement specifies the station at which start is made.
The .J command can set the same HIDAR conditions in stations 1 and 2.
- A statement can select both the classification part and histogram part, or only the classification part. However, the .J command cannot set the classification part only.
Neither statement nor .J command specifications can specify the histogram part only.

15. 2 Condition Setting with .J Command

Data on distribution of test results, and other items can be totaled in real time by a specified test number of a specified test program when the program is executed.

A maximum of 100 items can be specified by test number, and distribution data can be totaled independently of each other.

The .J command can verify and set previously set conditions again before it sets new conditions.

The procedure for specifying the .J command is described below.

```
.J ← Key in .J (this character string is not echoed back.)

SPECIFY HIDAR
  MODE 1 ... SPECIFY HIDAR CONDITIONS
  MODE 2 ... VERIFY CURRENT HIDAR CONDITIONS

SPECIFY MODE NUMBER THAT YOU WISH TO CHOOSE
  MODE = __ *1
```

**1 _ indicates the cursor position.*

Enter the .J command from the VKT when the controller is active.

Enter 1 or 2. (MODE1) is the mode in which a new HIDAR condition is set.

(MODE2) is the mode in which the previously set mode is displayed. Select this mode to verify the previously set conditions before setting new conditions.



When only the Return key is pressed, the .J command is terminated without processing.

◆ When MODE1 is Selected

When MODE1 is selected, the following message is displayed:

```
SPECIFY HIDAR CONDITIONS
STATION NO. = __ *1
```

**1 _ indicates the cursor position.*

Enter the target station number here.

When 1 is entered, conditions for station 1 are set.

When 2 is entered, conditions for station 2 are set.

When 0 is entered, the same conditions are set for stations 1 and 2.

When only the Return key is pressed, the initial menu is displayed again.

The following message is displayed after a station number is input:

```
STATION NO. n
PROGRAM NAME ? filename.
```

Enter the program name to be executed.

n indicates the specified station number.

Enter a file name with the file specifier PRO.

After the program name is entered, the menu for setting conditions is displayed.

- At automatic resolution :
When "AUTO" is input after a file name, the automatic resolution mode is set.
- At fixed resolution :
When "FIXED" is input after a file name, the fixed resolution mode is set.
- When the resolution mode is omitted, the automatic resolution mode is set.

```
1. STN - SPN          ?  1-10.
   NO. OF CELLS       ?  20.
   COMMENT            ?  VEE TEST.
   MAXIMUM DATA      ?  2A.
   MINIMUM DATA      ?  0.1A.

2. STN - SPN          ?  11-15.
   NO. OF CELLS       ?  20.
   COMMENT            ?  TEST P1.
   MAXIMUM DATA      ?  3A.
   MINIMUM DATA      ?  1A.

3. STN - SPN          ?  .

CHANGE (1) , CONTINUE (2) OR OK (CR) ?
```

In the conditions above a maximum of 100 conditions can be specified. (The underlined parts indicate data to be entered.)

STN-SPN

Set the start test number (STN) and stop test number (SPN) for totaling the distribution data.

The test results in the range between the start test number and stop test number are processed together.

If one of the test number has a different unit, an execution time error occurs.

If the range is different, data collection is continued.

A test number must be an integer within the range of 0 to 99999, and $STN \leq SPN$.

When the start test number and stop test number are the same number n, the following specification can be made: STN-SPN ? n.

When only the Return key is pressed, the condition setting is terminated.

NO. OF CELLS

Enter the number of divisions used in creating a histogram as "NO. OF CELLS."
Specify the range of allowable number of divisions as 5 to 200.

COMMENT

A comment can be made on each item specified by the start and stop test numbers with eight characters or less.

Enter the measurement item name, etc.

When only the Return key is pressed, the comment is omitted and the next item can be specified.

MAXIMUM DATA

Specify the maximum value of measurement data when distribution data is collected.

Measured data which exceeds this value is not collected as HIDAR distribution data, and is only counted in + OVER FLOW COUNT.

Specify MINIMUM DATA when abnormal data is not to be collected.

Pay attention to specify the minimum data value when the fixed resolution is specified.

MINIMUM DATA

Specify the minimum value of measured data when distribution data is collected.

Measured data which does not exceed this value is not collected as HIDAR distribution data, and is only counted in - OVER FLOW COUNT.

Specify MINIMUM DATA when abnormal data is not to be collected.

Pay attention to specify the minimum data value when the fixed resolution is specified.

One item consists of STN-SPN, NO. OF CELLS, COMMENT, MAXIMUM DATA and MINIMUM DATA. A maximum of 100 items can be specified.

When only the Return key is pressed for "STN-SPN," the following menu is displayed:

CHANGE (1) , CONTINUE (2) , OR OK (CR) ? ____

- To terminate setting conditions

Press only the Return key.

Previously set conditions are initialized, newly set conditions are written to a HIDAR ledger, ".J EXIT" is displayed on the VKT screen, and the .J command is terminated.

- To continue setting conditions

Enter "2↵".

The test number, NO. OF CELLS, and comment are displayed as in the previous message to enable the operator to continue setting conditions.

- To change any of the set conditions

Enter "1↵".

Station number and program name to be set and the start and stop test numbers and comment of the previously set items are displayed on the VKT screen.

```
STATION NO. n      PROGRAM NAME: filename

1. 1-10          2.11-15   to
VEE TEST          TEST P1
:                 :

SPECIFY ITEM NUMBER THAT YOU WISH TO CHANGE
NUMBER = ____
```

n: Station number to set conditions

A maximum of five sets of start and stop test numbers and comments can be displayed in a line, and a maximum of 40 lines can be displayed.

After the set conditions are displayed, enter the item number to be changed. Then, the message below is displayed on the VKT screen.

(For example, assume that 1↵ was entered.)

```
1. STN - SPN          = 1-10
   NO. OF CELLS       = 20
   COMMENT             = VEE TEST
   MAXIMUM DATA       = 2.000A
   MINIMUM DATA       = 0.100A

1. STN - SPN          1-8↵
   NO. OF CELLS       ? <LF>

   COMMENT            ? <LF>

   MAXIMUM DATA      ? √↵
   MINIMUM DATA      ? ↵

SPECIFY ITEM NUMBER THAT YOU WISH TO CHANGE
NUMBER = ____
```

The set conditions of the item numbers that were specified at first are displayed. Specify STN-SPN, NO. OF CELLS, COMMENT, MAXIMUM DATA, and MINIMUM DATA in this order.

If only the <LF> key is used for any of the three menus (STN-SPN, NO. OF CELLS, COMMENT, MAXIMUM DATA, and MINIMUM DATA), the contents of the conditions are not changed and the next menu is displayed. (As for MINIMUM DATA, the condition change session is terminated.)

When only the Return key is pressed, the contents of the conditions following the current conditions remain unchanged, and the system asks whether to change an item number. (The conditions set again before only the Return key is pressed are changed.)

When √↵ (Space and Return keys) are pressed in setting COMMENT, MAXIMUM DATA, or MINIMUM DATA, the previously set conditions are initialized.

```

1. STN - SPN      = 1-10
   NO. OF CELLS  = 20
   COMMENT       VEE TEST

   MAXIMUM DATA = 2.000A
   MINIMUM DATA = 0.100A

1. STN SPN      ? <LF>
   NO. OF CELLS ? √
   SPECIFY ITEM NUMBER THAT YOU WISH TO CHANGE.
   NUMBER = ____

```

The contents of the start and stop test numbers remain unchanged.

Finishes changing item number 1.
(The contents of NO. OF CELLS and COMMENT remain unchanged.)

When the change of the set conditions is terminated, press only the Return key at the "NUMBER=" message. The following message is displayed again:

```
CHANGE (1) , CONTINUE (2) , OR OK (CR) ? ____
```

When the Return key is pressed after the set conditions are verified, the previously set conditions are initialized, the currently set conditions are written to the HIDAR ledger, and .J is terminated.

◆ When MODE2 is Selected

When MODE2 is selected, the following message is displayed:

```

VERIFY CURRENT HIDAR CONDITIONS STATION NO. = n
LPT & VKT(1) OR VKT(CR) = ____

```

Specify a station number and output device to be verified.

Data specified in a specified station by .J is output.

(Data on the conditions set by a statement such as MEAS are not output.)

15. 2 Condition Setting with .J Command

Enter 1 to get LPT and VKT output at the same time. Press the Return key to get only VKT output.
Then, data is output to output devices as follows:

CURRENT HIDAR CONDITIONS BY '.J' COMMAND.					REV X.XX
STATION NO. 1					
PROGRAM NAME : PRO 100730					1998/10/27 17:11:42
1.1-1	2.2-2	3.3-3	4.4-4	5.5-5	
TEST 1	TEST 2	TEST 3	TEST 4	TEST 5	
10	10	10	10	10	
2.300UA	2.200UA	2.100UA	2.000UA	1.900UA	
200.0NA	300.0NA	400.0NA	500.0NA	600.0NA	
6.6-6	7.7-7	8.8-8	9.9-9	10.10-10	
TEST 6	TEST 7	TEST 8	TEST 9	TEST 10	
10	10	10	10	10	
1.800UA	1.700UA	1.600UA	1.500UA	1.400UA	
700.0NA	800.0NA	900.0NA	1.000UA	1.100UA	
11.11-11	12.12-12	13.13-13	14.14-14	15.15-15	
TEST 11	TEST 12	TEST 13	TEST 14	TEST 15	
10	10	10	10	10	
1.400UA	1.500UA	1.600UA	1.700NA	1.800UA	
1.100UA	1.000UA	900.0NA	800.0NA	700.0NA	
16.16-16	17.17-17	18.18-18	19.19-19	20.20-20	
TEST 16	TEST 17	TEST 18	TEST 19	TEST 20	
10	10	10	10	10	
1.900UA	2.000UA	2.100UA	2.200UA	2.300UA	
600.0NA	500.0NA	400.0NA	300.0NA	200.0NA	

The following information is output:

- Station number
- Program Name
- Revision number, date, and time
- Set conditions

The set conditions are displayed in the following format:

Item number	
1	1-1 (The start test number and stop test number are displayed in this order.)
	TEST1 (Comment specified by .J)
	10 (NO. OF CELLS data)
	2.3 UA (Maximum value of the collected data (MAXIMUM DATA))
	200.0 NA (Minimum value of the collected data (MINIMUM DATA))

After the conditions set in a specified station are output, the first menu is displayed.

```
SPECIFY HIDAR
  MODE 1 ... SPECIFY HIDAR CONDITIONS
  MODE 2 ... VERIFY CURRENT HIDAR CONDITIONS

SPECIFY MODE NUMBER THAT YOU WISH TO CHANGE
MODE = __
```

To set new conditions or set previous conditions again, enter 1. To terminate .J, press the Return key. When MODE2 has been selected previously and MODE1 is selected after the HIDAR conditions are verified, the following message is displayed:

```
SPECIFY HIDAR CONDITIONS.
NEW HIDAR(0) OR CURRENT HIDAR(1) = __
```

To set new conditions, enter 0. To set the previous conditions again, enter 1.

When 1 is entered, the following message is displayed: (When 0 is entered, the same thing as for ["When MODE1 is Selected" on page 15-6](#) happens.)

```
STATION NO.1 *1
PROGRAM NAME [100730] *2
CHANGE(1),CONTINUE(2), OR OK(CR) ? __
```

**1 This is the previously set station number.*

**2 This is the previously set program name.*

When the conditions are reset, distribution data collected under the previous conditions are initialized.

To change the previously set conditions, enter 1. To add some conditions, enter 2. To make no change or additions, press the Return key.

15. 3 Statement Specification

15. 3. 1 Syntax

Specifying HIDAR in the main program has the same effect as the .J command.

However, no station number can be specified when HIDAR is specified in a statement.

The following three statements are specified:

- MEAS DC statement
- MEAS VS statement
- JUDGE statement

```
MEAS DC (m, max, min, [mode]) : x [class name]
MEAS VSn (m, max, min, [mode]) : x [class name]
JUDGE (m, max, min, [mode]) data [class name]
```

m

NO. OF CELLS data

Specify the number of divisions used in creating a histogram.

Specify the allowable number of divisions using an integer-type constant or variable within the range of 5 to 200.

A class name has to be specified together with m.

[class name]

A class name can be given to each test item specified by the statement. Specify a class name using eight or less alphanumeric characters.

max

Maximum value at the time of collecting the measured data. (The same as MAXIMUM DATA of the .J command)

min

Minimum value at the time of collecting measurement data. (The same as MINIMUM DATA of the .J command)

[mode]

Sets a resolution mode.

Specify [AUTO] for the automatic resolution mode. Specify [FIXED] for the fixed resolution mode. When the resolution mode is omitted, the automatic resolution mode is set. (The automatic resolution mode is a default value.)

Data on distribution of test results and other items can be output in real time by specifying both NO. OF CELLS (m) and class name [class name] when the program is executed.

**Limitation**

Neither max nor min can be omitted in the above statements.

If [FIXED] is specified as the resolution mode, neither max nor min can be omitted.

15. 3. 2 Function

The following processing is performed by combining NO. OF CELLS (m) and class name [Class Name]:

Specifying both NO. OF CELLS and class name

```
MEAS DC (m) : X[Class Name]
```

Collects data on classification of mean value, variance, standard deviation, maximum, and minimum, and histogram.

Specifying only class name

```
MEAS DC : X[Class Name]
```

Collects data on classification of mean value, variance, standard deviation, maximum, and minimum. (Histogram data is not collected.)

Specifying only NO. OF CELLS

```
MEAS DC (m) : X
```

A compilation error occurs.

Not specifying NO. OF CELLS or class name

No data on distribution of test results, etc. are totaled.

**Tip**

- If there is a HIDAR ledger specified with another program name when the program in which HIDAR is specified is executed, an execution-time error occurs.
If the program name is the same, distribution data totaling continues.
- If there are the same class names specifying HIDAR in the MEAS or JUDGE statement and the HIDAR conditions (m, max, min, mode) are different, respectively, distribution data is totalled under the HIDAR condition of the first specified MEAS or JUDGE statement.

15. 4 Histogram Output (.H COMMAND)

15. 4. 1 Function

The .H command collects results of the test items specified by a MEAS statement, or .J command and outputs mean value, maximum, minimum, variance, standard deviation, and histogram by items. Specify an output device in advance by using the key command S.

The .H command is specified in the following procedure:

```
.H ← Key in .H (this character string is not echoed back.)
```

```
CALL HIDAR
```

```
STATION NO. = __ *1
```

**1 _ indicates the cursor.*

When the controller is active, enter the above .H command from the VKT.

You will first be prompted to give a station number. Then specify the station number to output the HIDAR information.

When 1 is specified, all HIDAR data in station number 1 is output.

When 2 is specified, all HIDAR data in station number 2 is output.

When 0 is specified, all HIDAR data in stations 1 and 2 is output.

When only the Return key is pressed, no processing is performed, and the .H command is terminated.

If the HIDAR information is not present in the specified station number, the following message is displayed on the VKT screen, and the .H command is terminated:

```
HIDAR TABLE DOES NOT EXIST! ; STATION NO. = n
```

n: Specified station number

The .H output format consists of the three parts of HEADER, CLASSIFICATION and HISTOGRAM display section.

First, the HIDAR information specified with the .J command is output. Then the HIDAR information specified with statements is output.

The HEADER part is output for every station numbers.

15. 4. 2 Output Information

- HEADER part

```

***** HIDAR REPORT *****
                                         REV 0.00
                                         DATE: 1998/08/30 16:58:53

DEVICE : SN74HS00      LOT NO.   : LT821028-12
DUT I/F: HANDLER      OPERATOR  : ADVANTEST
COMMENT: FINAL TEST    TESTER   : 0-1

```

The classification part displays the following data:

DEVICE

Device name

LOT NO.

Lot number

DUT I/F

DUT interface name

OPERATOR

Operator name

COMMENT

Comments

TESTER

Satellite and station numbers

0 - 1

↑ ↑

Station number

↑
Satellite number

DATE

Date and time

The HEADER part is output for each station number.

15. 4 Histogram Output (.H COMMAND)

- CLASSIFICATION part

```

TEST ITEM : TEST1
FILE NAME : PRO 100730      TEST NO. : 1 - 1

TOTAL TEST COUNT : 198          LOWER LIMIT      : 500.0NA
PASS COUNT       : 68           UPPER LIMIT      : 1.500UA
LOWER FAIL COUNT : 45           MINIMUM DATA   : 200.0NA
UPPER FAIL COUNT : 85           MAXIMUM DATA   : 2.400UA
-OVER FLOW COUNT : 17
+OVER FLOW COUNT : 6

MEAN      = 1.325UA  MAXIMUM = 2.400UA  MINIMUM = 220.0NA
STD DEV   = 737.6NA  3-SIGMA = 2.213UA  VARIANCE= 0.5442PA$

```

The classification part displays the following data:

TEST ITEM

Comment by the .J command

FILE NAME

Target program name

TEST NO.

Start and stop test numbers

$1 - 1$

 Start test number Stop test number

The above indicated data is output only when it is specified by the .J command. It is not output when specified by a statement.

TOTAL TEST COUNT

Total number of tests

PASS COUNT

Number of passes

LOWER FAIL COUNT

Number of lower fails

UPPER FAIL COUNT

Number of upper fails

LOWER LIMIT

Pass area lower limit

UPPER LIMIT

Pass area upper limit

- OVER FLOW COUNT

Number of -overflow measured data items

+ OVER FLOW COUNT

Number of +overflow measured data items

MINIMUM DATA

Minimum value for data collection

MAXIMUM DATA

Maximum value for data collection

MEAN =

Mean value

MAXIMUM =

Mean value

MINIMUM =

Minimum value

STD DEV =

Standard deviation

3-SIGME =

3 sigma

VARIANCE=

Variance

\$

Symbol \$ indicates A2 (ampere squared).

• **HISTOGRAM** part

(HISTOGRAM)				
NO. OF CELLS : 10		RESOLUTION : 640.0N		
CELL	% HISTOGRAM	COUNT	CUM%	
	8.586 ; #####	17	8.586	
⇒ 0.04UA	19.19 ; #####	38	27.78	
0.68UA	22.73 ; #####	45	50.51	
⇒ 1.32UA	29.29 ; #####	58	79.80	
1.96UA	17.17 ; #####	34	96.97	
2.60UA	0.000 ;	0	96.97	
3.24UA	0.000 ;	0	96.97	
3.88UA	0.000 ;	0	96.97	
4.52UA	0.000 ;	0	96.97	
5.16UA	0.000 ;	0	96.97	
5.80UA	0.000 ;	0	96.97	
	3.030 ;	6	100.0	
0.732% / PLOT CHARACTER		* :PASS # :FAIL		

The histogram part displays the following data:

NO. OF CELLS

Number of histogram divisions

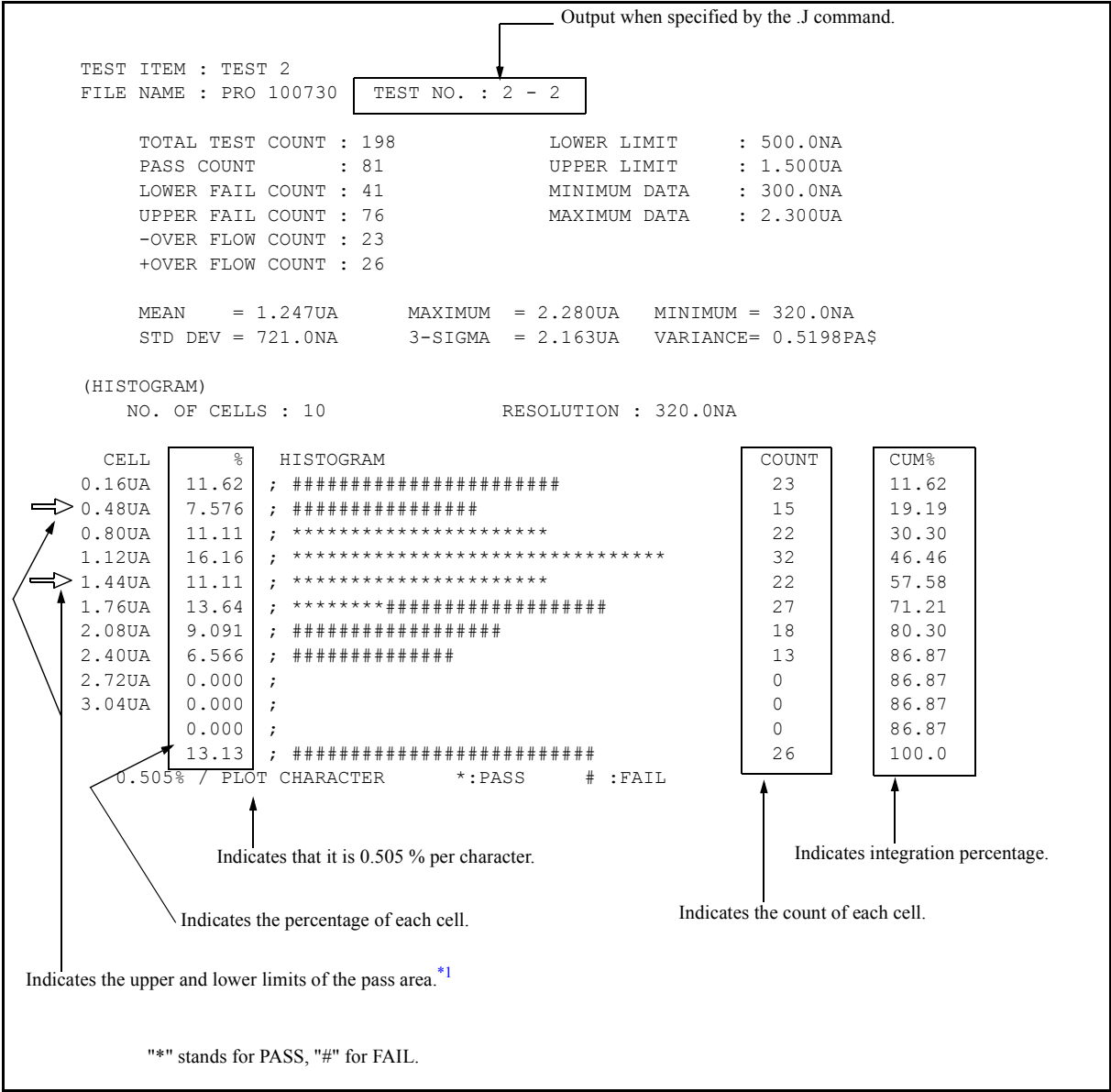
RESOLUTION

Histogram resolution

15. 4. 3 Histogram Output Example

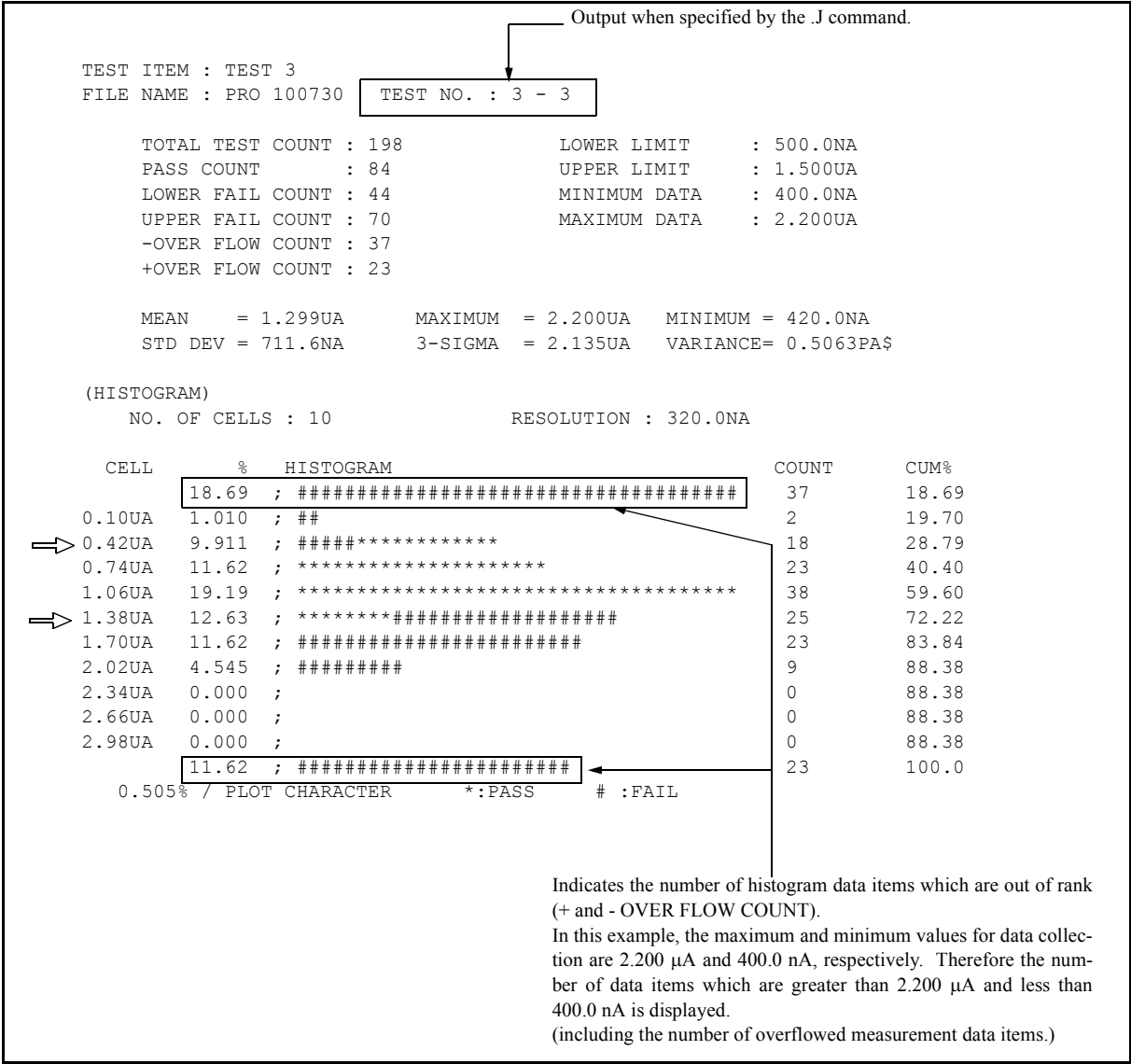
"Histogram Output Example (I)" on page 15-20 to "Histogram Output Example (I)" on page 15-20 to "Histogram Output Example (III)" on page 15-22 show histogram output examples.

◆ Histogram Output Example (I)

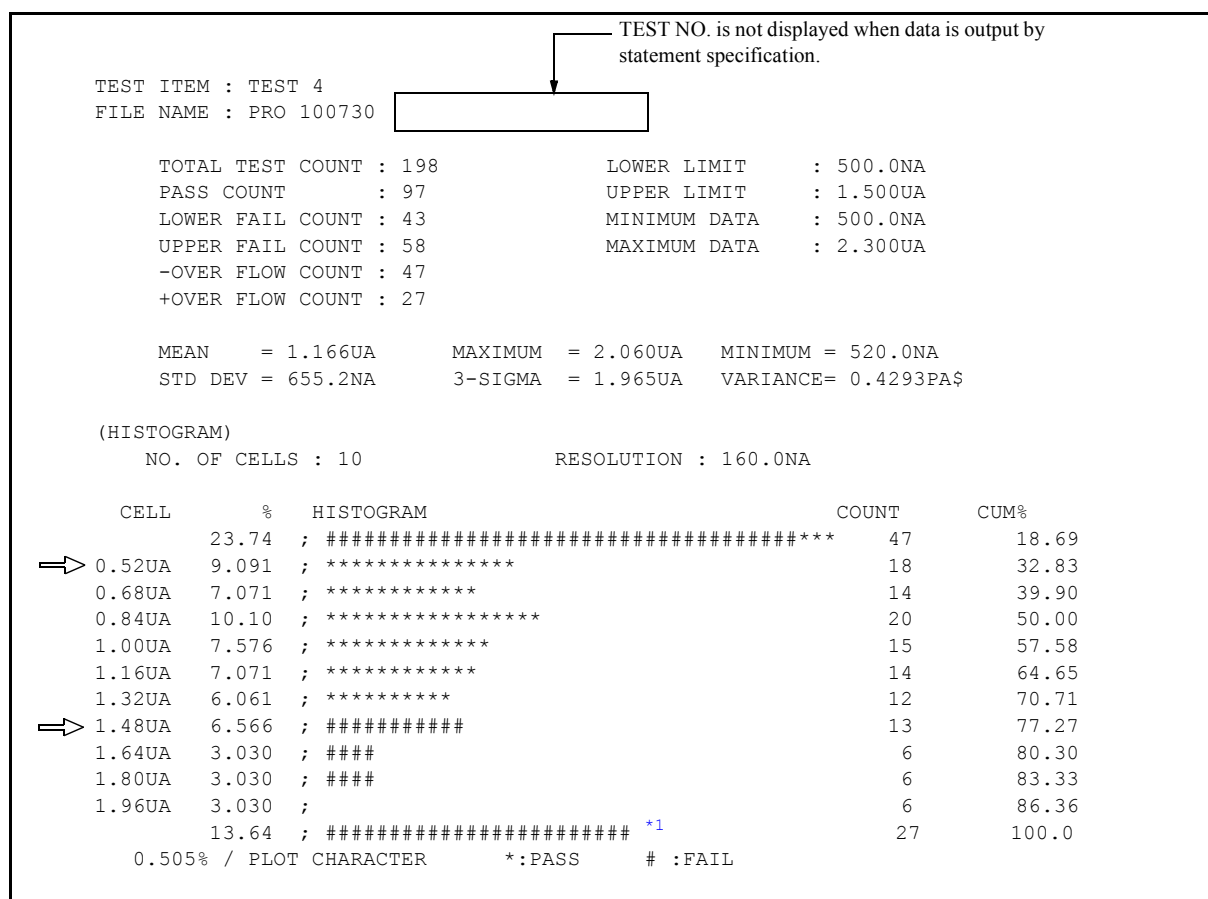


*1 An arrow mark is placed in the bottom cell if the specified upper limit value is out of the pertinent cell. Similarly, an arrow mark is placed in the top cell if the specified lower limit value is out of the pertinent cell.

◆ Histogram Output Example (II)



◆ Histogram Output Example (III)



*1 History display

When the maximum cell count exceeds 40, the number of cell display characters (# or *) is determined to be 40. The number of other cell display characters is calculated according to the ratio of the maximum count to each cell count.

(In the example above, the maximum count is 47, and the number of characters displayed is 40. The number of characters for another cell is calculated based on the ratio.)

When the maximum count is less than 40, each cell count indicates the number of characters displayed.

15. 5 HIDAR Release and Initialization (C Command)

The HIDAR specification with the .J command or statement can be canceled, and totaled distribution data can be initialized by using the C key command, as follows:

The C command initializes test failure counters and both releases and initializes HIDAR information.

```
C ← Key in C (This character is not echoed back.)

CLEAR COUNTER !
STATION NO. = n
CLEAR TEST FAIL INFO. YES(1) OR NO(0) ? 0 *1
CLEAR HIDAR INFO. YES(1) OR NO(0) ? 1 *2
```

When the controller is active, enter the above C command from the VKT.

**1 Message when information concerning the tester fail counter, etc., exists*

**2 Message concerning whether the HIDAR information should be cancelled or initialized*

To cancel and initialize the information, enter 1. Otherwise, enter 0.

When there is no HIDAR information in a specified station number, the message indicated in the footnote ^{*2} is not displayed, and the C command is terminated.

16. Wafer Map

This chapter describes wafer maps.

16. 1 Wafer Map Overview

The wafer map application displays the test results obtained in auto mode for up to 60×60 chips per wafer, in the following two ways:

- Classification according to pass and fail
- Classification according to category lamp

The test results can also be displayed in real time.

Furthermore, the test results can be output to a line printer at the end of each wafer.

Two types of data display are selectable. In one display type, the left uppermost end is the origin point, and in another display type, extra blanks are omitted.

Tip

*The auto mode is for starting the tester with the CALL statement for reading test results defined in the main program when the AUTO switch is pressed.
The entire wafer is scanned.*

Classification according to pass and fail

Symbol	Meaning
*	Pass
N	Fail

Classification according to category lamp

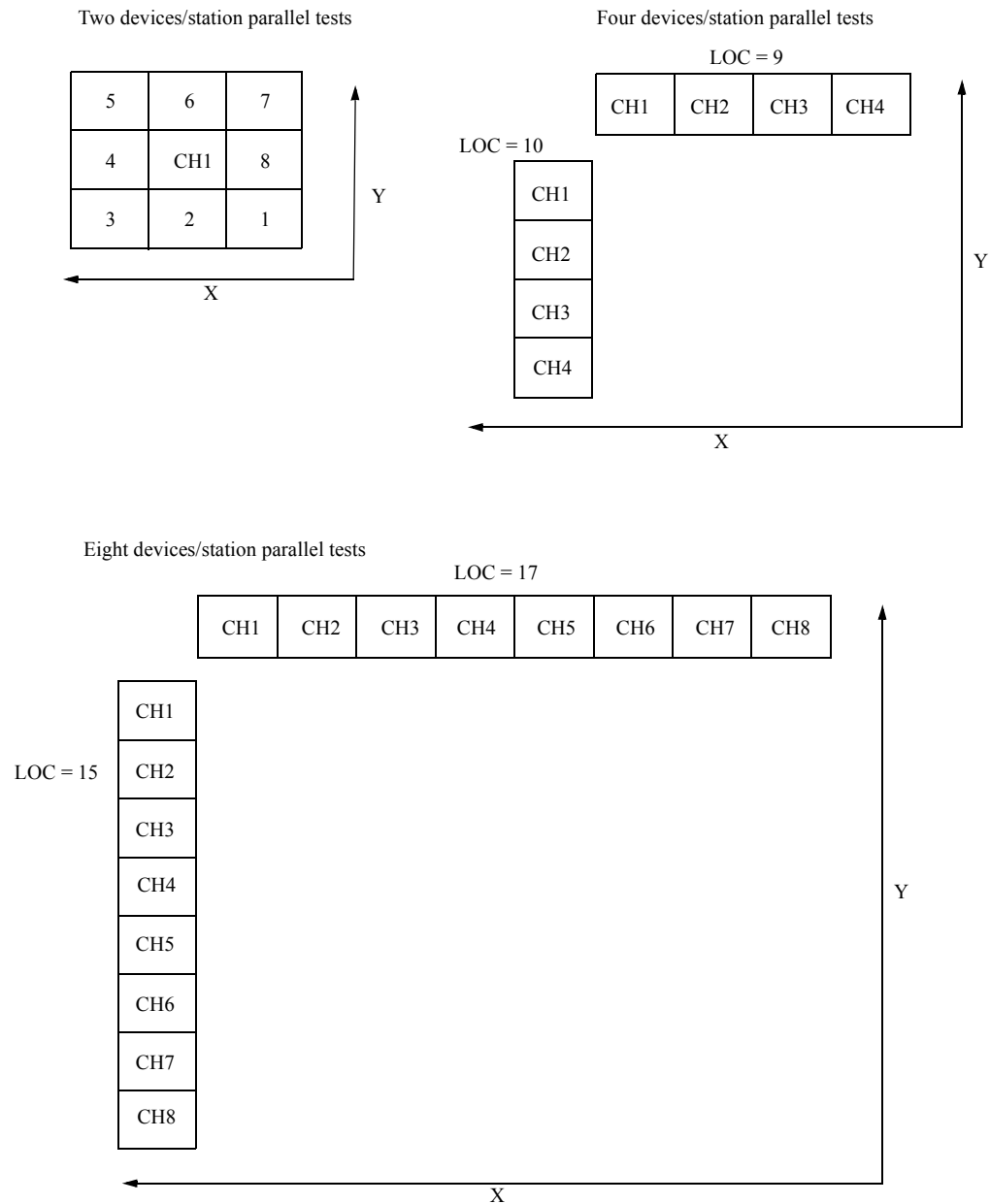
The three following classification methods are available:

- 27 classifications: Symbols A to Z and #
- 17 classifications: 0 to F and #
- 51 classifications: Symbol A to Z and a to y

Tip

Output of 51-classified wafer map to a line printer requires a printer capable of printing lower-case alphabets. When a printer which cannot print lower-case alphabetic characters is used, perform 51 classifications only on the screen display or specify another classification mode.

The following shows the relationship between prober location and wafer map in coordinates.



Tip

When the location of the prober in use is not on the origin of the location coordinates in this utility, revise the formula for determining the location of each DUT in ASC WFAUT, $STBF(n) = XCODI \pm m$, $STBF(n + 1) = YCODI \pm l$, for recompilation.

16. 2 Wafer Map Procedure in AUTO Mode (WFAUT)

In AUTO mode, the test plan program for the wafer test must have the following format:

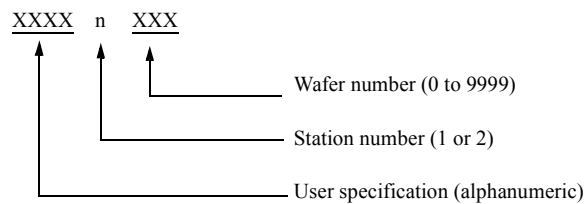
```

PRO      1
STRING V(8)
V="ABCD"
TEST 1
  DC test
  :
TEST 10
  FUNC test
  :
SET CATEGORY   #1   ----- (a)
CALL WFAUT(V, n1, n2) --- (b)
END
  
```

V

Specify a data file name in four or less characters.

The data file name has the following format:



n1

0/1/2/3/4/5/6/7/8/9/11

- 0 : Does not perform map output.
- 1 : Outputs map to the line printer at the end of each wafer. (Normal display)
- 2 : Performs real time map display for station 1.
- 3 : Performs real time map display for station 2.
- 4 : Performs real time map display for station 1, then outputs map to the line printer at the end of each wafer. (Normal display)
- 5 : Performs real time map display for station 2, then outputs map to the line printer at the end of each wafer. (Normal display)
- 6 : Performs real time map display for station 1.
The data file name is fixed to DAT \$WF.MAP.
- 7 : Performs real time map display for station 2.
The data file name is fixed to DAT \$WF.MAP.
- 8 : Performs real time map display for station 1, then outputs map to the line printer at the end of each wafer. (Displayed with blank filling)
- 9 : Performs real time map display for station 2, then outputs map to the line printer at the end of each wafer. (Displayed with blank filling)
- 11 : Outputs to the line printer. (Displayed with blank filling)

n2

0/1/2/3

- 0 : Category map (A to Z, # for 26 or more)
- 1 : Pass/fail map
- 2 : Category map (0 to F in hexadecimal notation, # for 16 or more)
- 3 : Category map (A to Z, and a to y)

According to the results of both the DC and FUNC tests, classify the test cells and set the CATEGORY lamp as shown on line (a).

Classification of up to 51 ranks is possible in wafer map display, and relationship with the SET CATEGORY statement is as follows:

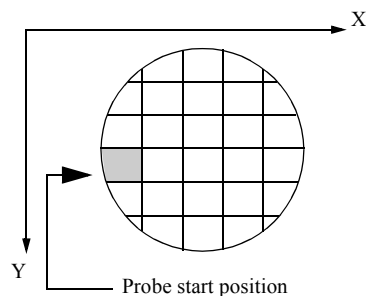
	n2 = 0	n2 = 2	n2 = 3
SET CATEGORY 0	A	0	A
SET CATEGORY 1	B	1	B
⋮			
⋮			
SET CATEGORY 15	P	F	P
SET CATEGORY 16	Q	#	Q
⋮			
⋮			
SET CATEGORY 25	Z	#	Z
SET CATEGORY 26	#	#	a
⋮			
⋮			
SET CATEGORY 50	#	#	Z

To create a wafer map, the statement shown on line (a) must be specified immediately before the END statement.

The results are stored into the data file by the CALL statement shown on line (b). This data file is used to store the results of both the DC and FUNC tests and the results of CATEGORY.

Enter the comment of the header part by the header command.

In the controller mode, move the probe to the cell at which the test is started.



Set the program number to the number generated in step (1) with the PRGM NAME switch on the control box. "1" is set in this example.

This completes preparations.

Press the AUTO switch on the probe.

The probe tests each cell in sequence, and test results are stored in the data file.

If setting has been done in (b) so that the map will be output to the printer, the wafer map is output to the line printer.

16. 3 Wafer Map Output (/WFMAP)

To display the wafer map after measurement, follow the procedure below.

1. Start the wafer map display program.

```
/WFMAP.␣
```

```
DATA FILE-NAME = ?
```

Press the Return key to terminate WFMAP.

When a file name (8 characters) is entered, the following prompt is displayed.

```
PASS, FAIL MAP          (1)
A - Z , #      CATEGORY MAP (0)
O - F , #      CATEGORY MAP (2)
A - Z , a - y  CATEGORY MAP (3)
SELECT MAP :
```

2. When a digit of 0 (Return key) to 3 is entered, the following prompt is displayed.

```
FORMAT SQUEEZED (1) /NORMAL (0) = ?
```

3. When 1 or 0 (Return key) is entered, the following prompt is displayed.

```
OUTPUT MAP TO LPT (1) /CRT (0) = ?
```

4. When 1 or 0 (Return key) is entered, the wafer map is output to the line printer or VKT, respectively, then the initial message is displayed again.

```
DATA FILE-NAME = ?
```

5. To terminate WFMAP, press the Return key.

◆ Output Example of Pass/Fail-Classified Wafer Map

DATE: 30.11.98 11:44:33 AM										STATUS: NORMAL																																						
DEVICE: 2732A										LOT: KSSS																																						
WAFFR#: 2										WAFFR: SAMPLE-1																																						
COMMENT: AUTO SCAN MODE																																																
										1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	3	3	3	3																
1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3																
										.											.											.																
1											.											.											.															
2											.											.											.															
3											.											.											.															
4											.											.											.															
5											.											.											.															
6											.											.											.															
7											.											.											.															
8											.											.											.															
9											.											.											.															
10	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.																
11											.											*	N	*	*	*	N	*	*	*	.																	
12											.											*	*	*	N	*	*	*	N	*	*	*	N	*	.													
13											.											N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	.											
14											.											*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*										
15											.											*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N								
16											.											*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*						
17											.											*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*						
18											.	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*							
19											.	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*							
20	.	.	.	.	.	.	.	.	.	.	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*												
21											.	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*							
22											.	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*							
23											.	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*							
24											.	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*							
25											.	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*							
26											.	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*							
27											.	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*								
28											.	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*								
29											.	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*									
30	.	.	.	.	.	.	.	.	.	.	.	.	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	*	N	*	.	.	.	.									
31											.											N	*	*	*	N	*	*	*	N	*	*	*	N	*	*	.											
32											.											*	*	*	N	*	*	*	N	*	*	*	N	*	*	.												
										TOTAL= 428										PERCENT (%)																												
* : PASS										320										74.77																												
N : DCT,FCT FAIL										108										25.23																												

◆ Output Example of Category Lamp-Classified Wafer Map

DATE: 30.11.98 11:44:03 AM										STATUS: NORMAL														
DEVICE: 273 2A										LOT: KSSS														
WAFFR#: 2										WAFFR: SAMPLE-1														
COMMENT: AUTO SCAN MODE																								
										1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 3 3 3 3														
1 2 3 4 5 6 7 8 9 0										1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3														
1										.										.				
2										.										.				
3										.										.				
4										.										.				
5										.										.				
6										.										.				
7										.										.				
8										.										.				
9										.										.				
10	. .																							

◆ Output Example of Blank Filling Display Wafer Map (50 Categories)

```
DATE: 1998/06/25 14:43:35      STATUS: NORMAL
DEVICE: 3308D                  LOT: L105.3
WAFER#: 7                      WAFER: SAMPLE-2
COMMENT: AUTO SCAN MODE
```

```
      1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 3 3 3 3
      9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3

      8      .                      y x w                      .
      9      .                      w h g c b A y x w          .
     10      . . . . . b A y x w h g c b A y x w . . . . .
     11      .          g c b A y x w h g c b A y x w      .
     12      .          w h g c b A y x w h g c b A y x w .
     13      .    y x w h g c b A y x w h g c b A y x w
     14      . b A y x w h g c b A y x w h g c b A y x w
     15      . b A y x w h g c b A y x w h g c b A y x w
     16      g c b A y x w h g c b A y x w h g c b A y x w
     17      g c b A y x w h g c b A y x w h g c b A y x w
     18      g c b A y x w h g c b A y x w h g c b A y x w
     19      w h g c b A y x w h g c b A y x w h g c b A y x w
     20      w h g c b A y x w h g c b A y x w h g c b A y x w
     21      w h g c b A y x w h g c b A y x w h g c b A y x w
     22      g c b A y x w h g c b A y x w h g c b A y x w
     23      g c b A y x w h g c b A y x w h g c b A y x w
     24      g c b A y x w h g c b A y x w h g c b A y x w
     25      . b A y x w h g c b A y x w h g c b A y x w
     26      . b A y x w h g c b A y x w h g c b A y x w
     27      .    y x w h g c b A y x w h g c b A y x w
     28      .          w h g c b A y x w h g c b A y x w .
     29      .          g c b A y x w h g c b A y x w      .
     30      . . . . . b A y x w h g c b A y x w . . . . .
     31      .                      w h g c b A y x w          .
     32      .                      y x w                      .
```

16. 4 Example 1 of Using Wafer Map Utility

16. 4. 1 Programming

- Use of SET CATEGORY statement

```
PRO MAP SMAP ;4DUT/1STN TEST PGM

IF INIT ENTRY *1
  RESET WFCNT ALL *2
  SET WFCNT 98 1 *3
EXIT

STRING V(8)
V="ABCD" *4

TEST m
.
.
.

IF PASS SET CATEGORY 1 *5
IF FAIL SET CATEGORY 2 *5
CALL WFAUT(V,0,0) *6

END
```

*1 This utility is executed only when the **[INIT]** key of a control box is set to ON.
The **[INIT]** key is automatically set to OFF if the program is executed once.

*2 Wafer data is all reset.

*3 The number of the wafer to be tested is set to 1.

*4 The name of a data file which stores wafer map data begins with ABCD.
For example, the data file name of the second wafer measured by station 1 is ABCD1002.

*5 Categories are set according to the measurement results.

*6 The wafer map utility is called.

- Use of CATEGORY TABLE

```

PRO MAP SMAP ;4DUT/1STN TEST PGM

IF INIT ENTRY
  RESET WFCNT ALL
  SET WFCNT 98 1
  EXIT

STRING V(8)
V="ABCD"

CATEGORY TABLE *1
  CATEGORY1 = PTm
  CATEGORY2 = FTm
CATEGORY END *1

SORT TABLE
  SORT1(PASS) = NC1
  SORT2(FAIL) = NC2
SORT END

TEST m
.
.
.

SET DECISION CATEGORY SORT *2
CALL WFAUT(V,0,0)

END

```

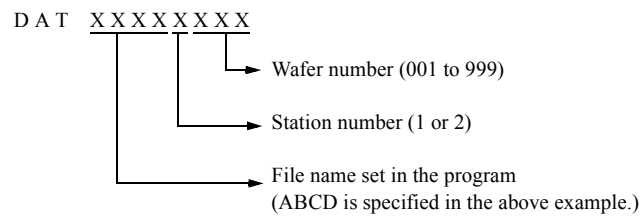
*1 A category table is set.

*2 A category number and sort number are determined before the wafer map utility is called.

16. 4. 2 Operation

1. Set up the auto prober.
2. Set the test program name and test conditions in the control box of the tester.
3. Press the [INIT] key of the control box.
4. Enter /HEADER↵ from the console to set the header information to the wafer map.
5. Execute start from the auto prober.
6. After one or more wafers have been tested, enter /WFMAP↵ from the console and output the wafer map of any wafer to VKT or LPT.

Note that format of the name of a data file which stores wafer map data is as follows:



16. 5 Example 2 of Using Wafer Map Utility

In the basic programming examples shown in the preceding subsection, you cannot change the data file name to store the wafer map information at your discretion. (The test program requires modification.)

Note that a wafer number always begins with 1.

Note that a wafer number always begins with 1. This section describes the use of the wafer map utility in which the following points are paid attention to:

- Arbitrarily setting and changing the name of the file which stores wafer map data
- Arbitrarily setting initial value of a wafer number
- Arbitrarily setting and changing wafer map header data without starting a header utility for simple and easier operations
- Creating a wafer map utility initializing program available for any test program commonly with which all preceding items can be executed.

16. 5. 1 Programming

- Creating a test program

The format of the test program (main program) is as follows:

```

PRO MAP  SMAP  ;PARALLEL TEST PGM

    ON WCNT :L1 *1

CATEGORY TABLE
CATEGORY1 = PTm
CATEGORY2 = FTm
CATEGORY END

SORT TABLE
SORT1(PASS) = NC1
SORT2(FAIL) = NC2
SORT END

TEST m
.
.
.

    SET DECISION CATEGORY SORT *2
L1:CALL WFAUT(GLOB1,0,0) *3

END

```

**1 A branch to label L1 occurs when one wafer has been measured.*

**2 A category number and sort number are determined before the wafer map utility is called.*

**3 The wafer map utility is called. A global argument is used for the argument of a file name in this case.*

- Creating a wafer map utility initializing program

The user can set and change data file names and wafer numbers arbitrarily by creating an initializing program as follow.

This initializing program allows the wafer map header data to be set and changed freely, thus eliminating the need for starting the header utility again.

```

PRO   IWFMAP                ;WFMAP  INITIAL PROGRAM
                                ;1986/08/11
                                ;

INPUT *,//,"*****",/
INPUT  " *      INITIALIZE PROGRAM START !!      *",/
INPUT  "*****",/
INPUT  "PLEASE INPUT WAFER-MAP INFORMATION",/

;---- READ STATION NUMBER ----;
READ STN TO STNM *1
IF STNM=1 INPUT //," ** STATION 1 **",/
IF STNM=2 INPUT //," ** STATION 2 **",/

;---- SET DATA-FILE NAME ----;
LL1:  STRING V1(8)="WMAP" *2
      INPUT[0,4,(4)] "    DATA-FILE NAME = ",V1,/ *3
      GLOB1=V1 *4

;---- SET HEADER INFORMATION ----;
INPUT[0, 2]"    DEVICE NAME    = ",DEVICE, *5
INPUT[0, 2]"    LOT NAME       = ",LOT, *5
INPUT[0, 2]"    WAFER NAME     = ",WAFER, *5
INPUT[0, 2]"    = ",COMMEN, *5

;---- SET FIRST WAFER # ----;
LL2: INPUT[0, 1]"    FIRST WAFER # = ",WN1, *6
      IF WN1<1 OR WN1>999 GOTO LL2

;---- RE-ENTER ROUTINE ----;
INPUT[0,4(1)]//,"DO YOU WANT CHANGE ? (YES,NO) ",CHN,/
IF CHN="Y" GOTO LL1

;---- RESET WAFER COUNTERS ----;
RESET WFCNT ALL *7
SET WFCNT 98 WN1 *8

INPUT "INITIALIZE END !!",/
END

```

*1 A data filename and wafer number can be set and changed freely only at the station which starts this initializing program.

*2 Area for the data file name is reserved to store the wafer map information for setting the initial data (WMAP in the above case).

*3 A data file name is set or changed.

- *4 A data file name is loaded into a global argument to enable the data file to be delivered to the test program.*
- *5 Wafer map header data is set or changed. Note that DEVICE, LOT, WAFER, and system arguments.*
- *6 The number of the wafer to be measured first is set or changed.*
- *7 Wafer data automatically counted by the system is reset.*
- *8 The wafer number set or changed in the footnote ^{*4} above is set in the system.*

16. 5. 2 Operation

This subsection describes the operation when the test program and the wafer map utility initializing program are prepared as shown in [16. 5. 1 "Programming" on page 16-13](#).

1. Set up the auto prober.
2. Set the test program name and test conditions in the control box of the tester.
3. Start the initializing program from the console as shown below:

For station 1

```
PRO/1 IWFMAP.┘
```

For station 2

```
PRO/2 IWFMAP.┘
```


4. When the initializing program is started, the questions as shown below are displayed in sequence. Enter the required answers from the console.

```
*****
*   INITIALIZE PROGRAM START !!   *
*****

PLEASE INPUT WAFER-MAP INFORMATION

** STATION 1 **
DATA-FILE NAME = YAMA
DEVICE NAME    = SRAM␣
LOT NAME       = 860811␣
WAFER NAME     = ABC-123␣
COMMENT        = ROOM
FIRST WAFER #  = 10␣

DO YOU WANT CHANGE ? (YES,NO) Y
INITIALIZE END !!
```

5. Execute start from the auto prober.
6. Input /WFMAP from the console after one or more wafers have been measured, and output the wafer map of an arbitrary selected wafer to VKT or LPT.
- Note that format of the name of a data file which stores wafer map data is as follows:

```
DAT XXXXX
```

Wafer number (001 to 999)
 Station number (1 or 2)
 File name set in 4.
 (YAMA in the above operation example)

◆ Output Examples

DATE	:1998/08/11	11:44:03	STATUS:	NORMAL
DEVICE	:SRAM		LOT:	860811
WAFR#	:10		WAFR:	ABC-123
COMMENT	:ROOM			
		1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 3 3 3 3		
	1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3			
1		.	.	.
2		.	.	.
3		.	.	.
4		.	.	.
5		.	.	.
6		.	.	.
7		.	.	.
8		.	.	.
9		.	.	.
10				
11		.	F I H G F I H G F	.
12		.	H G F I H G F I H G F I H	.
13		.	I H G F I H G F I H G F I H G	.
14		.	F I H G F I H G F I H G F I H G F	.
15		.	G F I H G F I H G F I H G F I H G F I	.
16		.	H G F I H G F I H G F I H G F I H G F I H	.
17		.	H G F I H G F I H G F I H G F I H G F I H	.
18		.	I H G F I H G F I H G F I H G F I H G F I H G	.
19		.	I H G F I H G F I H G F I H G F I H G F I H G	.
20		.	I H G F I H G F I H G F I H G F I H G F I H G	.
21		.	I H G F I H G F I H G F I H G F I H G F I H G	.
22		.	I H G F I H G F I H G F I H G F I H G F I H G	.
23		.	I H G F I H G F I H G F I H G F I H G F I H G	.
24		.	I H G F I H G F I H G F I H G F I H G F I H G	.
25		.	I H G F I H G F I H G F I H G F I H G F I H G	.
26		.	I H G F I H G F I H G F I H G F I H G F I H G	.
27		.	H G F I H G F I H G F I H G F I H G F I H	.
28		.	H G F I H G F I H G F I H G F I H G F I H	.
29		.	G F I H G F I H G F I H G F I H G F I	.
30		.	F I H G F I H G F I H G F I H G F I H G F	. . .
31		.	I H G F I H G F I H G F I H G F I H G	.
32		.	H G F I H G F I H G F I H G F I H	.

16. 6 Example 3 of Using Wafer Map Utility

This section describes using of the real time MAP output.

16. 6. 1 Programming

The format of the test program (main program) is as follows:

```
PRO  MAP  SMAP

      ON  WCNT : L1

      CHAR[1] NME = "ABCD"
      INTE[1] N1, N2
      N1 = 2 & N2 = 1
      CATEGORY TABLE
        CATEGORY1 = PTm
        CATEGORY2 = FTm
      CATEGORY END

      SORT TABLE
        SORT1 (PASS) = NC1
        SORT2 (FAIL) = NC2
      SORT END

      TEST m
      .
      .
      .
      SET DECISION CATEGORY SORT

      L1 : CALL WFAUT (NME, N1, N2)

      END
```

The ST2 pass/fail map is output in real time.

16. 6. 2 Output Examples

```

DATE:   1998/01/19   13:13:35                STATUS: NORMAL
DEVICE: 2732A                LOT:   NO1
WAFER#: 2                WAFER#: SAMPLE-1
COMMEN: REAL TIME WAFER MAP

                111111111122222222223333333333444444444455555555556
123456789012345678901234567890123456789012345678901234567890

17          |          |          |          |          |
18          |          |          |          |          |
19          |          |          |          |          |
20 -----|-----|-----|-----|-----|-----|-----|-----|
21          |  ****  |          |          |          |          |
22          |  ****  |          |          |          |          |
23          | NNNN   |          |          |          |          |
24          | NNNN   |          |          |          |          |
25          |  ****  |          |          |          |          |
26          |          |          |          |          |          |
          |          |          |          |          |          |

TOTAL = 20                PASS = 12

*** NOTE ***

B: BREAK                ^: DOWN SCROLL                V: UP SCROLL
C: CONTINUE            R: REFRESH

```

[B] key

Suspends the MAP output

[↑] key

Scrolls down the MAP display in the range of Y addresses 1 to 60 (effective after the B key is pressed).

[↓] key

Scrolls up the MAP display in the range of Y addresses 1 to 60 (effective after the B key is pressed).

[C] key

Resumes the MAP output (effective after the B key is pressed).

[R] key

Refreshes the display and starts display from Y-coordinate 1 (effective after the B key is pressed).

Tip

The above [B] key is effective only during WAFER MAP execution.

During WAFER MAP execution the STATUS indicator blinks.

17. FAIL TEST DISPLAY UTILITY (/FTDISP, /FTLIST)

/FTDISP and /FTLIST are used to display the sort numbers of the test results and the PASS/FAIL information for each test number and contact number.

```
/FTDISP␣ *1  
/FTLIST␣ *2
```

**1 Automatic activation at each test end*

**2 Activation by operator when output is required*

- Operation

Automatic activation at each test end

```
/FTDISP␣  
  
ENABLE(1)/DISABLE(0) = 0 1  
                        ↑  
                Current setting status
```

Enter 1 to place the utility in activation mode.

Enter 0 to release the utility.

Activation by operator when output is required

```
/FTLIST␣
```

- Output Examples

```
***** FAIL TEST DISPLAY *****      "P":PASS "*" :FAIL "-" :NO-TEST

TNO DDDD
  0000 ← DUT number (hundred's place)
  0000 ← DUT number (ten's place)
  1234 ← DUT number (unit's place)

001 PPPP
002 *PPP ← PASS
003 -PPP
100 -PP* ← FAIL
200 -P*- ← NO-TEST
↑
Test number

SRT 0000 ← Sort number (ten's place)
    1231 ← Sort number (unit's place)
```

18. Yield Report Output Utility (/YIELD)

This chapter describes the yield report output utility (/YIELD) that outputs yield reports.

18. 1 Function

- The yield report output utility outputs yield reports based on sorting and categorization.
The following information is output for each station:
Header: date and time, device name, lot name, handler name, operator name, and comment
Total yield
Yield for each sort and contact
Yield for each category and contact
- Enable the yield information to be filed as data.
(Only one data file can exist.)
The following operations are possible if a previously prepared data file exists:
Enable new information to be added to the previous data.
Enable the previous data to be rewritten to file.
- Enable the data file to be displayed.

18. 2 Syntax

18. 2. 1 /YIELD

Function: This utility is used to output yield reports.

Syntax

```
/YIELD↵
```

18. 3 Operation

- 1) Using the key command C, clear the counter area of the station to be used before starting tests. To test two stations in parallel, clear two stations.
- 2) Perform the tests.
- 3) Activate this utility.

Direct mode (displays only the yield information)

Executes /YIELD. Specify the following: Select each item and output the information to complete.

- Station to be output.
- Either VKT output or LPT output.
- Either VKT output or LPT output.

```
/YIELD.␣
```

```
SPECIFY STATION NO.,
  ST1(1) , ST2(2) OR ST1&ST2(B)           =  __
```

```
SPECIFY OUTPUT ITEM,
  SORT(1) , CATEGORY(2) OR SORT&CATEGORY(B) =  __
```

```
SPECIFY OUTPUT DEVICE,
  VKT(1) , LPT(2) OR VKT&LPT(B)           =  __
```

Interactive mode (displays the menu screen)

The menu screen is displayed when /YIELD/M is executed.

```
MENU 1 : OUTPUT YIELD (1)           --- Displays the YIELD information.
MENU 2 : OUTPUT DATA FILE (2)      --- Displays the data file information.
MENU 3 : OUTPUT DATA FILE (3)      --- Creates a data file to store the
                                     YIELD information.
MENU 4 : SELECT OUTPUT FORMAT (4)   --- Selects a display format.
```

Select a menu number.

YIELD/M.

YIELD REPORT MENU,

MENU 1 : OUTPUT YIELD (1)

MENU 2 : OUTPUT DATA FILE (2)

MENU 3 : CREATE DATA FILE (3)

MENU 4 : SELECT OUTPUT FORMAT (4)

(EXTRACT) ← The currently selected format is displayed.

SELECT MENU NUMBER MENU= __

When "MENU 1: DISPLAY YIELD" or "MENU 2: DISPLAY DATA FILE" is selected

Specify the following: Select each item and output the information to complete.

- Station to be output.
- Either for each sort or category.
- Either VKT output or LPT output.

SPECIFY STATION NO.,

ST1(1) , ST2(2) OR ST1&ST2(B) = __

SPECIFY OUTPUT ITEM,

SORT(1) , CATEGORY(2) OR SORT&CATEGORY(B) = __

SPECIFY OUTPUT DEVICE,

VKT(1) , LPT(2) OR VKT&LPT(B) = __

When "MENU 3 : CREATE DATA FILE" is selected

Select a station at which a data file is created.

SPECIFY STATION NO.,

ST1(1) , ST2(2) OR ST1&ST2(B) = __

Make the following selection if a data file already exists at the specified station:

DATA FILE(YIELD.BF) HAS ALREADY EXISTED

OVERWRITE(1) OR APPEND(2) = __

- Enable the previous data to be rewritten to file (select OVERWRITE(1)).

When the data file is created, the screen displays as follows to complete.

```
DATA FILE(YIELD.BF) HAS BEEN CREATED.
```

- Enable new information to be added to the previous data in the file (select APPEND(2)).

In the following cases, a prompt is displayed to ask whether the data is added or not. The window shown below is displayed.

When the number of DUTs and yield information to be added is differed.

When the header of the data file and the header of yield to be added is differed.

```
DIFFERENCE OF DUT NO. & HEADER INFORMATION !!

                                <DATA FILE HEADER>      <YIELD HEADER>
<<ST-1>>
DUT NO.           : 2                               : 1
---<HEADER>-----
DEVICE NAME       :                               : DRAM64K
SATELLITE NO.    : 0                               : 2

<<ST-2>>
DUT NO.           : 2                               : 1
---<HEADER>-----
DEVICE NAME       :                               : DRAM64K
LOT NAME          : LT841011-11                     : LT841011-12
DUT I/F          : T3998                             :
SATELLITE NO.    : 0                               : 2
OPERATOR         : TAKEDA                           : ADVAN

APPEND DATA FILE ?
YES (Y) OR NO (N)                               = Y
```

Select either of the following to make addition for different headers.

Make addition after rewriting the yield header (Y).

Add only the yield information without rewriting the yield header (N).

```
HEADER INFORMATION OVERWRITE ?
YES (Y) OR NO (N)                               =
```

When the data file is created, the screen displays as follows to complete.

```
DATA FILE(YIELD.BF) HAS BEEN CREATED.
```

When "MENU 4 : SELECT OUTPUT FORMAT" is selected

The following menu is displayed.

```
SELECT OUTPUT FORMAT, ALL(AL) EXTRACT(EX) OR EXTRACT2(E2) ?
```

Following three display formats are available to choose from.

- ALL

Counter value is displayed even if it is 0.

To choose this format, enter **AL**.

- EXTRACT

The category counters are divided into blocks, each consisting of 50 counters (1 to 50, 51 to 100, ..., 901 to 950, and 951 to 999). If all counter values in a block are zeros, the data of that block is not displayed.

The sort counters are divided into blocks, each consisting of 8 counters (1 to 8 and 9 to 16). If all counter values in a block are zeros, the data of that block is not displayed.

To choose this format, enter **EX**.

- EXTRACT2

The counter value is not displayed when all the data of horizontal one line are 0.

➔ For more information, refer to ["Output Example of Yield Reports \(for Each Sort\)" on page 18-9](#) or ["Output Example of Yield Reports \(for Each Category\)" on page 18-10](#).

To choose this format, enter **E2**.

After this selection, it returns to the menu of /YIELD/M again.

18. 4 Error Indication

- When outputting non-existed data file, the screen displays as follows to complete the program.

```
DATA FILE(YIELD.BF) DOES NOT EXIST !!
```

- When outputting the data file, the screen displays as follows if the data for the specified station is not existed.

```
STATION-1 DATA DOES NOT EXIST !!
```

or

```
STATION-2 DATA DOES NOT EXIST !!
```

- If the data file revision does not match with the OS revision when APPEND(2) is selected for MENU 3 : CREATE DATA FILE, the following message is displayed and the main menu is displayed.

```
DATA FILE(YIELD.BF) REV. UNMATCH
```

18. 5 Precautions

- The data in the header controls the output of system global variable contents.
The user can write system global variables by either using the /HEADER command or by writing them directly in a program.
- When executing the data filing, the header information is not added but overwritten by adding the previous file.
To overwrite the header information in the data file, use /YIELD/H.
The operation is same as the /HEADER↵ command operation.
- Percentages in the report represent the ratio with respect to the total number of DUTs tested.
- The APPEND function for a data file created in the OS (ASX/U-51 REV 6.02 or earlier) which contains up to 200 category numbers cannot be used in an OS (ASX/U-51 REV 6.03 or later) which contains up to 999 category numbers.

18. 6 Output Examples

◆ Output Example of Yield Reports (for Each Sort)

####

YIELD REPORT AT 2000/07/03 16:33:54 FOR T5500 SERIES

####

DEVICE: DRAM128M

LOT NO: LT841011-12

DUT I/F: M6541A

TESTER: 0-1

OPERATOR: R.ADVAN

COMMENT: FINAL TESTING

<STN-1>

TOTAL = 2000

PASS = 1900 (95.00%)

FAIL = 100 (5.000%)

↑

↑

↑

Total No. of tested DUTs

No. of passed DUTs

No. of failed DUTs

	TOTAL	DUT001	DUT002	DUT003	DUT004	← DUT number (Contact number)
SORT-01 =	1800	500	500	500	300	
	90.00%	25.00%	25.00%	25.00%	15.00%	
SORT-02 =	100	20	20	20	20	
	5.000%	1.000%	1.000%	1.000%	1.000%	
SORT-03 =	0	0	0	0	0	
:	0.000%	0.000%	0.000%	0.000%	0.000%	
:	:	:	:	:	:	
SORT-14 =	0	0	0	0	0	
	0.000%	0.000%	0.000%	0.000%	0.000%	
SORT-15 =	50	10	10	10	20	
	2.500%	0.500%	0.500%	0.500%	1.000%	
SORT-16 =	50	10	10	10	20	
	2.500%	0.500%	0.500%	0.500%	1.000%	

↑

Yield for each contactor

Total of all contactors

When output format is EXTRACT2, these data are not displayed.

◆ Output Example of Yield Reports (for Each Category)

```

#### YIELD REPORT AT 2000/07/03 16:33:54 FOR T5500 SERIES ####

DEVICE:  DRAM128M      LOT NO: LT841011-12
DUT I/F: M6541A      TESTER: 0-1      OPERATOR: R.ADVAN
COMMENT: FINAL TESTING

<STN-1>

TOTAL = 2000      PASS = 1900 (95.00%)      FAIL = 100 (5.000%)
  ↑              ↑              ↑
Total No. of tested DUTs  No. of passed DUTs      No. of failed DUTs


TOTAL  DUT001  DUT002  DUT003  DUT004 ← DUT number
                (Contact number)

CAT-001 =    50      10      10      10      20
          2.500%    0.500%    0.500%    0.500%    1.000%

CAT-002 =    50      10      10      10      20
          2.500%    0.500%    0.500%    0.500%    1.000%

CAT-003 =     0       0       0       0       0
          0.000%    0.000%    0.000%    0.000%    0.000%

CAT-004 =     0       0       0       0       0
          0.000%    0.000%    0.000%    0.000%    0.000%

CAT-005 =     0       0       0       0       0
          0.000%    0.000%    0.000%    0.000%    0.000%
:         :         :         :         :
:         :         :         :         :

CAT-999 =     0       0       0       0       0
          0.000%    0.000%    0.000%    0.000%    0.000%
          └──────────┘ └──────────┘ └──────────┘ └──────────┘
              ↑                      Yield for each contactor
Total of all contactors

```

19. Data Logging (/DTLOG)

This chapter describes the /DTLOG utility that is used to specify data logging.

19. 1 Syntax

19. 1. 1 /DTLOG

Function: /DTLOG specifies data logging. Basically, this has the same function as the M command.

Syntax

```
/DTLOG/n1 DCT RATEn4 TESTn5-n6
/DTLOG/n1 DCTP RATEn4 TESTn5-n6
/DTLOG/n1 DCTF RATEn4 TESTn5-n6
/DTLOG/n1 DCS RATEn4 TESTn5-n6
/DTLOG/n1 DCSP RATEn4 TESTn5-n6
/DTLOG/n1 DCSF RATEn4 TESTn5-n6
/DTLOG/n1 CLEAR RATEn4 TESTn5-n6
```

Argument: n1

Specifies station numbers. Enter 1 or 2.

When this argument is omitted

A prompt that asks the station number is displayed again.

DCT

Logs the data of DC parametric test results.

DCTP

Logs the data of DC parametric test results which have been checked as PASS.

DCTF

Logs the data of DC parametric test results which have been checked as FAIL.

DCS

Logs the data of DC parametric test results in a short-type format.

DCSP

Logs the data of DC parametric test results which have been checked as PASS in a short-type format.

DCSF

Logs the data of DC parametric test results which have been checked as FAIL in a short-type format.

CLEAR

Clears the data logging mode.

RATE n 4

Specify sample device data logging. Data logging is performed for each device specified by n 4. Specify n as a decimal number within the range of 1 to 32767.

When this argument is omitted

n is assumed to be 1 ($n = 1$).

TEST n 5- n 6

Specifies test number data logging. The data on the specified item is logged. n 5- n 6 indicates a range of values to be specified. The range must be between 0 to 99999 in decimal.

When this argument is omitted

Logging is performed for all tests.

Restrictions

- ➔ For the restrictions on functional test data logging, refer to the description of the M command in [T5500 Series Tester Controller Commands Reference Manual\(No. : 8256297\)](#).

Example

(1) When You Want to Log the Data of DC Parametric Test Results

```
/DTLOG/1 DCT↵
```

(2) When You Want to Log the Data of DC Parametric Test Results in Tests 10 to 20, 30, and 100 to 110

```
/DTLOG/1 DCT TEST10-20 TEST30 TEST100-110↵
```

20. Global Variables

This chapter describes how to display the contents of global variables and sample numbers and initialize them.

20. 1 Global Variable Type

- The global variables are unique to the system, and can be optionally treated in test programs. The global variable type is as follows:

SWITCH

Controls the test program sequence externally.

SWITCH is 3-word type, and can be specified with any type of data.

GLOB1 to 50

Used as the working area for data transfer between test programs, or from the outside to test programs. 50 variables are provided for GLOB.

GLOB1 to 9 are 5-word type variables and GLOB10 to 50 are 3-word type variables.

GLOB1-50 can be specified with any type of data.

DEVICE

Stores test device names attached to each report.

WAFER must be up to 12 characters.

LOT

Stores lot names attached to each report. WAFER must be up to 24 characters.

SUBLOT

Stores a number attached to a lot name (number following a hyphen "-"). The data in this variable is output next to the lot name attached to each report.

SUBLOT must be up to a 2-digit integer.

When using the initialization commands, no distinction is made between LOT and SUBLOT.

DUTIF

Stores station-type names attached to each report. For example, handler, probe or manual station is specified in DUTIF. WAFER must be up to 12 characters.

OPERATOR

Stores operator names attached each report.

WAFER must be up to 24 characters.

COMMENT

Stores comments to explain test conditions attached to each report. WAFER must be up to 40 characters.

WAFER

Stores wafer names attached to each report. WAFER must be up to 24 characters.

- The following three commands are used to initialize global variables:

/GLOBAL

This utility is used to display and initialize the data in all the global variables and sample numbers.

/SWITCH

This utility is used to display and initialize the data in SWITCH.

/HEADER

Initializes and displays the data in DEVICE, LOT, SUBLOT, DUTIF, OPERATOR, COMMENT, WAFER, and the sample number attached to each report as a header.



Tip

The sample number is not the global variable. The sample number, however, can be displayed and initialized by using the same commands as for the global variables since their usages are alike.

20. 2 Syntax

20. 2. 1 /GLOBAL

Function: This utility is used to display and initialize the data in all the global variables and sample numbers.

Syntax

```
/GLOBAL/n/P/L ALL      CLEAR␣
      SWITCH
      GLOB
      GLOB1
      GLOB2
      :
      GLOB50
      HEADER
      DEVICE
      LOT
      DUTIF
      OPERATOR
      COMMENT
      WAFER
      SAMPLE
```

Argument: /n

Specifies station numbers.

Enter 1 or 2 for n.

Default

A prompt that asks the station number is displayed again. The following is an example.

```
/GLOBAL SWITCH␣
STATION NO.=1
SWITCH=123  #11␣
              (# denotes a hexadecimal number.)
```

/P

Displays the data in HEADER.

/L can be omitted.

/L

Displays the data in HEADER on a line printer.

/L can be omitted.

ALL

Specifies all the global variables and one sample number.

GLOB

Specifies GLOB1 to 50.

HEADER

Specifies DEVICE, LOT, DUTIF, OPERATOR, COMMENT, WAFER and SAMPLE. Parameters concerned are displayed and/or set when such parameters are specified as SWITCH, GLOB1 to 50, DEVICE, LOT, DUTIF, OPERATOR, COMMENT, WAFER, or SAMPLE.

It is equivalent to designating ALL when ALL to SAMPLE are not specified.

CLEAR

Unconditionally clears the specified variables.

/L can be omitted.

Description

Two or more global variables can be specified at a time. The following is an example.

```
/GLOBAL SWITCH GLOB2.┘
```

Example

(1) When Displaying the Data in All the Global Variables and Initializing Some of Them

```

/GLOBAL.
STATION NO.=1 ← specification of a station number

SWITCH = 123  ↵
GLOB1  = 0  1.0.↵
GLOB2  = 0  100UA.↵
GLOB3  = 0  200UA.↵
GLOB4  GLOB variableGLOB4 = 0  10.↵
GLOB5  = 0  20.↵
GLOB6  = 0  ↵
GLOB7  = 0  ↵
GLOB8  = 0  ↵
GLOB9  = 0  ↵
GLOB10 = 0  ↵
:
:
:
GLOB47 = 0  ↵
GLOB48 = 0  ↵
GLOB49 = 0  ↵
GLOB50 = 0  ↵

DEVICE = 8256
LOT     = LT830108-12  ↵
DUTIF   = T3998  ↵
OPERATE= T.TAKEDA  ↵
COMMENT= FINAL TEST  ↵
WAFER   =  ↵
SAMPLE  = 1  2.↵

```

Where the names and data of the global variables are displayed. In response to the above display, specify new values. If nothing is changed, press only the Return key.

(2) When Displaying and Initializing the Data in GLOB1, 2, and 4

```

/GLOBAL/1 GLOB1 GLOB2 GLOB4.
STATION NO.=1

GLOB1  = 1.000  1.1.↵
GLOB2  = 100.0UA  200MA.↵
GLOB4  = 10  200.↵

```


(3) When Displaying Only the Data of All Global Variables:

```

/GLOBAL/P/1.↓
STATION NO.=1

SWITCH = 123
GLOB1  = 1.000      GLOB2  = 100.0UA      GLOB3  = 200.0UA
GLOB4  = 10         GLOB5  = 20         GLOB6  = 0
GLOB7  = 0          GLOB8  = 0
GLOB10 = 0          GLOB11 = 0          GLOB12 = 0          GLOB13 = 0
GLOB14 = 0          GLOB15 = 0          GLOB16 = 0          GLOB17 = 0
GLOB18 = 0          GLOB19 = 0          GLOB20 = 0          GLOB21 = 0
GLOB22 = 0          GLOB23 = 0          GLOB24 = 0          GLOB25 = 0
GLOB26 = 0          GLOB27 = 0          GLOB28 = 0          GLOB29 = 0
GLOB30 = 0          GLOB31 = 0          GLOB32 = 0          GLOB33 = 0
GLOB34 = 0          GLOB35 = 0          GLOB36 = 0          GLOB37 = 0
GLOB38 = 0          GLOB39 = 0          GLOB40 = 0          GLOB41 = 0
GLOB42 = 0          GLOB43 = 0          GLOB44 = 0          GLOB45 = 0
GLOB46 = 0          GLOB47 = 0          GLOB48 = 0          GLOB49 = 0
GLOB50 = 0

DEVICE = 8256              LOT      =LT830108-12
DUTIF  = T3998            OPERATOR =T. TAKEDA
COMMENT= FINAL TEST       WAFER    =

```

20. 2. 2 /SWITCH

Function: This utility is used to display and initialize the data in SWITCH.

Syntax

```
/SWITCH/n/P/L aaa↓
```

Argument: /n

Specifies station numbers.

Enter 1 or 2 for n.

Default

A prompt that asks the station number is displayed again. The following is an example.

```

/SWITCH/n/P/L.↓
STATION NO.=1

```

/P

Displays the data in HEADER.

/L can be omitted.

/L

Displays the data in HEADER on a line printer.

/L can be omitted.

aaa

Specifies the initial value. Specification of this argument is ignored when /P is specified.

/L can be omitted.

Example

(1) When Specifying New Data for Global Variable SWITCH

The following three examples are presented.

```
/SWITCH┐ ----- (a)
STATION NO.=1
PREVIOUS DATA=#00000011 #22┐
```

```
/SWITCH/1┐ ----- (b)
PREVIOUS DATA=#00000022 0┐
```

```
/SWITCH/1 #11┐ --- (c)
```

denotes a hexadecimal number.

Examples (a) and (b) show the specification of new values after the current data display.

If only the Return key is pressed, the data is not changed. In example (c), the specified value is directly set.

(2) When Displaying Only the Data of Global Variable SWITCH

The following two examples are presented.

```
/SWITCH/P.␣
STATION NO.=1
PREVIOUS DATA=#00000011
```

```
/SWITCH/P/1.␣
PREVIOUS DATA=123
```

20. 2. 3 /HEADER

Function: This utility is used to display and initialize the data in headers attached to each report. The header information referred to here mean the global variables of DEVICE, LOT, SUBLOT, DUTIF, OPERATOR, COMMENT, and WAFER, and the sample number.

Syntax

```
/HEADER/n/P/L.␣
```

Argument: /n

Specifies station numbers.

Enter 1 or 2 for n.

Default

A prompt that asks the station number is displayed again. The following is an example.

```
/HEADER.␣
HEADER INFORMATION SELECTION!
STATION NO.=1
```

/P

Displays the data in HEADER.

/L can be omitted.

/L

Displays the data in HEADER on a line printer.

/L can be omitted.

Example

(1) When Changing the Values in DEVICE and LOT (Including SUBLOT)

/HEADER.↓

HEADER INFORMATION SELECTION!

STATION No.=1 ← Enter a station number.

Currently specified state

↓

MODE 1 : DEVICE NAME =

MODE 2 : LOT NAME =

MODE 3 : DUT I/F =

MODE 4 : OPERATOR =

MODE 5 : WAFER NAME =

MODE 6 : COMMENT =

NO network operating systemMODE 7 : SAMPLE NO = 0

SPECIFY MODE NUMBER THAT YOU WISH TO CHANGE.

MODE= 1 ← Enter the mode number of the data to be changed.

DEVICE NAME = 8256.↓ ← Newly set value

↑

The current setting condition appears. At the initial condition, nothing appears.

Then, the newly specified state appears:

```
STATION NO.=1
```

```
MODE 1 : DEVICE NAME = 8256 ← This value is changed.
```

```
MODE 2 : LOT NAME =
```

```
MODE 3 : DUT I/F =
```

```
MODE 4 : OPERATOR =
```

```
MODE 5 : WAFER NAME =
```

```
MODE 6 : COMMENT =
```

```
NO network operating systemMODE 7 : SAMPLE NO = 0
```

```
SPECIFY MODE NUMBER THAT YOU WISH TO CHANGE.
```

```
MODE= 2 ← Enter the mode number of the data to be changed.
```

```
LOT NAME = LT830108-12. ← Newly set value
```

```
↑
```

The current setting condition appears. At the initial condition, nothing appears.

Then, the newly specified state appears:

```
STATION NO.=1
```

```
MODE 1 : DEVICE NAME = 8256
```

```
MODE 2 : LOT NAME = LT830108-12 ← This value is changed.
```

```
MODE 3 : DUT I/F =
```

```
MODE 4 : OPERATOR =
```

```
MODE 5 : WAFER NAME =
```

```
MODE 6 : COMMENT =
```

```
NO network operating systemMODE 7 : SAMPLE NO = 0
```

```
SPECIFY MODE NUMBER THAT YOU WISH TO CHANGE.
```

```
MODE= _ ← This command terminates when a value other than 1 to 7 is entered.
```

(2) When Changing the Sample Number:

```
/HEADER.↓
```

```
HEADER INFORMATION SELECTION!
```

```
STATION No. = 1 ← Enter a station number.
```

```
Currently specified state
```

```
↓
```

```
MODE 1 : DEVICE NAME = 8256
MODE 2 : LOT NAME    = LT830108-12
MODE 3 : DUT I/F     = T3998
MODE 4 : OPERATOR    = T.TAKEDA
MODE 5 : WAFER NAME   = WF8256
MODE 6 : COMMENT     = FINAL TEST
MODE 7 : SAMPLE NO   = 0
```

```
SPECIFY MODE NUMBER THAT YOU TO CHANGE.
```

```
MODE= 7 ← Enter the mode number of the data to be changed.
```

```
SAMPLE NO = 0 1.↓ ← Newly set value
```

```
Then, the newly specified state appears:
```

```
STATION NO.=1
```

```
MODE 1 : DEVICE NAME = 8256
MODE 2 : LOT NAME    = LT830108-12
MODE 3 : DUT I/F     = T3998
MODE 4 : OPERATOR    = T.TAKEDA
MODE 5 : WAFER NAME   = WF8256
MODE 6 : COMMENT     = FINAL TEST
MODE 7 : SAMPLE NO   = 1 ← Changed.
```

```
SPECIFY MODE NUMBER THAT YOU WISH TO CHANGE.
```

```
MODE= ↓ ← The command is terminated.
```

21. *Partial Test (/PART)*

This chapter describes the partial test (/PART) utility that is used to test particular test items.

21. 1 Partial Test Mode Overview

Usually, a test program consists of several test items and a device test is completed by testing all test items. Thus, the user must modify the test program spending a lot of effort when retesting a certain test item or when checking its data.

Even if debugger is used, the modifying operation still requires a great deal of labor. This test system is provided with a partial test mode to test a specified test item alone. This section describes the specification of the partial test mode and the function of this mode.

If partial test mode is specified to the test item, each statement in the test item can be executed. However, if partial test mode is not specified to the test item, condition setting statements, etc., are executed but test statements (MEAS/START/JUDGE statements) are not executed.

◆ When Test Items 1 and 3 are Specified in the Partial Test Mode

```

PRO 1

TEST 1
  Condition setting statement
  Test (MEAS/START/JUDGE)
  Test result processing statement

TEST 2
  Condition setting statement
  Test (MEAS/START/JUDGE)           ← Not executed.
  Test result processing statement ← (For more information, refer to the memo.)

TEST 3
  Condition setting statement
  Test (MEAS/START/JUDGE)
  Test result processing statement

TEST 4
  Condition setting statement
  Test (MEAS/START/JUDGE)           ← Not executed.
  Test result processing statement ← (For more information, refer to the memo.)

END

```

 Tip

In the test item which is not specified in the partial test mode, the object of the test result processing statement is the test result of the previous test item specified in the partial test mode.

For this reason, operation may be changed. The statement whose operation is changed depending on the test result is as follows.

- *IF PASS/FAIL/APASS/AFAIL statement*
- *READ FDUT/GFDUT statement*
- *READ RESULT statement*
- *READ FAIL statement*
- *READ DCT statement*
- *READ TPIN/FPIN statement*
- *READ FUNC/CUMFUNC statement*
- *READ DFM/AFM statement*

21. 2 Mode to Clear "REG MPAT"

Executing MEAS/START MPAT statement clears "REG MPAT" after its execution.

When MEAS/START MPAT statement is in the test item not specified in a partial test mode, "REG MPAT" in the test item is not cleared because MEAS/START MPAT statement is not executed.

It affects the next test item.

◆ When Test Item 2 is Specified:

```

PRO  1

TEST 1

    REG MPAT
    MEAS MPAT      ← REG MPAT is not cleared because it is not executed.

TEST 2
                ← REG MPAT remains in TEST 1
    REG MPAT      ← The settings with the same register name as TEST 1 are replaced
                    as the settings of TEST 2.
    MEAS MPAT

```

To avoid this effect, set a mode to clear REG MPAT.

➔ For more information, refer to [21. 3 "Syntax" on page 21-5](#).

Executing MEAS/START MPAT(3) statement does not clear "REG MPAT". When MEAS/START MPAT(3) is executed, "REG MPAT" is not cleared even if the mode to clear "REG MPAT" is specified.

- ◆ When the mode to Clear "REG MPAT" is Specified and the Test Item 3 is Specified in the Partial Test Mode.

```
PRO  1

TEST 1

  REG MPAT
  MEAS MPAT    ← REG MPAT is cleared

TEST 2

  REG MPAT    ← REG MPAT does not remain in TEST 1
  MEAS MPAT(3) ← REG MPAT is not cleared because (3) is specified.

TEST 3

  REG MPAT    ← REG MPAT remains in TEST 2
               ← The settings with the same register name as TEST 2 are replaced
               as the settings of TEST 3.
  MEAS MPAT
```

21. 3 Syntax

21. 3. 1 /PART

Function: This utility is used to enable you to select test items to be tested.

Syntax

```
/PART/n/P/L/R,␣
```

Argument: /n

Specifies station numbers.

Enter 1 or 2 for n.

Default

A prompt that asks the station number is displayed again. The following is an example.

```
/PART,␣
```

```
PART TEST MODE SELECTION !
STATION NO.=1
```

/P

Displays the data in HEADER.

/L can be omitted.

/L

Displays the data in HEADER on a line printer.

/L can be omitted.

/R

Sets the mode to clear "REG MPAT" in the test item which is not specified in the partial test mode.

Example

- (1) When Test Items 1to 6, 101 to 200, and 900 are to be Tested with Test Program 8256

```

/PART.␣

PART TEST MODE SELECTION !
STATION NO.=1      ← station number

PART TEST MODE      = DISABLE      ← current setting status
PRO NAME            =              ← current setting status
REG MPAT CLEAR MODE = DISABLE      ← current setting status
START-STOP TEST     =              ← current setting status

SPECIFY PART TEST MODE, ENABLE(E) OR DISABLE(D) ? E ← ENABLE
SPECIFY PRO NAME=   8256.␣
SPECIFY START TEST NO.& STOP TEST NO.
START-STOP TEST NO. = 1-6<LF>      ← Test items are specified
START-STOP TEST NO. = 101-200<LF> ← Test items are specified
START-STOP TEST NO. = 990.␣       ← A test item is specified

```

Where the part test mode is enabled if the specified program name matches that being executed. In specifying test items, when you press the <LF> (line feed) key, you can specify the next item. When you press the Return key, the command is terminated. Test items can be specified in up to 20 groups such as by single and continuous specification.

- (2) When Changing the Program Name

```

/PART/1.␣

PART TEST MODE SELECTION !
STATION NO.=1

PART TEST MODE      = ENABLE      ← current setting status
PRO NAME            = 8256        ← current setting status
REG MPAT CLEAR MODE = DISABLE      ← current setting status
START-STOP TEST     = 1-6         101-200 990 ← current setting status

SPECIFY PART TEST MODE, ENABLE(E) OR DISABLE(D) ? E ← You can press the Return key
instead.
SPECIFY PRO NAME =  SAMPLE.␣      ← new program name
SPECIFY START TEST NO.& STOP TEST NO.
START-STOP TEST NO. = ␣           ← When you press only the Return key, the test item
is not changed.

```

(3) When Changing Test Items

/PART/1.↓

PART TEST MODE SELECTION !
STATION NO.=1

PART TEST MODE	= ENABLE	← current setting status
PRO NAME	= SAMPLE	← current setting status
REG MPAT CLEAR MODE	= DISABLE	← current setting status
START-STOP TEST	= 1-6	101-200 990 ← current setting status

SPECIFY PART TEST MODE, ENABLE(E) OR DISABLE(D) ? ↓

SPECIFY PRO NAME = ↓

SPECIFY START TEST NO.& STOP TEST NO.

START-STOP TEST NO. = 1-5<LF>

START-STOP TEST NO. = 101.↓

↑

Specify the test items to be newly processed.

If no change is needed, press only the Return key.

(4) When the Mode to Clear "REG MPAT" is Set

/PART/1/R.↓ ← Specify /R.

PART TEST MODE SELECTION !
STATION NO.=1

PART TEST MODE	= DISABLE	← current setting status
PRO NAME	=	← current setting status
REG MPAT CLEAR MODE	= DISABLE	← current setting status
START-STOP TEST	=	← current setting status

SPECIFY PART TEST MODE, ENABLE(E) OR DISABLE(D) ? E

SPECIFY PRO NAME = SAMPLE.↓

SPECIFY REG MPAT CLEAR MODE, ENABLE(E) OR DISABLE(D) ? E ← (For more information, refer to the memo.)

SPECIFY START TEST NO.& STOP TEST NO.

START-STOP TEST NO. = 1-5.↓



Tip

When /R is specified, this message is displayed.

Enter E (ENABLE) or D (DISABLE).

(5) When Disabling the Partial Test mode

/PART/1.␣

PART TEST MODE SELECTION !
STATION NO.=1

PART TEST MODE	= ENABLE	← current setting status
PRO NAME	= SAMPLE	← current setting status
REG MPAT CLEAR MODE	= ENABLE	← current setting status
START-STOP TEST	= 1-5	← current setting status

SPECIFY PART TEST MODE, ENABLE (E) OR DISABLE (D) ? **D** ← DISABLE is specified.

In the disable mode, the system does not ask for a program name and test item.

22. Error Code Utility (/ERR)

This chapter describes the error code (/ERR) utility that is used to output the meaning of error codes to the VKT or a printer.

22. 1 Syntax

22. 1. 1 /ERR

Function: This utility is used to output the meaning of error codes to the VKT or a printer.

Syntax

```
/ERR/L error code␣
```

Argument: /L

Displays the data in HEADER on a line printer.

/L can be omitted.

error code

Specify an error code.

Default

A prompt that asks for an error code is displayed. Enter an error code.

Example

(1) When Entering Error Code 12035

```
/ERR 12035␣
```

```
.....ERR12.....
```

```
12035      In the SET BURST statement, the unit address set in the burst  
           block does not match that in out-buffer memory (OBM) in ALPG.
```

(2) When Entering Error Code 12035 in Response to the Prompt after Omitting Error Code Specification

```
/ERR.␣  
ERROR CODE ? 12035.␣  
  
.....ERR12.....  
  
12035      In the SET BURST statement, the unit address set in the burst  
           block does not match that in out-buffer memory (OBM) in ALPG.
```

(3) When Entering an Undefined Error Code

```
/ERR 357.␣  
  
Sorry no documentation on 357
```


23. Simplified Parallel Measurement Debugging Program (/MPDEBX)

This chapter describes the simplified parallel measurement debugging program (/MPDEBX) that is used to debug a test program.

23. 1 MPDEBX Overview

This utility program is used to debug test programs by simulating handler or prober operations during a device test.

The program's major functions include:

- Selection of the device to be connected.
- Setting of DUTs connected to the contactors (SDUT).
- Forcibly changing the test results.
- Inhibiting of test rejection.
- Inhibiting of exclusion.
- Prohibiting the counter from being updated.

Target counters:

Total/Pass/Fail counter

Category counter

Sort counter

Test Fail counter

- Setting of the prober state.

23. 2 Syntax

23. 2. 1 /MPDEBX

Function: This utility is used to simulate the operation of handlers and probers for debugging a test program.

Syntax

```
/MPDEBX.↓
```

23. 3 MPDEBX Operation

1) Start

Start this utility using the following procedure:

```
TESTtester simulation program## TEST␣

CONTROLLER READY

/MPDEBX␣

### MULTIPLE DUT MEASUREMENT DEBUG MODE ###

l s t   SPECIFY DEBUGGING MODE
m e n u  0: EXIT
s c r e e n 1: DEBUGGING MODE DISABLE
            2: HANDLER TESTING MODE      → Sets the mode for testing handlers.
            3: PROBER TESTING MODE       → Sets the mode for testing probers.
            4: MANUAL TESTING MODE       → Sets the mode for manual testing.
            MODE NO = 1
```

Enter the item number (1 to 4) to be set or confirmed after MODE NO =.

Entering 0 aborts the command.

Entering 1 disables the debug mode. (Specify 1 when debugging is completed.)

Details are explained in the following items.

When HANDLER TESTING MODE is selected

When HANDLER TESTING MODE is selected, the second menu screen shown below is displayed. Enter the desired item number for setting the item.

2 n d
menu
screen

```

SPECIFY DEBUGGING MODE FOR HANDLER TESTING
0: CLEAR
1: CHANGE THE SETTING DUT(ST1): #FFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
    FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
    CHANGE THE SETTING DUT(ST2): #FFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
    FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
2: CHANGE THE FAILURE DUT      : DISABLE
3: CHANGE NO-REJECTION MODE    : DISABLE
    NO-EXCLUSION MODE          : DISABLE
4: CHANGE NO-COUNT UP MODE     : DISABLE
MODE NO = 0      → The handler testing mode is cleared.

SPECIFY DEBUGGING MODE FOR HANDLER TESTING
0: CLEAR
1: CHANGE THE SETTING DUT(ST1): #FFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
    FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
    CHANGE THE SETTING DUT(ST2): #FFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
    FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
2: CHANGE THE FAILURE DUT      : DISABLE
3: CHANGE NO-REJECTION MODE    : DISABLE
    NO-EXCLUSION MODE          : DISABLE
4: CHANGE NO-COUNT UP MODE     : DISABLE
MODE NO = 1      → SET DUT setting is selected.

-----          FOR ST1          -----
SPECIFY THE SETTING DUT (DUT256-193) = #FFFFFFFFFFFFFFFF #0
SPECIFY THE SETTING DUT (DUT192-129) = #FFFFFFFFFFFFFFFF #0
SPECIFY THE SETTING DUT (DUT128- 65) = #FFFFFFFFFFFFFFFF #0
SPECIFY THE SETTING DUT (DUT 64-  1) = #FFFFFFFFFFFFFFFF #00000000FFFFFFFF
-----          FOR ST2          -----
SPECIFY THE SETTING DUT (DUT256-193) = #FFFFFFFFFFFFFFFF #0
SPECIFY THE SETTING DUT (DUT192-129) = #FFFFFFFFFFFFFFFF #0
SPECIFY THE SETTING DUT (DUT128- 65) = #FFFFFFFFFFFFFFFF #0
SPECIFY THE SETTING DUT (DUT 64-  1) = #FFFFFFFFFFFFFFFF #00000000FFFFFFFF
SDUT at station 1 is changed.
SDUT at station 2 is changed.

```

Current value

SPECIFY DEBUGGING MODE FOR HANDLER TESTING

0: CLEAR

1: CHANGE THE SETTING DUT(ST1): #00000000 00000000 00000000 00000000
00000000 00000000 00000000 FFFFFFFF

CHANGE THE SETTING DUT(ST2): #00000000 00000000 00000000 00000000
00000000 00000000 00000000 FFFFFFFF

2: CHANGE THE FAILURE DUT : DISABLE

3: CHANGE NO-REJECTION MODE : DISABLE

NO-EXCLUSION MODE : DISABLE

4: CHANGE NO-COUNT UP MODE : DISABLE

MODE NO = 2 → FAIL DUT setting is selected.

THE FAILURE DUT MODE !!

SPECIFY DISABLE(D) OR ENABLE(E) = DISABLE **E** → Changed to ENABLE.

----- FOR ST1 -----

SPECIFY THE FAILURE DUT (DUT256-193) = #0000000000000000 #0

SPECIFY THE FAILURE DUT (DUT192-129) = #0000000000000000 #0

SPECIFY THE FAILURE DUT (DUT128- 65) = #0000000000000000 #0

SPECIFY THE FAILURE DUT (DUT 64- 1) = #0000000000000000 #00000000FFFFFFFF

----- FOR ST2 -----

SPECIFY THE FAILURE DUT (DUT256-193) = #0000000000000000 #0

SPECIFY THE FAILURE DUT (DUT192-129) = #0000000000000000 #0

SPECIFY THE FAILURE DUT (DUT128- 65) = #0000000000000000 #0

SPECIFY THE FAILURE DUT (DUT 64- 1) = #0000000000000000 #00000000FFFFFFFF

↓

Current value

↓

New value

```

SPECIFY DEBUGGING MODE FOR HANDLER TESTING
0: CLEAR
1: CHANGE THE SETTING DUT(ST1): #00000000 00000000 00000000 00000000
                                00000000 00000000 00000000 FFFFFFFF
    CHANGE THE SETTING DUT(ST2): #00000000 00000000 00000000 00000000
                                00000000 00000000 00000000 FFFFFFFF
2: CHANGE THE FAILURE DUT(ST1): ENABLE
                                #00000000 00000000 00000000 00000000
                                00000000 00000000 00000000 FFFFFFFF
    CHANGE THE FAILURE DUT(ST2): ENABLE
                                #00000000 00000000 00000000 00000000
                                00000000 00000000 00000000 FFFFFFFF
3: CHANGE NO-REJECTION MODE    : DISABLE
    NO-EXCLUSION MODE          : DISABLE
4: CHANGE NO-COUNT UP MODE     : DISABLE
MODE NO = 3      → The NO-REJECTION MODE and NO-EXCLUSION MODE settings are selected.

SPECIFY NO-REJECTION MODE  DISABLE(D) OR ENABLE(E) = DISABLE  E → Changed to ENABLE.
SPECIFY NO-EXCLUSION MODE  DISABLE(D) OR ENABLE(E) = DISABLE  E → Changed to ENABLE.

SPECIFY DEBUGGING MODE FOR HANDLER TESTING
0: CLEAR
1: CHANGE THE SETTING DUT(ST1): #00000000 00000000 00000000 00000000
                                00000000 00000000 00000000 FFFFFFFF
    CHANGE THE SETTING DUT(ST2): #00000000 00000000 00000000 00000000
                                00000000 00000000 00000000 FFFFFFFF
2: CHANGE THE FAILURE DUT(ST1): ENABLE
                                #00000000 00000000 00000000 00000000
                                00000000 00000000 00000000 FFFFFFFF
    CHANGE THE FAILURE DUT(ST2): ENABLE
                                #00000000 00000000 00000000 00000000
                                00000000 00000000 00000000 FFFFFFFF
3: CHANGE NO-REJECTION MODE    : DISABLE
    NO-EXCLUSION MODE          : DISABLE
4: CHANGE NO-COUNT UP MODE     : DISABLE
MODE NO = 4      → The NO-COUNT UP MODE setting is selected.

SPECIFY DISABLE(D) OR ENABLE(E) = DISABLE  E      → Changed to ENABLE.

```

After this, a new 2nd screen is displayed for another DUT setting.

The FAILURE DUT setting in the item number 2 is used to forcibly set the test result. (For example, the test result is set to PASS or FAIL.)

When 1 is specified, DUT testing results in FAIL; when 0, DUT testing results in PASS.

The NO-REJECTION MODE setting in the item number 3 is used to enable or disable the SET REJECTION statement for the test plan program. When ENABLE is specified, the statement is made invalid. (Note that setting ENABLE means that no rejection is enabled and the SET REJECTION statement is made invalid.)

When DISABLE is specified, the SET REJECTION statement is valid.

As well as the NO-REJECTION MODE setting, the NO-EXCLUSION MODE setting is used to enable or disable the SET/RESET EXCLUSION statement (including SET/RESET EXCLUSION2) for the test plan program. When ENABLE is specified, the SET/RESET EXCLUSION statement is disabled.

When DISABLE is specified, the SET/RESET EXCLUSION statement is enabled.

When data is changed in item number 1 or 2, data at station 1 is changed for FOR ST1; data at station 2 for FOR ST2. Enter the data for the DUT numbers indicated by the SPECIFY THE SETTING DUT and SPECIFY THE FAILURE DUT messages. If no change is needed, press the Return key for each message.

CHANGE NO-COUNTUP MODE (item number 4) is used to set whether or not the counter is updated. The counter is not updated when ENABLE is set. The counter is updated when DISABLE is set.

When PROBER TESTING MODE is selected

When HANDLER TESTING MODE is selected, the second menu screen shown below is displayed. Enter the desired item number for setting the item.

- ➔ For the operation for item numbers 0, 1, 2 and 3, refer to ["When HANDLER TESTING MODE is selected" on page 23-4](#) because the operation is the same as for the HANDLER TESTING MODE.
- ➔ For the operation for item number 5, refer to item number 4 described in ["When HANDLER TESTING MODE is selected" on page 23-4](#) because the operation is the same as for item number 4 of the HANDLER TESTING MODE.

This section describes prober status setting.

2 n d
menu
screen

```

SPECIFY DEBUGGING MODE FOR PROBER TESTING
0: CLEAR
1: CHANGE THE SETTING DUT(ST1): #00000000 00000000 00000000 00000000
    00000000 00000000 00000000 00000001
    CHANGE THE SETTING DUT(ST2): #00000000 00000000 00000000 00000000
    00000000 00000000 00000000 00000001
2: CHANGE THE FAILURE DUT(ST1): ENABLE
    #00000000 00000000 00000000 00000000
    00000000 00000000 00000000 00000000
    CHANGE THE FAILURE DUT(ST2): ENABLE
    #00000000 00000000 00000000 00000000
    00000000 00000000 00000000 00000000
3: CHANGE NO-REJECTION MODE : ENABLE
    NO-EXCLUSION MODE : ENABLE
4: CHANGE PROBER STATUS : ST1 = NORMAL-START
    ST2 = NORMAL-START
5: CHANGE NO-COUNT UP MODE : DISABLE
MODE NO = 4 → The prober status setting is selected.

PROBER STATUS MODE !!
CURRENT STATUS ST1 = NORMAL-START → Current status
                ST2 = NORMAL-START → Current status

SPECIFY PROBER STATUS OF ST1 !!
0: NORMAL-START
1: WAFER-END
2: CARRIER-END
3: ALIGNMENT-ERROR
4: PROBER-DONE
                Status type to be set
STATUS NO = 1 → WAFER END is set (station 1).

SPECIFY PROBER STATUS OF ST2 !!
0: NORMAL-START
1: WAFER-END
2: CARRIER-END
3: ALIGNMENT-ERROR
4: PROBER-DONE
STATUS NO = 1 → WAFER END is set (station 2).
  
```



```

SPECIFY DEBUGGER MODE FOR PROBER TESTING
0: CLEAR
1: CHANGE THE SETTING DUT(ST1): #00000000 00000000 00000000 00000000
                                00000000 00000000 00000000 00000001
    CHANGE THE SETTING DUT(ST2): #00000000 00000000 00000000 00000000
                                00000000 00000000 00000000 00000001
2: CHANGE THE FAILURE DUT(ST1): ENABLE
                                #00000000 00000000 00000000 00000000
                                00000000 00000000 00000000 00000000
    CHANGE THE FAILURE DUT(ST2): ENABLE
                                #00000000 00000000 00000000 00000000
                                00000000 00000000 00000000 00000000
3: CHANGE NO-REJECTION MODE    : ENABLE
    NO-EXCLUSION MODE          : ENABLE
4: CHANGE PROBER STATUS        : ST1 = WAFER-END
                                ST2 = WAFER-END
5: CHANGE NO-COUNT UP MODE     : DISABLE
MODE NO =

```

When MANUAL TESTING MODE is selected

The second menu shown below is displayed. Enter the desired item number for setting the item.

To conduct manual testing, select this item from the first menu screen.

The items to be entered from the second screen are the same as those used in HANDLER TESTING MODE.

➔ For more information, refer to ["When HANDLER TESTING MODE is selected" on page 23-4.](#)

2 n d
menu
screen

SPECIFY DEBUGGING MODE FOR MANUAL TESTING

0: CLEAR

Current value

1: CHANGE THE SETTING DUT(ST1): #00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000

CHANGE THE SETTING DUT(ST2): #00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000

2: CHANGE THE FAILURE DUT : DISABLE

3: CHANGE NO-REJECTION MODE : DISABLE

NO-EXCLUSION MODE : DISABLE

4: CHANGE NO-COUNT UP MODE : DISABLE

MODE NO = 0

SPECIFY DEBUGGING MODE FOR MANUAL TESTING

0: CLEAR

1: CHANGE THE SETTING DUT(ST1): #FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF

CHANGE THE SETTING DUT(ST2): #FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF

2: CHANGE THE FAILURE DUT : DISABLE

3: CHANGE NO-REJECTION MODE : DISABLE

NO-EXCLUSION MODE : DISABLE

4: CHANGE NO-COUNT UP MODE : DISABLE

MODE NO = 1

```

----- FOR ST1 -----
SPECIFY THE SETTING DUT (DUT256-193) = #FFFFFFFFFFFFFFFF #0
SPECIFY THE SETTING DUT (DUT192-129) = #FFFFFFFFFFFFFFFF #FFFFFFFFFFFFFFFF
SPECIFY THE SETTING DUT (DUT128- 65) = #FFFFFFFFFFFFFFFF #FFFFFFFFFFFFFFFF
SPECIFY THE SETTING DUT (DUT 64-  1) = #FFFFFFFFFFFFFFFF #FFFFFFFFFFFFFFFF
----- FOR ST2 -----
SPECIFY THE SETTING DUT (DUT256-193) = #FFFFFFFFFFFFFFFF #0
SPECIFY THE SETTING DUT (DUT192-129) = #FFFFFFFFFFFFFFFF #FFFFFFFFFFFFFFFF
SPECIFY THE SETTING DUT (DUT128- 65) = #FFFFFFFFFFFFFFFF #FFFFFFFFFFFFFFFF
SPECIFY THE SETTING DUT (DUT 64-  1) = #FFFFFFFFFFFFFFFF #FFFFFFFFFFFFFFFF

```

```

SPECIFY DEBUGGING MODE FOR MANUAL TESTING
0: CLEAR
1: CHANGE THE SETTING DUT(ST1): #00000000 00000000 FFFFFFFF FFFFFFFF
    FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
    CHANGE THE SETTING DUT(ST2): #00000000 00000000 FFFFFFFF FFFFFFFF
    FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF

2: CHANGE THE FAILURE DUT      : DISABLE
3: CHANGE NO-REJECTION MODE    : DISABLE
    NO-EXCLUSION MODE          : DISABLE
4: CHANGE NO-COUNT UP MODE     : DISABLE
MODE NO = 2

THE FAILURE DUT MODE !!
SPECIFY DISABLE(D) OR ENABLE(E) = DISABLE  E

-----          FOR ST1          -----
SPECIFY THE FAILURE DUT (DUT256-193) = #0000000000000000 #0
SPECIFY THE FAILURE DUT (DUT192-129) = #0000000000000000 #FFFFFFFFFFFFFFFF
SPECIFY THE FAILURE DUT (DUT128- 65) = #0000000000000000 #FFFFFFFFFFFFFFFF
SPECIFY THE FAILURE DUT (DUT 64-  1) = #0000000000000000 #FFFFFFFFFFFFFFFF
-----          FOR ST2          -----
SPECIFY THE FAILURE DUT (DUT256-193) = #0000000000000000 #0
SPECIFY THE FAILURE DUT (DUT192-129) = #0000000000000000 #FFFFFFFFFFFFFFFF
SPECIFY THE FAILURE DUT (DUT128- 65) = #0000000000000000 #FFFFFFFFFFFFFFFF
SPECIFY THE FAILURE DUT (DUT 64-  1) = #0000000000000000 #FFFFFFFFFFFFFFFF
SPECIFY DEBUGGING MODE FOR MANUAL TESTING
0: CLEAR
1: CHANGE THE SETTING DUT(ST1): #00000000 00000000 FFFFFFFF FFFFFFFF
    FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
    CHANGE THE SETTING DUT(ST2): #00000000 00000000 FFFFFFFF FFFFFFFF
    FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF

2: CHANGE THE FAILURE DUT(ST1): ENABLE
    #00000000 00000000 FFFFFFFF FFFFFFFF
    FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
    CHANGE THE FAILURE DUT(ST2): ENABLE
    #00000000 00000000 FFFFFFFF FFFFFFFF
    FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF

3: CHANGE NO-REJECTION MODE    : DISABLE
    NO-EXCLUSION MODE          : DISABLE
4: CHANGE NO-COUNT UP MODE     : DISABLE
MODE NO =

```

24. Memory Redundancy Subroutine (FMSEARCH)

This chapter explains the memory redundancy subroutine (FMSEARCH) for quickly counting the failed bits in each ROW and COLUMN.

24. 1 FMSEARCH Overview

To enable recovery of failed bits with redundancy circuits used in large capacity memory, a function is required to quickly count the failed bits in each ROW, COLUMN.

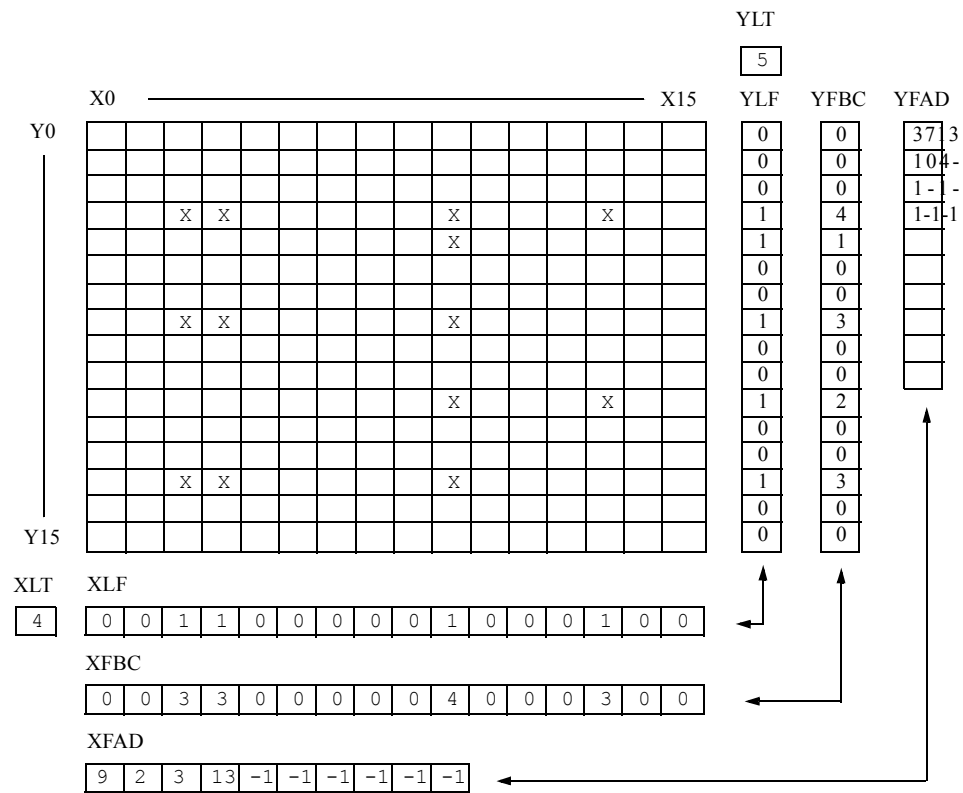
The SUB FMSEARCH utility program counts failed memory at high speed and then creates each counter containing failure presence or absence on X and Y address lines, the number of the failed bits, the number of the failed address lines and the sorted failed addresses. These fail data are shown in [Figure 24-1](#).

FMSEARCH is designed as a subroutine package to be included in a test plan program.

SUB FMSCAN, a utility program, is also available to check the failed address line transition, while recovering it using failure information created with FMSEARCH.

Limitation

This program cannot be executed on the T5377, T5336 and T5311.

Figure 24-1 Fail Data**XLF, YLF**

Data showing correspondence between failed bits on the X and Y address lines

XLT, YLT

Failure presence or absence on the X and Y address lines

XFBC, YFBC

Number of failed addresses on the X and Y lines

XFAD, YFAD

Failed addresses on the X and Y lines in descending order

24. 2 FMSEARCH Definitions

1) Variable definitions

Arguments to specify the operation mode and to store the processing results are defined with variables.

```

;                                < INPUT ARGUMENT >

      INTE [1] FM=1                ; FAI
      INTE [1] BIT=1               ; SEA
      INTE [1] AXIS(5)=1,0,511,0,511 ; SEA
      INTE [1] XLIN=5              ; XFA
      INTE [1] YLIN=7              ; YFA

;                                < OUTPUT ARGUMENT >

      INTE [1] XLT                  ; NUMBER
      INTE [1] YLT                  ; NUMBER
      LOGI [1] XLF(32)              ; FAIL AD
      LOGI [1] XLF(32)              ; FAIL AD
      INTE [1] XFBC(512)            ; FAIL BI
      INTE [1] XFBC(512)            ; FAIL BI
      INTE [1] XFAD(10)             ; FAIL AD
      INTE [1] XFAD(10)             ; FAIL AD
      INTE [1] IER                  ; ERROR

```

For the variable used in this program, specify 1 W type [1] as above examples.

However, specify 2 W type [2] in the following cases. A runtime error occurs when a 2 W type value is specified for variables other than those listed below.

- Setting of 32768 or more for search range (AXIS).
- The number of failed lines to be stored (XLT, YLT) is 32768 or more.
- The number of failed (XFBC, YFBC) is 32768 or more.
- Address stored to the failed address sort (XFAD, YFAD) is 32768 or more.

Arguments include:

FM

Specifies the mode of FM to be checked.(1 to 5)

FTFermmelde Technischen Zentralamt1 : FT

2 : FZ

3 : FO

4 : AF

5 : BF

BIT

Specifies the bit number.(1 to 144)

0 specification means an OR of each bit.

However, if one of the following conditions is met, BIT cannot be specified:

- On the T5593, T5592, T5591, T5586, T5585, T5581, T5581H, T5581P, and T5571P, n1 is 32 or larger in AFM BITSIZE = n1, n2.
- On the T5376, T5375, T5371, n2 is 32 or larger in "AFM BITSIZE = n1, n2".
- On the T5376, T5375, T5371, n1 is 32 or larger in "AFM BITSIZE = n1, n2" and SELECT COMPRESS CFM is specified.

The FMSEARCH can be executed for the OR data of arbitrary bit. In this case, define the variable "BIT" by one dimensional array variable (array element ≤ 72), and describe the bit number (1 to 144) to be ORed to each element.

AXIS(5)

Specifies the search range.

AXIS(1)

Switch to change the current search range.

0 : Does not change. (The range set in the tester is used.)

1 : Sets the new range.

AXIS(2)

Specifies the minimum value in the direction of the X address line when AXIS(1) is 1.

($0 \leq \text{AXIS}(2) \leq 262143$)

AXIS(3)

Specifies the maximum value in the direction of the X address line when AXIS(1) is 1.

($0 \leq \text{AXIS}(2) \leq \text{AXIS}(3) \leq 262143$)

AXIS(4)

Specifies the minimum value in the direction of the Y address line when AXIS(1) is 1.

($0 \leq \text{AXIS}(4) \leq 262143$)

AXIS(5)

Specifies the maximum value in the direction of the Y address line when AXIS(1) is 1.

($0 \leq \text{AXIS}(4) \leq \text{AXIS}(5) \leq 262143$)

 Tip

When $\text{AXIS}(1) = 0$, $\text{AXIS}(2)$ to $\text{AXIS}(5)$ are ignored. The program reads both the minimum and maximum values of the search range on the X and Y axes from the tester register, sets these values to $\text{AXIS}(2)$ to $\text{AXIS}(5)$, and executes FMSEARCH. If a read or write operation is performed for AFM before FMSEARCH is executed, the address range set when the read or write operation was performed is used.

XLIN

Specifies the number of failed address sorting lines on the X line.(0 to 10)

Tip

When 0 is specified, failed address sorting is not performed.

YLIN

Specifies the number of failed address sorting lines on the Y line.(0 to 10)

Tip

When 0 is specified, failed address sorting is not performed.

XLT

Specifies the variable to store the number of failed Y lines with X fixed.

YLT

Specifies the variable to store the number of failed X lines with Y fixed.

XLF

Specifies the variable to store the Y line failed bit data (array) with X fixed.

Indicates the position of Y line failed bit in relation with array element bit.

This variable must be defined as a logical type.

Specify a number of XLF elements that is equal to or greater than X0 to the maximum value on the X axis (number of XLF elements $\geq \text{AXIS}(3)/16 + 1$).

YLF

Specifies the variable to store the X line failed bit data (array) with Y fixed.

Indicates the position of X line failed bit in relation with array element bit.

This variable must be defined as a logical type.

Specify a number of YLF elements that is equal to or greater than Y0 to the maximum value on the Y axis (number of YLF elements $\geq \text{AXIS}(5)/16 + 1$).

XFBC

Specifies the number of Y line failed bits (array) with X fixed.

Stores the number of failed bits on the Y address line into the array elements.

Specify a number of XFBC elements that is equal to or greater than X0 to the maximum value on the X axis (number of XFBC elements $\geq \text{AXIS}(3) + 1$).

The relationships between addresses and array element numbers are as follows:

X address

0 → XFBC (1)
1 → XFBC (2)
2 → XFBC (3)

⋮ ⋮ ⋮
⋮ ⋮ ⋮

These relations are not affected by the setting arrangement of the search range (AXIS (2) to (5)).

- ◆ **When AXIS (2) = 2 (Minimum values of X) and AXIS (4) = 4 (Maximum value of X) are Provided, Each of the Following XFBC (n) is Respectively Stored in their Corresponding Addresses:**

X address

0	→	XFBC (1)	}	They are not to be stored.
1	→	XFBC (2)		
2	→	XFBC (3)	}	They are to be stored.
3	→	XFBC (4)		
4	→	XFBC (5)		
5	→	XFBC (6)	}	They are not to be stored.
⋮		⋮		
⋮		⋮		

Since a value equal to or greater than 65,536 cannot be specified for the number of array elements in the array declaration, define it with a two-dimensional array variable when the maximum value of the search range setting on the X axis is set to 65,535 or greater.

- ◆ **When INTE[2] XFBC(32768,2), the Array Element Numbers of XFBC for X Address are as Follows:**

X address

0	→	XFBC (1, 1)
1	→	XFBC (2, 1)
⋮		⋮
32767	→	XFBC (32768, 1)
32768	→	XFBC (1, 2)
32769	→	XFBC (2, 2)
⋮		⋮
65535	→	XFBC (32768, 2)

YFBC

Specifies the number of X line failed bits (array) with Y fixed.

Stores the number of failed bits on the X address line into the array elements.

Specify a number of YFBC elements that is equal to or greater than Y0 to the maximum value on the Y axis (number of YFBC elements $\geq$ AXIS(5) + 1).

The relationships between addresses and array element numbers are as follows:

Y address

0	→	YFBC (1)
1	→	YFBC (2)
2	→	YFBC (3)
⋮		⋮
⋮		⋮

These relations are not affected by the setting arrangement of the search range (AXIS (2) to (5)). Since a value equal to or greater than 65,536 cannot be specified for the number of array elements in the array declaration, define it with a two-dimensional array variable when the maximum value of the search range setting on the Y axis is set to 65,535 or greater.

XFAD

Specifies Y line failed address sorting (array) with X fixed.

This variable is loaded with the failed bit addresses sorted in descending order up to the number specified in XLIN. Specify 10 for the number of XFAD elements.

YFAD

Specifies the sorted X line failed addresses (array) with Y fixed.

This variable is loaded with the failed bit addresses sorted in descending order up to the number specified in YLIN. Specify 10 for the number of YFAD elements.

IER

Specifies the return code.

- When 0 is selected:

Normal return

- When 1 is selected:

Indicates that an invalid FM was specified.

- When 2 is selected:

The number of failed lines in the X or Y direction or the variable for the number of failures is specified as 1 W type and is 32768 or more.

The number of failures stored then turns into 0 or negative value.



Tip

If an error other than FM is detected, a run-time error will occur in execution.

➔ When a runtime error occurs, refer to [Error Code Reference Manual\(No. : 8141091\)](#).

24. 3 FMSEARCH Programming

24. 3. 1 FMSEARCH

Function: This subroutine is used to read the number of failed X or Y lines, the failed line bit data, the number of the failed bits on each address line and the sorted failed addresses.

Syntax

```
CALL FMSEARCH (FM, BIT, AXIS, XLIN, YLIN, XLT, YLT, XLF, YLF, XFBC, YFBC, XFAD, YFAD, IER)
```

Description

FMSCAN can be called in the same way as normal subroutine calling.

The failed line bit data is stored in array elements by designating the low address as MSB.

If no search range is specified in AXIS, or $AXIS(1) = 0$, the range set in the tester is used.

If a range is specified, or $AXIS(1) = 1$, it remains after executing the subroutine call. At AFM read/write, however, the range is changed to its address and the range must be reset.

In failed address sorting, the number of the lines to be sorted can be specified; that of the array elements must be set in 10.

If the number of the effective data is less than that of the lines to be sorted or the number of lines to be sorted is less than 10, -1 is set in ignored elements.

Limitation

- DUT numbers must be specified for parallel test. Specify the numbers with the AFM DUTSEL statement. This program handles DUTs specified with the AFM DUTSEL statement.
- When AFM PGSEL = MAIN, SUB is specified in AFM PGSEL statement, the failure memory on the MAIN or SUB side specified in AFM RWSEL statement is handled.
- When OR for more than one bit number is used to execute FMSEARCH, specify a one-dimensional array variable as BIT and assign bit numbers (1 to 144) to each array element.

However, there are the following restrictions.

On the T5593, T5592, T5591, T5586, T5585, T5581, T5581H, T5581P and T5571P

Specify the bit number within one I/O block. If bit numbers belonging to different blocks are specified, a run time error occurs.

➔ For more information on the I/O block, refer to the overview of AFM in [ATL-51 Test Plan Program Reference Manual\(No. : 8431498\)](#).

On the T5376, T5375 and T5371

When n2 or n1 is 32 or larger in AFM BITSIZE = n1, n2 and SELECT COMPRESS CFM is specified, specify bit numbers belonging to one block.

If bit numbers belonging to different blocks are specified, a run time error occurs.

➔ For more information on the I/O block, refer to the overview of AFM in [ATL-51 Test Plan Program Reference Manual\(No. : 8431498\)](#).

When zero is assigned in some array elements, zero is ignored. However, if zero is assigned to all array elements, OR is carried out for all bits.

The FMSEARCH recognizes up to 72 array elements. More than 72 array elements are ignored.

The example of executing the FMSEARCH for OR data of arbitrary bit is shown below.

- 1) Execute the FMSEARCH for the OR data of 1st, 3rd, 5th and 7th bits.
- 2) Execute the FMSEARCH for the OR data of 2nd, 4th, 6th and 8th bits.

PRO EXAMPLE1

```
;      <INPUT ARGUMENT>

INTE[1] FM=1
INTE[1] BIT (4)=1, 3, 5, 7
INTE[1] AXIS(5)=1, 0, 511, 0, 511
INTE[1] XLIN=5
INTE[1] YLIN=7

;      <OUTPUT ARGUMENT>

INTE[1] XLT
INTE[1] YLT
LOGI[1] XLF(32)
LOGI[1] YLF(32)
INTE[1] XFBC(512)
INTE[1] YFBC(512)
INTE[1] XFAD(10)
INTE[1] YFAD(10)
INTE[1] IER

CALL FMSEARCH(FM,BIT,XLIN,YLIN,XLT,YLT,XLF,YLF,@
              XFBC,YFBC,XFAD,YFAD,IER)

DIM BIT=2, 4, 6, 8

CALL FMSEARCH(FM,BIT,XLIN,YLIN,XLT,YLT,XLF,YLF,@
              XFBC,YFBC,XFAD,YFAD,IER)
```

25. *Memory Redundancy Subroutine (FMSCAN)*

This chapter explains memory redundancy subroutine (FMSCAN) for checking failed memory for its transition when a failed address line is recovered using data created by SUB FMSEARCH.

25. 1 FMSCAN Overview

The SUB FMSCAN utility program checks failed memory for its transition when a failed address line is recovered, using data which is created with SUB FMSEARCH, that is, failure presence or absence on the X and Y address lines, the number of the failed bits, the number of the failed address lines and the sorted failed addresses.

 Limitation

This program cannot be executed on the T5377, T5336 and T5311.

25. 2 FMSCAN Definitions

1) Variable definitions

Arguments to specify the operation mode and to store the processing results are defined with variables.

```

;                                < INPUT ARGUMENT >

INTE [1] FM=1                    ; FAIL
INTE [1] BIT=1                    ; SEARC
INTE [1] AXIS (5)=1,0,15,0,7    ; SEARC
INTE [1] XLIN=5                  ; XFAD,
INTE [1] YLIN=7                  ; YFAD,

INTE [1] SEL=0                    ; X OR Y
INTE [1] ADR=3                    ; RERAI

;                                < OUTPUT ARGUMENT >

INTE [1] XLT                      ; NUMBER
INTE [1] YLT                      ; NUMBER
LOGI [1] XLF (64)                 ; FAIL ADDR
LOGI [1] YLF (64)                 ; FAIL ADDR
INTE [1] XFBC (512)               ; FAIL BIT
INTE [1] YFBC (512)               ; FAIL BIT
INTE [1] XFAD (10)                ; FAIL ADD
INTE [1] YFAD (10)                ; FAIL ADD
INTE [1] IER                      ; ERROR

```

In the example above, SEL and ADR variables are not used in SUB FMSEARCH. Other variables are same as those used in FMSEARCH. Variables that are used only in this program are explained here.

SEL

Specifies the X and Y address lines to be checked.

Specifies which of X and Y addresses is to be recovered.

When specifying 0 (X address).

When specifying 1 (Y address).

ADR

Specifies the address to be recovered.

When any of 32768 to 262143 is specified in retrieval address, 2 W type specification [2] is required.

25. 3 FMSCAN Programming

25. 3. 1 FMSCAN

Function: This subroutine is used, assuming that a failed address line is recovered, to change each counter containing the number of the failed X or Y line, the failed line bit data, the number of the failed bits on each address line and the sorted failed addresses.

Syntax

```
CALL FMSCAN (FM,BIT,SEL,ADR,XLIN,YLIN,XLT,YLT,XLF,YLF,XFBC,YFBC,XFAD,YFAD,IER)
```

Description

FMSCAN can be called in the same way as normal subroutine calling.

Specify the address to be recovered in the input arguments, SEL and ADR, and set up each counter, then execute the program. Each counter is changed to the same value as when the memory address is PASS.

The search is handled in the range set by AXIS.

AXIS is variable which is used in the utility program FMSEARCH and specifies search range.

The search range is set to tester by calling FMSEACH. (only at AXIS (1) = 1)

The number of XFAD or YFAD lines can be specified in XLIN and YLIN.

Tip

- DUT numbers must be specified for parallel test. Specify the numbers with the AFM DUTSEL statement.
This program handles DUTs specified with the statement.
- When AFM PGSEL = MAIN, SUB is specified in AFM PGSEL statement, the failure memory on the MAIN or SUB side specified in AFM RWSEL statement is handled.
- When the retrieval address is other than the search range, a run time error is occurred.
- When the search range in FMSEARCH is divided, FMSCAN is executed if necessary.

26. CTRL/C Disable Setting Program (/CKBRCNT)

This chapter explains the CTRL/C disable setting program (/CKBRCNT) that disables the forced termination.

26. 1 Overview

This program is used to disable forced termination of the program caused by misoperation, etc. The CTRL/C key and the reset button on the control box are used to force the program in operation to terminate; this program is used to disable such termination.

26. 2 Syntax

26. 2. 1 /CKBRCNT

Function: This command is used to disable the forced termination with the CTRL/C key or the reset button on the control box.

Syntax

```
/CKBRCNT↵
```

26. 3 Method of Operation

From the controller READY state, set as follows:

```
/CKBRCNT.┘  
MODE SELECTION!  
ENABLE(0) OR DISABLE(1)=0 1.┘ --- (A)  
      ↓  
      Current mode  
  
CTRL/C & RESET BUTTON ARE DISABLED  
      ↓  
Program start
```

When DISABLE is input in the message (A), RESET buttons on both control boxes 1 and 2 are disabled.

If no setting is performed, they are set to ENABLE.

27. CLRCAL

This chapter explains CLRCAL that increments the system history number by one.

27. 1 Syntax

27. 1. 1 /CLRCAL

Function: By this command, all the existing CAL files will be invalidated. Thus, at the first program execution after this command executed, calibration is to be performed forcibly.

Syntax

```
/CLRCAL↵
```

27. 2 Method of Operation

From the controller READY state, set as follows:

```
/CLRCAL.␣  
CAL FILE IN CURRENT DIRECTORY WILL BE UPDATED!!
```

Use to nullify the CAL file in the directory.

```
/CLRCAL/A.␣  
CAL FILE IN EVERY DIRECTORY WILL BE UPDATED!!
```

Use to nullify all the CAL files in the directories, for example, when the board is replaced.

28. CTLCTEST Effective Set Program (/CSUBMODE)

This chapter explains the CTLCTEST effective set program for specifying whether or not to activate "SUB CTLCTEST."

28. 1 Overview

Specifies whether or not "SUB CTLCTEST" is activated using the CTRL/C key or the reset button on the control box during the tester controller operation.

As for the units inside the tester, reference-off is performed when the CTRL/C or the reset button is pressed during test execution. However, if the tester is connected to the external measurement devices by GPIB, etc., sometimes those external devices also require reset processing. Such a case is just for the program use.

28. 2 Syntax

28. 2. 1 /CSUBMODE

Function: This command is used to specify whether or not "SUB CTLCTEST" is activated using the CTRL/C key or the reset button on the control box during the tester controller operation.

Syntax

`/CSUBMODE↵`

28. 3 Method of Operation

From the controller READY state, set as follows:

```

/CSUBMODE.
MODE SELECTION!
ENABLE(1) OR DISABLE(0) = 0 1. --- (A)
      ↓
      Current mode

CTLCTEST OR CTLCRUN IS ENABLED
      ↓
Program start
  
```

SUB CTLCTEST starts by using the reset button on the control box or the CTRL/C key when ENABLE is input in the above message (A).

Limitation

Precautions on creating SUB CTLCTEST

SUB CTLCTEST is called by the tester controller in user subroutine call format. Those programs must be terminated by RETURN. Note that device test cannot be executed in the SUB CTLC TEST.

Example

```

SUBR CTLCTEST
:
RETURN
END
  
```

Store the SUB CTLCTEST file in the customer DP0 or the current customer.

29. Clock Setting Program (/TGSET)

This chapter explains the clock setting program (/TGSET) for checking for DUT PASS/FAIL transition during debugging a device measurement program.

29. 1 TGSET Overview

This program displays and changes clock timing values on the VKT screen during debugging a device measurement program, then starts the pattern program to check for DUT PASS/FAIL transition.

This program is a combination of the conventional pin monitor and data set clock section.

29. 2 Syntax

29. 2. 1 /TGSET

Function: The TGSET utility program consists of two major functions.

1. Change to display TG setting values and execute the pattern program to check for PASS/FAIL transition.
2. Display only contents of TG and ALPG.

Syntax

```
/TGSET,↓
```

Description

The table below summarizes program functions.

	Description of functions	Remarks
1.	Displays clock timing values currently set in the TG. Changes clock timing values currently set in the TG. Then starts the pattern program to check the timing value for each clock currently set on TG for PASS/FAIL transition. (This function is used for functional tests.)	Main function. Execute this function on the first screen.
2.	Displays the name of the clock for which the value specified with the SET OFFSET2 statement is valid. (The clock names specified with the SELECT OFFSET statement.)	Subfunctions. Execute these functions on the second screen.
	Displays OFFSET (1) and OFFSET2 values being currently set.	
	Displays the name of the pattern file being transferred to the ALPG.	
	Displays the file name of the scramble program being transferred to the address descrambler or prescrambler.	

29. 3 TGSET Function Operation

This section describes the main function operation. Further, its flowchart is shown in the next page.

1) ##Execute TEST(return key).

Select the test mode (MANUAL/RECYCLE/HOLD) from the CONTROLLER READY state and start the test plan program.

2) Set the controller to the HOLD state and input the message as follows:

`/TGSET↵`

The clock timing values currently set at the VKT in the format to be described later (1st screen) are displayed.

PASS/FAIL information displayed in the lower column shows the test result executed just before the controller is set to the hold state. DUTs not to be tested are displayed as "-."

Change the clock timing values and execute START MPAT. The PASS/FAIL information is updated. To execute START MPAT, press the [PE2] key.

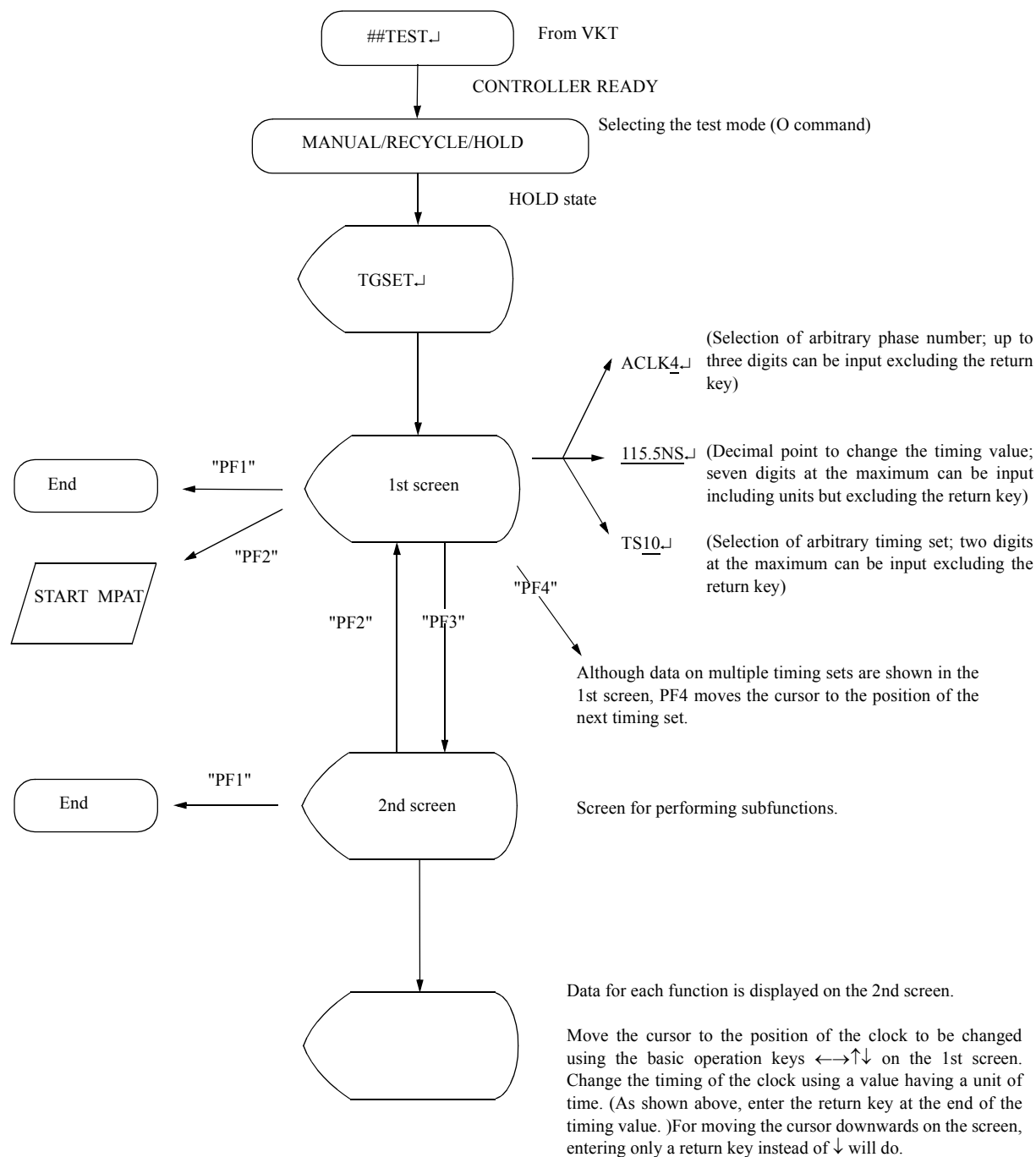
If TGSET is actuated after executing MEAS MPAT(4), a START MPAT(4,V) statement, for MEAS MPAT(5), a START MPAT(5) statement is executed, and otherwise, a START MPAT(3) statement is executed.

➔ For more information on underlined statements, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498\)](#).

Limitation

Before executing START MPAT, the pattern program must be transferred to the ALPG. If only timing values are changed, the pattern program does not need to be transferred.

29. 3 TGSET Function Operation



29. 4 Notes for TGSET Usage

- Before executing START MPAT, manual/RECYCLE/HOLD must be set to HOLD. Otherwise, this statement cannot be executed.
- If a SET OFFSETn statement is described in the test plan program, the offset values are displayed as they are actually set on the TG unit. (These values are displayed if any offset timing value is changed.)
- If there is any input keying mistakes made while a timing set or a clock phase number is selected, or when a timing value is changed on the first screen, the ! sign is highlighted and the VKT buzzer is sounded. Then the original value is displayed.
The restrictions on key input on the 1st screen are shown below.
- If the number of parallel tests exceeds 128, the number of rows displaying PASS/FAIL increases. Expand the window size to 80 columns × 36 rows before running this program in order to display the rows used by PASS/FAIL.

Table 29-1 Key Input Restriction List

Key operation	MODE	Timing data range	Remarks
Selection of timing set required	NORMAL	Timing set 1 to 16; any other set causes an an error	The value other than 1 of SET TGTS causes an error.

Key operation	MODE	Timing data range	Remarks
Selection of a phase number for the clock required	NORMAL	On the T5593 BCLK1 to 128, CCLK1 to 128, STRB1 to 256 DREL1 to 256, DRET1 to 256 Any other sets cause an error.	The number of phases of the clock is from the minimum to the maximum. If a socket program is used, the maximum of the phase number is the maximum DUT pin number written in the socket program.
		On the T5592 Odd-numbered phase of BCLK1 to 128, CCLK1 to 128, STRB1 to 255 DREL1 to 256, DRET1 to 256 Any other sets cause an error.	
		On T5591, T5586, T5585, T5581, T5581H, T5581P and T5571P ACLK1 to 128, BCLK1 to 128, CCLK1 to 128, STRB1 to 256, DREL1 to 128, DRET1 to 128, WSTRBL1 to 128, WSTRBT1 to 128 Any other sets cause an error.	
		On the T5336 and T5311 ACLK1 to 4, BCLK1 to 8, CCLK1 to 8, STRB1 to 2, DREL1, DRET1, WSTRBL1 to 4, WSTRBT1 to 4 Any other sets cause an error.	--
		On the T5371 ACLK1 to 128, BCLK1 to 128, CCLK1 to 128, DREL1 to 128, DRET1 to 128, STRB1 to 4, WSTRBL1 to 4, WSTRBT1 to 4 Any other sets cause an error.	If a socket program is used, ACLK, BCLK, CCLK, DRE and DRET numbers can be specified up to the maximum DUT pin number written in the socket program.
		On the T5377, T5376 and T5375 ACLK1 to 128, BCLK1 to 128, CCLK1 to 128, DREL1 to 128, DRET1 to 128, STRB1 to 4, WSTRBL1 to 3, WSTRBT1 to 3 Any other sets cause an error.	

29. 5 Example of TGSET Display

In the first screen, NORMAL MODE is displayed.

On the T5592

<<<<<<<< CONDITION OF CLOCKS >>>>>>>>									
					*2 SET TGTS = 1		*1 MODE = NORMAL		
	TS01 *3		TS02		TS03		TS04		
RATE	7.679US		7.679US *4		7.500US		7.000US		
B,CCLK1 *5	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK2	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK3	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK4	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK5	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK6	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK7	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK8	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK9	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK10	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
DRE1	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
DRE2	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
STRB1	0.000S		0.000S		0.000S		0.000S		
STRB3	0.000S		0.000S		0.000S		0.000S		
STRB5	0.000S		0.000S		0.000S		0.000S		
STRB7	0.000S		0.000S		0.000S		0.000S		
P:PASS *: FAIL -:NOT TESTING									
PF1:EXIT	ST1 #01 02 03 04 05 06 07 08								
PF2:START MPAT	*6 P P P - - - - -								
PF3:OTHERS FUNC	ST2 #01 02 03 04 05 06 07 08								
PF4:NEXT TS	* * * P P P P -								

*1 Displays TG operation modes, NORMAL.

*2 Displays SET TGTS data.

*3 Inputs a timing set.
1 to 16 can be input.

*4 The timing value of each clock and RATE. Up to 7 digits (including decimal point and unit) can be input in common.
When an offset value is set, the input data and the echoed back data are different.
Taking into account of an offset value, the actual value set on the TG units is echoed back.

*5 Inputs the clock phase number. Up to 3 digits can be input.

*6 When [PF2] is pressed (START MPAT), if an error has occurred, the VKT buzzer rings. A typical reason for an error is that there are no patterns may in the ALPG.
When the pattern program is completed normally, the PASS/FAIL information is updated. However, if P→* or *→P are shown, the characters "P, *" flash, then the last character of P, * is displayed.



• If OPEN is specified, "-----" is displayed.

29. 5 Example of TGSET Display

- On the DRE display, the left side of the ITS is a leading edge timing and the right side is a trailing edge timing.

On the T5591, T5591R, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5377, T5376, T5375, T5371, T5336 and T5331

<<<<<<<< CONDITION OF CLOCKS >>>>>>>>									
TS5↓ ^{*3}				TS02		TS03		TS04	
RATE	1.000MS			1.000MS↓ ^{*4}		900.0US		800.0US	
ACLK1↓ ^{*5}	0.000S			1.550US↓ ^{*4}		0.000S		0.000S	
ACLK2	0.000S			0.000S		0.000S		0.000S	
ACLK3	0.000S			0.000S		0.000S		0.000S	
ACLK4	0.000S			0.000S		0.000S		0.000S	
B,CCLK1	0.000S	0.000S		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK2	0.000S	0.000S		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK3	0.000S	0.000S		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK4	0.000S	0.000S		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK5	0.000S	0.000S		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK6	0.000S	0.000S		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
B,CCLK7	0.000S	0.000S		0.000S	0.000S	0.000S	0.000S		
B,CCLK8	0.000S	0.000S		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
DRE1	0.000S	0.000S		0.000S	0.000S	0.000S	0.000S	0.000S	0.000S
STRB1	0.000S			0.000S		0.000S		0.000S	
STRB2	0.000S			0.000S		0.000S		0.000S	
P:PASS *: FAIL -:NOT TESTING									
PF1:EXIT	ST1 #01 02 03 04 05 06 07 08								
PF2:START MPAT	^{*6} P P P - - - - -								
PF3:OTHERS FUNC	ST2 #01 02 03 04 05 06 07 08								
PF4:NEXT TS	* * * P P P P -								

^{*1} Displays TG operation modes, NORMAL.

^{*2} Displays SET TGTS data.

^{*3} Inputs a timing set.
1 to 16 can be input.

^{*4} The timing value of each clock and RATE. Up to 7 digits (including decimal point and unit) can be input in common.

When an offset value is set, the input data and the echoed back data are different.

Taking into account of an offset value, the actual value set on the TG units is echoed back.

^{*5} Inputs the clock phase number. Up to 3 digits can be input.

^{*6} When [PF2] is pressed (START MPAT), if an error has occurred, the VKT buzzer rings. A typical reason for an error is that there are no patterns may in the ALPG.

When the pattern program is completed normally, the PASS/FAIL information is updated. However, if P→* or *→P are shown, the characters "P, *" flash, then the last character of P, * is displayed.

(from the left side, it means bit 1 and clock 1)

On the T5586, T5585, T5581, T5581H, T5581P and T5571P

<<<<<<<< OTHERS FUNCTION >>>>>>>>	
1 : DISPLAY OFFSET & MPAT	SELECT NO. : [1] (For more information, refer to the memo)
**SELECT OFFSET ** (For more information, refer to the memo) ACLK=1-5,8,10 BCLK=1,2,5,12,50,51,120,121 CCLK=12 DREL=1,3 DRET=1 STRB=1, 2, 4 (For more information, refer to the memo)	
** OFFSET(1) & OFFSET2 ** OFFSET(1) = 10NS OFFSET2 = 5NS	
** MPAT,SCR FILE NAME ** MPAT = File name or NONE SCR (MAIN) = File name or NONE SCR (SUB) = File name or NONE PRESCR(MAIN) = File name or NONE PRESCR(SUB) = File name or NONE	
PF1:EXIT PF2:POP UP MENU	

The values and file names displayed in the second screen cannot be changed.

Tip

- *SELECT NO. is fixed to 1.*
- *If SELECT OFFSET cannot be displayed within the area, data is displayed as hexadecimal notation as follows.*
(Example) #AAAA AAAA AAAA AAAA AAAA AAAA AAAA AAAA
In the above example, every other number up to 127 is specified as 1, 3, 5, 7, 9 ··· 125, 127.
(from the left side, it means bit 1 and clock 1)
- *On the T5586 and T5585, STRB1-128 and STRB129-255 are displayed.*

On the T5377, T5376, T5375, T5371, T5336 and T5311

<<<<<<<< OTHERS FUNCTION >>>>>>>>	
1 : DISPLAY OFFSET & MPAT	SELECT NO. : [1] (For more information, refer to the memo)
**SELECT OFFSET ** ACLK=1 BCLK=1, 2, 5 CCLK=8 DREL=1 DRET=1 STRB=1, 2	
** OFFSET(1) & OFFSET2 ** OFFSET(1) = 10NS OFFSET2 = 5NS	
** MPAT,SCR FILE NAME ** MPAT = File name or NONE SCR = File name or NONE PRESCR = File name or NONE (For more information, refer to the memo)	
PF1:EXIT PF2:POP UP MENU	

The values and file names displayed in the second screen cannot be changed.

Tip

- *SELECT NO. is fixed to 1.*
- *This information is not displayed on the T5336 and T5311.*

30. *Time Measurement Program (TMEAS)*

This chapter explains the time measurement program (TMEAS) that makes the time measurement at the high speeds possible.

30. 1 TMEAS Overview

This utility program is used to measure times at high speeds using the binary search. Resolution can be specified and the available values are shown below.

Target system	Resolution
T5593	7.32421875 ps
T5592	14.6484375 ps
T5591, T5591R, T5586, T5585, T5581, T5581H, T5581P and T5571P	15.625 ps
T5377, T5376 and T5375	27.34375 ps
T5371, T5336 and T5311	62.5 ps

The resolution can be specified, too. In addition, times for all the specified DUTs can be tested in parallel in a parallel test.

30. 2 TMEAS Syntax

Two statements are available for TMEAS program execution and must be used as shown below.

30. 2. 1 TMEAS

Function: This syntax is used to specify time measurement.

Syntax

(1) To Insert into the Object File of the Test Plan Program

```
ASM TMEAS (ARG)
ASM TMEAS (ARG, ARG2)
ASM TMEAS (ARG, ARG2, ARG3)
ASM TMEAS (ARG, ARG2, ARG3, VAR, ERR) (specified in parallel test)
```

(2) To Call at Execution of the Test Program

```
EXEC TMEAS (ARG)
EXEC TMEAS (ARG, ARG2)
EXEC TMEAS (ARG, ARG2, ARG3)
EXEC TMEAS (ARG, ARG2, ARG3, VAR, ERR) (specified in parallel test)
```

Description

To insert into the object file of the test plan program

Since the TMEAS program uses the ASM statement, it is built into the object file of a test program at compilation. (To reduce overhead time and speed up calling the TMEAS program)

Therefore, the test program must be recompiled to replace the old TMEAS program in the object file of the test program with a new TMEAS program. When the TMEAS program is modified, the test program which uses the TMEAS program must be recompiled.

To call at execution of the test program

The EXEC statement has the same functions as the CALL statement. The test program need not be recompiled when the TMEAS program is modified, and is called by the EXEC statement.

30. 3 TMEAS Argument

For the TMEAS argument, 11 three-word array variables must be specified. Specify the variables as follows:

```
DIM ARG (11) = 0 (or DIM ARG (56) = 0)
DIM ARG2 (10) = 0
DIM ARG3 (1) = 0 (1 = 1 to 9)
DIM VAR (m) = 0 (m = 1 to 512)
DIM ERR (m) = 0
```

- ARG(11) or ARG(56)

The contents of the 11 or 56 array elements are as follows:

ARG (1)

Measurement start value (Unit of value is time)

ARG (2)

Measurement stop value (Unit of value is time)

ARG (3)

Specifies timing pulse edge for time measurement.

Specify the UNIT or TGCLK number.

➔ See [Table 30-1](#) through [Table 30-5](#).

ARG (4)

Stores time measurement results.

ARG (5)

Stores error codes. In parallel test, stores error code indicating the total result.

➔ See [Table 30-6](#).

ARG (6)

Specifies timing set of timing pulse edge for time measurement.(1 to 16)

ARG (7)

ALPG PC start address

ARG(8) --- TRACKING #1

Specifies timing pulse edge to be traced.

Specify the UNIT or TGCLK number.

➔ See [Table 30-1](#) through [Table 30-5](#).

ARG(9) --- TRACKING #1

Specifies the tracing value. (Unit: time) Specifies the increment for tracing.

(Tracing unit) value = (Measurement unit) value + ARG (9)

ARG(10) --- TRACKING #1

Specifies timing set of timing pulse edge to be traced.(1 to 16)

ARG (11)

DUT No. (1 to 512) Specifies DUT number of the device to be measured.

When 0 is specified, 1 is used.

Specify DUT numbers in accordance with the socket program.

When ARG (56) is specified, the tracing timing pulse edges can be specified by up to 16 digits.

ARG(12) --- TRACKING #2

Specifies the second tracing timing pulse edge.

ARG(13) --- TRACKING #2

Specifies the second tracing value.

ARG(14) --- TRACKING #2

Specifies the timing set of the second tracing timing pulse edge.

:

ARG(54) --- TRACKING #16

Specifies the 16th tracing timing pulse edge.

ARG(55) --- TRACKING #16

Specifies the 16th tracing value.

ARG(56) --- TRACKING #16

Specifies the timing set of the 16th tracing timing pulse edge.

- ARG2 (10)

ARG2 (1)

Specifies a resolution.

	Resolution range	Notes
T5593	7.32421875 ps ≤ resolution < 960 ns (If ARG2 is not specified, 7.32421875 ps is used.)	If a resolution not exceeding 7.32421875 ps is specified, times are always measured with a resolution of 7.32421875 ps. When resolution ≥ 960 ns or resolution < 0 s, an error occurs.
T5592	14.6484375 ps ≤ resolution < 960 ns (When ARG2 is not specified, measures with 14.6484375 ps.)	If a resolution not exceeding 14.6484375 ps is specified, times are always measured with a resolution of 14.6484375 ps. When resolution ≥ 960 ns or resolution < 0 s, an error occurs.
T5591 T5586 T5585 T5581 T5581H T5581P T5571P	15.625 ps ≤ resolution ≤ 1 ms (When ARG2 is not specified, measures with 15.625 ps.)	If the start value ARG(1) or the stop value ARG(2) is equal to or larger than 1.024 μs, a resolution of 2 ns is used. If a resolution equal to or smaller than 15.625 ps is specified, 15.625 ps is used. When resolution > 1 ms or resolution < 0 s, an error occurs.
T5377 T5376 T5375	27.34375 ps ≤ resolution ≤ 14 ms (When ARG2 is not specified, measures with 27.34375 ps.)	If a resolution not exceeding 27.34375 ps is specified, times are always measured with a resolution of 27.34375 ps. When resolution > 14 ms or resolution < 0 s, an error occurs.
T5371 T5336 T5311	62.5 ps ≤ resolution ≤ 16 ms (When ARG2 is not specified, measures with 125 ps.)	If a resolution not exceeding 62.5 ps is specified, times are always measured with a resolution of 62.5 ps. When resolution > 16 ms or resolution < 0 s, an error occurs.

ARG2(2) to ARG2(10)

Spare

The ARG2 can be omitted.

- ARG3(1)**

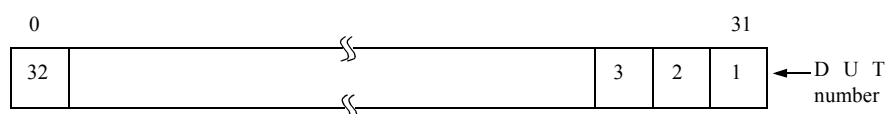
Specifies the required digit for the number of an array element ().

ARG3(1)

OFFSET data available flag (=0 : Not Available, =1 : Available)

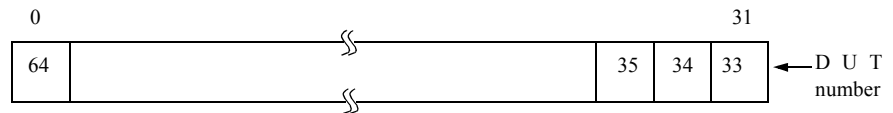
ARG3(2)

Specifies the DUT number (1 to 32) in parallel test.

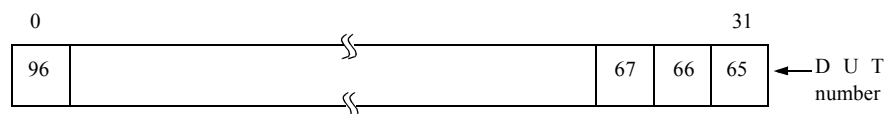


ARG3(3)

Specifies the DUT number (33 to 64) in parallel test.

**ARG3 (4)**

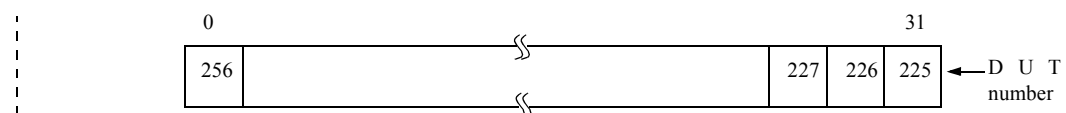
Specifies the DUT number (65 to 96) in parallel test.

**ARG3 (5)**

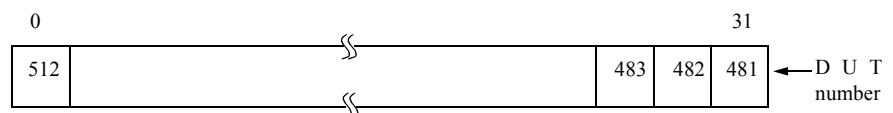
Specifies the DUT number (97 to 128) in parallel test.

**ARG3 (9)**

Specifies the DUT number (225 to 256) in parallel test.

**ARG3 (17)**

Specifies the DUT number (481 to 512) in parallel test.



For ARG3 (2) to ARG3 (17), specify "1" for the bit corresponding to each DUT number in parallel test.

- **VAR (m)**

Specifies the maximum number of the DUT to be measured for an array element (m).

- **VAR (1)**

Stores the time measurement results of the DUT number 1 in parallel test.

:

VAR (256)

Stores the time measurement results of the DUT number 256 in parallel test.

:

VAR (512)

Stores the time measurement results of the DUT number 512 in parallel test.

- **ERR (m)**

Specifies the maximum number of the DUT to be measured for an array element (m).

ERR (1)

Stores the error code of the DUT number 1 in parallel test.

:

ERR (256)

Stores the error code of the DUT number 256 in parallel test.

:

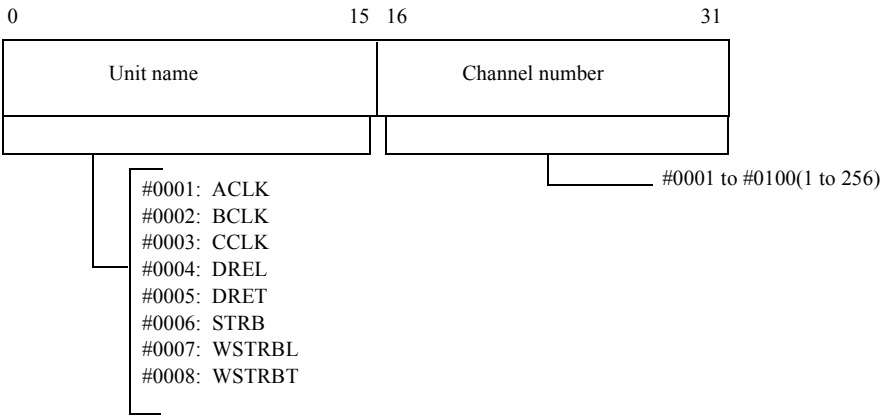
ERR (512)

Stores the error code of the DUT number 512 in parallel test.

 Tip

- Specify the start point of the test area as the ARG (1) value.
- Specify the start point of the test area as the ARG (2) value.
There must be only one boundary point of the pass area and fail area between ARG (1) and ARG (2).
- Specify the value of ARG(3) by the TGCLK number or the unit number.
→ See [Table 30-1](#) through [Table 30-5](#).
ACLK1 to 16, BCLK1 to 20, CCLK1 to 20, DREL1 to 2, DRET1 to 2, STRB1 to 4, WSTRBL1 to 2, and WSTRBT1 to 2 can specified by either the unit number or the TGCLK number.
Method for specifying with the unit number.
Specification method by unit number Unit name and channel number are represented by hexadecimal notation.
0 to 15 bits: Unit name

16 to 31 bits: Channel number



- Result of time measurement is stored in ARG (4). In parallel test, the result is stored in VAR as the same element number as DUT. This value is the last functional test result which is a pass.
- If an error occurs during execution, that error code is stored in ARG (5). In parallel test, the error code is stored in ERR as the same element number as DUT.

➔ See [Table 30-6](#).

- ARG (6) sets the timing set of timing pulse edge ARG (3) for time measurement.1 Specify within the range of 1 to 16.
- Specify ALPG PC start address as ARG (7).
- The tracing units can be specified by up to 16 digits.
The settings required for tracing are as follows;
ARG (n): The tracing unit number.
ARG (n+1): Specifies the tracing value. (added value)
ARG (n+2): Specifies the timing set of the tracing timing pulse edge.

Here, "n" mean the following digits.

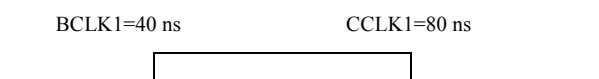
Here, n = 8, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48, 51, 54.
Specify as (Number of tracing units 16) × (Number of data 3).

When the number of tracing units is below 16, specify 0 to the unit number next to the last tracing unit specified.
In the TMEAS utility, until the tracing unit Nos. reach 0, the units are regarded as tracing units to be processed.
The arguments after the tracing unit No. reaches 0 are ignored.

- When a time is measured being traced to the measuring timing pulse edge, specify the timing pulse edge, which it is desired to trace, for the ARG (n) by the UNIT number or the TGCLK number.
If the tracing timing pulse edge is not required, specify 0 for the ARG (n).
Here, n = 8, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48, 51, 54.
- When the tracing timing pulse edge is specified, set an increment for tracing ARG(n+1). (Unit: Time)

The increment is a positive value when the measured timing pulse edge is in front of the tracing timing pulse edge, and is a negative value when the measured timing pulse edge is behind the tracking timing pulse edge. When no tracking timing pulse edge is specified, set 0 (second).

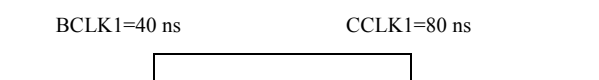
- ◆ **When the Measured Timing Pulse Edge is BCLK1 and the Tracing Timing Pulse Edge is CCLK1**



The tracing value ARG(9) is as follows:

$$\begin{aligned} \text{ARG}(9) &= (\text{Tracing timing pulse edge}) - (\text{Measured timing pulse edge}) \\ &= 80 \text{ ns} - 40 \text{ ns} \\ &= 40 \text{ ns} \end{aligned}$$

- ◆ **When the Measured Timing Pulse Edge is CCLK1 and the Tracing Timing Pulse Edge is BCLK1**



The tracing value ARG(9) is as follows:

$$\begin{aligned} \text{ARG}(9) &= (\text{Tracing timing pulse edge}) - (\text{Measured timing pulse edge}) \\ &= 40 \text{ ns} - 80 \text{ ns} \\ &= -40 \text{ ns} \end{aligned}$$

- The timing set of the tracing timing pulse edge ARG (n) is specified as ARG (n+2).1 Specify within the range of 1 to 16.
When no timing pulse edge is traced, specify 0.
- The TMEAS argument cannot be specified directly by a constant. It must always be specified by 11 or 56 three-word array variables.
- When ARG (1) make the offset data set in the OFFSET statement apply in TMEAS, specify 1.

Table 30-1 TGCLK Number Correspondence

TGCLK number	Timing pulse edge name
1	BCLK1
2	BCLK2
3	BCLK3
4	BCLK4
5	BCLK5
6	BCLK6
7	BCLK7
8	BCLK8

TGCLK number	Timing pulse edge name
9	CCLK1
10	CCLK2
11	CCLK3
12	CCLK4
13	CCLK5
14	CCLK6
15	CCLK7
16	CCLK8
17	ACLK1
18	ACLK2
19	ACLK3
20	ACLK4
21	STRB1
22	STRB2
23	STRB3
24	STRB4
25	ACLK9
26	ACLK10
27	ACLK13
28	ACLK14
29	DREL1
30	DREL2
31	DRET1
32	DRET2
33	BCLK9
34	BCLK10
35	BCLK11
36	BCLK12
37	BCLK13
38	BCLK14
39	BCLK15
40	BCLK16
41	CCLK9
42	CCLK10
43	CCLK11
44	CCLK12
45	CCLK13
46	CCLK14
47	CCLK15
48	CCLK16
49	ACLK5
50	ACLK6
51	ACLK7
52	ACLK8

TGCLK number	Timing pulse edge name
53	BCLK17
54	BCLK18
55	BCLK19
56	BCLK20
57	CCLK17
58	CCLK18
59	CCLK19
60	CCLK20
61	ACLK11
62	ACLK12
63	ACLK15
64	ACLK16
100	WSTRBL1
101	WSTRBL2
102	WSTRBL3
103	WSTRBL4
104	WSTRBT1
105	WSTRBT2
106	WSTRBT3
107	WSTRBT4

 Tip

- *TGCLK numbers 100 to 107 indicate the clocks for TMEAS. WSTRBL1 to 4 specify the leading edge of WSTRB, and WSTRBT1 to 4 specify the trailing edge of WSTRB.*
- *ACLK cannot be used on the T5593 and T5592.*
- *STRB restrictions are shown below.*

Type/Model	Restrictions
T5592	STRB2 or STRB4 cannot be used.
T5591	STRB2 or STRB4 cannot be used for 4 -Way Interleaved Mode.
T5593, T5586, T5585, T5581, T5581H and T5581P	STRB2 or STRB4 cannot be used for 2 -Way Interleaved Mode.

- *WSTRB restrictions are shown below.*

Type/Model	Restrictions
T5593 and T5592	WSTRBL and WSTRBT cannot be used.
T5591	WSTRBL and WSTRBT cannot be used for 4-Way Interleaved Mode.
T5586, T5585, T5581, T5581H and T5581P	WSTRBL and WSTRBT cannot be used for 2-Way Interleaved Mode.
T5377, T5376 and T5375	WSTRBL4 and WSTRBT4 cannot be used.

- *These parameters cannot be specified on the T5336 and T5311.
STRB3, STRB4, ACLK5 to ACLK16, BCLK9 to BCLK20, CCLK9 to CCLK20, DREL1, DREL2,
DRET1, DRET2*

Table 30-2 Unit Number Correspondence (For the T5500)

Unit number	Unit name
#00000001 ⋮ #0000006B	TGCLK1 ⋮ TGCLK107
#00010001 ⋮ #00010080	ACLK1 ⋮ ACLK128
#00020001 ⋮ #00020080	BCLK1 ⋮ BCLK128
#00030001 ⋮ #00030080	CCLK1 ⋮ CCLK128
#00040001 ⋮ #00040100	DREL1 ⋮ DREL256
#00050001 ⋮ #00050100	DRET1 ⋮ DRET256
#00060001 ⋮ #00060100	STRB1 ⋮ STRB256
#00070001 ⋮ #00070080	WSTRBL1 ⋮ WSTRBL128

#00080001	WSTRBT1
⋮	⋮
#00080080	WSTRBT128



Tip

- *ACLK cannot be specified on the T5593 and T5592.*
- *STRB restrictions are shown below.*

Type/Model	Restrictions
T5593	In 1-WAY Interleaved Mode, odd number STRB1 to 256; in 2-WAY Interleaved Mode, odd number STRB1 to 128
T5592	Odd number STRB1 to STRB256 can be used.
T5591	In 1-WAY/2-WAY Interleaved Mode, odd number STRB1 to 256; in 4-WAY Interleaved Mode, odd number STRB1 to 256
T5586 and T5585	In 1-WAY Interleaved Mode, odd number STRB1 to 128 and 129 to 256; in 2-WAY Interleaved Mode, odd number STRB1 to 128
T5581, T5581H and T5581P	In 1-WAY Interleaved Mode, odd number STRB1 to 128; in 2-WAY Interleaved Mode, odd number STRB1 to 128
T5571P	STRB1 to 128

- *WSTRB restrictions are shown below.*

Type/Model	Restrictions
T5593 and T5592	WSTRBL and WSTRBT cannot be used.
T5591	In 1-WAY/2-WAY Interleaved Mode, WSTRBL/WSTRBT1 to 128 cannot be used; in 4-WAY Interleaved Mode, WSTRBL/WSTRBT cannot be used.
T5586, T5585, T5581, T5581H and T5581P	In 1-WAY Interleaved Mode, WSTRBL/WSTRBT1 to 64 cannot be used; in 2-WAY Interleaved Mode, WSTRBL/WSTRBT cannot be used.
T5571P	WSTRBL/WSTRBT1 to 64

- *For ACLK, BCLK and CCLK, a unit number is used for the specification and a socket program is used, up to the maximum DUT pin number written in the socket program can be specified as the phase number. The maximum value is 128.*

➔ For more information on TGCLK, see [Table 30-1](#).

Table 30-3 Unit Number Correspondence (For the T5336 and T5311)

Unit number	Unit name
#00000001 ⋮ #0000006B	TGCLK1 ⋮ TGCLK107
#00010001 ⋮ #00010004	ACLK1 ⋮ ACLK4
#00020001 ⋮ #00020008	BCLK1 ⋮ BCLK8
#00030001 ⋮ #00030008	CCLK1 ⋮ CCLK8
#00040001	DREL1
#00050001	DRET1
#00060001 ⋮ #00060002	STRB1 ⋮ STRB2
#00070001 ⋮ #00070004	WSTRBL1 ⋮ WSTRBL4
#00080001 ⋮ #00080004	WSTRBT1 ⋮ WSTRBT4

➔ For more information on TGCLK, see [Table 30-1](#).

Table 30-4 Unit Number Correspondence (For the T5371)

Unit number	Unit name
#00000001 ⋮ #0000006B	TGCLK1 ⋮ TGCLK107
#00010001 ⋮ #00010080	ACLK1 ⋮ ACLK128
#00020001 ⋮ #00020080	BCLK1 ⋮ BCLK128
#00030001 ⋮ #00030080	CCLK1 ⋮ CCLK128

#00040001 ⋮ #00040080	DREL1 ⋮ DREL128
#00050001 ⋮ #00050080	DRET1 ⋮ DRET128
#00060001 ⋮ #00060004	STRB1 ⋮ STRB4
#00070001 ⋮ #00070004	WSTRBL1 ⋮ WSTRBL4
#00080001 ⋮ #00080004	WSTRBT1 ⋮ WSTRBT4



Tip

For *ACLK*, *BCLK* and *CCLK* on the *T5371*, a unit number is used for the specification and a socket program is used; up to the maximum DUT pin number written in the socket program can be specified as the phase number. The maximum value is 128.

➔ For more information on TGCLK, see [Table 30-1](#).

Table 30-5 Unit Number Correspondence (For the T5377, T5376 and T5375)

Unit number	Unit name
#00000001 ⋮ #0000006A	TGCLK1 ⋮ TGCLK106
#00010001 ⋮ #00010080	ACLK1 ⋮ ACLK128
#00020001 ⋮ #00020080	BCLK1 ⋮ BCLK128
#00030001 ⋮ #00030080	CCLK1 ⋮ CCLK128
#00040001 ⋮ #00040080	DREL1 ⋮ DREL128
#00050001 ⋮ #00050080	DRET1 ⋮ DRET128

#00060001 : #00060004	STRB1 : STRB4
#00070001 : #00070003	WSTRBL1 : WSTRBL3
#00080001 : #00080003	WSTRBT1 : WSTRBT3

 Tip

For ACLK, BCLK and CCLK on the T5377, T5376 and T5375, a unit number is used for the specification and a socket program is used; up to the maximum DUT pin number written in the socket program can be specified as the phase number. The maximum value is 128.

➔ For more information on TGCLK, see [Table 30-1](#).

30. 4 TMEAS Error Codes

If an error occurs during execution of the TMEAS program, an error code is stored in argument ARG (5). In parallel test, the code is stored in ERR as the same element number as DUT, further the code indicating the total result is stored in ARG (5). Error codes and their descriptions are shown in [Table 30-6](#).

Table 30-6 TMEAS Error Codes

Error code	Description
1	There is an error in the specified TGCLK number or timing set. (ARG(3), ARG(6), ARG(8) or ARG(10))
2	Both ARG (1) and ARG (2) passed.
3	Both ARG (1) and ARG (2) failed.
4	ALPG stack overflowed.
5	ALPG TIME or OUT error.
6	There is no MPAT file in ALPG. Transfer the pattern file to ALPG.
7	Specified DUT NO. is not DUT for parallel test.
8	Specified DUT NO. is not DUT for test.
9	In this system, a specified timing pulse edge is not added optionally.
10	A resolution is specified incorrectly. (ARG2(1))
11	Some DUTs for parallel test occurs errors.
12	Array element number of VAR or ERR is not enough.
13	VMEAS cannot be executed because of PLL clock restrictions.
14	A setting value is specified incorrectly. (ARG(1), ARG(2))

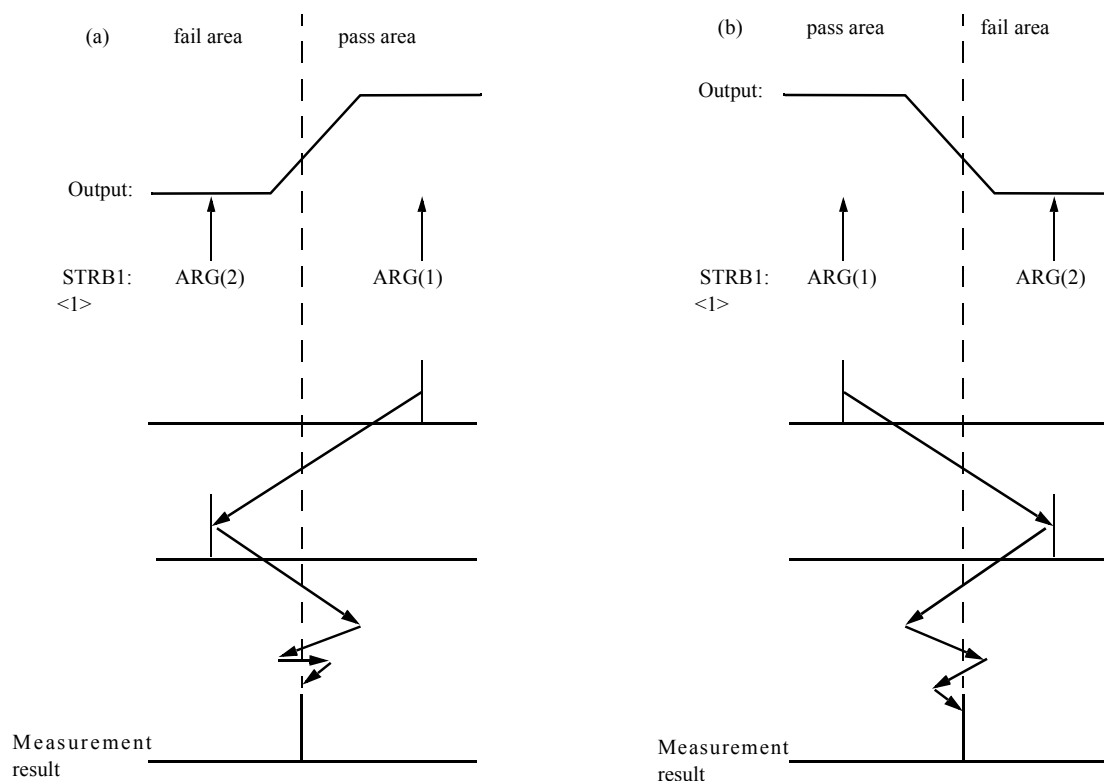


Tip

In the parallel test, the error code is ARG (5)=11 if some DUT errors are occurred during execution of the TMEAS program. When all DUTs indicate one sort of error code, the code is stored in ARG (5). Also, error code stored in ERR is any one of 2, 3, 4, 5, 7 and 8.

30. 5 TMEAS Use

- To measure the rise time of the waveform shown in [Figure 30-1\(a\)](#),
Set the TMEAS argument as follows:
Set DIM ARG(11) = 0 and then set as follows:
ARG (1)
Sets the value in the pass area. (Or sets the value in the fail area.)
ARG (2)
Sets the value in the fail area. (Or sets the value in the pass area.)
ARG(3) to ARG(6)
Since the timing pulse edge is STRB1<1>, TGCLK number is 21 (#00060001 when it is specified by the unit number), and the timing set is 1. Therefore, set ARG(3) = 21 (ARG(3) = #00060001 when it is specified by the unit number), and ARG(6) = 1.
ARG (7)
Sets the ALPG start address by testing the MPAT pattern program.
ARG(8) to ARG(10) need not be set because there is no tracing edge.
Also, ARG (4) and ARG (5) need not be set. They are initialized to 0 by DIM ARG (11)=0.
- To measure the fall time of the waveform shown in [Figure 30-1\(b\)](#),
The fall time can be measured with the same argument as described in (a) by changing ARG(1) and ARG(2) values.

Figure 30-1 TMEAS When the Tracing Timing Pulse Edge is Not Specified

If there is a timing pulse edge (CCLK1) to be traced simultaneously when the timing pulse edge (BCLK1) is measured as shown in [Figure 30-2\(a\)](#), the TMEAS argument is set as follows:

Set DIM ARG(11) = 0 and then set as follows:

ARG (1)

Sets a value in the pass area (or fail area). The measured timing pulse edge is ARG (1), and the tracing timing pulse edge is ARG (1)+ARG (9). Under this condition, set ARG (1) so that the test passes (or fails).

ARG (2)

Sets a value in the fail area (or pass area). The measured timing pulse edge is ARG (2), and the tracing timing pulse edge is ARG (2)+ARG (9). Under this condition, set ARG (2) so that the test fails (or passes).

ARG(3) to ARG(6)

Since measured timing pulse edge is BCLK1<1>, ARG(3) = 1 (ARG(3) = #00020001 when it is specified by the unit number) and the timing set is 1 based on [Table 30-1](#), set ARG(6) = 1.

ARG (8)

Since measured timing pulse edge is CCLK1<1>, ARG(8) = 9 (ARG(8) = #00030001 when it is specified by the unit number) and the timing set is 1 based on [Table 30-1](#), set ARG(10) = 1.

ARG (9)

Sets a tracing value. Increment for tracing is specified: It is a positive value.

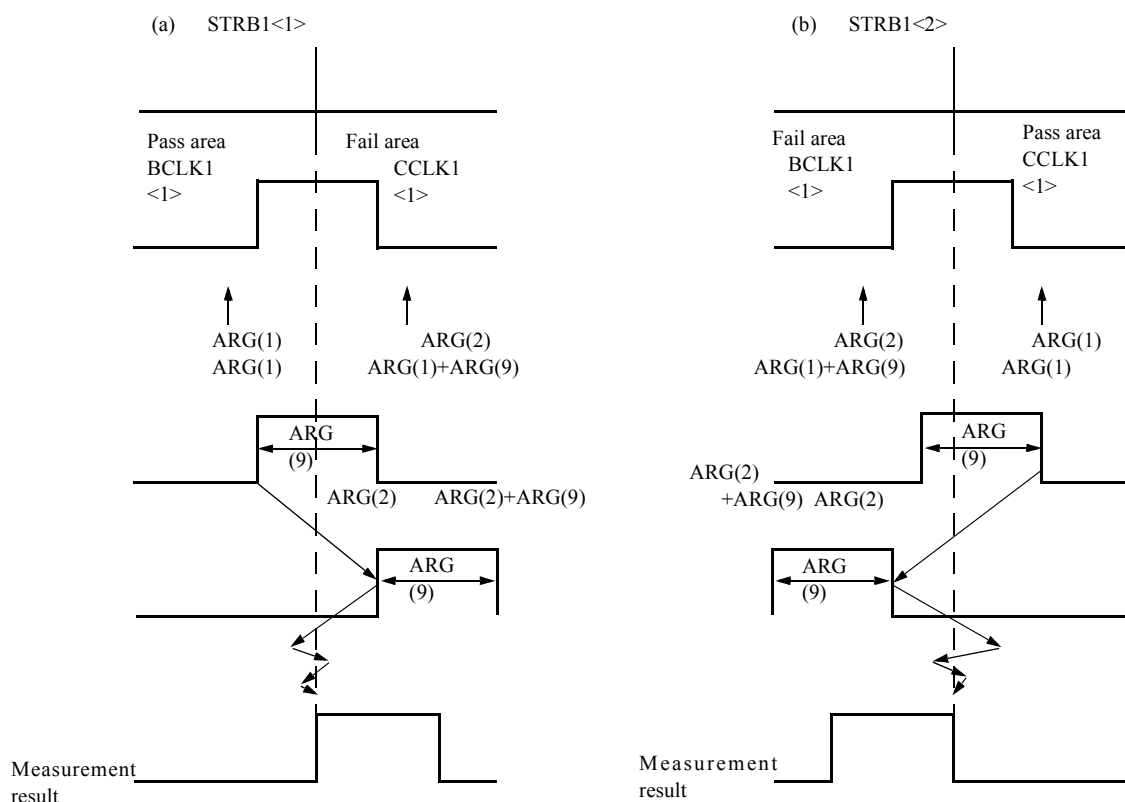
ARG (7)

Sets a PC start address by the testing MPAT pattern program.

No values need to be specified for ARG (4) and ARG (5).

- In [Figure 30-2\(b\)](#), when CCLK1<1> is specified for the measurement timing pulse edge instead of BCLK1<1> in [Figure 30-1\(b\)](#) above and BCLK1<1> is specified for the tracing timing pulse edge, set the argument of TMEAS same as in [Figure 30-1\(b\)](#). Set ARG(3) = 9 (ARG(3) = #00030001 when it is specified by the unit number) and ARG(6) = 1 since the measured timing pulse edge is CCLK1<1>. Set ARG(8) = 1 (ARG(8) = #00020001 when it is specified by the unit number) and ARG(10) = 1 since the tracing timing pulse edge is BCLK1<1>. The tracing ARG(10) is negative since the measured timing pulse edge is behind the tracing timing edge.

Figure 30-2 TMEAS When the Tracing Timing Pulse Edge is Not Specified



- Parallel test

When DUT for parallel test in ARG3 (2) and ARG3 (3) is specified and VAR or ERR is set in the argument, any DUT time specified in one time TMEAS execution is measured. In this case, ARG (11) is ignored and the measured result of error codes is stored in each DUT corresponding number of VAR or ERR.

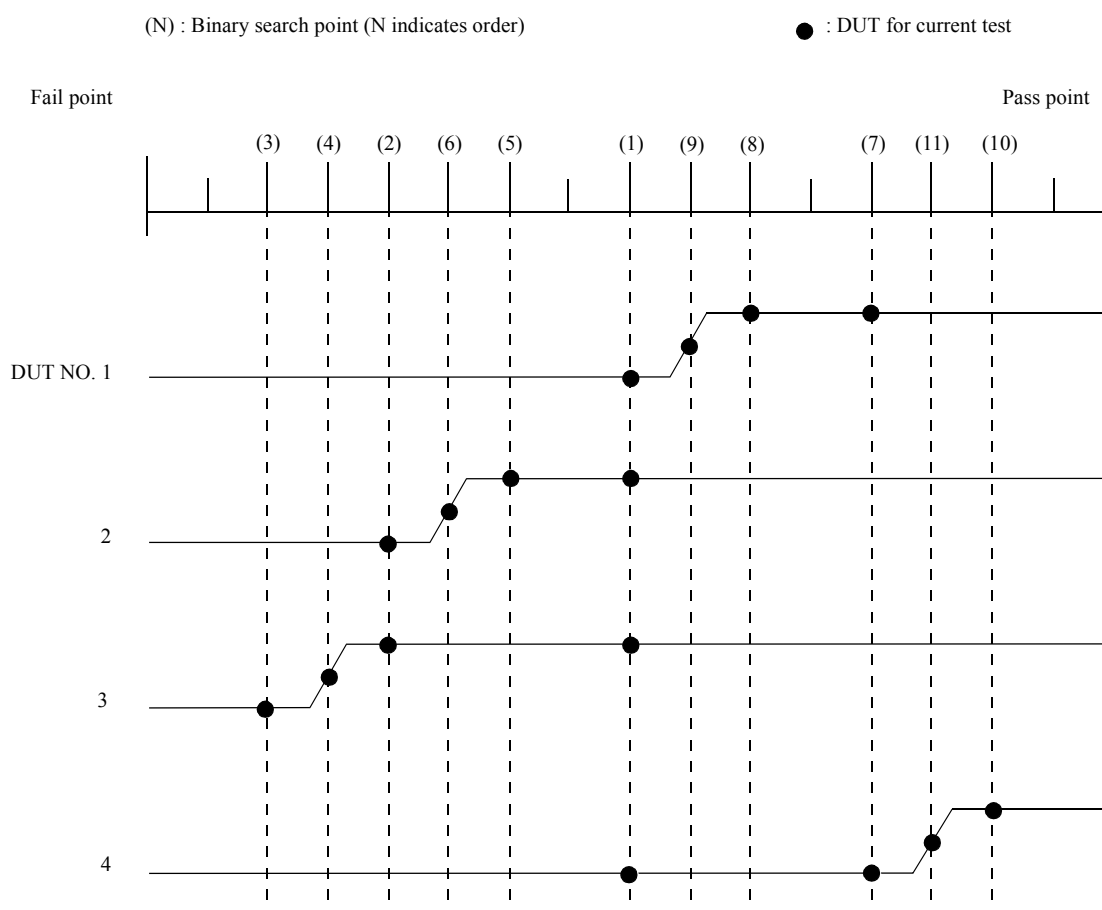
Any result is not stored in ARG (4). The total result of error codes is stored in ARG (5).

In parallel test, the binary search measures all DUTs at first.

Failed DUTs are removed from the measuring target and the binary search is performed so that one DUT may pass. When a final time is decided, set a removed DUT group to the removed time as to repeat the binary search.

The times of MEAS will be certainly decreased, comparing with some separate TMEAS executions, and the time for multiple time measurement will be shortened.

Figure 30-3 Binary Search Point Movement in 4-DUT Parallel Test



Consider the rising time of four DUTs of the waveform as shown in [Figure 30-3](#) is measured with a strobe moved. Pass point and fail point indicate a time set in ARG (1) or ARG (2). Also, each DUT is allowed to decide a transit point by four-time MEAS execution.

When MEAS is executed at point (1) the first time, DUT NO.=1, 4 fail and are removed from the next test target. Since MEAS is executed at point (2) the second time, DUT NO.=2 fails and is removed from the next test target. DUT NO. = 3 fails at the third time, however, since there will be no test target DUT if it is removed, (3) is taken as the fail point and MEAS is continued. At fourth time, DUT NO. = 3 transit point is decided.

Next, set the last removed DUT NO. = 2 to pass point (1) and fail point (2) at the removed condition to execute MEAS. At 6th time, DUT NO. = 2 transit point is decided. This time, set the removed DUT NO. = 1,4 to fail point (1) and pass point at the removed condition to execute MEAS. As the same, DUT NO.4 is removed at 7th time, DUT NO. = 1 transit point is decided at 9th time, and if DUT NO. = 4 is set to fail point (7) and pass point to execute MEAS, at 11th time a transit point is decided.

The total 16-times MEAS is executed when each separate DUT is tested in TMEAS, however, this case will be executed at 11 times. Also, when DUT as the same transit point or closing DUT is measured, the times to execute MEAS will be decreased.

30. 6 TMEAS Programming Notes

- Set conditions and transfer MPAT files to be used to the ALPG before executing this program.
- There must be a pass area and a fail area between ARG (1) and ARG (2) when this program is executed.
- Execute the SRON statement before executing the ASM TMEAS. (REFERENCE is not turned on in TMEAS.)
- The following statements cannot be written until ASM TMEAS is specified after the MPAT file used in the SEND MPAT statement is transferred.

"CALL CALB()"

"MEAS MPAT"

"STOP MPAT"

"RESET MPAT"

Note the case given below:

```
REG MPAT PC=#0
SEND MPAT
REG MPAT PC=#20
SEND MPAT x
```

At this stage, the SEND MPAT x statement is required. Because, even if a pattern file name to be executed is the same as the previous name, the difference in the PC start address (module of the pattern program) requires data retransfer of MODE, REGISTER, ARIRAM, SCRAM, etc., which can be specified in each module. (Data is not retransferred in the TMEAS program.)

```
EXEC TMEAS (ARG)
```

- 11 or 56 three-word array variables must be specified as the argument of this program in advance.
- If either measurement start value (ARG(1)) or measurement stop value (ARG(2)) is set to 1 μ s or more, the time measurement is executed at the resolution of 2 ns. However, the specified resolution is used on the T5592.
- This program cannot be executed in the following cases:
When using a pattern program that uses the burst data in the main program
When using a pattern program that is designed for MEAS MPAT(5)-based fast erasure test
- On the T5591 and T5591R, STRB129 to 256, WSTRBL65 to 128 and WSTRBT65 to 128 can be used.

30. 7 Examples of Using TMEAS

◆ When the Tracing Timing Pulse Edge is Not Used

```

PRO 100

P1 = IN1, RZO, BCLK1, CCLK1, <FH>          *1
P33 = OUT1, STRB1, <FH>                     *1

IN1 = 5V, 1V                                *2
OUT1= 3V, 2V                                *2

RATE =100NS                                 *3
BCLK1= 40NS                                 *3
CCLK1= 80NS                                 *3
STRB1= 60NS                                 *3

SEND MPAT ALL1                               *4

DIM ARG(11)= 0                               *5
ARG(1) =60NS ; PASS                          *5
ARG(2) = 0NS ; FAIL                          *5
ARG(3) = 21 ; TGCLK NO. (STRB1)              *5
          ; ARG(4) : RESULT
          ; ARG(5) : ERR CODE
ARG(6) = 1 ; TIMING SET                       *5
ATG(7) = 0 ; ALPG START ADDRESS              *5
ARG(8) = 0 ; NO-TRACKING                     *5
ARG(9) = 0NS ; TRACKING DATA                *5
ARG(10)= 0 ; TRACKING                        *5
          ; TIMING SET
ARG(11)= 1 ; DUT NO.                          *5

SRON ; REFERENCE ON!                         *6

EXEC TMEAS(ARG)                              *7

IF ARG(5)=0 THEN WRITE "MEASURE RESULT = " ARG(4) / *8
ELSE WRITE "ERROR STATUS = " ARG(5) /        *8

END

```

An RZO mode waveform of BCLK1 and CCLK1 is output. The rise time by BCLK1 is measured. Pins 1 and 33 on the performance board are jumpered. The measurement procedure is explained below with an example.

**1 Write set conditions of pins.*

**2 Description of voltage conditions for the driver and comparator*

*3 Sets timing information.

*4 Transfer the MPAT file to be used in the test to the ALPG. The pattern of this example is explained in ["MPAT Pattern Program Used" on page 30-29](#).

*5 Sets the argument.

The measurement timing pulse edge at STRB1<1> is ARG(3) = 21 (ARG(3) = #00060001 when specified with a unit number), and ARG(6) = 1.

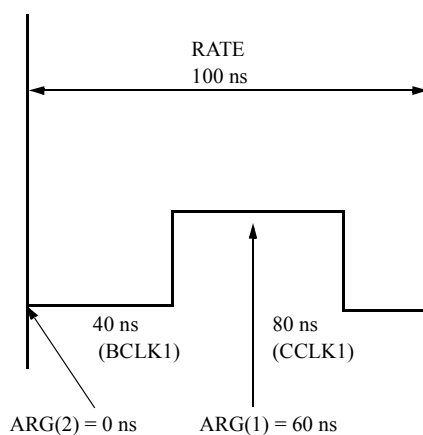
After setting ARG(1) to 60[ns] and ARG(2) to 0[ns], set ARG(8) = 0, ARG(9) = 0 [ns] and ARG(10) = 0 since the tracing timing pulse edge is not specified.

*6 Performs REFERENCE ON

*7 Measures time.

*8 Displays measurement results or error code.

Figure 30-4 Output Waveform of Example of Using TMEAS (1)



The BCLK1<1> rise time is measured under these conditions by moving STRB1<1> from 0 ns to 60 ns.

➔ For more information on used MPAT patterns, refer to ["MPAT Pattern Program Used" on page 30-29](#).

◆ When Tracing Timing Edge is Specified

```

PRO 200

P1  = IN1, RZO, BCLK1, CCLK1, <FH>
P33 = OUT1, STRB1, <FH>

IN1 = 5V, 1V
OUT1= 3V, 2V

RATE  =100NS
BCLK1 = 40NS
CCLK1 = 80NS
STRB1 = 60NS

SEND MPAT ALL1

DIM ARG(11) = 0
ARG(1) =30NS ; FAIL
ARG(2) =70NS ; PASS
ARG(3) =1      ; TGCLK NO. (BCLK1)
                ; ARG(4) : RESULT
                ; ARG(5) : ERR CODE
ARG(6) = 1      ; TIMING SET
ATG(7) = 0      ; ALPG START ADDRESS
ARG(8) = 9      ; TRACKING(CCLK1)
ARG(9) = 40NS; TRACKING DATA
ARG(10)= 1      ; TRACKING TIMING SET
ARG(11)= 1      ; DUT NO.

SRON          ; REFERENCE ON!

EXEC TMEAS (ARG)

IF ARG(5)=0 THEN WRITE "MEASURE RESULT = " ARG(4) /
                    ELSE WRITE "ERROR STATUS   = " ARG(5) /

END

```

The example above is a program with measured timing pulse edge BCLK1<1> and tracking timing pulse edge CCLK1<1>. Pins, voltage, timing conditions, and patterns are the same as those in the example in ["When the Tracing Timing Pulse Edge is Not Used" on page 30-24](#).

The TMEAS argument is explained below.

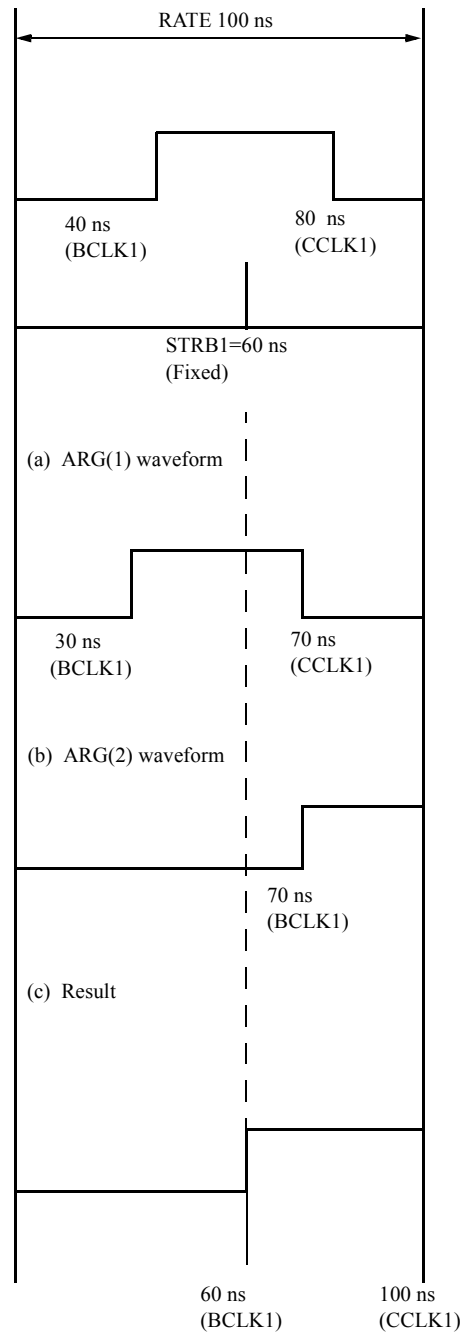
ARG(1) and ARG(2) are 30 ns and 70 ns, and their output waveforms are shown in [Figure 30-5 \(a\) and \(b\)](#).

When BCLK1<1> is in the state of ARG(1) = 30 ns, the tracking timing edge CCLK1<1> is ARG(1) + ARG(9) = 30 ns + 40 ns = 70 ns and the test passes ([Figure 30-5\(a\)](#)). When BCLK1<1> is in the state of ARG(2) = 70 ns, the tracking timing edge CCLK1<1> is ARG(2) +

$ARG(10) = 70 \text{ ns} + 40 \text{ ns} = 110 \text{ ns}$ and the test fails ([Figure 30-5\(b\)](#)). Set BCLK1<1> with $ARG(3) = 1$ ($ARG(3) = \#00020001$ when it is specified by the unit number) and $ARG(6) = 1$, and set CCLK1<1> with $ARG(8) = 9$ ($ARG(8) = \#00030001$ when it is specified by the unit number) and $ARG(10) = 1$.

$ARG(9)$ is the increment for tracking, and 40 ns ($= 80 \text{ ns} - 40 \text{ ns}$) is set as its value.

The measurement result shows a waveform rising at BCLK1<1> 60 ns as shown in [Figure 30-5\(c\)](#) where $ARG(4) = 60 \text{ ns}$ and $ARG(5) = 0$.

Figure 30-5 Output Waveform of Example of Using TMEAS (2)

- ◆ **MPAT Pattern Program Used**

The MPAT pattern program used in examples ["When the Tracing Timing Pulse Edge is Not Used"](#) on page 30-24 and ["When Tracing Timing Edge is Specified"](#) on page 30-26 is shown below.

```
MPAT ALL1

REGISTER
  PC = #0
  TPH = #3FFF

START    #0
NOP      TPH>TP
NOP      W R TP>TP
NOP      W R TP>TP
NOP      W R TP>TP
STPS

END
```

◆ 4 DUTs Parallel Test

```

PRO 300 S4DUT

P1  = IN1, RZO, BCLK1, CCLK1, <FH>
P33 = OUT1, STRB1, <FH>
IN1 = 5V, 1V
OUT1= 3V, 2V

RATE  =100NS
BCLK1 = 40NS
CCLK1 = 80NS
STRB1 = 60NS

SEND MPAT ALL1

DIM ARG(11)  = 0          *1
DIM ARG2(10) = 0          *2
DIM ARG3(2)  = 0          *2
DIM VAR(4)   = 0          *3
DIM ERR(4)   = 0          *3

ARG(1) = 30NS ; PASS      *4
ARG(2) = 70NS ; FAIL      *4
ARG(3) = 1    ; TGCLK NO. (BCLK1) *4
                ; ARG(4) : (RESULT)
                ; ARG(5) : ERR CODE
ARG(6) = 1    ; TIMING SET *4
ARG(7) = 0    ; ALPG START ADDRESS *4
ARG(8) = 0    ; NO-TRACKING *4
ARG(9) = 0NS  ; TRACKING DATA *4
ARG(10)= 0    ; TRACKING TIMING SET *4
                ; ARG(11): (DUT NO.)

ARG3(1)= 0    ; NO-EFFECT
ARG3(2)= #F    ; DUT NO. = 1 - 4

SRON          ; REFERENCE ON!

EXEC TMEAS(ARG, ARG2, ARG3, VAL, ERR) *5

WRITE [0, 1,] "TOTAL ERROR STATUS =" ARG (5)/

DO LOOP I=1,4
WRITE "DUT NO. = " I
IF ERR(I)=0 THEN WRITE " MEASURE RESULT = " VAR(I)/
                ELSE WRITE " ERROR STATUS   = " ERR(I)/
    LOOP:
END

```

**1 Prepare ARG2 even when the resolution is not specified.*

**2 Enough for up to ARG3(2) because 4-DUT parallel test.*

**3 Enough for 4 array elements as four DUTs parallel test. Error occurs when array are provided elements less than 4.*

**4 Set ARG the same as in ["When the Tracing Timing Pulse Edge is Not Used" on page 30-24](#).*

**5 Perform parallel test when 5 arguments are correctly set to TMEAS.*

◆ **Socket Program Used**

The socket program used in the previous example (["4 DUTs Parallel Test" on page 30-30](#)) is shown below.

```
SOCKET S4DUT

DEFINE MULTIPLE 4DUT 1STN

    BIT = 1
    PPS = 1
    PIN MODE = A
    CHTYPE 4

DUT#    = VS1 PD1-12    P33

DUT1    = VS1 PD1-12A1 P33A
DUT2    = VS2 PD1-12A2 P161A
DUT3    = VS3 PD1-12B1 P33B
DUT4    = VS4 PD1-12B2 P161B

END
```

31. Voltage Measurement Program (VMEAS)

This chapter describes the voltage measurement program (VMEAS) for measuring the boundary point between the pass area and fail area.

31. 1 VMEAS Overview

The VMEAS utility program measures the boundary point between the pass area and fail area for the OUT, IN, VS, and IL units by performing the function test. Also, in parallel test, all specified DUTs are measured once.

31. 2 VMEAS Syntax

Two statements are available for VMEAS program execution and must be used as shown below.

31. 2. 1 VMEAS

Function: This syntax is used to measure the boundary point between the pass area and fail area for the OUT, IN, VS, and IL units by performing the function test.

Syntax

(1) To Insert into the Object File of the Test Plan Program

```
ASM VMEAS (ARG)
ASM VMEAS (ARG, ARG2, VAR, ERR)                (specified in parallel test)
```

(2) To Call at Program Execution

```
EXEC VMEAS (ARG)
EXEC VMEAS (ARG, ARG2, VAR, ERR)                (specified in parallel test)
```

Description

To insert into the object file of the test plan program

Since the TMEAS program uses the ASM statement, it is built into the object file of a test program at compilation. (To reduce overhead time and speed up calling the VMEAS program)

When the VMEAS program is modified, any test program that uses the VMEAS program must be recompiled. The following EXEC statement is used so that it is not necessary to incorporate a VMEAS program into the object file of the test program:

To call at program execution

The EXEC statement has the same functions as the CALL statement. The test program need not be recompiled when the VMEAS program is modified, and is called by the EXEC statement.

31. 3 VMEAS Argument

For the VMEAS argument, 10 three-word array variables must be specified. Specify the variables as follows:

```
DIM ARG (10)  =0
DIM ARG2 (1)  = 0 (1 = 1 to 8)
DIM VAR (m)   = 0 (m = 1 to 512)
DIM ERR (m)   = 0
```

- ARG (10)

The contents of the 10 array elements are as follows:

ARG (1)

Measurement start value (in volts)

ARG (2)

Measurement stop value (in volts)

ARG (3)

Specifies the unit name for this measurement. Specify the unit name using a unit number in [Table 31-1](#) or [Table 31-2](#).

ARG (4)

Stores measurement results.

ARG (5)

Stores the error codes of the errors that occurred during execution.

➔ For more information, refer to [Table 31-4](#).

ARG (6)

Specifies the interval to be waited after the voltage is set for the unit.

If 0 is specified, this is equivalent to 3 ms. (Unit:Time)

ARG (7)

ALPG start address

ARG (8)

Specifies tracing unit name. Specify the unit name using a unit number in [Table 31-1](#) or [Table 31-2](#). When 0 is specified, it is assumed that there is no tracing unit.

ARG (9)

Specifies the tracing value (increment).

Tracing unit value = Measurement unit value + ARG (9)

ARG (10)

Specifies DUT number of the device to be measured.(1 to 512)

When 0 is specified, 1 is used.

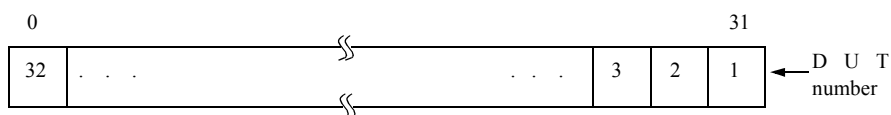
Specify DUT numbers in accordance with the socket program.

- ARG2(l)

Specify the required number for the number of array elements.

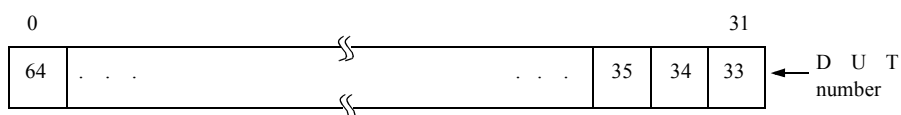
ARG2 (1):

Specifies the DUT number (1 to 32) in parallel tests.



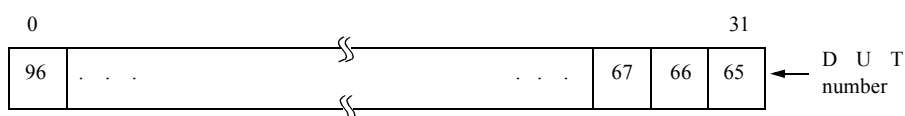
ARG2 (2):

Specifies the DUT number (33 to 64) in parallel tests.



ARG2 (3):

Specifies the DUT number (65 to 96) in parallel tests.



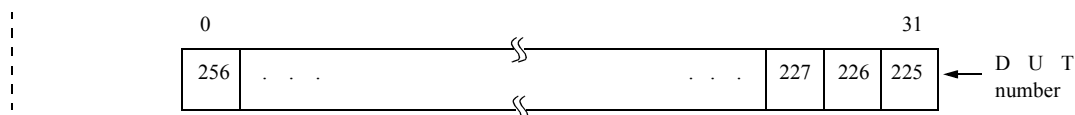
ARG2 (4):

Specifies the DUT number (97 to 128) in parallel tests.



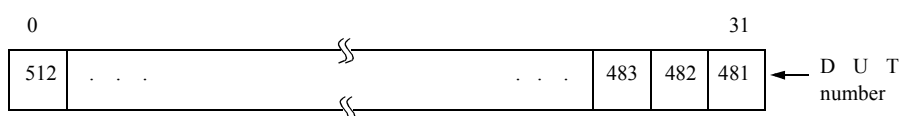
ARG2 (8):

Specifies the DUT number (225 to 256) in parallel tests.



ARG2 (16):

Specifies the DUT number (481 to 512) in parallel tests.



For ARG2 specify "1" for the bit corresponding to each DUT number in parallel test.

- VAR (m)

Specifies the required digit for the number of an array element m.

VAR (1):

Stores the time measurement results of the DUT number 1 in parallel test.

:

VAR (256):

Stores the time measurement results of the DUT number 256 in parallel test.

:

VAR (512):

Stores the time measurement results of the DUT number 512 in parallel test.

- ERR (m)

Specifies the required digit for the number of an array element m.

ERR (1):

Stores the error code of the DUT number 1 in parallel test.

:

ERR (256):

Stores the error code of the DUT number 256 in parallel test.

:

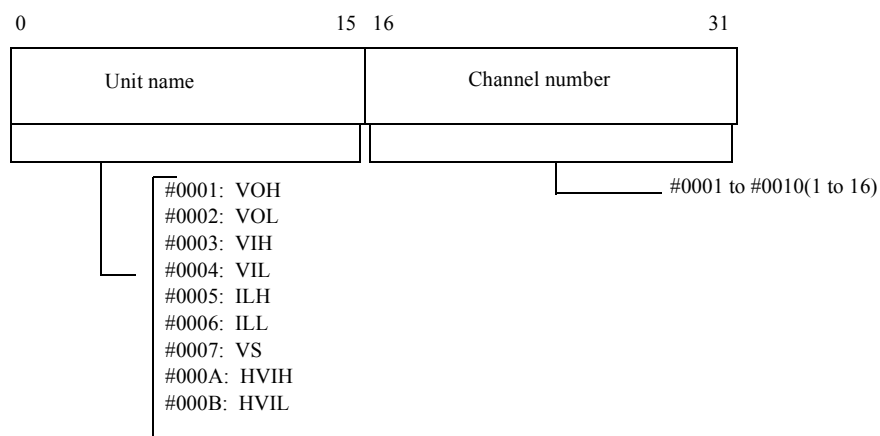
ERR (512):

Stores the error code of the DUT number 512 in parallel test.

Tip

- Specify the start point of the test area as the ARG (1) value.
➔ For more information on the setting range for the value, see [Table 31-3](#).
- Specify the start point of the test area as the ARG (2) value.
➔ For more information on the setting range for the value, see [Table 31-3](#).
There must be only one boundary point of the pass area and fail area between ARG (1) and ARG (2).
- Specify the unit number in [Table 31-1](#) or [Table 31-2](#) as the ARG(3) value. (See [Table 31-3](#) for the setting range.)
Specify the unit number in [Table 31-1](#) or [Table 31-2](#) for channels 1 to 8 in the VOH, VOL, VIH, VIL, ILH, ILL, HVIH, and HVIL unit, and channels 1 to 12 in the VS unit.
The specification method in [Table 31-2](#).
Specification method by unit number Unit name and channel number are represented by hexadecimal notation.
0 to 15 bits: Unit name

16 to 31 bits: Channel number



- The result of the measurement is stored in ARG (4). This value is the last functional test result which is a pass.
- If an error occurs during VMEAS execution, its code is stored in ARG (5).
→ For more information, refer to [Table 31-4](#).
- ARG (6) specifies the interval to be waited after the voltage is set (rise and fall time). When 0 is specified, the waiting time is set to 3 ms.
- Specify the ALPG start address as ARG (7). Specify the ALPG start address to start pattern generation.
- Specify the number of a tracing unit to perform the measurement as ARG (8) using a unit number from [Table 31-1](#) or [Table 31-2](#). When no tracing unit is needed, specify 0 as ARG (9).
- Specify the tracing value (increment) as ARG (9). (Unit :volts) When no tracing unit is specified, specify 0.
- The VMEAS argument cannot be specified directly by a constant. It must always be specified by 10 three-word array variables.

Table 31-1 VMEAS Unit Number Correspondence

Unit number	Unit name
1	V _{OH1}
2	V _{OH2}
3	V _{OH3}
4	V _{OH4}
5	V _{OH5}
6	V _{OH6}
7	V _{OH7}
8	V _{OH8}

Unit number	Unit name
11	V _{OL1}
12	V _{OL2}
13	V _{OL3}
14	V _{OL4}
15	V _{OL5}
16	V _{OL6}
17	V _{OL7}
18	V _{OL8}
21	V _{IH1}
22	V _{IH2}
23	V _{IH3}
24	V _{IH4}
25	V _{IH5}
26	V _{IH6}
27	V _{IH7}
28	V _{IH8}
31	V _{IL1}
32	V _{IL2}
33	V _{IL3}
34	V _{IL4}
35	V _{IL5}
36	V _{IL6}
37	V _{IL7}
38	V _{IL8}
51	I _{LH1}
52	I _{LH2}
53	I _{LH3}
54	I _{LH4}
55	I _{LH5}
56	I _{LH6}
57	I _{LH7}
58	I _{LH8}
61	I _{LL1}
62	I _{LL2}
63	I _{LL3}
64	I _{LL4}
65	I _{LL5}
66	I _{LL6}
67	I _{LL7}
68	I _{LL8}

Unit number	Unit name
71	VS1
72	VS2
73	VS3
74	VS4
75	VS5
76	VS6
77	VS7
78	VS8
79	VS9
80	VS10
81	VS11
82	VS12
101	HV _{IH1}
102	HV _{IH2}
103	HV _{IH3}
104	HV _{IH4}
105	HV _{IH5}
106	HV _{IH6}
107	HV _{IH7}
108	HV _{IH8}
111	HV _{IL1}
112	HV _{IL2}
113	HV _{IL3}
114	HV _{IL4}
115	HV _{IL5}
116	HV _{IL6}
117	HV _{IL7}
118	HV _{IL8}



Tip

For VS unit, T5581: VS1, VS2, VS9 to VS12, for other than T5581: VS1 to VS12 can be specified.

Table 31-2 VMEAS Unit Number Correspondence

Unit number	Unit name
#00010001	VOH1
⋮	⋮
#00010010	VOH16
#00020001	VOL1
⋮	⋮
#00020010	VOL16
#00030001	VIH1
⋮	⋮
#00030010	VIH16

#00040001 : #00040010	VIL1 : VIL16
#00050001 : #00050010	ILH1 : ILH16
#00060001 : #00060010	LL1 : ILL16
#00070001 : #0007000C	VS1 : VS12
#000A0001 : #000A0010	HVIH1 : HVIH16
#000B0001 : #000B0010	HVIL1 : HVIL16

Table 31-3 Setting Data Range and Resolution in VMEAS

Unit		T5593			
		Driver channel	I/O channel	High-speed clock driver channel	Resolution
IN	VIH	-1.8 V ≤ Vih ≤ 5.0 V		-0.5 V ≤ Vih ≤ 3.0 V	2 mV
	VIL	-2.0 V ≤ Vil ≤ 3.3 V		-0.5 V ≤ Vil ≤ 3.0 V	
	--	0.2 V ≤ (Vih-Vil) ≤ 7.0 V		0.2 V ≤ (Vih-Vil) ≤ 2.0 V	
OUT	VOH	--	-2.0 V ≤ Voh ≤ 5.0 V		2 mV
	VOL		-2.0 V ≤ Vol ≤ 5.0 V		

Unit		T5592 and T5591	T5581, T5581H, T5581P and T5571P	T5371, T5336 and T5311	Resolution
IN	VIH	$-1.8 \text{ V} \leq \text{Vih} \leq 5.5 \text{ V}$	$-1.5 \text{ V} \leq \text{Vih} \leq 8.0 \text{ V}$	$-1.5 \text{ V} \leq \text{Vih} \leq 8.0 \text{ V}$	2 mV
	VIL	$-2.5 \text{ V} \leq \text{Vil} \leq 5.0 \text{ V}$	$-3.0 \text{ V} \leq \text{Vil} \leq 5.0 \text{ V}$	$-3.0 \text{ V} \leq \text{Vil} \leq 5.0 \text{ V}$	
	--	$0.2 \text{ V} \leq (\text{Vih-Vil}) \leq 8.0 \text{ V}$	$0.2 \text{ V} \leq (\text{Vih-Vil}) \leq 8.0 \text{ V}$ $ \text{Vih-Vt} \leq 8.0 \text{ V}$	$0.5 \text{ V} \leq (\text{Vih-Vil}) \leq 8.0 \text{ V}$	
INH	HVIH	--	--	$1 \text{ V} \leq \text{Vihh} \leq 15.0 \text{ V}$	4 mV
	HVIL			$0 \text{ V} \leq \text{Vilh} \leq 10.0 \text{ V}$	
	--			$1.0 \text{ V} \leq (\text{Vihh-Vilh}) \leq 15.0 \text{ V}$	
OUT	VOH	$-2.5 \text{ V} \leq \text{Voh} \leq 5.5 \text{ V}$	$-3.0 \text{ V} \leq \text{Voh} \leq 8.0 \text{ V}$	$-3.0 \text{ V} \leq \text{Voh} \leq 8.0 \text{ V}$	2 mV
	VOL	$-2.5 \text{ V} \leq \text{Vol} \leq 5.5 \text{ V}$	$-3.0 \text{ V} \leq \text{Vol} \leq 8.0 \text{ V}$	$-3.0 \text{ V} \leq \text{Vol} \leq 8.0 \text{ V}$	
IL	ILH	--	$-25 \text{ mA} \leq \text{Ilh} \leq 0 \text{ mA}$	$-20 \text{ mA} \leq \text{Ilh} \leq 0 \text{ mA}$	20 μA
	ILL		$0 \text{ mA} \leq \text{Ill} \leq 25 \text{ mA}$	$0 \text{ mA} \leq \text{Ill} \leq 20 \text{ mA}$	

Unit		T5586 and T5585		
		Driver channel	I/O channel	Resolution
IN	VIH	$-1.5 \text{ V} \leq \text{Vih} \leq 8.0 \text{ V}$	$-1.5 \text{ V} \leq \text{Vih} \leq 7.0 \text{ V}$	2 mV
	VIL	$-3.0 \text{ V} \leq \text{Vil} \leq 5.0 \text{ V}$	$-2.0 \text{ V} \leq \text{Vil} \leq 5.0 \text{ V}$	
	--	$0.2 \text{ V} \leq (\text{Vih-Vil}) \leq 8.0 \text{ V}$ $ \text{Vih-Vt} \leq 8.0 \text{ V}$	$0.2 \text{ V} \leq (\text{Vih-Vil}) \leq 8.0 \text{ V}$ $ \text{Vih-Vt} \leq 8.0 \text{ V}$	
OUT	VOH	--	$-2.0 \text{ V} \leq \text{Voh} \leq 7.0 \text{ V}$	2 mV
	VOL		$-2.0 \text{ V} \leq \text{Vol} \leq 7.0 \text{ V}$	

Unit		T5586 and T5585		
		Driver channel	I/O channel	Resolution
IL	ILH	--	$-25\text{ mA} \leq \text{Ilh} \leq 0\text{ mA}$ $0\text{ mA} \leq \text{Ill} \leq 25\text{ mA}$	20 μA
	ILL			

Unit		T5377					
		Driver channel	I/O channel	HV driver channel	HC driver channel	LV driver channel	Resolution
IN	VIH	-1.5 V ≤ Vih ≤ 8.0 V	-1.5 V ≤ Vih ≤ 8.0 V	-1.5 V ≤ Vih ≤ 8.0 V	-1.5 V ≤ Vih ≤ 8.0 V	--	2 mV
	VIL	-3.0 V ≤ Vil ≤ 5.0 V	-2.0 V ≤ Vil ≤ 5.0V	-3.0 V ≤ Vil ≤ 5.0 V	-3.0 V ≤ Vil ≤ 5.0 V		
	--	0.2 V ≤ (Vih-Vil) ≤ 8.0 V	0.2 V ≤ (Vih-Vil) ≤ 8.0 V -8.0 V ≤ Vih-Vt ≤ 8.0 V -8.0 V ≤ Vil-Vt ≤ 8.0 V	0.2 V ≤ (Vih-Vil) ≤ 8.0 V	0.2 V ≤ (Vih-Vil) ≤ 8.0 V		
INH	HVIH	--	--	0.5 V ≤ Vihh ≤ 15.0 V	-3.0 V ≤ Vihh ≤ 15.0 V	0 V ≤ Vihh ≤ 15.0 V	4 mV
	HVIL			0 V ≤ Vilh ≤ 10.0 V	-3.5 V ≤ Vilh ≤ 10.0 V	--	
	--			0.5 V ≤ (Vihh-Vilh) ≤ 15.0 V	0.5 V ≤ (Vihh-Vilh) ≤ 15.0 V		
OUT	VOH	--	-2.0 V ≤ Voh ≤ 8.0 V	--	--	--	2 mV
	VOL		-2.0 V ≤ Vol ≤ 8.0 V				
IL	ILH	--	-2.5 mA ≤ Ilh ≤ 0 mA	--	--	--	20 μA
	ILL		0 mA ≤ Ill ≤ 2.5 mA				

Unit		T5376			
		I/O channel	HC driver channel	LV driver channel	Resolution
IN	VIH	-1.5 V ≤ Vih ≤ 6.0 V	-1.5 V ≤ Vih ≤ 8.0 V		2 mV
	VIL	-2.0 V ≤ Vil ≤ 5.0 V	-3.0 V ≤ Vil ≤ 5.0 V		
	--	0.2 V ≤ (Vih-Vil) ≤ 7.0 V Vih-Vt ≤ 7.0 V Vil-Vt ≤ 7.0 V	0.2 V ≤ (Vih-Vil) ≤ 8.0 V		
INH	HVIH	--	0.5 V ≤ Vihhl ≤ 5.0 V	0 V ≤ Vihh ≤ 15.0 V	4 mV
	HVIL		0 V ≤ Vilh ≤ 10.0 V	--	
			0.5 V ≤ (Vihh-Vilh) ≤ 15.0 V		

Unit		T5376			
		I/O channel	HC driver channel	LV driver channel	Resolution
OUT	VOH	$-2.0 \text{ V} \leq \text{Voh} \leq 6.0 \text{ V}$	--	--	2 mV
	VOL	$-2.0 \text{ V} \leq \text{Vol} \leq 6.0 \text{ V}$			
IL	ILH	$-25 \text{ mA} \leq \text{Ilh} \leq 0 \text{ mA}$	--	--	20 μA
	ILL	$0 \text{ mA} \leq \text{Ill} \leq 25 \text{ mA}$			

Unit		T5375				
		Driver channel	I/O channel	HV driver channel HC driver channel	LV driver channel	Resolu- tion
IN	VIH	$-1.5\text{ V} \leq \text{Vih} \leq 8.0\text{ V}$	$-1.5\text{ V} \leq \text{Vih} \leq 6.0\text{ V}$	$-1.5\text{ V} \leq \text{Vih} \leq 8.0\text{ V}$	--	2 mV
	VIL	$-3.0\text{ V} \leq \text{Vil} \leq 5.0\text{ V}$	$-2.0\text{ V} \leq \text{Vil} \leq 5.0\text{ V}$	$-3.0\text{ V} \leq \text{Vil} \leq 5.0\text{ V}$		
	--	$0.2\text{ V} \leq (\text{Vih}-\text{Vil}) \leq 8.0\text{ V}$	$0.2\text{ V} \leq (\text{Vih}-\text{Vil}) \leq 7.0\text{ V}$ $ \text{Vih}-\text{Vt} \leq 7.0\text{ V}$ $ \text{Vil}-\text{Vt} \leq 7.0\text{ V}$	$0.2\text{ V} \leq (\text{Vih}-\text{Vil}) \leq 8.0\text{ V}$		
INH	HVIH	--	--	$0.5\text{ V} \leq \text{Vihh} \leq 15.0\text{ V}$	$0\text{ V} \leq \text{Vihh} \leq 15.0\text{ V}$	4 mV
	HVIL			$0\text{ V} \leq \text{Vilh} \leq 10.0\text{ V}$	--	
	--			$0.5\text{ V} \leq (\text{Vih}-\text{Vil}) \leq 15.0\text{ V}$		
OUT	VOH	--	$-2.0\text{ V} \leq \text{Voh} \leq 6.0\text{ V}$ $-2.0\text{ V} \leq \text{Vol} \leq 6.0\text{ V}$	--	--	2 mV
	VOL					
IL	ILH	--	$-25\text{ mA} \leq \text{Ilh} \leq 0\text{ mA}$ $0\text{ mA} \leq \text{Ill} \leq 25\text{ mA}$	--	--	20 μA
	ILL					

On the T5581 and T5581H

Unit	Range		Setting range	Resolution
VS1, 2	R10V	Other than 800 mA range	$-10 \text{ V} \leq \text{VS} \leq 10 \text{ V}$	2 mV
		800 mA range	$-2.5 \text{ V} \leq \text{VS} \leq 5 \text{ V}$	
VS9 to 12 R16V	R16V	--	$0 \text{ V} \leq \text{VS} \leq 16 \text{ V}$	4 mV
	R8V	--	$-8 \text{ V} \leq \text{VS} \leq 8 \text{ V}$	2 mV
	R16V	--	$-16 \text{ V} \leq \text{VS} \leq 16 \text{ V}$	4 mV

On the T5593

Unit	Range	Setting range	Resolution
VS1 to 12	R16V	$-5\text{ V} \leq \text{VS} \leq 5\text{ V}$	2 mV

On the T5592, T5591, T5586, T5585, T5581P, T5571P, T5377, T5376, T5375, T5371, T5336 and T5311

Unit	Range	Setting range	Resolution
VS1 to 12	R5V	$0\text{ V} \leq \text{VS} \leq 5\text{ V}$	2 mV
	R16V	$-6\text{ V} \leq \text{VS} \leq 16\text{ V}$	2 mV
	2mV	$-20\text{ V} \leq \text{VS} \leq 20\text{ V}$	2 mV



Tip

- The R20V can be specified on the T5377, T5376 and T5375.
- The VS range is determined by the VS statement setting for the T5377, T5376 and T5375 as shown in the table below.

VS statement setting		Setting range
Applied voltage value	Range	
$0\text{ V} \leq \text{VS} \leq 5\text{ V}$	5V	$0\text{ V} \leq \text{VS} \leq 5\text{ V}$
$-6\text{ V} \leq \text{VS} \leq -2\text{ mV}$	16V	$-6\text{ V} \leq \text{VS} \leq -2\text{ mV}$
$0\text{ V} \leq \text{VS} \leq 16\text{ V}$	16V	$0\text{ V} \leq \text{VS} \leq 16\text{ V}$
$-20\text{ V} \leq \text{VS} \leq -2\text{ mV}$	20V	$-20\text{ V} \leq \text{VS} \leq -2\text{ mV}$
$0\text{ V} \leq \text{VS} \leq 20\text{ V}$	20V	$0\text{ V} \leq \text{VS} \leq 20\text{ V}$

Relationships between the source data, measurement ranges, and clamp values on the T5593

	Current measurement range	Clamp value		Setting range
		positive range	negative range	
R16V	4UA, 40UA, 400UA, 4MA,	0 mA to 400 mA	-400 mA to 0 mA	$-3\text{ V} \leq \text{VS} \leq 5\text{ V}$
		0 mA to 40 0mA	-400 mA to 0 mA	$-5\text{ V} \leq \text{VS} \leq -3\text{ V}$
	80MA	0 mA to 80 mA	-80 mA to 0 mA	$-5\text{ V} \leq \text{VS} \leq 3\text{ V}$
		0 mA to 800 mA	-800 mA to 0 mA	$0\text{ V} \leq \text{VS} \leq 5\text{ V}$
		0 mA to 400 mA	-400 mA to 0 mA	$-5\text{ V} \leq \text{VS} \leq 0\text{ V}$

Relationships between the source data, measurement ranges, and clamp values on the T5592, T5591, T5586, T5585, T5581P, T5571P, T5377, T5376, T5375, T5371, T5336 and T5311

	Current measurement range	Clamp value		Setting range
		positive range	negative range	
R5V	4 UA, 40 UA, 400 UA, 4 MA, 80 MA	0 mA to 400 mA	-400 mA to 0 mA	$0\text{ V} \leq \text{VS} \leq 5\text{ V}$
	800 MA	0 mA to 800 mA	-800 mA to 0 mA	
R16V	4 UA, 40 UA, 400 UA, 4 MA,	0 mA to 400 mA	-400 mA to 0 mA	$-3\text{ V} \leq \text{VS} \leq 8\text{ V}$
		0 mA to 200 mA	-200 mA to 0 mA	$-3\text{ V} \leq \text{VS} \leq 13\text{ V}$
	80 MA	0 mA to 400 mA	-400 mA to 0 mA	$-3\text{ V} \leq \text{VS} \leq 8\text{ V}$
		0 mA to 200 mA	-200 mA to 0 mA	$-3\text{ V} \leq \text{VS} \leq 13\text{ V}$
		0 mA to 80 mA	-80 mA to 0 mA	$-6\text{ V} \leq \text{VS} \leq 16\text{ V}$
	800 MA	0 mA to 800 mA	-800 mA to 0 mA	$0\text{ V} \leq \text{VS} \leq 5\text{ V}$
		0 mA to 400 mA	-400 mA to 0 mA	$-6\text{ V} \leq \text{VS} \leq 13\text{ V}$
		0 mA to 200 mA	-200 mA to 0 mA	$-6\text{ V} \leq \text{VS} \leq 16\text{ V}$

31. 4 VMEAS Error Codes

If an error occurs during execution of the VMEAS program, its code is stored in argument ARG (5). In parallel tests, the code is stored in ERR as the same element number as DUT, further the code indicating the total result is stored in ARG (5). Error codes and their descriptions are shown in [Table 31-4](#).

Table 31-4 VMEAS Error Code Table

Error Code	Description
0	Normal end
1	There is an error in the specified unit number (ARG (3), ARG (8)). (ARG(3), ARG(8)) A negative value is specified as the ALPG start address. (ARG(7)) A negative value or a value over 512 is specified as DUT number. (ARG(10))
2	Test results passed at both the start value (ARG (1)) and stop value (ARG (2)). (There is no pass ↔ fail boundary point between the start and stop values.)
3	Test results failed at both start and stop values.
4	The ALPG stack overflowed.
5	ALPG TIME or OUT error.
6	There is no pattern file in the ALPG. Transfer the pattern file to ALPG.
7	The specified DUT number is not that of a DUT for simultaneous measurement.
8	The specified DUT number is not that of a test DUT.
9	Measurement unit and tracing unit are the same. (ARG(3) = ARG(8))
10	The start value (ARG (1)) or stop value (ARG (2)) of the measurement unit is out of the setting range.
11	The tracing unit value (ARG (1) +ARG (9), ARG (2) + ARG (9)) is out of the setting range.
12	--
13	When VSs are specified as the measurement unit and tracing unit, the setting range does not include the unit numbers of the specified VSs. ➔ For more information, refer to Table 31-3 .
14	When IN is specified for the measurement unit and tracing unit, $V_{IH}-V_{IL}$ is out of the setting range.
15	Some DUTs for parallel test occurs errors.
16	Array element number of VAR or ERR is not enough.
17	When VS is specified to the measurement unit and tracking unit, VS exceeds the clamp value or measurement range setting.
18	VMEAS cannot be executed because of PLL clock restrictions.

 Tip

In multiple simultaneous measurement, errors are occurred in some DUTs, ARG (5)=15 is provided. When all DUTs indicate one sort of error code, the code is stored in ARG (5). Also, error code stored in ERR is any one of 2, 3, 4, 5, 7 and 8.

31. 5 VMEAS Use

- To measure the rise time of the waveform shown in [Figure 31-1](#), set the VMEAS argument as follows:

Set DIM ARG(10) = 0 and then set as follows:

ARG (1)

Sets the start value of the measurement range.

ARG (2)

Sets the stop value of the measurement range.

ARG (3)

Since the unit to be measured is V_{OH1} , the unit number is 1 (or #00010001), set ARG (3)=1 (or ARG (3) =#00010001).

ARG (4)

No setting needed.

ARG (5)

No setting needed.

ARG (6)

Set the waiting time considered necessary. (Unit: Time)

When 0 is set, 3 ms is used.

ARG (7)

Sets a ALPG start address by the testing MPAT pattern program. (A negative value is considered as an error.)

ARG (8)

Since there is no tracing unit, no setting is necessary.

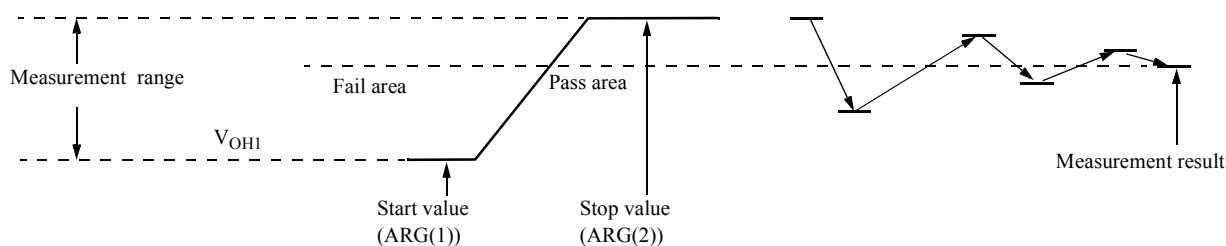
ARG (9)

Since there is no tracing unit, no setting is necessary.

ARG (10)

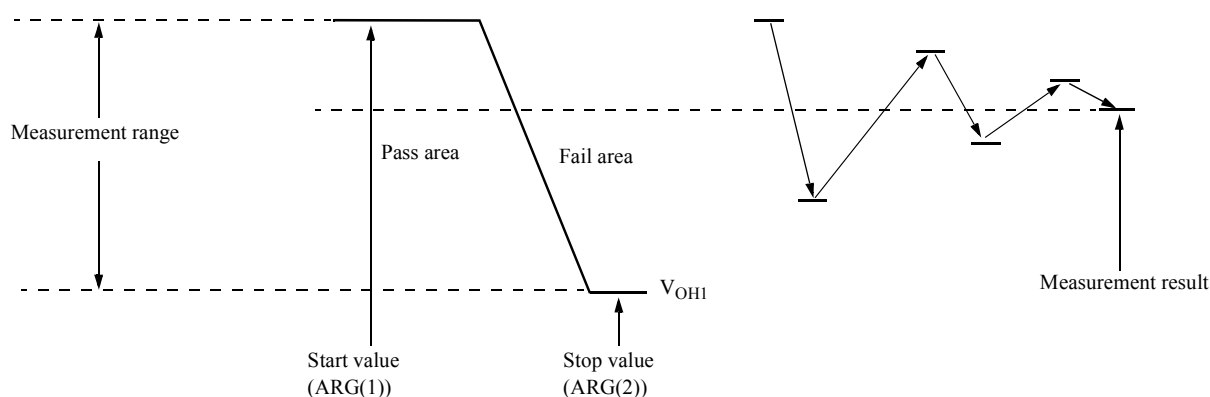
Specify DUT number of the device to be tested.

Figure 31-1 Example of VMEAS Setting (1)



- To measure the fall of the waveform shown in [Figure 31-2](#), use the same argument as given above by changing the values of ARG (1) and ARG (2).

Figure 31-2 Example of VMEAS Setting (2)



- If there is a tracing unit, the VMEAS argument is set as follows in addition to the above:

ARG (8)

Sets the unit number of the tracing unit.

➔ For more information, refer to [Table 31-1](#) and [Table 31-2](#).

ARG (9)

Sets the increment for tracing (volts). (With unit)

If the level of the tracing unit is higher than the measurement unit, a positive value is set and, if lower, a negative value is set.

- Parallel test

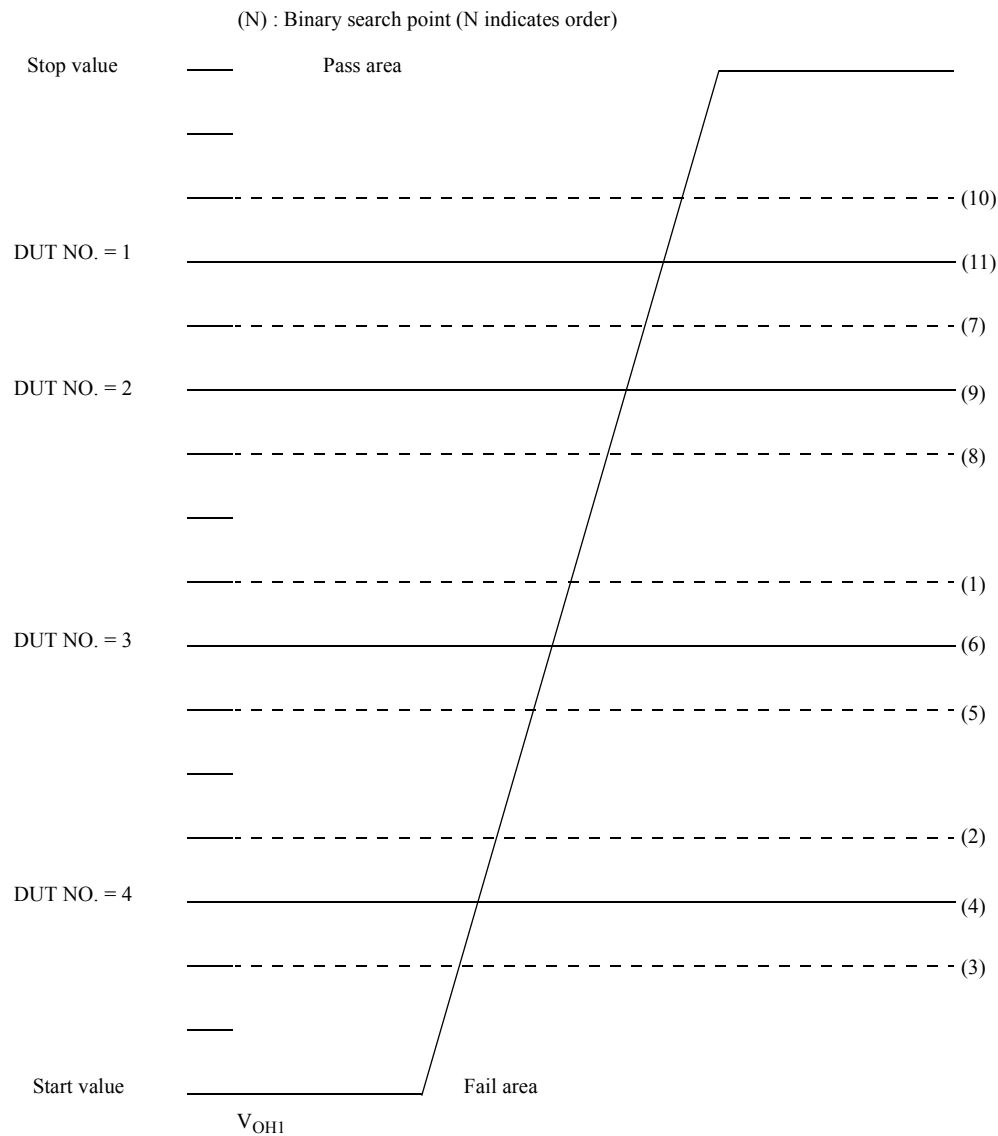
When DUT for parallel test in ARG2 (1) and ARG2 (2) is specified and VAR or ERR is set as the argument, all DUTs specified in first time VMEAS execution is executed. In this case, ARG (10) is ignored and the measured result of error codes is stored in each DUT corresponding number of VAR or ERR.

Any result is not stored in ARG (4). The total results of error code are stored in ARG (5).

In multiple simultaneous measurement, the binary search measures all DUTs at first.

Failed DUTs are removed from the measuring target and the binary search is performed so that one DUT may pass. When the final results are given, DUTs group which were removed (failed DUTs) are set to the previous value and repeat the binary search. The times of MEAS will be certainly decreased, comparing with some separate VMEAS executions, and the time for multiple time measurement will be shortened.

Figure 31-3 Binary Search Point Movement in 4-DUT Parallel Test



Consider measuring the rising time when four DUTs include boundary points as shown in the figure above in the condition unit shown in [Figure 31-3](#). Start value and stop value indicate a value set in ARG (1) or ARG (2). Also, each DUT is allowed to decide a boundary point by four-time MEAS execution.

When MEAS is executed at point (1) the first time, DUT NO.=1, 2 fail and are removed from the next test target. Since MEAS is executed at point (2) the second time, DUT NO.=3 fails and is removed from the next test target. DUT NO. = 4 fails at the third time, however, since there will be no test target DUT if it is removed, (3) is taken as the fail point and MEAS is continued. At fourth time, DUT NO.4 boundary point is decided.

Next, set the last removed DUT NO.=3 to pass point (1) and fail point (2) at the removed condition to execute MEAS. At 6th time, DUT NO.=3 boundary point is decided. This time, set the removed DUT NO.=1 and 2 to stop value and fail point (1) at the removed condition to execute MEAS. Similarly, at the 7th time DUT NO. = 1 is removed, and the DUT NO. = 2 boundary point is decided at the 9th time. DUT No. = 1 is set for the stop value and fail point (7) and MEAS is executed to decide the boundary point at the 11th time.

MEAS is executed 16 times in total when each separate DUT is tested in VMEAS, however, in this case it will be executed 11 times. Also, when DUT as the same boundary point or closing DUT is measured, the times to execute MEAS will be decreased.

31. 6 VMEAS Programming Notes

- Set conditions and transfer MPAT files to be used to the ALPG before executing this program.
- There must be a pass area and a fail area between ARG (1) and ARG (2) when this program is executed.
- The setting range for each unit is shown in [Table 31-3](#). Set ARG(1) and ARG(2) within the appropriate range. Set ARG (9) so that ARG (1)+ARG (9) and ARG (2)+ARG (9) are within the range shown in [Table 31-3](#).
- Execute the SRON statement before executing "EXEC (ASM) VMEAS". (REFERENCE is not turned on in VMEAS.)
- The following statements cannot be written until "EXEC VMEAS" is specified after the MPAT file used in the SEND MPAT statement is transferred.

"CALL CALB()"

"MEAS MPAT"

"STOP MPAT"

"RESET MPAT"

Note the case given below:

```
ARG (6) =
ARG (7) = #B0 ;ALPG start value
ARG (8) =
:
REG MPAT PC=#B0 ; necessary to retransfer REG statement, etc., in #B0
SEND MPAT

VMEAS (ARG)
:
```

At this stage, the SEND MPAT x statement is required. Because, even if a pattern file name to be executed is the same as the previous name, the difference in the PC start address (module of the pattern program) requires data retransfer of MODE, REGISTER, ARIRAM, SCRAM, etc., which can be specified in each module. (Data is not retransferred in the VMEAS program.)

```
EXEC VMEAS (ARG)
```

- 10 three-word array variables must be specified as the argument of this program in advance.
- This program cannot be executed in the following cases:
When using a pattern program that uses the burst data in the main program
When using a pattern program that is designed for MEAS MPAT(5)-based fast erasure test

31. 7 Examples of Using VMEAS

◆ When the Tracked Unit is Not Used

```

PRO      1

      RATE      = 200NS                      *1
      BCLK1     = 40NS                       *1
      CCLK1     = 60NS                       *1
      STRB1     = 50NS                       *1

      IN1       =4.0V, 0.0V                  *2
      OUT1      =3.5V, 1.5V                  *2

      P33       =IN1,BCLK1,CCLK1,RZO,OUT1,STRB1,IOC,DRECLK1,<D0> *3
      DIM ARG(10)=0                          *4
      ARG(1) = 5.0V                          ;; START VOH1 (PASS) *4
      ARG(2) = 3.0V                          ;; STOP VOH1 (FAIL) *4
      ARG(3) = 1                            ;; UNIT VOH1         *4
                                          ;; ARG (4):RESULT
                                          ;; ARG (5):ERROR STATUS
                                          ;; ARG (6):SETTLING TIME
                                          ;;          DEFAULT=3MS
      ARG(7) =#0                            ;; ALPG PC START ADDRESS *4
      ARG(8) = 0                            ;; TRACKING UNIT      *4
      ARG(9) = 0                            ;; TRACKING VALUE     *4
      ARG(10)= 1                            ;; DUT NO.            *4
      SEND MPAT ALL1                        *5

      SRON                                     *6

      EXEC VMEAS (ARG)                       *7

      IF ARG(5)=0 THEN WRITE "MEASURE RESULT = " ARG (4) /      *8
      ELSE WRITE "ERROR STATUS = " ARG (5) /                    *8

END

```

An RZO waveform by BCLK1 and CCLK1 is output.

Pin 33 is I/O common and used to measure the fall time of V_{OH1} .

The measurement procedure is explained below with an example.

*1 Sets timing information.

*2 Sets voltage conditions for the driver and comparator.

*3 Sets pin conditions.

**4 Sets the argument.*

ARG (3)=1 (or ARG (3)=#00010001) because V_{OH1} is to be measured.

Set ARG (1)=5V (pass) and ARG (2)=3V (fail).

No tracing is performed.

**5 Transfers the MPAT file.*

**6 Performs REFERENCE ON.*

**7 Executes VMEAS.*

**8 Displays measurement results or an error code after measurement.*

◆ **An Example of a Program in Which a Tracing Unit is Specified is Shown Below**

```

PRO      2

      RATE      = 200NS
      BCLK1     = 40NS
      CCLK1     = 60NS
      STRB1     = 45NS

      IN1       = 4.0V, 0.0V
      OUT1      = 3.5V, 1.5V

      P33       = IN1,BCLK1,CCLK1,RZO,OUT1,STRB1,IOC,DRECLK1,<D0>
      DIM ARG(10) =0
      ARG (1) = 5.0V           ;; START VOH1 (PASS)
      ARG (2) = 3.0V           ;; STOP VOH1 (FAIL)
      ARG (3) = 1              ;; UNIT VOH1
                                ;; ARG (4):RESULT
                                ;; ARG (5):ERROR STATUS
                                ;; ARG (6):SETTLING TIME
      ARG (7) = 0              ;;          DEFAULT=3MS
      ARG (8) = 31             ;; TRACKING UNIT (VIL 1)
      ARG (9) = -1V            ;; TRACKING VALUE
      ARG(10) = 1              ;; DUT NO.
      SEND MPAT ALL1          ;;

      SRON

      EXEC VMEAS (ARG)

      IF ARG(5)=0 THEN WRITE "MEASURE RESULT = " ARG (4) /
      ELSE WRITE "ERROR STATUS = " ARG (5) /

      END

```

This example above is a program in which the unit being measured is specified as V_{OH1} and the tracing unit as V_{IL1} .

Measures relatively stable level (high level) of V_{OH1}

◆ **MPAT Pattern Program Used**

MPAT Pattern Program Used The MPAT pattern program used in examples ["When the Tracked Unit is Not Used"](#) on page 31-23 and ["An Example of a Program in Which a Tracing Unit is Specified is Shown Below"](#) on page 31-24 is shown below.

```
MPAT    ALL1
REGISTER
  PC    = #0
  TPH   = #3FFF
START
  NOP    TPH>TP
  NOP    W R TP>TP
  NOP    W R TP>TP
  NOP    W R TP>TP
  STPS
END
```

◆ For Simultaneous Measurement of 4 Devices:

```

PRO 3      S4DUT

RATE      = 200NS
BCLK1     = 40NS
CCLK1     = 60NS
STRB1     = 50NS

IN1       = 4.0V, 0.0V
OUT1      = 3.5V, 1.5V

P33       = IN1, BCLK1, CCLK1, RZO, OUT1, STRB1, IOC, DRECLK1, <D0>

DIM ARG(10)=0                                *1
DIM ARG2(1)=0                                *1
DIM VAR(4) =0                                *2
DIM ERR(4) =0                                *2

ARG(1) = 5.0V                                ;; START VOH1 (PASS)          *3
ARG(2) = 3.0V                                ;; STOP VOH1 (FAIL)             *3
ARG(3) = 1                                    ;; UNIT VOH1                  *3
                                           ;; ARG(4): (RESULT)
                                           ;; ARG(5): ERROR STATUS
                                           ;; ARG(6): SETTLING TIME
                                           ;;          DEFAULT=3MS
ARG(7) =#0                                    ;; ALPG PC START ADDRESS      *3
ARG(8) = 0                                    ;; TRACKING UNIT             *3
ARG(9) = 0                                    ;; TRACKING VALUE            *3
                                           ;; ARG(10): (DUT NO.)
ARG2(1)=#F                                    ;; DUT NO. = 1-4             *3

SEND MPAT ALL1

SRON

EXEC VMEAS(ARG, ARG2, VAL, ERR)                *4

WRITE "ARG(5)=" ARG(5) /

DO LOOP I=1, 4

WRITE "DUT NO. = " I
IF ERR(I)=0 THEN WRITE " MEASURE RESULT = " VAR(I) /
ELSE WRITE " ERROR STATUS    = " ERR(I) /

LOOP:

END

```

**1 Enough for only ARG2 (1) as 4-DUTs simultaneous measurement.*

**2 Enough for 4 array elements as 4-DUTs simultaneous measurement. Error occurs when array are provided elements less than 4.*

**3 Set ARG the same as in ["When the Tracked Unit is Not Used" on page 31-23](#).*

**4 Perform multiple simultaneous measurement when 4 arguments are correctly set to VMEAS.*

◆ **Socket Program Used**

The socket program used in the previous example (["For Simultaneous Measurement of 4 Devices:" on page 31-26](#)) is shown below.

```
SOCKET S4DUT

DEFINE MULTIPLE 4DUT 1STN

    BIT      = 1
    PPS      = 1
    PIN MODE = A
    CHTYPE 4

DUT#      = VS1 PD1-12    P33

DUT1      = VS1 PD1-12A1 P33A
DUT2      = VS2 PD1-12A2 P161A
DUT3      = VS3 PD1-12B1 P33B
DUT4      = VS4 PD1-12B2 P161B

END
```

32. *Extension Pin Monitor (/EPM)*

Displays the operating condition of the tester and the setting condition of the unit on the Video Keyboard Terminal (VKT) or the printer.

32. 1 Start of Extension Pin Monitor

32. 1. 1 Start from the Main Program

◆ Function

Call the extension pin monitor through the main program.

◆ Syntax

```
CALL LIB EPM("{Global switch}{Page name}")
```

32. 1. 2 Start by Command Input

◆ Function

Start the extension pin monitor when the controller is operating.

Actually, the extension pin monitor shall be used when the unit is being stopped in "MANUAL HOLD" mode, being looped in "MANUAL RECYCLE HOLD" mode, or being stopped on the break point of the debugger.

◆ Syntax

```
/EPM{Global switch}{Page name}↵
```

32. 1. 3 Global Switch

The following switches can be specified globally:

The global switch can be omitted.

/V

Sets the output unit on the VKT.

/L

Sets the output unit on the LPT.

/V/L

Sets the output unit on both the VKT and the LPT.

/H

Displays the tester pin after the end of socket conversion when the setting condition of the pin is displayed. (If nothing is specified, displays the DUT pin).

/G

Creates the generation file in the specified condition. Or sets the condition by conversational format.

/I

Displays each PPS setting used by VS at the time of VS setting display. On the T5581 and T5581H, this switch is effective only when the 16 V range is specified for VS1 and VS2.

/A

Displays data of all cycle palette after the end of socket conversion when the setting condition of the pin is displayed.

32. 1. 4 Page Name

The following nine page names, and several additional names can be specified.

The page name can be omitted.

PIN

Displays the setting condition of a pin.

LEVEL

Displays the input/output level.

TGt

Displays the set condition of timing generator.

DELAYADJUST Pm:

The STRB compensation values which are currently set are displayed. Can be specified on the T5586 and T5585.

ALPGu

Displays the Operating Condition of ALPG.

FM

Displays the FM operation mode.

OBMn

Displays the contents of the OUT BUFFER MEMORY.

SOFT

Displays the Software Status.

DEBUG

Displays the debugging condition of the debugger.

t means a timing set number.1 Specify within the range of 1 to 16. If t is omitted, the default value is 1. t can be specified continuously or several ts can be specified, however, if the number of timing sets exceeds the range, an error is caused.

m means a DUT pin number.1 Specify within the range of 1 to 512. If omitted, all DUT pin numbers that can be specified are specified. m can be specified continuously or respecified multiple times, however, if the DUT pin number exceeds the range, an error is caused.

n means a OBM block number.0 Specify within the range of 1 to 15. If omitted, 0 is specified for the block number. n can be specified continuously or respecified multiple times, however, if the number of blocks exceeds the range, an error is caused.

U is the ALPG UNIT number. Specify within the bounds of number of ALPG used in the system. When omitting, all ALPG number are specified. Continuous numbers and multiple numbers can be specified though an error occurs if the ALPG UNIT number exceeds the allowable area.

/EPM TG1, 2, 3

/EPM TG1-3

Both inputs above display the setting condition of timing sets 1 to 3 of TG.

/EPM DELAYADJUST P33-34,36,37

/EPM DELAYADJUST P33-34,36-37

Both inputs above display the setting condition of DUT pin number 33 to 34 and 36 to 37.

/EPM ALPG1-2

/EPM ALPG1,2

The above two entries show operation condition of ALPG UNIT1 to UNIT2.

/EPM OBM0-2,4,5

/EPM OBM0-2,4-5

Both inputs above display the contents of blocks 0 to 2 and blocks 4 and 5 of OBM.

32. 1. 5 Explanation of Commands

When a page name is specified, the extension pin monitor displays the page, however, if a page name is omitted, the display is performed according to the generation file.

The generation file is the data file where the output unit or the page to be displayed is set. If the global switch/G is specified, the file can change the contents.

32. 1. 5. 1 Display by Generation File

/EPM.J

When the above command is input, the display by the generation file is performed.

A global switch cannot be specified. If specified, it causes a command error.

If no generation files exist in the directory, the extension pin monitor defaults creating a generation file, and performance a display according to the condition.

For default condition, each command creates the following condition.

PIN	ENABLE (DUT PIN) (CYP1)
LEVEL	ENABLE (No PPS display)
TG	ENABLE (Timing set 1)
DELAYADJUST:	DISABLE
ALPG	ENABLE (All ALPG unit numbers which can be specified)
FM	ENABLE
OBM	DISABLE
SOFT	ENABLE
DEBUG	ENABLE
OUTPUT DEVICE	VKT

32. 1. 5. 2 Update Generation File

```
/EPM/G↵
```

Entering the above command displays the menu and allows you to set conditions in an interactive manner. The global switch in the command is used to sets conditions for the generation file.

Though the menu displays the contents of the generation file, if no files exist in the directory, default is displayed.

◆ Example for the T5581

```

***** GENERATION OF EXTENDED PIN MONITOR *****

1. SELECT MONITOR ITEM
PIN          ( PIN ) -----: ENABLE
  SELECT PIN  DUT=0,TESTER=1 -----: DUT
LEVEL        ( LEVEL ) -----: ENABLE
  SELECT LEVEL=0, LEVEL+PPS=1 -----: LEVEL
TIMING GENERATOR ( TG ) -----: ENABLE
  TIMING SET NO.-----: 1
ALPG         ( ALPG ) -----: ENABLE
  ALPG UNIT NO.-----: 1-2
FM MODE      ( FM ) -----: ENABLE
OUT BUFFER MEMORY ( OBM ) -----: DISABLE
  OBM BLOCK NO.-----:
SOFTWARE STATUS ( SOFT ) -----: ENABLE
DEBUG        ( DEBUG ) -----: ENABLE
2. SELECT OUTPUT DEVICE VKT=0,LPT=1,VKT & LPT=2 : VKT
3. SET DEFAULT

[WORK]          SELECT NUMBER ____

[NOTE] <CR> : MONITOR      PF1 : EXIT

```

◆ Example for the T5585

```

***** GENERATION OF EXTENDED PIN MONITOR *****

1. SELECT MONITOR ITEM
PIN          ( PIN ) -----: ENABLE
      SELECT PIN  DUT=0,TESTER=1 -----: DUT
LEVEL        ( LEVEL ) -----: ENABLE
      SELECT LEVEL=0, LEVEL+PPS=1 -----: LEVEL
TIMING GENERATER ( TG ) -----: ENABLE
      TIMING SET NO.-----: 1
STRB DELAYADJUST ( DELAYADJUST )-----: DISABLE
      PIN NO. -----:
ALPG         ( ALPG ) -----: ENABLE
      ALPG UNIT NO.-----: 1-2
FM MODE      ( FM ) -----: ENABLE
OUT BUFFER MEMORY ( OBM ) -----: DISABLE
      OBM BLOCK NO.-----:
SOFTWARE STATUS ( SOFT ) -----: ENABLE
DEBUG        ( DEBUG ) -----: ENABLE
2. SELECT OUTPUT DEVICE VKT=0,LPT=1,VKT & LPT=2 : VKT
3. SET DEFAULT

[WORK]          SELECT NUMBER ____

[NOTE] <CR> : MONITOR      PF1 : EXIT

```

The menu setting method is explained below.

If the menu is displayed, the system waits for key input. Enter one of the following:

- <CR> Creates a generation file to start a display.
- PF1 Creates a generation file to return the control to the original condition.
- 1 Selects the page to be displayed. (Selection of Output Page)
- 2 Selects the output unit.
- 3 Returns to the default.

Selection of Output Page

Since the system requests ENABLE/DISABLE in the displayed sequence by menu, input “E” for the page to be displayed, and input “D” for the page where no display is required. If a key other than “E” or “D” is input, the system skips without changing the setting condition.

- When a pin is set to ENABLE, select the pin type and cycle palette to be displayed next.
When 0 is input, DUT PIN is displayed, and when 1 is input, TESTER PIN is displayed.
Then, on the T5593, T5592, T5377, T5376, T5375 and T5371, when "0" is input, the cycle palette 1 is displayed. When "1" is input, all cycle palettes are displayed.
When a pin is set to DISABLE, the system skips unconditionally.
- When LEVEL is set to ENABLE, select the PPS display next.

When 0 is input, PPS settings are not displayed. When 1 is input, VS settings are displayed and continuously PPS settings used by VS are displayed. On the T5581 and T5581H, this setting is effective only when the 16 V range is specified for VS1 and VS2. When LEVEL is set to DISABLE, this setting is skipped unconditionally.

- When TG is set to ENABLE, input the timing set number within the range of 1 to 16 next. Continuous numbers and multiple numbers can be specified.
When TG is set to DISABLE, the system skips unconditionally.
- When DELAYADJUST is set to ENABLE, input the DUT pin number within the range of 1 to 512 next. Continuous numbers and multiple numbers can be specified.
The minimum pin number to maximum pin number are displayed.
If DELAYADJUST is set to DISABLE, the system skips unconditionally.
- When ALPG is set to ENABLE, input the number of ALPG units existing in the system for the ALPG UNIT number to be displayed next. Continuous numbers and multiple numbers can be specified.
If ALPG is set to DISABLE, the system skips unconditionally.
- When OBM is set to ENABLE, input the timing set number within the range of 0 to 15 next. Continuous numbers and multiple numbers can be specified.
When OBM is set to DISABLE, the system skips unconditionally.

```
/EPM/G page name↵
```

When page names are specified, the menu is not displayed and the generation file is created on condition that only specified page names are enabled and the specified pages are displayed simultaneously.

Displays the specified page only, without changing the contents of the generation file.

```
/EPM page name↵
```

For the command shown above, global switches other than /G can be specified.

Displays the specified page only without changing the contents of the generation file.

32. 2 Explanation of Operation

VKT

When the output unit is VKT or VKT & LPT, the screen uses 132 columns mode, and displays the page specified after the header has been output. Every time one page display has been completed, "STRIKE ANY KEY TO CONTINUE" is displayed and the system waits for a key input.

When an optioned key is input, the display of the next page is started.

After the last page has been displayed, the message END OF EXTENDED PIN MONITOR is displayed, indicating that the extension pin monitor has been completed.

LPT

When the output unit is LPT only, nothing is displayed on the screen. When the output to the printer is completed, the control is returned to the original condition.

The display is output with the header on the top of each page.

Limitation

Even if the output unit is specified by means of an S command from the controller, it will be ignored.

32. 3 Function

32. 3. 1 Header

When a program is output on the VKT, the header is displayed only once when a display is started.
When it is output to the printer, the header is printed on every page.

EXTENDED PIN MONITOR	PRO SAMPLE	AT 1998/03/30 09:18:41	
DEVICE :	SAMPLE : 351	LOT NO:	TEST:
DUT I/F:	TESTER : 0-1	OPERATOR:	LINE:
COMMENT:			

The following program names are displayed.

Program name

Date

DEVICE: Device name

SAMPLE: Sample number

LOT NO: Lot name

DUT I/F: Type of measurement

TESTER: Tester No. (Satellite station No. - Station No.)

OPERATOR: Operator name

TEST: Test No. under execution

LINE: Line No. under execution

COMMENT: Comment

32. 3. 2 PIN

Displays the setting condition of a pin. The display is the same as that of the pin of the pin monitor.

Displays the connection state for Pn.

BANK

Bank number (This is not displayed on the T5377, T5376, T5375, T5371, T5336 or T5311.)

Displayed messages and their meanings are as follows.

"A"	The same setting has been made for each bank.
"1"	Bank1
"2"	Bank2
"3"	Bank3 (2WAY or 4WAY on the T5592 or T5591)
"4"	Bank4 (2WAY or 4WAY on the T5592 or T5591)

When "!" is displayed to the left of a number, A, B, C and DRE clocks stored in banks do not work. (ACLK is not used on the T5593 and T5592.)

DC

Connection of the DC test unit.

TMDC

Specification of connecting DC measurement unit to terminal channel. (This is displayed on the T5593.)

GND

Connection of the GND relay. (This is not displayed on the T5593 and T5592.)

IN

VI unit number connected.

On the T5377, T5376, T5375, T5371, T5336 and T5311, INHn is displayed as Hn. (n indicates the INH number.)

HSC

Specification of a high-speed clock driver. (This is displayed only on the T5593.)

ACLK**BCLK****CCLK**

Clock number used. (ACLK is not displayed on the T5593 and T5592.)

PCLK

PCLK specification (This is not displayed on the T5593, T5592, T5377, T5376, T5375, T5371, T5336 or T5311.)

MODE

Waveform mode.

LEVEL

Driver initial level specified by the PIN statement.

The current level of the hardware unit driver is shown in parentheses.

(This is not displayed on the T5591.)

RINV

Specification of real-time invert mode.

DMX

Specification of data multiplex mode (This is displayed only on the T5336 and T5311.)

SDM

Specification of data swap mode (This is not displayed on the T5593 and T5592.)

PSM

Specification of pattern swap mode (This is not displayed on the T5336 and T5311.)

SMC

Specification of swap mode control (This is not displayed on the T5336 and T5311.)

RDSM

Specification of real-time data swap mode.

UTL

Connection of the UTL relay (This is not displayed on the T5593, T5592, T5377, T5376, T5375, T5371, T5336 or T5311.)

PPAT

Specification of the DBM as a pattern to shape the driver waveform (This is not displayed on the T5377, T5376, T5375, T5371, T5336 or T5311.)

DRDLY

Number of delay cycles for the driver pattern (This is not displayed on the T5336 and T5311.)

DRPA

Specification of dummy cycle driver pattern for port A (This is displayed only on the T5593.)

DRPB

Specification of dummy cycle driver pattern for port B (This is displayed only on the T5593.)

SCFIX

Specification of the same cycle fix (This is displayed only on the T5336 and T5311.)

RINHA

Specification of real-time inhibit ACLK (This is displayed only on the T5336 and T5311.)

RINHB

Specification of real-time inhibit BCLK (This is displayed only on the T5336 and T5311.)

RINHC

Specification of real-time inhibit CCLK (This is displayed only on the T5336 and T5311.)

IOC

Specification of the common I/O pin.

IOCP

Specification of the mode in which to use a comparator channel as an I/O common channel (This is displayed only on the T5593.)

DRE

DRE signal used.

Displayed messages and their meanings are as follows.

"1"	DRE1
"2"	DRE2
"C12"	C12 (This is displayed only on the T5593 and T5592.)
"C13"	C13 (This is displayed only on the T5593 and T5592.)
"C14"	C14 (This is displayed only on the T5593 and T5592.)
"C15"	C15 (This is displayed only on the T5593 and T5592.)
"P"	(This is not displayed on the T5593, T5592, T5377, T5376, T5375, T5371, T5336 and T5311.)

PDRE

Specification of PDRE (This is displayed only on the T5593 and T5592.)

Displayed messages and their meanings are as follows.

"P" PDRE

DRENrz

Specification of driver-enabled area takes place in NRZ or DNRZ mode.

Displayed messages and their meanings are as follows.

"-" DRERZ

"1" DRENrz

"D" DREDNRZ (This is displayed only on the T5593, T5592, T5377, T5376 and T5375.)

DRON

Initial state of the common I/O pin is the driver.

DREDLY

Number of delay cycles for the DRE pattern (This is not displayed on the T5336 and T5311.)

DREPA

Specification of dummy cycle DRE pattern for DREA. (This is displayed only on the T5593.)

DREPB

Specification of dummy cycle DRE pattern for DREB. (This is displayed only on the T5593.)

DRECLK

DRE clock number used.

DBMfix

Specification of the mode for disabling the INHDBM signal. (This is displayed only on the T5593.)

OUT

VO unit number connected.

TERM

Connection of the terminator using the OUT instruction.

Displayed messages and their meanings are as follows.

"1" Connects the TERM relay. (Excluding the T5593, T5592 and T5591)
Connects the comparator-side TERM relay and driver-side TERM relay. (On the T5593, T5592 and T5591)

"C" Connects the comparator side TERM relay only. (On the T5593, T5592 and T5591)

"D" Connects the driver side TERM relay only. (On the T5592, T5591)

VT

Terminator level number used

IL

Programmable load number used (This is not displayed on the T5593 and T5592.)

For ILALLn, "An" is displayed. (n indicates the ILALL number.)

On the T5592

STRB

Strobe number used.

CPE

Control pattern number used in strobe number.

HZ

Specification of Hi-Z used in strobe number.

On the T5377, T5376, T5375, T5371, T5336 and T5311

STRB

Strobe number used.

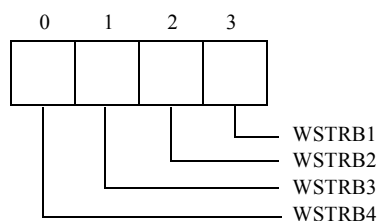
Displayed messages and their meanings are as follows.

"1"	STRB1
"2"	STRB2
"3"	STRB3
"4"	STRB4
"A"	STRB1, STRB2
"B"	STRB1, STRB4
"C"	STRB3, STRB2
"D"	STRB3, STRB4

WSTRBn is displayed as "Wn." (n indicates the WSTRB number.)

However, when MWS mode is specified, it is displayed as Mn. The n is displayed in hexadecimal notation and corresponds to each bit as shown below.

WSTRB4 is not displayed on the T5377, T5376 and T5375.



CPE

Control pattern number used in strobe number.

Displayed messages and their meanings are as follows.

"1"	CPE1
"2"	CPE2
"3"	CPE3
"4"	CPE4
"A"	CPE1, CPE2
"B"	CPE1, CPE4
"C"	CPE3, CPE2
"D"	CPE3, CPE4

HZ2

Specification of Hi-Z used in strobe number. (This is not displayed on the T5336 and T5311.)

On test systems other than the T5593, T5592, T5377, T5376, T5375, T5371, T5336 and T5311

STRB1

Odd-numbered phase strobe number used.

CPE

Control pattern number used in odd-numbered phase strobe.

HZ

Specification of Hi-Z used in odd-numbered phase strobe.

STRB2

Even-numbered phase strobe number used.

CPE

Control pattern number used in even-numbered phase strobe.

HZ

Specification of Hi-Z used in even-numbered phase strobe.

WSTRBn is displayed as "Wn." (n indicates the WSTRB number.)

On the T5593

STRB1

Odd-numbered phase strobe number used.

CPE

Control pattern number used in odd-numbered phase strobe.

HZ

Specification of Hi-Z or real time Hi-Z used in odd-numbered phase strobe.

"1"	HZ
"A"	RHZA
"B"	RHZA

STRB2

Even-numbered phase strobe number used.

CPE

Control pattern number used in even-numbered phase strobe.

HZ

Specification of Hi-Z or real time Hi-Z used in odd-numbered phase strobe.

"1"	HZ
"A"	RHZA
"B"	RHZB

MATCH

Specification of the match mode (This is not displayed on the T5593 and T5592.)

PCPE

Specification of PCPE, CCPE, ACPE (This is not displayed on the T5377, T5376, T5375, T5371, T5336 or T5311.)

Displayed messages and their meanings are as follows.

"P"	PCPE
"C"	CCPE (This is not displayed on the T5593 and T5592.)
"A"	ACPE (This is not displayed on the T5593 and T5592.)

EINV

Specification of inversion mode for the expected data (This is not displayed on the T5336 and T5311.)

ARID

Specification of the ARIRAM enable mode (This is not displayed on the T5593.)

LOAD

Connection of the LOAD relay (This is not displayed on the T5593 and T5592.)

DCLP

Specification of the dynamic clamp. (This is not displayed on the T5593.)

EXP

Specification of the expected pattern output (Port A or Port B), FIXL or FIXH (This is not displayed on the T5336 and T5311.)

EXPDLY

Number of delay cycle for the expected pattern. (This is not displayed on the T5336 and T5311.)

CPEDLY

Number of delay cycle for the CPE pattern. (This is not displayed on the T5336 and T5311.)

ESM

Specification of the swap mode of the expected data (This is not displayed on the T5593, T5592, T5377, T5376, T5375, T5371, T5336 and T5311.)

BMUX

Specification of main bank and sub-bank (This is not displayed on the T5592, T5377, T5376, T5375, T5371, T5336 and T5311.)

Displayed messages and their meanings are as follows.

"1"	BMAIN
"2"	BSUB
"3"	BMUX

MSM

Specification of the mode used for the SUBPG option in 2 WAY interleave mode. (This is displayed only on the T5593.)

DSS

Specification of data (This is not displayed on the T5377, T5376, T5375, T5371, T5336 and T5311.)

CYP

Cycle palette number (This is displayed only on the T5593, T5592, T5377, T5376, T5375 and T5371.)

Displayed messages and their meanings on the T5593 are as follows:

"An"	Cycle palette A
"Bn"	Cycle palette B

(n represents a cycle palette number)

RESET DRIVER

Execution state of the RESET DRIVER statement immediately before the MEAS/START MPAT statement is executed, or of driver output level initialization when the RESET DRIVER ENABLE/DISABLE setting program (/RSTDRV) specifies ENABLE.

/RSTDRV

The RESET DRIVER ENABLE/DISABLE setting program (/RSTDRV) setting

Display example for the T5591

B DG I A B C P M	LRSPSRUPD IDDDD	D O T V I S CH	S CH MPEALDEECE BD
A CN N C C C C O	EIDSMOTPR ORRRR	R U E T L T PZ	T PZ ACIROCXXPS MS
N D L L L L D	VNMMCSLAD CEEOE	E T R R E	R E TPNIALPPEM US
K K K K K E	EV M TL NND	C M B	B CEVDDP DD X
	L Y R L L	1 2	H LL
	Z Y K		YY
33 1 -- 1 - 1 1 - FIXL	0-----0 ----	- 1 - - - 1 1-	- -- -----B00- 31 <FL,-,-,FL>
		129 1-	- -- --- A
2 -- 1 - 2 1 - FIXL	0-----0 ----	- 1 - - - - -	2 2- -----B00- 31 <FL,-,-,FL>
		- --	130 2- --- A
3 -- 1 - 1 1 - FIXL	0-----0 ----	- 1 - - - 1 1-	- -- -----B00- 31 <FL,-,-,FL>
		129 1-	- -- --- A
4 -- 1 - 2 1 - FIXL	0-----0 ----	- 1 - - - - -	2 2- -----B00- 31 <FL,-,-,FL>
		- --	130 2- --- A
RESET DRIVER : EXECUTED			
/RSTDRV : DISABLE			

Tip

- On the T5593 and T5592, pin data information is displayed in two lines.
In this case, DRE ,DRECLK, STRB, CPE, HZ, PCPE, EINV, ARID and EXP on T5592 are displayed as follows:
First line: Displays the settings for odd-numbered DRECLK1 through DRECLK128 and STRB1 through STRB127.
Second line: Displays the settings for odd-numbered DRECLK129 through DRECLK256 and STRB129 through STRB255.
In this case, DRE, DRECLK, STRB1, CPE, HZ, STRB2, CPE, HZ, PCPE, EINV and EXP on the 5593 are displayed as follows:
1st line: displays the settings of DRECLK1 through DRECLK128 and STRB1 through STRB128.
2nd line: displays the settings of DRECLK129 through DRECLK256 and STRB129 through STRB256.
- For the I/O pin in the system where STRB129 through STRB256 are available, the pin condition is always displayed in two lines. The STRB1, CPE, HZ, STRB2, CPE, HZ, PCPE, EINV, ARID and EXP displays have the following meanings.
The first line displays the setting of STRB1-128.
The second line displays the setting of STRB129-256.

Example

```

. . . . A . . . . I S CH S CH MPEALDEECE . . . .
C . . . . L T PZ T PZ ACIROCXXPS
L R E R E TPNIALPPEM
K B B CEVDDP DD
1 2 H LL
YY

33 . . . . !1 . . . . - 1 1- 2 2- -----B--- . . . .
129 1- 130 2- --- A ← setting of STRB129-256

```

- When both *ACLK* and *STRB129-256* are set on the *T5591*, "!" is displayed to the left of the disabled clock number or strobe number. In the example above, *ACLK* is disabled.
- On the *T5586* or *T5585* used in 1 WAY, when both the odd-numbered phase strobe numbers of the *STRB129-STRB256* group and the even-numbered phase strobe numbers of the *STRB1-STRB128* group or *WSTRB1-WSTRB64* are set simultaneously, "!" is displayed to the left of each strobe number.
- When *HSCMAIN* is specified on the *T5593*, pin data information is displayed always in two lines. The first line displays *HSCMAIN*, and the second displays *HSCSUB*.

32. 3. 3 LEVEL

Displays the input/output level.

IN1-16

Driver level setting

INH1-16

Driver level setting (This is displayed only on the *T5376*.)

OUT1-16

Set value of voltage level of comparator

VT1-16

Set value of terminal voltage level

IL1-16

Set value of load current of programmable load (This is not displayed on the *T5593*, *T5592* and *T5591*.)

DCLP

Set value of the dynamic clamp (This is not displayed on the *T5593*.)

DC UNIT

Set value of DC unit

VS1-12

Set value of power supply for device (only the available VS is displayed)

LIMIT

Displays limit value of DC Test by unit

VCON

Control state of power supply for device

PCON

Control state of relay for bypass capacitor

LCON

Control state of loading relay

CW1-2

Control word

PBID

Performance board ID number of the station that is currently in action (On the T5593, T5586, T5585, T5377, T5376 and T5375, PBID1 or PBID2 is displayed.)

SETTLING TIME

Set time for DC test

RETRY

AUTO range measuring mode

DCMODE

Control mode of reference ON/OFF for DC measuring unit

WETMODE

Wet mode, DC met mode

REF ON

Referenced-on units (On the T5581 and T5581H, VS3 to VS8 are not present.)

TIME1-11

Sequence control of reference ON/OFF (On the T5581 and T5581H, VS3 to VS8 are not present.)

◆ Display Examples

On the T5592, T5591, T5581, T5581H, T5581P, T5571P, T5371, T5336 and T5311, the pin condition is displayed as shown below.

```
[LEVEL]
IN1 = 0.000V 0.000V      OUT1 = 0.000V 0.000V      VT1 = 0.000V      IL1 = 0.000A 0.000A      DCLP = 8.480V -3.488V
IN2 = 0.000V 0.000V      OUT2 = 0.000V 0.000V      VT2 = 0.000V      IL2 = 0.000A 0.000A
IN3 = 0.000V 0.000V      OUT3 = 0.000V 0.000V      VT3 = 0.000V      IL3 = 0.000A 0.000A
IN4 = 0.000V 0.000V      OUT4 = 0.000V 0.000V      VT4 = 0.000V      IL4 = 0.000A 0.000A
IN5 = 0.000V 0.000V      OUT5 = 0.000V 0.000V      VT5 = 0.000V      IL5 = 0.000A 0.000A
IN6 = 0.000V 0.000V      OUT6 = 0.000V 0.000V      VT6 = 0.000V      IL6 = 0.000A 0.000A
IN7 = 0.000V 0.000V      OUT7 = 0.000V 0.000V      VT7 = 0.000V      IL7 = 0.000A 0.000A
IN8 = 0.000V 0.000V      OUT8 = 0.000V 0.000V      VT8 = 0.000V      IL8 = 0.000A 0.000A
IN9 = 0.000V 0.000V      OUT9 = 0.000V 0.000V      VT9 = 0.000V      IL9 = 0.000A 0.000A
IN10 = 0.000V 0.000V     OUT10 = 0.000V 0.000V     VT10 = 0.000V     IL10 = 0.000A 0.000A
IN11 = 0.000V 0.000V     OUT11 = 0.000V 0.000V     VT11 = 0.000V     IL11 = 0.000A 0.000A
IN12 = 0.000V 0.000V     OUT12 = 0.000V 0.000V     VT12 = 0.000V     IL12 = 0.000A 0.000A
IN13 = 0.000V 0.000V     OUT13 = 0.000V 0.000V     VT13 = 0.000V     IL13 = 0.000A 0.000A
IN14 = 0.000V 0.000V     OUT14 = 0.000V 0.000V     VT14 = 0.000V     IL14 = 0.000A 0.000A
IN15 = 0.000V 0.000V     OUT15 = 0.000V 0.000V     VT15 = 0.000V     IL15 = 0.000A 0.000A
IN16 = 0.000V 0.000V     OUT16 = 0.000V 0.000V     VT16 = 0.000V     IL16 = 0.000A 0.000A
VSIM = 1.000V R10V M80MA 79.92MA -79.92MA T17.50US      LIMIT DC = NEGLECT NEGLECT
VS1 = 5.000V R5V M0.8A 800.0MA -800.0MA T499.2US      LIMIT VS1 = NEGLECT NEGLECT
VS2 = 1.000V R5V M0.8A 800.0MA -800.0MA T39.00US C2 FLT1 LIMIT VS2 = NEGLECT NEGLECT
VS3 = 5.000V R5V M400UA 400.0MA -400.0MA T39.00US      LIMIT VS3 = NEGLECT NEGLECT
VS4 = VSVM 5.000V R16V      LIMIT VS4 = NEGLECT NEGLECT
VS9 = MVM FLT3          LIMIT VS9 = NEGLECT NEGLECT
VS10 = 5.000V R16V M4MA 400.0MA -400.0MA T39.00US      LIMIT VS10 = NEGLECT NEGLECT
VS11 = 5.000V R16V M80MA 400.0MA -400.0MA T39.00US      LIMIT VS11 = NEGLECT NEGLECT
VS12 = 5.000V R16V M0.8A 400.0MA -400.0MA T39.00US      LIMIT VS12 = NEGLECT NEGLECT
          LIMIT JUDGE = NEGLECT NEGLECT

VCON = VS1 VS2 VS3 VS4 VS9 VS10 VS11 VS12
PCON = VS1 VS2 VS3 VS4 VS9 VS10 VS11 VS12
LCON = L1 L2 L3
CW1 = #00000000 CW2 = #00000000 PBID: STN1 = #0000
SETTLING TIME = 3.000MS 3.000MS  RETRY: DISABLE  DCMODE: DCON  WETMODE: RESET

[REF ON]      DC 1  VS 1234----9ABC IN 123456789ABCDEF0 VT 123456789ABCDEF0 IL 123456789ABCDEF0 DCLP 1
TIME1 = 3.000MS :DC - VS 1234----9ABC IN 12-----VT -----IL -----DCLP 1
TIME2 = 3.000MS :DC - VS -----IN --34-----VT -----IL -----DCLP -
TIME3 = 3.000MS :DC - VS -----IN ---56-----VT -----IL -----DCLP -
TIME4 = 3.000MS :DC - VS -----IN ----78-----VT -----IL -----DCLP -
TIME5 = 3.000MS :DC - VS -----IN -----9ABC-----VT -----IL -----DCLP -
TIME6 = 3.000MS :DC - VS -----IN -----DEF0-----VT -----IL -----DCLP -
TIME7 = 3.000MS :DC - VS -----IN -----12345678-----VT -----IL -----DCLP -
TIME8 = 3.000MS :DC - VS -----IN -----9ABCDEF0-----VT -----IL -----DCLP -
TIME9 = 3.000MS :DC - VS -----IN -----IL 12345678-----DCLP -
TIME10 = 3.000MS :DC - VS -----IN -----IL -----9ABCDEF0-----DCLP -
TIME11 = 3.000MS :DC 1 VS -----IN -----VT -----IL -----DCLP -
```

On T5376, displayed as shown below.

```
[LEVEL]
IN1 = 0.000V 0.000V    INH1 = 0.000V 0.000V    OUT1 = 0.000V 0.000V    VT1 = 0.000V    IL1 = 0.000A 0.000A
IN2 = 0.000V 0.000V    INH2 = 0.000V 0.000V    OUT2 = 0.000V 0.000V    VT2 = 0.000V    IL2 = 0.000A 0.000A
IN3 = 0.000V 0.000V    INH3 = 0.000V 0.000V    OUT3 = 0.000V 0.000V    VT3 = 0.000V    IL3 = 0.000A 0.000A
IN4 = 0.000V 0.000V    INH4 = 0.000V 0.000V    OUT4 = 0.000V 0.000V    VT4 = 0.000V    IL4 = 0.000A 0.000A
IN5 = 0.000V 0.000V    INH5 = 0.000V 0.000V    OUT5 = 0.000V 0.000V    VT5 = 0.000V    IL5 = 0.000A 0.000A
IN6 = 0.000V 0.000V    INH6 = 0.000V 0.000V    OUT6 = 0.000V 0.000V    VT6 = 0.000V    IL6 = 0.000A 0.000A
IN7 = 0.000V 0.000V    INH7 = 0.000V 0.000V    OUT7 = 0.000V 0.000V    VT7 = 0.000V    IL7 = 0.000A 0.000A
IN8 = 0.000V 0.000V    INH8 = 0.000V 0.000V    OUT8 = 0.000V 0.000V    VT8 = 0.000V    IL8 = 0.000A 0.000A
IN9 = 0.000V 0.000V    INH9 = 0.000V 0.000V    OUT9 = 0.000V 0.000V    VT9 = 0.000V    IL9 = 0.000A 0.000A
IN10 = 0.000V 0.000V   INH10 = 0.000V 0.000V    OUT10 = 0.000V 0.000V   VT10 = 0.000V    IL10 = 0.000A 0.000A
IN11 = 0.000V 0.000V   INH11 = 0.000V 0.000V    OUT11 = 0.000V 0.000V   VT11 = 0.000V    IL11 = 0.000A 0.000A
IN12 = 0.000V 0.000V   INH12 = 0.000V 0.000V    OUT12 = 0.000V 0.000V   VT12 = 0.000V    IL12 = 0.000A 0.000A
IN13 = 0.000V 0.000V   INH13 = 0.000V 0.000V    OUT13 = 0.000V 0.000V   VT13 = 0.000V    IL13 = 0.000A 0.000A
IN14 = 0.000V 0.000V   INH14 = 0.000V 0.000V    OUT14 = 0.000V 0.000V   VT14 = 0.000V    IL14 = 0.000A 0.000A
IN15 = 0.000V 0.000V   INH15 = 0.000V 0.000V    OUT15 = 0.000V 0.000V   VT15 = 0.000V    IL15 = 0.000A 0.000A
IN16 = 0.000V 0.000V   INH16 = 0.000V 0.000V    OUT16 = 0.000V 0.000V   VT16 = 0.000V    IL16 = 0.000A 0.000A
DCLP = 6.000V -2.000V
VSIM = 1.000V R20V M128MA 128.0MA -128.0MA T39.40US
VS1 = 1.000V R5V M0.8A 800.0MA -800.0MA T39.00US
VS2 = 2.000V R16V M0.8A 400.0MA -400.0MA T39.00US
VS3 = 3.000V R5V M0.8A 800.0MA -800.0MA T39.00US
VS4 = 4.000V R5V M0.8A 800.0MA -800.0MA T39.00US
VS9 = MVM M16V FLT3
VS10 = 10.00V R16V M0.8A 400.0MA -400.0MA T39.00US
VS11 = 11.00V R16V M0.8A 400.0MA -400.0MA T39.00US
VS12 = 12.00V R16V M0.8A 400.0MA -400.0MA T39.00US
LIMIT DC = NEGLECT NEGLECT
LIMIT VS1 = NEGLECT NEGLECT
LIMIT VS2 = NEGLECT NEGLECT
LIMIT VS3 = NEGLECT NEGLECT
LIMIT VS4 = NEGLECT NEGLECT
LIMIT VS9 = NEGLECT NEGLECT
LIMIT VS10 = NEGLECT NEGLECT
LIMIT VS11 = NEGLECT NEGLECT
LIMIT VS12 = NEGLECT NEGLECT
LIMIT JUDGE = NEGLECT NEGLECT
VCON = VS1 VS2 VS3 VS4 VS9 VS10 VS11 VS12
PCON = VS1 VS2 VS3 VS4 VS9 VS10 VS11 VS12
LCON = L1 L2 L3
CW1 = #00000000 CW2 = #00000000 PBID1: STN1 = #00000000 STN2 = #00000000
PBID2: STN1 = #00000000 STN2 = #00000000
SETTLING TIME = 3.000MS 3.000MS RETRY: DISABLE DCMODE: DCON WETMODE: RESET

[REF ON]          DC 1  VS 1234----9ABC IN 123456789ABCDEF0 VT 123456789ABCDEF0 IL 123456789ABCDEF0 DCLP 1
TIME1 = 3.000MS :DC - VS 1234----9ABC IN 12----- VT ----- IL ----- DCLP 1
TIME2 = 3.000MS :DC - VS ----- IN --34----- VT ----- IL ----- DCLP -
TIME3 = 3.000MS :DC - VS ----- IN ---56----- VT ----- IL ----- DCLP -
TIME4 = 3.000MS :DC - VS ----- IN -----78----- VT ----- IL ----- DCLP -
TIME5 = 3.000MS :DC - VS ----- IN -----9ABC----- VT ----- IL ----- DCLP -
TIME6 = 3.000MS :DC - VS ----- IN -----DEF0----- VT ----- IL ----- DCLP -
TIME7 = 3.000MS :DC - VS ----- IN ----- VT 12345678----- IL ----- DCLP -
TIME8 = 3.000MS :DC - VS ----- IN ----- VT -----9ABCDEF0----- IL ----- DCLP -
TIME9 = 3.000MS :DC - VS ----- IN ----- VT ----- IL 12345678----- DCLP -
TIME10 = 3.000MS :DC - VS ----- IN ----- VT ----- IL -----9ABCDEF0----- DCLP -
TIME11 = 3.000MS :DC 1 VS ----- IN ----- VT ----- IL ----- DCLP -
```

32. 3. 4 TG

Displays the set condition of timing generator.

DROFFTIME

Displays the minimum ON/OFF time interval for the driver (DRE).

SELECT RATE

Displays the mode specified by the SELECT RATE statement

This is displayed on the T5591, T5586, T5585, T5581, T5581H, T5581P and T5571P.

SELECT OFFSET

Displays clock name specified by SELECT OFFSET statement

OFFSET1

Displays set data for OFFSET 1

OFFSET2

Displays set data for OFFSET 2

PCLK 1 to 4

On the system with PCLK, these are displayed following OFFSET.

The following names are displayed for every timing set No. specified.

TIMING SET

Displays timing set No. specified

RATE

Displays rate set value

ACLK1-128**BCLK1-128****CCLK1-128**

Displays the set value of timing pulse edge of input clock

DREL1-256**DRET1-256**

Displays driver enable clock

STRB1-256**WSTRB1-128**

Displays comparison timing

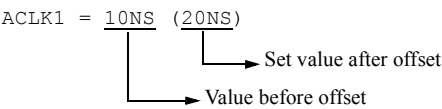
◆ **Display Examples**

```
[TIMING GENERATER]
DROFFTIME: NORMAL   SELECT RATE: NORMAL
SELECT OFFSET: NONE
OFFSET1 = 0.000S OFFSET2 = 0.000S

TIMING SET 1
RATE      = 1.000MS
ACLK1     = 100.0US
ACLK2     = 110.0US
ACLK3     = 120.0US
ACLK4     = 130.0US
BCLK1     = 200.0US
BCLK2     = 210.0US
BCLK3     = 220.0US
BCLK4     = 250.0US
CCLK1     = 500.0US
CCLK2     = 510.0US
CCLK3     = 520.0US
CCLK4     = 530.0US
CCLK5     = 540.0US
CCLK6     = 550.0US
CCLK7     = 560.0US
CCLK8     = 570.0US
DREL1     = 10.00US
DREL2     = 11.00US
DRET1     = 900.0US
DRET2     = 910.0US
STRB1     = 400.0US
STRB2     = 410.0US
STRB3     = 420.0US
STRB4     = 430.0US
```

Tip

- On the T5593, T5592, T5591, T5586, T5585, T5581, T5581H, T5581P and T5571P, the TG numbers are displayed up to the maximum number being used in each.
On the T5336 and T5311, all of the TG numbers that can be used in each are displayed.
All the ACLK, BCLK, CCLK, DREL and DRET numbers that are currently being used are displayed when used on the T5377, T5376, T5375 and T5371. All available STRB and WSTRB numbers are displayed.
 - The clock to which off-set is added displays two values: values set by the program and, in parentheses, those set by the hardware.
- Example



32. 3. 5 DELAYADJUST

The STRB compensation values which are currently set are displayed.

DUTn

All DUT numbers specified by the socket program.

PIN

DUT pin number specified by the menu.

◆ Display Examples

[STRB DELAYADJUST DATA]								
	DUT1	DUT2	DUT3	DUT4	DUT5	DUT6	DUT7	DUT8
	DUT9	DUT10	DUT11	DUT12	DUT13	DUT14	DUT15	DUT16
	DUT17	DUT18	DUT19	DUT20	DUT21	DUT22	DUT23	DUT24
[PIN]	DUT25	DUT26	DUT27	DUT28	DUT29	DUT30	DUT31	DUT32
P33	000.0PS	000.0PS	000.0PS	000.0PS	000.0PS	000.0PS	000.0PS	000.0PS
	15.63PS	15.63PS	15.63PS	15.63PS	15.63PS	15.63PS	15.63PS	15.63PS
	62.50PS	62.50PS	62.50PS	62.50PS	62.50PS	62.50PS	62.50PS	62.50PS
	125.0PS	125.0PS	125.0PS	125.0PS	125.0PS	125.0PS	125.0PS	125.0PS
P34	000.0PS	000.0PS	000.0PS	000.0PS	000.0PS	000.0PS	000.0PS	000.0PS
	15.63PS	15.63PS	15.63PS	15.63PS	15.63PS	15.63PS	15.63PS	15.63PS
	62.50PS	62.50PS	62.50PS	62.50PS	62.50PS	62.50PS	62.50PS	62.50PS
	125.0PS	125.0PS	125.0PS	125.0PS	125.0PS	125.0PS	125.0PS	125.0PS

 Tip

- *STRB compensation values are displayed only on the T5586 and T5585.*
 - *Only DUT1 is displayed when no socket program is specified.*
 - *When the socket program is specified, the default pin numbers are all the comparator pin numbers that are written on the line indicating DUT#. When no socket program is specified, all the comparator pin numbers are displayed.*
-

32. 3. 6 ALPG

Displays the Operating Condition of ALPG.

Test systems other than T5571P, T5377, T5376, T5375, T5371, T5336 and T5311

[ALPG UNIT #1]

Displays the operation condition of ALPG UNIT1.

[REGISTER UNIT #1]

Displays the contents of each register on the ALPG UNIT1.

[ALPG UNIT #2]

Displays the operating status of ALPG UNIT2. Displayed contents is the same as ALPG UNIT1.

[REGISTER UNIT #2]

Displays the contents of each register on the ALPG UNIT2.

When the ALPG unit 3 and 4 exist, their operating status is displayed as well.

On the T5571P, T5377, T5376, T5375, T5371, T5336 and T5311

[ALPG UNIT #1]

Displays the operation condition of ALPG UNIT.

[REGISTER UNIT #1]

Displays the contents of each register on the ALPG UNIT.

For [ALPG UNIT #1]

ILMODE

Displays interleaved mode.

"1WAY" is displayed on the T5377, T5376, T5375, T5371, T5336 and T5311.

MPA

Displays pattern file names transferred to ALPG.

PCC

Displays the PCC setting mode.

(This is not displayed on the T5593, T5377, T5376, T5375, T5371, T5336 and T5311.)

SCR

Displays scram file names used in the ALPG.

When the T5377 is used and SCRAM MODE is set to SEPARATE, scram file names for both DUT and FM are displayed in order.

PDS DATA

Displays the test data to be output to PDS.

PRESCR

Displays scramble file name transferred to the prescrambler.

(This is not displayed on the T5336 and T5311.)

PATTERN MODE

Displays the ALPG operation mode specified in the pattern program.

ALPG MODE

Displays the operation mode of ALPG.

SCRAM

Displays ENABLE or DISABLE of the address descrambler.

SCRAM MODE

Displays DIVIDE or SEPARATE mode of the address descrambler.

(This is not displayed on the T5376, T5375, T5371, T5336 and T5311.)

ARIRAM

Displays the operation mode of the ARIRAM.

PRESCRAM

Displays ENABLE/DISABLE of the prescrambler.

(This is not displayed on the T5336 and T5311.)

PRESCRAM MODE

Displays the operation mode of the prescrambler.

(This is not displayed on the T5336 and T5311.)

PRESCRAM ADDRESS

Displays the control mode used to determine whether the address output from the prescrambler or ALPG should be sent to DUT, FM FP function or the address used for auto page mode testing.

(This is not displayed on the T5336 and T5311.)

WRAP ADDRESS

Displays valid bit of address output from the prescrambler.

(This is not displayed on the T5336 and T5311.)

LEVELHOLD

Displays the initialized mode of the driver output level.

HZSENSE

Displays the HZ mode. (This is not displayed on the T5593, T5377, T5376 and T5375.)

SDMODE

Displays the multi strobing mode.

PATDELAY

Displays expected pattern control data, X address, Y address, and the number of TP data delay cycles. (This is not displayed on the T5371.)

On the T5336 or T5311, only expected pattern control data is displayed.

TP data is displayed only on the T5593, T5377, T5376, and T5375.

WINDOW MODE

Displays the window mode. (This is not displayed on the T5593.)

WINDOW XADRS

Displays the window Xaddress. (This is not displayed on the T5593.)

WINDOW YADRS

Displays the window Yaddress. (This is not displayed on the T5593.)

◆ Display Examples

On the T5592, T5591, T5586, T5585, T5581, T5581H, T5581P, and T5571P, the pin condition is displayed as shown below.

[ALPG UNIT #1]		ILMODE 1WAY			
MPA	UEPMM050	PATTERN MODE	:	JETM TFM18 DFCA1 DFCB1	
PCC	: DISABLE	ALPG MODE	:	NORMAL	LEVELHOLD: DISABLE
SCR		SCRAM	:	DISABLE	HZSENSE : DISABLE
		SCRAM MODE	:	NORMAL	SDMODE : DISABLE
		ARIRAM	:	DISABLE	PATDELAY : X=0 Y=0
		PRESRAM	:	DISABLE	
		PRESRAM MODE	:	NORMAL	
		PRESRAM ADDRESS:	:	DUT	
PDS DATA: D/D		WRAP ADDRESS	:	X0-15 Y0-15	
[REGISTER UNIT #1]					
PC	= #09944007	CFLG	= #00000000	XMAX	= #00000000
ISP	= #4F7F7FF2	OUTBLK	= #00000000	YMAX	= #00000000
BAR	= #00E46015	CPE1	= FIXL	ZMAX	= #00000000
IDX1	= #00000000	CPE2	= FIXL	LMAX	= #00000000
IDX2	= #00000000	CPE3	= FIXL	HMAX	= #00000000
IDX3	= #00000000	CPE4	= FIXL	XMASK	= #00000000
IDX4	= #00000000	DRE1	= FIXL	YMASK	= #00000000
IDX5	= #00000000	DRE2	= FIXL	RLMAX	= #0000003F
IDX6	= #00000000	RINV	= FIXL		
IDX7	= #00000000	REFRES	= #00000000		
IDX8	= #00000000	PAUSE	= #00000000		
[ALPG UNIT #2]					
MPA	UEPMM050	PATTERN MODE	:	TFM18 DFCA1 DFCB1	
PCC	: DISABLE	ALPG MODE	:	NORMAL	LEVELHOLD: DISABLE
SCR		SCRAM	:	DISABLE	HZSENSE : DISABLE
		SCRAM MODE	:	NORMAL	SDMODE : DISABLE
		ARIRAM	:	DISABLE	PATDELAY : X=0 Y=0
		PRESRAM	:	DISABLE	
		PRESRAM MODE	:	NORMAL	
		PRESRAM ADDRESS:	:	DUT	
PDS DATA: D/D		WRAP ADDRESS	:	X0-15 Y0-15	
[REGISTER UNIT #2]					
PC	= #09944007	CFLG	= #00000000	XMAX	= #00000000
ISP	= #4F7F7FF2	OUTBLK	= #00000000	YMAX	= #00000000
BAR	= #00E46015	CPE1	= FIXL	ZMAX	= #00000000
IDX1	= #00000000	CPE2	= FIXL	LMAX	= #00000000
IDX2	= #00000000	CPE3	= FIXL	HMAX	= #00000000
IDX3	= #00000000	CPE4	= FIXL	XMASK	= #00000000
IDX4	= #00000000	DRE1	= FIXL	YMASK	= #00000000
IDX5	= #00000000	DRE2	= FIXL	RLMAX	= #0000003F
IDX6	= #00000000	RINV	= FIXL		
IDX7	= #00000000	REFRES	= #00000000		
IDX8	= #00000000	PAUSE	= #00000000		

DBMLPB and DBMLPE are displayed only on the T5593 and T5592.



In PATTERN MODE, the modes are displayed in mnemonic form of the pattern program. JETM is displayed as the timer operation mode aside from PAUSE, REFRESHA or REFRESHB. The JETM mode means that the timer is in ENABLE (normal) state.

If AUTOSCAN COUNTER is enabled by using the AUTOSCAN utility, the ASCNT register value is stored in the RHZA and RHZB registers every time the MEAS MPAT or START MPAT statement is executed.

The ASCNT register value cannot be stored by MEAS MPAT(1) or START MPAT(1).

32. 3. 7 FM

Displays the FM operation mode.

Display of operating condition of AFM

MULTIMODE

Displays the AFM mode. After this display, data following DUT: shows the DUT assigned to the current AFM. The format is the same as that of the data read by the READ DDUT statement.

For the T5336 and T5311, this is always displayed as 0.

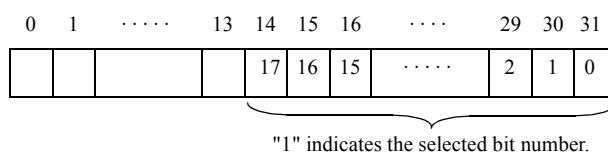
BITSIZE

Displays the number of bits of DUT and the number of bits loaded to AFM.

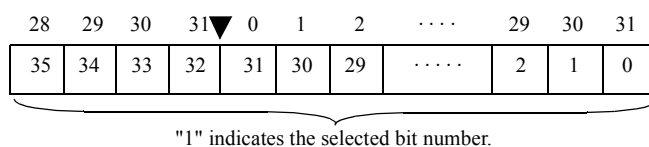
BITSEL

Displays which pin is selected to the bit number of AFM side.

On test systems other than the T5377, T5376, T5375, T5371, T5336 and T5311, pin data from 0 to 17 is displayed in bits as shown below using hexadecimal notation.



On the T5377, T5376, T5375 and T5371, pin data 0 to 35 is displayed in bits as shown below using hexadecimal notation.



For the T5336 and T5311, this is always displayed as NONE.

DUTSEL

Displays the number of notable DUT.

RATE

Displays the DUT test cycle.

PGSEL

Displays the ALPG connecting to the fail memory.

MAIN ADDRESS

Assignment of the address signal of MAIN.

(This is displayed for test systems other than the T5377, T5376, T5375, T5371, T5336 and T5311.)

Displayed here for the T5591 in the two-speed storing mode.

SUB ADDRESS

Assignment of the address signal of SUB.

(This is displayed for test systems other than the T5377, T5376, T5375, T5371, T5336 and T5311.)

ADDRESS

Displays an assigned address signal.

(This is displayed only for the T5377, T5376, T5375, T5371, T5336 and T5311.)

EXP ADDRESS

Displays the address of the EXP mode.

ACTION

Displays the AFM operation mode.

DBMBASE

Displays the DBMBASE. (This is displayed only on the T5377, T5376, T5375, T5371, T5336 and T5311.)

On the T5336 and T5311, however, this is displayed only when compatibility version B of AFM CONT (BGR-023301) has been installed on these systems.

➔ For more information on compatibility versions, refer to the SET DBMBASE statement in [ATL-51 Test Plan Program Reference Manual\(No. : 8431498 \)](#).

Display of operating condition of DFM

MAIN MODE

Displays the DFM mode of MAIN. (This is displayed for test systems other than the T5377, T5376, T5375, T5371, T5336 and T5311.)

SUB MODE

Displays the DFM mode of SUB. (This is displayed for test systems other than the T5377, T5376, T5375, T5371, T5336 and T5311.)

MODE

Displays the DFM mode. (This is displayed only on the T5393, T5377, T5376, T5375, T5371, T5336 and T5311.)

◆ Display Examples

(This applies to the T5592, T5591, T5586, T5585, T5581, T5581H, T5581P and T5571P.)

[AFM]

MULTIMODE : SINGLE DUT: #00000001

BITSIZE : 18 18

BITSEL : 0 1 2 3 4 5 6 7 8

: #00001 #00002 #00004 #00008 #00010 #00020 #00040 #00080 #00100

: 9 10 11 12 13 14 15 16 17

: #00200 #00400 #00800 #01000 #02000 #04000 #08000 #10000 #20000

DUTSEL : 1

RATE : HS

PGSEL : MAIN, SUB(RWSEL : MAIN)

MAIN ADDRESS: X0-9 Y0-11 DELAY0

SUB ADDRESS: X0-9 Y0-11 DELAY0

EXP ADDRESS: X0-9 Y0-11

ACTION : FT>FMX

Pin data

[DFM]

MAIN MODE : ALL UP TO FCNT(#00000000)

SUB MODE : ALL UP TO FCNT(#00000000)

For the T5377, T5376 and T5375, the pin condition is displayed as shown below.

[AFM]

MULTIMODE : SINGLE DUT: #00000001

BITSIZE : 36 36

BITSEL : 0 1 2 3 4 5 6 7 8

: #000000001 #000000002 #000000004 #000000008 #000000010 #000000020 #000000040 #000000080 #000000100

: 9 10 11 12 13 14 15 16 17

: #000000200 #000000400 #000000800 #000001000 #000002000 #000004000 #FFFFFF8000 DISABLE DISABLE

: 18 19 20 21 22 23 24 25 26

: DISABLE DISABLE DISABLE DISABLE DISABLE DISABLE DISABLE DISABLE DISABLE

: 27 28 29 30 31 32 33 34 35

: DISABLE DISABLE DISABLE DISABLE DISABLE DISABLE DISABLE DISABLE DISABLE

DUTSEL : 1

RATE : HS

PGSEL : MAIN

ADDRESS : X0-9 Y0-11 DELAY0 SCRAM

EXP ADDRESS: X0-9 Y0-11

ACTION : FM>EXP

DEMBASE : BLOCK NO.(1) BLOCK SIZE(16M)

[DFM]

MODE : ALL UP TO FCNT(#00000000)

Only the block number is displayed for DBMBASE when using test systems other than the T5377, T5376 and T5375.

32. 3. 8 OBM

Displays the contents of the block where the OUT BUFFER MEMORY is specified.

There are 0 to 15 blocks in the OBM. For the blocks not set, nothing is displayed. The data to be displayed is VS and WAIT TIME.

◆ Display Examples

```
[OBM ]
BURST BLOCK 0
    VS1 = 10.00V R(10.00V)
    WAIT TIME = 10.00US

BURST BLOCK 1
    VS2 = 2.000V R(10.00V)
    WAIT TIME = 20.00US
```

32. 3. 9 SOFT

Displays the Software Status.

ON JUMP

Display the event name used by the ON jump instruction.

"*" is added to the top of effective event names.

OVERRIDE

Displays the name of the event where On Jump is inhibited.

FKMODE

Displays the set condition of the function key mode.

ASMODE

Displays the set condition of the action selector mode.

FK

Displays the set condition of the function key.

ASEL

Displays the set condition of the function selector switch.

MCT

Displays the control status of DUT. The state of MCT on the performance board is shown.

PPS

Displays the PPS mask status. The relay used for the displayed PPS is open.

MULTIPLE DUT

Displays the simultaneous measuring mode.

DDUT

Displays 1 for bits corresponding to devices targeted for parallel tests (as defined in the socket program) and 0 for the other bits.

SDUT

Displays 1 for bits corresponding to DUTs set for the handler or prober, and 0 for the other bits. However, jammed devices under test or devices which are not ON WAFER are excluded.

MDUT

Displays 1 for bits corresponding to fail-stopped DUTs or devices that are excluded from testing by the SET REJECTION instruction, and 0 for the other bits.

EDUT

Displays 1 for bits corresponding to fail-stopped DUTs, or devices that are excluded from testing by the SET REJECTION or SET EXCLUSION instruction, and 0 for the other bits.

CDUT

Normally, the data displayed is the same as the data displayed for EDUT. For the interval, however, from the MULTIPLE DUT PARALLEL, SINGLE, DOUBLE or VS PARALLEL instruction to the MULTIPLE DUT END instructions, 1 is displayed for bits that correspond to target DUTs sequentially in order of DUT numbers, and 0 is displayed for the other bits.

PDUT

Displays 1 for bits corresponding to devices that have passed the last test and 0 for the other bits.

FDUT

Displays 1 for bits corresponding to devices that have failed the last test and 0 for the other bits.

GFDUT

Displays 1 for bits corresponding to devices that have failed in the tests executed so far and 0 for the other bits.

◆ Display Example 1

The following screen is an example of the complete SOFT display.

```
[SOFTWARE STATUS ]
ON  JUMP:  ERR FCT DCT OVERFLOW KEY
OVERRIDE:  ERR STOP TEST FCT DCT

FKMODE: 4 6 12 16 22 24
ASMODE: 9 10 11 12 13 14

STATION 1
    FK      : 1  2  3
    ASEL    : 2  4  6  8 10 12 14 16
    MCT     : 1
    PPS     : 2  3  4  5  6  7  8  9 10

MULTIPLE DUT:
      ! 00000000 ! 00111111 ! 11112222 ! 22222233 !
      ! 01234567 ! 89012345 ! 67890123 ! 45678901 !
      ! ----- ! ----- ! ----- ! ----- !
DDUT ! 00000000 ! 00000000 ! 00000000 ! 00000001 !
SDUT ! 00000000 ! 00000000 ! 00000000 ! 00000001 !
MDUT ! 00000000 ! 00000000 ! 00000000 ! 00000001 !
EDUT ! 00000000 ! 00000000 ! 00000000 ! 00000001 !
CDUT ! 00000000 ! 00000000 ! 00000000 ! 00000001 !
PDUT ! 00000000 ! 00000000 ! 00000000 ! 00000001 !
FDUT ! 00000000 ! 00000000 ! 00000000 ! 00000000 !
GFDUT ! 00000000 ! 00000000 ! 00000000 ! 00000000 !
```

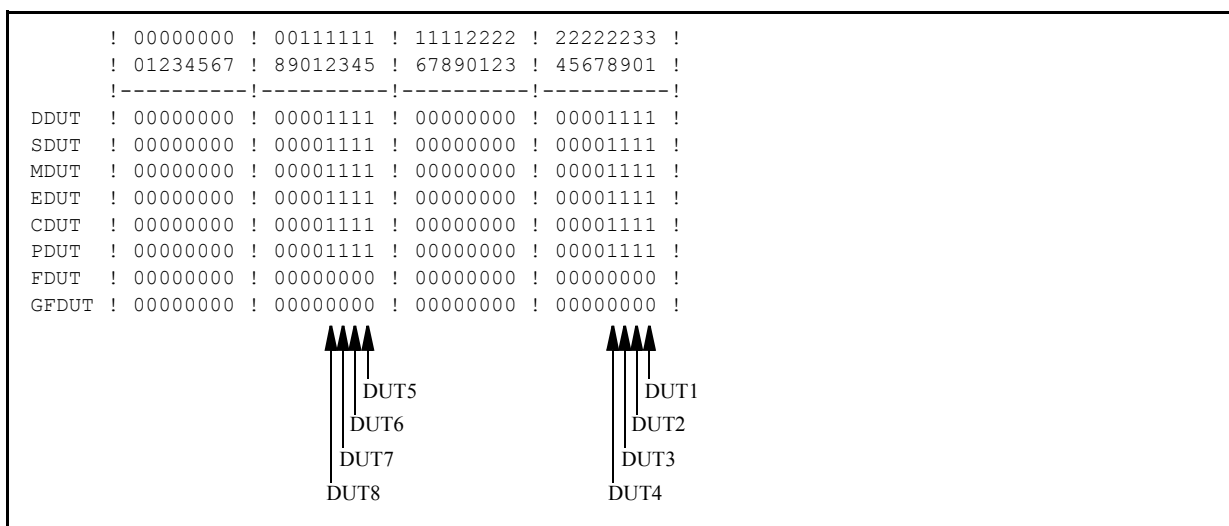
◆ Display Example 2

This section provides several examples of the DDUT, SDUT, MSUT, EDUT, CDUT, PDUT, FDUT and GFDUT displays for each parallel test mode.

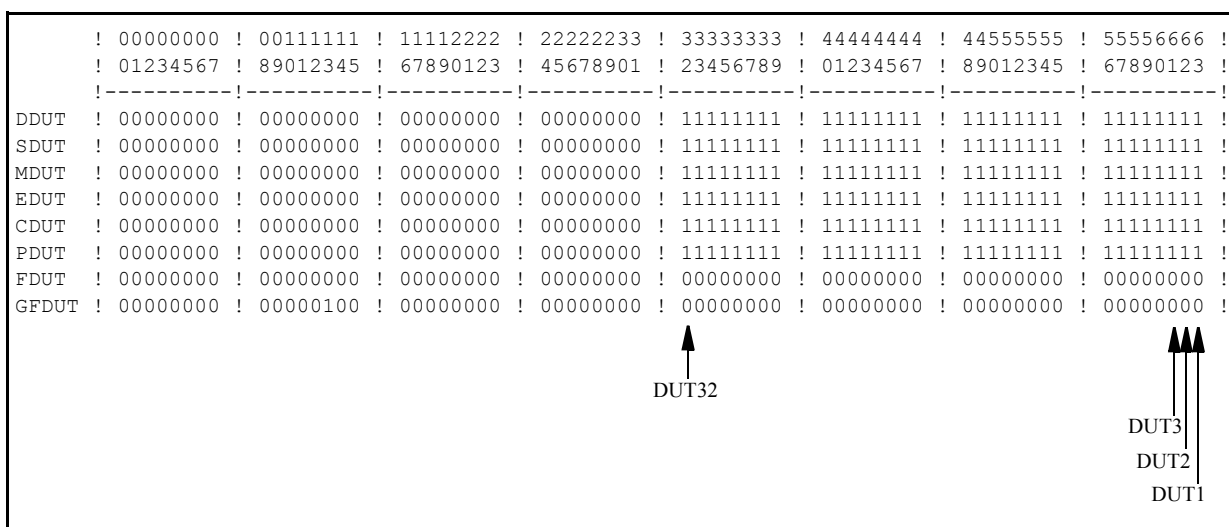
- For 1-device parallel tests using one test station

```
      ! 00000000 ! 00111111 ! 11112222 ! 22222233 !
      ! 01234567 ! 89012345 ! 67890123 ! 45678901 !
      ! ----- ! ----- ! ----- ! ----- !
DDUT ! 00000000 ! 00000000 ! 00000000 ! 00000001 !
SDUT ! 00000000 ! 00000000 ! 00000000 ! 00000001 !
MDUT ! 00000000 ! 00000000 ! 00000000 ! 00000001 !
EDUT ! 00000000 ! 00000000 ! 00000000 ! 00000001 !
CDUT ! 00000000 ! 00000000 ! 00000000 ! 00000001 !
PDUT ! 00000000 ! 00000000 ! 00000000 ! 00000001 !
FDUT ! 00000000 ! 00000000 ! 00000000 ! 00000000 !
GFDUT ! 00000000 ! 00000000 ! 00000000 ! 00000000 !
```

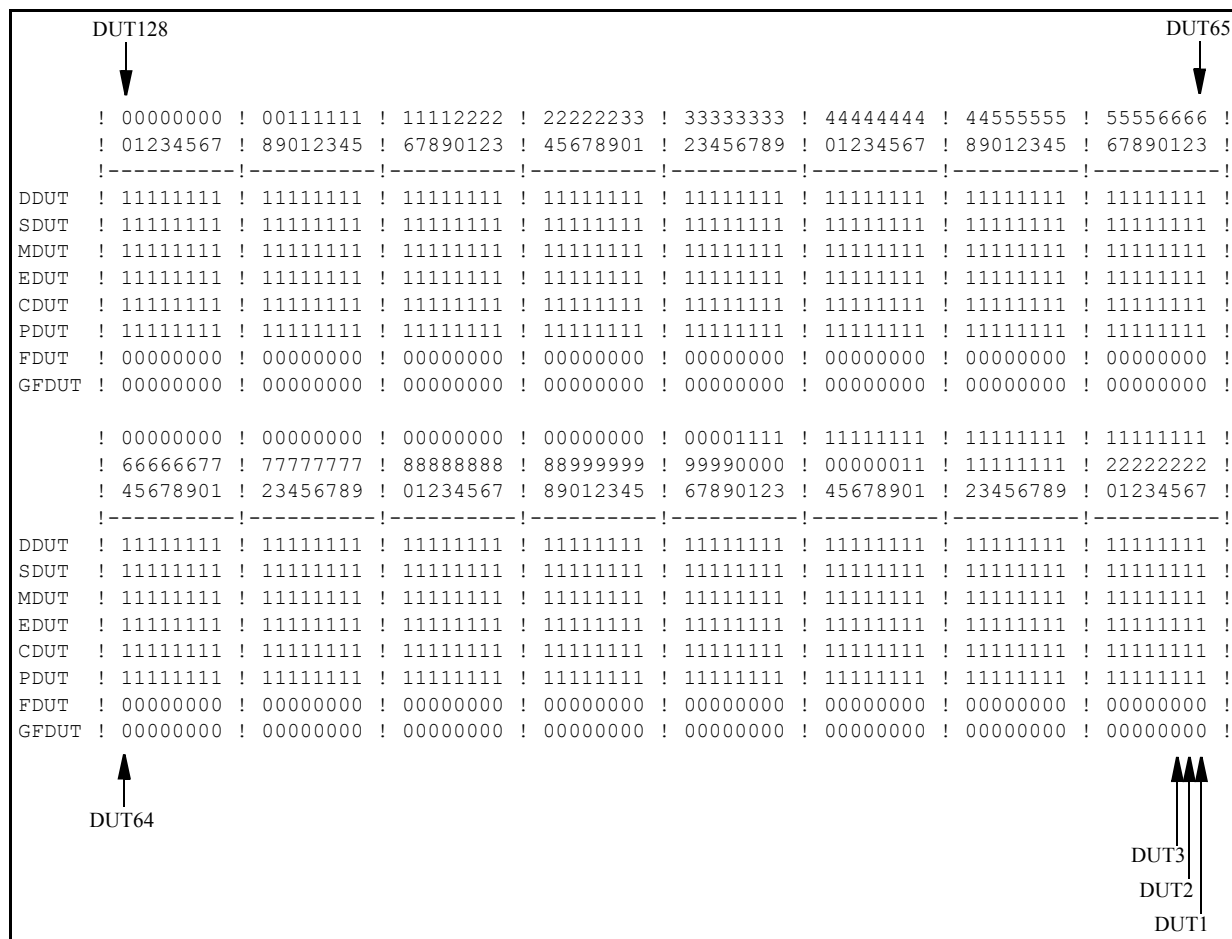

- For 8-device parallel tests using two test station



- For 32-device parallel tests using one test station



- For 128-device parallel tests using two test station



- For 256-device parallel tests using two test station

	DUT256									DUT193								
																		
	!	00000000	!	00111111	!	11112222	!	22222233	!	33333333	!	44444444	!	44555555	!	55556666	!	
	!	01234567	!	89012345	!	67890123	!	45678901	!	23456789	!	01234567	!	89012345	!	67890123	!	
	!	-----		!	-----		!	-----		!	-----		!	-----		!	-----	
DDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
SDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
MDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
EDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
CDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
PDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
FDUT	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	
GFDUT	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	
	!	00000000	!	00000000	!	00000000	!	00000000	!	00001111	!	11111111	!	11111111	!	11111111	!	
	!	66666677	!	77777777	!	88888888	!	88999999	!	99990000	!	00000011	!	11111111	!	22222222	!	
	!	45678901	!	23456789	!	01234567	!	89012345	!	67890123	!	45678901	!	23456789	!	01234567	!	
	!	-----		!	-----		!	-----		!	-----		!	-----		!	-----	
DDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
SDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
MDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
EDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
CDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
PDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
FDUT	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	
GFDUT	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	
	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
	!	22333333	!	33334444	!	44444455	!	55555555	!	66666666	!	66777777	!	77778888	!	88888899	!	
	!	89012345	!	67890123	!	45678901	!	23456789	!	01234567	!	89012345	!	67890123	!	45678901	!	
	!	-----		!	-----		!	-----		!	-----		!	-----		!	-----	
DDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
SDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
MDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
EDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
CDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
PDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
FDUT	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	
GFDUT	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	
	!	11111111	!	22222222	!	22222222	!	22222222	!	22222222	!	22222222	!	22222222	!	22222222	!	
	!	99999999	!	00000000	!	00111111	!	11112222	!	22222233	!	33333333	!	44444444	!	44555555	!	
	!	23456789	!	01234567	!	89012345	!	67890123	!	45678901	!	23456789	!	01234567	!	89012345	!	
	!	-----		!	-----		!	-----		!	-----		!	-----		!	-----	
DDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
SDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
MDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
EDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
CDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
PDUT	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	11111111	!	
FDUT	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	
GFDUT	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	00000000	!	
																		
	DUT64																	DUT3
																		DUT2
																		DUT1

BREAK

When a break point is set, this command displays the call point Nos, program names, line Nos, customer names, and source file names.

INSERT

When a statement is added, this command displays the program names, line Nos., call point Nos., statement added, and the statement before adding.

CHANGE

When a statement is changed, this command displays the program names, line Nos., call point Nos., the statement after changing, and the statement before changing.

DELETE

When a statement is deleted, this command displays the program names, line Nos., call point Nos., and the deleted statements.

◆ Display Examples

```
[DEBUG]
```

```
BREAK:
```

```
04B PRO UEPM0001 13 [ EPM:ASC UEPM0001-1 ]
05B PRO UEPM0001 14 [ EPM:ASC UEPM0001-1 ]
06B PRO UEPM0001 15 [ EPM:ASC UEPM0001-1 ]
```

```
INSERT: PRO UEPM0001
```

```
00010 001I          ACLK1 = 40NS
---ORIGINAL--->    ACLK2 = 50NS
```

```
CHANGE: PRO UEPM0001
```

```
00011 002C          P2 = IN1, ACLK1
---ORIGINAL--->    P2 = IN2, ACLK2
```

```
DELETE: PRO UEPM0001
```

```
00012 003Z          PC33 = DC, OUT1, STRB1
```

33. *THTYPE Utility (/THTYPE)*

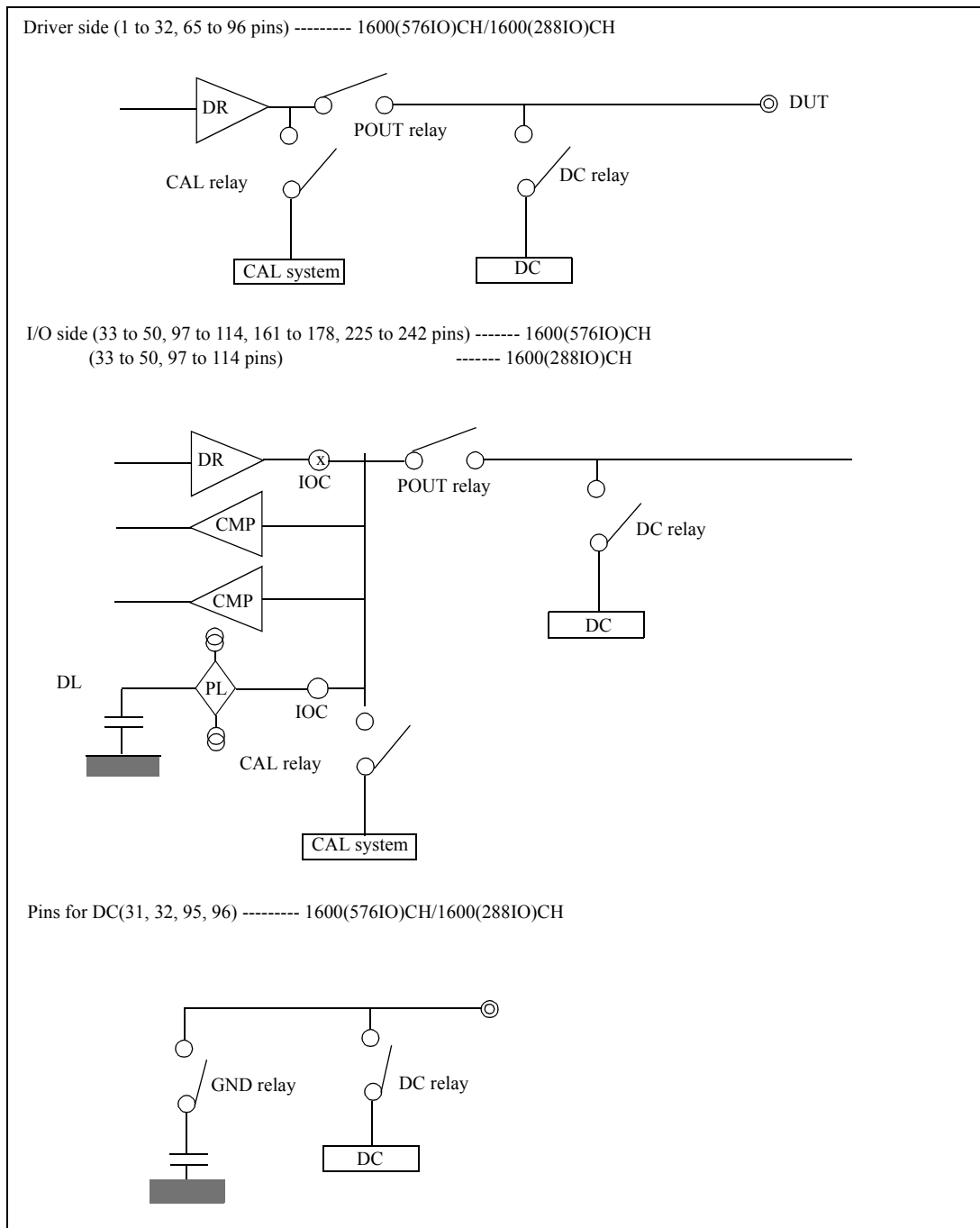
This chapter describes the THTYPE utility (/THTYPE) for displaying the status of various relays or signals for the test head.

33. 1 Overview

This utility is used to display the status of various relays or signals for the test head. The object to be displayed are as follows:

- DC relay
- CAL relay
- POUT relay
- DR (Driver operation)
- CMP (COMParator operation)
- IOC (I/O switch operation)
- PPS mask signal (including PCON mask)
- MCT Signal (parallel test control signal)
- LIMITER FAIL
- GND

For reference, the figure below shows the conceptual drawing of the pin card.



33. 2 Syntax

33. 2. 1 /THTYPE

Function: This utility is used to display the status of various relays or signals for the test head.

Syntax

```
/THTYPE↵
```


33. 3 Start Up

Input through the VKT as follows:

/THTYPE/J	-----	Displays on VKT. (For more information, refer to the memo)
/THTYPE/L/J	-----	Displays on LPT.
/THTYPE/S/J@	-----	Displays on the output unit specified with the S command.
/THTYPE/1/J	-----	Outputs only the pin conditions of the STN 1 side.
/THTYPE/2/J	-----	Outputs only the pin conditions of the STN 2 side.

Tip

For 2 stations, the ST1 and ST2 are output.

For 1 station, the current executing ST is output.

33. 4 Output Example 1 (For 1600(576IO)CH Test Head)

### PIN / FPS MASK / MCT RELAY DISPLAY ###									
	000000000000000000000000000000	00	000000000000000000000000000000	000000000000000000000000000000	00	00011111111111111111			
	000000000111111111222222222223	33	3333333444444444445	6666677777777777888888888899999	99	999000000000011111			
	123456789012345678901234567890	12	345678901234567890	56789012345678901234	56	789012345678901234			
ST1	A1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	A2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
	B1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	B2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
	C1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	C2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
	D1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	D2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
	E1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	E2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
	F1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	F2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
	G1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	G2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
	H1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	H2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
ST2	A1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	A2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
	B1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	B2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
	C1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	C2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
	D1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	D2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
	E1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	E2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
	F1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	F2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
	G1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	G2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
	H1	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG	OOOOYYYY-----		GG	-----		
	H2	DDDDDDDDDDDDDDDDDDDDDDDDDDDDDD	GG						
		11111111111111111111	2222222222222222						
		666666666777777777	2222333333333444						
		123456789012345678	567890123456789012						
ST1	A1	OOOOYYYY-----			<PRS32-1>	<VS12-9>			
	A2				#00000000	#0			
	B1	OOOOYYYY-----			<PRS64-33>				
	B2				#FFFFFFF				
	C1	OOOOYYYY-----			<MCT32-1>				
	C2				#FFFFFFF				
	D1	OOOOYYYY-----			<MCT64-33>				
	D2				#FFFFFFF				
	E1	OOOOYYYY-----							
	E2								
	F1	OOOOYYYY-----							
	F2								
	G1	OOOOYYYY-----							
	G2								
	H1	OOOOYYYY-----							
	H2								
ST2	A1	OOOOYYYY-----			<PRS32-1>	<VS12-9>			
	A2				#00000000	#0			
	B1	OOOOYYYY-----			<PRS64-33>				
	B2				#FFFFFFF				
	C1	OOOOYYYY-----			<MCT32-1>				
	C2				#FFFFFFF				
	D1	OOOOYYYY-----			<MCT64-33>				
	D2				#FFFFFFF				
	E1	OOOOYYYY-----							
	E2								
	F1	OOOOYYYY-----							
	F2								
	G1	OOOOYYYY-----							
	G2								
	H1	OOOOYYYY-----							
	H2								
<NOTE> "D"=DC "X"=DR&DC "A"=CAL "S"=CMP&DR "##"=OPEN "I"=DR "Y"=CMP&DC "B"=CAL&DR&DC "T"=CMP&DR&DC "***"=LMT-FAIL "O"=CMP "Z"=IOC&CMP&DR&DC "C"=IOC&CMP&DR "G"="GND" "- " =MASK									

 Tip

- *<PPS32-1> shows the state of the mask relay of PPS.
It is displayed in hexadecimal notation and PPS which is displayed with 1 is electrically masked.*
- *<PPS64-33> shows the state of the mask relay of PPS.
It is displayed in hexadecimal notation and PPS which is displayed with 1 is electrically masked.
This is displayed when the second DPU is mounted.*
- *<VS12-9> shows the state of the mask relay of VS9 to VS12 in hexadecimal notation.*
- *<MCT32-1> shows the state of MCT1 to MCT32 on the performance board in hexadecimal notation.*
- *<MCT64-33> shows the state of MCT33 to 64 on the performance board in hexadecimal notation.*
- *LMT-FAIL shows the state that a limiter fail is generated.*
- *MASK shows the state that all relays are electrically masked and opened.*

33. 5 Output Example 2 (For 1600(288IO)CH Test Head)

### PIN / PPS MASK / MCT RELAY DISPLAY ###									
	00000000000000000000000000000000	00	00000000000000000000000000000000	00	00000000000000000000000000000000	00	00011111111111111111		
	00000000011111111122222222222223	33	333333344444444444444445	66666777777777777888888888899999	99	99900000000000011111			
	123456789012345678901234567890	12	345678901234567890	567890123456789012345678901234	56	789012345678901234			
ST1 A1	GG	GG	GG	GG	GG	GG	<PPS32-1>	<VS12-9>	
A2	GG	GG	GG	GG	GG	GG	#00000000	#0	
B1	GG	GG	GG	GG	GG	GG	<PPS64-33>		
B2	GG	GG	GG	GG	GG	GG	#FFFFFFF		
C1	GG	GG	GG	GG	GG	GG	<MCT32-1>		
C2	GG	GG	GG	GG	GG	GG	#FFFFFFF		
D1	GG	GG	GG	GG	GG	GG	<MCT64-33>		
D2	GG	GG	GG	GG	GG	GG	#FFFFFFF		
E1	GG	GG	GG	GG	GG	GG			
E2	GG	GG	GG	GG	GG	GG			
F1	GG	GG	GG	GG	GG	GG			
F2	GG	GG	GG	GG	GG	GG			
G1	GG	GG	GG	GG	GG	GG			
G2	GG	GG	GG	GG	GG	GG			
H1	GG	GG	GG	GG	GG	GG			
H2	GG	GG	GG	GG	GG	GG			
ST2 A1	GG	GG	GG	GG	GG	GG	<PPS32-1>	<VS12-9>	
A2	GG	GG	GG	GG	GG	GG	#00000000	#0	
B1	GG	GG	GG	GG	GG	GG	<PPS64-33>		
B2	GG	GG	GG	GG	GG	GG	#FFFFFFF		
C1	GG	GG	GG	GG	GG	GG	<MCT32-1>		
C2	GG	GG	GG	GG	GG	GG	#FFFFFFF		
D1	GG	GG	GG	GG	GG	GG	<MCT64-33>		
D2	GG	GG	GG	GG	GG	GG	#FFFFFFF		
E1	GG	GG	GG	GG	GG	GG			
E2	GG	GG	GG	GG	GG	GG			
F1	GG	GG	GG	GG	GG	GG			
F2	GG	GG	GG	GG	GG	GG			
G1	GG	GG	GG	GG	GG	GG			
G2	GG	GG	GG	GG	GG	GG			
H1	GG	GG	GG	GG	GG	GG			
H2	GG	GG	GG	GG	GG	GG			

<NOTE> "D"=DC "X"=DR&DC "A"=CAL "S"=CMP&DR "##"=OPEN
 "I"=DR "Y"=CMP&DC "B"=CAL&DR&DC "T"=CMP&DR&DC "LMT-FAIL"
 "O"=CMP "Z"=IOC&CMP&DR&DC "C"=IOC&CMP&DR "G"="GND" "- "=MASK

Tip

- <PPS32-1> shows the state of the mask relay of PPS.
It is displayed in hexadecimal notation and PPS which is displayed with 1 is electrically masked.
- <PPS64-33> shows the state of the mask relay of PPS.
It is displayed in hexadecimal notation and PPS which is displayed with 1 is electrically masked.
This is displayed when the second DPU is mounted.
- <VS12-9> shows the state of the mask relay of VS9 to VS12 in hexadecimal notation.
- <MCT32-1> shows the state of MCT1 to MCT32 on the performance board in hexadecimal notation.
- <MCT64-33> shows the state of MCT33 to 64 on the performance board in hexadecimal notation.
- LMT-FAIL shows the state that a limiter fail is generated.
- MASK shows the state that all relays are electrically masked and opened.

34. *MARGIN Program (/MARGIN)*

This chapter describes the MARGIN program (/MARGIN) used for obtaining information on the operational margin and defective items.

34. 1 Overview

This utility program aims to do the following:

- Checking of the operational margin of DUT
- Examine the defective items (level and timing)

This utility program can be started by inputting the command from the VKT. Also, this utility program allows you to write it into a test program using CALL statements.

Checking of the operational margin of DUT

Even if the DUT has been tested and passed, it is very difficult to know with what margin the DUT has been passed. However, if this utility program is used, the operational margin of the pin being used can quickly and easily be obtained by changing the total level systems and total timing systems automatically, based on the set values of present level and timing.

Defect Items (Level and Timing)

When the DUT has been tested and failed, finding the cause of the failure is most time consuming. However, this utility program can be used to find the kinds of levels, timing and pins involved in the failure by changing total level systems or total timing systems automatically.

34. 2 Target Level or Timing

The target items of margin test are shown in the table below.

Item	Parameter	Default when range is not specified.
Voltage	IN, OUT, VT	± 0.5 V
	INH	± 0.48 V
Load current	IL	± 1 mA
Timing	ACLK, BCLK, CCLK, STRB, WSTRB, DREL, DRET	± 10 nS

When no items are specified as margin test targets, the margin test is performed for all items. The margin test is performed by varying the level setting and timing data for a focused pin.

Tip

- The timing data output in the MARGIN utility is the value which is actually set to each unit. This means that the offset data in the test plan program has been added to this utility.
- When the same unit is selected for two or more pins for a margin test which targets timing data, the timing data is varied as follows:
Timing data ACLK, BCLK, CCLK, DREL and DRET are varied only for a focused pin on the T5500, T5377, T5376, T5375 and T5371.
The manner used to vary timing data STRB and WSTRB depends on the test system used.
T5500
Timing data is varied for a focused pin only.
T5377, T5376, T5375 and T5371
Timing data is varied simultaneously for all pins for which the same unit is selected.
On the T5336 and T5311, timing data is varied simultaneously for all pins for which the same unit is selected.
- In the T5593, T5592, and T5591, if the VT connected to I/O channels is specified, the comparator VT value specified in the test plan program is set to both the driver VT and comparator VT in the measurement.

34. 3 Method to Execute Margin Utility by Inputting Commands

The command input is very important to check the specified information of the specified pin.

The command input consists of the following three formats:

1. Command Input Format 1

Specification of device number, pin number, and parameter. Information on the target pin can be obtained by these specifications.

2. Command Input Format 2

Specification for editing output result. The data obtained in format 1 is output in more detail in this format.

3. Command Input Format 3

Specification of resolution. In this format, the data is output by the resolution of the tester.

For stopping execution, press the return key or [B] key.

34. 3. 1 Command Input Format 1

34. 3. 1. 1 /MARGIN

Function: This format is used to obtain information on the target pin.

34. 3 Method to Execute Margin Utility by Inputting Commands

Syntax

```

/MARGIN  DUT=2    PD1, P35      IN          = 0.5V
                                         HIN
                                         LIN
                                         INH          = 0.48V
                                         HINH
                                         LINH
                                         OUT          = 0.48V
                                         HOUT
                                         LOUT
                                         VT          = 0.5V
                                         IL          = 1MA
                                         HIL
                                         LIL
                                         CLK<T1>      = 10NS
                                         ACLK<T1>
                                         BCLK<T1>
                                         CCLK<T1>
                                         STRB          = 10NS
                                         STRBL
                                         STRBT
                                         WSTRB
                                         WSTRBL
                                         WSTRBT
                                         DRE          = 10NS
                                         DREL
                                         DRET

```

Margin test
target device

Margin test
target pin

Margin test
target item

Margin test range
specification

Argument: DUT=2

Specifies the margin test target device.

In specifying the target device, the following rules apply:

- Only one device can be specified.
- When the device number is not specified, 1 is set as the target.
- The minimum device number is 1 and the maximum is 256.

Specify DUT numbers in accordance with the socket program.

PD1,P35

Specifies the margin test target device.

In specifying the target device, the following rules apply:

34. 3 Method to Execute Margin Utility by Inputting Commands

- When the pin number is unspecified, all the pins are the target.
- The tests are performed in the setting sequence of the pin numbers in the driver items, and in the setting sequence of the pin numbers of the comparator items.

For example, when specified as P35, PC33, PD3-4, P40, tests are performed in the following order:

P35 → PD3 → PD4 → P40 → P35 → PC33 → P40

Test of IN, CLK and DRE

Test of OUT, STRB, WSTRB, VT and IL

- The pin number which is used in the program can be specified.

T1

Specify the T number (timing set)

In specifying the target device, the following rules apply: (Used for timing-related items only)

- Specify the T number within the range of T1 to 16. A T number outside the range causes an error.
- When the T number is not specified, T1 to 4 will be tested.
- T1, T2, and T3 will be tested, when T numbers are specified in the following manner: $ACLK < T1-2,3 \geq 10NS$.

IN=0.5V

Specify margin test target items and test range

The following rules apply:

- When the test range is not specified, the value will be as shown in the table below.

Test target items	Values when the test range is omitted.
IN	± 0.5 V
INH	± 0.48 V
OUT	± 0.5 V
VT	± 0.5 V
IL	± 1.0 MA
STRB	± 10 NS
WSTRB	± 10 NS
ACLK, BCLK, CCLK	± 10 NS
DREL, DRET	± 10 NS

- The test range is specified as follows:

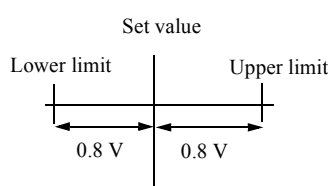
34. 3 Method to Execute Margin Utility by Inputting Commands

Test target item = Measurement value upper limit, measurement value lower limit
or

Test target item = Measurement value lower limit, measurement value upper limit

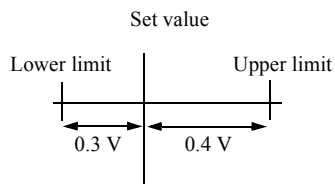
When the measurement value upper limit or the measurement value lower limit setting is omitted, the measurement range is set to the setting value $\pm$ the measurement value.

◆ **IN=0.8 V**



In this case, for the test range of IN, ± 0.8 V of the setting values is set for both upper and lower limits.

◆ **IN = -0.3 V, 0.4 V or IN = 0.4 V, -0.3 V**



In this case, for the test range of IN, the upper limit is set to the setting value $+0.4$ V, and the lower limit is set to the setting value -0.3 V.

- For value, if the setting of a unit is forgotten or if a unit is different, an error results.

IL=10%

IL can be input in % values. In this case, $\pm 10\%$ of the current value will be the upper and lower limits.

34. 3 Method to Execute Margin Utility by Inputting Commands

Description

Before performing a margin test, the range of values are checked for the following items:

Unit		T5593				
		Driver channel		I/O channel	High-speed clock driver channel	Resolution
		Normal range	High-voltage range			
IN	VIH	-1.8 V ≤ Vih ≤ 5.0 V			-0.5 V ≤ Vih ≤ 3.0 V	2 mV
	VIL	-2.0 V ≤ Vil ≤ 3.3 V			-0.5 V ≤ Vil ≤ 3.0 V	
	--	0.2 V ≤ (Vih-Vil) ≤ 7.0 V			0.2 V ≤ (Vih-Vil) ≤ 2.0 V	
OUT	VOH	--			-2.0 V ≤ Voh ≤ 5.0 V	2 mV
	VOL				-2.0 V ≤ Vol ≤ 5.0 V	
VT		0 V ≤ VT ≤ 2.0 V	0 V ≤ VT ≤ 7.0 V	-2.0 V ≤ VT ≤ 5.0 V		2 mV

Unit	T5592 and T5591	T5581, T5581H, T5581P and T5571P	T5371, T5336 and T5311	Resolution
IN	$-1.8 \text{ V} \leq \text{Vih} \leq 5.5 \text{ V}$ $-2.5 \text{ V} \leq \text{Vil} \leq 5.0 \text{ V}$ $0.2 \text{ V} \leq (\text{Vih}-\text{Vil}) \leq 8.0 \text{ V}$	$-1.5 \text{ V} \leq \text{Vih} \leq 8.0 \text{ V}$ $-3.0 \text{ V} \leq \text{Vil} \leq 5.0 \text{ V}$ $0.2 \text{ V} \leq (\text{Vih}-\text{Vil}) \leq 8.0 \text{ V}$ $ \text{Vih}-\text{Vt} \leq 8.0 \text{ V}$	$-1.5 \text{ V} \leq \text{Vih} \leq 8.0 \text{ V}$ $-3.0 \text{ V} \leq \text{Vil} \leq 5.0 \text{ V}$ $0.5 \text{ V} \leq (\text{Vih}-\text{Vil}) \leq 8.0 \text{ V}$	2 mV
INH	--	--	$1 \text{ V} \leq \text{Vihh} \leq 15.0 \text{ V}$ $0 \text{ V} \leq \text{Vilh} \leq 10.0 \text{ V}$ $1.0 \text{ V} \leq (\text{Vihh}-\text{Vilh}) \leq 15.0 \text{ V}$	4 mV
OUT	$-2.5 \text{ V} \leq \text{Voh} \leq 5.5 \text{ V}$ $-2.5 \text{ V} \leq \text{Vol} \leq 5.5 \text{ V}$	$-3.0 \text{ V} \leq \text{Voh} \leq 8.0 \text{ V}$ $-3.0 \text{ V} \leq \text{Vol} \leq 8.0 \text{ V}$	$-3.0 \text{ V} \leq \text{Voh} \leq 8.0 \text{ V}$ $-3.0 \text{ V} \leq \text{Vol} \leq 8.0 \text{ V}$	2 mV
IL	--	$-25 \text{ mA} \leq \text{Ilh} \leq 0 \text{ mA}$ $0 \text{ mA} \leq \text{Ill} \leq 25 \text{ mA}$	$-20 \text{ mA} \leq \text{Ilh} \leq 0 \text{ mA}$ $0 \text{ mA} \leq \text{Ill} \leq 20 \text{ mA}$	20 μA
VT	$-2.5 \text{ V} \leq \text{Vt} \leq 5.5 \text{ V}$	$-3.0 \text{ V} \leq \text{Vt} \leq 4.0 \text{ V}$	$-2.5 \text{ V} \leq \text{Vt} \leq 6.0 \text{ V}$	2 mV

Unit		T5586 and T5585		
		Driver channel	I/O channel	Resolution
IN	VIH	$-1.5 \text{ V} \leq \text{Vih} \leq 8.0 \text{ V}$	$-1.5 \text{ V} \leq \text{Vih} \leq 7.0 \text{ V}$	2 mV
	VIL	$-3.0 \text{ V} \leq \text{Vil} \leq 5.0 \text{ V}$	$-2.0 \text{ V} \leq \text{Vil} \leq 5.0 \text{ V}$	
	--	$0.2 \text{ V} \leq (\text{Vih}-\text{Vil}) \leq 8.0 \text{ V}$ $ \text{Vih}-\text{Vt} \leq 8.0 \text{ V}$	$0.2 \text{ V} \leq (\text{Vih}-\text{Vil}) \leq 8.0 \text{ V}$ $ \text{Vih}-\text{Vt} \leq 8.0 \text{ V}$	

34. 3 Method to Execute Margin Utility by Inputting Commands

Unit		T5586 and T5585		
		Driver channel	I/O channel	Resolution
OUT	VOH	--	$-2.0\text{ V} \leq \text{Voh} \leq 7.0\text{ V}$ $-2.0\text{ V} \leq \text{Vol} \leq 7.0\text{ V}$	2 mV
	VOL			
IL	ILH	--	$-25\text{ mA} \leq \text{Ilh} \leq 0\text{ mA}$ $0\text{ mA} \leq \text{Ill} \leq 25\text{ mA}$	20 μA
	ILL			
VT		--	$-2.0\text{ V} \leq \text{Vt} \leq 4.0\text{ V}$	2 mV

34. 3 Method to Execute Margin Utility by Inputting Commands

Unit		T5377					
		Driver channel	I/O channel	HV driver channel	HC driver channel	LV driver channel	Resolution
IN	VIH	-1.5 V ≤ Vih ≤ 8.0 V	-1.5 V ≤ Vih ≤ 8.0 V	-1.5 V ≤ Vih ≤ 8.0 V	-1.5 V ≤ Vih ≤ 8.0 V	--	2 mV
	VIL	-3.0 V ≤ Vil ≤ 5.0 V	-2.0 V ≤ Vil ≤ 5.0 V	-3.0 V ≤ Vil ≤ 5.0 V	-3.0 V ≤ Vil ≤ 5.0 V		
	--	0.2 V ≤ (Vih-Vil) ≤ 8.0 V	0.2 V ≤ (Vih-Vil) ≤ 8.0 V -8.0 V ≤ Vih-Vt ≤ 8.0 V -8.0 V ≤ Vil-Vt ≤ 8.0 V	0.2 V ≤ (Vih-Vil) ≤ 8.0 V	0.2 V ≤ (Vih-Vil) ≤ 8.0 V		
INH	HVIH	--	--	0.5 V ≤ Vihh ≤ 15.0 V	-3.0 V ≤ Vihh ≤ 15.0 V	0 V ≤ Vihh ≤ 15.0 V	4 mV
	HVIL			0 V ≤ Vilh ≤ 10.0 V	-3.5 V ≤ Vilh ≤ 10.0 V	--	
	--			0.5 V ≤ (Vihh-Vilh) ≤ 15.0 V	0.5 V ≤ (Vihh-Vilh) ≤ 15.0 V		
OUT	VOH	--	-2.0 V ≤ Voh ≤ 8.0 V	--	--	--	2 mA
	VOL		-2.0 V ≤ Vol ≤ 8.0 V				
IL	ILH	--	-25 mA ≤ Ilh ≤ 0 mA	--	--	--	20 μA
	ILL		0 mA ≤ Ill ≤ 25 mA				
VT		--	-2.0 V ≤ Vt ≤ 6.0 V	--	--	--	2 mV

Unit		T5376			
		I/O channel	HC driver channel	LV driver channel	Resolution
IN	VIH	-1.5 V ≤ Vih ≤ 6.0 V	-1.5 V ≤ Vih ≤ 8.0 V		2 mV
	VIL	-2.0 V ≤ Vil ≤ 5.0 V	-3.0 V ≤ Vil ≤ 5.0 V		
	--	0.2 V ≤ (Vih-Vil) ≤ 7.0 V Vih-Vt ≤ 7.0 V Vil-Vt ≤ 7.0 V	0.2 V ≤ (Vih-Vil) ≤ 8.0 V		

34. 3 Method to Execute Margin Utility by Inputting Commands

Unit		T5376			
		I/O channel	HC driver channel	LV driver channel	Resolution
INH	HVIH	--	$0.5\text{ V} \leq \text{Vihhl} \leq 5.0\text{ V}$	$0\text{ V} \leq \text{Vihh} \leq 15.0\text{ V}$	4 mV
	HVIL		$0\text{ V} \leq \text{Vilh} \leq 10.0\text{ V}$	--	
			$0.5\text{ V} \leq (\text{Vihh}-\text{Vilh}) \leq 15.0\text{ V}$		
OUT	VOH	$-2.0\text{ V} \leq \text{Voh} \leq 6.0\text{ V}$	--	--	2 mV
	VOL	$-2.0\text{ V} \leq \text{Vol} \leq 6.0\text{ V}$			
IL	ILH	$-25\text{ mA} \leq \text{Ilh} \leq 0\text{ mA}$	--	--	20 μA
	ILL	$0\text{ mA} \leq \text{Ill} \leq 25\text{ mA}$			
VT		$-2.0\text{ V} \leq \text{Vt} \leq 6.0\text{ V}$	--	--	2 mV

34. 3 Method to Execute Margin Utility by Inputting Commands

Unit		T5375				
		Driver channel	I/O channel	HV driver channel HC driver channel	LV driver channel	Resolution
IN	VIH	-1.5 V ≤ Vih ≤ 8.0 V	-1.5 V ≤ Vih ≤ 6.0 V	-1.5 V ≤ Vih ≤ 8.0 V	--	2 mV
	VIL	-3.0 V ≤ Vil ≤ 5.0 V	-2.0 V ≤ Vil ≤ 5.0 V	-3.0 V ≤ Vil ≤ 5.0 V		
	--	0.2 V ≤ (Vih-Vil) ≤ 8.0 V	0.2 V ≤ (Vih-Vil) ≤ 7.0 V Vih-Vt ≤ 7.0 V Vil-Vt ≤ 7.0 V	0.2 V ≤ (Vih-Vil) ≤ 8.0 V		
INH	VIHH	--	--	0.5 V ≤ Vihh ≤ 15.0 V	0 V ≤ Vihh ≤ 15.0 V	4 mV
	VILH			0 V ≤ Vilh ≤ 10.0 V	--	
				0.5 V ≤ (Vihh-Vilh) ≤ 15.0 V		
OUT	VOH	--	-2.0 V ≤ Voh ≤ 6.0 V -2.0 V ≤ Vol ≤ 6.0 V	--	--	2 mV
	VOL					
IL	ILH	--	-25 mA ≤ Ilh ≤ 0 mA 0 mA ≤ Ill ≤ 25 mA	--	--	20 μA
	ILL					

--	T5593 and T5592	T5591, T5586, T5585, T5581, T5581H, T5581P and T5571P	T5377, T5376 and T5375	T5371, T5336 and T5311
ACLK, BCLK, CCLK, DREL, DRET, STRB, WSTRB	$0 \text{ ns} \leq t < 960 \text{ ns}$	$0 \text{ ns} \leq t \leq 1 \text{ ms}$	$0 \text{ ns} \leq t \leq 14 \text{ ms}$	$0 \text{ ns} \leq t \leq 16 \text{ ms}$

When the data that margin test is executed is specified over the measurement possible range that system has, the following error messages are output.

THE VALUE IS TOO LARGE

When the specified resolution is too small,

THE DELTA IS TOO SMALL

34. 3. 2 Command Input Format 2

34. 3. 2. 1 /MARGIN

Function: This command is used to output the data obtained in the command input format 1 in more detail.

Syntax

```
/MARGIN/L INVERT, TRANSIT, CEL=60, FAIL, CAL
```

Argument: /L

Outputs the result to the line printer.

INVERT

Normally, when this is not specified, outputs pass with "." and fail with "F." If this is specified, pass is output with "P" and fail is output with ".."

TRANSIT

Outputs the margin in the test condition. This margin is output only when the test condition changes from fail to pass and from pass to fail.

CEL=60

Specify the number of cells. Normally when this specification is not present, 20 is set. For the range of CEL, 4 to 61 can be specified. However, when an odd number is set, the processing is the same as when 1 is subtracted from that value.

For example, the processing for CEL=4 and CEL=5, or CEL=60 and CEL=61 are the same.

FAIL

Outputs the address of the failed pattern program. The address is output from the failed portion only when the test condition is changed from fail to pass and from pass to fail.

CAL

This command is used when the calibration condition is input.

(The calibration setting is disabled on the T5593, T5592 and T5591.)

Inputting Calibration Conditions

As the menu shown below is displayed when CAL is specified and the margin utility is executed, input calibration conditions one by one.

34. 3 Method to Execute Margin Utility by Inputting Commands

- 1) When calibration is to be performed, input "1".

```
CAL ENABLE(1) OR DISABLE(0) = 0      1.↵
```

- 2) Specify the string variable name where the calibration conditions are stored when CALB is specified in the test plan program.

```
VAL NAME =      ABC.↵
```

- 3) Specify the first two letters of the file name of the CAL attribute when calibration is performed in the margin utility.

```
FILE TAG =      CF.↵
```

- 4) Input the calibration condition to be added or changed.

```
CAL STRING =  
DRREF = 2.4V, .....↵
```

When the calibration condition is described directly without displaying the menu, input the calibration condition in the following format:

```
/MARGIN CAL      *(ABC, CF, DRREF = 2.4V, ...) * , ...
```

_____ CAL condition
_____ CAL file name
_____ String variable name

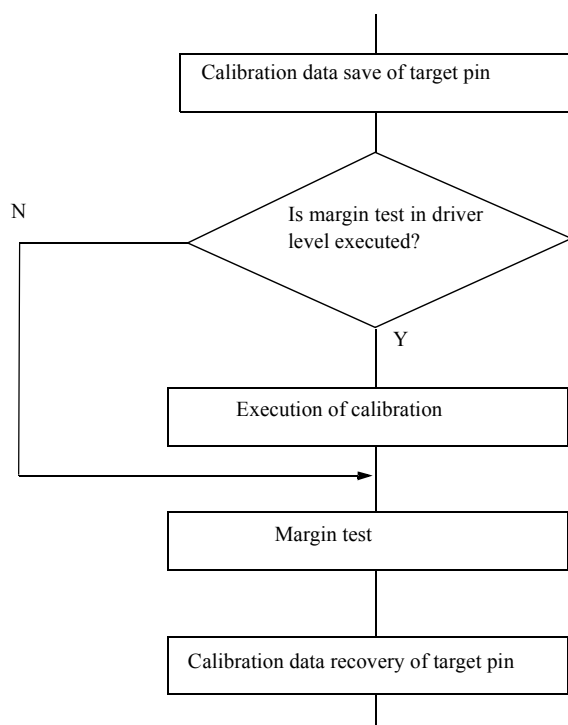
[Calibration in Margin Utility]

Timing calibration carried out in the test plan program is nullified when the margin utility, which changes level inputs, is executed. Therefore, to execute the margin test in the timing set by the test plan program, it is necessary to execute the calibration again in the calibration condition set by the test plan program for every target pin.

34. 3 Method to Execute Margin Utility by Inputting Commands

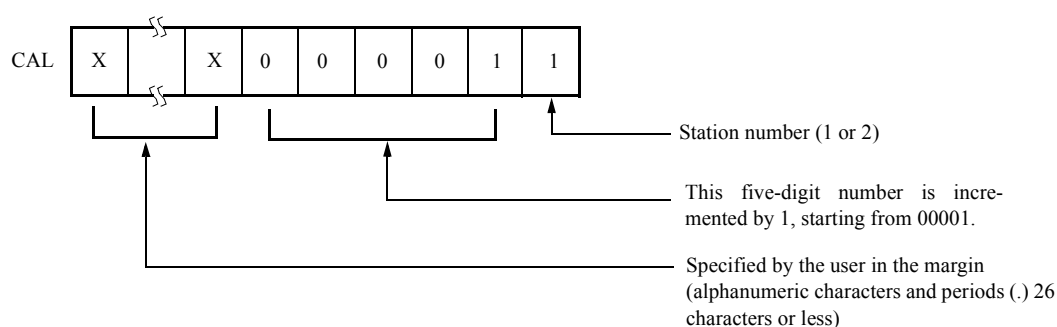
Calibration Execution Sequence

In margin utility, calibration is performed in the following sequence:



When "CAL ENABLE" is displayed, each point is recalibrated for the driver amplitude margin test.

File names with CAL attributes



The previously set CAL attribute names are not deleted.

❗ Important

- When the margin with calibration is run, the file of the CAL attribute is created in DISK for every execution of calibration. Be very careful when executing calibration for the file control (for maintenance, deletion, and name of the file).

34. 3 Method to Execute Margin Utility by Inputting Commands

- The calibration setting is disabled on the T5593, T5592 and T5591.
-

Specifying the string variable name of the calibration facility

CALB specification of main program shall be performed using the string variable names as shown below.

Correct

```
STRING JJ(1000)
```

```
JJ="DRREF=2.4V,..."
```

```
CALL CALB("64",JJ)
```

When the character string is specified directly as follows, conditions for the program cannot be delivered correctly to the character string of the margin.

Incorrect

```
CALL CALB("64K","DRREF=2.4V,...")
```

Additional character string specified with margin

The additional character strings specified by the margin are used in the form to be added to the string variable in the test plan program. When the same specifications are present, the one specified last becomes effective.

- ➔ For more information on specification, refer to [T5500 Series Memory Test Systems Calibration Program Reference Manual\(No. : 8256296\)](#).

34. 3. 3 Command Input Format 3

34. 3. 3. 1 /MARGIN DELTA

Function: This command is used to output data by the resolution of the tester.

Syntax

```
/MARGIN DELTA HIN=3V, 2mV, STRB=10NS, 125PS
```

Argument: DELTA

The resolution can be specified.

34. 3 Method to Execute Margin Utility by Inputting Commands

Format for specifying the resolution.

IN	}	= measured value, resolution
HIN		
LIN		
INH		
HINH		
LINH		
OUT		
HOUT		
LOUT		
VT		
IL		
HIL		
LIL		
CLK<T1>		
ACLK<T1>		
BCLK<T1>		
CCLK<T1>		
STRB		
STRBL		
STRBT		
WSTRB		
WSTRBL		
WSTRBT		
DRE		
DREL		
DRET		

For the initial value to be measured, the level to start the measurement is set. That is, in the example mentioned above, the test is performed by the resolution from 3 V to 2 mV for HIN.

34. 3 Method to Execute Margin Utility by Inputting Commands

Method of Calculating Resolution

- When DELTA Is Specified

--	T5593	T5592	T5591, T5586, T5585, T5581, T5581H, T5581P and T5571P	T5377, T5376 and T5375
IN, OUT, VT	(Set value/2 mV) × 2 mV	(Set value/2 mV) × 2 mV	(Set value/2 mV) × 2 mV	(Set value/2 mV) × 2 mV
INH	--	--	--	(Set value/4 mV) × 4 mV
ACLK, BCLK, CCLK, DREL, DRET, STRB, WSTRB	(Set value/ 7.32421875 ps) × 7.32421875 ps	(Set value/ 14.6484375 ps) × 14.6484375 ps	(Set value/15.625 ps) × 15.625 ps	(Set value/27.34375 ps) × 27.34375 ps

--	T5371, T5336 and T5311
IN, OUT, VT	(Set value/2 mV) × 2 mV
INH	(Set value/4 mV) × 4 mV
ACLK, BCLK, CCLK, DREL, DRET, STRB, WSTRB	(Set value/62.5 ps) × 62.5ps

- When DELTA Is Not Specified

(Measurement value upper limit + measurement value lower limit)/Number of CELs.

A value obtained from the above expression is a set value for each unit or timing data. Resolutions can be obtained using this set value in the same manner used when DELTA is specified.

34. 4 To Check Margin

This is used to test voltage and timing in a pin. The test method is shown below.

- Command Format

➔ For more information on command format, refer to [34. 3. 1 "Command Input Format 1" on page 34-3](#) and [34. 3. 2 "Command Input Format 2" on page 34-12](#).

- Output Examples

Here, the explanation is given for outputs of the two ways shown below.

Checking margin of voltage

Checking margin of timing

- ◆ **When Margin of Voltage is Checked**

In the program, the following messages are set.

```
IN3 = 1.2V, 0V
PD1 = IN3, NRZA, ACLK1, <X0>
```

Next, input the message as follows:

```
/MARGIN TRANSIT PD1 HIN = 0.5V.␣
```

The following will be displayed:

DUT	ITEM	STARTING	TRANSIT	
PIN		VALUE	F-->P	
			I+++++++S++++++I	
1	HIN3	700.0MV	1.100V	FFFFFFFF..*.....FFF
		TRANSIT	ENDING	DELTA
		P-->F	VALUE	SETTING
		1.550V	1.700V	(50.00MV) 1.200V

The above output result shows that the Pass range of the DUT driver pin 1 is 1.00 V to 1.550 V.

- ◆ **When Margin of Timing is Checked**

In the program, the following messages are set.

```
STRB1 = 52NSPC33 = OUT1, STRB1
```

Next, when the message is input as follows,

```
/MARGIN TRANSIT PC33 STRB<1> = 10NS.↓
```

The following will be displayed:

DUT PIN	ITEM	STARTING VALUE	TRANSIT F-->P		
				I+++++++S+++++I	
33	STRB1-01	42.00NS	44.00NS	FF.....*FFFFFFFFF	
		TRANSIT P-->F	ENDING VALUE	DELTA	SETTING VALUE
		52.00NS	62.00NS	(1.00NS)	52.00NS

The above output result shows that the Pass range of the DUT comparator pin 33 is 44.00 ns to 52.00 ns.

Here, when the margin of timing is checked by the fine resolution, specification of CEL is performed.

```
/MARGIN CEL=60 TRANSIT PC33 STRB<1> = 10NS.↓
```

The following will be displayed:

DUT PIN	ITEM	STARTING VALUE	TRANSIT F-->P		
33	STRB1-01	42.625NS			
				I+++++++I+++++++I+++++++S+++++I+++++++I+++++++I	
				*.FFFFFFFFFFFFFFFFFFFFFFFFF	
		TRANSIT P-->F	ENDING VALUE	DELTA	SETTING VALUE
		52.625NS	61.375NS	(312.5PS)	52.00NS

By specifying the CEL, the resolution, which was 1.000 ns so far, becomes 312.5 ps.

34. 5 To Search for Cause of Failed DUT

When the DUT has many pins, this is used to examine the failed pins and the failed items. The test method is shown below.

- Command Format

➔ For more information on command format, refer to [34. 3. 1 "Command Input Format 1" on page 34-3](#), [34. 3. 2 "Command Input Format 2" on page 34-12](#) and [34. 3. 3 "Command Input Format 3" on page 34-15](#).

- Output Examples

Here, the method to search the course of the failed device is explained. At first, when a device fails, the processing below is executed. In the program, the following messages are set.

```
RATE=1US
IN1=1.4V,0V
OUT1=1.3V,1.0V
OUT2=1.0V,0.8V
VT1=1V
IL1=-5MA,5MA
BCLK1=45NS
CCLK1=60NS
DREL1=45NS
DRET1=60NS
STRB1=50NS

P33=IN1,BCLK1,CCLK1,RZX,IOC,OUTL1,STRB1,IL1,<D0>
P34=IN1,BCLK1,CCLK1,NRZBC,IOC,OUTL2,STRB1,IL1,<D1>
```

Next, when the message is input as follows,

```
/MARGIN TRANSIT FAIL.␣
```


To perform further fine measurement, specify the portion using the maximum resolution (2 mV).

```
/MARGIN DELTA CEL=60  TRANSIT P33 HOUT=1.100V, 2MV.┘
```

Next, messages are output as follows:

```
DUT ITEM      STARTING  TRANSIT
PIN          VALUE      F-->P

33  HOUT1     1.100V

I+++++++I+++++++I+++++++I+++++++I+++++++I+++++++I
.....FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF

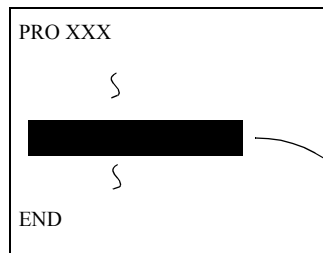
          TRANSIT      ENDING      DELTA      SETTING
          P-->F        VALUE              VALUE
          1.158V      1.220V      (2.000MV)  1.300V
```

From this result, it is verified that the HOUT of the pin 33 in this device is the cause of failure.

34. 6 Method to Incorporate Utility Program in Test Program

This utility program can be incorporated in the test plan program.
The method of use is shown below.

Test plan program



```
CALL MARGIN("/L PD1-5, TRANSIT")
```

Message can be described as above. The parameters in parentheses are same as input format 1, 2 and 3. Also, the following specification is possible.

```
STRING PARM(20) = "/L PD1-5, TRANSIT"
      :
      :
CALL MARGIN(PARM)
```

34. 7 Command Error

- When an error is found in the parameter to be input, the following display appears.

```
/MARGIN STROB = 10NS, 20NS.↓
```

The following will be displayed:

```
COMMAND ERROR!!  
STROB = 10NS, 20NS  
----- ^
```

This case shows that an error was present when reading the parameter one place before "^". This means "STROB" is the error. Change this to "STRB" and input again.

- When the pattern file name is different or absent, the following message is displayed.

```
UNDEFINE PATTERN FILE
```

35. LOG ON Utility (/LOG ON, /LOG OFF)

This chapter describes the LOG ON utility (/LOG ON, /LOG OFF) for creating log files.

35. 1 Overview

This LOG ON utility allows the user to write any characters displayed on the VKT in a log file.

35. 2 Syntax

35. 2. 1 /LOG ON

Function: This command is used to start logging.

Syntax

```
/LOG ON file name↵
```

Argument: File name

Specify the log file name. Neither customer name nor file attributes can be specified.
The customer is the current customer, and the file attribute is "ASC".

Default

ASC LOG. CM

35. 2. 2 /LOG OFF

Function: This command is used to end logging.

Syntax

```
/LOG OFF↵
```

35. 3 Precautions

- The LOG ON and LOG OFF commands of the controller are executed by the same functions as the LOG ON and LOG OFF commands of CLI. Therefore, these commands can be mixed. For example, the user can logs on using CLI and can logs off using the controller.
- Once the LOG ON command is issued, data write continues until the LOG OFF command is issued or the system is stopped.
- If the log file already exists during log on, the log data is updated from its end.
- The system file name such as "\$LPT" can be used as the file name. However, the <LF> (Line Feed) code is not output and, therefore, the lines are not fed.

35. 4 Error Message

The following error messages may be output:

35. 4. 1 Error Message

ILLEGAL LOG SPECIFIER : XXXX

A command argument other than "ON" or "OFF" was specified after "LOG".

NOT ENOUGH ARGUMENT : LOG

The "LOG" command argument was not followed by "ON" or "OFF".

ILLEGAL LOG FILENAME : XXXXI

An invalid file name was specified.

CUSTOMER NAME CAN NOT SPECIFY : XXXX

The customer name was specified for the file name.

36. *VKTSPEED Utility*

This chapter describes the VKTSPEED utility that controls the display speed of the VKT.

36. 1 Overview

The ADVANsite has the VKT display co-processor and its display speed is relatively high when compared with the conventional MOST systems. However, the excessive display speed may cause the user to difficult to observe the display information or log data on the screen. Also the screen of data may be scrolled even in the hold mode. This utility controls the VKT display speed to solve these problems. The LGSPEED utility is also provided to control only the data log display speed.

➔ For more information on the LGSPEED utility, refer to [37. "LGSPEED Utility" on page 37-1](#).

36. 2 Syntax

36. 2. 1 /VKTSPEED

Function: This command is used to control the VKT display speed.

Syntax

```
/VKTSPEED speed
```

Argument: speed

HIGH

High-speed mode

Input of three characters at the top is also valid.

MIDDLE

Medium-speed mode

Input of three characters at the top is also valid.

LOW

Low-speed mode

n(Baudrate)

Baud rate mode

"n" can be a value between 100 and 38400.

0

Release of the selected mode

Default

As the speed is asked, specify the speed.

```
/VKTSPEED
How much fast do you want High, Middle, Low
or Baudrate [speed]? speed
```

36. 3 Function

The VKTSPEED utility holds a single line of data on the screen for the specified wait time as follows:

- When the high-speed mode is released, the display speed is almost the same as the mode is released.
- In the medium-speed mode, a single line of data is held on the screen for approximately 200 msec.
- In the low-speed mode, a single line of data is held on the screen for approximately 400 msec.
- If a baud rate is specified in the baud rate mode, the display data is held on the screen according to the appropriate baud rate speed.
- If the mode is omitted or it is released, data is displayed at the highest display speed.

36. 4 Precautions

- The actual baud rate depends on the CPU speed and others.
- Once the display speed mode is selected by the LGSPEED utility, it remains valid when control returns to the tester driver. However, this specification is ignored in the different directory.
- If the lower display speed is selected by using the LGSPEED utility, the data processing speed is also reduced.

37. *LGSPEED Utility*

This chapter describes the LGSPEED utility that controls data log display speed.

37. 1 Overview

The ADVANsite has the VKT display co-processor and its display speed is relatively high when compared with the conventional MOST systems. However, the excessive display speed may cause the user to difficult to observe the display information including pin monitor and FBMAP data as well as log data on the screen. The screen of data may be scrolled even in the hold mode. Also, if data is logged in the manual, recycle, or hold mode, the utility cannot start quickly because the data remains on the screen. This utility controls the data log display speed to solve these problems.

The VKTSPEED utility is also provided to control the display speed of the VKT itself.

➔ For more information on the VKTSPEED utility, refer to [36. "VKTSPEED Utility" on page 36-1](#).

37. 2 Syntax

37. 2. 1 /LGSPEED

Function: This command is used to control the data log display speed.

Syntax

```
/LGSPEED speed
```

Argument: speed

HIGH

High-speed mode

Input of three characters at the top is also valid.

MIDDLE

Medium-speed mode

Input of three characters at the top is also valid.

LOW

Low-speed mode

n(Baudrate)

Baud rate mode

"n" can be a value between 100 and 38400.

0

Release of the selected mode

Default

As the speed is asked, specify the speed.

```
/LGSPEED  
How much fast do you want High, Middle, Low  
or Baudrate [speed]? speed
```

37. 3 Function

The LGSPEED utility holds a single line of data on the screen for the specified wait time as follows:

- When the high-speed mode is released, the display speed is almost the same as the mode is released.
- In the medium-speed mode, a single test of data is held on the screen for approximately 200 msec.
- In the low-speed mode, a single test of data is held on the screen for approximately 400 msec.
- If a baud rate is specified in the baud rate mode, the display data is held on the screen according to the appropriate baud rate speed.
- If the mode is omitted or it is released, data is displayed at the highest display speed.

37. 4 Precautions

- The actual baud rate depends on the CPU speed and others.
- Once the display speed mode is selected by the LGSPEED utility, it remains valid when control returns to the tester driver. However, this specification is ignored in the different directory.
- If the lower display speed is selected by using the LGSPEED utility, the data processing speed is also reduced.
- The specification by the LGSPEED utility is also valid for the function data log.
- If the LGSPEED and VKTSPEED utilities are used with together, the display speed of data log is reduced two times.

38. Handler Control Program with GPIB Interface (/STHAND)

This chapter describes the handler control program with GPIB Interface (/STHAND).

38. 1 Overview

This utility program controls the handler with GPIB interface.

When the "/STHAND" command is input from the VKT, the utility program can be operated. The use only modifies the device program except he describes "HANDLER" in the socket program parallel test definition section.

This utility program can control the following handlers:

- SYMTEK BY-16 series
- ADVANTEST M3700/M6800/M6700/M6500 series (Referred to as ADVANTEST handler hereafter.)
- Handlers available from other manufacturers based on standard ADVANTEST handler control specifications (For more information, refer to the memo.) (These are hereafter called the other handlers based on standard ADVANTEST specifications.)

When using handlers other than those listed above (i.e., handlers with different protocols than the above), it is necessary to create a control program matched to the protocol of the handler used.

Tip

The standard ADVANTEST handler control specifications refer to those written in the "T5300/T5500+M3700 Series GPIB Handler Control Utility Program Specifications".

38. 2 Syntax

38. 2. 1 (/STHAND)

Function: This command is used to set handler control.

Syntax

```
/STHAND↵
```

38. 3 Operating Method of "/STHAND" Command

Input the following command from the VKT in the controller mode:

```
/STHAND␣
```

When /STHAND is started up first, the following default menu is displayed on VKT.

The menu varies with the specified HANDLER TYPE (the type of handler used)

```
***STHAND *** (SELECT GP-IB I/F OF HANDLER)
```

```
- GP-IB ADDRESS OF STN1 HANDLER ..... DISABLE
- GP-IB ADDRESS OF STN2 HANDLER ..... DISABLE
- HANDLER WAIT TIME ..... 15 SEC
- HANDLER TYPE ..... ADVANTEST PRODUCTS
- DUT NUMBER SELECT OF STN1 HANDLER .. DISABLE
- DUT NUMBER SELECT OF STN2 HANDLER .. DISABLE
- DUMMY START ..... ENABLE
- SEND ALARM ..... DISABLE
- TIMER PAUSE ON CALIBRATION ..... DISABLE
- TESTER ACTION
```

```
[WORK]
```

```
[NOTE]-----
      PF1:EXIT
```

Settings of /STHAND are saved to a file when /STHAND is terminated. Therefore, settings are retained intact even when the controller is exited or the test system's power supply is shut down.

The file name to which these settings are saved is normally fixed (DP0:SYS STHAND). However, the settings can be saved to any other file by specifying a desired file name following /STHAND. This will be convenient when settings of STHAND need to be changed for each model such as when measuring multiple models using one test system.

There are following two methods for specifying a file name. (Available for only the handlers made by ADVANTEST)

```
/STHAND file name
/STHAND/R file name
```

- When the command line "/STHAND file name" is executed, data is read from the specified file and displayed on the screen. If the specified file is nonexistent, the contents previously set are displayed.

The contents of settings are saved to a specified file when terminating /STHAND and handler control is set up.

- When the command line "/STHAND/R file name" is executed, the specified file is read in and handler control is set up according to the contents of the file.
The file that can be specified here is one that was created in "/STHAND file name." The setup contents of a specified file cannot be modified.



Limitation

- When specifying a "/STHAND file name", note that nothing but ADVANTEST PRODUCTS (handlers made by ADVANTEST) can be specified for HANDLER TYPE (the type of handler used).
Although the previously set contents are displayed when a specified file is nonexistent, note that if anything but ADVANTEST PRODUCTS was set for HANDLER TYPE previously, the HANDLER TYPE is changed to ADVANTEST PRODUCTS as data is displayed.
- Even when a file name is specified, the setup contents are reflected in the entire test system. Since settings are saved to DP0:SYS STHAND also, they become effective for all customers.
- If a file name is specified, data is read from and saved to the specified file that exists in the current directory. The file attribute of a specified file is DAT (data file).

38. 3. 1 Address Specification for GPIB Interface (GP-IB ADDRESS OF STN1/2 HANDLER)

Move the cursor to "GP-IB ADDRESS OF STN1 HANDLER" or "GP-IB ADDRESS OF STN2 HANDLER" in the above menu, then, press the RETURN key. The following message will be displayed in the [WORK] area.

- If "GP-IB ADDRESS OF STN1 HANDLER" is selected;

[WORK] SPECIFY GP-IB ADDRESS OF STN1 HANDLER (D,0-30) _

- If "GP-IB ADDRESS OF STN2 HANDLER" is selected;

[WORK] SPECIFY GP-IB ADDRESS OF STN2 HANDLER (D,0-30) _

This time, specify the GPIB interface address of the station 1 (STN1) or station 2 (STN2), then, press the RETURN key. The designation range is 0 to 30 in decimal numeral.

If "D" is specified, the designation of GPIB interface address is disabled in the station.



Limitation

- Do not specify the same address in the station 1 (STN1) and station 2 (STN2) if the parallel test of multiple devices in two stations is carried out.
- The default address of the GPIB interface in the M2216D/M2816J/M2816Z series made by SYMTEK is 17.

- The GPIB interface and handler interface cannot be used simultaneously, if the parallel test of multiple devices in two stations is carried out.

38. 3. 2 Specification of Handler Wait Time (HANDLER WAIT TIME)

The period from the time when a test ends to the time when the next test starts is specified for the handler. When the test system is used to test more than one device using two test stations in parallel and received a test start signal sent from one handler, the test system waits for another handler to send a test start signal within an allowed time period (wait time). If the test system does not receive the test start signal within the allowed time, testing is started with the handler which sent the test start signal.

When both handlers are in one of the following states and one handler sends a test start signal to the test system, testing is started.

- Lot start
- Retest lot start
- Lot end
- Final lot end
- Query of HIFIX PBTYP No.
- Another handler is in a stopped state.
- Devices cannot be set because of a jam or alarm.

Move the cursor to "HANDLER WAIT TIME" in the menu screen and press the return key. The following message will be displayed in the [WORK] area.

```
[WORK]
SPECIFY HANDLER WAIT TIME (SEC)
```

Set the wait time in units of seconds. Round up any fraction of a second to the next second.
The set range is 0 to 3600 second.

38. 3. 3 Specification of Handler Type to be Used (HANDLER TYPE)

Several handler types are available to choose from, including the BY-16 series handlers made by SYMTEK, the M3700, M6800, M6700, and M6500 series handlers made by ADVANTEST, other handlers based on standard ADVANTEST specifications, and those that do not belong to any of the above (OTHERS).

Move the cursor to "HANDLER TYPE" in the menu screen and press the return key. The following message will be displayed in the [WORK] area.

```
[WORK]
SPECIFY HANDLER TYPE 1:SYMTEK      BY-16 SERIES
                      2:ADVANTEST PRODUCTS
                      3:ADVANTEST STANDARD
                      O:OTHERS
SELECT HANDLER TYPE:
```

At this stage, specify 1, 2, 3 or O and press the return key.

SYMTEK BY-16 SERIES

Choose this entry when using the BY-16 series handlers made by SYMTEK.

ADVANTEST PRODUCTS

Choose this entry when using handlers made by ADVANTEST.

When ADVANTEST PRODUCTS is selected, the following six items are displayed to the menu. These menu items will be described later.

```
- DUT NUMBER SELECT OF STN1 HANDLER .. DISABLE
- DUT NUMBER SELECT OF STN2 HANDLER .. DISABLE
- DUMMY START ..... ENABLE
- SEND ALARM ..... DISABLE
- TIMER PAUSE ON CALIBRATION ..... DISABLE
- TESTER ACTION
```

ADVANTEST STANDARD

Choose this entry when using other handlers based on standard ADVANTEST specifications.

The other handlers based on standard ADVANTEST specifications refer to those that operate under GPIB handler control that conform to the "T5300/T5500+M3700 Series GPIB Handler Control Utility Program Specifications".

OTHERS

Choose this entry when the handler used is not the BY-16 series made by SYMTEK, the ADVANTEST handlers, or other handlers based on ADVANTEST specifications.

If OTHERS is selected, the following items must be set according to the messages displayed on the screen.

- 1) When the "OTHERS" is selected as the handler type, the following message is displayed on the WORK area. At this stage, enter the name of the handler type with up to 24 characters. When the name of the handler type is omitted, nothing is displayed on the screen.

```
[WORK]
SPECIFY HANDLER NAME (MAX 24BYTES):  Handler name_
```

When the name of the handler type is entered, the following items are added in the menu:

```
- INITIALIZE SUB NAME .....
- BEGIN SUB NAME .....
- END SUB NAME .....
```

When using a handler that has none of the protocols of the BY-16 series made by SYMTEK, the ADVANTEST handlers, or other handlers based on ADVANTEST specifications, note that handler control by the utility program cannot be supported.

Therefore, the control programs are required according to the protocol of the handler to be used.

The control programs required are shown as follows:

INITIALIZE SUB

Program to initialize the GPIB and handler

BEGIN SUB

Program to read out the target DUT information from a handler before a device measurement

END SUB

Program to transfer the measured results to a handler after a device measurement.

➔ For more information, refer to [38. 6 "Control Outline of Handler Other Than SYMTEK BY-16 Series and ADVANTEST" on page 38-39](#).

2) Input the three program names described above.

When three program names are entered, the menu screen is terminated.

If the [PF1] is entered without entering the program names, the warning messages will be displayed as follows.

[WORK]

```
INITIALIZE SUB NAME ..... NOT DEFINED!
BEGIN SUB NAME ..... NOT DEFINED!
END SUB NAME ..... NOT DEFINED!
SPECIFY THE PROGRAM NAME!
```

38. 3. 4 Specification of the Contactor Number and DUT Number (DUT NUMBER SELECT OF STN1/2 HANDLER)



Tip

Only for handlers made by ADVANTEST

The relationship between contactor and DUT numbers normally is predetermined by the number of devices to be tested in parallel.

This specification allows the user to set any desired relationship between contactor and DUT numbers irrespective of the number of devices to be tested in parallel.

However, the relationship between contactor and DUT numbers is automatically set when PBTYPE is specified for HIFIX. Consequently, the display "DUT NUMBER SELECT OF STN1 HANDLER" is removed from the menu when PBTYPE is specified for the station 1 HIFIX and "DUT NUMBER SELECT OF STN2 HANDLER" is removed from the menu when PBTYPE is specified for the station 2 HIFIX. As a result, this function cannot be used. At this time, be sure to set "DUT-NUMSEL Function" of the handler of the station where PBTYPE is set for HIFIX to "Automatic Recognition". If the handler's "DUTNUSEL Function" is set to "Handler side" or "Tester side", PBTYPE of HIFIX is not specified for the handler.

➔ For more information on the correspondence between contactor numbers and DUT numbers by PBTYPE, refer to "HF xxxxx HIFIX SPECIFICATION" attached to HIFIX or [Manual Performance Board PBTYPE Specifications\(No. : 8292985\)](#).

When PBTYPE is not specified for HIFIX, move the cursor to "DUT NUMBER SELECT OF STN1 HANDLER" or "DUT NUMBER SELECT OF STN2 HANDLER" on the menu screen and press the RETURN key. The following message will be displayed in the [WORK] area.

- When "DUT NUMBER SELECT OF STN1 HANDLER" is selected

[WORK]

SPECIFY DUT NUMBER SELECT OF STN1 HANDLER (D,E,F) _

- When "DUT NUMBER SELECT OF STN2 HANDLER" is selected

[WORK]

SPECIFY DUT NUMBER SELECT OF STN2 HANDLER (D,E,F) _

At this stage, specify D, E or F and press the return key. The function of each mode is described below.

When D is specified

If D is specified, the relationship between contactor and DUT numbers is disabled.

In this case, the relationship between contactor and DUT numbers is fixed as shown below.

- For parallel test of 16 Devices:
Contactors 1 to 8 and 17 to 24 correspond to DUTs 1 to 16.

- For parallel test of 8 Devices:
Contactors 1, 18, 3, 20, 5, 22, 7, and 24 correspond to DUTs 1 to 8.
- For measurements other than the above
Contactor numbers and DUT numbers correspond one for one.

When E is specified

If E is specified, the relationship between contactor and DUT numbers is enabled.

The information shown below is displayed in the [WORK] area and set the relationship between contactor and DUT numbers. Any desired relationship between contactor and DUT numbers can be specified by setting it up on this screen.

The number of contactors is limited to 32 or 64 if ENABLE is selected. Choose F and then select DATAFILE mode so that relationship between the contactor and DUT numbers can be set if the number of contactors in the handler is not 32 or 64.

- When E (Enable) is selected in "DUT NUMBER SELECT OF STN1 HANDLER"

```
[WORK]
DUT NUMBER SELECT OF STN1 HANDLER

CONTACTOR NO. : 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16
DUT NO.       : _1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16

CONTACTOR NO. : 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
DUT NO.       : 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 [33-64]
```

- When E (Enable) is selected in "DUT NUMBER SELECT OF STN2 HANDLER"

```
[WORK]
DUT NUMBER SELECT OF STN2 HANDLER

CONTACTOR NO. : 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16
DUT NO.       : _1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16

CONTACTOR NO. : 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
DUT NO.       : 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 [33-64]
```

Here, input the DUT numbers that correspond to each contactor number at station 1 (STN1) or station 2 (STN2). The DUT number that has been input is confirmed when you press the RETURN key and the cursor moves to the next contactor position. The cursor also can be moved to the immediately preceding contactor position by using the ← key or the immediately following contactor position by using the → key.

Placing the cursor over [33-64] and pressing the Return key displays the screen shown below.
You can now enter DUT numbers for contactor numbers between 33 and 64.

```
[WORK]
DUT NUMBER SELECT OF STN1 HANDLER

CONTACTOR NO. : 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48
DUT NO.       : _33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48

CONTACTOR NO. : 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64
DUT NO.       : 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 [1-32]
```

DUT numbers between 1 to 64 are available. However, if the test system is connected to a handler which is not capable of 64-DUT parallel tests, DUT numbers have to be the same as the corresponding contactor numbers for the contactor numbers between 33 to 64.

Specify DUT numbers for all contactors. The DUT numbers assigned to each contactor must all be unique.

When you have finished entering all DUT numbers, press the [PF1] key to exit this setup screen. If two or more contactors are found to have been assigned the same DUT number at this time, an error message is displayed as shown below and this setup screen cannot be exited. Change the overlapped DUT numbers.

◆ Example of Error Message

```
[WORK]
DUT NUMBER SELECT OF STN1 HANDLER

CONTACTOR NO. : 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
DUT NO.       : _2 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

CONTACTOR NO. : 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
DUT NO.       : 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 [33-64]
                ERROR!! DUT NO=2 IS OVERLAPPED
```

Tip

Contactor and DUT numbers are set when the GPIB command "DUTNUMSEL" is transferred from the test controller to the handler. When the used handler does not correspond to the "DUTNUMSEL" GPIB command, disable the specification.

The following shows examples for the specification of "DUT NUMBER SELECT OF STN1/2 HANDLER" and the resultant set of the DUT numbers on HIFIX. (In this example, an ADVANTEST M3700 series handler is used.)

◆ Example of Parallel Test of 32 Devices

[WORK]

DUT NUMBER SELECT OF STN1 HANDLER

CONTACTOR NO. : 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

DUT NO. : _1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

CONTACTOR NO. : 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32

DUT NO. : 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 [33-64]

The DUT numbers on HIFIX are selected as shown below.

Numbers in () represent the contactor numbers on the handler side.

Lane	@ 1	2	3	4	5	6	7	8
4th line	25 (25)	26 (26)	27 (27)	28 (28)	29 (29)	30 (30)	31 (31)	32 (32)
3rd line	17 (17)	18 (18)	19 (19)	20 (20)	21 (21)	22 (22)	23 (23)	24 (24)
2nd line	9 (9)	10 (10)	11 (11)	12 (12)	13 (13)	14 (14)	15 (15)	16 (16)
1st line	1 (1)	2 (2)	3 (3)	4 (4)	5 (5)	6 (6)	7 (7)	8 (8)

◆ Example of Parallel Test of 16 Devices

[WORK]

DUT NUMBER SELECT OF STN1 HANDLER

CONTACTOR NO. : 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

DUT NO. : _1 2 3 4 5 6 7 8 17 18 19 20 21 22 23 24

CONTACTOR NO. : 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32

DUT NO. : 9 10 11 12 13 14 15 16 25 26 27 28 29 30 31 32 [33-64]

The DUT numbers on HIFIX are selected as shown below.

Numbers in () represent the contactor numbers on the handler side.

Lane	@ 1	2	3	4	5	6	7	8
4th line	-- --	-- --	-- --	-- --	-- --	-- --	-- --	-- --
3rd line	9 (17)	10 (18)	11 (19)	12 (20)	13 (21)	14 (22)	15 (23)	16 (24)
2nd line	-- --	-- --	-- --	-- --	-- --	-- --	-- --	-- --
1st line	1 (1)	2 (2)	3 (3)	4 (4)	5 (5)	6 (6)	7 (7)	8 (8)

◆ Example of Parallel Test of 8 Devices

[WORK]

DUT NUMBER SELECT OF STN1 HANDLER

CONTACTOR NO. : 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

DUT NO. : _1 9 2 10 3 11 4 12 13 14 15 16 17 18 19 20

CONTACTOR NO. : 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32

DUT NO. : 21 5 22 6 23 7 24 8 25 26 27 28 29 30 31 32 [33-64]

The DUT numbers on HIFIX are selected as shown below.

Numbers in () represent the contactor numbers on the handler side.

Lane	@ 1	2	3	4	5	6	7	8
4th line	-- --	-- --	-- --	-- --	-- --	-- --	-- --	-- --
3rd line	-- --	5 (18)	-- --	6 (20)	-- --	7 (22)	-- --	8 (24)
2nd line	-- --	-- --	-- --	-- --	-- --	-- --	-- --	-- --
1st line	1 (1)	-- --	2 (3)	-- --	3 (5)	-- --	4 (7)	-- --

When F is specified

If F is specified, a new screen is displayed and you are prompted to specify a data file.

The data file is used to set the relationship between contactor and DUT numbers.

- If F is selected for DUT NUMBER SELECT OF STN1 HANDLER, the following screen is displayed:

```

DUT NUMBER SELECT OF STN1 HANDLER

CONT NO. : 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16
DUT NO.  : 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16

CONT NO. : 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
DUT NO.  : 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32

SPECIFY DUT NUMBER SELECT DATA FILE(STN1): ASC _

```

Specify the name of the file that includes the relationship between contactor and DUT numbers for Station 1 (STN1) or Station 2 (STN2).

If the return key is pressed without entering the file name, "DATAFILE" mode is set in the column of "DUT NUMBER SELECT OF STN1 HANDLER" or "DUT NUMBER SELECT OF STN2 HANDLER." The correspondence between the contactor number and the DUT number is as shown in the screen.

Up to 32 characters and ASC file specifier can be used for a file name.

Contents of the files are read only up to 1000 characters for a line and up to 4000 characters as a whole. When the total characters read exceeds 4000, the message "FILE SIZE OVER" is displayed and a prompt is displayed.

If the specified file name does not exist in the current directory or DP0, the message "FILE DOES NOT EXIST" is displayed and you are prompted to enter another file name.

The structure of the file specified as the DUT NUMBER SELECT DATA FILE is as follows:

- The number of contactors of handler must be specified just after C. Next, specify DUT numbers corresponding to the contactor numbers (1, 2, 3,..., n). An error, which indicates that the specified number of contactors does not match the number of contactors installed in the handler, occurs when /STHAND ends.
- A comma, space, tab or line feed must be placed between numbers.
- However, if the file contents satisfy one of the conditions described below, an error message is displayed and you are prompted to change the file.
 - The number of contactors does not match the number of specified DUTs.
 - The DUT number is larger than the maximum number of contactors.
 - The same DUT number is specified more than once.
- Comments can be written between ";" and the line feed character.

◆ DUT NUMBER SELECT DATA FILE Example

```

C64,                ; NUMBER OF CONTACTORS
1, 9,17,25,33,41,49,57, ; CONTACTOR 1- 8
2,10,18,26,34,42,50,58, ; CONTACTOR 9-16
3,11,19,27,35,43,51,59, ; CONTACTOR 17-24
4,12,20,28,36,44,52,60, ; CONTACTOR 25-32
5,13,21,29,37,45,53,61, ; CONTACTOR 33-40
6,14,22,30,38,46,54,62, ; CONTACTOR 41-48
7,15,23,31,39,47,55,63, ; CONTACTOR 48-56
8,16,24,32,40,48,56,64  ; CONTACTOR 57-64

```

If a file name is input, the file is opened and the relationship between contactor and DUT numbers is displayed.

```

DUT NUMBER SELECT OF STN1 HANDLER

CONT NO. : 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16
DUT NO.  : 1  9 17 25 33 41 49 57  2 10 18 26 34 42 50 58

CONT NO. : 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
DUT NO.  :  3 11 19 27 35 43 51 59  4 12 20 28 36 44 52 60

CONT NO. : 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48
DUT NO.  :  5 13 21 29 37 45 53 61  6 14 22 30 38 46 54 62

CONT NO. : 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64
DUT NO.  :  7 15 23 31 39 47 55 63  8 16 24 32 40 48 56 64

SPECIFY DUT NUMBER SELECT DATA FILE(STN1): ASC _

```

If the return key is entered without inputting the file name, the initial screen is displayed.

```

***STHAND *** (SELECT GP-IB I/F OF HANDLER)

- GP-IB ADDRESS OF STN1 HANDLER ..... 1
- GP-IB ADDRESS OF STN2 HANDLER ..... DISABLE
- HANDLER WAIT TIME ..... 15 SEC
- HANDLER TYPE ..... ADVANTEST PRODUCTS
- DUT NUMBER SELECT OF STN1 HANDLER .. DATAFILE
- DUT NUMBER SELECT OF STN2 HANDLER .. DISABLE
- DUMMY START ..... ENABLE
- SEND ALARM ..... DISABLE
- TIMER PAUSE ON CALIBRATION ..... DISABLE
- TESTER ACTION

[WORK]

[NOTE]-----
      PF1:EXIT

```

When DUT NUMBER SELECT OF STNn HANDLER is set to DATAFILE, specifying F for DUT NUMBER SELECT OF STNn HANDLER displays the relationship between contactor and DUT numbers and you can confirm the current setting.

Tip

- After a data file is entered, the file contents are displayed on the DUT NUMBER SELECT screen. After the data file name is entered, changes made to the file are not reflected on the DUT NUMBER SELECT screen.
- Even when the DUT NUMBER SELECT menu selection is changed to DISABLE, ENABLE or DATAFILE is changed for, the DUT NUMBER SELECT screen contents are not stored.
- The data file used for the DUT NUMBER SELECT screen cannot be modified using the DUT NUMBER SELECT ENABLE mode.

38. 3. 5 Specification of Dummy Start (DUMMY START)

Tip

Only for handlers made by ADVANTEST

After performing preprocessing before device measurement (e.g., analysis of service request sources, determination of the DUT to be measured, and special processing), specify whether or not to execute the test program when the DUT to be measured is nonexistent.

DUMMY START refers to executing the test program while the DUT to be measured is nonexistent.

Move the cursor to "DUMMY START" in the menu screen and press the return key. The following message will be displayed in the [WORK] area.

```
[WORK]
SPECIFY DUMMY START D:DISABLE
                        E:ENABLE
                        P:ENABLE & PRIOR TO NORMAL
SELECT DUMMY START MODE:
```

Here, specify D (DISABLE), E (ENABLE) or P (ENABLE & PRIOR TO NORMAL) for DUMMY START and press the RETURN key.

When D is specified

DUMMY START is disabled when D is specified.

After performing preprocessing before device measurement, the test program is STOP'd without being executed.

When E is specified

DUMMY START is enabled when E is specified.

After performing preprocessing before device measurement, the test program is executed while the DUT to be measured is nonexistent (SDUT = 0).

When P is specified

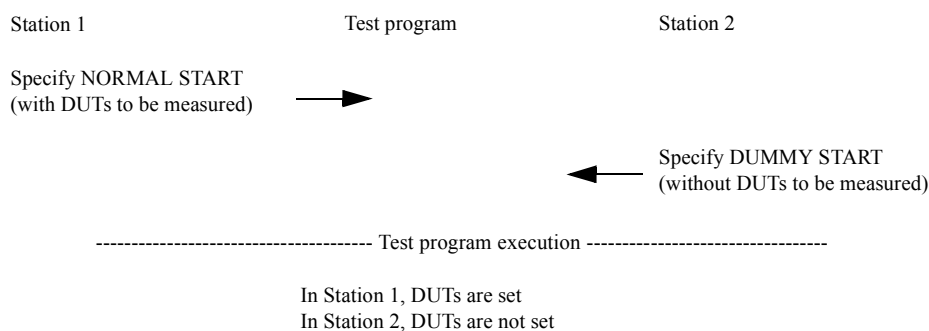
DUMMY START is set to the ENABLE & PRIOR TO NORMAL mode.

ENABLE & PRIOR TO NORMA is used so that DUMMY START has priority when NORMAL START (SDUT<>0) is specified for one station and DUMMY START (SDUT=0) is specified for the other station for an inter-station parallel test. When the test program is executed for both stations, it is assumed that DUMMY START is specified for both stations (SDUT=0).

Next, the test program is executed by NORMAL START from the waiting station. In this case, if NORMAL START (SDUT<>0) is also specified for the other station (where DUMMY START is specified), the test program is executed for both stations assuming that NORMAL START (SDUT<>0) is specified for both stations. Using this mode instead of ENABLE mode decreases the amount of time necessary to complete tests because both stations can be used simultaneously. The ENABLE & PRIOR TO NORMAL mode is effective at increasing the test efficiency when the test program execution time is longer than the HANDLER WAIT TIME.

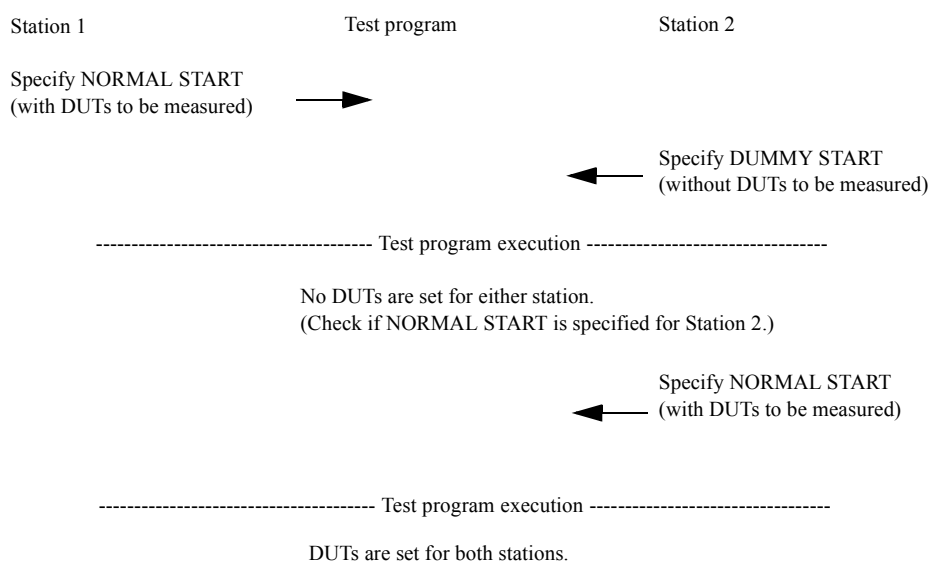
The difference between the ENABLE and ENABLE & PRIOR TO NORMAL modes is shown below.

[ENABLE mode]



In this case, the test efficiency is decreased because the test program for Station 2 cannot be executed till the program for Station 1 is completed.

[ENABLE & PRIOR TO NORMAL mode]



In this case, the test efficiency is increased because both station 1 and 2 can be executed at the same time provided that Station 2 is set with DUTs after the test program is executed once without DUTs.

38. 3. 6 Specifying Signal Tower Red Flash in Case of Error (SEND ALARM)



Only for handlers made by ADVANTEST

Specify whether the signal tower at a station to be tested in parallel is made to flash red when a run-time error or a system error occurs in the test controller.

Move the cursor to "SEND ALARM" in the menu screen and press the return key. The following message will be displayed in the [WORK] area.

```
[WORK]
SPECIFY SEND ALARM (D,E) _
```

At this stage, specify DISABLE or ENABLE with D or E and press the return key.

When D is specified

If D is specified, the signal tower is not made to flash red when a run-time error or a system error occurs.

When E is specified

If E is specified, the signal tower at a station to be tested in parallel is made to flash red when a run-time error or a system error occurs. The signal towers also are made to flash red when calibration has failed.



The signal tower is made to flash red by sending a GPIB command "TESTALARM" from the test controller to the handler. Therefore, when the used handler does not correspond to the "TESTALARM" GPIB command, disable the specification.

38. 3. 7 Stopping Timer when Executing Calibration (TIMER PAUSE ON CALIBRATION)



Only for handlers made by ADVANTEST

Specify whether the timer for the handler's maximum test time is made to stop when executing calibration.

If the calibration execution time is long, the handler's maximum test time may be exceeded. To avoid this problem, the timer for the handler's maximum test time need to be stopped before executing calibration and restarted after completion of the calibration.

Move the cursor to "TIMER PAUSE ON CALIBRATION" in the menu screen and press the return key. The following message will be displayed in the [WORK] area.

```
[WORK]
SPECIFY TIMER PAUSE ON CALIBRATION (D,E) _
```

At this stage, specify DISABLE or ENABLE with D or E and press the return key.

When D is specified

If D is specified, the timer for the handler's maximum test time is not stopped when executing calibration.

When E is specified

If E is specified, the timer for the handler's maximum test time is stopped before executing calibration and restarted after completion of the calibration.

Tip

- The timer for the handler's maximum test time is stopped and restarted by sending a GPIB command "TESTPEND1/0" from the test controller to the handler. For this reason, if the handler used does not support the GPIB command "TESTPEND1/0," disable this specification.
- If settings of the handler's maximum test time are invalidated, this specification does not have any effect.

38. 3. 8 Specifying Special Processing Operation Mode (TESTER ACTION)

Tip

Only for handlers made by ADVANTEST

Specify the operation mode used for special processing. The operation mode on the test system side can be selected according to the special processing performed on the handler side.

There are two operating modes used for special processing: PRETEST MODE and AUTO RESTART MODE.

Move the cursor to "TESTER ACTION" in the menu screen and press the return key. The currently displayed menu will change to the TESTER ACTION menu shown below.

```
**** SPECIFY TESTER ACTION ****
```

```
      . RETEST MODE ..... 1
      . AUTO RESTART MODE ..... DISABLE
```

```
[WORK]
```

```
[NOTE]-----
```

```
PF1:EXIT FOR TESTER ACTION MENU
```

To exit the TESTER ACTION menu to return to the main menu, press the [PF1] key here.

38. 3. 8. 1 RETEST MODE

To select RETEST MODE from the TESTER ACTION menu, move the cursor to "RETEST MODE" and press the RETURN key. The following message will be displayed in the [WORK] area.

```
[WORK]
```

```
SPECIFY RETEST MODE 1: CLEAR SORT DATA
                   2: SAVE & CLEAR SORT,CATEGORY DATA
SELECT RETEST MODE:
```

Here, the operation mode on the test system side in cases when lot management is exercised by the handler can be selected.

Specify 1 or 2 and press the return key.

When lot management is exercised by the handler, the following processing is performed on the test system side according to the selected mode.

When 1 is specified (RETEST MODE 1)

If a remeasurement request is received from the handler, the following remeasurement processing is performed.

The sort counter for the sort number that has been specified for remeasurement is cleared to 0, while at the same time subtracting remeasurement counts from the total and fail counters. This processing is performed individually for each station. The same processing is performed for the handler counters also. (This does not includes clearing the category counter.)

When 2 is specified (RETEST MODE 2)

In addition to the remeasurement processing that is performed when RETEST MODE 1 is specified, several other tasks are performed that include saving category and sort data to a file and clearing the category data.

- Specifying directory in which to store data files

If RETEST MODE 2 is specified, the following message is displayed in the [WORK] area.

[WORK]

SPECIFY DIRECTORY (MAX 32CHARACTER) :

Specify the directory where you want the save file for the category and sort data on the test system side to be stored.

If "ADI\$DP1" is specified (default specification), the file is stored in the current directory.

 Important

If the specified directory cannot be found when a directory name is specified, an error is assumed.

- Processing

In addition to the remeasurement processing that is performed when RETEST MODE 1 is specified, the following processing is performed:

The category and sort data on the test system side are saved to a file after completion of matrix measurement.

All of the test system category data is cleared after saving the data to a file.

The category and sort data on the test system side are saved to a file after completion of remeasurement.

This processing does not includes clearing the test system category data after completing the remeasurement.

The file that contains the saved data of the station to be measured is cleared before starting the next matrix measurement.

Processing is performed solely for the data of the station to be measured.

The following shows timings at which the file is saved and cleared.

Lot sequence	Saving file	Clearing saved data file and category data
Lot start Matrix measurement Lot end	Matrix measurement data is saved.	Saved data file is cleared. Category data is cleared.
Remeasurement lot start, 1st First remeasurement Lot end	Remeasurement data is saved.	Category data is cleared.
Remeasurement lot start, 2nd Second remeasurement Final lot end	Remeasurement data is saved.	



Tip

*When a saved data file is cleared, only the contents of the file are cleared.
The file itself is retained intact, with its file size set to 0.*

- Saved data file name

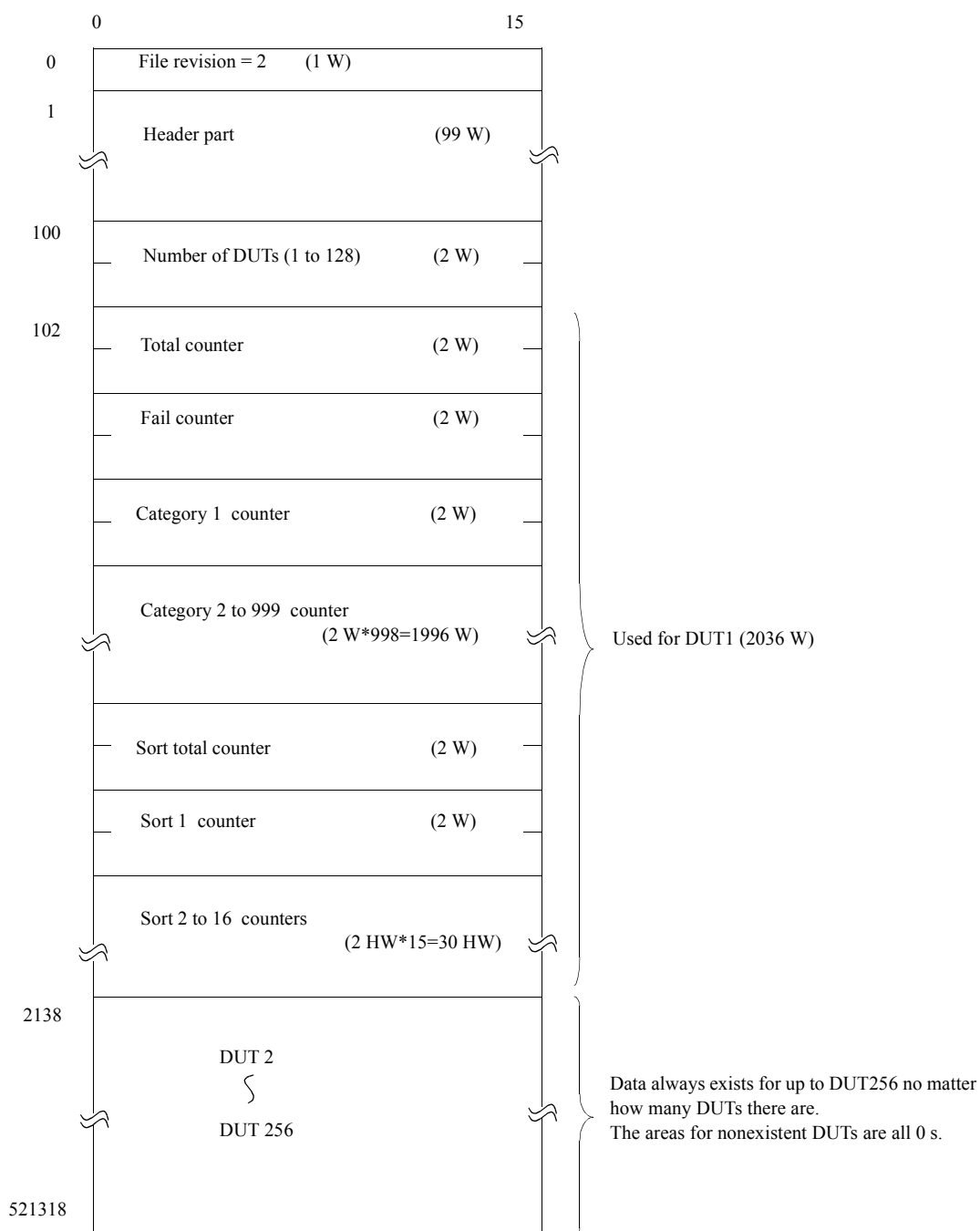
The data are saved to a specified directory in the file names listed below.

Data	File name
Matrix measurement data measured at station 1	DAT MTESTD1
First remeasurement data measured at station 1	DAT RTESTD11
Second remeasurement data measured at station 1	DAT RTESTD12
Matrix measurement data measured at station 2	DAT MTESTD2
First remeasurement data measured at station 2	DAT RTESTD21
Second remeasurement data measured at station 2	DAT RTESTD22

- Format of saved data file

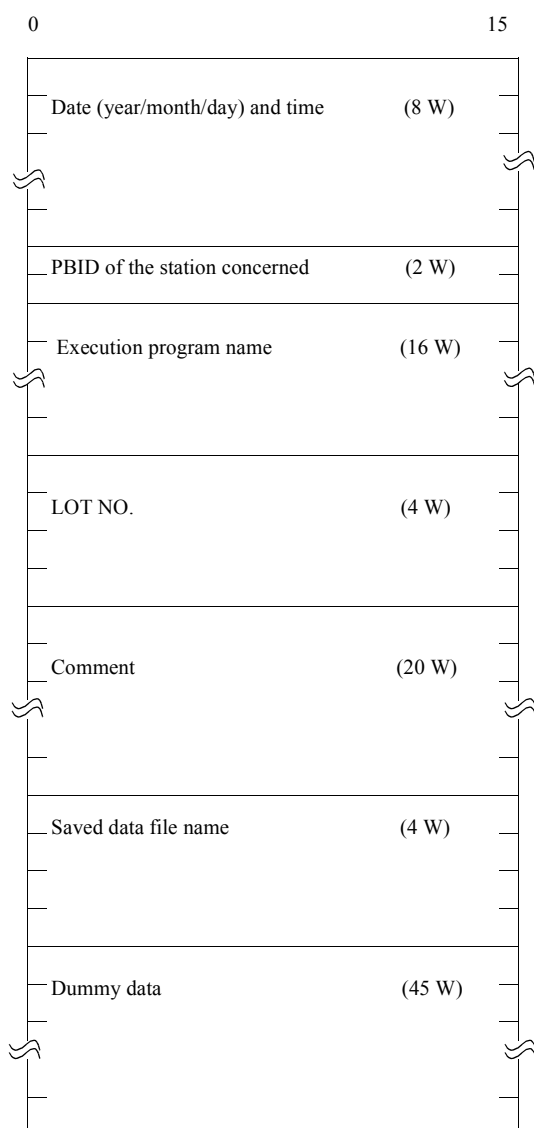
The files are saved in the format shown below.

Save format : Binary (Header part in ASCII)
 The data of each DUT is saved.
 File size : 521318 words (1 word = 16 bits)



Format of heater part

Save format : ASCII
 File size : 99 words (1 word = 16 bits)



Each information when data was saved is stored here.

Date (year/month/day) and time

Indicates the date and time when the data was saved.

yyyy/mm/dd HH:MM

yyyy for calendar year (4 digits)

mm for month

dd for day

HH for hours (24-hour system)

MM for minutes

(Example)1997/06/25 12:00

PBID of the station concerned

Stores the ID number (hex) that was set by the ID switch of the performance board in ASCII.

Execution program name

Execution program name that was executed when saving data.

LOT NO.

Indicates the content of system variable SUBLOT (decimal) stored in ASCII.

Comments

Indicates the content of system variable COMMEN.

Saved file name

Indicates the file name in which the data was saved.

MTESTD1/RTESTD11/RTESTD12/MTESTD2/RTESTD21/RTESTD22

Dummy data

This is a dummy area (all 0 s).

The saved data files can be called up by the lcommand utility for displaying the data contents.

38. 3. 8. 2 AUTO RESTART MODE

◆ Menu Operation

To select AUTO RETEST MODE from the TESTER ACTION menu, move the cursor to "AUTO RETEST MODE" and press the RETURN key. The following message will be displayed in the [WORK] area.

```
AUTO RESTART MODE (D:DISABLE/E:ENABLE) :
```

At this stage, specify D or E and press the return key.

The meaning for each mode is as follows:

When E is specified

AUTO RESTART MODE is enabled, and the following message will be displayed in the [WORK] area.

```
AUTO RESTART MODE (D:DISABLE/E:ENABLE) :E
AUTO RESTART MAXIMUM WAIT TIME (SEC)   :
```

Specify the maximum wait time counted from the start of the test program execution (AUTO RESTART MAXIMUM WAIT TIME) (sec) in a range between 0 to 3600.

The time entered will be displayed in the "AUTO RESTART MODE" field as shown below.

```

**** SPECIFY TESTER ACTION ****

      . RETEST MODE ..... 1
      . AUTO RESTART MODE ..... 1000SEC

[WORK]

[NOTE]-----
      PF1:EXIT FOR TESTER ACTION MENU
  
```

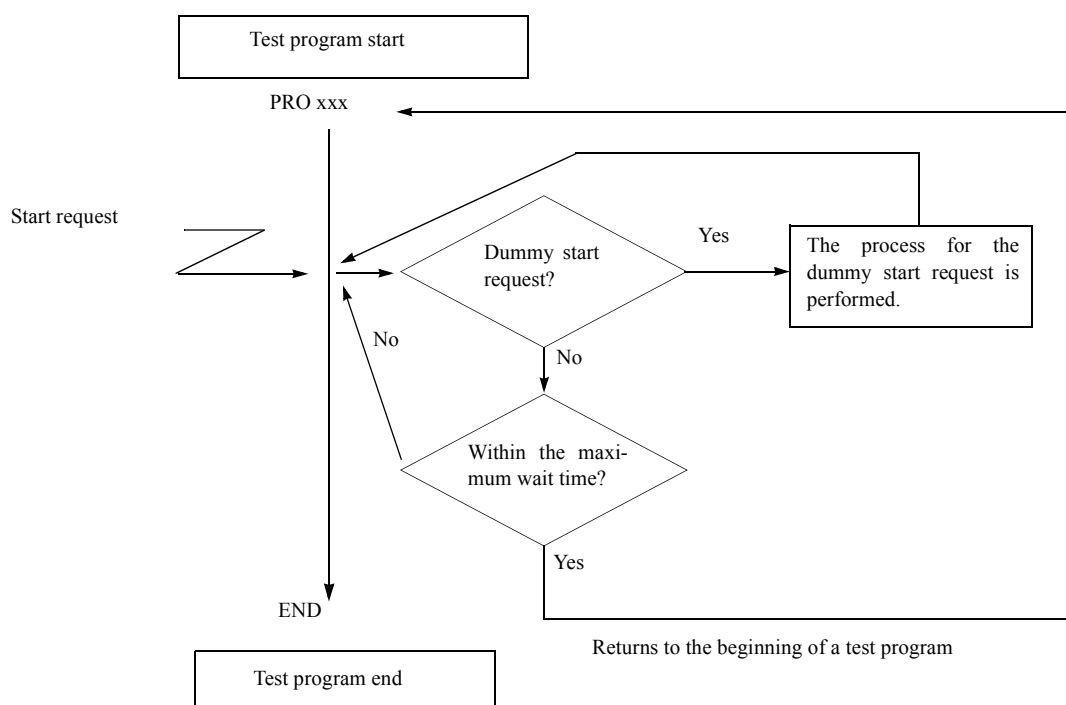
When "D" is specified

When "D" is specified, AUTO RESTART MODE is disabled and "DISABLE" is displayed in the "AUTO RESTART MODE" field as shown below.

◆ Operation in AUTO RESTART MODE

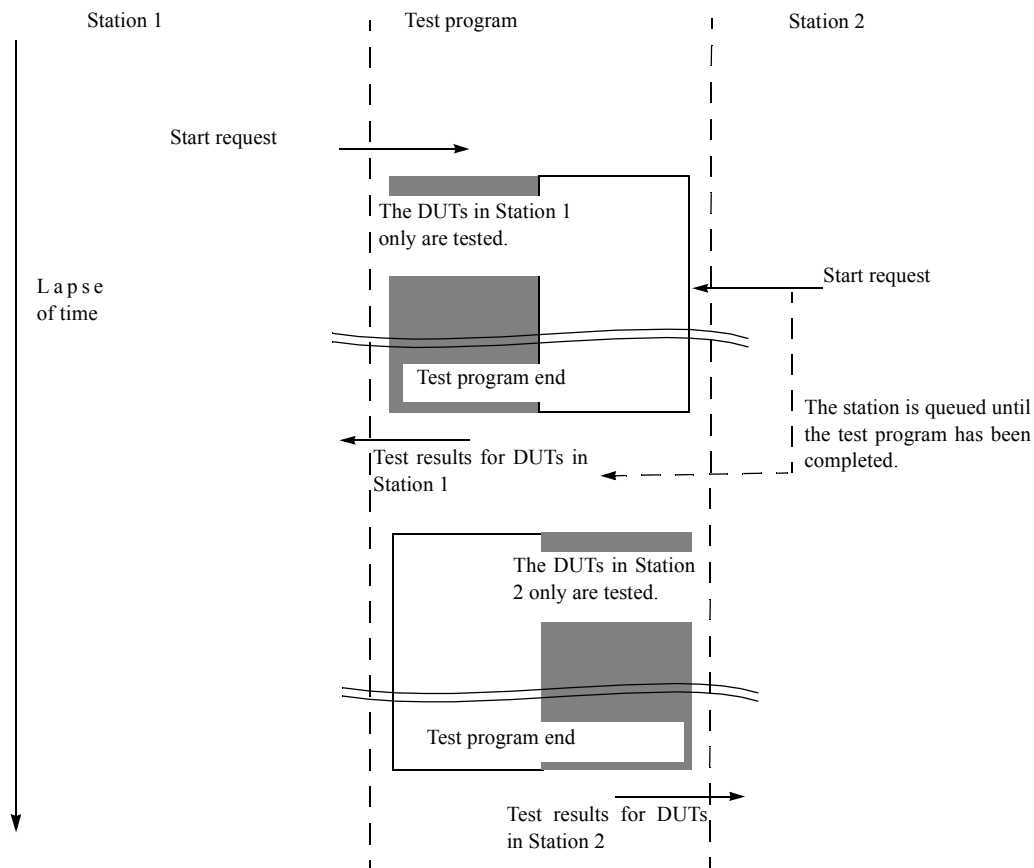
While the tester is executing a test program based on the start request from one of two stations in AUTO RESTART MODE, if a start request from the other test station comes to the tester, the tester responds to that request.

Figure 38-1 Operation in AUTO RESTART MODE Overview



Parallel test using two stations in AUTO RESTART MODE set to DISABLE: When the tester tests devices in parallel between two stations and the AUTO RESTART MODE is set to DISABLE: While the tester is executing a test program based on the start request from one of two stations, if a start request from the other test station comes to the tester, the tester holds the response to that request until the current test is complete.

Figure 38-2 Operation when AUTO RESTART MODE Is Set to DISABLE



If one of two stations sends a start request to the tester, and if the other test station does not send a start request within the specified HANDLER WAIT TIME, the tester starts testing with only one station. The HANDLER WAIT TIME starts when the tester receives the first start request.

When the tester tests devices in parallel between two stations and the AUTO RESTART MODE is set to ENABLE: While the tester is executing a test program based on the start request from one of two stations, if a start request from the other test station comes to the tester within the AUTO RESTART MAXIMUM WAIT TIME, the tester cancels the current testing and restarts testing with two stations. The AUTO RESTART MAXIMUM WAIT TIME starts when the tester begins testing.

Figure 38-3 Operation when the AUTO RESTART MODE is Set to ENABLE and the Other Test Station Sends a Start Request within the AUTO RESTART MAXIMUM WAIT TIME

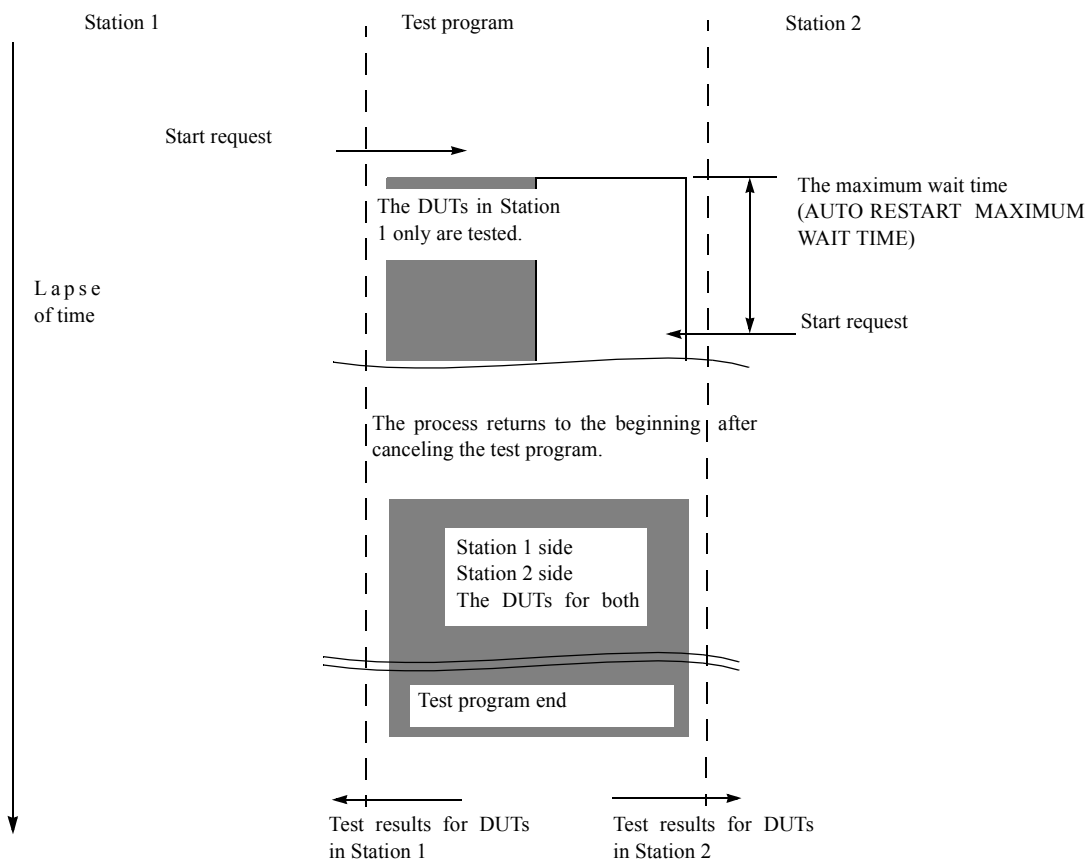
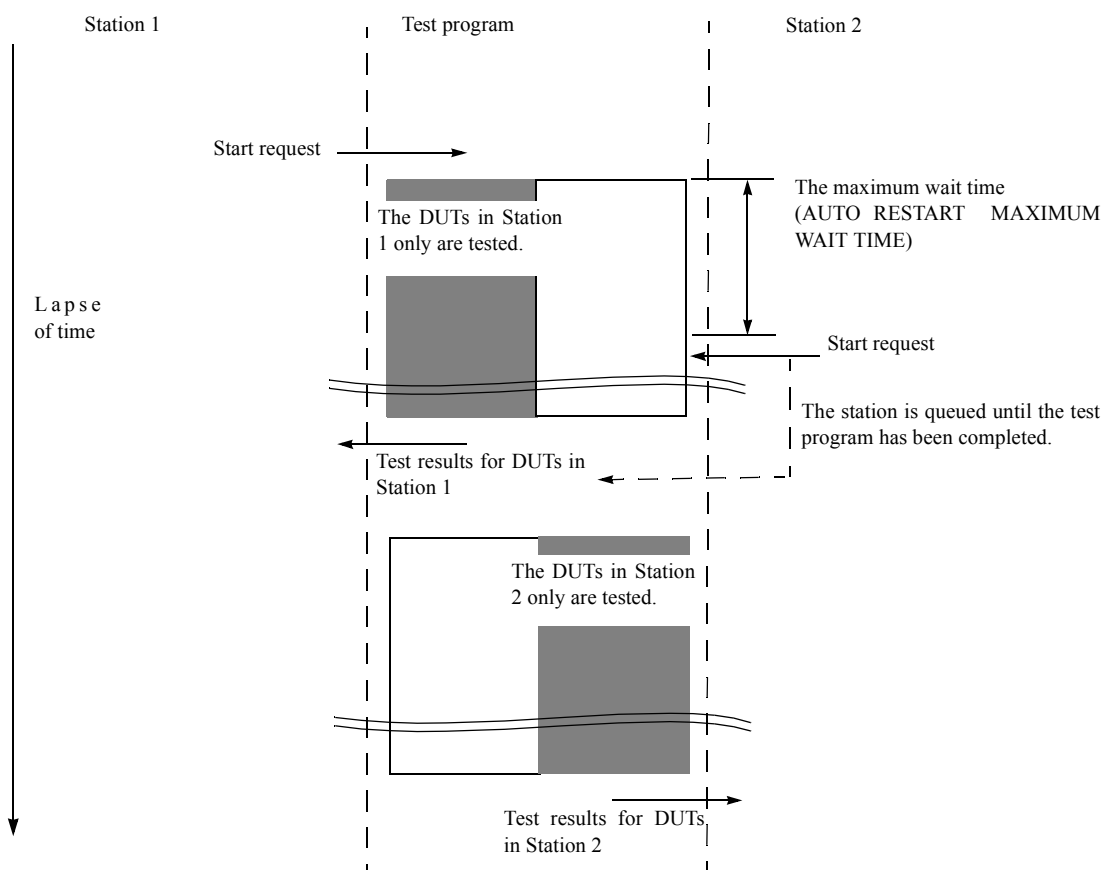


Figure 38-4 When the Other Side of the Two Stations Starts after the Maximum Wait Time (in AUTO RESTART MODE Set to ENABLE)



By using the AUTO RESTART MODE, the number of tests using only one station is reduced and the test efficiency will increase.

When the AUTO START MODE is set to ENABLE, a dummy start request is immediately executed without waiting for the test end.

👉 Limitation

- AUTO RESTART MODE can be enabled when the test program using two stations is used.
- When a start request, which occurs while a MEAS statement is being executed, is queued until the MEAS statement is completed.

38. 3. 9 Completion of "/STHAND" Command

Press the [PF1] key to end the "/STHAND" command. The following message is displayed before the "/STHAND" command ends.

```
*** END OF JOB (STHAND) ***
```

Tip

If "ADVANTEST PRODUCTS" is selected as the handler type, make sure the handler power is turned on when the /STHAND utility has finished.

(The specified contactor number and DUT number are sent to the handler when the utility is finished.)

If the handler power is still turned off when the utility has finished, the following message will be displayed.

When the GPIB interface address for either GP-IB ADDRESS OF STN1 HANDLER or GP-IB ADDRESS OF STN2 HANDLER is specified

```
STNn HANDLER NOT RESPONSE, HANDLER POWER ON ? (n represents a STN number)
```

When the GPIB interface addresses for both GP-IB ADDRESS OF STN1 HANDLER and GP-IB ADDRESS OF STN2 HANDLER are specified

```
STN1 OR STN2 HANDLER NOT RESPONSE, HANDLER POWER ON ?
```

Turn on the power supply for the handler if the power supply is off and execute /STHAND again.

If /STHAND is not executed here again, the following error messages are given when the handler is started and the system stops without performing measurement.

```
ERR:/STHAND NOT COMPLETED, RE-EXECUTE /STHAND
```

The number of contactors in the DUT NUMBER SELECT DATA FILE used for Station n is not specified for the handler.
Check the DUT NUMBER SELECT DATA FILE contents and fix accordingly.

The number of contactors specified in DUT NUMBER SELECT DATA FILE on the station n side does not coincide with the number of the contactors of the handler.

Check the contents of DUT NUMBER SELECT DATA FILE.

When the handler type OTHERS is selected, the program for initializing (INITIALIZE SUB) is started. If the program for initializing cannot be executed, the following message is displayed.

```
FILE DOSE NOT EXIT: SUB program name GP-IB CAN NOT INITIALIZE!  
*** END OF JOB (STHAND) ***
```

This message is for giving an alarm. Information which has been entered is set correctly.

The data set once is valid until a new setting is performed. Also, they are valid for other customers.

38. 3.10 Error Messages

When "3. ADVANTEST PRODUCTS" is specified as the handler type to be used, an error message may be displayed. Each error message and possible solutions are described below.

➔ For more information on handler types, refer to [38. 3. 3 "Specification of Handler Type to be Used \(HANDLER TYPE\)" on page 38-5.](#)

DUT NUMBER SELECT setting on the tester does not match the setting on the STNn handler.
Change DUTNUMSEL Function to Handler Side on the STNn handler, or check DUT NUMBER SELECT STNn Handler for the correct setting on the tester.

The station n DUT NUMBER SELECT settings for the handler and tester do not match. Do one of the following:

- Change the "DUTNUMSEL Function" on the side of the handler connected to station n to Handler Side.
- Change the setting of "DUT NUMBER SELECT STNn Handler" of station n to ENABLE on the tester side.

DUT NUMBER SELECT setting on the tester does not match the setting on the STNn handler.
Change DUTNUMSEL Function to Tester Side on the STNn handler, or check DUT NUMBER SELECT STNn Handler for the correct setting on the tester.

The station n DUT NUMBER SELECT settings for the handler and tester do not match. Do one of the following:

- Change the "DUTNUMSEL Function" on the side of the handler connected to station n to Tester Side.
- Change the setting of "DUT NUMBER SELECT STNn Handler" of station n to DISABLE on the tester side.
- Set the correspondence between the contactor number and DUT number correctly.

DUT NUMBER SELECT setting on the tester does not match the setting on the STNn handler.
Change DUTNUMSEL Function to Auto Confirmation Side on the STNn handler.
Check HIFIX TYPE setting on the STNn handler is correct.

The station n DUT NUMBER SELECT settings for the handler and tester do not match. Do one of the following:

38. 3 Operating Method of "/STHAND" Command

- Change the "DUTNUMSEL Function" on the side of the handler connected to station n to Auto Confirmation.
- Set HIFIX TYPE correctly for the handler.

In the parallel measurement, the number of DUTs under test that is set in the test program does not match the number set for HIFIX. Make sure the number set in the test program or in STNn handler's HIFIX TYPE is correct.

The number of devices tested in parallel in station n is different between the handler and tester. Set HIFIX TYPE correctly for the handler.

DUT NUMBER SELECT setting on the tester does not match the setting on the STNn handler.
Change DUTNUMSEL Function to Tester or Handler Side on the STNn handler.

DUT NUMBER SELECT is set to Auto Confirmation mode on the side of the handler connected to station n. Change the setting of DUT NUMBER SELECT on the handler side to Tester Side/Handler Side.

The number of DUTs under test that is set in the test program does not match the number of DUTs setting on STNn Handler.

The handler in station n cannot handle the number of DUTs specified for parallel tests in the test program.

Change the number of DUTs to be tested in parallel in the test program.

DUT NUMBER SELECT OF STNn setting does not match the number of DUTs under test that is set in the test program.

The test program cannot be executed with the current DUT NUMBER SELECT OF STNn setting as made by the /STHAND utility. For the contactor numbers between 33 to 64, DUT numbers have to be the same as the corresponding contactor numbers.

The number of DUTs under test that is set in the test program does not match the number of DUTs setting on STNn HIFIX.

HIFIX in station n cannot handle the number of DUTs specified for parallel test in the test program. Change the number of DUTs specified for the test program into the number of DUTs which HIFIX can hold.

PBTYPE No. settings on DSA1 and DSA2 does not match on STNn HIFIX.

The PBTYPE No. settings are different between STNn HIFIX DSA1 and DSA2.

Use the same PBTYPE number for DSA1 and DSA2.

There is an incorrect DUT number specified
in the DUT NUMBER SELECT DATA FILE used for Station n.
Check the DUT number and fix accordingly.

An incorrect DUT number is specified in DUT NUMBER SELECT DATA FILE on the station n side.

Check the DUT number specified in DUT NUMBER SELECT DATA FILE.

The number of contactors does not match
the number of specified DUTs
in the DUT NUMBER SELECT DATA FILE used for Station n.
Check the number specified
in the DUT NUMBER SELECT DATA FILE and fix accordingly.

The number of contactors set in DUT NUMBER SELECT DATA FILE on the station n side does not coincide with the number of specified DUT numbers.

Check the numbers specified in DUT NUMBER SELECT DATA FILE.

38. 4 Control of SYMTEK BY-16 Series and ADVANTEST Handlers

38. 4. 1 Overview

GPIB in ADVANTEST handlers and the SYMTEK BY-16 series handlers are controlled by the following five programs:

SUB STHAND

Program to setup the handler type and GPIB address

SUB INHAND

Program to initialize the GPIB

SUB RGPBSRQ

Program to read the SRQ status byte

SUB WTHAND

Program to wait the start request of another handler, setup the SDUT, program to process remeasurements and double-tests.

SUB SRTHAND

Program to transfer the sort data

The remeasurement and double-test processing are performed only when a request for remeasurement or double-test is received from the handler.

Remeasurement processing

This processing is performed according to the RETEST MODE specified in the TESTER ACTION menu.

➔ For more information, refer to [38. 3. 8 "Specifying Special Processing Operation Mode \(TESTER ACTION\)" on page 38-19](#).

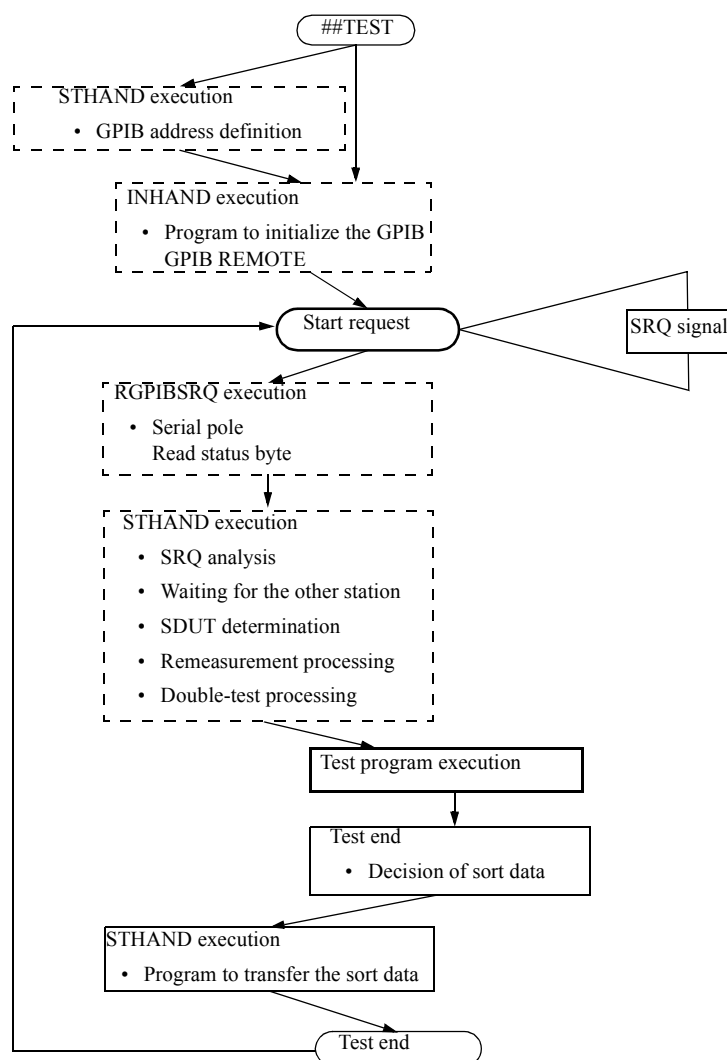
Double-test processing

The number of DUTs that have been classified into a double-test sort number by the immediately preceding test is subtracted from the sort counter, while at the same time subtracting a double-test count from the total and fail counters. (Refer to the memo.) This processing is performed individually for each station.

Limitation

The double-test sort numbers that can be used on the tester side are 1 to 16.

The relationship of these programs is as follows:



38. 4. 2 Precautions on Using Control Programs

- The contents set by /STHAND normally are saved to SYS MCONT.BF and SYS STHAND. Therefore, the setup contents are reflected in the entire system and are retained intact even after the power supply is shut down.
The contents set by a command line "/STHAND file name" are saved to a specified file, as well as to the above files.
- When the GPIB control is not performed, turn all the GPIB addresses to DISABLE.
- The wait time should be set few seconds longer than the index time of handler.
- When the test is executed only in 1 station, it is efficient to turn the GPIB address of unused stations to DISABLE.

38. 4 Control of SYMTEK BY-16 Series and ADVANTEST Handlers

- When DISABLE is selected from the SEND ALARM menu, observe the following note.
If a runtime error occurred in user program when a handler is controlled by the GPIB, the handler becomes in condition of testing.
Reset the I/F in the handler side to restart or enter the next program on the console of the tester side.

`/SRTHAND RETEST ALL.`

Also, enable the device of the contactor to execute the RETEST from the device program as follows:

```
ON ERR:RTERR

RTERR: SET SDUT #0
        CALL SUB SRTHAND("RETEST")
        STOP
```

- When starting from the VKT or C-BOX, the control cannot be performed by the GPIB. The device program is executed as none of DUT (SDUT) to be measured.

38. 5 Control Outline of Other Handlers Based on Standard ADVANTEST Specifications

38. 5. 1 Overview

GPIB in other handlers based on standard ADVANTEST specifications are controlled by the following five programs:

SUB STHAND

Program to setup the handler type and GPIB address

SUB GPINT.AT

Program to initialize the GPIB (equivalent to SUB INHAND)

SUB RGPBSRQ

Program to read the SRQ status byte

SUB GPBGN.AT

Program to wait the start request of another handler, setup the SDUT (equivalent to SUB WTHAND)

SUB GPEND.AT

Program to transfer the sort data (equivalent to SUB SRTHAND)

Program operations are almost the same as those of the ADVANTEST handlers except for a few functions such as remeasurement and double-test processing.

➔ For more information on the program specifications, refer to [T5300/T5500+M3700 Series GPIB Handler Control Utility Program Specifications\(No. : 8269079\)](#).

38. 5. 2 Precautions on Using Control Programs

If the protocol of the handler used matches the control program specifications in every detail, the control programs can be used directly as they are. However, if specifications differ partly or the user's exclusive functions are added, modify the control programs following the procedure below and choose HANDLER TYPE = OTHERS.

- 1) This is necessary to ensure that the control programs are matched to the handler used. Copy the following three handler control programs from \$AT_ADP0R to the user directory. (For all of the three programs regardless of whether they require a modification)

SUB GPINT.AT

SUB GPBGN.AT

SUB GPEND.AT

38. 5 Control Outline of Other Handlers Based on Standard ADVANTEST Specifications

- 2) Change the handler control program names. (For all of the three programs regardless of whether they require a modification)
- 3) Change the handler control programs as necessary and translate.
- 4) Select HANDLER TYPE=OTHERS.
- 5) For the handler control programs to be specified in OTHERS, specify the three handler control programs that have been modified in 2) above.

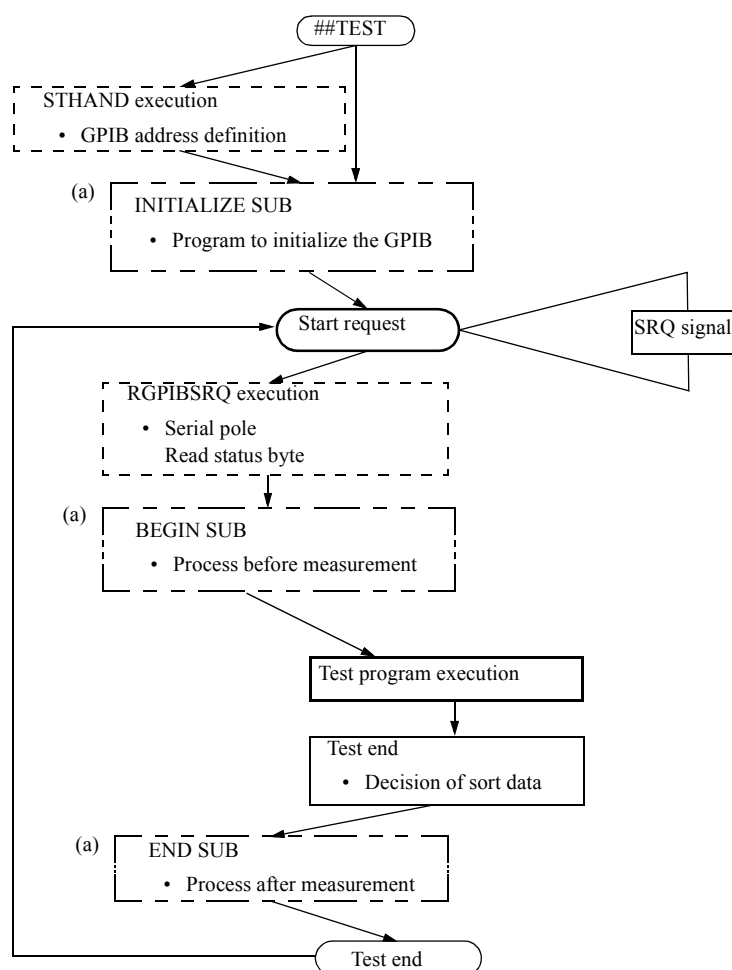
Always be sure to follow the above procedure when modifying the control programs (SUB GPINT.AT, SUB GPBGN.AT, and SUB GPEND.AT).

Note that control program specifications may be changed without prior notice.

38. 6 Control Outline of Handler Other Than SYMTEK BY-16 Series and ADVANTEST

38. 6. 1 Overview

When using a handler other than SYMTEK BY-16 series or ADVANTEST, program the contents inside of (a) according to the protocol of each handler. The system recognizes the created control program by the /STHAND.



38. 6. 2 Precautions on Using Control Programs

- When the STHAND is terminated or the controller is activated, the program for initializing (INITIAL SUB) starts. Therefore, if the program for initializing is not provided, a warning message or runtime error occurs.

- In the program for process before measurement (BEGIN SUB), the DUT to be measured should be set by the SET SDUT statement.
- The program for process after measurement (END SUB) is only once activated when the following statement in the device program is executed.
SET DECISION statement
STOP statement
END statement

38. 6. 3 Instruction for GPIB Control

The ATL language provides the following statements as to send or receive data by the GPIB. The control programs are created using these statements as follows:

- SEND GPIB function name [Data]

Function name

CLEAR, REMOTE, LOCAL, GTL, SDC, GET, LLO, DCL, SPE, SPD, UNL, UNT

- SEND GPIB MLA listener address
- SEND GPIB MTA Talker address
- SEND GPIB DATA Data
- SEND GPIBn TO v1 v2 ... v36
- TIME OUT t :GPIB:Label
- ON GPSRQ: Label

➔ For more information, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498 \)](#).

39. *FORK*

This chapter describes the FORK utility for executing a shell command.

39. 1 FORK Overview

The FORK utility program generates a subprocess on the environment processor, based on the main program, and executes a shell command.

The shell commands are called as the commands in this manual.

Using FORK, FORKLIST, FORKWAIT, and FORKKILL commands, you can call up, and then execute, cp, ls, and other commands from the main program.

The descriptive formats, arguments, etc. of these commands are described in order below.

Limitation

- MOVE, GDUMP, GLOAD, XTRANS, or other commands that use a tester processor cannot be used as the arguments of the FORK command.
- The tool "/FORK" allows to activate the environment processor command independent of test controller.

Note that the following when /FORK is used in combination with the test controller command, especially [U7] (key command U and mode 7).

The command executed with /FORK inherits the environment in which the test controller was started.

Therefore, the object directory of the command execution is the same as that when the test controller was started even through the [U7] command (change of customer) is executed.

For example, if the customer is changed with the [U7] command then a data file created with the device program is deleted by executing '/FORK rm data-file-name', an error "The file does not exist." arises.

```
% setenv WRK1 /home/asx/work1
% pwd
/home/asx
% test
/FORK pwd
/home/asx
U7 WRK1 ← Changes the controller environment to WRK1.
/FORK pwd ← The UNIX environment of FORK does not change because it succeeded the test execution environment.
/home/asx
% pwd
/home/asx
% test
U7 WRK1
/FORK rm workfile.dat
No such file or directory
```

Therefore, 'workfile.dat' cannot be deleted with this procedure even if 'workfile.dat' exists in /home/asx/work 1.

39. 2 FORK

FORK can be called up using one of the following two methods:

Tip

When you use the NO WAIT mode, up to a maximum of 24 processes of process status table data will be reserved to retain the completion status of the current command.

Thus, up to a maximum of 24 subprocesses can be executed at the same time during this mode. The status table will be released when the completion status is confirmed using the LIB FORKWAIT(or /FORKLIST) command.

Before using the NO WAIT mode in excess of 24 times, therefore, you must execute the LIB FORKWAIT(or /FORKLIST) command to check that the current command has been executed to completion.

39. 2. 1 /FORK

Function: This command is used to execute commands from the test controller.

Syntax

```
/FORK [Command string]
```

Description

On completion of execution of the entered command character string, the active subprocess terminates and control is returned to the controller.

If the command string is omitted, a prompt of the shell command is output.

When an exit is entered, the controller is returned to ready.

```
% TEST

CONTROLLER READY

=
/FORK ps      ! If the command string is specified:

  PID TT STAT TIME COMMAND
  125 p0 S    0:00 -csh (csh)
  536 p0 S    0:00 /bin/csh -f /home/atsys/all/system/execute_on_tp1 stest
  569 p0 S    0:00 tt stest
  570 p0 S    0:00 sh /tmp/forkfile_569.9
  571 p0 R    0:00 ps
  148 p1 IW   0:00 -csh (csh)

=
/FORK date
Sat Sep 26 09:48:39 JST 1992
=
/FORK ls *.asc
a.asc      demod.asc  fk24a.asc  fk291.asc  fk50.asc
atls spawn.asc e.asc      fk25.asc  fk292.asc  fk51.asc
demo.asc    fk20.asc   fk26.asc  fk30.asc   fk52.asc
demo2.asc   fk21.asc   fk26a.asc fk31.asc   hawk.asc
demo3.asc   fk22.asc   fk27.asc  fk311.asc  kill.asc
demo4.asc   fk23.asc   fk28.asc  fk32.asc   mouse.asc
demoa.asc   fk24.asc   fk29.asc  fk33.asc   wait.asc

=
/FORK vi b.asc

----- vi b.asc -----
This file name is b.asc.

-----

=
/TYPE B
This file name is b.asc.

=
/FORK rm b.asc && echo OK
OK

=
```

```
% TEST

CONTROLLER READY

=
/FORK                                ! If the command string is omitted:

Type exit for return

% vi a.asc
----- vi a.asc -----
ASX/U                                1992-9-26
FORK library routines
1.1

-----

% ps
  PID TT  STAT TIME COMMAND
  125 p0  IW   0:00 -csh (csh)
  467 p0  IW   0:00 /bin/csh -f /home/atsys/all/system/execute_on_tp1 stest
  500 p0  IW   0:00 tt stest
  501 p0  IW   0:00 sh /tmp/forkfile_500.9
  502 p0  S    0:00 -sh (csh)
  519 p0  R    0:00 ps
  148 p1  IW   0:00 -csh (csh)
%

% who
ho  tty0    Sep 26 09:04  (crow)
ho  tty1    Sep 26 09:07  (crow)
%

% pwd
/home/ho/spawn
%

% exit
%

/TYPE A
ASX/U                                1992-9-26
FORK library routines
1.1

=
```

39. 2. 2 CALL LIB FORK

Function: This is used to execute any DCL command from the main program by calling up LIB FORK.

Syntax

```
CALL LIB FORK(command,mode,process_name,output_file,input_file,
               process_id,error_code,error_status,completion_code)
```

Argument: command

A character string of the shell command to execute. It must be up to 132 characters.

Type

STRING

Access

Input

mode

This argument specifies the command execution mode. Specify the WAIT or NO WAIT mode character string. If you use the WAIT mode, execution of the command will be followed by the return of control from LIB FORK. If you use the NO WAIT mode, control will return to the controller when call for command execution is completed.

Type

STRING

Access

Input

Limitation

You must use the WAIT mode when executing the editor or other commands that need to be entered from the keyboard. If such commands are executed in the NO WAIT mode, neither keyboard entry nor echoback will be correctly performed since contention for the higher priority of input will occur between the subprocess and the main process. Also, as mentioned previously, after having used the NO WAIT mode, you must execute the LIB FORK (or /FORK- LIST) command to ensure that the current command has been executed to completion.

process_name

Process name string of the sub process. It must be up to 15 characters.

Type

STRING

Access

Input

output_file

This argument specifies an output file name character string.

Type

STRING

Access

Input

input_file

This argument specifies an input file name character string.

After the desired command and this input file have been given, that command is executed first and then the command procedure that has been specified by input file is executed.

Type

STRING

Access

Input

process_id

This argument returns the process ID (identification) code of the subprocess which has been generated using the NO WAIT mode.

For the WAIT mode, this code is not set.

Type

INTEGER[2]

Access

Output

error_code

This argument returns the error code corresponding to the particular error code of MOST.

The error code -1 will be returned when call for subprocess generation comes to a normal end.

Type

INTEGER[1]

Access

Output

error_status

This argument returns the status code that will occur if execution of the subprocess generation request from the environment processor fails.

Type

LOGICAL[2]

Access

Output

completion_code

This argument returns the completion status of the executed command.

Type

LOGICAL[2]

Access

Output

Description

All the arguments listed above can be omitted. Omission of an argument is followed by the prompt display, which requests for entry of a command.

However, since the order of argument specification is predetermined, specification of part of the arguments cannot be skipped in the middle of the program.

For example, if you specify arguments as follows, the third argument will not be correctly processed:

```
CALL LIB FORK(COMMAND, MODE, INFILE)
```

When omitting the string argument, use the variable to be omitted as the NULL string.

```
PNAME=" "CALL LIB FORK(COMMAND, MODE, PNAME, INFILE)
```

If arguments are to be placed after those not of the string-variable type, those arguments cannot be omitted.

 Tip

When executing a command in the NOWAIT mode, you must follow the instructions given below.

- *The controller terminates even when a process started in the NOWAIT mode is still running. Therefore, if you need to check the command termination, always use the FORKLIST or FORKWAIT option.*
- *If a command is executed in the NOWAIT mode, the fork_error.log error log file is created in the directory of AT ALOG.*

This AT ALOG is the current directory (default). If an error has occurred, the message is appended to this file. We recommend you to periodically check and delete the messages.

Use this function to check errors.

39. 2. 3 Error Codes (error_code, error_status, completion_code)

The occurrence of an error during use of LIB FORK causes the appropriate error code to be set in the error_code, error_status, or completion_code argument, depending on the particular type of processing phase.

The following lists major error codes and causes:

Argument name and error codes	Error cause and corrective action
error_code 10	The fork_error.log file cannot be opened. In this case, delete fork_error.log.
11	This code denotes the absence of a free process status table. If this code occurs, execute the LIB FORK (or /FORKLIST) command to check for the end of the immediately preceding process.
24	The contents of the LIB FORK argument are not correct. Or, sub process cannot be created. If this code occurs, error_status will be set. error status
error_status	A UNIX error will be set if error code 24 of error_code argument occurs. If vfork, execlp or waitpid has failed, its error is set.
completion_code	The error_code and error_status arguments indicate that an error has occurred during command-based issuance of an execution request after a subprocess was generated, whereas the completion_code argument denotes the results of execution of that command. This status, therefore, will be set when the command is executed in the WAIT mode. The status returned by waitpid is set.

39. 2. 4 Sample Program for Using LIB FORK

The following gives a programming example to execute `ls *.lib` and `a`, and outputs its result to the `sample.log`:

```
----- sample.asc -----
PRO SAMPLE

STRING DCL(132)           ; ARG1 INPUT
STRING MODE (6)           ; ARG2 INPUT
STRING PNAM(16)           ; ARG3 INPUT
STRING OUTF(20)           ; ARG4 INPUT
STRING INF (20)           ; ARG5 INPUT
LOGICAL[2] PID=-1         ; ARG6 OUTPUT
INTEGER[1] ECD=-1         ; ARG7 OUTPUT
LOGICAL[2] STS=-1         ; ARG8 OUTPUT
LOGICAL[2] CMP            ; ARG9 OUTPUT

DCL="ls *.lib"
MODE="WAIT"
PNAM="SAMPLE"
OUTF="sample.log"
INF="a"

CALL LIB FORK (DCL,MODE,PNAM,OUTF,INF,PID,ECD,STS,CMP)

WRITE "      PID          STATUS          ERROR", /
WRITE "                                CODE", /
WRITE[1,5] PID
WRITE[1,5] "      ",STS
WRITE[1,5] "      ",ECD
WRITE/
END
```

39. 3 FORKLIST

39. 3. 1 /FORKLIST

Function: This command is used to display the information about the corresponding subprocess when the FORK command is executed in the NOWAIT mode.

Syntax

`/FORKLIST`

Information Displayed is Listed Below

Process ID

Denotes the ID code of the FORK-generated subprocess. This value is the same as the Pid code that is output by PS command.

Parent PRO

Denotes the name of the main program which called up FORK.

Comment

A process name, if specified, is displayed in this field.

Status

Represents the completion status of the executed command.

If the command is uncompleted, '-' will be displayed.

Command

The executed command is displayed here.

The following shows an example of execution results of the /FORKLIST command:

```
=
/FORKLIST
Process ID  Parent PRO  Comment      Status      Command
000001B8    DEMO      Sleep 200    00000009    sleep 200    ..... *1
000001C1    DEMO      ls lib       00000000    ls *.lib
000001C3    DEMO      rm gen       00000000    rm *.gen
000001BE    DEMO      sleep long   -           sleep 2000   ..... *2
000001C5    DEMO      command miss 00000100    moer a.asc    ..... *3

=

/FORKLIST
Process ID  Parent PRO  Comment      Status      Command
000001BE    DEMO      sleep long   -           sleep 2000
=
```

- *1 The process has been deleted from the status by the kill command.
The status returned by waitpid () is displayed. Check the details with the man waitpid command.
- *2 This process has not yet been terminated, and the "-" status is indicated.
- *3 An invalid command name has been entered.
- *4 Once you have checked the status, its process information is cleared.
Therefore, the previous result is not displayed by the subsequent /FORKLIST.

39. 4 FORKWAIT

Tip

When you execute FORK in the NO WAIT mode, a process status table will be reserved for each process generated. 24 slots are provided to accommodate these tables.

Only after the completion status has been confirmed using the FORKWAIT (or /FORKLIST) command, will the corresponding process status table be released. Therefore, when repeatedly executing FORK 24 times or more in the NO WAIT mode, you must execute the FORKWAIT (or /FORKLIST) command to confirm the completion status of the current command.

39. 4. 1 CALL LIB FORKWAIT

Function: This command is used to wait for the completion of the process that has been started by FORK in the NOWAIT mode.

Syntax

```
CALL LIB FORKWAIT(process_id,time,error_code,completion_code)
```

Argument: process_id

This argument denotes the ID code of the process whose completion is to be awaited. The value that has been returned by FORK is set.

This argument cannot be omitted.

Type

LOGICAL[2]

Access

Input

time

This argument specifies a waiting time. One count denotes one second.

This argument cannot be omitted.

Type

LOGICAL[2]

Access

Input

error_code

This argument denotes an error code. The error code (value larger than 0) denoted by this argument will be set if the entry of the process corresponding to the process ID denoted by `process_id` is not present in the process table. If this entry is present, -1 will be set.

This argument cannot be omitted.

Type

INTEGER[1]

Access

Input

completion_code

This argument denotes a completion status code. The completion status of the selected process is set. The bit of #FFFFFFFF is set if the process has not been completed within the waiting time specified in the time argument.

This argument cannot be omitted.

Type

LOGICAL[2]

Access

Output

39. 4. 2 Sample Program for Using FORKWAIT

```
PRO DEMO4

LOGICAL[2] PID          ; ARG1 INPUT
LOGICAL[2] DELAY=0      ; ARG2 INPUT
INTEGER[1] ERCODE=-1    ; ARG3 OUTPUT
LOGICAL[2] CMPSTS=0     ; ARG4 OUTPUT

WRITE "DELAY = "
INPUT DELAY
WRITE "PID = "
INPUT[1,10] PID
CALL LIB FORKWAIT(PID,DELAY,ERCODE,CMPSTS)
WRITE "Process ID      Error  Completion",/
WRITE "                code    status",/
WRITE[1,5] PID
WRITE[1,5] "          ",ERCODE
WRITE[1,5] "          ",CMPSTS,/
WRITE/
END
```

39. 5 FORKKILL

Use the FORKKILL command to abort the active subprocess that has been generated using FORK.

39. 5. 1 /FORKKILL

Function: This command is used to abort the active subprocess from the test controller.

Syntax

```
/FORKKILL process ID_1 [process ID_2 process ID_3...]
```

Argument: Process ID

Specify the hexadecimal ID code of the intended process.

Example

(1) /FORKKILL Execution Example

```
=
/FORKKILL 21E 220 222          ..... *1
Process ID   Error
              code
0000021E     -1                ..... *2
00000220     -1
00000222     -1

=                                ..... *3
/FORKLIST
Process ID   Parent PRO   Comment      Status      Command
0000021E     DEMO        1s           00000000     1s          ..... *4
00000220     DEMO        sleep long 00000009     sleep 1000  ..... *4
00000222     DEMO        date       00000000     date

=
```

*1 Specify the desired ID code.

*2 The forcible termination has been executed normally.

*3 Check the command termination after execution of FORKKILL command.

*4 The ls and other commands have already terminated before FORKKILL is executed. However, sleep 1000 has been terminated forcibly by FORKKILL.

39. 5. 2 CALL LIB FORKKILL

Function: This command is used to abort the active subprocess from the main program.

Syntax

```
CALL LIB FORKKILL (process_id,error_code)
```

Argument: process_id

This argument denotes a process ID code.

Type

LOGICAL[2]

Access

Input

error_code

This argument denotes an error code. The appropriate error code will be set if the process specified by process_id is not that which has been generated using FORK.

If the process is that which has been generated using FORK, then -1 will be set. This argument denotes a process ID code.

Type

INTEGER[1]

Access

Output

Example

(1) CALL LIB FORKKILL Execution Example

```
PRO DEMO3

LOGICAL[2] PID          ; ARG1 INPUT
INTEGER[1] ERCODE=-1    ; ARG2 OUTPUT

INPUT[1,10] "PID = ",PID
CALL LIB FORKKILL(PID,ERCODE)
WRITE "ERROR CODE = ",ERCODE,/
END
```


40. Prober Selection and Initialization Program (/PROBSEL)

This chapter describes the PROBSEL utility to select a wafer prober and set initial values.

40. 1 Outline of PROBSEL

The PROBSEL is a utility to set the wafer prober relation.

This system automatically controls wafer probers with the "PROBER" specification in a socket program. To control all functions of a prober via the IEEE-488 (GPIB) bus, you should register the kind of the prober in the system and specify talker and listener addresses and a timeout.

This registration is required also when the control method is changed from the IEEE-488 (GPIB) bus to the prober interface.

This program can set the prober control of an external subroutine, or the optional functions.

40. 2 Syntax

40. 2. 1 /PROBSEL

Function: This statement is used to set a wafer prober.

Syntax

```
/PROBSEL↵
```


40. 4 Quitting PROBSEL

When you press the [PF1] key in the menu, the system stores current parameters displayed on the screen and terminates the /PROBSEL program. At this time, the following message appears:

```
NOW PARAMETER FILES ARE BEING MODIFIED.
```

This message means that the new parameter values are being written in the parameter file.

```
PROBER SELECT UTILITY END
```

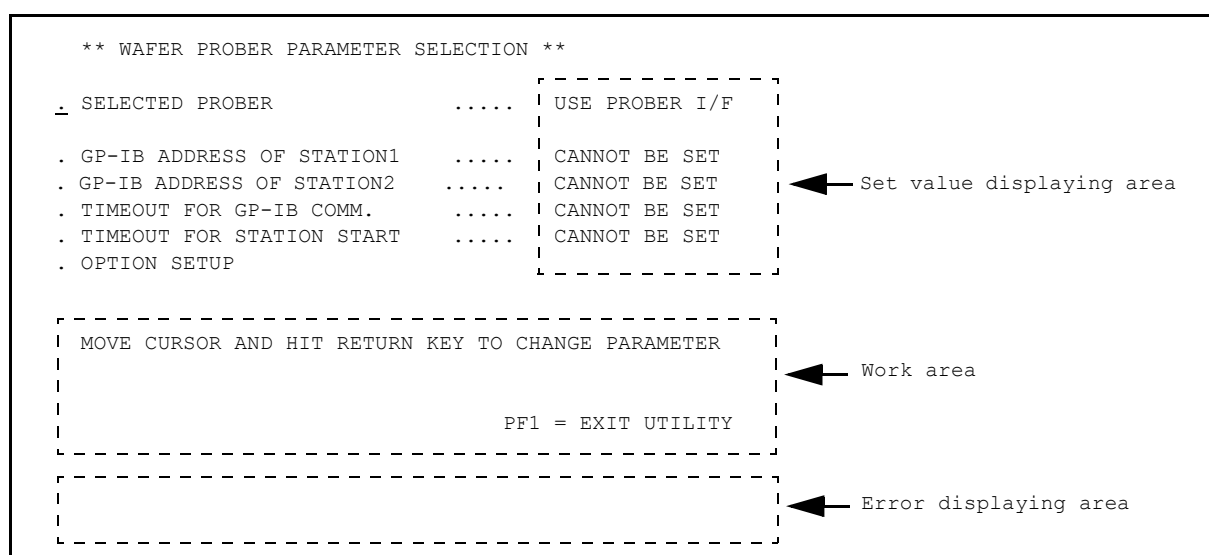
With this message, the /PROBSEL program ends.

40. 5 Menu Screen

The menu of PROBSEL consists of a main menu for basic settings and an option menu for optional function settings. The option menu varies with a selected prober. When PROBSEL starts up, the main menu appears.

Each parameter of the probe is displayed on the left side of each menu and current set values are displayed on its right side. If specifying an invalid parameter to the probe, "CANNOT BE SET" appears. A cursor displayed at the left end of the screen indicates the current parameter.

40. 5. 1 Main Menu Display



- . **SELECTED PROBER**
Select this to specify a desirable prober.
- . **GP-IB ADDRESS OF STATION1, 2**
Select this to change station addresses.
- . **TIMEOUT FOR GP-IB COMM.**
Select this to change the timeout period of the IEEE-488 (GPIB) bus.
- . **TIMEOUT FOR STATION START**
Select this to change the timeout period of a START request sent from the prober.
- . **OPTION SETUP**
Select this to move to the option menu.

40. 5. 2 Option Menu Display

- When some prober other than "USE SUBPROGRAM" is selected.

```

** WAFER PROBER PARAMETER SELECTION **

. SELECTED PROBER          ..... MODEL80

. NUMBER OF NIDLES FOR ST.1 ..... EQUAL TO SOCKET
. NUMBER OF NIDLES FOR ST.2 ..... EQUAL TO SOCKET
. LOOKUP MODE              ..... DISABLED
. SEND RESULT DATA        ..... PASS/FAIL
. EXIT TO PREVIOUS MENU

MOVE CURSOR AND HIT RETURN KEY TO CHANGE PARAMETER

                                PF1 = EXIT UTILITY

```

← Set value displaying area

← Work area

← Error displaying area

. SELECTED PROBER

Displays the selected prober. This parameter cannot be changed in the option menu.

. NUMBER OF NIDLES FOR ST.1, 2

Select this when the number of devices specified in the socket program for parallel test is different from the number of needles of the prober.

➔ For more information, refer to [40. 10 "Setting at the Time when the Number of Devices Specified in the Socket Program for Parallel Test Differs from the Number of Prober Needles"](#) on page 40-12.

. LOOKUP MODE

Select this to change the setting of the LOOKUP mode. (This item is displayed only when MODEL80 is selected as a prober.)

➔ For more information, refer to [40. 11 "LOOKUP Mode"](#) on page 40-13.

. SEND RESULT DATA

After the test program is executed, determine whether to send the test data to the Prober.

➔ For more information, refer to [40. 12 "Test Result to be Sent to a Prober"](#) on page 40-15.

. EXIT TO PREVIOUS MENU

Select this to move to the main menu.

- When "USE SUBPROGRAM" is selected for the prober.

```

** WAFER PROBER PARAMETER SELECTION **

. SELECTED PROBER          .....  USE SUBPROGRAM
. FORMER SUBPROGRAM NAME   .....
. LATTER SUBPROGRAM NAME   .....
. EXIT TO PREVIOUS MENU    .....

MOVE CURSOR AND HIT RETURN KEY TO CHANGE PARAMETER

                                PF1 = EXIT UTILITY


```

Set value displaying area

Work area

Error displaying area

. SELECTED PROBER

Displays the selected prober. This parameter cannot be changed in the option menu.

. FORMER SUBPROGRAM NAME

Select this to specify a subprogram that starts up before the execution of a test program.

. LATTER SUBPROGRAM NAME

Select this to specify a subprogram that starts up after the execution of a test program.

. EXIT TO PREVIOUS MENU

Select this to move to the main menu.

40. 6 Selection of Items to be Changed

Move the position of the cursor to an item that you want to change in the menu and press the Return key.

40. 7 Prober Specification

When you select "SELECTED PROBER", the following messages appear in the work area and the system waits for an input.

```
1 : PROBER CONTROLLED BY PROBER I/F
2 : PROBER CONTROLLED BY GP-IB (SELECT PROBER)
3 : PROBER CONTROLLED BY GP-IB SUBPROGRAM
   SPECIFY NUMBER WHICH YOU WISH TO USE. = __
```

- To control a prober via the prober interface, select 1.
- To control each prober via the IEEE-488 (GPIB) bus, select 2.
- To control a prober via the IEEE-488 (GPIB) bus with the use of a subprogram, select 3.

➔ For more information, refer to [40. 13 "Using Subprograms" on page 40-16](#).

Here, if you select "2", the following messages appear in the work area and the system waits for an input.

```
1 : MODEL80 SERIES      FULL GP-IB CONTROL
2 : A-PM-80/90 SERIES  FULL GP-IB CONTROL
3 : EG4000X SERIES     FULL GP-IB CONTROL
   SPECIFY NUMBER WHICH YOU WISH TO USE. = __
```

- Select a prober number to be used.

When you select the number of the desirable prober, the prober's name is displayed beside the item "SELECTED PROBER".

Further, the address of each station and the default value of timeout is displayed in the main menu because the prober is controlled via the IEEE-488 (GPIB) bus. (See the figure below.) If the parameters have already been set, however, they are displayed.

```
** WAFER PROBER PARAMETER SELECTION **

┌ SELECTED PROBER          .....  MODEL80

┌ GP-IB ADDRESS OF STATION1 .....  DISABLED
┌ GP-IB ADDRESS OF STATION2 .....  DISABLED
┌ TIMEOUT FOR GP-IB COMM.   .....  2.00S
┌ TIMEOUT FOR STATION START .....  10.00S
┌ OPTION SETUP

MOVE CURSOR AND HIT RETURN KEY TO CHANGE PARAMETER

PF1 = EXIT UTILITY
```

40. 8 Setting the IEEE-488 (GPIB) Address of Each Station

When you select GP-IB ADDRESS OF STATION1 to 2, the following message appears and the system waits for you to enter data. Enter the IEEE-488(GPIB) address of an appropriate prober.

The example below sets the address of station1 to 3.

```
SPECIFY GP-IB ADDRESS (D, 0-30) OF STATION1 = 3↵
```

If "D" is entered, the prober is not intended for the station and "DISABLE" is displayed at its display area. An entered numeric value appears at the display area of the main menu.

Limitation

If this program starts up after setting addresses with the standard utility, the station address becomes "DISABLE", even though it has formerly been specified.

40. 9 Setting of Timeout Values

When you select "TIMEOUT FOR GP-IB COMM.", the following message appears and the system waits for an input. Enter a timeout period in seconds. The value should be followed by the unit indication S.

The example below sets 2 seconds as a timeout period.

```
SPECIFY TIME FOR GP-IB COMMUNICATION TIMEOUT WITH UNIT (S) = 2S↵
```

When you select "TIMEOUT FOR STATION START", the following message appears and the system waits for an input. Enter a timeout period in seconds. The value should be followed by the unit indication S.

In the two-station multiple parallel test, when either of probers does not start within the specified time, for example, when one station results in wafer end, only one station operates for test.

The example below sets 10 seconds as a timeout period.

```
SPECIFY TIME FOR STATION START TIMEOUT WITH UNIT (S) = 10S↵
```

In either case, enter a value that falls within a range of 0 to 60 seconds. The control returns to the menu after entering time. Then the entered time is displayed.

Tip

"TIMEOUT FOR STATION START" and "START TIMER" statements use the same timer. So, programs that specify the "START TIMER" statement at the end of test and that read out time at the head of the next test do not operate properly.

40. 10 Setting at the Time when the Number of Devices Specified in the Socket Program for Parallel Test Differs from

40.10 Setting at the Time when the Number of Devices Specified in the Socket Program for Parallel Test Differs from the Number of Prober Needles

When you select NUMBER OF DUT FOR STATION1to2, the following message appears and the system waits for you to enter data. Enter the number of prober needles.

The example below sets 16.

SPECIFY THE NUMBER OF NIEDLES (D, 1-32) FOR STATION1 = 16↵

If you enter **D** here, the system assumes that the number of needles for that station is equal to the number of devices specified to be tested in parallel in a socket program and displays "EQUAL TO SOCKET" next to the station in the set value displaying area.

If you enter the number of needles, the control returns to the menu. Then the entered time is displayed at the display area of set values.

40.11 LOOKUP Mode

LOOKUP mode is a function that controls probes according to wafers and their x and y addresses specified beforehand in order, instead of the auto probing function. Especially, under the multiple devices' parallel test, the number of test for each wafer can be less than at the use of the auto probing function.

You can select the LOOKUP mode with a prober, MODEL80 manufactured by TOKYO ELECTRON Co. Ltd.

40.11. 1 Setting LOOKUP Mode

When you select LOOKUP MODE, the following message appears and the system waits for you to enter data. Enter **D** or **E**.

```
SPECIFY LOOKUP MODE TO BE ENABLED(E) OR DISABLED(D)  _
```

To use the LOOKUP mode, enter **E**. Enter **D** when you do not use the LOOKUP mode. Then "ENABLE" or "DISABLE" is displayed in the display area for set values and the control returns to the menu.

40.11. 2 Creating LOOKUP Files

To use the LOOKUP mode, it is necessary to prepare two kinds of files described below.

- A file describing the correspondence between an execution program and a LOOKUP file to be used
- A file (LOOKUP file) that contains addresses to be followed by the prober and on-wafer data

The following is the explanation of these files.

File describing the correspondence between execution programs and LOOKUP files

There may be cases where LOOKUP files vary with execution programs at a directory or on a customer. Therefore it is necessary to prepare a file describing the correspondence between an execution program and a LOOKUP file. Each work directory or customer requires two or three files like this. The number of files differs according to whether the LOOKUP files to be used at station are specified separately.

- When the same LOOKUP file is used for Station 1 and Station 2
Contain appropriate data in DAT LKUPFILE(lkupfile.dat).
- When different LOOKUP files are used separately for Station 1 and Station 2
Enter data for Station 1 in DAT LKUPFIL1(lkupfil1.dat) and enter data for Station 2 in DAT LKUPFIL2(lkupfil2.dat).

The file names for use in ASX are described in angle brackets. The format of the file is shown below.

```
<Program name 1>, <LOOKUP file name 1>,
<Program name 2>, <LOOKUP file name 2>,
<Program name 3>, <LOOKUP file name 3>,
      :
<Program name 10>, <LOOKUP file name 10>,
```

Enter all test programs to be controlled by the LOOKUP mode. The number of correspondence data that can be described is up to 10 a file. A comma must be put at the end of each line. Enter a file name in a MOST compliant format. Be sure to enter an extension too. The following is an example.

```
PRO MODETEST, DAT MOVEDATA,
```

File (LOOKUP file) containing addresses and on-wafer data

The LOOKUP file stores prober's movement of a wafer's first chip to the wafer end, and on-wafer data. Each work directory or customer requires a file minimum, and 10 files maximum. The file name must be the same as a name written in the file describing the correspondence between an execution program and a LOOKUP file.

The format of the file is shown below.

```
<Total number of indices>
<Dummy number>,<X address>,<Y address>,<On-wafer data>,
      :
```

The total number of index specifies the number of times a wafer is indexed. The dummy number is the line number of data and is not used in the system. Enter a natural number in ascending order.

Enter X and Y addresses and on-wafer data as appropriate.

Decimal and hexadecimal notations can be used. X and Y addresses must fall within a range of 0 to 999. An on-wafer value must fall within a range of #0 to # FFFFFFFF (on-wafers of DUT numbers 1 to 32).

The following is an example.

```
32,1, 4,5,#F,2, 8,5,#E, :32, 8,20,#6,
```

40.12 Test Result to be Sent to a Prober

After a test program is executed, the tester controller sends the test results (PASS/FAIL information) to the prober. When category information is sent from the test program to the prober, however, PASS/FAIL information sent after the category information causes a malfunction in the prober.

To avoid this problem, it is necessary to make the Tester Controller send nothing.

40.12. 1 Output Selection of Test Result to be Sent to a Prober

When selecting "SEND RESULT DATA", the following messages and prompt appear in the work area. When you enter **1**, PASS/FAIL information is sent to the prober. When you enter **2**, no data is sent.

The following example shows an example sending nothing.

```
1 : PASS/FAIL INFORMATION
2 : NOTHING
    SPECIFY NUMBER WHICH YOU WISH TO USE. = 2
```

The set value of SEND RESULT DATA varies from the input number. Using Subprograms

40.13 Using Subprograms

Generally, the prober control via the IEEE-488 (GPIB) bus is executed according to ADVANTEST's standard sequence. Since the standard sequence, however, brings minimum control, there may be cases where all request cannot be supported. In such a case, it becomes possible to realize each request by preparing a prober's control routine as a subprogram.

Necessary programs as subprograms are the following two.

- Subprogram for before-testing process (FORMER SUBPROGRAM)
- Subprogram for the after-measurement process (LATTER SUBPROGRAM)

These files are described in detail in [40. 13. 2 "Subprogram Files" on page 40-16](#).

40.13. 1 Setting Subprograms

When you select FORMER (LATTER) SUBPROGRAM NAME, the following message appears and the system waits for an input. Enter a subprogram name. Up to 32 characters can be used. The file extension is unnecessary.

The example below specifies "PROBSUB 1" as a subprogram for before-testing process.

SPECIFY THE NAME OF FORMER SUBPROGRAM = PROBSUB1

When you specify the subprogram, its name is displayed in the display area for set values and the control returns to the menu.

Tip

To specify the subprogram name, probers to be used must be specified to "USE SUBPROGRAM".

➔ For more information on the specification of a prober to be used, refer to [40. 7 "Prober Specification" on page 40-9](#).

40.13. 2 Subprogram Files

This section explains the outline and calling of subprograms.

Subprogram for before-testing process (FORMER SUBPROGRAM)

This is a subprogram started to inform the tester of the current condition of the prober before a test. Data such as status, on-wafer data, X and Y addresses, and locations is set. Further, under certain prober's conditions, an appropriate command must be given to the prober.

The format of the subprogram is shown below.

```
SUBR XXXX (STNO, TLAD, STB, PBST, ONWF, XYAD, LOCT)
:
RETURN
END
```

STNO

Integer type variable consisting of one word. The station number of the currently-aimed prober is assigned.

TLAD

Integer type variable consisting of one word. The IEEE-488 (GPIB) address of the currently-aimed prober is assigned.

STB

Integer type variable consisting of one word. The SRQ status byte that is output from the currently-aimed prober at its start is assigned.

PBST

Logical type variable consisting of two word. It sets the status of the prober.

ONWF

Logical type variable consisting of two word. It sets the on-wafer of the prober.

XYAD

Logical type variable consisting of two word. It sets the XY address of the prober.

LOCT

Logical type variable consisting of two word. It sets the location of the prober.

➔ For more information on formats in which data such as status, on-wafer data, X and Y addresses, and locations are stored, refer to the description of the READ PROBER statement in [ATL-51 Test Plan Program Reference Manual\(No. : 8431498 \)](#).

Subprogram for the after-measurement process (LATTER SUBPROGRAM)

This subprogram is started to notify the prober of the sort data after the measurement. The subprogram reads and formats the measurement result then sends it to the prober. To measure the next chip, an instruction is required.

The format of the subprogram is shown below.

```
SUBR YYYY (STNO, TLAD)
:
RETURN
END
```

STNO

Integer type variable consisting of one word. The station number of the currently-aimed prober is assigned.

TLAD

Integer type variable consisting of one word. The IEEE-488 (GPIB) address of the currently-aimed probe is assigned.

 Limitation

The START TIMER statement in a subprogram affects the correct execution of a parallel test in two stations. So, do not use the START TIMER statement.

40.13. 3 Control Commands

The ATL language provides the following statements so that data can be sent or received through the IEEE-488(GPIB). The control programs are created using these statements as follows:

- SEND GPIB function name [Data]

Function name

CLEAR, REMOTE, LOCAL, GTL, SDC, GET, LLO, DCL, SPE, SPD, UNL

- SEND GPIB MLA listener address
- SEND GPIB MTA Talker address
- SEND GPIB DATA Data
- SEND GPIBn TO v1 v2 ... v36
- TIME OUT t :GPIB:Label
- ON GPSRQ: Label

➔ For more information, refer to I/O statements for GPIB in [ATL-51 Test Plan Program Reference Manual\(No. : 8431498 \)](#).

40.14 Error Message

When an error arises while this program is running, an error message appears in the error display area. Error messages and their contents are shown below.

- Setting Value Input Error

```
PARDON?
```

The above message appears when you enter incorrect characters or numerics for each parameter. After displaying the message, the system waits for an input.

- Case of Selecting an Item which Modification was Prohibited

```
SORRY, THIS ITEM CANNOT BE SPECIFIED.
```

The above message appears when you select an item next which "CANNOT BE SET" is displayed in the menu screen. After displaying the message, the system waits for an input.

- Address Setting Error

```
ILLEGAL ASSIGNMENT! PLURAL STATIONS HAVE SAME ADDRESS!!  
PLEASE PRESS ANY KEY TO RETRY.
```

The above message appears when you compare each address of plural stations that have been specified to the same address and where probbers controlled via the IEEE-488 are selected, just before the program terminates. This error does not arise only when you specify plural systems to the same address.

After displaying the message, press any key to return the menu.

- Parameter Fail Error

```
UNKNOWN ERROR OCCURRED! CODE : xxxx  
ON DP0:<File>  
[xxxx is an error code and <File> is the name of the file in which an error  
occurred.]
```

The above message appears if an error arises while the parameter file is read or written, and then the program is forcibly terminated. In this case, start and set the program again after removing the cause of the error.

41. CALCHECK

This chapter describes the LIB CALCHECK, a program to skip external subroutines for calibration.

41. 1 CALCHECK Overview

When the calibration function is executed during debugging, much time is required and efficiency is low. Therefore, the LIB CALCHECK program is used to skip external subroutine programs such as calibration.

LIB CALCHECK includes the following functions:

- Specification of programs whose execution is to be skipped.
- Execution of other programs in place of skipped programs.
- Cancellation of specification of skipped programs.
- Display of specified skipped program name.
- Display of help screen.

41. 2 Syntax

Input the following command when the controller is operating:

41. 2. 1 /CALCHECK

Function: This statement is used to skip external subroutines.

Syntax

```
/CALCHECK/global_switch mode filename1 filename2
```

Argument: global_switch (global switch)

/A

Subsequent programs to be skipped are skipped until the specification is canceled.

/A is effective only when the subsequent specified program to be skipped is SUB CALB.

/H

A help screen (input format) is displayed.

Default

Once the device program is executed, the program specified to be skipped is deselected.

mode (mode)

ON

The file to be specified next is skipped.

OFF

The skipping of the specified file is canceled. If the file name is not specified, all files to be skipped are canceled.

LIST

The specified program name to be skipped is displayed.

Default

The file to be specified next is skipped.

Same as ON mode.

filename1

The file name of the program to be skipped or the file name specification is to be canceled is specified.

When mode is OFF, the filename may be omitted.

filename2

Instead of filename1 to be skipped, the file name of the program to be executed is specified. May be omitted.

Precautions

- Global switch

To specify a global switch, specify either /A or /H.

/A is effective only when the subsequent specified program to be skipped is SUB CALB.

For other programs, /A is specified automatically.

When the ON, OFF, or LIST mode is specified along with the /H global switch, that mode does not work and a help screen is displayed.

- ◆ **The Following Specification does not List Program Names, But Displays a Help Screen:**

```
/CALCHECK/H LIST↓
```

- filename

For filename, the file identifier is written and a space is opened between the identifier and the file name.

Up to 32 characters can be used for file name.

Programs other than "SUB," "UTL" or "LIB" cannot be skipped.

If the "SUB" program or the "LIB" program is skipped, the "UTL" program cannot be executed.

Conversely, when a "UTL" program is skipped, "SUB" and "LIB" programs cannot be executed for the UTL program.

All 8 programs can be skipped.

41. 3 Error Indication

41. 3. 1 Error Indication

Command error

Input format different.

Illegal file specifier

Program to be skipped is other than "SUB", "UTL", or "LIB"

Illegal file specifier at filename2

Program of "SUB" or "LIB" is skipped and "UTL" is executed instead or on the contrary, program of "UTL" is skipped and "SUB" or "LIB" is executed in stead.

Table overflow

8 external subroutines to be skipped were already specified.

No such file

File whose skipping is to be canceled does not exist.

41. 4 Example of Execution

- 1) To skip an external subroutine (SUB CALB or UTL ABC), enter the following commands when the controller is in operation:

```
/CALCHECK/A ON SUB CALB↵  
/CALCHECK ON UTL ABC↵
```

- 2) For the execution of other programs (LIB GHI) in place of skipped programs (SUB DEF), input the following command when the controller is operating:

```
/CALCHECK ON SUB DEF LIB GHI↵
```

- 3) Display the skipped program name.

After the above procedures 1) and 2), enter the following command:

```
/CALCHECK LIST↵
```

The following will be displayed:

```
1  A  SUB CALB  -----  
2      UTL ABC  -----  
3      SUB DEF  LIB GHI
```

Number			
Global switch			
File name to be skipped			
			File name to be executed instead

Skipped program names are displayed.

- 4) Skipped program (SUB DEF) is canceled.

After the above procedures 1) and 2), enter the following command:

```
/CALCHECK OFF SUB DEF↵
```


The list indication is displayed.

```
/CALCHECK LIST↵
```

The following will be displayed:

```
1  A  SUB CALB  -----
2      UTL ABC  -----
```

SUB DEF is canceled.

5) All skipped programs are canceled.

After the above procedures 1) and 2), execute the following command:

```
/CALCHECK OFF↵
```

The list indication is displayed.

```
/CALCHECK LIST↵
```

Nothing is displayed on the VKT. This indicates that all programs specified to be skipped in 1) and 2) were deselected.

6) Help screen is displayed.

```
/CALCHECK/H↵
```

The following will be displayed:

```
COMMAND FORMAT
  /CALCHECK[/A][ON] filename
  /CALCHECK[/A][ON] filename1 filename2
  /CALCHECK OFF filename
  /CALCHECK OFF
  /CALCHECK LIST
  /CALCHECK/H
```

42. GETHIDAR Utility

This chapter describes the GETHIDAR utility to read HIDAR information.

42. 1 GETHIDAR Overview

GETHIDAR can read the max., min., average, standard deviation, 3SIGMA, and variance values from the HIDAR information of device test programs installed by HIDAR (a histogram creation program).

42. 2 GETHIDAR Call

42. 2. 1 CALL LIB GETHIDAR

Function: This statement is used to read the max., min., average, standard deviation, 3SIGMA, and variance values from HIDAR information.

Syntax

```
CALL LIB GETHIDAR(stno,item,max,min,mean,svd,sigma,varia,ier)
```

Argument: stno

Specifies station numbers.

stno = 1: Station 1

stno = 2: Station 2

stno = 0: Station 1 and Station 2

(However, in TESTING, this parameter is ignored and the station number during execution is set.)

Tip

HIDAR table exists for each station. Therefore, when HIDAR conditions are set for a test parameter for a parallel test in two stations, HIDAR information for this test parameter is stored separately for the HIDAR table for Station 1 and the HIDAR table for Station 2. When 0 is specified for stno, GETHIDAR determines values from max to varia based on data in the tables for Station 1 and Station 2.

➔ For more information, refer to [42. 3 "Information Stored in Parallel Test between 2 Stations" on page 42-5.](#)

However, when HIDAR conditions such as start and stop test numbers and the maximum and minimum values of measurements to be collected are different between Station 1 and Station 2, an error code is stored.

Type

INTEGER[1]

Access

Input

item

Specifies test parameter. Specify a test number or a class name.

**Tip**

Use a string of up to 8 characters to specify a test number or an INTEGER[2] class name.

Type

INTEGER[2] or STRING

Access

Input

max

Sets max. value of HIDAR information.

Type

REAL[3]

Access

Output

min

Sets min. value of HIDAR information.

Type

REAL[3]

Access

Output

mean

Sets average value of HIDAR information.

Type

REAL[3]

Access

Output

sdv

Sets standard deviation value of HIDAR information.

Type

REAL[3]

Access

Output

sigma

Sets 3SIGMA of HIDAR information.

Type

REAL[3]

Access

Output

varia

Sets variance value of HIDAR information.

Tip

A variance value is followed by the unit indication "\$", which means the square of the measurement data unit.

Type

REAL[3]

Access

Output

ier

Sets error code.

Type

INTEGER[1]

Access

Output

Description

Contain this statement in a main program to call a subroutine. The stno and item are arguments for input and the max to ier are arguments for output. For example, GETHIDAR is called by specifying the station number and the test number or class name respectively in stno and item. HIDAR information for the test parameter specified by the item argument is read from the HIDAR table of the station specified by the stno argument and stored in max to varia. Also, if the information cannot be read, an error code is stored in ier.

42. 3 Information Stored in Parallel Test between 2 Stations

- Maximum value (max)
High values in the max. values of station 1 and station 2
- Minimum value (min)
Low values in the min. values of station 1 and station 2
- Average value (mean)
Average value of collection data of station 1 and station 2

$$\text{mean} = \frac{\text{smean1} + \text{smean2}}{\text{tcnt1} + \text{tcnt2}}$$

smean1 total of collection data in station 1

smean2 total of collection data in station 2

tcnt1 number of tests in station 1 minus number of times of overflow

tcnt2 number of tests in station 2 minus number of times of overflow

- Standard deviation value (sdv)
Standard deviation data of collection data in station 1 and station 2
 $\text{sdv} = \sqrt{\text{varia}}$
Positive square root of a variance (varia) (Refer to the following variance.)
- 3SIGMA (sigma)
3SIGMA value of collection data in station 1 and station 2
 $\text{sigma} = 3 \times \text{sdv}$
3 times standard deviation value
- Variance (varia)
Variance of collection data in station 1 and station 2.

$$\text{varia} = \frac{1}{\text{tcnt1} + \text{tcnt2}} \times \left(\text{ssdv1} + \text{ssdv2} - \frac{(\text{smean1} + \text{smean2})^2}{\text{tcnt1} + \text{tcnt2}} \right)$$

smean1 total of collection data in station 1

smean2 total of collection data in station 2

ssdv1 total of (collection data in station 1) 2

ssdv2 total of (collection data in station 2) 2

tcnt1 number of tests in station 1 minus number of times of overflow

tcnt2 number of tests in station 2 minus number of times of overflow

42. 4 Error Codes

If the HIDAR information is not read, the following error codes is stored in ier:

Error Codes	Description
0	Normal end
1	Specification of station no. is different.
2	Area for a HIDAR table has not been secured (HIDAR conditions have not been set by a statement or the “.J” command).
3	Argument (item) is not INTEGER or STRING type.
4	A key command (.J) could not find a table (search by test no. cannot be performed).
5	Test no. specified by argument (item) does not exist.
6	Class name specified by argument (item) does not exist.
7	Not a single test has been performed (HIDAR information to read does not exist).
8	Measurement data does not exist in HIDAR measurement range.
9	Test conditions in HIDAR are different between Station 1 and Station 2 when 0 is specified for stno (a parallel test in two stations).
10	Start and stop test numbers in HIDAR are different between Station 1 and Station 2 when 0 is specified for stno (a parallel test in two stations).
11	The units of HIDAR information are different between Station 1 and Station 2 when 0 is specified for stno (a parallel test in two stations).

43. Write and Read of MCW

This chapter describes the WMCW utility to write data to MCW terminals and the RMCW utility to read data from MCW terminals.

43. 1 Overview

WMCW and RMCW are utility programs to output (write) data to MCW terminals on the performance board or to read data from these terminals.

Using the MCW terminals, the user can set the desired DUT conditions and read the DUT status.

An MCW terminal is independent on each station.

The number of MCW terminals on each station is shown below:

Test head type	Number of MCW terminals on a station
3712CH test head 3456CH test head	256 (MCW0 to 255)
2880CH test head 1664CH test head 1600CH test head 1344CH test head	64 (MCW0 to 63)
800CH test head 352CH test head	32 (MCW0 to 31)

43. 2 Syntax

43. 2. 1 WMCW

Function: This statement is used to output (write) data to MCW terminals.

Syntax

```
EXEC WMCW (STN number, Data1, Data2, Data3, Data4, Data5, Data6, Data7, Data8)
ASM WMCW (STN number, Data1, Data2, Data3, Data4, Data5, Data6, Data7, Data8)
```

Argument: STN number

Specifies station numbers.

Enter an integer or logical constant or a variable.

0: Enter 0 to specify Stations 1 and 2.

1: Enter 1 to specify Station 2.

2: Enter 2 to specify Station 2.

When a variable is used to specify a station number, enter a 1 or 2 word variable.

Data n (n=1 to 8)

Specifies the data to be written.

Enter an integer or logical constant or a variable.

The data size is 32 bits.

When a variable is used to specify data to be written, enter a 2-word variable.

Data 3 through data 8 can be omitted.

However, data with a number in the middle of consecutive data numbers cannot be omitted.

Description

Station 1 and 2 exist independently. Therefore, different values can be set separately for Station 1 and Station 2. (for a parallel test in two stations).

43. 2. 2 RMCW

Function: This statement is used to read data from MCW terminals.

Syntax

```
EXEC RMCW (STN number, Data1, Data2, Data3, Data4, Data5, Data6, Data7, Data8)
ASM RMCW (STN number, Data1, Data2, Data3, Data4, Data5, Data6, Data7, Data8)
```

Argument: STN number

Specifies station numbers.

Enter an integer or logical constant or a variable.

0: Enter 0 to specify Stations 1 and 2.

1: Enter 1 to specify Station 2.

2: Enter 2 to specify Station 2.

When a variable is used to specify a station number, enter a 1 or 2 word variable.

If "0" is specified, the read-out data is the logical sum of the data of stations 1 and 2.

Data n (n=1 to 8)

Specifies a variable used to store the read-out data. Enter a logical type variable.

The data size is 32 bits. Enter a 2-word variable.

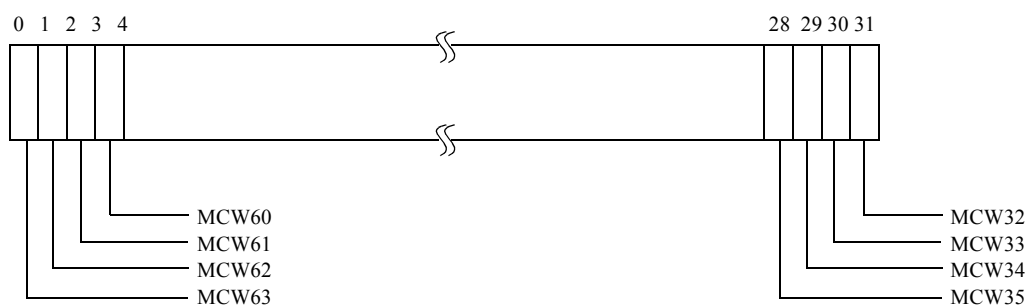
Data 3 through data 8 can be omitted.

However, data with a number in the middle of consecutive data numbers cannot be omitted.

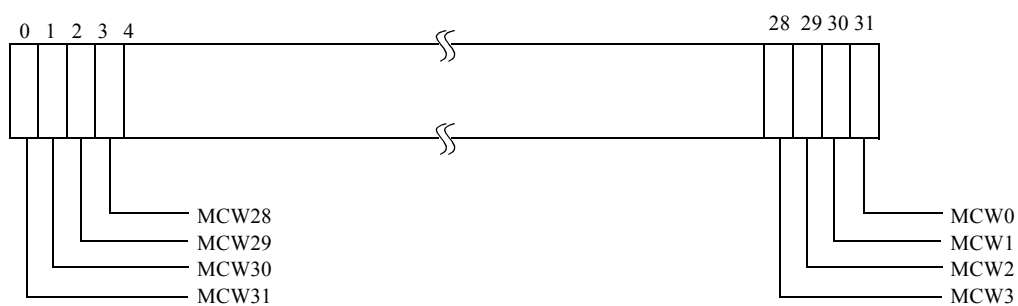
43. 3 Data Format

The format of the data n is as shown below.

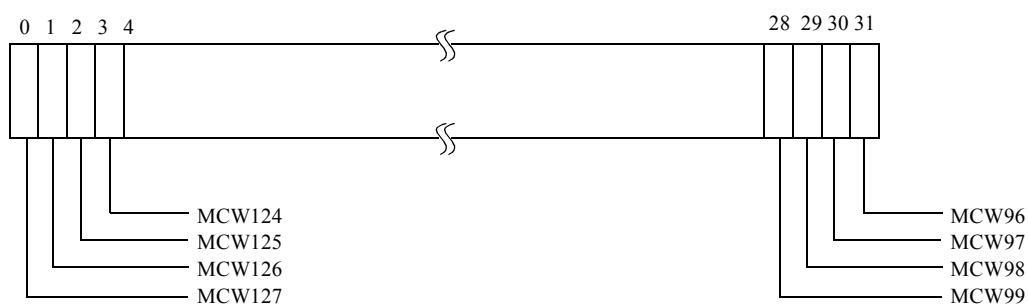
Data1



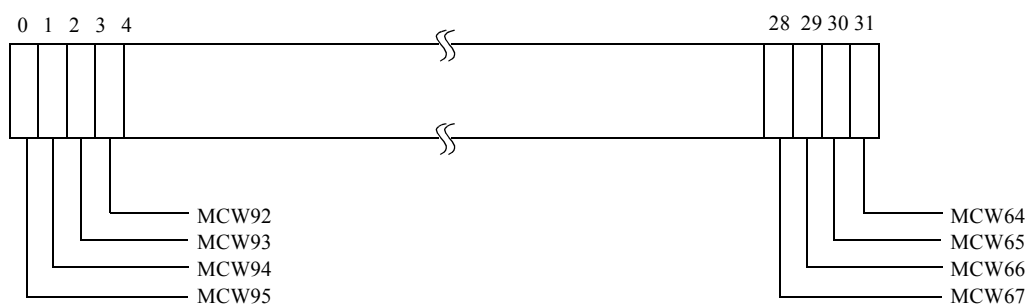
Data2



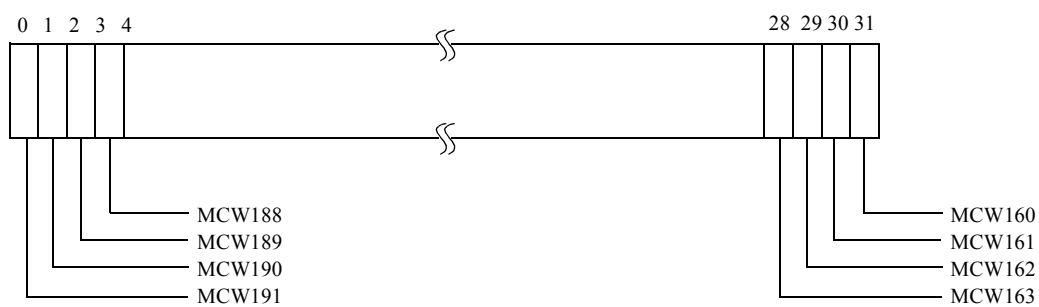
Data3



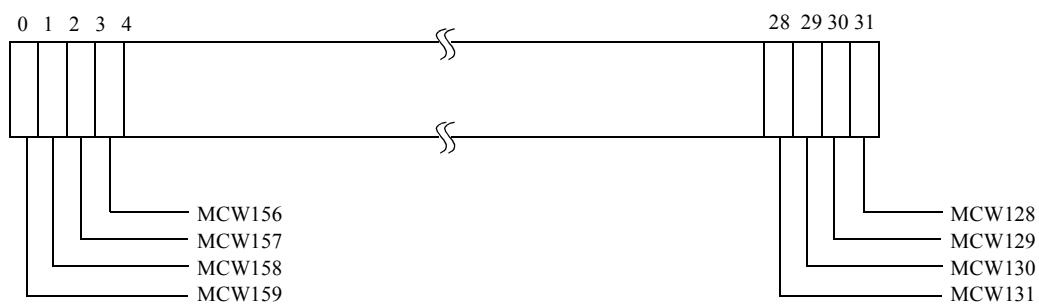
Data4



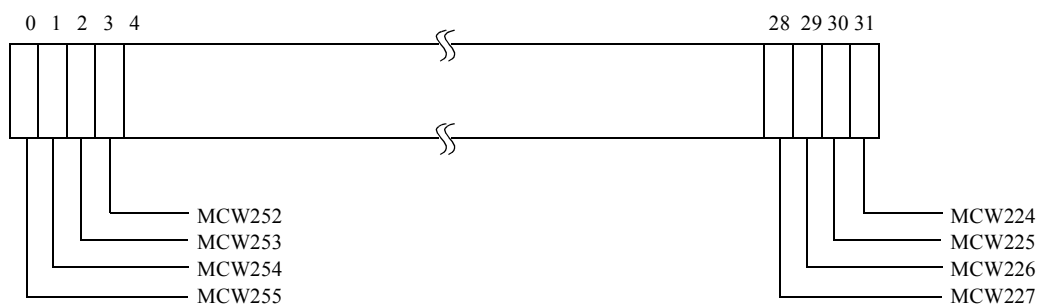
Data5



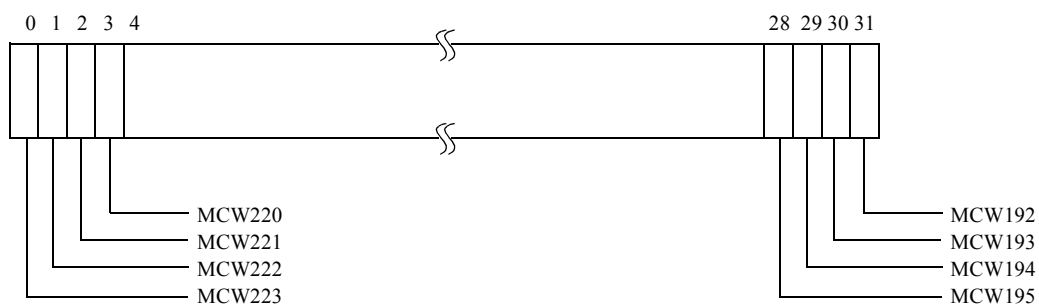
Data6



Data7



Data 8



44. DIRLINK

This chapter describes the DIRLINK utility to specify directories to search for a file.

44. 1 Outline of DIRLINK

The DIRLINK utility allows to search a file in the specified directory.

The file search sequence is changed when this utility is used.

The conventional file search sequence is:

- 1) CURRENT DIRECTORY
- 2) DP0 file

When this utility is used, the file search sequence is changed to the following:

- 1) CURRENT DIRECTORY
- 2) DIRECTORY specified by DIRLINK
- 3) DP0 file

 Tip

When this utility is used in the ADVANsite environment, the directory to be specified has had to been defined.

➔ For more information, refer to [ADVANsite System Introduction\(No. : 8227727\)](#).

44. 2 Functions of DIRLINK

The DIRLINK utility specifies a directory in the directory search list and refers to the search list. This utility provides the following primary functions:

- specify up to 4 directory names in the search list.

#	Directory name
1	
2	
3	
4	

Searches the file in the sequence defined in the search list.

- Can specify any location of directory in the search list.
- Can specify the directory names in the space areas in the ascending order if there are unused space areas in the search list.
- Usually, when a different directory is used, "INIT" should be performed. This utility automatically performs "INIT" when you specify a directory.
- Can use the following three calling types:
 1. When the utility is started by "/DIRLINK parameter" in the single-line interactive mode, you can directly specify directories and their position in the search list.
 2. If the utility is started by "CALL LIB DIRLINK(arg)" in the programmable mode, the user can search files by directly specifying the sequential number of the search list and the directory name with in the program.
 3. When the utility is started by "/DIRLINK" in the semigraphic interactive mode, you can use your key pad to specify directory names in a screen where all other entries in the search list are displayed.
- If an information is set in the search list by this utility, it is saved in the executed current directory.
- Although a directory name can be specified to 32 characters in ASX/U-51, the directory name which can be specified by the DIRLINK function is to eight characters.

44. 3 DIRLINK Syntax

The user can start the DIRLINK utility by using one of the following formats:

44. 3. 1 /DIRLINK Directory Name

Function: This syntax is used to specify or release the directory name directly on the controller.

Syntax

```
/DIRLINK ... , Directory name/n↓      (a)
/DIRLINK/R ... , n↓                    (b)
```

Specify a Directory Name (a)

n

The sequence number of the directory names in the search list.

It may be 1, 2, 3 or 4.

If n=0 is specified, the directory names are automatically set in the space areas in the ascending order.

Remove a Directory Name from the Search List (b)

n

The sequence number of the directory names to be deleted from the search list.

It may be 1, 2, 3 or 4.

44. 3. 2 CALL LIB DIRLINK

Function: Specify or release the directory name in the program.

Syntax

```
CALL LIB DIRLINK("... , Directory name/n")      (c)
CALL LIB DIRLINK("/R ... , n")                    (d)
```

Specify a Directory Name in a Program (a)

n

The sequence number of the directory names in the search list.

It may be 1, 2, 3 or 4.

If n=0 is specified, the directory names are automatically set in the space areas in the ascending order.

Remove a Directory Name from the Search List in a Program (b)

n
The sequence number of the directory names to be deleted from the search list.
It may be 1, 2, 3 or 4.

44. 3. 3 /DIRLINK

Function: This syntax is used to select the interactive semigraphic mode.

Syntax

```
/DIRLINK↓
```

Description

The following semigraphic screen is displayed to help you specify directories.

System: T5500ASX/U-51 LIB DIRLINK Utility1998/01/01 00:00:00

Search List for Directory Linkon directory WORK

#	Directory	Information
1		
2		
3		
4		

Help:

DELETEDelete character

CTRL/AChange Insert/Change mode

CTRL/EMove cursor to E.O.L.

CTRL/HMove cursor to B.O.L

CTRL/LReset all data

CTRL/WRefresh screen

CTRL/ZExit and update

PF4Delete DIR. name

↑UP

DOWN↓

←LEFT

→RIGHT

Status:Insert

- The system name, OS name, and date are displayed.
- You can enter directory names in Directory #1 to #4.
When you enter "**CTRL/Z**" after entering directory names, they are entered in the search list.
- You can use the ↑, ↓, ← and → keys to move the cursor up or down or side to side.
You can also use the the Enter or Return key to move the cursor down.
The Enter and Return keys scroll up and down the page of screen.

- The following explains the input from the keyboard:

DELETE

Deletes the character(s).

CTRL/A

Switches the Insert and Change modes.

PF1, PF2

Switches the Insert and Change modes.

CTRL/E

Shifts the cursor to the end of the character string of the directory name.

CTRL/H

Shifts the cursor to the beginning of the character string of the directory name.

CTRL/L

Deletes all data.

CTRL/W

Redisplays the screen.

CTRL/Z

Exits from DIRLINK and sets the directories in the search list.

PF4

Deletes the specified directory name.

- A message is displayed in the information field. One of the following may be displayed according to the input of directory names:

Initialized

The specified directory has been initialized.

Not found

The specified directory does not exist.

Multi defined with #n

An attempt was made to specify the directory which has already been specified as directories #1 to #4.

Already in system

An already existing directory name was specified again.

Unsolicited error

An error other than Not found, Multi defined with #n, or Already in system occurred.

- One of the following messages may be displayed in the "Status" field according to directory names you entered:

"When you enter **CTRL/L**

The message "Are you sure to reset all data [Y(yes) or N(no)]?" is displayed. Enter **Y** to reset data. Enter **N** when you do not want to reset data.

"When you enter **Y**, the the following message is displayed: "All data have been initialized"

"When you enter **N**, the following message is displayed: "Command is canceled"

When the specified directory has been initialized normally, the following message is output:

"DIR directory name at #n has been successfully INITed"

If the specified directory has already been initialized

"Warning : DIR Directory name at #n is already in system"

If the specified directory does not exist

"Warning : DIR Directory name at #n does not exist"

If an attempt was made to specify the directory name which has already been specified as directories #1 to #4.

"Warning : DIR Directory name at #n is already specified at #m"

When an error other than the above "Warnings" occurs

"Unsolicited error occurs"

Tip

- *If one of these errors occurs, your entries are accepted by the system. However, when searches are performed according to the search list, an error directory is skipped and the next directory is searched.*
- *On the T5500 series where ASX/U-51 Rev. 4.02 or later is installed, the SEND DBM statement cannot use the directory specified by DIRLINK.*
 - ➔ For more information, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498\)](#).

44. 4 Error Message

The following messages are displayed:

44. 4. 1 Error Message

Illegal global switch(es)

A switch other than "/R" was specified.

Illegal sequence number

A value other than 0 to 4 was specified for "n" when a directory name was specified.

Directory name table is full

0 was specified for n when the search list was full of directory names. There is no space to enter a directory name.

Illegal argument (s)

You specified an argument incorrectly.

44. 5 Examples of DIRLINK

The following gives an example if "DIRLINK" is executed in the interactive single-line mode.

44. 5. 1 Directory Specification Using Search List Number

- Execute the following command on the controller:

```
/DIRLINK WORK1/1 WORK4/4
```

- The following will be set in the search list:

#	Directory name
1	WORK1 file
2	
3	
4	WORK4 file

- The files are searched in the search list:

- 1) CURRENT DIRECTORY
- 2) WORK1 file
- 3) WORK4 file
- 4) DP0 file

44. 5. 2 Directory Specification Using Space Area of the Search List

- Execute the following command on the controller:

```
/DIRLINK WORK2/0
```

- The following will be set in the search list:

#	Directory name
1	WORK1 file
2	WORK2 file
3	
4	WORK4 file

- The files are searched in the search list:

- 1) CURRENT DIRECTORY
- 2) WORK1 file
- 3) WORK2 file
- 4) WORK4 file
- 5) DP0 file

44. 5. 3 Release of Directories

- Execute the following command on the controller:

```
/DIRLINK/R 1,4↵
```

- The following will be set in the search list:

#	Directory name
1	
2	WORK2 file
3	
4	

- The files are searched in the search list:

- 1) CURRENT DIRECTORY
- 2) WORK2 file
- 3) DP0 file

45. RTEMASK

RTEMASK is used to disable (MASK ENABLE) or enable (MASK DISABLED) the error check for specified items in a test plan program.

No runtime error is output for the item where the error check is disabled to execute.

The setting is effective until the tester driver controller is stopped. Items are Default when the tester driver controller is operating.

Two setting methods of using menu and using argument are provided. Select one method depending on the existence of the argument when this utility is started up. These methods are described in detail later.

45. 1 Applicable Items

IN

The setting values in the IN statement (IN1-16) can be specified to check or not.

Specifically, you can specify whether or not to check the range of Vih, Vil and Vih-Vil in INn = Vih-Vil.

These are also effective for the IN statement written in the BURST BLOCK statement.

DCGRDALM, PSGRDALM

- DCGRDALM

You can specify whether or not to check the DC guard alarm (error code 3240).

- PSGRDALM

You can specify whether or not to check the DC guard alarm (error code 3240).

However, HVPPS used for VS9 to 12 on the T5581 or T5581H cannot be specified.

Tip

Guard Alarm is used to alert users of DC test units and device power supply failures.

Set MASK to ENABLE only when the guard alarm is activated and then remove the cause of the error.

If Guard Alarm is activated, applied voltage, applied current and measurement values may change.

After the cause of the error is removed, set MASK to DISABLE to allow the guard alarm to detect errors again.

DPUTYPE1

You can specify whether or not to check whether a pattern program designed on the T5581 or T5581H test system with a VS statement (DPUTYPE1) specified for a burst test condition is transferred to the T5592, T5591, T5591R, T5586, T5585, T5581P, or T5571P test system (error code 12375).

When MASK is set to ENABLE, a pattern program designed on the T5581 or T5581H test system with a VS statement (DPUTYPE1) specified for a burst test condition can be transferred to the T5592, T5591, T5591R, T5586, T5585, T5581P, or T5571P test system.

Tip

Note the following points when setting MASK ENABLE to transfer a pattern program designed on the T5581 or T5581H test system with a VS statement (DPUTYPE1) specified for a burst test condition to the T5592, T5591, T5591R, T5586, T5585, T5581P, or T5571P test system.

- *When a VS statement is used as a burst test condition, the voltage range specified in the VS statement of the pattern program is disabled.
The VS statement of the test plan program determines the voltage range of the burst test.*
- *When a VS statement is used as a burst test condition, the voltage range specified in the VS statement of the pattern program is not checked. Specify a voltage that falls within the range specified in the VS statement of the test plan program.*
- *The SEND MPAT statement must come after the VS statement of the test plan program, as shown below.*

Example

- *Correct specification*

```
VS9=6V,R16V
SEND MPAT MPAT1
MEAS MPAT
```

- *Incorrect specification*

```
SEND MPAT MPAT1
VS1=6V,R16V
MEAS MPAT
```

- *When the pattern program used in the previous test is used again, the MEAS MPAT or START MPAT statement does not transfer the VS statement for a burst test condition. Therefore, use the SEND MPAT statement to transfer the VS statement for a burst test condition when you change the voltage range in the VS statement.*

Example

- *Correct specification*

```
TEST 10
VS9=6V,R16V
SEND MPAT MPAT1
MEAS MPAT
TEST 20
VS9=4V,R5V
SEND MPAT MPAT1
MEAS MPAT
```


- *Incorrect specification*

```
TEST 10
  VS9=6V,R16V
  SEND MPAT MPAT1
  MEAS MPAT
TEST 20
  VS9=4V,R5V
  MEAS MPAT MPAT1 ;The range is set to R16V
```

AFMCAPA

The AFM size is checked (The error code is 12464).

Checks if the addresses specified using the AFM ADDRESS statement match with the installed AFM memory or not.

AFMSCRAM

The relation between SCRAM specified in the AFM ADDRESS statement and the transferred scramble program is checked.

When MASK is set to DISABLE when SCRAM has been specified in the AFM ADDRESS statement, it is checked whether a scramble program has been transferred to the address descrambler. When a scramble program has not been transferred, a runtime error (error code: 12473) occurs.

LMTFAIL

A test head error code of 12531 is the target for LMTFAIL.

LMTFAILs (limiter failures) are checked in the following statements:

- MEAS/START DC statement
- MEAS/START VS statement
- MEAS/START MPAT statement
- JUDGE statement
- STOP statement
- END statement

A runtime error will occur when a limiter failure is detected while LMTFAIL is set to DISABLE. A runtime error will not occur even when a limiter failure is detected while LMTFAIL is set to ENABLE.

MBLOCK

An error code of 12532 occurring when the program is started with the HIFIX unlocked is the target for MBLOCK.

The following will occur when the program is started with MBLOCK set to DISABLE:

- For a 1-station parallel test

When the test station contains DUTs to be tested, it is checked whether HIFIX has been locked^{*1} or not.

When the test station does not contain DUTs to be tested, no check is made.

- For a 2-station parallel test

When Station 1 contains DUTs to be tested, it is checked whether HIFIX has been locked^{*1} for Station 1 or not.

After that, when Station 2 contains DUTs to be tested, it is checked whether HIFIX has been locked^{*1} for Station 2 or not.

**1 A HIFIX is locked while the LOCK button (mounted in the test head's Mother Board Lock Free switch) is being pressed*

A runtime error will occur if the HIFIX locked state check mentioned above determines that the target HIFIX is unlocked.

When MBLOCK is set to ENABLE, a runtime error will not occur whether HIFIX has been locked or not at the start of the program.

45. 2 ENABLE/DISABLE

ENABLE

MASK ENABLE is set. Error check is not executed.

DISABLE

MASK DISABLE is set.

The default settings are as follows.

IN	DISABLE
DCGRDALM	DISABLE
PSGRDALM	DISABLE
DPUTYPE1	DISABLE
AFMCAPA	ENABLE
AFMSCRAM	ENABLE
LMTFAIL	ENABLE
MBLOCK	ENABLE

➔ For more information on how to disable or enable check functions, refer to [45. 3 "Setting Using Menu" on page 45-6](#) and [45. 4 "Setting Using Argument" on page 45-7](#).

45. 3 Setting Using Menu

Setting can be made with referring to the current settings.

45. 3. 1 Start Up

Start RTEMASK with no argument specified from the TCLI or a program.

- Start up through the TCLI.

```
/RTEMASK.␣
```

- Start up through the program.

```
CALL LIB RTEMASK
```

45. 3. 2 Menu

ENABLE and DISABLE are switched at every time the return key is entered with positioning the cursor on the objective item. You can also enter the [E] key to set the selected item to ENABLE and the [D] key to set the selected item to DISABLE.

```
*** R.T.E MASK UTILITY ***
.IN -----: DISABLE
.DCGRDALM --: DISABLE
.PSGRDALM --: DISABLE
.DPUTYPE1  --: DISABLE
.AFMCAPA   ---: ENABLE
.AFMSCRAM  --: ENABLE
.LMFAIL    ----: ENABLE
.MBLOCK    ----: ENABLE
```

```
[NOTE]
  PF1:EXIT
```

Enter the [PF1] key to terminate the menu. After the PF1 key is entered, the display of [NOTE] is changed to EXIT! to show the menu is ended.

Limitation

The setting are effective only when the menu is entered by the PF1 key. If it is forcibly ended by CTRL/C, it is ineffective.

45. 4 Setting Using Argument

ENABLE or DISABLE of the error check function can be set without opening the menu.

This setting method is effective to change ENABLE or DISABLE in the test plan program.

- Start up through the TCLI.

```
/RTEMASK IN=ENABLE.␣  
/RTEMASK IN=DISABLE.␣  
/RTEMASK DCGRDALM=ENABLE PSGRDALM=DISABLE.␣
```

- Start up through the program.

```
CALL LIB RTEMASK ("IN=ENABLE")  
CALL LIB RTEMASK ("IN=DISABLE")  
CALL LIB RTEMASK ("DCGRDALM=ENABLE PSGRDALM=DISABLE")
```

If an incorrect argument is used, the error message "ILLEGAL ARGUMENT!!" is displayed.

If this occurs, all settings become invalid.

46. Pin Monitor by Each MEAS Statement (/PMSET)

This utility is the program to start the pin monitor by each MEAS statement.

46. 1 Operation of PMSET

46. 1. 1 Conversational Mode

◆ Start

Enter the following command when the controller is operating.

```
/PMSET↵
```

The following menu is displayed. Select a MENU item (1 to 7) to set the condition.

```

***** MENU OF REAL TIME PIN MONITOR ON THE MEAS *****

1.  SELECT ENABLE/DISABLE ----- ENABLE
2.  SELECT TEST NUMBER  ----- 0-99999
3.  SELECT UNIT NAMES  ----- EPM
    GLOBAL SWITCH ----- NO SELECT
4.  SELECT OUTPUT DEVICE ----- VKT
5.  LOOP CHECK MODE (E/D) ----- DISABLE
6.  DISPLAY MODE ----- DEFAULT
7.  SOCKET TRANSLATION (E/D) -- DISABLE

[WORK]
SELECT NUMBER = [ ]

[NOTE] NUMBER(*1-7) : SELECT PARAMATER    RETURN : EXIT

```

In this screen when the information is formerly set, this information is displayed.

When the information is not formerly set, the information above mentioned is displayed.

◆ Description for Each Item

ENABLE/DISABLE

```
SELECT NUMBER = [1]
```

The following message is displayed in the WORK area by entering the above number.

```
[WORK]
1. SELECT ENABLE/DISABLE (1:ENABLE 2:DISABLE) = 1.
                                     ↑
                                     Entry example
[NOTE] NUMBER(#1,2) : SELECT ENABLE/DISABLE RETURN : EXIT
```

1:ENABLE Enable the pin monitor by each MEAS statement.
2:DISABLE Disable the pin monitor by each MEAS statement.

Enter 1 or 2 and press the Return key.

When you press the Return key with no value specified, the previous setting is retained.

TEST NUMBER

```
SELECT NUMBER = [2]
```

The following message is displayed in the WORK area by entering the above number.

```
[WORK]
2. SELECT TEST NUMBER (0-99999) = 1-10,100,200,300.
                                     ↑
                                     Entry example
[NOTE] NUMBER(#0-99999) : SELECT TEST NUMBER      RETURN : EXIT
```

Enter the test number to be monitored by each test program.

When you press the Return key with no value specified, the previously set number is retained.

(1-10 in the above example means the same as 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10.)

Tip

When entering 141 characters or more to the test number, the following message is displayed to enter them again.

```
TEST NUMBER OVERFLOW
```

UNIT NAMES

```
SELECT NUMBER = [3]
```

The following message is displayed in the WORK area by entering the above number.

```
[WORK]
LIST OF UNIT NAMES (Pn      ,PDn      ,PCn      ,VSn      ,DC      ,
                     INn      ,OUTn     ,VTn      ,ILn      ,VSIM     ,ISVM     ,
                     VCON     ,PCON     ,LCON     ,TSn      ,REF      ,          ,
                     RATE<m> ,TGCLKn<m>,ACLKn<m> ,BCLKn<m> ,CCLKn<m> ,DRELn<m> ,
                     DREtn<m>,STRBn<m> ,WSTRBn<m> ,          ,SETTLING ,ALL      ,
                     EPM      )

SELECT UNIT NAMES = BCLK<1> ,CCLK<1> ,IN1 ,OUT1.␣
                    ↑
                    Entry example

[NOTE] RETURN : EXIT
```

Enter a unit name to be monitored and press the Return key.

When you press the Return key with no value specified, the previously set unit is retained.

When entering EPM to the unit name, the following message is displayed in the WORK area.

```
[WORK]
3. SELECTED UNIT NAME = EPM

SELECT GLOBAL SWITCH (1:/L 2:/H 3:NO SELECT 4:/A*1) = 2.␣
                                                    ↑
                                                    Entry example

[NOTE] RETURN : EXIT
```

**1 Displayed on the T5371 test system.*

Set the global switch appended to the extended pin monitor.

Select a value from 1 to 4 and press the Return key.

Tip

When entering 180 characters or more to the unit name, the following message is displayed to enter them again.

```
UNIT NAME OVERFLOW
```

ALL and EPM cannot be set with another unit combined.

ALL and EPM cannot be combined for use together.

When you enter ALL, the output format when "/PM" for the the pin monitor is entered is used.

OUTPUT DEVICE

```
SELECT NUMBER = [4]
```

The following message is displayed in the WORK area by entering the above number.

```
[WORK]
  4. SELECT OUTPUT DEVICE (1:VKT 2:LPT 3:FILE) = 1.
                                     ↑
                                   Entry example
```

```
[NOTE] NUMBER(#1-3) : SELECT OUTPUT DEVICE RETURN : EXIT
```

- 1:VKT Output to VKT.
- 2:LPT Output to printer.
- 3:FILE Output to specified file.

Select a number and press the Return key.

When you press the Return key with no value specified, the previous setting is retained.

More than one number can be selected. However, when both LPT and FILE are selected, LPT is not set.

When entering "3 : FILE" to the output device, the following message is displayed in the WORK area.

```
[WORK]
  4. SELECT OUTPUT DEVICE (1:VKT 2:LPT 3:FILE) = 3.
                                     ↑
                                   Entry example

  OUTPUT FILE NAME = FILENAME.
                                     ↑
                                   Entry example
```

```
[NOTE] RETURN : EXIT
```

Set the file name.

Enter a file name and press the Return key.

LOOP CHECK MODE (E/D)

```
SELECT NUMBER = [5]
```

The following message is displayed in the WORK area by entering the above number.

[WORK]

5. LOOP CHECK MODE ENABLE/DISABLE (1:ENABLE 2:DISABLE) = 1↵

↑

Entry example

[NOTE] NUMBER(#1, 2) : SELECT LOOP CHECK ENABLE/DISABLE RETURN : EXIT

- | | |
|------------|---|
| 1: ENABLE | When using the MEAS statement of the same test number in loop, conduct the pin monitor only first once. |
| 2: DISABLE | Execute the pin monitor to all MEAS statements. |

Enter 1 or 2 and press the Return key.

When you press the Return key with no value specified, the previous setting is retained.

DISPLAY MODE

SELECT NUMBER = [6]

The following message is displayed in the WORK area by entering the above number.

[WORK]

6. SELECT NORMAL(80)/WIDE(132) (1:NORMAL 2: WIDE 3: DEFAULT) = 1↵

↑

Entry example

[NOTE] NUMBER(#1-3) : SELECT NORMAL/WIDE/DEFAULT RETURN : EXIT

- | | |
|------------|---|
| 1: NORMAL | Output in 80-column mode. |
| 2: WIDE | Output in 132-column mode. |
| 3: DEFAULT | Output in the column mode for the tester default value. |

Select a value from 1 to 3 and press the Return key.

When you press the Return key with no value specified, the previous setting is retained.

SOCKET TRANSLATION (E/D)

SELECT NUMBER = [7]

The following message is displayed in the WORK area by entering the above number.

[WORK]

```
7. SELECT SOCKET TRANSLATION (1:ENABLE 2:DISABLE ) = 1.↓
                                     ↑
                               Entry example
```

[NOTE] NUMBER(#1,2) : SELECT ENABLE/DISABLE RETURN : EXIT

- | | |
|------------|--|
| 1: ENABLE | The pin number is set according to the socket program. |
| 2: DISABLE | The pin number represents the pin number for the tester. |

Enter 1 or 2 and press the Return key.

When you press the Return key with no value specified, the previous setting is retained.

When all conditions are set, exit the menu by pressing the Return key.

The following message is displayed.

```
***** PMSET EXIT !! *****
```

The following files are created to the using customer.

- ASC PM.SE1 (Test number information)
- ASC PM.SE2 (Information on the number of output columns, socket conversion information, unit information)
- ASC PM.SE3 (Output device information)
- ASC PM.SE4 (ENABLE/DISABLE, global switch information of EPM)
- ASC PM.SE5 (Loop check mode information)

46. 1. 2 Direct Setting Mode

Enter during the controller is operating.

Setting the direct information

The information can be set by using the following method other than the direct mode.

```
/PMSET Global switch Mode Test number Unit name Number of output columns Socket
conversion
Loop check mode
```



Limitation

The global switch, number of output column, and socket conversion can be omitted.

- Global switch

Specify /V, /L or /F. Otherwise, data is output to the VKT.

More than one global switch can be specified. However, when both /L and /F are specified, /L is not set.

Specifying something other than the following four global switches causes an error to occur.

/V	Outputs to VKT.
/L	Outputs to printer.
/F	Outputs to the specified file. The output file name leaves one column space to be appended in the end of the command.



Tip

When the log turns on, the information is output to the specified file.

/A	Outputs cycle palette data.
----	-----------------------------

- Mode

ENABLE	Enable the pin monitor for each MEAS statement (Abbreviation "E").
DISABLE	Disable the pin monitor for each MEAS statement (Abbreviation "D").

- Test number

Enter the test number to be monitored.

- Unit name

Enter the unit name to be monitored.

- Number of output column

H	Output in 80-column mode.
W	Output in 132-column mode.

When this argument is omitted, the default number of columns for the tester is used.

- Socket conversion

H	The pin number represents the pin number for the tester.
---	--

When this argument is omitted, the pin number is set according to the socket program.

- Loop check mode

ENABLE	Enable the loop check mode (Abbreviation "E").
DISABLE	Disable the loop check mode (Abbreviation "D").

When all conditions are set, press the Return key.

When the setting makes an error, the following error message is displayed.

```
ILLEGAL ARGUMENT !! TRY AGAIN
*** PMSET EXIT !! ***
```

Default information direct setting

Use this method when monitoring after the default information or the information that a user set formerly is enabled.

```
/PMSET/D
```

Display the previous setting condition as follows to exit by entering the above mentioned command.

When the previous set information is not existed, the default value is displayed.

```
***** MENU OF REAL TIME PIN MONITOR ON THE MEAS *****

1.  SELECT ENABLE/DISABLE ----- ENABLE
2.  SELECT TEST NUMBER  ----- 0-99999
3.  SELECT UNIT NAMES  ----- EPM
    GLOBAL SWITCH ----- NO SELECT
4.  SELECT OUTPUT DEVICE ----- VKT
5.  LOOP CHECK MODE (E/D) ----- DISABLE
6.  DISPLAY MODE ----- DEFAULT
7.  SOCKET TRANSRATION (E/D) -- DISABLE

***** PMSET EXIT !! *****
```

47. Operation Panel

Operation panel is the tool operated on the work station to realize functions equivalent to the control box of T5500 series by using software.

The setting of device program to be executed, start, reset and the setting of test mode can be specified.

Utility menu enables the functions equivalent to the controller commands.

The OS shown below can be used.

ASX/U-51 REV 1.01 to REV 4.03-1

Sun OS Release 5.4 or later

ASX/U-51 REV 4.03 A or later

Sun OS Release 5.5.1 or later

Window manager is operated on the open look or Motif.

In this manual, the screen of the Motif window manager is used.

Tip

Operation panel is compatible with a maximum of 64 DUTs for one station, or a maximum of 128 DUTs for two stations.

Category number is compatible with 1 to 200. Test number is compatible with 0 to 999.

47. 1 Environmental Setting of Operation Panel

Before starting the operation panel, set the following environmental variables in the work station where the test controller is started.

Path of library

Set the path of library when the operation panel is executed.

```
% setenv LD_LIBRARY_PATH /usr/openwin/lib:/usr/dt/lib
```

Path of device program

Set the path where the device program exists.

```
% setenv SOFTPANL /home/asx:${HOME}/device
```

Above example sets that the device program exists in the /home/ASX and the device under the log in path.

When the device program to be executed, the path which activates the test controller and the path which activates the operation panel are the same, it is operated even if there is no setting.

 Tip

*SOFTPANL is the same setting as the /DIRLINK of the tester utility program.
When there is no space in the DIRLINK setting, it is ignored.*

➔ For more information on /DIRLINK, refer to [44. "DIRLINK" on page 44-1](#).

Tester name

Set the tester name displayed in the title part of the operation panel.

```
% setenv TESTER_NAME testername
```

Set a tester name in the testername argument.

Up to 30 alphanumeric characters can be used to specify a tester name.

 Tip

*You cannot omit specifying the path to the library.
When the path of device program is not set, current directory is regarded as the path.
When the tester name is not specified, the host name of the work station is regarded as the tester name.*

When you remotely log in to the tester, you need to specify an additional two work stations as follows:

- Display side (Remote side)
Set the work station of the host.

```
% xhost hostname
```

Set a work station name (host name) to execute the program for the hostname argument.

- Execution side (Host side)
Set the work station of the remote.

```
% setenv DISPLAY remote name:0
```

Set a work station name (remote name) to display data for the remote name argument.

47. 2 Activation and Termination of Operation Panel

◆ Start

To start the operation panel, enter the following command:

```
% opanel [-time 500ms]
```

You can specify an optional time period in the brackets.

opanel

This is a command to start the operation panel.

-time 500ms

Specify a time interval at which the operation panel is displayed.

Enter a time interval equal to or greater than 1 ms after -time.

Only ms and s can be specified as the unit.

Otherwise, 500 ms is assumed.

If the option is omitted, 500 ms is specified.

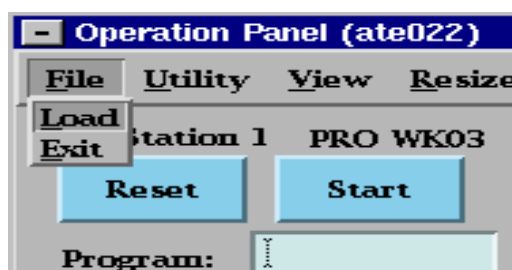


Tip

*Time interval to display means that the time interval from the display to the next display.
It does not mean that the display is executed in the specified time interval.*

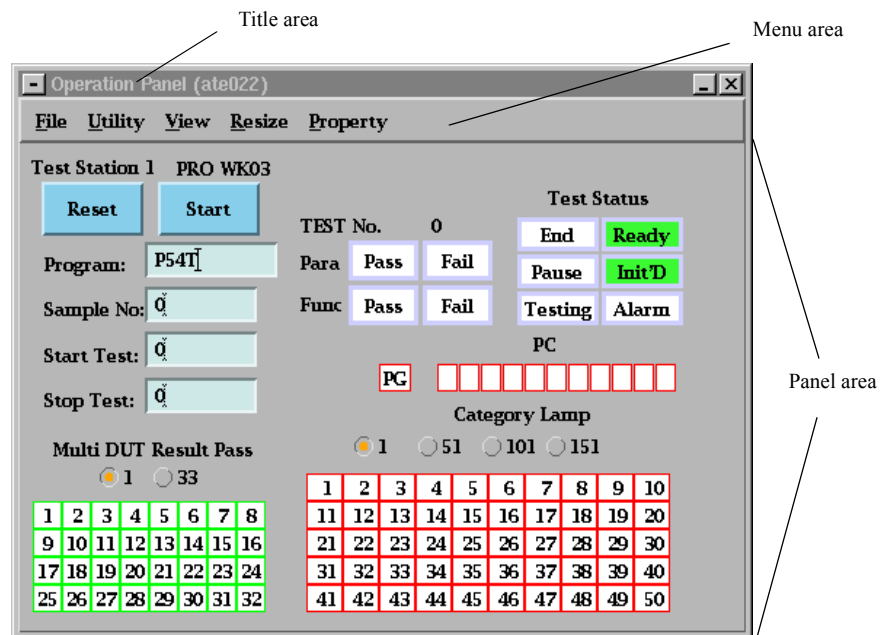
◆ Terminating Method

Select **[Exit]** from the **[File]** menu in the operation panel.



47. 3 Basic Operation of Operation Panel

The following screen is displayed when the operation panel is activated.



Title area

A tester name or host name is displayed.

Menu area

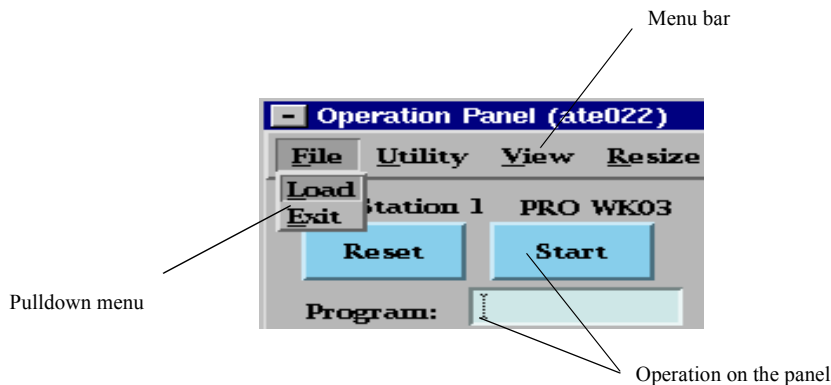
Execute the operation of the operation panel.

Panel area

Start/reset, setting of program and various status are displayed.

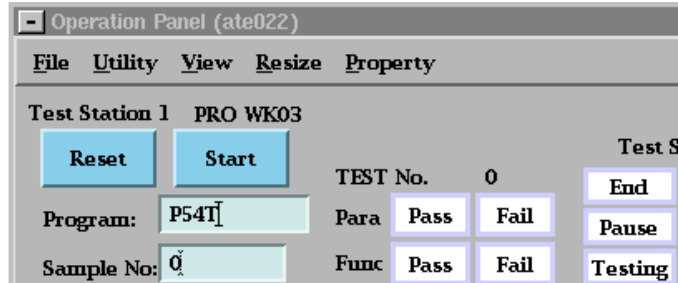
There are two ways to operate the operation panel; one is to use buttons on the panel and another is to select pulldown menus from the menu bar.

When you select a pulldown menu, a sub panel may be displayed. In that case, various setting is specified by the sub panel.



47. 3. 1 Operation of Menu

Menus include **[File]**, **[Utility]**, **[View]**, **[Resize]**, and **[Property]**.



[File] menu

Sets execution program and terminates operation panel.

[Utility] menu

Selects this menu when you want to display data such as counters.

[View] menu

Selects the station to be specified to the panel area.

[Resize] menu

Changes the size of the panel area.

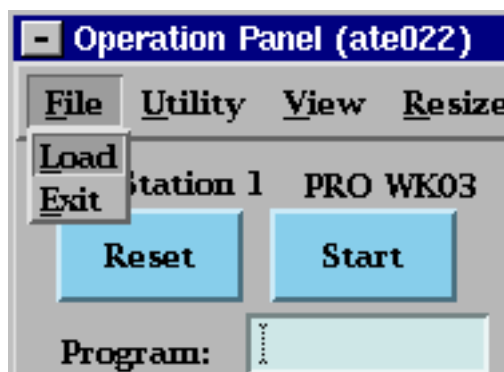
[Property] menu

Sets the operation mode of the test controller.

47. 3. 1. 1 File Menu

The **[File]** menu is used to set an execution program or terminate the operation panel.

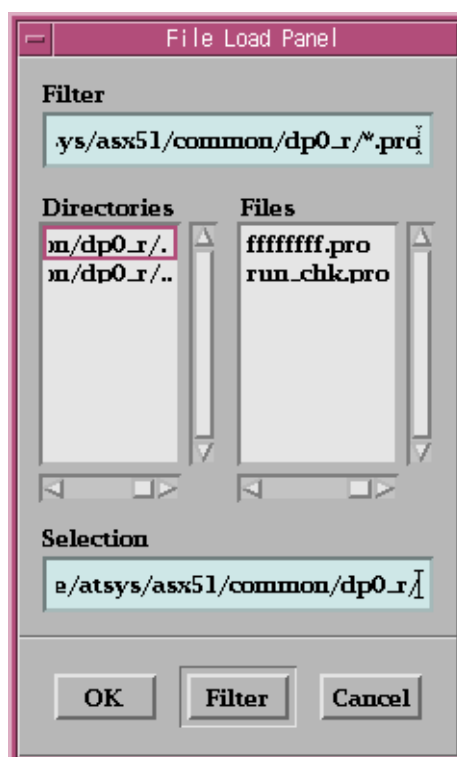
When you click the [File] menu, a pulldown menu containing **[Load]** and **[Exit]** is displayed.



[Load]

Sets the program to be executed.

When you select **[File]**, the Load sub window is displayed.



The program name to be executed can be set in the Load sub window.

When you select a file name to be executed in the Files field and press the **[OK]** button, the specified file name is displayed in the Program field of the current panel.

When you do not want to change the execution program, press the **[Cancel]** button.

When you want to change the directory, select the directory to be changed from the list of Directories.

Limitation

Only the directories set in the "SOFTPANL" environmental variable can be set here. If the file is set by using other directory, the program cannot be executed.

[Exit]

Terminates the operation panel.

47. 3. 1. 2 Utility Menu

Select this menu when you want to display data such as counters.

This menu allows you to execute functions equivalent to the controller commands L, N, and C in real time.

Utility menu has the following menu.

[L command]

Displays Category 1 to 200, Sort 1 to 16, and Test Fail.

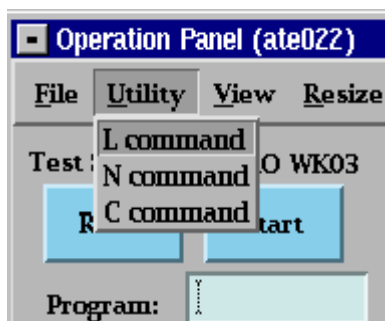
[N command]

Displays the fail test number.

[C command]

Selects the counter you want to clear.

The selected counter is cleared at the test end.

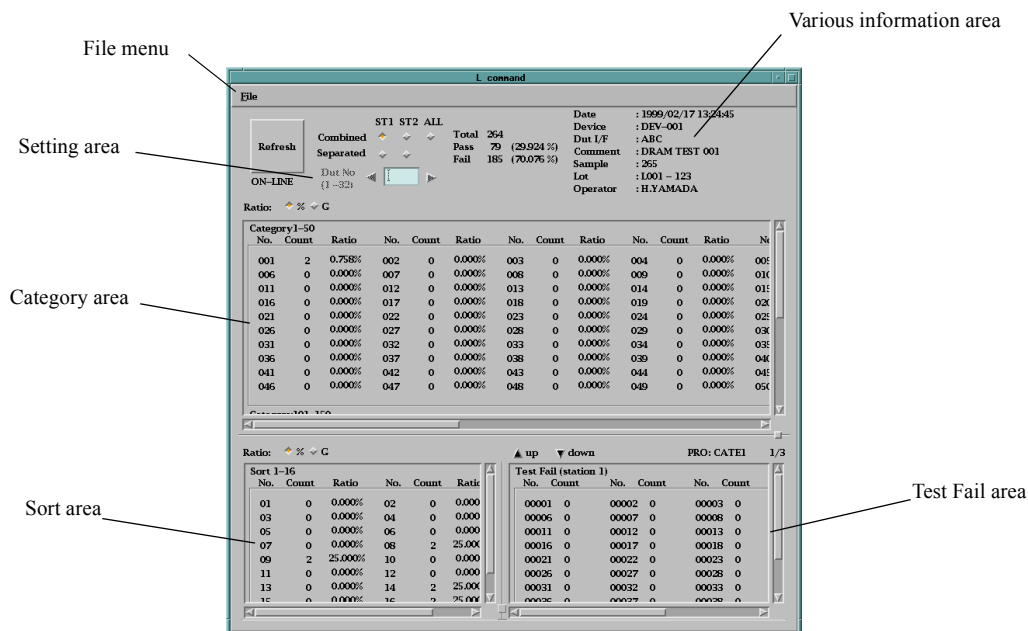


◆ [L Command]

The following screen is displayed when the L command is activated.

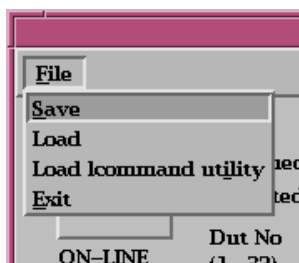
47. 3 Basic Operation of Operation Panel

This screen displays Category 1 to 200, Sort 1 to 16 and Test Fail.



[File] menu

When you click **[File]** in the L command panel, the following pulldown menu is displayed.



The **[File]** menu contains the following sub menu.

- **[Save]**
Saves the contents of the L command currently displayed in a file.
Selecting this menu displays the file name selection window.
- **[Load]**
Displays the contents of the selected file.
Selecting this menu displays the file name selection window.
- **[Load lcommand utility]**
Loads a file in the lcommand utility format.
For files in this format, the matrix measurement obtained by RETEST MODE2 (select using / STHAND utility) and retest category and sort data for a test system which is connected with a handler are saved.

The same file as the file displayed in [52. "Category and Sort Data Output Program \(lcommand\)" on page 52-1](#) is used.

- **[Exit]**

Terminates the L command panel.

File selection window

When you select **[Save]** or **[Load]** from the **[File]** menu in the L command panel, the following file selection window is displayed.



This window has the following contents.

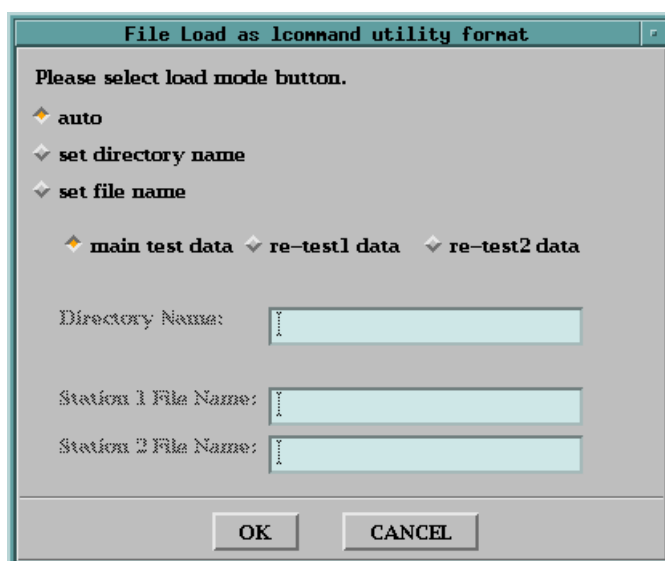
- **[Filter]**
Sets file name attributes.
When Save is selected, you can specify *.dat or *.asc.
When Load is selected, you can specify *.dat only.
- **[Directories]**
Selects the directory that contains the file you want to select.
- **[Files]**
Selects the file to execute Save or Load.
- **[Selection]**
Displays the file name currently selected.
- **[OK]** button
Press this button when the correct file is selected in the Selection field.
This button confirms the selected file and closes the window.
- **[Filter]** button
Press this button when you want to change a setting in the Filter field.

- **[Cancel]** button

Press this button when you want to cancel a setting.

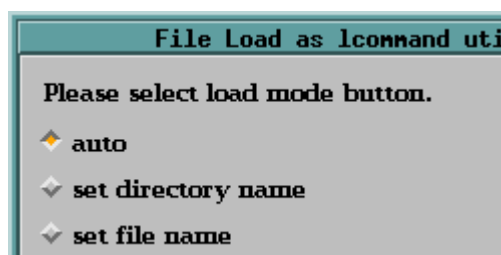
lcommand utility file selection window

When you select **[lcommand utility]** from the **[File]** menu in the L command panel, the following screen is displayed.

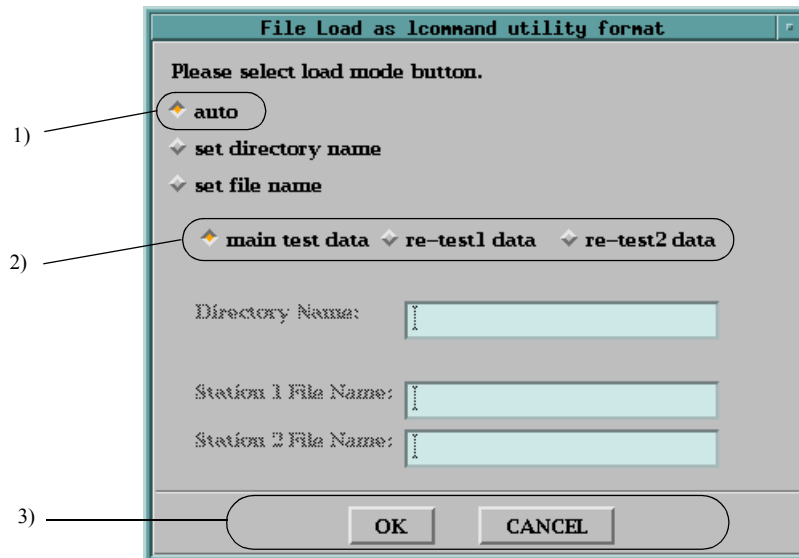


You can load a file in three different ways: by selecting the **[auto]**, **[set directory name]**, or **[set file name]** button.

1. When the **[auto]** button is selected
The file saved using the /STHAND utility is located in the current directory, that is, the directory when opanel is activated.
2. When the **[set directory name]** button is selected
The file saved using the /STHAND utility is not located in the current directory.
3. When the **[set file name]** button is selected
The file saved using the /STHAND utility has been renamed, that is, the file name has been changed.



1. When the **[auto]** button is selected



Take the following steps.

- 1) Select the **[auto]** button.

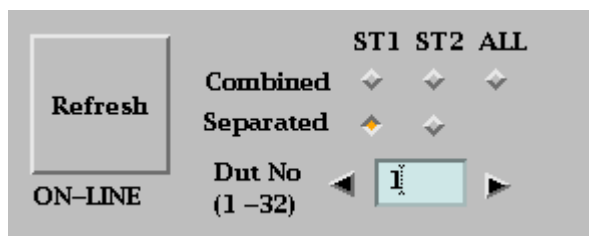
- 2) Select a data type.

main test data	Matrix measurement data
re-test1 data	Data of 1st remeasurement
re-test2 data	Data of 2nd remeasurement

◀ main test data ▶ re-test1 data ▶ re-test2 data

Make sure that the selector button is properly set in the L command panel.

The selector buttons of L command panel are as below.



Combined ST1	The integrated value of all DUTs in station 1 is displayed.
Combined ST2	The integrated value of all DUTs in station 2 is displayed.
Combined ALL	The integrated value of all DUTs is displayed.
Separated ST1	Information on a DUT selected from Station 1
Separated ST2	Information on a DUT selected from Station 2

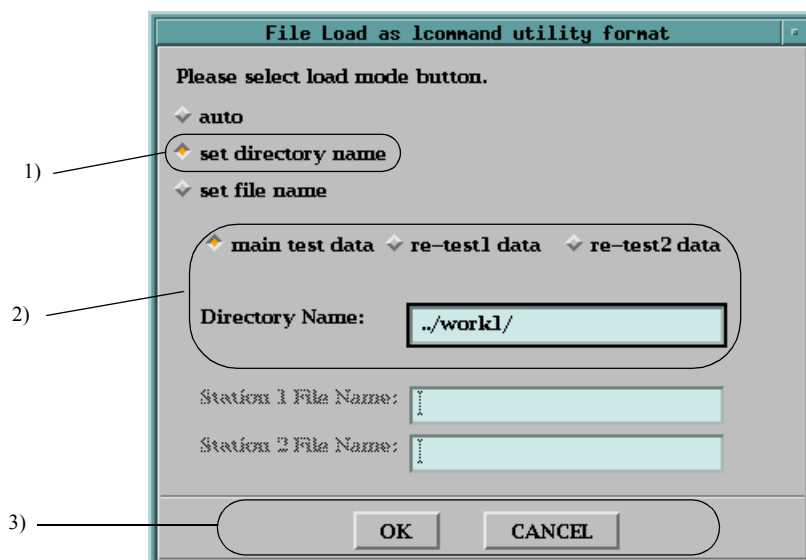
You also need to select a DUT number for Separated.

- 3) Click the **[OK]** button. To cancel the settings, click the **[CANCEL]** button.

The loaded file names will be as shown in the table below.

Selector button of L command panel	main test data	re-test1 data	re-test2 data
Combinated st1 Separated st1	mtestd1.dat	retestd11.dat	retestd12.dat
Combinated st2 Separated st2	mtestd2.dat	retestd21.dat	retestd22.dat
Combinated all	mtestd1.dat mtestd2.dat	retestd11.dat retestd21.dat	retestd12.dat retestd22.dat

2. When the **[set directory name]** button is selected



Take the following steps.

- 1) Select the **[set directory name]** button.
- 2) Select the desired data type and enter a directory name.

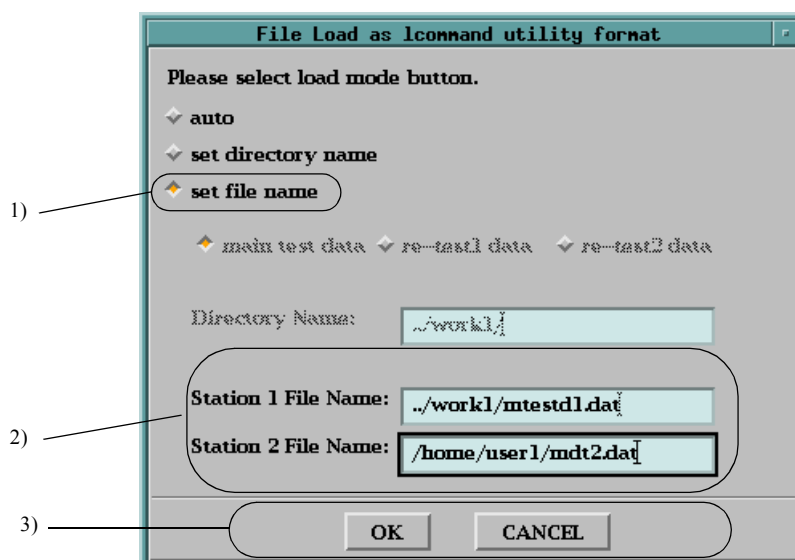
Select a data type. Alternatives are the same as those when the **[auto]** button is selected.

Enter a directory name in the Directory Name input area using the keyboard, by specifying either the absolute or relative path.

Check that buttons are selected properly in the L command panel in the same way as when the **[auto]** button is selected.

- 3) Click the **[OK]** button. File names to be loaded are the same as those loaded when the **[auto]** button is selected. To cancel the settings, click the **[CANCEL]** button.

3. When the **[set file name]** button is selected



Take the following steps.

1) Select the **[set file name]** button.

2) Enter a file name.

Enter a file name in the Station 1 or Station 2 File Name input area using the keyboard, by specifying either the absolute or relative path.

To apply your settings of the buttons in the L command panel to Station 1

Enter a file name in the Station 1 File Name field. If you enter a file name in the Station 2 File Name input area, the data is not valid.

To apply your settings of the buttons in the L command panel to Station 2

Enter a file name in the Station 2 File Name field. If you enter a file name in the Station 1 File Name input area, the data is not valid.

To apply your settings of the buttons in the L command panel to ALL

Enter a file name in both the Station 1 File Name and Station 2 File Name fields.

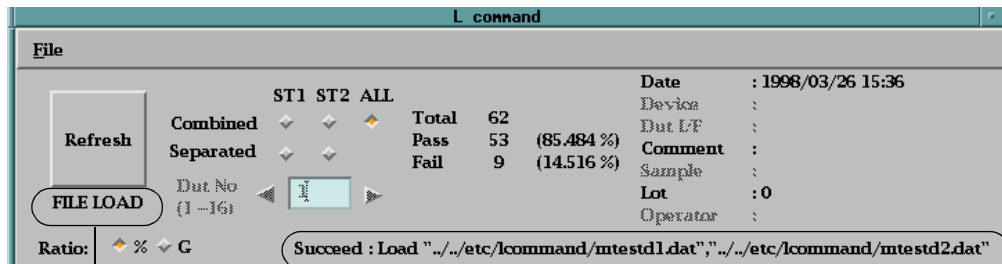
3) Click the **[OK]** button. To cancel the settings, click the **[CANCEL]** button.

◆ **When a File is Loaded Successfully**

When the data has been loaded successfully, the L command panel display changes as shown below.

The status display changes to FILE LOAD and the loaded file name appears.

In this example, the set directory name button is selected and ../../etc/lcommand is specified as a directory name.



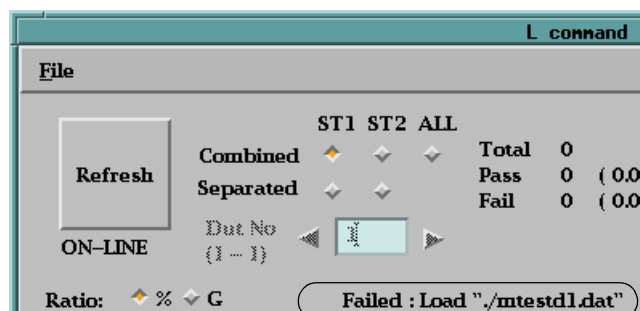
The status display shows FILE LOAD.

The loaded file name is displayed.

◆ When a File is not Loaded Successfully

When the data loading has failed, the L command panel display remains unchanged and the fail file name appears.

In this example, the L command panel is set to Combined ST1 and the lcommand utility panel is set to auto.

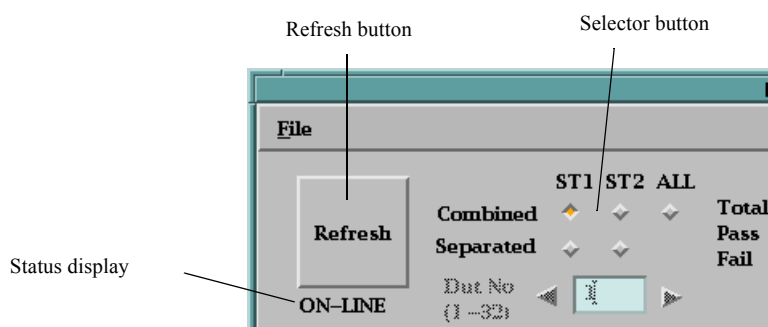


The fail file name is displayed.

Setting area

This area is used to select the display status of the L command panel.

To read new data from the controller, select appropriate buttons and click the **[Refresh]** button. This will refresh the L command panel and display ON-LINE or OFF-LINE to indicate the status.



- **[Refresh]** button
Refreshes the contents to be taken from the controller and displays it on the screen.
- Status display
Indicates the display status. There are three kinds of display status.

ON-LINE	Displays the data taken in when the controller is active.
OFF-LINE	Displays the data taken in when the controller is disconnected.
FILE-LOAD	Displays the data taken from a file.
- Selector button
Selects the contents to be taken from the controller and displays them on the screen. You can select ST1, ST2 or ALL in the Combined section.

ST1	The integrated value of all DUTs in Station 1 is displayed.
ST2	The integrated value of all DUTs in Station 2 is displayed.
ALL	The integrated value of all DUTs is displayed.

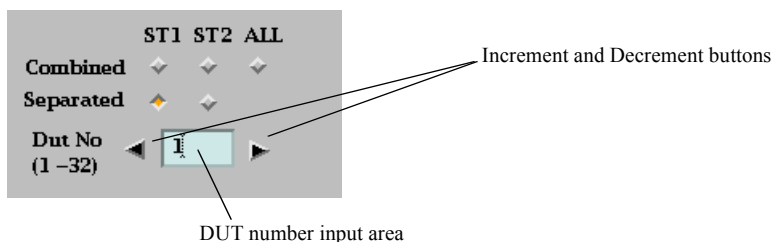
You can select ST1 or ST2 in the Separated section. You can also select a DUT number.

- | | |
|-----|--|
| ST1 | Information on a DUT selected from Station 1 is displayed. |
| ST2 | Information on a DUT selected from Station 2 is displayed. |

Select a DUT number by using one of the following.

Entering the appropriate number in the input area below the selector button.

The arrow buttons are used to increase or decrease the DUT number. Every time you click the arrow buttons, the DUT number increases or decreases by 1. The numbers displayed in the parentheses below "Dut No" show the currently selectable DUT numbers.



Various information area

Total, Pass, and Fail counter values are displayed at the top of the L command panel.

On the right-hand side, you can see the various header information.

Date shows the time when the data was refreshed.

Device, Dut I/F, and other remaining items indicate the values specified for DEVICE, DUTIF, and other respective system variables.

When the selector button is set to ST1 or ALL, the header information for Station 1 is displayed.

When the selector button is set to ST2, the header information for Station 2 is displayed.

		Date	: 1999/02/03 17:09:49
		Device	: DEV-001
Total	168	Dut I/F	: ABC
Pass	79 (47.024 %)	Comment	: DRAM TEST 001
Fail	89 (52.976 %)	Sample	: 169
		Lot	: L001 - 123
		Operator	: H.YAMADA

Category area

Categories 1 to 200 are displayed.

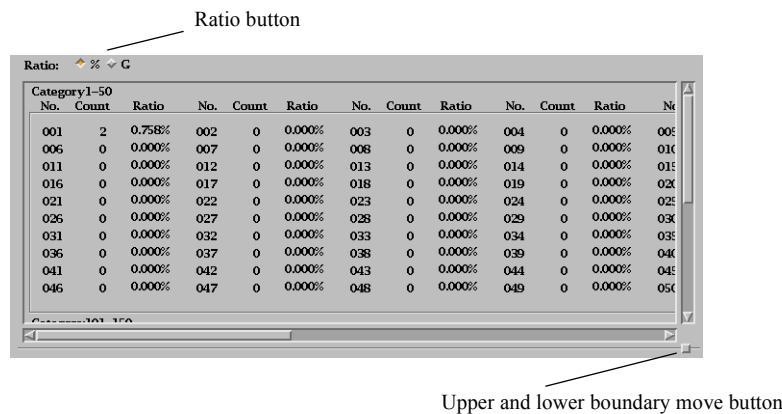
Categories are displayed in blocks of 50.

To move the display up and down, use the scroll bar located on the right-hand side.

To move the display to the right and left, use the scroll bar located at the bottom.

When you click % in the **[Ratio]** button, figures are displayed as percentages (%). When you click G (abbreviation of Graph), figures are converted into a bar graph. The following shows an example of figures displayed in percentages (%). (For an example of a bar graph, see the Sort area.)

To extend the Category display area, drag the upper and lower boundary move button using the mouse.



Sort area

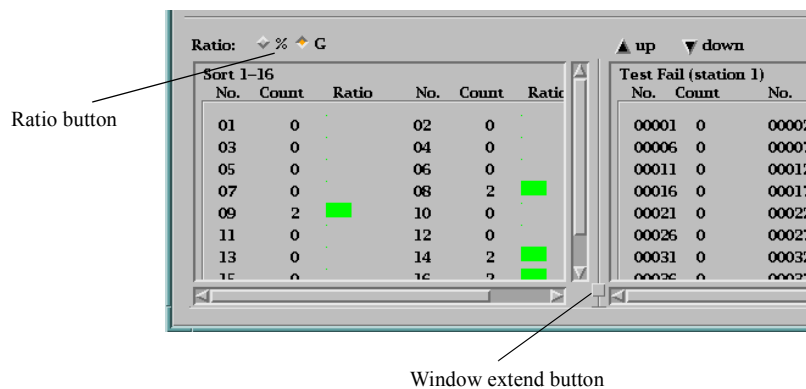
Sorts 1 to 16 are displayed.

To move the display up and down, use the scroll bar located on the right-hand side.

To move the display to the right and left, use the scroll bar located at the bottom.

When you click % in the Ratio button, figures are displayed as percentages (%). When you click G (abbreviation of Graph), figures are converted into a bar graph. An example of the bar graph display is shown below. (For an example of figures displayed in %, see the Category area.)

To extend the Sort display area, drag the window extend button using the mouse.

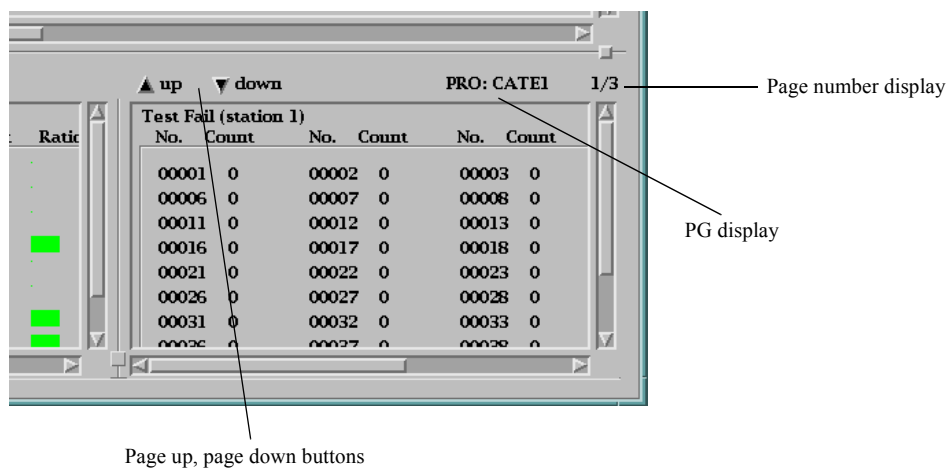


Test Fail area

The values set by the .T controller command for test numbers are displayed.

To move the display up and down, use the scroll bar located on the right-hand side.

To move the display to the right and left, use the scroll bar located at the bottom.

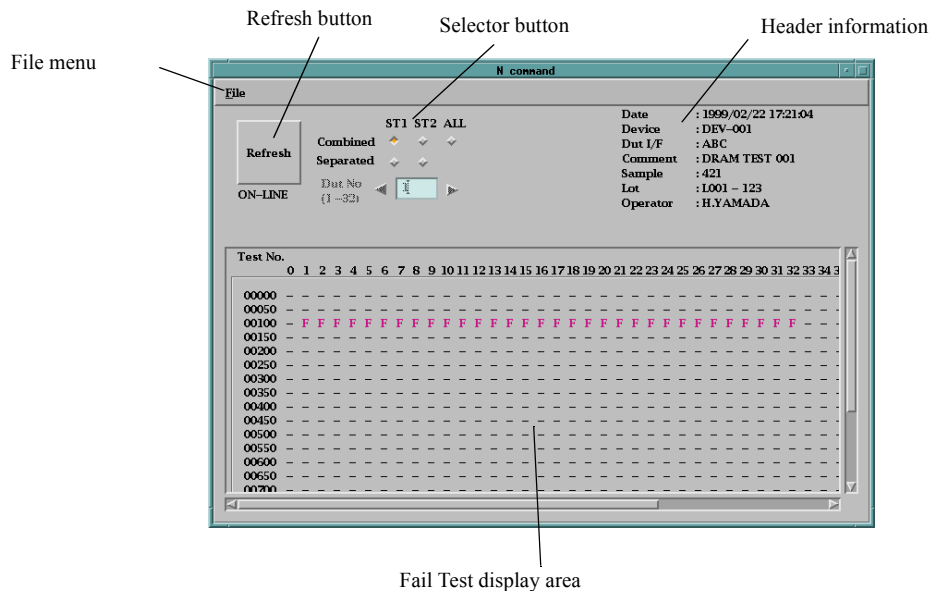


- **[up]** and **[down]** buttons
Click the **[up]** button to go back to the previous page.
Click the **[down]** button to display the next page.
- PG display
Displays the name of the program being executed.
- Page number display
Displays the current page number and the total number of pages.
1/3 in the above example indicates that page 1 of a total of 3 pages is currently displayed.

◆ N Command

The following screen is displayed when the N command is activated.
This screen shows the fail test number.

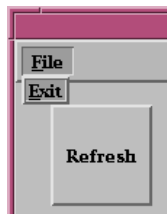
Fail tests are displayed as F in pink, Pass tests are P in green, and NO-TEST is - in black.



This screen has the following areas.

[File] menu

The **[File]** menu contains the following sub menu.



- **[Exit]**

Terminates the N command panel.

[Refresh] button

Refreshes the contents to be taken from the controller and displays it on the screen.

Selector button

Selects the contents to be taken from the controller and displays them on the screen.

You can select ST1, ST2 or ALL in the Combined section.

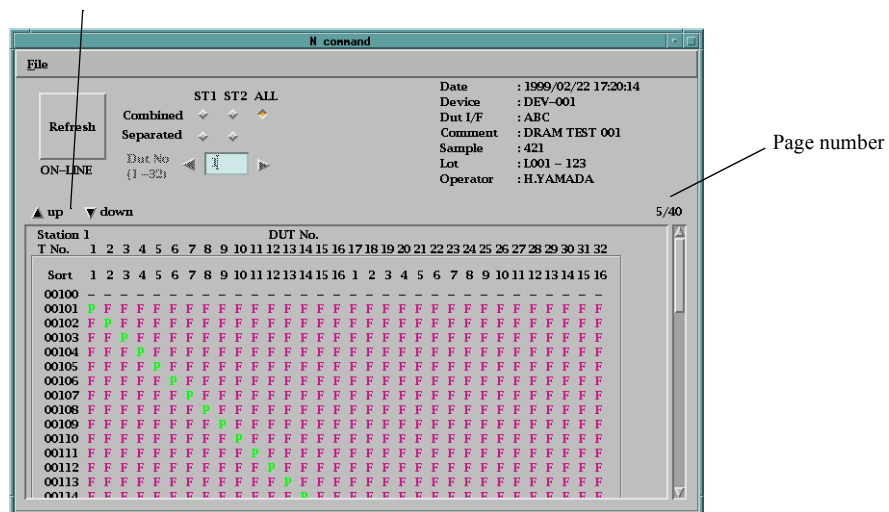
ST1	Test numbers failed by one or more DUTs in Station 1
ST2	Test numbers failed by one or more DUTs in Station 1
ALL	Test numbers failed by all DUTs

You can select ST1 or ST2 in the Separated section. In the Dut No. input area, enter the DUT number you want to display. The number numbers in parentheses shows the currently selectable DUT number.

- ST1 Test numbers failed by a DUT selected from Station 1
- ST2 Test numbers failed by a DUT selected from Station 2

When the selector button is set to ALL, the display looks as shown below.

Test number change buttons



- Test number change buttons
Increment and decrement the fail test number page being displayed.
Click the **[up]** button to go back to the previous page.
Click the **[down]** button to display the next page.

Header information

Displays the header information. Date shows the time when the data was refreshed.
When the ALL button is selected, the header information shows the contents of ST1.

Fail Test display area

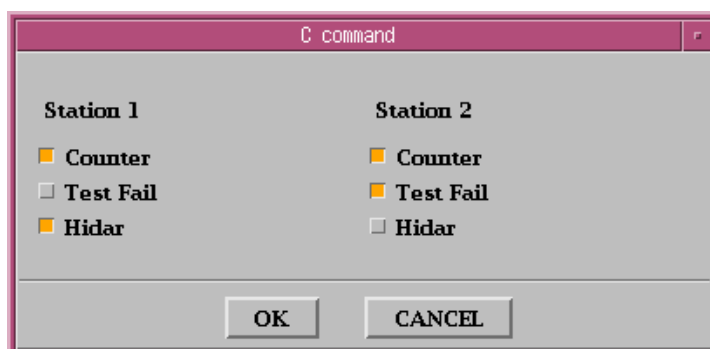
Displays the fail tests.
Click the **[Refresh]** button to update data.

◆ C Command

Activating the C command displays the window as shown below.
In this window you can select the counters you want to clear.
The button currently set appears highlighted.
The selected counter will be cleared at the TEST END.

 Tip

The counters of a station that contains DUTs (SDUTs) to be tested are not cleared.



This window has the following contents.

Counter

You can set the following counters here:

- total counter
- fail counter
- category counter
- sort counter

 Tip

*By cancellation of the Station 1 Counter, Station 1 Test Fail or Hidar will also be disabled.
The same is the case with Station 2.*

Test Fail

You can select this only when the test fail counter is specified using the .T command.

 Limitation

You can select Station 1 Test Fail only when Station 1 Counter is selected.
The same is the case with Station 2.

Hidar

You can select this only when Hidar is specified using the .J command.

 Limitation

You can select Station 1 Hidar only when Station 1 Counter is selected.
The same is the case with Station 2.

[OK] button

Press this button when the setting is correct.

This button confirms the selected counters and dismisses this window.

The selected counters will be cleared at the next TEST END.

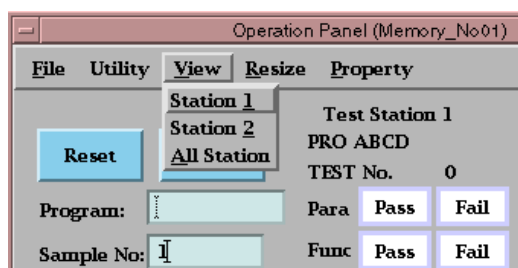
[Cancel] button

Close window without selecting the counters.

The counters will not be cleared at the TEST END.

47. 3. 1. 3 View Menu

When you click the **[View]** menu, a pulldown menu containing **[Station 1]**, **[Station 2]**, and **[All Station]** is displayed.



Selects the station to be displayed.

[Station 1]

The panel of station 1 side is displayed.

[Station 2]

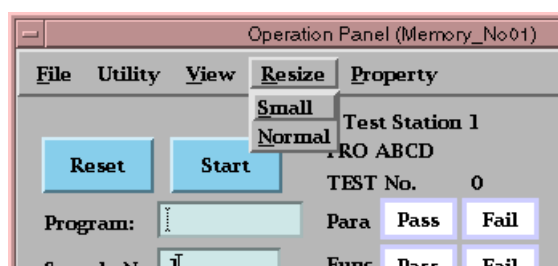
The panel of station 2 side is displayed.

[All Station]

The panel of station 1 side and station 2 side are displayed.

47. 3. 1. 4 Resize Menu

When you click the **[Resize]** menu, a pulldown menu containing **[Small]** and **[Normal]** is displayed.



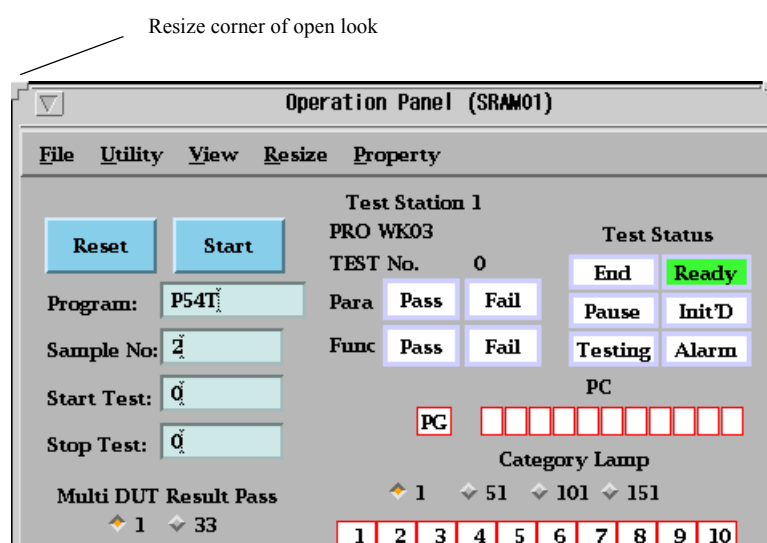
The size of the panel is changed here.

When the panel size is decreased, DUT pass lamp and category lamp are not displayed.

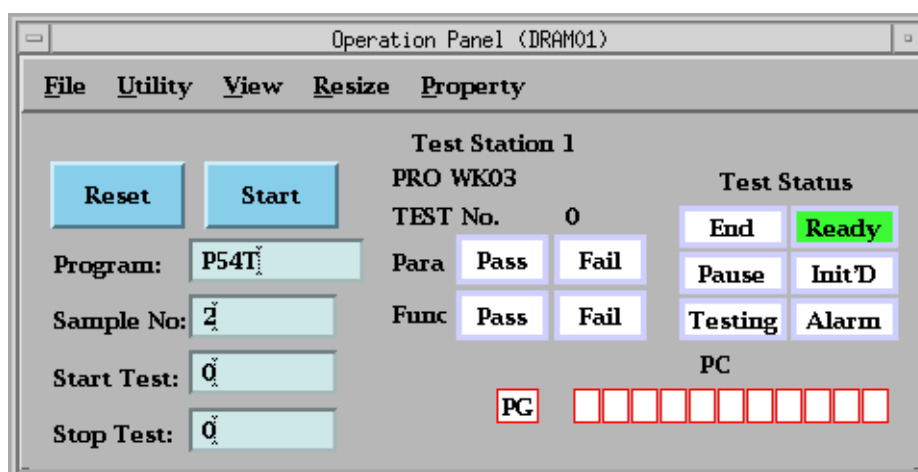
Tip

Use this menu to change the size of the panel.

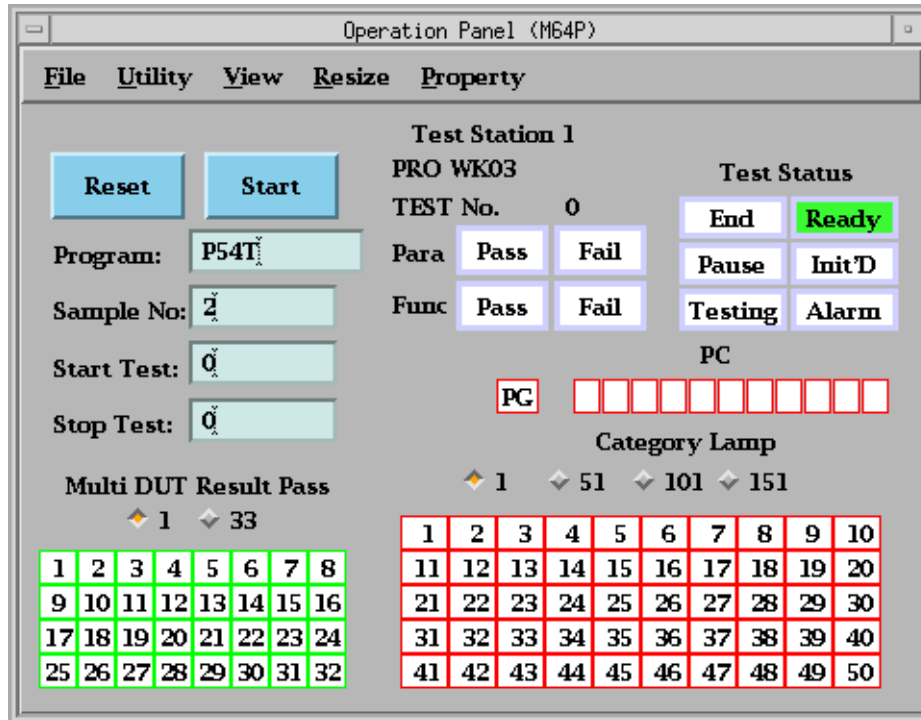
When the open look is used as the window manager, enlarging the panel size with a resize corner after selecting [Small] as the panel size does not display DUT Result Pass and Category Lamp.



When [Small] is selected

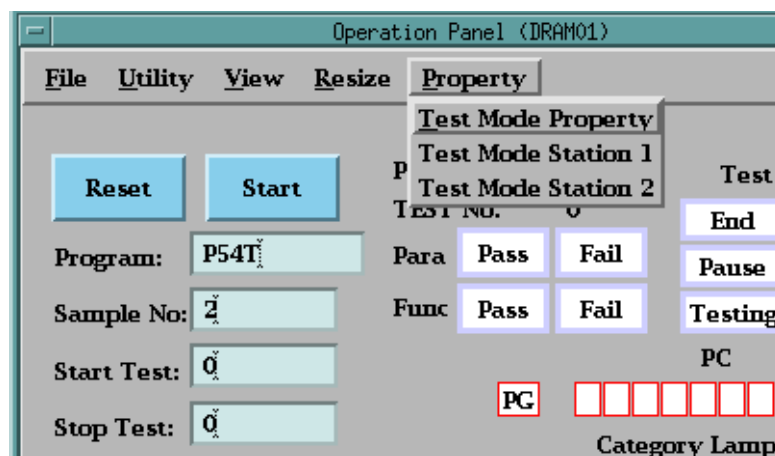


When **[Normal]** is selected



47. 3. 1. 5 Property Menu

When you click the **[Property]** menu, a pulldown menu containing **[Test Mode Property]**, **[Test Mode Station 1]**, and **[Test Mode Station 2]** is displayed.



The operation mode of each station can be set here.

The sub window which sets the operation mode of each station is displayed.

You can click buttons displayed here to set the test controller.

[Test Mode Property]

The property panel of displayed station side is displayed.

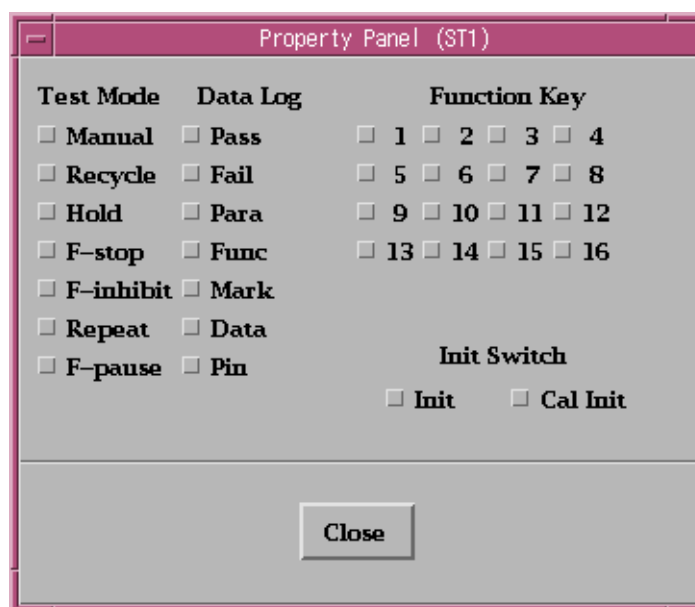
[Test Mode Station 1]

The property panel of station 1 side is displayed.

[Test Mode Station 2]

The property panel of station 2 side is displayed.

The property panel of station 1 side is shown below.



You can click buttons displayed here to set the operating mode of the test controller.
 The operating mode currently set is displayed in the state that the button is depressed.

[Test Mode]

Sets the operating mode of the test.

[Data Log]

Sets data log mode.

[Function Key]

Sets function key.

[Init Switch]

Set the **[INIT]** key that is referred to by IF INIT or IF CALINIT of the ATL statement.

➔ For more information on IF INIT and IF CALINIT, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498\)](#).

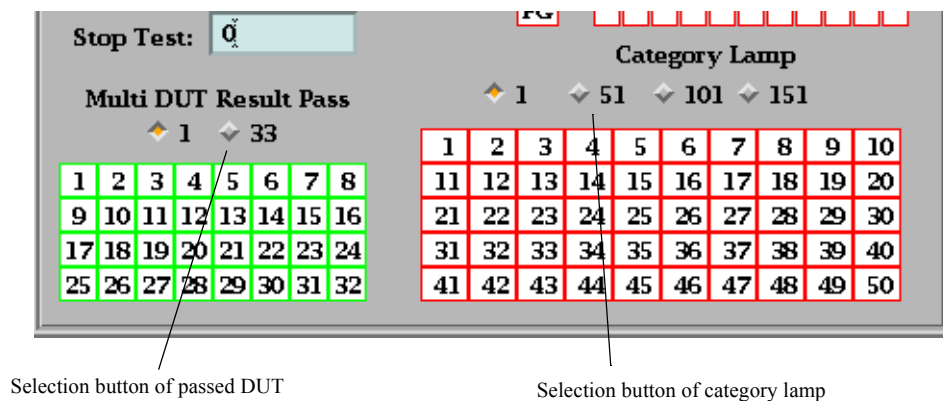
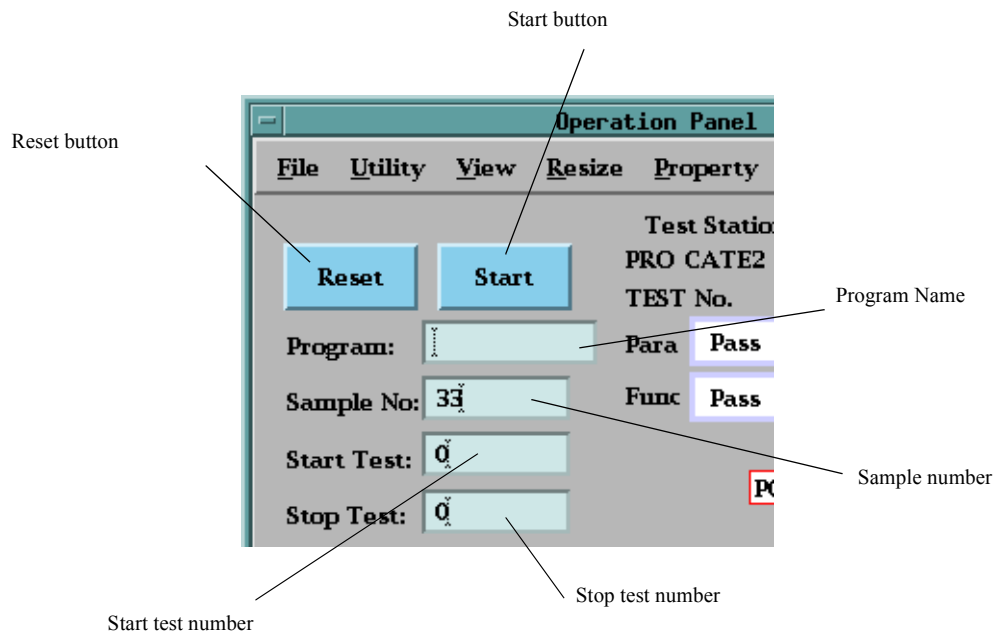
[Close]

Closes the property panel.

47. 3. 2 Operating Method of Panel Area

Panel area has the operating part and the display part.

◆ Operating Part



Reset button

Resets the station that the button exists.

Start button

Executes a program in the station that the button exists.

Program Name

Input the program name to be executed.

Setting a program name here is the same as setting a program name by the O command.

Sample number

Sets the sample number.

Start test number

Sets the start test number.

Stop test number

Sets the stop test number.

Selection button of passed DUT

Select the category lamp to be displayed.

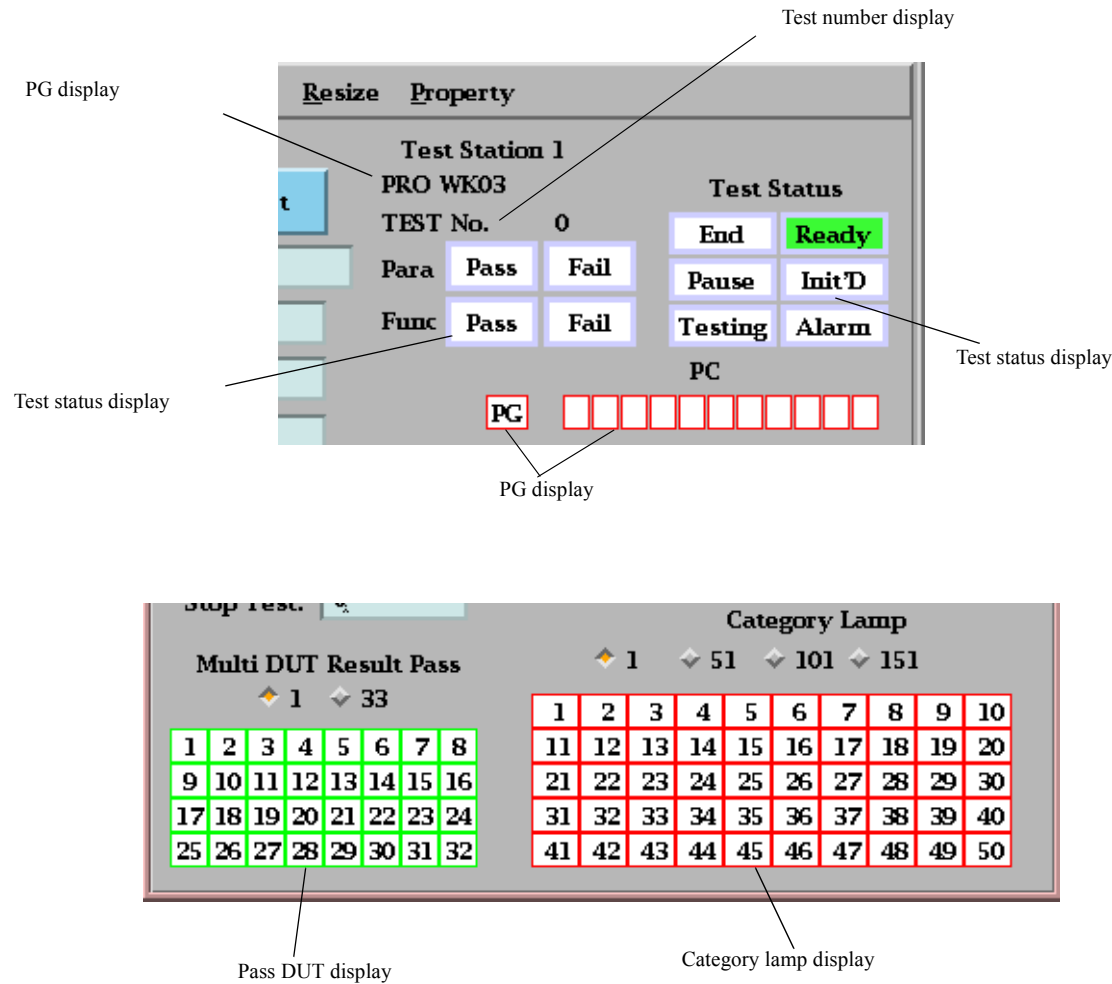
- | | |
|----|------------------------------|
| 1 | DUTs 1 to 32 are displayed. |
| 33 | DUTs 33 to 64 are displayed. |

Selection button of category lamp

Select the category lamp to be displayed.

- | | |
|-----|--------------------------------------|
| 1 | Categories 1 to 50 are displayed. |
| 51 | Categories 51 to 100 are displayed. |
| 101 | Categories 101 to 150 are displayed. |
| 151 | Categories 151 to 200 are displayed. |

◆ Display Part



- PG display**
 Executing program name or the thing specified by SET DISPLAY statement is displayed.
- Test number display**
 Test number specified by test statement is displayed.
- Test status display**
 The total number of passed and failed DUTs is displayed.
 It is displayed at the test end or the manual mode.
- PG display**
- PG When the PG is operating, it is displayed with red.
 - PC The program counter that PG is operated is displayed.

Test status display

The state of the test controller is displayed.

Pass DUT display

Pass DUT is displayed.

Category lamp display

The state of the category lamp selected by the button of category display is displayed.

47. 4 Error Message

Error messages output when the operation panel is activated are shown below.

Error: Can't open display:

The environmental variable DISPLAY is not set.

Could not connect TEST Controller.

Tester does not turn the power on, or the system cannot be used.

Could not execute. (Unsupported.)

The test controller which does not support the operation panel is used.

48. Fail Bit Map (FBMAP)

The program which outputs a fail bit map used for evaluation of a memory device is described.

The utility program which outputs a fail bit map used for evaluation of a memory device is described.

- A fail bit map with expected value can be output.
- The failure bitmap can shrink to a 64-dot square and all addresses can be displayed on a single screen. This function is useful for mass storage memory devices.
- The mode in which scaled down and failed blocks (128 × 128) are displayed is available (ZOOM & ALL).
- The number of devices which failed the specified mode test.
 - ➔ For details about mode, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498 \)](#).
- Fail Bit Map is executed by a keyboard access and also can be executed as a subroutine of the main program.
- Possible to output a result in rectangular form.
- Possible to output an ORed result for multi-bit configuration.
- Failed address can be output by X and Y addresses. In this case, the number of outputting failure addresses can be specified.
- The mode (START ADDRESS) which indicates one (1) block from the specified address permitted in the address area of X and Y addresses in the AFM ADDRESS statement, is prepared.
- Both main PG and sub-PG can be specified for the failure memory.



Limitation

This utility program cannot be executed on the T5336 and T5311.

48. 1 FBMAP Operation

48. 1. 1 Execution by Key Input

- 1) Set the user-specified test plan program to the MANUAL HOLD test mode and pause with the MEAS statement of the specified test number.

If necessary, set the program to the MANUAL RECYCLE HOLD and move to the point at which failed bit map is displayed, through data scanning.

2) Then, enter the following command from the VKT:

```

/FBMAP.␣

**** FBMAP(T5500 SERIES) ****          <PARAMETER FILE NAME IS AQ2.FB>

. FAIL SELECT-----:FT
. AFM BITSIZE-----:4
. BIT # -----:ALL
. DISPLAY TYPE -----:ZOOM & ALL
. START MPAT -----:YES
. DUT # -----:1
. OUTPUT DEVICE-----:LPT
. AFM PGSEL -----:MAIN
. AFM ADDRESS -----:X0-10 Y0-10  DELAY0
. AFM RATE -----:HS
. CHANGE MENU-PARAMETER FILE

[WORK]                                     [WORK]

[NOTE] ----- [NOTE]
      PF1:EXIT   PF2:DISPLAY   ENTER:CHANGE-PARAMETER

```

To modify this menu, position the cursor at the item to be modified using the [↑] and [↓] keys, and press the Return key. It can be modified in the work area. After menu modification is complete, press the [PF2] key. A map is then output.

To terminate this utility program, press the [PF1] key.

◆ FAIL SELECT (Display Data Specification)

When the cursor is positioned at AFM ADDRESS and the return key is pressed, the following message is displayed in the work area:

```

[WORK]
FAIL SELECT      *0=FAIL      ,    1=FZ      ,    2=FO      ,    3=FT      *
                  *4=FAIL1    ,    5=FZSB1    ,    6=FOSB1    ,    7=FTSB1    *
                  *8=FAIL2    ,    9=FZSB2    ,   10=FOSB2    ,   11=FTSB2    *
                  *12=FAIL(v) ,   13=FZSB(v) ,   14=FSB(v)   ,   15=FTSB(v)  *
                SELECT # = __ SB ( # )

```

When 0 is selected:

FZ and FO are read.

A fail bit map with an expected value is output.

When 4 is selected:

FZSB1 and FOSB1 are read.

A fail bit map with expected values of odd-numbered phase STRBs on the STRB 1 to STRB128 side is output.

When 8 is selected:

FZSB2 and FOSB2 are read.

A fail bit map with expected values of even-numbered phase STRBs on the STRB 1 to STRB128 side is output.

When 12 is selected:

The FZSB(v) and FOSB(v) contents are read.

A fail bit map with an expected value of any STRB is output.

. =PASS

0=0FAIL

1=1FAIL

X=0/1FAIL

When 1 is selected:

The FZ contents are output.

When 5 is selected:

The FZSB1 contents are output.

When 9 is selected:

The FZSB2 contents are output.

When 13 is selected:

The FZSB(v) contents are output.

. =PASS

0=0FAIL

When 2 is selected:

The FO contents are output.

When 6 is selected:

The FOSB1 contents are output.

When 10 is selected:

The FOSB2 contents are output.

When 14 is selected:

The FOSB(v) contents are output.

. =PASS

1=1FAIL

When 3 is selected:

The FT contents are output.

When 7 is selected:

The FTSB1 contents are output.

When 11 is selected:

The FTSB2 contents are output.

When 15 is selected:

The FTSB(v) contents are output.

. =PASS

X=0/1FAIL

Make a selection from the 16 choices (0 to 15) shown above.

12 to 15 is entered, the cursor is positioned as shown below.

```
[WORK]
FAIL SELECT      * 0=FAIL      , 1=FZ      , 2=FO      , 3=FT      *
                  * 4=FAIL1     , 5=FZSB1    , 6=FOSB1    , 7=FTSB1    *
                  * 8=FAIL2     , 9=FZSB2    , 10=FOSB2   , 11=FTSB2   *
                  * 12=FAIL(v)  , 13=FZSB(v) , 14=FSB(v)  , 15=FTSB(v) *
SELECT # = 12 SB( #F*1 #__ )
```

**1 The previously specified STRB is displayed.*

STRB is not displayed when a mode (0 to 3) without the specified STRB has been previously specified.

During the FBMAP activation, the set STRB is displayed when a STRB is set in the mode specified by the AFM ACTION statement in the test plan program. When no STRB is specified, it is not displayed.

Enter 1 to F in the v parameter representing a strobe signal such as FAIL(v), FZSB(v), FOSB(v) or FTSB(v).

➔ For details, refer to ["Note 2" on page 48-12](#).

◆ AFM BITSIZE (Measurement Device Bit Number Specification)

When the cursor is positioned at AFM ADDRESS and the return key is pressed, the following message is displayed in the work area:

```
[WORK]
AFM BITSIZE      (1=1BIT, 2=4BIT, 3=8BIT, 4=9BIT, 5=16BIT, 6=18BIT
                  7=32BIT, 8=36BIT, 9=64BIT, 10=72BIT, 11=128BIT, 12=144BIT)
SELECT # = ____
```

Enter one of 1 to 12 according to the number of DUT bits.

◆ BIT # (Bit Number to be Displayed)

When the cursor is positioned at AFM ADDRESS and the return key is pressed, the following message is displayed in the work area:

```
[WORK]
BIT # (0=ALL, 1-144, O=ORDATA)
      SELECT # = ____
```

Enter any bit number (0 to 144, or O) to be displayed.

➔ For details, refer to ["Note 8" on page 48-23](#) or ["Note 10" on page 48-24](#).

◆ DISPLAY TYPE (Output Format Specification)

When the cursor is positioned at AFM ADDRESS and the return key is pressed, the following message is displayed in the work area:

```
[WORK]
DISPLAY TYPE      * 0=ALL, 1-262144, Z=ZOOM, ZA=ZOOM & ALL *
                  * FA=FAIL ADRS DISP., S=START ADDRESS *
      SELECT # = ____ FAIL COUNT (0=ALL, 1-4294967296)
                  SELECT # =
```

DISPLAY TYPE has the following two forms of output:

1. Fail Bit Map Display (0, 1-262144, Z, ZA, S)

Divide MUT SIZE into blocks, and output bit map.

➔ For details, refer to ["Note 7" on page 48-22](#).

A block is a display unit of 128 rows × 128 columns. This block number is ignored when X and Y addresses are equal to or less than seven bits, because they are stored in a block. When 0 is input, all blocks are displayed.

➔ For information on the correspondence between each block and DUT part, refer to ["Note 3" on page 48-14](#).

When Z is specified as a block number, the block is scaled down to a 64 × 64 matrix or smaller one.

For example, when the MUT SIZE is 4 M, the fail status of the 32 × 32 portion is expressed by one bit.

When ZA is specified, failed blocks are displayed after the blocks have been scaled down.

Select a value between 0 or 1 and 262144, Z, ZA or S, and press the Return key.

➔ For details, refer to ["Note 3" on page 48-14](#).

When S is specified in this work area, the cursor moves as follows:

```
[WORK]
DISPLAY TYPE      * 0=ALL, 1-262144, Z=ZOOM, ZA=ZOOM & ALL *
                  * FA=FAIL ADRS DISP., S=START ADDRESS   *
SELECT # = S      START ADDRESS (X:#__ Y:#__)
```

In this case, specify the start address of outputting map with the allowable number in address space indicated by X and Y addresses of AFM ADDRESS.

➔ For details, refer to ["Note 6" on page 48-22](#).

0 to Input using 3FFFF

2. Failure Address Display (FA)

Outputs the failed address in the whole MUT SIZE.

➔ For details, refer to ["Note 9" on page 48-24](#).

Increase the failure address from the minimum (Y = 0, X = 0) to X address in the whole DUT SIZE. When the X address increases to the maximum, increase/output Y address.

Input FA, and press the Return key.

When the FA is specified, the cursor moves as follows:

```
[WORK]
DISPLAY TYPE      * 0=ALL, 1-262144, Z=ZOOM, ZA=ZOOM & ALL *
                  * FA=FAIL ADRS DISP., S=START ADDRESS   *
SELECT # =        FAIL COUNT (0=ALL, 1-4294967296)
                  SELECT # = __
```

In this case, specify the number of outputting fail address with the allowable number in DUT SIZE (number of address indicated by X and Y addresses).

➔ For details, refer to ["Note 5" on page 48-22](#).

Input using from 1 to 4294967296

A Default is ALL.

◆ START MPAT (Pattern Generator Operation)

When the cursor is positioned at AFM ADDRESS and the return key is pressed, the following message is displayed in the work area:

```
[WORK]
START MPAT? (0 = YES, 1 = NO, 2=YES (WITHOUT AFM ADDRESS SETTING))
SELECT # = __
```

When 0 is selected:

Runs test patterns after setting AFM modes and then reads out the AFM data.

When 1 is selected:

Reads data contained in the current AFM, and processes the data.

When 2 is selected:

Runs test patterns after setting each AFM mode other than AFM ADDRESS and then reads out the AFM data.

Make a selection from 0 to 2.

➔ For details, refer to ["Note 4" on page 48-20](#).

◆ DUT # (DUT Number to be Displayed)

When the cursor is positioned at AFM ADDRESS and the return key is pressed, the following message is displayed in the work area:

```
[WORK]
DUT # (1-n)
SELECT # = __
```

n

Maximum value of specifiable DUT number.

Enter the DUT number to be displayed, and press the ENTER key.

◆ OUTPUT DEVICE (Output Device)

When the cursor is positioned at AFM ADDRESS and the return key is pressed, the following message is displayed in the work area:

```
[WORK]
OUTPUT DEVICE (0=VKT, 1=LPT, 2=VKT & LPT, 3=SCOMMAND)
SELECT # = __
```

Input the output device number to be displayed (0, 1 or 2).

If 0 (VKT) or 2 (VKT & LPT) is selected here, the following message is displayed before pausing each time the header and map are displayed.

```
STRIKE ANY KEY TO CONTINUE
```

◆ AFM PGSEL (Specification of PG to be Connected)

When the cursor is positioned at AFM ADDRESS and the return key is pressed, the following message is displayed in the work area:

```
[WORK]
AFM PGSEL (0=MAIN, 1=MAIN & SUB)
  SELECT # = ____          AFM RWSEL (0=MAIN, 1=SUB)
                               SELECT # =
```

Select either MAIN PG or both PG to be connected to AFM with entering one of 0 to 1.

Valid when START MPAT is in "YES".

When 1 (MAIN & SUB) for AFM PGSEL is selected, or the Return key is pressed while AFM PGSEL is in the MAIN & SUB state, the cursor indicates as follows:

```
[WORK]
AFM PGSEL (0=MAIN, 1=MAIN & SUB)
  SELECT # = 1          AFM RWSEL (0=MAIN, 1=SUB)
                               SELECT # = ____
```

Select MAP of either MAIN or SUB side for display with entering 0 or 1 for AFM RWSEL.

Tip

The menu screen for AFM PGSEL is shown as follows:

```
.AFM PGSEL ----- : MAIN & SUB (MAIN)
```

The message in parentheses () indicates the setting value of RWSEL. This message is displayed only when MAIN & SUB for AFM PGSEL is selected, however, this is not displayed when MAIN for AFM PGSEL is selected.

◆ AFM ADDRESS (Specification of DUT SIZE Address)

When the cursor is positioned at AFM ADDRESS and the return key is pressed, the following message is displayed in the work area:

```
[WORK]
AFM MAIN ADDRESS = X0-l Y0-m SCRAM DELAYp
AFM SUB ADDRESS = X0-s Y0-t SCRAM DELAYq
  ADDRESS (0=MAIN, 1=SUB, 2=MAIN & SUB)
  SELECT # = ____
```

Select either MAIN address or SUB address with entering one of 0 to 2 for the specification of DUT size, SCRAM address or DELAY cycle.

When a specification from 0 to 2 is made, the following message is displayed in the work area.

When 0 (MAIN) is selected (Main address setting):

```
[WORK]
AFM MAIN ADDRESS = X0-l Y0-m SCRAM DELAYp
ADDRESS (0 = X, 1 = Y, 2 = SCRAM, 3 = DELAY)
SELECT # = __
```

Enter a value from 0 to 3 to specify DUT size, SCRAM address, or DELAY cycle.

When 1 (SUB) is selected (SUB address setting):

```
[WORK]
AFM SUB ADDRESS = X0-s Y0-t SCRAM DELAYq
ADDRESS (0 = X, 1 = Y, 2 = SCRAM, 3 = DELAY)
SELECT # = __
```

Enter a value from 0 to 3 to specify DUT size, SCRAM address, or DELAY cycle.

When 2 (MAIN to SUB) is selected (MAIN/SUB address setting):

```
[WORK]
AFM MAIN ADDRESS = X0-l Y0-m SCRAM DELAYp
AFM SUB ADDRESS = X0-s Y0-t SCRAM DELAYq
ADDRESS (0 = X, 1 = Y, 2 = SCRAM, 3 = DELAY)
SELECT # = __
```

Enter a value from 0 to 3 to specify DUT size, SCRAM address, or DELAY cycle.

- When specifying 0 (X address),
The following message is displayed:

```
X0 - __
```

Input a selection from 1 to 17.

Part of s or l takes on the inputted value.

- When specifying 1 (Y address),
The following message is displayed:

```
Y0 - __
```

Input a selection from 1 to 17. Part of m or t takes on the inputted value.

- When specifying 2 (SCRAM):
The following message is displayed:

```
SCRAM (0=YES, 1=NO) __
```

Selects whether to assign addresses converted using the address descrambler to FM.

If the value is 0, "SCRAM" to be displayed; if 1, nothing is displayed.

Valid when START MPAT is in "YES".

- When specifying 3 (DELAY):
The following message is displayed:

```
DELAY (0-31) ____
```

Specifies the number of delay cycles from 0 to 7 for the address assigned to FM (0 to 31 for the T5377, T5376, T5375 and T5371). The value assigned to p or q is displayed.

This setting is valid when START MPAT is set to 0.

Tip

The menu screen for AFM ADDRESS is shown as follows:

```
.AFM ADDRESS ----- : (MAIN) X0-10 Y0-10 DELAY0
```

The figure in parentheses indicates whether the AFM ADDRESS setting on the menu screen is for MAIN or SUB. AFM PGSEL determines whether the AFM ADDRESS setting displayed on the menu screen is for MAIN or SUB.

"MAIN" is displayed when MAIN for AFM PGSEL is selected. "MAIN & SUB" is selected to be displayed according to AFM RWSEL state. In other words, when RWSEL is MAIN, MAIN is displayed; when it is SUB, SUB is displayed.

This display changes automatically according to how RGSEL is specified.

Note that SCRAM of NO is not displayed in the main menu.

◆ AFM RATE (Specifying AF Operation Cycle)

When the cursor is positioned at AFM ADDRESS and the return key is pressed, the following message is displayed in the work area:

```
[WORK]
AFM RATE (1 = HS, 2 = MS, 3 = LS)
SELECT # = ____
```

Enter any one of 1 to 3 for AFM operational.

◆ CHANGE MENU-PARAMETER FILE (Creating a Parameter File)

When the cursor is positioned at AFM ADDRESS and the return key is pressed, the following message is displayed in the work area:

```
[WORK]
PARAMETER FILE NAME = ____
```

Specify the file name to be changed, and press the ENTER key.

Consists of up to six alphanumeric characters beginning with an alphabetical character.

If a file has already been specified, a parameter is input from the file. If no file is specified, a new parameter file is created from the current parameter.

◆ Precautions

Note 1

A check of AFM capacity is not performed. When BIT #, DUT #, or FAIL SELECT specification exceeds the capacity, or when an error selection occurs, the error message below is displayed during program execution.

◆ When FAIL SELECT ERROR Occurs (Error in Selecting ACTION Mode)

```
***** READ AFM ERROR *****
FAIL SELECT ERROR
***** BREAK MAP OUTPUT ROUTINE *****
        ERROR CODE IS    12014
- STRIKE ANY KEY TO CONTINUE -
```

◆ When Bit Specification Exceeds the Range

```
***** READ AFM ERROR *****
BIT SELECT ERROR
***** BREAK MAP OUTPUT ROUTINE *****
        ERROR CODE IS    12013
- STRIKE ANY KEY TO CONTINUE -
```

◆ When DUT Specification Exceeds the Range

```
***** READ AFM ERROR *****
DUT SELECT ERROR
***** BREAK MAP OUTPUT ROUTINE *****
        ERROR CODE IS    12012
- STRIKE ANY KEY TO CONTINUE -
```

Note 2

If START MPAT is not selected, FAIL SELECT must be set as shown below according to the test program AFM setting.

When FT > FMX:

Selects FT

When FZ > FMX:

Selects FZ

When FO > FMX:

Selects F0

When FZ > FMX and FO > FMX

Select FAIL, FZ, or FO

When FTSBn > FMX:

- When n is 1:
Specify STRB as 1 using either FTSB1 or FTSB(v).
- When n is 2:
Specify STRB as 2 using either FTSB2 or FTSB(v).
- When n is 129:
Specify STRB as 4 using FTSB(v).
- When n is 130:
Specify STRB as 8 using FTSB(v).

When FZSBn > FMX:

- When n is 1:
Specify STRB as 1 using either FZSB1 or FZSB(v).
- When n is 2:
Specify STRB as 2 using either FZSB2 or FZSB(v).
- When n is 129:
Specify STRB as 4 using FZSB(v).
- When n is 130:
Specify STRB as 8 using FZSB(v).

When FOSBn > FMX:

- When n is 1:
Specify STRB as 1 using either FOSB1 or FOSB(v).
- When n is 2:
Specify STRB as 2 using either FOSB2 or FOSB(v).
- When n is 129:
Specify STRB as 4 using FOSB(v).
- When n is 130:

Specify STRB as 8 using FOSB(v).

When FZSBn > FMX and FOSBn > FM:

- When n is 1:
STRB is specified as 1 using FAIL1 and FAIL(v), FZSB1 and FZSB(v), or FOSB1 and FOSB(v).
- When n is 2:
FAIL2 or FAIL(v) and STRB specified as 2, or FZSB2 or FZSB(v) and STRB specified as 2, or FOSB2 or FOSB(v) and STRB specified as 2.
- When n is 129:
FAIL(v) and STRB specified as 4, or FZSB(v) and STRB specified as 4, or FOSB(v) and STRB specified as 4.
- When n is 130:
FAIL(v) and STRB specified as 8, or FZSB(v) and STRB specified as 8, or FOSB(v) and STRB specified as 8.

When FTSB(v) > FMX

- When v is #1:
Specify STRB as 1 using either FTSB1 or FTSB(v).
- When v is #2:
Specify STRB as 2 using either FTSB2 or FTSB(v).
- When v is neither #1 nor #2:
Specify STRB as v using FTSB(v).

When FZSB(v) > FMX:

- When v is #1:
Specify STRB as 1 using either FZSB1 or FZSB(v).
- When v is #2:
Specify STRB as 2 using either FZSB2 or FZSB(v).
- When v is neither #1 nor #2:
Specify STRB as v using FZSB(v).

When FOSB(v) > FMX

- When v is #1:
Specify STRB as 1 using either FOSB1 or FOSB(v).
- When v is #2:
Specify STRB as 2 using either FOSB2 or FOSB(v).
- When v is neither #1 nor #2:
Specify STRB as v using FOSB(v).

When FZSB(v) > FMX and FOSB(v) > FMX:

- When v is #1:

STRB is specified as 1 using FAIL1 and FAIL(v), FZSB1 and FZSB(v), or FOSB1 and FOSB(v).

- When v is #2:
FAIL2 or FAIL(v) and STRB specified as 2, or FZSB2 or FZSB(v) and STRB specified as 2, or FOSB2 or FOSB(v) and STRB specified as 2.
- When v is neither #1 nor #2:
FAIL(v) and STRB specified as v, FZSB(v) and STRB specified as v, or FOSB(v) and STRB specified as v.

Note 3

The part of DUT which each block corresponds example to is shown below.

◆ When 64 K (Square)

	0	80	100	← Hexadecimal display
	1	2		
80				
	3	4		
100				

◆ When 256 K (Square)

	0	80	100	180	200	← Hexadecimal display
	1	2	3	4		
80						
	5	6	7	8		
100						
	9	10	11	12		
180						
	13	14	15	16		
200						

◆ When 1 M (Square)

	0	80	100	180	200	280	300	380	400
80	1	2	3	4	5	6	7	8	
100	9	10	11	12	13	14	15	16	
180	17	18	19	20	21	22	23	24	
200	25	26	27	28	29	30	31	32	
280	33	34	35	36	37	38	39	40	
300	41	42	43	44	45	46	47	48	
380	49	50	51	52	53	54	55	56	
400	57	58	59	60	61	62	63	64	

← Hexadecimal display

◆ When 4 M (Square)

	0	80	100	180	200	280	300	380	400	480	500	580	600	680	700	780	800	←Hexadecimal display
80	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		
100	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32		
180	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48		
200	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64		
280	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80		
300	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96		
380	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112		
400	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128		
480	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144		
500	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160		
580	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176		
600	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192		
680	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208		
700	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224		
780	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240		
800	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256		

◆ When 16 M (Square)

0													#1000
	1	2	3	4	5	6	27	28	29	30	31	32	
	33	34	35	36	37	38	59	60	61	62	63	64	
	65	66	67	68	69			92	93	94	95	96	
	97	98	99	100	101			124	125	126	127	128	
	129	130	131	132	133			156	157	158	159	160	
	161	162									191	192	
							Omitted						
	833	834									863	864	
	865	866	867	868	869			892	893	894	895	896	
	897	898	899	900	901			924	925	926	927	928	
	929	930	931	932	933			956	957	958	959	960	
	961	962	963	964	965			988	989	990	991	992	
#1000	993	994	995	996	997			1020	1021	1022	1023	1024	

↑
Max BLOCK = 1024

◆ When 256 M (Square)

0						#4000					
1	2	3	4	5	6	123	124	125	126	127	128
129	130	131	132	133	134	251	252	253	254	255	256
257	258	259	260	261	262	379	380	381	382	383	384
385	386	387	388	389	390	507	508	509	510	511	512
513	514	515	516	517	518	635	636	637	638	639	640
641	642	643	644	645	646	763	764	765	766	767	768
						Omitted					
15617	15618	15619	15620	15621	15622	15739	15740	15741	15742	15743	15744
15745	15746	15747	15748	15749	15750	15867	15868	15869	15870	15871	15872
15873	15874	15875	15876	15877	15878	15995	15996	15997	15998	15999	16000
16001	16002	16003	16004	16005	16006	16123	16124	16125	16126	16127	16128
16129	16130	16131	16132	16133	16134	16251	16252	16253	16254	16255	16256
16257	16258	16259	16260	16261	16262	16379	16380	16381	16382	16383	16384
#4000											

↑
Max BLOCK = 16384

Specifying the Block No.

Regard to addresses (X and Y axes).

Obtain first the number of blocks for the X axis.

X axis	The number of blocks in the X direction
X0-1	1 device
X0-2	1 device
X0-3	1 device
X0-4	1 device
X0-5	1 device
X0-6	1 device
X0-7	2 devices
X0-8	4 devices
X0-9	8 devices
X0-10	16 devices
X0-11	32 devices
X0-12	64 devices
X0-13	128 devices
X0-14	256 devices
X0-15	512 devices
X0-16	1024 devices
X0-17	2048 devices

Then, obtain the number of blocks for the Y axis.

X axis	The number of blocks in the X direction
Y0-1	1 device
Y0-2	1 device
Y0-3	1 device
Y0-4	1 device
Y0-5	1 device
Y0-6	1 device
Y0-7	2 devices
Y0-8	4 devices
Y0-9	8 devices
Y0-10	16 devices
Y0-11	32 devices
Y0-12	64 devices
Y0-13	128 devices
Y0-14	256 devices
Y0-15	512 devices
Y0-16	1024 devices
Y0-17	2048 devices

Count up the number of blocks in the X direction.

Notes on specifying ZOOM

In the setting of AFM ADDRESS, when the difference between X address and Y address is 7 bits or more, DISPLAY TYPE cannot be specified as ZOOM.

If specified, the following message is displayed in the command execution.

```
***** BREAK MAP OUTPUT ROUTINE *****
*** DISPLAY TYPE ERROR ***
      ADDRESS ----> ZOOM
- STRIKE ANY KEY TO CONTINUE -
```

Note 4

When YES (0 or 1) is selected for START MPAT, the utility FBMAP operates as described below.

1. The operation mode of the AFM is set as follows after resetting the AFM.

AFM MULTIMODE:

SINGLE

AFM RATE

Depends on the conditions set by the argument or MENU.

AFM BITSIZE

Depends on the conditions set by the argument or MENU.

AFM DUTSEL

Depends on the conditions set by the argument or MENU.

AFM ADDRESS

Depends on the conditions set by the argument or MENU when START MPAT is 0.

FBMAP does not execute the AFM ADDRESS statement when START MPAT is 2.

AFM PGSEL

Depends on the conditions set by the argument or MENU.

When START MPAT is 0, FBMAP executes the AFM ADDRESS statement on the conditions set by the argument or MENU.

When START MPAT is 2, FBMAP does not execute the AFM ADDRESS statement.

This selection is effective when FBMAP issues a start instruction to the pattern to output a fail bit map in a two-speed mode or an address random specification.

FBMAP will not execute the AFM ADDRESS statement after setting the two-speed mode or random address of the AFM ADDRESS statement in the test plan program with FBMAP stopped, setting START MPAT to 2, and executing START MPAT using FBMAP. Consequently, the AFM ADDRESS statement specified by the test plan program will be enabled and a fail bit map in the two-speed mode or address random specification can be output.

However, the output range of the fail bit map depends on the conditions set by the FBMAP argument or MENU.

Therefore, when the random address is specified, the same bit length addresses as used for the specified X and Y addresses must be set using the FBMAP argument or MENU. When the two-speed is specified, the same bit length addresses as used for the X and Y addresses (the first specified address inside the ◇ part, and the addresses specified outside the ◇ part) must be set using the FBMAP argument or MENU.

Once executed with START MPAT set to 0, the AFM ADDRESS statement specified by the test plan program will be disabled because the AFM ADDRESS statement has already been executed under the conditions set by the FBMAP argument or MENU. Subsequent executions of the AFM ADDRESS statement will be controlled by the same condition as those established in the previous execution.

2. The ALPG mode is set as shown below.

SELECT MPAT INHIBIT

The following mode is selected by default when quitting this utility.

SELECT MPAT NORMAL

3. This utility only issues a start instruction to the pattern. Therefore, the pattern must be transferred to ALPG and in a selected state and reference must be turned on before this utility can be used.

A START MPAT statement is executed in this utility to issue a start instruction to the pattern.

When MEAS MPAT (4):

START MPAT (4, v) is executed.

When MEAS MPAT (5):

START MPAT (5) is executed.

Otherwise:

START MPAT (3) is executed.

➔ For information about test rates (HS, MS, and LS) and capacities, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498 \)](#)

Note 5

When the number of output failures is greater than the number of addresses of DUT SIZE (indicated by X and Y addressed), the following messages are displayed.)

```
***** BREAK MAP OUTPUT ROUTINE *****
*** FAIL ADRS DISP. OVER COUNT ERROR ***
- STRIKE ANY KEY TO CONTINUE -
```

Note 6

When START ADDRESS exceeds the allowable range in address space indicated by X and Y addresses for AFM ADDRESS, the message below is output at the execution.

```
***** BREAK MAP OUTPUT ROUTINE *****
*** ERROR : "START ADDRESS" OUT OF RANGE ***
- STRIKE ANY KEY TO CONTINUE -
```

Note 7

When a bit map is output on the fail bit map display, if Max Y address of output block is given 4 digits (#1000 or more), care must be taken that a space (boundary) between a Y address section and a bit map section is not provided. (When Max Y address of output block is given 3 digits or less, a space is provided as a boundary between a Y address section and a bit map section.)

Note 9

When FAIL SELECT is FAIL, FAIL 1, FAIL 2 or FAIL (v), FA (Fail Address display) cannot be specified to DISPLAY TYPE. When FAIL SELECT is FAIL, FAIL1, FAIL 2 or FAIL (v) and DISPLAY TYPE is FA, the following message is output at execution.

```
***** BREAK MAP OUTPUT ROUTINE *****
*** DISPLAY TYPE ERROR ***
"FA" (FAIL ADRS DISP.) cannot be designated.
```

Note 10

When a map is displayed by FBMAP in AFM BITSIZE=n1, n2 state, specification range of BIT NO. to be displayed is from 0 to n2 (0 to n1 when CFM is selected in the SELECT COMPRESS statement.). When a value out of the range is specified, an error message (refer to ["Note 1" on page 48-11](#)) is displayed.

48. 1. 2 Call from Program

Outputting a fail bit map by incorporating it into the test plan program The subroutine is called in the following format:

```
CALL FBMAP("character argument")
```

The character argument is specified by the following two methods: Specifying parameters directly
Specifying parameter file

◆ Specifying Parameters Directly

The character argument must be specified by a total of 21 numerals or alphanumeric characters by these methods.

The parameter N14 (Number of fail address) is available only when N5 (DISPLAY TYPE) is "FA" (FAIL ADRS DISP.), and the parameter N15 (START ADDRESS X-axis) and N16 (START ADDRESS Y-axis) are available only when N5 is "S" (START ADDRESS).

The parameter N17 (AFM RWSEL) is available only when N9 (AFM PGSEL) is "1" (MAIN & SUB), and the parameters N18 to N21 (AFM SUB ADDRESS specification) are available only when AFM MAIN ADDRESS and AFM SUB ADDRESS are separately set.

The parameter N22 (FAIL SELECT STRB specification) is effective only when N2 (FAIL SELECT) is either "12" (FAIL(v)), "13" (FZSB(v)), "14" (FOSB(v)), or "15" (FTSB(v)).

Accordingly, the direct specification is performed as the following formats:

- When the parameter N2 is one of "0" to "11", N5 is anything other than "FA" or anything other than "S", and N9 is "0", 13 arguments (N1 to N13) are applicable.

```
CALL FBMAP ("N1,N2,N3,N4,N5,N6,N7,N8,N9,N10,N11,N12,N13")
```

- When the parameter N2 is one of "0" to "11", N5 is "FA", and N9 is "0", 14 arguments (N1 to N14) are applicable.

```
CALL FBMAP ("N1,N2,N3,N4,N5,N6,N7,N8,N9,N10,N11,N12,N13,N14")
```

- Select 16 arguments (N1 to N16) when the parameter N2 is one of "0" to "11", N5 is "S", and N9 is "0." In this case, specify 0 as N14.

```
CALL FBMAP ("N1,N2,N3,N4,N5,N6,N7,N8,N9,N10,N11,N12,N13,N14,N15,N16")
```

- When parameter N2 is between "0" to "11", N9 is "1", and AFM MAIN ADDRESS and AFM SUB ADDRESS are simultaneously set (for the same setting of AFM MAIN ADDRESS and AFM SUB ADDRESS), specify 17 arguments (N1 to N17). When the parameter N5 is "FA" in this case, specify 0 as N15 and N16, and when the parameter N5 is "S", specify 0 as N14.

```
CALL FBMAP ("N1,N2,N3,N4,N5,N6,N7,N8,N9,N10,N11,N12,N13,N14,N15,N16,N17")
```

- When parameter N2 is between "0" to "11", N9 is "1", and AFM MAIN ADDRESS and AFM SUB ADDRESS settings are changed, specify 21 arguments (N1 to N21). When the parameter N5 is "FA" in this case, specify 0 as N15 and N16, and when the parameter N5 is "S", specify 0 as N14.

```
CALL FBMAP ("N1,N2,N3,N4,N5,N6,N7,N8,N9,N10,N11,N12,N13,N14,N15,N16,N17,N18,N19,
N20,N21")
```

- When parameter N2 is between "12" to "15", specify 22 arguments (N1 to N22). In this case, when the parameter N5 is "FA", specify N15 and N16 as 0, when the parameter N5 is "S", specify N14 as 0, when N9 is 0, specify N17 to N21 as 0, and when N9 is 1 and AFM MAIN ADDRESS and AFM SUB ADDRESS are simultaneously set, specify N18 to N21 as 0.

```
CALL FBMAP ("N1,N2,N3,N4,N5,N6,N7,N8,N9,N10,N11,N12,N13,N14,N15,N16,N17,N18,N19,
N20,N21,N22")
```

N1

Specifies AFM RATE. (MENU item(10))
Make a selection from 1 to 3.

N2

Specifies FAIL, FZ, FO, FT, FAIL1, FZSB1, FOSB1, FTSB1, FAIL2, FZSB2, FOSB2, FTSB2, FAIL(v), FZSB(v), FOSB(v), FTSB(v) (MENU item (1)).

Make a selection from 0 to 15.

N3

Specifies AFM BITSIZE (MENU item(2)).

Make a selection from 1 to 12.

N4

Specifies DISPLAY BIT No (MENU item(3)).

Select 0, a value from 1 to 144, or a value from O.

When N3 (AFM BITSIZE) is 7 (32BIT) or more, O(ORDATA) cannot be specified.

However, when AFM bit compression is performed and N6 (START MPAT) is 1 (NO) on the T5377, T5376, T5375 or T5371, O (ORDATA) can be specified.

N5

Specifies DISPLAY TYPE (MENU item(4)).

Make a selection from among 0 to 262144, Z, ZA, FA, S.

N6

Selects either START MPAT or READ ONLY (MENU item (5)).

Make a selection from 0 to 2

N7

Specifies DUT No (MENU item (6)).

Make a selection from 1 to 512.

N8

Specifies OUTPUT DEVICE (MENU item(7)).

Make a selection from 1 to 3.

N9

Specifies AFM PGSEL (MENU item(8)).

Select 0 or 1.

N10

Specifies last address of X in AFM ADDRESS statement (MENU item (9)).

Make a selection from 1 to 17.

N11

Specifies last address of Y in AFM ADDRESS statement (MENU item (9)).

Make a selection from 1 to 17.

N12

Specifies SCRAM in AFM ADDRESS statement (MENU item (9)).

Specify 0 or 1.

N13

Specifies DELAY in AFM ADDRESS statement (MENU item (9)).

Make one selection from 0 to 7 (0 to 31 for the T5377, T5376, T5375 or T5371).

N14

Specifies the number of output failure addresses at FAIL ADRS DISP (MENU item (4)).

Make a selection from 0 to 4294967296.

(Care must be taken for AFM ADDRESS size indicated by X and Y.)

N15

Specifies start X address of bit map outputting at START ADDRESS.

(MENU item(4))

Make a selection from #0 to #3FFFF.

(Care must be taken for AFM ADDRESS size indicated by X and Y. The same is true for N16.)

N16

Specifies the start Y address of bit map outputting at START ADDRESS.

(MENU item(4))

Make a selection from #0 to #3FFFF.

N17

Specifies AFM RWSEL (MENU item(8)).

Specify 0 or 1.

N18

Specifies last address of X in AFM SUB ADDRESS statement (MENU item (9)).

Make a selection from 1 to 17.

N19

Specifies last address of Y in AFM SUB ADDRESS statement (MENU item (9)).

Make a selection from 1 to 17.

N20

Specifies SCRAM in AFM SUB ADDRESS statement (MENU item (9)).

Specify 0 or 1.

N21

Specifies DELAY in AFM SUB ADDRESS statement (MENU item (9)).

Make a selection from 0 to 7.

N22

Specifies STRB in FAIL SELECT (MENU item (1)).

Make a selection from #1 to #F.

The following can be used to call this utility.

Examples of each call are shown below.

◆ When Outputting Maps after Running Patterns

```
PRO xxx
TEST xx
  Level setting
  Timing setting
SEND MPAT filename
REG PC = xx
SRON
CALL FBMAP ("3,3,2,1,ZA,0,1,0,0,10,10,0,0")
```

In this example, FAIL BIT MAP is output for a MUT with 4 M × 4.

◆ When Outputting Only Maps

```
PRO xxx
TEST xx
  Level setting
  Timing setting
  AFM setting
REG PC = xx
MEAS MPAT filename
CALL FBMAP ("3,3,2,1,ZA,1,1,0,0,10,10,0,0")
```

In this example, FAIL BIT MAP is also output for a MUT with 4 M × 4. Care must be taken when setting the AFM operation mode and FBMAP FAIL SELECT mode.

If a number exceeds the range when a character argument is specified by one of 13, 14, 16, 17, 21, and 22 numbers, the message below is output and the CALL statement is ended.

```

      ILLEGAL ARGUMENT! TRY AGAIN

/FBMAP N1  N2  N3  N4  N5  N6  N7  N8  N9  N10 N11
      N12 N13 N14 N15 N16 N17 N18 N19 N20 N21

N 1 : AFM RATE           (1 - 3)
N 2 : FAIL/FZ/FO/FT... (0 - 15)
N 3 : BIT SIZE           (1 - 12)
N 4 : BIT #              (0,1-144,0)
N 5 : DISPLAY TYPE       (0,1-262144,Z,ZA,FA,S)
N 6 : START MPAT         (0 - 2)
N 7 : DUT #              (1 - 512*1)
N 8 : OUTPUT DEVICE      (0 - 3)
N 9 : PGSEL              (0 - 3)
N10 : X-STOP ADRS        (1 - 17)
N11 : Y-STOP ADRS        (1 - 17)
N12 : SCRAM              (0 , 1)
N13 : DELAY              (0 - 31)
N14 : FAIL COUNT         (0,1-4294967296)
N15 : START X ADDRESS    (#0 - #3FFFF)
N16 : START Y ADDRESS    (#0 - #3FFFF)
N17 : RWSEL              (0 , 1)
N18 : X-STOP ADRS (SUB) (1 - 17)
N19 : X-STOP ADRS (SUB) (1 - 17)
N20 : SCRAM              (0 , 1)
N21 : DELAY              (0 , 7)
N22 : STRB               (#1-#F)

      ILLEGAL ARGUMENT! >> N2 *2

FBMAP EXIT!!

```

**1 The number of DUTs being tested in parallel is displayed.*

**2 The number of the argument that is out of the range is displayed.*

There are two ways to enter data used for FAIL BIT MAP; the interactive way (where each condition is entered one by one, as shown above) or the all-in-one way (where all arguments are entered at one time as shown below).

```
/FBMAP 3,3,2,1,ZA,1,1,0,0,10,10,0,0
```

In this case, the same result is obtained as when the previously described CALL statement is executed. That is, FAIL BIT MAP with a 4 M × 4 DUT is output.

If an error occurs when the number of the character argument is specified, the error message above is output.

◆ Specifying Parameter File

The character argument must be specified by an alphanumeric character and a number in this method.

```
CALL FBMAP ("FNAM,N1")
```

FNAM

Specifies a parameter file name. Consists of up to six alphanumeric characters beginning with an alphabetical character. If the specified file number is not present, the "AQ2.FB" parameter file is used.

N1

Selects whether or not a parameter is specified in the conversational mode. Specify 0, 1, or 2. When 1 is specified, a parameter can be selected in interactive mode. When 2 is specified, no menu is displayed and outputs the fail bit map directly to the output unit selected by the key command "S".

When specifying numbers, no parts can be omitted even though they are unnecessary.

Call this utility in the same way as in ["Specifying Parameters Directly" on page 48-24](#) of the previous section.

For example, specify as shown below to output a failed bit map after running patterns.

```
PRO xxx
TEST xx
  Level setting
  Timing setting
SEND MPAT filename
REG PC = xx
SRON
CALL FBMAP (AQ2.FB,1)
```

If a number exceeds the range when a character argument is specified by two numbers, the message below is output, and the CALL statement is ended.

```
ILLEGAL ARGUMENT ! TRY AGAIN
/FBMAP FNAM N1

FNAM      : PARAMETER FILE NAME
N1        : MENU (YES=1, NO=0)

FBMAP EXIT!!
```

In executing by keying in, not only can conditions be entered sequentially in the interactive mode, they can also be entered by specifying the parameter file as follows:

```
/FBMAP AQ2.FB,0↵
```

In this case, the same result is obtained as when the previously described CALL statement is executed. To modify output conditions, specify as follows:

```
/FBMAP AQ2.FB,1
```

When this is specified, output conditions can be modified in the interactive mode by using parameter file AQ2.FB.

➔ For details, refer to [48. 1. 1 "Execution by Key Input" on page 48-1](#).

If an error occurs in specifying the character argument, the previously described error message is output.

Limitation

A parameter file created on the ASX/U-50 cannot be used on the ASX/U-51. If the specified parameter file has been created on the ASX/U-50, an error message like the following will appear:

```
ILLEGAL ARGUMENT! TRY AGAIN  
PARAMETER FILE (TEST) IS NOT T5500 SERIES FORMAT  
FBMAP EXIT!!
```

48. 2 FBMAP Output Example

When the display statement is executed, the following message is displayed:

```
*****
*          T5500 SERIES FBMAP UTILITY          *
*****
```

48. 2. 1 Fail Bit Map Display

After display instruction is executed, a fail bit map of specified device is displayed. When suppressing the execution of the instruction, press the [B] key.

◆ Scaled Down Output (DISPLAY TYPE = Z) Mode

A fail bit map with a 4 M × 1 configuration

```
*****          FBMAP (T5500 SERIES)          ***** FBMAP REV # 1.00

1998/04/08  11:59:30

DEVICE  XXXXXXXX  *1
LOT #    20800-8  *1
SAMPLE #         6  *1
TEST #         7  *1
OPERATOR : ADVAN

PARAMETER FILE NAME IS AQ2.FB
BIT SIZE      =   1
BIT #         =   1
DISPLAY TYPE  =   1
DUT# = 1
PGSEL         =  MAIN
ADDRESS       =  (MAIN) X0-10, Y0-10,      DELAY0
RATE          =  HS

+++++ FAIL DATA ONLY +++++
'.' :          PASS
'0' :  0        FAIL
'1' :  1        FAIL
'X' :  0 OR 1   FAIL

FZ COUNT =      the count value for 253 --- 0  fail  *2
FO COUNT =      the count value for 345 --- 1  fail  *2
```

*1 Header output.

**2 When DISPLAY TYPE is specified as 1 to 262144 and S, the count in that block is displayed as the count for the entire DUT when 0, ZA, and Z.*

	11111111222222223333333344444444555555556666666677777777
	2468ACE02468ACE02468ACE02468ACE02468ACE02468ACE02468ACE02468ACE
	00
0	XX.....
20	110XX111.....
40	
60	
80	
A0	
C0	
E0	
100	
120	
140	
160	
180	
1A0	
1C0	
1E0	

Tip

Scaled down indication means that when DUT SIZE is 16 M (square), the logical sum of the 64 × 64-bit portion is expressed by one element.

(BIT# = 1 and DISPLAY TYPE = 1 are specified.)

```
*****          FBMAP (T5500 SERIES)          *****  FBMAP REV #   1.00

1998/04/08 11:59:30

DEVICE      XXXXXXXX
LOT #       20800-8
SAMPLE #    6
TEST #      7
OPERATOR    :ADVAN

PARAMETER FILE NAME IS *1
BIT SIZE    =    1
BIT #       =    1
DISPLAY TYPE =    1
DUT# = 1
PGSEL       =    MAIN
ADDRESS     =    (MAIN) X0-10  Y0-10          DELAY0
RATE        =    HS

+++++ FAIL DATA ONLY +++++
'.' :          PASS
'0' :    0      FAIL
'1' :    1      FAIL
'X' :    0 OR 1  FAIL

FZ COUNT =  53 ----- 0 fail count value *2
FO COUNT = 145 ----- 1 fail count value *2
```

*1 Not displayed when parameters are specified directly.

*2 Indicates a fail counter in one block since DISPLAY TYPE is 1.

	11111111111111112222222222222222333333333333333344444
	0123456789ABCDEF0123456789ABCDEF0123456789ABCDEF01234
0	1.....
1	1.....1.....
2	1.....
3	1.....
4	1.....1.....
5	1.....
6	1.....
7	1.....1.....
8	1.....
9	1.....
A	1.....1.....
B	1.....
C	1.....
D	1.....1.....
E	1.....
F	1.....
10	1.....1.....
11	1.....
12	1.....
13	1.....1.....
14	1.....
15	1.....
16	1.....1.....
17	1.....
18	1.....
19	1.....1.....
1A	1.....0.000000...00000.....
1B	.0..00..0..0..0.0.0..00.0.0.0.....1.....
1C	1.....000.....0..0.0.....1.....
1D	1.....
1E	1.....
1F	1.....1.....
20	1.....
21	1.....
22	1.....1.....
23	1.....
24	1.....
25	1.....1.....
26	1.....
27	1.....
28	1.....1.....
29	1..0..0.....0.....
2A	00..0..01..0.0.....00.....
2B	1.....1.....

◆ FZ Display Mode

FZ-mode data with a $4\text{ M} \times 1$ configuration

/FBMAP.␣

When the input wait state of interactive mode occurs, specify the items as shown below.

(BIT# = 1 and DISPLAY TYPE = 1 are specified.)

```

**** FBMAP(T5500 SERIES) ****          <PARAMETER FILE NAME IS AQ2.FB>

. FAIL SELECT -----: FZ
. AFM BITSIZE -----: 1
. BIT # -----: 1
. DISPLAY TYPE -----: 1
. START MPAT -----: YES
. DUT # -----: 1
. OUTPUT DEVICE -----: LPT
. AFM PGSEL -----: MAIN
. AFM ADDRESS -----: (MAIN)X0-10, Y0-10,   DELAY0
. AFM RATE -----: HS
CHANGE MENU-PARAMETER FILE

[WORK]                                     [WORK]

[NOTE] ----- [NOTE]
      PF1:EXIT   PF2:DISPLAY   ENTER:CHANGE-PARAMETER

```

Press the [PF2] key to execute the program.

```
*****          FBMAP (T5500 SERIES)          *****  FBMAP REV # 1.00

1998/04/08 11:59:30

DEVICE      XXXXXXXX
LOT #       20800-8
SAMPLE #    6
TEST #      7
OPERATOR :  ADVAN

PARAMETER FILE NAME IS AQ2.FB
BIT SIZE    =    1
BIT #       =    1
DISPLAY TYPE =    1
DUT#= 1
PGSEL       =    MAIN
ADDRESS     =    (MAIN) X0-10  Y0-10      DELAY0
RATE        =    HS

+++++ FAIL DATA ONLY +++++
'.' :          PASS
'0' : 0        FAIL

FZ COUNT = 137  *1
```

**1 Indicates 0-FAIL count value in a block since DISPLAY TYPE is 1.*

◆ FO Display Mode

FO data of a 4 M × 1 configuration

/FBMAP.↓

When the input wait state of interactive mode occurs, specify the items as shown below.

(BIT# = 1 and DISPLAY TYPE = 1 are specified.)

```

**** FBMAP(T5500 SERIES) ****          <PARAMETER FILE NAME IS AQ2.FB>

. FAIL SELECT -----: FO
. AFM BITSIZE -----: 1
. BIT # -----: 1
. DISPLAY TYPE -----: 1
. START MPAT -----: YES
. DUT # -----: 1
. OUTPUT DEVICE -----: LPT
. AFM PGSEL -----: MAIN
. AFM ADDRESS -----: (MAIN) X0-10, Y0-10,    DELAY0
. AFM RATE -----: HS
CHANGE MENU-PARAMETER FILE

[WORK]                                     [WORK]

[NOTE] ----- [NOTE]
      PF1:EXIT   PF2:DISPLAY   ENTER:CHANGE-PARAMETER

```

Press the [PF2] key to execute the program.

```
*****          FBMAP (T5500 SERIES)          *****  FBMAP REV # 1.00

1998/04/08 11:59:30

DEVICE      XXXXXXXX
LOT #       20800-8
SAMPLE #    6
TEST #      7
OPERATOR :  ADVAN

PARAMETER FILE NAME IS AQ2.FB
BIT SIZE    =    1
BIT #       =    1
DISPLAY TYPE =    1
DUT#= 1
PGSEL       =    MAIN
ADDRESS     =    (MAIN) X0-10  Y0-10      DELAY0
RATE        =    HS

+++++ FAIL DATA ONLY +++++
'.' :          PASS
'1' :    1      FAIL

FO COUNT = 137  *1
```

48. 2 FBMAP Output Example

*1 Indicates 1-FAIL count value in a block since DISPLAY TYPE is 1.

```

0123456789ABCDEF0123456789ABCDEF0123456789ABCDEF0123456789ABCDEF01234
0 .....1.....
1 .....1.....1.....
2 .....1.....
3 .....1.....
4 .....1.....1.....
5 .....1.....
6 .....1.....
7 .....1.....1.....
8 .....1.....
9 .....1.....
A .....1.....1.....
B .....1.....
C .....1.....
D .....1.....1.....
E .....1.....
F .....1.....
10 .....1.....1.....
11 .....1.....
12 .....1.....
13 .....1.....1.....
14 .....1.....
15 .....1.....
16 .....1.....1.....
17 .....1.....
18 .....1.....
19 .....1.....1.....
1A .....1.....
1B .....1.....
1C .....1.....1.....
1D .....1.....
1E .....1.....
1F .....1.....1.....
20 .....1.....
21 .....1.....
22 .....1.....1.....
23 .....1.....
24 .....1.....
25 .....1.....1.....
26 .....1.....
27 .....1.....
28 .....1.....1.....
29 .....1.....
2A .....1.....
2B .....1.....1.....

```

◆ FT Display Mode

FT data of a 4 M × 4 configuration

/FBMAP.␣

When the input wait state of interactive mode occurs, specify the items as shown below.

(BIT# = 1 and DISPLAY TYPE = 1 are specified.)

```

**** FBMAP(T5500 SERIES) ****          <PARAMETER FILE NAME IS AQ2.FB>

. FAIL SELECT -----: FT
. AFM BITSIZE -----: 4
. BIT # -----: ORDATA
. DISPLAY TYPE -----: 1
. START MPAT -----: YES
. DUT # -----: 1
. OUTPUT DEVICE -----: LPT
. AFM PGSEL -----: MAIN
. AFM ADDRESS -----: (MAIN) X0-10, Y0-10,   DELAY0
. AFM RATE -----: HS
CHANGE MENU-PARAMETER FILE

[WORK]                                     [WORK]

[NOTE] ----- [NOTE]
      PF1:EXIT   PF2:DISPLAY   ENTER:CHANGE-PARAMETER

```

Press the [PF2] key to execute the program.

```
*****          FBMAP(T5500 SERIES)          *****  FBMAP REV # 1.00

1998/04/08 11:59:30

DEVICE      XXXXXXXX
LOT #       20800-8
SAMPLE #    6
TEST #      7
OPERATOR :  ADVAN

PARAMETER FILE NAME IS AQ2.FB
BIT SIZE    =    4
BIT #       =   ORDATA
DISPLAY TYPE =    1
DUT#= 1
PGSEL       =   MAIN
ADDRESS     = (MAIN) X0-10  Y0-10      DELAY0
RATE        =   HS

+++++ FAIL DATA ONLY +++++
'.' :          PASS
'X' : 0 OR 1   FAIL

FT COUNT = 37    *1
```


**1 Indicates 0/1-FAIL count value in a block since DISPLAY TYPE is 1.*

		11111111111111112222222222222222333333333333333344444
	0123456789ABCDEF0123456789ABCDEF0123456789ABCDEF0123456789ABCDEF01234	
0	X.....	
1	X.....X.....	
2	X.....	
3	X.....	
4	X.....X.....	
5	X.....	
6	X.....	
7	X.....X.....	
8	X.....	
9	X.....	
A	X.....X.....	
B	X.....	
C	X.....	
D	X.....X.....	
E	X.....	
F	X.....	
10	X.....X.....	
11	X.....	
12	X.....	
13	X.....X.....	
14	X.....	
15	X.....	
16	X.....X.....	
17	X.....	
18	X.....	
19	X.....X.....	
1A	X.....	
1B	X.....	
1C	X.....X.....	
1D	X.....	
1E	X.....	
1F	X.....X.....	
20	X.....	
21	X.....	
22	X.....X.....	
23	X.....	
24	X.....	
25	X.....X.....	
26	X.....	
27	X.....	
28	X.....X.....	
29	X.....	
2A	X.....	
2B	X.....X.....	

◆ Output Example of any Address Display (DISPLAY TYPE = S)

A fail bit map for one block from the specified address is displayed.

- When START ADDRESS (X: #0030, Y: #0040) is specified.

```

*****          FBMAP (T5500 SERIES)          *****  FBMAP REV # 1.00

1998/04/08 11:59:30

DEVICE      XXXXXXXX
LOT #       20800-8
SAMPLE #    6
TEST #      7
OPERATOR : ADVAN

PARAMETER FILE NAME IS AQ2.FB
BIT SIZE    = 1
BIT #       = 1
DISPLAY TYPE = START ADDRESS (X:#0040, Y=#0030)
DUT#= 1
PGSEL       = MAIN
ADDRESS      = (MAIN) X0-10  Y0-10          DELAY0
RATE        = HS

+++++ FAIL DATA ONLY +++++
'.' : PASS
'0' : 0    FAIL
'1' : 1    FAIL
'X' : 0 OR 1 FAIL

FZ COUNT = 37  *1
FO COUNT = 17  *1

```


- When START ADDRESS (X: #0FF0, Y: #0FFA) is specified.

```

*****          FBMAP (T5500 SERIES)          *****  FBMAP REV # 1.00

1998/04/08 11:59:30

DEVICE      XXXXXXXX
LOT #       20800-8
SAMPLE #    6
TEST #      7
OPERATOR : ADVAN

PARAMETER FILE NAME IS AQ2.FB
BIT SIZE    = 1
BIT #       = 1
DISPLAY TYPE = START ADDRESS (X:#0FF0, Y:#0FFA)
DUT#= 1
PGSEL       = MAIN
ADDRESS     = (MAIN) X0-10  Y0-10          DELAY0
RATE        = HS

+++++ FAIL DATA ONLY +++++
'. ' :      PASS
'0 ' : 0      FAIL
'1 ' : 1      FAIL
'X ' : 0 OR 1  FAIL

FZ COUNT = 5 *1
FO COUNT = 5 *1

```

**1 Indicates a fail counter in one block since DISPLAY TYPE is S.*

```

7777777777777777
FFFFFFFFFFFFFFFF
0123456789ABCDEF

7FA .....000...
7FB .....1110011...
7FC .....
7FD .....
7FE .....
7FF .....

```

A map for one block from (X: #07F0, Y: #07FA) is output. In the above figure, the address for one block from the specified position exceeds the range of address space, therefore the displaying map is displayed in the range of address space.

48. 2. 2 Fail Address Display

After display instruction is executed, fail address of specified device is displayed. When suppressing the execution of the instruction, press the [B] key.

◆ FAIL Display Mode

A fail address with a 4 M × 1 configuration is displayed.

```
*****          FBMAP (T5500 SERIES)          *****  FBMAP REV # 1.00

1998/04/08 11:59:30

DEVICE      XXXXXXXX
LOT #       20800-8
SAMPLE #    6
TEST #      7
OPERATOR :  ADVAN

PARAMETER FILE NAME IS AQ2.FB
BIT SIZE    =    1
BIT #       =    1
DISPLAY TYPE =  FAIL ADDRESS DISPLAY
DUT#= 1
PGSEL       =  MAIN
ADDRESS     =  (MAIN) X0-10  Y0-10          DELAY0
RATE        =  HS

+++++  FAIL DATA  +++++
TOTAL FAIL COUNT = 4194303  *1
OUTPUT COUNT = ALL          *2
```

*1 The number of all failures shown by the X and Y addresses.

*2 The specified output number is displayed. When it is ALL, ALL is displayed. When it is not ALL, the specified value is displayed.

F-CNT	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD
1	#00000	#00000	#00000	#00001	#00000	#00002	#00000	#00003	#00000	#00004	#00000	#00005	#00000	#00006	#00000	#00007
9	#00000	#00008	#00000	#00009	#00000	#0000A	#00000	#0000B	#00000	#0000C	#00000	#0000D	#00000	#0000E	#00000	#0000F
17	#00000	#00010	#00000	#00011	#00000	#00012	#00000	#00013	#00000	#00014	#00000	#00015	#00000	#00016	#00000	#00017
24	#00000	#00018	#00000	#00019	#00000	#0001A	#00000	#0001B	#00000	#0001C	#00000	#0001D	#00000	#0001E	#00000	#0001F
33	#00000	#00020	#00000	#00021	#00000	#00022	#00000	#00023	#00000	#00024	#00000	#00025	#00000	#00026	#00000	#00027
41	#00000	#00028	#00000	#00029	#00000	#0002A	#00000	#0002B	#00000	#0002C	#00000	#0002D	#00000	#0002E	#00000	#0002F
39	#00000	#00030	#00000	#00031	#00000	#00032	#00000	#00033	#00000	#00034	#00000	#00035	#00000	#00036	#00000	#00037
4194296	#007FF	#007F8	#007FF	#00FF9	#007FF	#007FA	#007FF	#007FB	#007FF	#007FC	#007FF	#007FD	#007FF	#007FE	#007FF	#007FF

◆ FZ Display Mode

FAIL-mode data with a 4 M × 1 configuration

/FBMAP.␣

The debugger waits for interactive-mode input. Items are set as follows:

(BIT# = 1 and DISPLAY TYPE = FA are specified.)

```

**** FBMAP(T5500 SERIES) ****          <PARAMETER FILE NAME IS AQ2.FB>

. FAIL SELECT -----: FZ
. AFM BITSIZE -----: 1
. BIT # -----: 1
. DISPLAY TYPE -----: FAIL ADRS DISP. 100
. START MPAT -----: YES
. DUT # -----: 1
. OUTPUT DEVICE -----: LPT
. AFM PGSEL -----: MAIN
. AFM ADDRESS -----: X0-10, Y0-10, DELAY0
. AFM RATE -----: HS
CHANGE MENU-PARAMETER FILE

[WORK]                                     [WORK]

[NOTE] ----- [NOTE]
      PF1:EXIT   PF2:DISPLAY   ENTER:CHANGE-PARAMETER

```

Press the [PF2] key to execute the program.

```

*****          FBMAP (T5500 SERIES)          *****  FBMAP REV # 1.00

1998/04/08 11:59:30

DEVICE      XXXXXXXX
LOT #       20800-8
SAMPLE #    6
TEST #      7
OPERATOR :  ADVAN

PARAMETER FILE NAME IS AQ2.FB
BIT SIZE    =    1
BIT #       =    1
DISPLAY TYPE =  FAIL ADDRESS DISPLAY
DUT# = 1
PGSEL       =  MAIN
ADDRESS     =  (MAIN) X0-10  Y0-10          DELAY0
RATE        =  HS

+++++ FZ DATA  +++++
TOTAL FZ COUNT = 4194304  *1
OUTPUT COUNT = 100      *2

```

*1 The number of all failures shown by the X and Y addresses.

*2 The specified output number is displayed. When it is ALL, ALL is displayed. When it is not ALL, the specified value is displayed.

F-CNT	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD
1	#00000	#00000	#00000	#00001	#00000	#00002	#00000	#00003	#00000	#00004	#00000	#00005	#00000	#00006	#00000	#00007
9	#00000	#00008	#00000	#00009	#00000	#0000A	#00000	#0000B	#00000	#0000C	#00000	#0000D	#00000	#0000E	#00000	#0000F
17	#00000	#00010	#00000	#00011	#00000	#00012	#00000	#00013	#00000	#00014	#00000	#00015	#00000	#00016	#00000	#00017
24	#00000	#00018	#00000	#00019	#00000	#0001A	#00000	#0001B	#00000	#0001C	#00000	#0001D	#00000	#0001E	#00000	#0001F
33	#00000	#00020	#00000	#00021	#00000	#00022	#00000	#00023	#00000	#00024	#00000	#00025	#00000	#00026	#00000	#00027
41	#00000	#00028	#00000	#00029	#00000	#0002A	#00000	#0002B	#00000	#0002C	#00000	#0002D	#00000	#0002E	#00000	#0002F
39	#00000	#00030	#00000	#00031	#00000	#00032	#00000	#00033	#00000	#00034	#00000	#00035	#00000	#00036	#00000	#00037
96	#00000	#00061	#00000	#00062	#00000	#00063	#00000	#00064								

- 60 lines/1 page
- The number of failure addresses is 100.

◆ FT Display Mode

A fail address with a 16 M × 1 configuration

/FBMAP 1,3,1,1,FA,0,1,1,0,10,10,0,0,0,┘

(BIT# = 1 and DISPLAY TYPE = FA are specified.)

```

*****          FBMAP (T5500 SERIES)          *****  FBMAP REV #   1.00

1998/04/08 11:59:30

DEVICE      XXXXXXXX
LOT #       20800-8
SAMPLE #    6
TEST #      7
OPERATOR    :ADVAN

PARAMETER FILE NAME IS *1
BIT SIZE    =    1
BIT #       =    1
DISPLAY TYPE =  FALL ADDRESS DISPLAY
DUT# = 1
PGSEL       =  MAIN
ADDRESS     =  (MAIN) X0-10  Y0-10          DELAY0
RATE        =  HS

+++++ FT DATA +++++
TOTAL FT COUNT = 1000 *2
OUTPUT COUNT = ALL *3

```

*1 Not displayed when parameters are specified directly.

*2 The number of all failures shown by the X and Y addresses.

*3 The specified output number is displayed. When it is ALL, ALL is displayed. When it is not ALL, the specified value is displayed.

F-CNT	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD	Y-AD	X-AD
1	#00000	#00000	#00000	#00001	#00000	#00002	#00000	#00003	#00003	#00004	#00000	#00005	#00000	#00006	#00000	#00007
9	#00000	#00008	#00000	#00009	#00000	#0000A	#00000	#0000B	#00003	#0000C	#00000	#0000D	#00000	#0000E	#00000	#0000F
17	#00000	#00010	#00000	#00011	#00000	#00012	#00000	#00013	#00003	#00014	#00000	#00015	#00000	#00016	#00000	#00017
24	#00000	#00018	#00000	#00019	#00000	#0001A	#00000	#0001B	#00003	#0001C	#00000	#0001D	#00000	#0001E	#00000	#0001F
33	#00000	#00020	#00000	#00021	#00000	#00022	#00000	#00023	#00003	#00024	#00000	#00025	#00000	#00026	#00000	#00027
41	#00000	#00028	#00000	#00029	#00000	#0002A	#00000	#0002B	#00003	#0002C	#00000	#0002D	#00000	#0002E	#00000	#0002F
39	#00000	#00030	#00000	#00031	#00000	#00032	#00000	#00033	#00003	#00034	#00000	#00035	#00000	#00036	#00000	#00037
993	#00000	#003E0	#00000	#003E1	#00000	#003E2	#00000	#003E3	#00003	#003E4	#00000	#003E5	#00000	#003E6	#00000	#003E7

60 lines/1 page

49. RESET DRIVER ENABLE/DISABLE Setting Program (/RSTDRV)

It is a utility program to specify if the initialization of driver output level is performed or not (ENABLE/DISABLE) at pattern generation for the I/O pins in which driver pin and DRONINIT are specified.

The initial value (HI/LOW) of driver output at pattern generation is not steady under the influence of last state of the previous pattern. By specifying to initialize automatically with this utility program, the initial value (HI/LOW) of driver output at pattern generation always becomes steady.

Tip

On the T5376/T5375

Driver output levels are automatically initialized every time a pattern is generated, regardless of whether or not they are specified in the utility program.

49. 1 Method of Operation

There are two ways. One is to specify with controller mode. The other is to specify with device program.

When specifying using controller mode:

Input as follows:

```
/RSTDRV E.  ----- (a)
/RSTDRV D.  ----- (b)
or
/RSTDRV.  ----- (c)
```

- For (a):

The following message is displayed, and ENABLE mode is turned on. (The initialization is performed automatically.)

```
RESET DRIVER MODE = ENABLE
```

```
*** EXIT ! ***
```

- For (b):

The following message is displayed, and DISABLE is turned on. (The initialization is not performed automatically.)

```
RESET DRIVER MODE = DISABLE

*** EXIT ! ***
```

- For (c):

The mode is specified interactively. The following menu is displayed.

```
*** SPECIFY RESET DRIVER MODE, ENABLE OR DISABLE ***

RESET DRIVER MODE = DISABLE ← The current mode is displayed.

SPECIFY RESET DRIVER MODE, ENABLE (E) OR DISABLE (D) :__
```

Enter E↵ or D↵ here.

The mode (ENABLE or DISABLE) is input.

The following message is displayed to end.

```
RESET DRIVER MODE = ENABLE ← The specified mode is displayed.

*** EXIT ! ***
```

In the event of ending without specification, enter ↵ only.

The following message is displayed to end.

```
*** EXIT ! ***
```

When specifying using a device program:

Input as follows:

```
CALL LIB RSTDRV("E", IER) ----- (a)
or
CALL LIB RSTDRV("D", IER) ----- (b)
```

When (a), ENABLE is turned on. (The initialization is performed automatically.)

When (b), DISABLE is turned on. (The initialization is not performed automatically.)

IER

The argument for identifying input argument errors.

When the argument specified is correct, IER=0.

When the argument specified is incorrect, IER=1.

IER can be omitted.

49. 2 Function

When the driver initialization ENABLE/DISABLE setting is set to ENABLE, the driver output level (HI/LOW) of the I/O pins for which driver pin and "DRONINIT" are specified is automatically initialized immediately before the pattern is started by the MEAS/START MPAT statement. The driver output level (HI/LOW), however, is not automatically initialized if the MEAS/START MPAT (1) statement is used.

The operation of the initialization is the same as the RESET DRIVER statement.

(Example) When using the /NRZA waveform, it always starts with HI level.

When set to DISABLE, the initialization is not performed automatically.

49. 3 About Keeping the Specified Mode

Mode default value is DISABLE.

The specified mode is kept even if the controller mode is restarted (it is saved in the SYS CONT.BF file).

The mode does not change even if CTRL/C is input.

49. 4 About the Influence on Other Statements

RESET DRIVER statement and /RSTDRV setting have no influence each other.

Also, SELECT LEVELHOLD ENABLE/DISABLE statement and /RSTDRV setting have no influence each other.

This statement is valid only at pin statement execution so that the initialization at pattern start is not influenced.

50. Fail Memory Data Transfer (/FMXFER)

The MASK ROM is usually measured using the ROM data pattern which has been transferred to the PM (Pattern memory) and is assumed to be an expected valued pattern. In this utility, FMXFER, the expected value pattern data of ROM needed for the MASK ROM test can be filed, transferred to PM, and PM/AFM (Address failure memory) can be checked by displaying the contents. Also, the pattern data of DBM (Data buffer memory) /CBM (Command buffer memory) can be displayed.

50. 1 FMXFER on T5500

50. 1. 1 Creating File/Display/Transfer of ROM Test Pattern Data

This section describes about the usage of creating files of ROM test pattern data and display confirmation of the file, transferring to PM and display confirmation of the contents of the PM/AFM. Functions described in this section can be used in the system which has AFM and PM.

50. 1. 1. 1 Method of Use

- 1) Execute the test plan program specified by the user, and pause the program when fail data has been stored in the AFM, or the AFM mode has been set.

To pause the program, use the PAUSE statement or set MANUAL and HOLD for the test mode.

- 2) Enter the utility start command from VKT.

```
/FMXFER.]
```

The following menu is displayed. Select the item to execute using the [↑], [↓] keys, and press the Return key.

```

***  FUNCTION MENU  ***

__SEND FM (DISK ---> FM)
.READ FM (FM ---> DISK)
.DISPLAY FM
.DISPLAY DBM
.DISPLAY CBM
.DISPLAY ROM PAT FILE

*** NOTE ***
PF1 :  EXIT

```

However, when n1 of AFM BITSIZE=n1,n2 exceeds 36-bit, the following menu is displayed.

```

***  FUNCTION MENU  ***

.SEND FM (DISK ---> FM)    NO USE THIS MENU
.READ FM (FM ---> DISK)    NO USE THIS MENU
__DISPLAY FM
.DISPLAY DBM
.DISPLAY CBM
.DISPLAY ROM PAT FILE

*** NOTE ***
PF1 :  EXIT

```

.SEND FM

Transfers the expected data file to PM where the EXP mode is set. The applicable file is an MPAT file.



Limitation

This command cannot be specified when n1 of AFM BITSIZE=n1,n2 exceeds 36-bit.

.READ FM

This command reads out the data stored in AFM, and creates an MPAT file using this data.

Limitation

This command cannot be specified when n1 of AFM BITSIZE=n1,n2 exceeds 36-bit.

.DISPLAY FM

This command outputs data stored in PM or AFM.

.DISPLAY DBM

This command outputs data stored in the contents of DBM memory.

.DISPLAY CBM

This command outputs data stored in the contents of CBM memory.

.DISPLAY ROM PAT FILE

This command outputs data stored in the contents of expected data file.

PF1:

The [PF1] key is used to exit from this utility, FMXFER.

◆ Expected Data File Transfer to PM (.SEND FM)

This command is used to transfer the MPAT file of the pattern program where the MPAT file created by selecting the .READ FM or the ROM test pattern data is stored, to the PM where EXP mode is set.

```
*** MENU FOR SEND MODE ***
```

```
__SEND FILE NAME
.SEND
```

```
*** NOTE ***
PF1 : LAST MENU
```

.SEND FILE NAME

Select this command when the MPAT file name to be transferred is entered. Enter a customer name and a file name by following the message below.

```
CUSTOMER NAME = RMUTL __NEWCST.↓
```

**I "RMUTL" is a current customer name, and "NEWCST" is the name of the customer to be entered this time.*

If a customer name is changed, enter the new customer name using up to 32 characters. If a present customer name is used, press the Return key.

```
FILE NAME = MPA ROMPAT1 ____ ROMPAT2
```

**1 "MPA ROMPAT1" is the name of the previous file, while "ROMPAT2" is the name of the file to be transferred this time.*

If a file name is changed, enter the new file name using up to exceed 32 characters. If a file name is not changed, press the Return key. The file specified here is the MPAT file only.

.SEND

This command starts transmission to PM.

```
WORKING!
```

When the transmission is completed, the above message is cleared.

PF1:

Pressing the [PF1] causes the system to return to the *** FUNCTION MENU ***.

◆ Fail Store Data Filing (.READ FM)

This command is used to read out the data stored in AFM to create the MPAT file by executing the test plan program specified by a user. The MPAT file can be transferred using the SEND FM statement.

If the data storing in AFM is bit compressed data, it is read and MPAT file can be created. When this MPAT file is transferred using the SEND FM statement, compressed data is stored in the PM.

```
***   MENU FOR READ MODE   ***

__MAKE A FILE
.READ AFM MODE -----: FT
.AFM DUTSEL -----: 1
.START ADDRESS -----: X:#0      Y:#0
.STOP ADDRESS  -----: X:#3FF    Y:#FFF
.SELECT FM (MAIN/SUB=M/S) -----: M
```

```
***   NOTE   ***
PF1 : LAST MENU
'B' : BREAK
```

.MAKE A FILE

This command specifies for which customer and with what file name the MPAT file is created.
Enter a customer name and a file name following the message below.

```
CUSTOMER NAME = RMUTL __NEWCSTR.↓
```

**I "RMUTL" is a current customer name, and "NEWCSTR" is the name of the customer to be entered this time.*

```
FILE NAME = MPA RFIL1 __RFIL2.↓
```

**I "MPA ROMPAT1" is the name of the previously created file, while "RFIL2" is the name of the file created this time.*

- When the same file name does not occur among the customers specified this time
The following message is displayed and filing begins:

```
WORKING!
```

After filing is completed, the following message is displayed.

```
COMPLETE NEWCST : MPA RFIL2
```

**I "NEWCST" is a customer name, and "MPA RFIL2" is a file name.*

To interrupt filing, press the [B] key. Then, the following message is displayed and filing is interrupted.

```
BREAK! EXECUTION ABORTED!!?
```

A file until interruption has been created.

- When the same file name already exists among the customers selected this time:
The following message is displayed:

```
**  FILE ALREADY EXIST  **

*  CUSTOMER : RMUTL / FILE NAME : MPA RFIL1

      ---->   DELETE ? *

'Y'   :  YES (=DELETE)
'N'   :  NO  (=EXIT)

* Push 'Y' or 'N' key =[ ]*
```

**I "RMUTL" is a customer name, and "MPA RFIL1" is a file name.*

Press [Y] key to delete the MPAT file existing in the currently specified customer. Press [N] key if you do not want to delete it.

When the [N] key is pressed:

The existing MPAT file is not deleted but exits.

When the [Y] key is pressed:

The following message is displayed:

```
DELETE! :: RMUTL : MPA RFIL1      WORKING!
```

**1 "RMUTL" is the specified customer name, and "MPA RFIL1" is the customer name to be deleted.*

After deletion, "WORKING !" is cleared and the following message is displayed.

```
DELETE! OK!!      :: RMUTL : MPA RFIL1
```

**1 "RMUTL" is a customer name, and "MPA RFIL1" is the name of a deleted file.*

The message above is cleared approximately 2 seconds after the display.

Limitation

Only the current customer (used to start this utility) can be deleted.

If the customer to be deleted is not on the current customer list, the following message is displayed, and no deletion takes place:

```
ERCODE : 3772      Can't! DELETE!!
```

Only the files having the current customer name can be deleted.

If the file can be deleted, the following message is displayed.

```
** MAKE A FILE **

* CUSTOMER:RMUTL / FILE NAME:MPA RFIL1

      ----> MAKE A FILE ?

'Y'   : YES (=MAKE A FILE)
'N'   : NO  (=EXIT)

* Push 'Y' or 'N' key =[ ]*
```

**1 "RMUTL" is a customer name, and "MPA RFIL1" is a file name.*

- When the [Y] key is pressed:

Filing starts for the customer specified this time with the deleted file name. The operation and message display after filing starts are same as those of item (a) for the

- ["When the same file name does not occur among the customers specified this time" on page 50-5](#) command.

- When the [N] key is pressed:
The file deleted now is not created for the customer specified this time but exits. Only the deletion of the existing MPAT file is possible by selecting the MAKE A FILE command.

Tip

For creating file of bit compressed fail store data, file of BITSIZE of ROM test pattern data is created by n2 of AFM BITSIZE=n1,n2 (EXP mode is n1.). However, when n1 is 32 or 36, the file of BITSIZE is created by 36 without regard to n2 (except EXP mode).

READ AFM MODE

Specify the operation mode of AFM to be read. The mode specified by this command must be set by the user-specified test plan program before it can be specified here.

When this utility starts, data entered in the AFM program statement in the test plan program specified by a user is set. If the multiple modes are specified by the AFM ACTION statement in the test plan program specified by a user, the mode written first will be set. Enter a device number when the message shown below is displayed.

```
READ AFM MODE = __ FZ ↵
```

- Specifiable modes
EXP, FT, FZ, FO, AF, BF, MASK
FTSBn, FZSBn, FOSBn, AFSBn, BFSBn, MASKSBn
FTSB(v), FZSB(v), FOSB(v), AFSB(v), BFSB(v), MASKSB(v)
n
1, 2, 129, or 130 can be specified.
v
Can be specified in the range #1 to #F.

.AFM DUTSEL

Specify a device number whose data is stored in the AFM. This utility starts according to the AFM DUTSEL statement. Enter a device number when the message shown below is displayed.

```
AFM DUTSEL = __ 4↵
```

Note that the specifiable range is 1 to 256.

.START ADDRESS

This command is used to set the start address of the AFM data to be filed. Enter the command in hexadecimal following the message below.

```
START ADDRESS = X : # __F↵ Y : # F↵
```

The cursor indicates the X address position during display of the above message.

If no change is made, press the Return key. Then, the cursor indicates the Y address position. If no change is made, press the Return key again.

Note that the default is X :#0 and Y :#0. A value greater than STOP ADDRESS cannot be set.

.STOP ADDRESS

This command is used to set the stop address of the AFM data to be filed. Enter the command in hexadecimal following the message below.

```
STOP ADDRESS = X : # __3F↵ Y : # FF↵
```

The cursor indicates the X address position during display of the above message.

If no change is made, press the Return key. Then, the cursor indicates the Y address position. If no change is made, press the Return key.

The default is the maximum of the AFM ADDRESS specified in the test plan program specified by a user. A value larger than the default and smaller than the START ADDRESS cannot be specified.

Tip

When READ AFM MODE = EXP, the default value is the maximum number of AFM EXP ADDRESS.

.SELECT FM (MAIN/SUB=M/S)

This command is used to set AFM to be filed.

Enter a device number when the message shown below is displayed.

```
MAIN/SUB (M/S) = __M
```

Enter "M" when specifying the MAIN AFM. Enter "S" when specifying the SUB AFM. Default is provided for M.

Limitation

Only when (MAIN, SUB) is set in AFM PGSEL statement, SELECT FM can be specified.

When MAIN is set in AFM PGSEL statement, the following message is displayed and SELECT FM can not be specified (fixed at MAIN).

```
. SELECT FM (MAIN=M)
```

PF1:

Pressing the [PF1] causes the system to return to the *** FUNCTION MENU ***.

◆ Output of Fail Store Data (.DISPLAY FM)

Outputs the contents of AFM specified in the READ AFM mode in ["Fail Store Data Filing \(.READ FM\)" on page 50-4](#).

```
*** MENU FOR DISPLAY FM MODE ***

_DISPLAY
__START ADDRESS----- : X:#0 Y:#0
.DATA TYPE (HEXA/BINARY=H/B) ----- : H
.FORMAT (VERTICAL/LATERAL=V/L) ----- : V
.OUTPUT (VKT/VKT+LPT/S COMMAND=V/L/S) ----- : V
.READ AFM MODE----- : FT
.SELECT FM (MAIN/SUB=M/S) ----- : M

*** NOTE ***
PF1 : LAST MENU
```

.DISPLAY

Starts displaying the data output according to the conditions specified by START ADDRESS, DATA TYPE, FORMAT, OUTPUT, and READ AFM MODE.

.START ADDRESS

Enter the address to begin output. Enter a device number when the message shown below is displayed.

```
START ADDRESS = X:#__ 5↓ Y:# 100↓
```

This setting ranges from 0 to the maximum value of the AFM ADDRESS specified by the user in the test plan program. Its default value is X: #0, Y: #0.

Tip

When READ AFM MODE = EXP, the available range is from 0 to the maximum value of AFM EXP ADDRESS.

.DATA TYPE (HEXA/BINARY=H/B)

Specifies the type of output data. Enter a device number when the message shown below is displayed.

```
HEXA/BINARY (H/B) = __B
```

Input "H" for the hexadecimal type. Input "B" for the binary type. When omitted, H is assumed.

 Tip

When "B" (the binary type) is input, the data type is changed to FORMAT (VERTICAL/LATERAL=V/L) =V unconditionally.

.FORMAT (VERTICAL/LATERAL=V/L)

Input the output data format according to the following message. Enter a device number when the message shown below is displayed.

```
VERTICAL/LATERAL (V/L) = __ L
```

Input "V" when the data is output in the format to advance the address in the vertical direction, or "L" when it is output in the format to advance the address in the lateral direction. When omitted, V is assumed.

 Tip

When L (the address is advanced in the lateral direction) is input, the format is changed to DATA TYPE (HEXA/BINARY) = H unconditionally. In executing the DISPLAY, the VKT screen becomes 132 columns automatically. When execution ends, it returns to 80 columns.

.OUTPUT (VKT/VKT+LPT/S COMMAND=V/L/S)

Select the output device. Enter a device number when the message shown below is displayed.

```
VKT/VKT+LPT/S COMMAND (V/L/S) = __ L
```

Input V when the output device is VKT. Input "L" when the output devices are VKT and LPT. Input S if it obeys S command. When omitted, V is assumed to be input.

READ AFM MODE

Same as ["READ AFM MODE" on page 50-7](#) of ["Fail Store Data Filing \(.READ FM\)" on page 50-4](#). Consequently, the AFM mode is selected in two places (the READ FM menu and the DISPLAY FM menu). The last one of these two modes that is set is set in common for the two.

.SELECT FM DATA (MAIN/SUB=M/S)

Specify AFM to be output.

The specification mode is the same as [".SELECT FM \(MAIN/SUB=M/S\)" on page 50-8](#) of ["Fail Store Data Filing \(.READ FM\)" on page 50-4](#).

Therefore, AFM board is selected in two positions.

(READ FM menu and DISPLAY FM menu)

The last mode set in two positions is commonly set.

Output Format

The format depends on the specification of DATA TYPE, FORMAT and bit size. Compressed data is displayed for the bit compressed data.

The displaying data format when DATA TYPE is HEXA is shown below.

- When VERTICAL is selected (For 1 to 36 bits)

Y	X	BIT=8	Y	X	BIT=8	Y	X	BIT=8	Y	X	BIT=8
00000	00000	AA	00000	00010	AA	00000	00020	AA	00000	00030	AA
00000	00001	55	00000	00011	55	00000	00021	55	00000	00031	55
00000	00002	AA	00000	00012	AA	00000	00022	AA	00000	00032	AA
00000	00003	55	00000	00013	55	00000	00023	55	00000	00033	55
00000	00004	AA	00000	00014	AA	00000	00024	AA	00000	00034	AA
:			:			:			:		
00000	0000E	55	00000	0001E	AA	00000	0002E	AA	00000	0003E	AA
00000	0000F	AA	00000	0001F	55	00000	0002F	55	00000	0003F	55
*** NOTE ***			*	*** MEMO ***							
			I	SELECT FM = MAIN							
PF1 : LAST MENU			I	DATA TYPE = HEXA							
PF2 : NEXT PAGE			I	FORMAT = VERTICAL							
			*	OUTPUT = VKT							

*1 "Y" is the Y address (hexadecimal), and "X" is the X address (hexadecimal).

"BIT=8" is the number of data bits (displayed as a decimal number, number of bits ^{*2} captured to the AFM).

*2 The number of bits is specified by n_2 of AFM BITSIZE= n_1, n_2 (EXP mode is n_1) when ALL or AFM is specified in the SELECT COMPRESS statement.

When SELECT COMPRESS=CFM, AFM BITSIZE==the number of bits specified in n_1 of n_1 and n_2 .

- When VERTICAL is selected (For 64 or 72 bits)

Y	X	BIT=72		Y	X	BIT=72	
00000	00000	FFFFFFFF	FFFFFFFF	00001	00000	FFFFFFFF	FFFFFFFF
00000	00001	00000000	00000000	00001	00001	00000000	00000000
:				:			
00000	0000E	FFFFFFFF	FFFFFFFF	00001	0000E	FFFFFFFF	FFFFFFFF
00000	0000F	00000000	00000000	00001	0000F	00000000	00000000
*** NOTE ***				*** MEMO ***			
				SELECT FM = MAIN			
PF1:LAST MENU				DATA TYPE = HEXA			
PF2:NEXT PAGE				FORMAT = VERTICAL			
				OUTPUT = VKT			

- When VERTICAL is selected (For 128 or 144 bits)

Y	X	BIT=144	
00000	00000	FFFFFFFF	FFFFFFFF
00000	00001	00000000	00000000
:			
00000	0000E	FFFFFFFF	FFFFFFFF
00000	0000F	00000000	00000000
*** NOTE ***		*** MEMO ***	
		SELECT FM = MAIN	
PF1:LAST MENU		DATA TYPE = HEXA	
PF2:NEXT PAGE		FORMAT = VERTICAL	
		OUTPUT = VKT	

Pressing the [PF1] key returns the screen display to *** MENU FOR DISPLAY FM MODE ***.

When the [PF2] key is pressed, the next 64-address (32-address for 64 or 72 bits, 16-address for 128 or 144 bits) data is output.

When data for up to the maximum address has been output, the data output restarts from address 0.

The bit configuration of the output data is as follows:

(Example) When BIT = 8

BIT=1 2 3 4 5 6 7 8
 ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
 AA (hexadecimal)=1 0 1 0 1 0 1 0 (binary)

For 64 or 72 bits, data is output in two lines.
(When 64 bits)

00000 0000E FFFFFFF FFFFFFF
 1 to 32 bits 33 to 64 bits

(When 72 bits)

00000 0000E FFFFFFF FFFFFFF
 1 to 36 bits 37 to 72 bits

For 128 or 144 bits, data is output in four lines.
(When 128 bits)

00000 0000E FFFFFFF FFFFFFF FFFFFFF FFFFFFF
 1 to 32 bits 33 to 64 bits 65 to 96 bits 97 to 128 bits

(When 144 bits)

00000 0000E FFFFFFF FFFFFFF FFFFFFF FFFFFFF
 1 to 36 bits 37 to 72 bits 73 to 108 bits 109 to 144 bits

- When LATERAL is selected (For 1 to 18 bits)

Y	X	BIT=8															
00000	00000	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55
00000	00010	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55
00000	00020	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55
00000	00030	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55
:																	
*** NOTE ***				*	*** MEMO ***												
				I	SELECT FM = MAIN												
PF1 : LAST MENU				I	DATA TYPE = HEXA												
'B' : BREAK				I	FORMAT = LATERAL												
PF2 : RESET BREAK				*	OUTPUT = VKT												

- When LATERAL is selected (For 32, 36 bits)

Y	X	BIT=36							
00000	00000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000
00000	00008	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000
00000	00010	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000
00000	00018	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000
:									
*** NOTE ***			*	*** MEMO ***					
			I	SELECT FM = MAIN					
PF1 :	LAST MENU		I	DATA TYPE = HEXA					
'B' :	BREAK		I	FORMAT = LATERAL					
PF2 :	RESET BREAK		*	OUTPUT = VKT					

- When LATERAL is selected (For 64, 72 bits)

Y	X	BIT=72							
00000	00000	FFFFFFFF	FFFFFFFF	000000000	000000000	FFFFFFFF	FFFFFFFF	000000000	000000000
00000	00004	FFFFFFFF	FFFFFFFF	000000000	000000000	FFFFFFFF	FFFFFFFF	000000000	000000000
00000	00008	FFFFFFFF	FFFFFFFF	000000000	000000000	FFFFFFFF	FFFFFFFF	000000000	000000000
00000	0000C	FFFFFFFF	FFFFFFFF	000000000	000000000	FFFFFFFF	FFFFFFFF	000000000	000000000
00000	00010	FFFFFFFF	FFFFFFFF	000000000	000000000	FFFFFFFF	FFFFFFFF	000000000	000000000
:									
*** NOTE ***			*	*** MEMO ***					
			I	SELECT FM = MAIN					
PF1 :	LAST MENU		I	DATA TYPE = HEXA					
'B' :	BREAK		I	FORMAT = LATERAL					
PF2 :	RESET BREAK		*	OUTPUT = VKT					

- When LATERAL is selected (For 128, 144 bits)

Y	X	BIT=144							
00000	00000	FFFFFFFF	FFFFFFFF	FFFFFFFF	FFFFFFFF	000000000	000000000	000000000	000000000
00000	00002	FFFFFFFF	FFFFFFFF	FFFFFFFF	FFFFFFFF	000000000	000000000	000000000	000000000
00000	00004	FFFFFFFF	FFFFFFFF	FFFFFFFF	FFFFFFFF	000000000	000000000	000000000	000000000
00000	00006	FFFFFFFF	FFFFFFFF	FFFFFFFF	FFFFFFFF	000000000	000000000	000000000	000000000
00000	00008	FFFFFFFF	FFFFFFFF	FFFFFFFF	FFFFFFFF	000000000	000000000	000000000	000000000
:									
*** NOTE ***			*	*** MEMO ***					
			I	SELECT FM = MAIN					
PF1 :	LAST MENU		I	DATA TYPE = HEXA					
'B' :	BREAK		I	FORMAT = LATERAL					
PF2 :	RESET BREAK		*	OUTPUT = VKT					

For 64 or 72 bits, data is output in two lines.

(When 64 bits)

00000 0000E	<u>FFFFFFFF</u>	<u>FFFFFFFF</u>
	1 to 32 bits	33 to 64 bits

(When 72 bits)

00000 0000E	<u>FFFFFFFFF</u>	<u>FFFFFFFFF</u>
	1 to 36 bits	37 to 72 bits

For 128 or 144 bits, data is output in four lines.

(When 128 bits)

0000 000E	<u>FFFFFFFF</u>	<u>FFFFFFFF</u>	<u>FFFFFFFF</u>	<u>FFFFFFFF</u>
	1 to 32 bits	33 to 64 bits	65 to 96 bits	97 to 128 bits

(When 144 bits)

0000 000E	<u>FFFFFFFFF</u>	<u>FFFFFFFFF</u>	<u>FFFFFFFFF</u>	<u>FFFFFFFFF</u>
	1 to 36 bits	37 to 72 bits	73 to 108 bits	109 to 144 bits

The data from the specified START ADDRESS to the largest address to be decided by the AFM ADDRESS or AFM EXP ADDRESS of the user specified test plan program, continues to be output in hexadecimal for 16 addresses (8 addresses for 32 or 36 bits, 4 addresses for 64 or 72 bits, 2 addresses for 128 or 144 bits) in one line. When data for up to the maximum address has been output, the data output restarts from address 0.

For the VKT display, scrolling takes place between lines 3 and 18, when output to LPT, line feeding is carried out every 60 lines. In the latter case, a page No. is output on the right upper part of the printing paper.

Pressing the [B] key stops the output. Restart by pressing the [PF2] key. Pressing the [PF1] key returns the screen display to *** MENU FOR DISPLAY FM MODE ***.

Tip

*When DISPLAY is selected, *** MEMO *** is displayed under VKT. It indicates the status of SELECT FM, DATA TYPE, FORMAT, and OUTPUT specified in the *** MENU FOR DISPLAY FM MODE ***.*

◆ Output of Expected Value Data File (DISPLAY ROM PAT FILE)

Output of Expected Value Data File (.DISPLAY DATA FILE)

Also, data contents of MPAT file of the pattern program in which ROM test pattern data is described can be output. In this case, the following data is output.

- When ROM test pattern data is described in the common part:
Data from the common part is output.

- When ROM test pattern data is not described in the common part:
Data from the module where ROM test pattern data is first described is output.
- When multiple FMAs are described in the ROM test pattern data:
Only data corresponding to the first FMA is output.

```

*** MENU FOR DISPLAY ROM PAT FILE MODE ***
__DISPLAY
.CUSTOMER/FILE NAME           :????:MPA ?????
.DATA TYPE (HEXA/BINARY=H/B)  :H
.FORMAT (VERTICAL/LATERAL=V/L):V
.PAGE PAUSE (YES/NO=Y/N)      :Y
.OUTPUT (VKT/LPT=V/L)         :V

```

*** NOTE ***

PF1 : LAST MENU

.DISPLAY

Data output begins in accordance with the conditions specified using CUSTOMER/FILE NAME, DATA TYPE, FORMAT, PAGE PAUSE, and OUTPUT.

.CUSTOMER/FILE NAME

Specifies the customer names and file names.

➔ For details, refer to [".SEND FILE NAME" on page 50-3](#) of ["Expected Data File Transfer to PM \(.SEND FM\)" on page 50-3](#).

.DATE TYPE(HEXA/BINARY=H/B)

Specifies the type of output data.

Types are hexadecimal number and binary number. Specify these types using H or B.

➔ For details, refer to [".DATA TYPE \(HEXA/BINARY=H/B\)" on page 50-10](#) of ["Output of Fail Store Data \(.DISPLAY FM\)" on page 50-9](#).

.FORMAT(VERTICAL/LATERAL=V/L)

Specifies the format for output data.

There are two types of format, one to proceed an address vertically, the other to proceed an address laterally. Specify these types using V or L.

➔ For details, refer to [".FORMAT\(VERTICAL/LATERAL=V/L\)" on page 50-10](#) of ["Output of Fail Store Data \(.DISPLAY FM\)" on page 50-9](#).

.PAGE PAUSE(YES/NO=Y/N)

Specifies pause or no pause after outputting one page of VKT.

Enter a device number when the message shown below is displayed.

```
PAGE PAUSE (YES/NO=Y/N) = __N
```

Inputs "Y" when pausing for each page (16 lines), and "N" when not making a pause. The default is incorporated in Y.

.OUTPUT (VKT/VKT+LPT/S COMMAND=V/L/S)

Specifies the output equipment.

There are three types of specifications according to VKT, VKT+LPT, and S command. Specify each using V, L, or S.

Output Format

The format depends on the specification of DATA TYPE, FORMAT, and bit size.

The displaying data format when DATA TYPE is HEXA is shown below.

- When VERTICAL is selected (For 1 to 36 bits)

MPAT FILE NAME = Customer name: MPA File name Date Time											
Y	X	BIT=8	Y	X	BIT=8	Y	X	BIT=8	Y	X	BIT=8
00000	00000	AA	00000	00010	AA	00000	00020	AA	00000	00030	AA
00000	00001	55	00000	00011	55	00000	00021	55	00000	00031	55
00000	00002	AA	00000	00012	AA	00000	00022	AA	00000	00032	AA
00000	00003	55	00000	00013	55	00000	00023	55	00000	00033	55
00000	00004	AA	00000	00014	AA	00000	00024	AA	00000	00034	AA
:			:			:			:		
00000	0000E	55	00000	0001E	AA	00000	0002E	AA	00000	0003E	AA
00000	0000F	AA	00000	0001F	55	00000	0002F	55	00000	0003F	55
*** NOTE ***											
'B' : LAST MENU											
'SP' : NEXT PAGE											

When all the data is output, return to *** MENU FOR DISPLAY ROM PAT FILE ***.

Return to *** MENU FOR DISPLAY ROM PAT FILE *** using the [B] key.

When the PAGE PAUSE is 'YES,' proceed to the next page (for 16 lines) using the [SP] key.

When the PAGE PAUSE is 'NO,' the message "SP: NEXT PAGE" is not output.

The data format is the same as that for DISPLAY FM.

When the data output is being processed, the message Just a moment is displayed in the center of the lower part of VKT.

- When LATERAL is selected (For 1 to 18 bits)

MPAT FILE NAME = Customer name: MPA File name Date Time															
Y	X	BIT=8													
00000	00000	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55
00000	00010	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55
00000	00020	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55
00000	00030	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55	AA	55
:															
*** NOTE ***															
		'B' : LAST MENU													
		'SP' : NEXT PAGE													

- When LATERAL is selected (For 32, 36 bits)

MPAT FILE NAME = Customer name : MPA File name Date Time															
Y	X	BIT=36													
00000	00000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000
00000	00008	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000
00000	00010	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000
00000	00018	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000	FFFFFFFF	000000000
:															
*** NOTE ***															
		'B' : LAST MENU													
		'SP' : NEXT PAGE													

When all the data is output, return to *** MENU FOR DISPLAY ROM PAT FILE ***.

Return to *** MENU FOR DISPLAY ROM PAT FILE *** using the [B] key.

When the PAGE PAUSE is 'YES', proceed to the next page (for 16 lines) using the [SP] key.

When the PAGE PAUSE is 'NO,' the message "SP: NEXT PAGE is not output.

The format of the data is the same as that for DISPLAY FM.

When the data output is being processed, the message Just a moment is displayed in the center of the lower part of VKT.

For the VKT, scrolling is performed between line No. 5 and No. 21, and for the LPT, page ejection is performed for every 60 lines. For the LPT, the number of pages is output at the upper right of the printer paper.

50. 1. 1. 2 Usage Examples

Copying Method from K.G.D (known Good Device)

1) Generates a program (main program/MPAT program) to be required for copying from K.G.D.

```

PRO 2418          ; 8 * 128k  MOS MASK ROM

VS1   = 5.000V          ; VCC
IN1   = 4.000V, 0.000V  ; VIH, VIL *1
OUT1  = 1.500V, 1.500V  ; VOH, VOL *2
RATE  = 1000NS
ACLK1 = 50NS
BCLK1 = 100NS
CCLK1 = 950NS
STRB1 = 700NS *1

PD1-10 = IN1, NRZA, ACLK1, <X0-9> ; A0-9
PD11-17 = IN1, NRZA, ACLK1, <Y0-6> ; A10-16
PD20   = IN1, /RZZ, BCLK1, <FL>   ; CE
PC33-40 = OUT1, STRB1, <D0-7>     ; 01 to 08

RESET AFM ALL
AFM ADDRESS = X0-9, Y0-6
AFM BITSIZE = 8
AFM MULTIMODE = SINGLE
AFM ACTION = FT>FMX

SELECT MPAT INHIBIT
MEAS MPAT ROMD *3

PAUSE

END

```



```
MPAT ROMD
```

```
REGISTER
```

```
XMAX = #3FF
```

```
YMAX = #7F
```

```
TPH = #0 *4
```

```
IDX1 = #1FFFFFF
```

```
PC = #0
```

```
START #0
```

```
ST0 : NOP YB<0 TP<TPH
```

```
ST1 : JN11 ST1 R XB<XB+1 YB<YB+1^BX X<XB Y<YB *5
```

```
STPS
```

```
END
```

**1 Set the voltage condition and timing condition with a margin.*

**2 In the ROM data, "1" is copied at high output voltage and "0" is copied at low output voltage on the basis of the V_{OH} voltage.*

**3 The MPAT file must be transferred for execution.*

**4 The expected value pattern is assumed to be set to 0.*

**5 The address data must always be 1 data increment pattern.*

- 2) /FMXFER is activated when the device stops upon receipt of PAUSE. Next, select READ FM through the *** FUNCTION MENU ***, and specify the file name using the MAKE A FILE command through the *** MENU FOR READ MODE ***, then the K.G.D. data of 8*128 K is made into a file.

Transfer to PM

Generates a Program (Main Program) Required for Transfer to FM. The following is an example of a program in which the expected value data file is transferred to PM for which EXP mode is set in the 1st execution and the expected value is generated from PM from and after the 2nd test.

```

PRO SAMPLE S1B8BN4 ; 1M ROM, DUT TOTAL = 8/1ST

:

AFM ADDRESS = X0-9, Y0-6
AFM BITSIZE = 8
AFM MULTIMODE = ALL
AFM ACTION = PM>EXP

IF INIT THEN ENTRY

    SEND FM ROMDAT ----- (a)

EXIT

:

MEAS MPAT PFIL

:

END

```

When the FMXFER utility is used, transfer the file according to the following procedure.

- 1) Stop the device with the PAUSE command in place of the SEND FM statement execution.
- 2) Start /FMXFER.
- 3) Select SEND FM using *** FUNCTION MENU ***.
- 4) Input the file name using *** MENU FOR SEND MODE ***.
- 5) Execute the SEND command.
- 6) Then, the expected value data file is transferred to the PM for which EXP mode is set.

50. 1. 2 Pattern Data Display of DBM Memory

This section describes about the usage of contents display confirmation of DBM test pattern data. Functions described in this section can be used in the system which has DBM.

50. 1. 2. 1 Method of Use

Display contents of DBM memory. (.DISPLAY DBM)

```

***      MENU FOR DISPLAY DBM MODE      ***

__DISPLAY
.SELECT DBMFILE----- :MPA SAMPLE1 PC(#000) *1
.START ADDRESS-----  :#00000
.STOP ADDRESS-----   :#3FFFF
.OUTPUT (VKT/VKT+LPT/S COMMAND=V/L/S)----- : V

***      NOTE      ***
PF1 : LAST MENU

```

*1 The display only appears when DBM test pattern data has been sent to the DBM memory using the SEND DBMR statement.

.DISPLAY

Data output is started according to the conditions specified by .SELECT DBMFILE, .START ADDRESS, .STOP ADDRESS and .OUTPUT.

.SELECT DBMFILE

Selects the pattern data to display when multiple DBM pattern data are sent to the DBM memory. Enter a file name when the message shown below is displayed.

```
SELECT DBMFILE = MPA SAMPLE1 __ SAMPLE2
```

SAMPLE1 refers to the previously selected file name, and SAMPLE2 refers to the current one.

If a file name is changed, enter the new file name using up to exceed 32 characters. If a file name is not changed, enter the CR key. The file specified here is the MPAT file only.

The following message appears when the selected file has not been sent to the DBM memory:

```
ERROR : SPECIFIED FILE IS NOT TRANSFERRED TO THE DBM
```

The PC specification is selected when when the selected file has been sent to the DBM memory: The PC value is input in accordance with the following message:

```

SELECT PC (-1/#000/#010/#020/#030/#040/#050/#060/#070/#080/
          #090/#100/#110/#120/#130/#140/#150/#160/#170/#180/
          #190/#200/#210/#220/#230/#240/#250/#260/#270/#280)
          = -1 *1 __ #100 *2

```

*1 Previously selected PC selection

*2 PC selection to be selected this time

When changing the specification for a selected PC, specifiable PC are displayed in parentheses; the specification is selected from among them. If a file name is not changed, enter the CR key.

Tip

"-1" is displayed if there are no PC specifications.

To select no PC specification, enter "-1."

.START ADDRESS

Input DBM output start address.

Enter a device number when the message shown below is displayed.

```
START ADDRESS = # 5
```

The setting range is as follows. The default is #0.

DBM board capacity	Transfer method	Absence/presence of RDBMA specification	Setting range
64KW	--	--	#0 to #FFFF
256KW	--	--	#0 to #3FFFF
1 MW	--	--	#0 to #FFFFFF
2 MW	SEND DBM	Present	#0 to #FFFFFF
		Absent	#0 to #1FFFFFF
	SEND DBMR	--	Scope of DBM test pattern data selected using SELECT DBMFILE

Tip

START ADDRESS in this case is not DBMA. It is the ADDRESS of hardware-like.

When the specification is out of the setting range, the following message is displayed.

```
ERROR : START ADDRESS RANGE OVER!!
```

.STOP ADDRESS

Input DBM output stop address.

Enter a device number when the message shown below is displayed.

```
STOP ADDRESS = # 100
```

The setting range is as follows. The default is the maximum value available for the system.

DBM board capacity	Transfer method	Absence/presence of RDBMA specification	Setting range
64KW	--	--	#0 to #FFFF
256KW	--	--	#0 to #3FFFF
1 MW	--	--	#0 to #FFFFFF
2 MW	SEND DBM	Present	#0 to #FFFFFF
		Absent	#0 to #1FFFFFF
	SEND DBMR	--	Scope of DBM test pattern data selected using SELECT DBMFILE



Tip

STOP ADDRESS in this case is not DBMA. It is the ADDRESS of hardware-like.

When the specification is out of the setting range, the following message is displayed.

ERROR : STOP ADDRESS RANGE OVER!!

.OUTPUT (VKT/VKT+LPT/S COMMAND=V/L/S)

Select the output device. Enter a device number when the message shown below is displayed.

VKT/VKT+LPT/S COMMAND (V/L/S)=__ L

Input V when the output device is VKT.

Input L when the output devices are VKT and LPT.

Input S if it obeys S command.

When omitted, V is assumed to be input.

◆ .DISPLAY DBM Output Format

DBM pattern data

The data read out from DBM-A(PPAT), DBM-B(PDRE) and DBM-C(PCPE) is converted and displayed as shown in the following table.

Display	Applied pattern (PPAT)	DRE pattern (PDRE)	CPE pattern (PCPE)
0	0	1	0
1	1	1	0
L	0	0	1
H	1	0	1
X	1	0	0
-	For combinations other than those listed above		



Tip

When the DBM bit length is 144 bit, 0 is displayed when the applied pattern (PPAT) is 0, 1 is displayed when it is 1 for bit numbers 96 to 143.

Display format

- When the output specification is V (VKT)

When the DBM contains 80 bits, the display format is as follows.

E-ADRS.	ADRS.	UNIT	ILMODE	2WAY	OUTPUT	= VKT
			D39			D0
			D79			D40
000000	000000	1	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
000001	000000	2	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
000002	000001	1	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
000003	000001	2	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
000004	000002	1	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
000005	000002	2	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
000006	000003	1	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
000007	000003	2	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
*** NOTE ***						
PF1 : LAST MENU			PF2 : FORWARD		PF3 : BACK	

*1 "E-ADRS" corresponds to DBMA, "ADRS" is the hardware address, and "UNIT" is the DBM unit.

Data order of each unit is as follows.

Dn is DBM physical bit No.

D39				D0
↓				↓
0000000000	0000000000	0000000000	0000000000	
0000000000	0000000000	0000000000	0000000000	
↑				↑
D79				D40

When the DBM contains 96 bits, the display format is as follows.

E-ADRS.	ADRS.	UNIT	ILMODE	1WAY	OUTPUT = VKT		
			D49				D0
			D95				D50
000000	000000	1	0000000000	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000	0000000000
000000	000000	2	0000000000	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000	0000000000
000000	000000	3	0000000000	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000	0000000000
000000	000000	4	0000000000	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000	0000000000
000001	000001	1	0000000000	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000	0000000000
000001	000001	2	0000000000	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000	0000000000
000001	000001	3	0000000000	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000	0000000000
000001	000001	4	0000000000	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000	0000000000
*** NOTE ***							
PF1 : LAST MENU		PF2 : FORWARD		PF3 : BACK			

Data order of each unit is as follows.

Dn is DBM physical bit No.

D49					D0
↓					↓
0000000000	0000000000	0000000000	0000000000	0000000000	
	000000	0000000000	0000000000	0000000000	
	↑				↑
	D95				D50

When the DBM contains 144 bits, the display format is as follows.

E-ADRS.	ADRS.	UNIT	ILMODE	IWAY	OUTPUT = VKT									
			D79										D0	D80
000000	000000	1	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
				D143										
			0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
000000	000000	2	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
			0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
000001	000001	1	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
			0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
000001	000001	2	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
			0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
000002	000002	1	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
			0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
000002	000002	2	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
			0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
000003	000003	1	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
			0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
000003	000003	2	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
			0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
			*** NOTE ***											
PF1 : LAST MENU			PF2 : FORWARD				PF3 : BACK							

Data order of each unit is as follows.

Dn is DBM physical bit No.

D79														D0
↓														↓
0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
			0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
			↑											↑
			D143											D80

When the [PF2] key is pressed, the next 8-address data is displayed.

When the [PF3] key is pressed, the data 8-address before is displayed.

For the VKT display, scrolling takes place between lines 4 and 19.

When START ADDRESS>STOP ADDRESS, the displayed data appears in the following order: START ADDRESS.→ maximum specifiable value →#0→STOP ADDRESS.

Pressing the [PF1] key returns the screen display to *** MENU FOR DISPLAY FM MODE ***.

- When the output specification is L (VKT+LPT) or S (S command):

When the DBM contains 80 bits, the display format is as follows.

E-ADRS.	ADRS.	UNIT	ILMODE	2WAY	OUTPUT	= VKT&LPT
			D39			D0
			D79			D40
000000	000000	1	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
000001	000000	2	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
000002	000001	1	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
000003	000001	2	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
000004	000002	1	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
000005	000002	2	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
000006	000003	1	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
000007	000003	2	0000000000	0000000000	0000000000	0000000000
			0000000000	0000000000	0000000000	0000000000
*** NOTE ***						
PF1 : LAST MENU			'B' : BREAK		PF2 : RESET BREAK	

When the DBM contains 96 bits, the display format is as follows.

E-ADRS.	ADRS.	UNIT	ILMODE	1WAY	OUTPUT	= VKT&LPT
			D49			D0
			D95			D50
000000	000000	1	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000
000000	000000	2	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000
000000	000000	3	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000
000000	000000	4	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000
000001	000001	1	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000
000001	000001	2	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000
000001	000001	3	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000
000001	000001	4	0000000000	0000000000	0000000000	0000000000
			000000	0000000000	0000000000	0000000000
*** NOTE ***						
PF1 : LAST MENU			'B' : BREAK		PF2 : RESET BREAK	

When the DBM contains 144 bits, the display format is as follows.

E-ADRS.	ADRS.	UNIT	IIMODE	1WAY	OUTPUT = VKT&LPT										D0
			D79												D80
					D143										
000000	000000	1	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
					0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
000000	000000	2	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
					0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
000001	000001	1	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
					0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
000001	000001	2	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
					0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
000002	000002	1	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
					0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
000002	000002	2	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
					0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
000003	000003	1	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
					0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
000003	000003	2	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
					0000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000	0000000000
PF1 : LAST MENU			*** NOTE ***			'B' : BREAK			PF2 : RESET BREAK						

Data from START ADDRESS to STOP ADDRESS is displayed sequentially.

Scrolling on the screen takes place between lines 4 and 19. In the latter case, a page No. is output on the right upper part of the printing paper.

Pressing the [B] key stops the output. Restart by pressing the [PF2] key.

Pressing the [PF1] key returns the screen display to *** MENU FOR DISPLAY FM MODE ***.

50. 1. 3 Pattern Data Display of CBM Memory

This section describes about the usage of contents display confirmation of CBM test pattern data.

50. 1. 3. 1 Method of Use

Display contents of CBM memory. (.DISPLAY CBM)

```

***      MENU FOR DISPLAY CBM MODE      ***

__DISPLAY
.START ADDRESS----- : #0000
.STOP ADDRESS----- : #7FFF
.OUTPUT (VKT/VKT+LPT/S COMMAND=V/L/S) ----- : V

***      NOTE      ***
PF1 : LAST MENU

```

.DISPLAY

Data output is started according to the conditions specified by .START ADDRESS, .STOP ADDRESS, and .OUTPUT.

.START ADDRESS

Input CBM output start address.

Enter a device number when the message shown below is displayed.

```
START ADDRESS = # __ 5↓
```

The setting range is from #0 to #7FFF. The default is #0.



Tip

START ADDRESS in this case is not CBMA. It is the ADDRESS of hardware-like.

When the specification is out of the setting range, the following message is displayed.

```
ERROR : START ADDRESS RANGE OVER!!
```

.STOP ADDRESS

Input CBM output stop address.

Enter a device number when the message shown below is displayed.

```
STOP ADDRESS = # __ 100↓
```

The setting range is from #0 to #7FFF. The default is #7FFF.



Tip

STOP ADDRESS in this case is not CBMA. It is the ADDRESS of hardware-like.

When the specification is out of the setting range, the following message is displayed.

ERROR : STOP ADDRESS RANGE OVER!!

.OUTPUT (VKT/VKT+LPT/S COMMAND=V/L/S)

Select the output device. Enter a device number when the message shown below is displayed.

VKT/VKT+LPT/S COMMAND (V/L/S)=__ L

Input V when the output device is VKT.

Input L when the output devices are VKT and LPT.

Input S if it obeys S command.

When omitted, V is assumed to be input.

◆ .DISPLAY CBM Output Format

When the output specification is V (VKT)

E-ADRS.	ADRS.	UNIT	ILMODE	2WAY	OUTPUT	= VKT
			D35			D0
0000	0000	1	000000000	000000000	000000000	000000000
0001	0000	2	000000000	000000000	000000000	000000000
0002	0001	1	000000000	000000000	000000000	000000000
0003	0001	2	000000000	000000000	000000000	000000000
0004	0002	1	000000000	000000000	000000000	000000000
0005	0002	2	000000000	000000000	000000000	000000000
0006	0003	1	000000000	000000000	000000000	000000000
0007	0003	2	000000000	000000000	000000000	000000000
0008	0004	1	000000000	000000000	000000000	000000000
0009	0004	2	000000000	000000000	000000000	000000000
000A	0005	1	000000000	000000000	000000000	000000000
000B	0005	2	000000000	000000000	000000000	000000000
000C	0006	1	000000000	000000000	000000000	000000000
000D	0006	2	000000000	000000000	000000000	000000000
000E	0007	1	000000000	000000000	000000000	000000000
000F	0007	2	000000000	000000000	000000000	000000000
*** NOTE ***						
PF1 : LAST MENU			PF2 : FORWARD		PF3 : BACK	

*1 "E-ADRS" corresponds to CBMA^{*2}, "ADRS" is the hardware address, and "UNIT" is the CBM unit.

*2 E-ADRS. displayed here corresponds to CBMA in ECBM mode and 1WAY.

However, in 2WAY and except ECBM mode (normal mode), E-ADRS. is not matched with CBMA.

When the [PF2] key is pressed, the next 16-address data is displayed.

When the [PF3] key is pressed, the data 16-address before is displayed.

For the VKT display, scrolling takes place between lines 3 and 18.

Data displayed when START ADDRESS > STOP ADDRESS

appear in the following order: START ADDRESS.→7FFF→#0→STOP ADDRESS.

Pressing the [PF1] key returns the screen display to *** MENU FOR DISPLAY CBM MODE ***.

When the output specification is L (VKT+LPT) or S (S command):

E-ADRS.	ADRS.	UNIT	ILMODE	2WAY	OUTPUT	= VKT&LPT
			D35			D0
0000	0000	1	000000000	000000000	000000000	000000000
0001	0000	2	000000000	000000000	000000000	000000000
0002	0001	1	000000000	000000000	000000000	000000000
0003	0001	2	000000000	000000000	000000000	000000000
0004	0002	1	000000000	000000000	000000000	000000000
0005	0002	2	000000000	000000000	000000000	000000000
0006	0003	1	000000000	000000000	000000000	000000000
0007	0003	2	000000000	000000000	000000000	000000000
0008	0004	1	000000000	000000000	000000000	000000000
0009	0004	2	000000000	000000000	000000000	000000000
000A	0005	1	000000000	000000000	000000000	000000000
000B	0005	2	000000000	000000000	000000000	000000000
000C	0006	1	000000000	000000000	000000000	000000000
000D	0006	2	000000000	000000000	000000000	000000000
000E	0007	1	000000000	000000000	000000000	000000000
000F	0007	2	000000000	000000000	000000000	000000000
*** NOTE ***						
PF1 : LAST MENU			'B' : BREAK		PF2 : RESET BREAK	

Data from START ADDRESS to STOP ADDRESS is displayed sequentially.

Scrolling on the screen takes place between lines 3 and 18.

Pressing the [B] key stops the output. Restart by pressing the [PF2] key.

Pressing the [PF1] key returns the screen display to *** MENU FOR DISPLAY CBM MODE ***.

50. 1. 4 Restrictions

Table 50-1 Restrictions with Menu Common Items

Item	Menu					
	SEND FM	READ FM	DISPLAY FM	DISPLAY ROM PAT FILE	DISPLAY DBM	DISPLAY CBM
READ AFM MODE		✓	✓			

Item	Menu					
	SEND FM	READ FM	DISPLAY FM	DISPLAY ROM PAT FILE	DISPLAY DBM	DISPLAY CBM
DATA TYPE			✓	✓		
FORMAT			✓	✓		
OUTPUT			✓	✓	✓	✓
CUSTOMER/FILE SPECIFICATION	✓	✓		✓		
SELECT FM		✓	✓			

(Description)

Item of READ AFM is in "READ FM menu" and "DISPLAY FM menu"

The later setting of these two modes is validated.

Similarly for DATA TYPE, FORMAT, OUTPUT and SELECT FM, each item exists in multiple menus. Contents of the latest setting menu is validated.

Customer/file specification exists in "SEND FM menu", "READ FM menu" and "DISPLAY ROM PAT FILE menu". The later setting of customer/file name of "SEND FM menu" and "READ FM menu" is valid for "DISPLAY ROM PAT FILE menu". However, if a file is deleted in "READ FM menu" and enter to "DISPLAY ROM PAT FILE menu", note that the deleted file name is displayed.

50. 2 FMXFER on the T5377, T5376, T5375, T5371, T5336 or T5311

This utility can run on a system which has address failure memory (AFM) and pattern memory (PM) installed.

50. 2. 1 Conversational Mode

- 1) Execute the test plan program specified by the user, and pause the program when the AFM mode setting has been made.

To pause the program, use the PAUSE statement or specify MANUAL and HOLD for the test mode.

- 2) Enter the utility start command from the VKT.

```
/FMXFER ↵
```

The following menu is output. Select the item to execute using the [↑], [↓] keys, and press the Return key.

```
***  FUNCTION MENU  ***

__SEND FM (DISK ---> FM)
.READ FM (FM ---> DISK)
.DISPLAY FM
.DISPLAY DATA FILE

*** NOTE ***
PF1 :  EXIT
```

.SEND FM

Transfers the expected data file to PM where the EXP mode is set. The applicable file is an MPAT file.

.READ FM

This command reads out either expected data stored in PM where EXP mode is set, or fail data stored in AFM, and creates an MPAT file using this data.

.DISPLAY FM

This command outputs data stored in PM or AFM.

.DISPLAY DATA FILE

This command outputs data stored in the contents of the expected value data file created in the FMXFER utility is displayed.

PF1

The [PF1] key is used to exit from this utility, FMXFER.

◆ Expected Data File Transfer to PM (.SEND FM)

This command is used to transfer the MPAT file of the pattern program where the MPAT file created by selecting the .READ FM or the ROM test pattern data is stored, to the PM where EXP mode is set.

```
*** MENU FOR SEND MODE ***
```

```
__SEND FILE NAME
.SEND
```

```
*** NOTE ***
PF1 : LAST MENU
```

.SEND FILE NAME

Select this command when the MPAT file name to be transferred is entered. Enter a customer name and a file name following the message below.

```
CUSTOMER NAME = RMUTL __NEWCST.↓
```

**I "RMUTL" is a current customer name, and "NEWCST" is the name of the customer to be entered this time.*

If a customer name is changed, enter the new customer name using up to 32 characters. If a present customer name is used, press the Return key.

```
FILE NAME = MPA ROMPAT1 __ ROMPAT2
```

**I "MPA ROMPAT1" is the name of the previous file, while "ROMPAT2" is the name of the file to be transferred this time.*

If a file name is changed, enter the new file name using up to exceed 32 characters. If a file name is not changed, press the Return key. The file specified here is the MPAT file only.

.SEND

This command starts transmission to PM.

```
WORKING!
```

When the transmission is completed, the above message is cleared.

PF1:

Pressing the [PF1] causes the system to return to the *** FUNCTION MENU ***.

◆ Creating a File Using Expected Data or Failure Data (.READ FM)

A MPAT file can be created by executing a specified test plan program and reading expected data stored in the PM or AFM data. The MPAT file can be transferred using the SEND FM statement.

The compressed AFM data bits can be read to create a MPAT file. When this MPAT file is transferred using the SEND FM statement, the compressed data is stored in the PM.

```

***      MENU FOR READ MODE      ***

__MAKE A FILE
.READ AFM MODE -----: FT  *1
.AFM DUTSEL  -----: 1  *1
.START ADDRESS -----: X:#0   Y:#0
.STOP ADDRESS -----: X:#3FF  Y:#FFF

***      NOTE      ***
PF1   : LAST MENU
'B'   : BREAK

```

*1 This line is not displayed on the T5336 and T5311.

.MAKE A FILE

This command specifies for which customer and with what file name the MPAT file is created. Enter a customer name and a file name following the message below.

```
CUSTOMER NAME = RMUTL __NEWCSTR_
```

*1 "RMUTL" is a current customer name, and "NEWCSTR" is the name of the customer to be entered this time.

```
FILE NAME = MPA RFIL1 __RFIL2_
```

*1 "MPA ROMPAT1" is the name of the previously created file, while "RFIL2" is the name of the file created this time.

- When the same file name does not occur among the customers specified this time
The following message is displayed and filing begins:

```
WORKING!
```

50. 2 FMXFER on the T5377, T5376, T5375, T5371, T5336 or T5311

After filing is completed, the following message is displayed.

```
COMPLETE NEWCST : MPA RFIL2
```

**I "NEWCST" is a customer name, and "MPA RFIL2" is a file name.*

To interrupt filing, press the [B] key. Then, the following message is displayed and filing is interrupted.

```
BREAK! EXECUTION ABORTED!!?
```

A file until interruption has been created.

- When the same file name already exists among the customers selected this time:

The following message is displayed:

```
**  FILE ALREADY EXIST  **

*  CUSTOMER : RMUTL / FILE NAME : MPA RFIL1

      ---->  DELETE ? *

'Y'   :  YES  (=DELETE)
'N'   :  NO   (=EXIT)

*  Push 'Y' or 'N' key =[ ]*
```

**I "RMUTL" is a customer name, and "MPA RFIL1" is a file name.*

Press [Y] key to delete the MPAT file existing in the currently specified customer. Press [N] key if you do not want to delete it.

When the [N] key is pressed:

The existing MPAT file is not deleted but exits.

When the [Y] key is pressed:

The following message is displayed:

```
DELETE! :: RMUTL : MPA RFIL1      WORKING!
```

**I "RMUTL" is the specified customer name, and "MPA RFIL1" is the customer name to be deleted.*

After deletion, "WORKING !" is cleared and the following message is displayed.

```
DELETE! OK!!      :: RMUTL : MPA RFIL1
```

**I "RMUTL" is a customer name, and "MPA RFIL1" is the name of a deleted file.*

The message above is cleared approximately 2 seconds after the display.



Limitation

Only the current customer (used to start this utility) can be deleted.

If the customer to be deleted is not on the current customer list, the following message is displayed, and no deletion can take place:

```
ERCODE : 3772      Can't! DELETE!!
```

Only the files having the current customer name can be deleted.

If the file can be deleted, the following message is displayed.

```

**  MAKE A FILE  **

*  CUSTOMER:RMUTL /  FILE NAME:MPA RFIL1

      ---->  MAKE A FILE ?

'Y'   :  YES (=MAKE A FILE)
'N'   :  NO  (=EXIT)

*  Push 'Y' or 'N' key =[ ]*
```

*1 "RMUTL" is a customer name, and "MPA RFIL1" is a file name.

- When the [Y] key is pressed:

Filing starts for the customer specified this time with the deleted file name. The operation and message display after filing starts are same as those of item (a) for the

- ["When the same file name does not occur among the customers specified this time" on page 50-37](#) command.

- When the [N] key is pressed:

The file deleted now is not created for the customer specified this time but exits. Only the deletion of the existing MPAT file is possible by selecting the MAKE A FILE command. Note that the only current customer (for starting this utility) can be deleted.



Tip

When compressed failure data is filed, ROM test pattern data is filed according to bit length n_1 in AFM BIT SIZE = n_1, n_2 . However, if n_1 is 32 or 36, a bit length of 36 is applied regardless of n_2 .

READ AFM MODE

Specify the operation mode of AFM to be read. The mode specified by this command must be set by the user-specified test plan program before it can be specified here.

This utility starts according to the AFM ACTION statement specified in the test plan program. When more than one mode is specified in the AFM ACTION statement, the first specified mode is used. Enter a device number when the message shown below is displayed.

```
READ AFM MODE = __ FZ ↵
```

- Specifiable modes

EXP, FT, FZ, FO, AF, BF, MASK

FTSBn, FZSBn, FOSBn, AFSBn, BFSBn, MASKSBn

FTSB(v), FZSB(v), FOSB(v), AFSB(v), BFSB(v), MASKSB(v)

n

1 or 2 can be specified.

v

Can be specified in the range #1 to #3.

.AFM DUTSEL

Specify a device number whose data is stored in the AFM. This utility starts according to the AFM DUTSEL statement. Enter a device number when the message shown below is displayed.

```
AFM DUTSEL = __ 4↵
```

Note that the specifiable range is 1 to 512.

.START ADDRESS

This command is used to set the start address of the expected value data or fail store data to be filed. Enter the command in hexadecimal following the message below.

```
START ADDRESS = X : # __F↵ Y : # __F↵
```

The cursor indicates the X address position during display of the above message.

If no change is made, press the Return key. Then, the cursor indicates the Y address position. If no change is made, press the Return key again.

Note that the default is X :#0 and Y :#0. A value greater than STOP ADDRESS cannot be set.

.STOP ADDRESS

This command is used to set the stop address of the expected value data to be filed. Enter the command in hexadecimal following the message below.

```
STOP ADDRESS = X :# __3F↵ Y :# __ FF↵
```

The cursor indicates the X address position during display of the above message.

If no change is made, press the Return key. Then, the cursor indicates the Y address position. If no change is made, press the Return key.

The default is the maximum of the AFM ADDRESS specified in the test plan program specified by a user. A value larger than the default and smaller than the START ADDRESS cannot be specified.

 Tip

When READ AFM MODE = EXP, the default value is the maximum number of AFM EXP ADDRESS.

PF1:

Pressing the [PF1] causes the system to return to the *** FUNCTION MENU ***.

◆ Output of PM or AFM Data (.DISPLAY FM)

Outputs the contents of AFM or PM specified in the READ AFM mode in ["Creating a File Using Expected Data or Failure Data \(.READ FM\)" on page 50-37](#).

```

***      MENU FOR DISPLAY FM MODE      ***

_DISPLAY
.EXP START ADDRESS----- : X:#0 Y:#0 *1
.DATA TYPE (HEXA/BINARY=H/B)----- : H:
.FORMAT (VERTICAL/LATERAL=V/L) ----- : V
.OUTPUT (VKT/VKT+LPT/S COMMAND=V/L/S)----- : V
.READ AFM MODE----- : EXP
.DBM START ADDRESS ----- : #0
.DBMBASE ----- : 1 *2

***      NOTE      ***
PF1 : LAST MENU

```

*1 On the T5377, T5376, T5375 or T5371, ".START ADDRESS" is displayed.

*2 On the T5336 and T5311, however, this is displayed only when version B of AFM CONT (BGR-023301) has been installed on these systems.

→ For compatibility, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498\)](#).

DISPLAY

Starts outputting the data according to the conditions specified by .EXP START ADDRESS or .START ADDRESS, .DATA TYPE, .FORMAT, .OUTPUT, .READ AFM MODE, .DBM START ADDRESS, and .DBMBASE.

.EXP START ADDRESS or .START ADDRESS

Enter the address to begin output. Enter the first address when the message shown below is displayed. (On the T5377, T5376, T5375 or T5371, "START ADDRESS" is displayed instead.)

```
EXP START ADDRESS = X:# 5J Y:# 100J
```

This setting ranges from 0 to the maximum value of the AFM ADDRESS specified by the user in the test plan program. Its default value is X: #0, Y: #0.

 Tip

When READ AFM MODE = EXP, the available range is from 0 to the maximum value of AFM EXP ADDRESS.

.DATA TYPE (HEXA/BINARY=H/B)

Specifies the type of output data. Input a value according to the following message.

HEXA/BINARY (H/B) = __B

Input "H" for the hexadecimal type. Input "B" for the binary type. When omitted, H is assumed.

 Tip

When "B" (the decimal type) is input, the data type is changed to FORMAT (VERTICAL/LATERAL=V/L) =V unconditionally.

.FORMAT (VERTICAL/LATERAL=V/L)

Input the output data format according to the following message. Enter a device number when the message shown below is displayed.

VERTICAL/LATERAL (V/L) = __ L

Input V when the data is output in the format to advance the address in the vertical direction, or L when it is output in the format to advance the address in the lateral direction. When omitted, V is assumed.

 Tip

When "L" (the address is advanced in the lateral direction) is input, the format is changed to DATA TYPE (HEXA/BINARY) = H unconditionally. In executing the DISPLAY, the VKT screen becomes 132 columns automatically. When execution ends, it returns to 80 columns.

.OUTPUT (VKT/VKT+LPT/S COMMAND=V/L/S)

Select the output device. Enter a device number when the message shown below is displayed.

VKT/VKT+LPT/S COMMAND (V/L/S) = __ L

Input V when the output device is VKT. Input L when the output devices are VKT and LPT. Input S if it obeys S command. When omitted, V is assumed to be input.

READ AFM MODE

Specify the operation mode of AFM to be read. The mode specified by this command must be set by the user-specified test plan program before it can be specified here.

After this utility is started up, the mode written in the AFM ACTION statement of the user-specified test plan program is set.

```
READ AFM MODE =__ EXP↵
```

- Specifiable modes

DBM, EXP, FT, FZ, FO, AF, BF, MASK

FTSBn, FZSBn, FOSBn, AFSBn, BFSBn, MASKSBn

FTSB(v), FZSB(v), FOSB(v), AFSB(v), BFSB(v), MASKSB(v)

n

1 or 2 can be specified.

v

Can be specified in the range #1 to #3.

 Tip

*If DBM is specified, the output start address specified by DBM START ADDRESS becomes valid.
When something other than DBM is specified, the address specified by EXP START ADDRESS (START ADDRESS) will be used.*

.DBM START ADDRESS

This command is effective if DBM is specified in the READ AFM MODE, and specifies an output start address.

Input a value according to the following message.

```
DBM START ADDRESS = #__10↵
```

The available ranges are as follows:

DBM block size	Range
64KW	#0 to #FFFF
16 MW	#0 to #FFFFFF

The default value is #0.

.DBMBASE

This command is used when DBM is specified in the READ AFM MODE, and specifies a DBM-BASE value to be output.

Input a value according to the following message.

```
DBMBASE = #__2↵
```


50. 2 FMXFER on the T5377, T5376, T5375, T5371, T5336 or T5311

The available ranges are shown below.

System name	PM Board Capacity	AFM RATE	DBM block Variable size	Range
T5336 and T5311	144 M bit	LS, MS and HS	64KW	1 to 64
	288 M bit	LS, MS and HS	64KW	1 to 128
	576 M bit	LS, MS and HS	64KW	1 to 256
T5371	576 M bit	LS	64KW	1 to 256
		MS/HS	64KW	1 to 128
T5377, T5376 and T5375	1152 M bit	LS	64KW	1 to 512
			16 MW	1 to 2
		MS/HS	64KW	1 to 256
			16 MW	1

The default value depends on the SET DBMBASE statement in the test plan program.

Output Format

The format will change depending on the specification of DATA TYPE, FORMAT and READ AFM MODE, and bit size.

The displaying data format when DATA TYPE is HEXA is shown below.

- VERTICAL, EXP or failure store mode

Y	X	BIT=8	Y	X	BIT=8	Y	X	BIT=8	Y	X	BIT=8
00000	00000	AA	00000	00010	AA	00000	00020	AA	00000	00030	AA
00000	00001	55	00000	00011	55	00000	00021	55	00000	00031	55
00000	00002	AA	00000	00012	AA	00000	00022	AA	00000	00032	AA
00000	00003	55	00000	00013	55	00000	00023	55	00000	00033	55
00000	00004	AA	00000	00014	AA	00000	00024	AA	00000	00034	AA
:			:			:			:		
00000	0000E	55	00000	0001E	AA	00000	0002E	AA	00000	0003E	AA
00000	0000F	AA	00000	0001F	55	00000	0002F	55	00000	0003F	55
*** NOTE ***			*	*** MEMO ***							
			I								
PF1 : LAST MENU			I	DATA TYPE = HEXA							
PF2 : NEXT PAGE			I	FORMAT = VERTICAL							
			*	OUTPUT = VKT							

50. 2 FMXFER on the T5377, T5376, T5375, T5371, T5336 or T5311

*1 "Y" is the Y address (hexadecimal), "X" is the X address (hexadecimal), and "BIT=8" is the number of data bits (decimal number; displayed as the number of bits specified using AFM BIT-SIZE*2).

*2 AFM BITSIZE= n_1, n_2 of $_2(EXP \text{ mode is } n_1)$ when specified using the SELECT COMPRESS ALL or AFM statement.

For SELECT COMPRESS CFM, AFM BITSIZE=the number of bits specified using n_1 of n_1 and n_2 .

- When VERTICAL and DBM is selected

ADRS	BIT=36	ADRS	BIT=36	ADRS	BIT=36	ADRS	BIT=36
000000	000000000	000010	000000000	000020	000000000	000030	000000000
000001	000040001	000011	000040001	000021	000040001	000031	000040001
000002	000080002	000012	000080002	000022	000080002	000032	000080002
:	:	:	:	:	:	:	:
00000E	000400040	00001E	000400040	00002E	000400040	00003E	000400040
00000F	000800080	00001F	000800080	00002F	000400040	00003F	000800080
*** NOTE ***		*	*** MEMO ***				
		I					
PF1 : LAST MENU		I	DATA TYPE = HEXA				
PF2 : NEXT PAGE		I	FORMAT = VERTICAL				
		*	OUTPUT = VKT				

Pressing the [PF1] key returns the screen display to *** MENU FOR DISPAY FM MODE ***. When [PF2] is pressed, the next 32-address data is output. When data for up to the maximum address has been output, the data output restarts from address 0.

The bit configuration of the output data is as follows:

(Example) When BIT = 8

BIT=1 2 3 4 5 6 7 8
 ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
 AA(16 hexadecimal)=1 0 1 0 1 0 1 0 (binary)

50. 2 FMXFER on the T5377, T5376, T5375, T5371, T5336 or T5311

- LATERAL, EXP or failure store mode (For up to 18 bits)

Y	X	BIT=8						
00000	00000	AA	55	AA	55	-----	AA	55
00000	00010	AA	55	AA	55	-----	AA	55
00000	00020	AA	55	AA	55	-----	AA	55
00000	00030	AA	55	AA	55	-----	AA	55
:								
*** NOTE ***		*	*** MEMO ***					
		I						
PF1 : LAST MENU		I	DATA TYPE = HEXA					
'B' : BREAK		I	FORMAT = LATERAL					
PF2 : RESET BREAK		*	OUTPUT = VKT					

- LATERAL, EXP or failure store mode (For 32, 36 bits)

Y	X	BIT=36						
00000	00000	FFFFFFFF	000000000	FFFFFFFF	-----	FFFFFFFF	000000000	
00000	00008	FFFFFFFF	000000000	FFFFFFFF	-----	FFFFFFFF	000000000	
00000	00010	FFFFFFFF	000000000	FFFFFFFF	-----	FFFFFFFF	000000000	
00000	00018	FFFFFFFF	000000000	FFFFFFFF	-----	FFFFFFFF	000000000	
:								
*** NOTE ***		*	*** MEMO ***					
		I						
PF1 : LAST MENU		I	DATA TYPE = HEXA					
'B' : BREAK		I	FORMAT = LATERAL					
PF2 : RESET BREAK		*	OUTPUT = VKT					

- When LATERAL, DBM are selected

ADRS BIT=36

```
000000 000040001 000080002 000100004 ----- 002000080
000008 004000100 008000200 010000400 ----- 200008000
000010 000040001 0000C0003 0001C0007 ----- 001FC007F
000018 003FC00FF 007FC01FF 00FFC03FF ----- 3FFFCFFFF
000020 000000000 FFFFFFFF 000000000 ----- FFFFFFFF
000028 000000000 FFFFFFFF 000000000 ----- FFFFFFFF
```

:

*** NOTE ***

*

*** MEMO ***

I

PF1 : LAST MENU

I

DATA TYPE = HEXA

'B' : BREAK

I

FORMAT = LATERAL

PF2 : RESET BREAK

*

OUTPUT = VKT

When READ AFM MODE is not DBM:

The AFM data from the start address (determined by EXP START ADDRESS (or START ADDRESS on the T5377, T5376, T5375 or T5371)) to the maximum address (determined by the AFM ADDRESS statement (or the AFM EXP ADDRESS statement in EXP mode) in the test plan program) is output in hexadecimal, for 16 addresses per line (or 8 addresses in 32 or 36 bit mode).

When READ AFM MODE is DBM:

Data from the start address specified by DBM START ADDRESS to #FFFF (or #FFFFFF when the DBM block size is 16 MW) is output in hexadecimal, for 8 addresses per line.

When data for up to the maximum address has been output, the data output restarts from address 0.

For the VKT display, scrolling takes place between lines 3 and 18, when output to LPT, line feeding is carried out every 60 lines. In the latter case, a page No. is output on the right upper part of the printing paper.

Pressing the [B] key stops the output.

Restart by pressing the [PF2] key.

Pressing the [PF1] key returns the screen display to *** MENU FOR DISPLAY FM MODE ***.



Tip

When DISPLAY is selected, *** MEMO *** is displayed under VKT. It indicates the status of DATA TYPE, FORMAT, and OUTPUT specified in the *** MENU FOR DISPLAY FM MODE ***.

◆ It Indicates the Status of DATA TYPE, FORMAT, and OUTPUT Specified in the *** MENU FOR DISPLAY MODE ***

Output of Expected Value Data File (.DISPLAY DATA FILE)

```
*** MENU FOR READ DATA FILE MODE ***
  _DISPLAY
  .CUSTOMER/FILE NAME-----:????=MPA ?????
  .DATA TYPE (HEXA/BINARY=H/B) -----:H
  .FORMAT (VERTICAL/LATERAL=V/L) -----:V
  .PAGE PAUSE (YES/NO=Y/N) -----:Y
  .OUTPUT (VKT/LPT=V/L) -----:V

*** NOTE ***
PF1 : LAST MENU
```

DISPLAY

Data output begins in accordance with the conditions specified using CUSTOMER/FILE NAME, FORMAT, PAGE PAUSE, and OUTPUT.

CUSTOMER/FILE NAME

Specifies the customer names and file names.

➔ For details, refer to [".SEND FILE NAME" on page 50-36](#) of ["Expected Data File Transfer to PM \(.SEND FM\)" on page 50-36](#).

DATE TYPE(HEXA/BINARY=H/B)

Specifies the type of output data.

Types are hexadecimal number and binary number. Specify these types using H or B.

➔ For details, refer to [".DATA TYPE \(HEXA/BINARY=H/B\)" on page 50-42](#) of ["Output of PM or AFM Data \(.DISPLAY FM\)" on page 50-41](#).

FORMAT(VERTICAL/LATERAL=V/L)

Specifies the format for output data.

There are two types of format, one to proceed an address vertically, the other to proceed an address laterally. Specify these types using V or L.

➔ For details, refer to [".FORMAT\(VERTICAL/LATERAL=V/L\)" on page 50-42](#) of ["Output of PM or AFM Data \(.DISPLAY FM\)" on page 50-41](#).

PAGE PAUSE(YES/NO=Y/N)

Specifies pause or no pause after outputting one page of VKT.

Enter a device number when the message shown below is displayed.

```
PAGE PAUSE (YES/NO=Y/N) = __N
```

Inputs "Y" when pausing for each page (16 lines), and "N" when not making a pause.

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The default is incorporated in Y.

OUTPUT (VKT/VKT+LPT/S COMMAND=V/L/S)

Specifies the output equipment.

There are three types of specifications according to VKT, VKT+LPT, and S command. Specify each using V, L, or S.

Output Format

- When VERTICAL is selected

MPAT FILE NAME = Customer name: MPA File name Date Time											
Y	X	BIT=8	Y	X	BIT=8	Y	X	BIT=8	Y	X	BIT=8
00000	00000	AA	00000	00010	AA	00000	00020	AA	00000	00030	AA
00000	00001	55	00000	00011	55	00000	00021	55	00000	00031	55
00000	00002	AA	00000	00012	AA	00000	00022	AA	00000	00032	AA
00000	00003	55	00000	00013	55	00000	00023	55	00000	00033	55
00000	00004	AA	00000	00014	AA	00000	00024	AA	00000	00034	AA
00000	00005	55	00000	00015	55	00000	00025	55	00000	00035	55
:			:			:			:		
00000	0000E	AA	00000	0001E	AA	00000	0002E	AA	00000	0003E	AA
00000	0000F	55	00000	0001F	55	00000	0002F	55	00000	0003F	55
*** NOTE ***											
'B' : LAST MENU											
'SP' : NEXT PAGE											

When all the data is output, return to *** MENU FOR READ DATA FILE ***.

Return to *** MENU FOR READ DATA FILE *** using the [B] key.

When the PAGE PAUSE is 'YES,' proceed to the next page (for 16 lines) using the [SP] key.

When the PAGE PAUSE is 'NO,' the message 'SP': NEXT PAGE is not output.

The data format is the same as that for DISPLAY FM.

When the data output is being processed, the message Just a moment is displayed in the center of the lower part of VKT.

50. 2 FMXFER on the T5377, T5376, T5375, T5371, T5336 or T5311

- When LATERAL is selected (For up to 18 bits)

MPAT FILE NAME = Customer name: MPA File name Date Time									
Y	X	BITSIZE=8							
00000	00000	AA	55	AA	55	AA	-----	AA	55
00000	00010	AA	55	AA	55	AA	-----	AA	55
00000	00020	AA	55	AA	55	AA	-----	AA	55
00000	00030	AA	55	AA	55	AA	-----	AA	55
00000	00040	AA	55	AA	55	AA	-----	AA	55
:									
*** NOTE ***									
'B' : LAST MENU									
'SP' : NEXT PAGE									

- When LATERAL is selected (For 32, 36 bits)

MPAT FILE NAME = Customer name: MPA File name Date Time									
Y	X	BITSIZE=36							
00000	00000	FFFFFFFF	000000000	FFFFFFFF	000000000	----	FFFFFFFF	000000000	
00000	00008	FFFFFFFF	000000000	FFFFFFFF	000000000	----	FFFFFFFF	000000000	
00000	00010	FFFFFFFF	000000000	FFFFFFFF	000000000	----	FFFFFFFF	000000000	
:									
*** NOTE ***									
'B' : BREAK									
'SP' : NEXT PAGE									

When all the data is output, return to *** MENU FOR READ DATA FILE ***.

Return to *** MENU FOR READ DATA FILE *** using the [B] key.

When the PAGE PAUSE is 'YES,' proceed to the next page (for 16 lines) using the [SP] key.

When the PAGE PAUSE is 'NO', the message 'SP': NEXT PAGE is not output.

The format of the data is the same as that for DISPLAY FM.

When the data output is being processed, the message Just a moment is displayed in the center of the lower part of VKT.

For the VKT, scrolling is performed between line No. 5 and No. 21, and for the LPT, page ejection is performed for every 60 lines. For the LPT, the number of pages is output at the upper right of the printer paper.

50. 2. 2 Restrictions

Table 50-2 Restrictions with Menu Common Items

Item	Menu			
	SEND FM	READ FM	DISPLAY FM	DISPLAY DATA FILE
READ AFM MODE		✓ (Only T5377, T5376, T5375 or T5371)	✓	
DATA TYPE			✓	✓
FORMAT			✓	✓
OUTPUT			✓	✓
CUSTOMER/FILE SPECIFICATION	✓	✓		✓

(Description)

The item of DATA TYPE, FORMAT, and OUTPUT is present in "DISPLAY FM MENU" and "DISPLAY DATA FILE MENU".

For the T5377, T5376, T5375 or T5371, READ AFM MODE is located in the READ FM and DISPLAY FM menus.

These items are enabled in the last used menu.

Although customer and file names can be specified in the SEND FM, READ FM or DISPLAY DATA FILE menus, only the customer and file name specified last in either the SEND FM or READ FM menu is applied to the DISPLAY DATA FILE menu. Even though a file is deleted in the READ FM menu, it is still displayed in the DISPLAY DATA FILE menu.

50. 2. 3 Usage Examples for Transfer to FM

Generates a Program (Main Program) Required for Transfer to FM. The following is an example of a program in which the expected value data file is transferred to PM for which EXP mode is set in the 1st execution and the expected value is generated from PM from and after the 2nd test.

```

PRO SAMPLE S1B8BN4 ; 1M ROM, DUT TOTAL = 8/1ST

:

AFM ADDRESS = X0-9, Y0-6
AFM BITSIZE = 8
AFM MULTIMODE = SINGLE
AFM ACTION = PM>EXP

IF INIT THEN ENTRY

    SEND FM ROMDAT ----- (a)

EXIT

:

MEAS MPAT PFIL

:

END

```

When the FMXFER utility is used, transfer the file according to the following procedure.

- 1) Stop the device with the PAUSE command in place of the SEND FM statement execution.
- 2) Start /FMXFER.
- 3) Select SEND FM using *** FUNCTION MENU ***.
- 4) Input the file name using *** MENU FOR SEND MODE ***.
- 5) Execute the SEND command.
- 6) Then, the expected value data file is transferred to the PM for which EXP mode is set.

51. Random Test Number Mode (*TRANDOM*)

To operate a program in which the test number is specified randomly, the random test number mode has been created.

Previously, the test number have had to be numbered in ascending order. See sample below.

```
PRO SAMPLE

    GOTO TEST 30

TEST 10

TEST 20

TEST 50  ← Branches to TEST 50.

TEST 30

TEST 40
```

The program above compares the TEST statement number with the branch destination (GOTO) number.

In a case like the example above, a mode allowing branching to TEST 30 is prepared.

```
PRO SAMPLE

    CALL TRANDOM  ← insert this line at the beginnings.

    GOTO TEST 30

TEST 10

TEST 20

TEST 50

TEST 30  ← Branches to TEST 30.

TEST 40
```

In the above sample, the addition of the CALL TRANDOM statement line which switches the program to TRANDOM mode causes the program to compare the TEST statement number with the branch destination (GOTO) number, and start testing only when they are equal.

When using the TRANDOM mode, the START TEST NO. and STOP TEST NO. settings perform the same function as the GOTO statement.

In other words, size is not compared; start test number and stop test numbers are compared with the test statement numbers.

Tip

Handling test number 0 (zero)

Zero can be prepared as a testing option.

The program can branch to TEST 0 (zero) using the GOTO TEST statement.

It is not possible to start from 0 (zero) using START TEST NO. or to be stopped with 0 (zero) using STOP TEST NO. This is why setting the start test number and the stop test number to 0 is regarded as no specification.

51. 1 Using TRANDOM

TRANDOM is used in the same way as the normal subroutine in the test plan program.

Syntax

```
CALL TRANDOM
```

Function

The GOTO TEST statement is executed or START TEST NO. and STOP TEST NO. are set only when the test number and the branch destination number specified by the GOTO TEST statement or START TEST NO. and STOP TEST NO. are equal.

This function is available only during the test plan program execution after the "CALL TRANDOM" statement line has been entered.

52. Category and Sort Data Output Program (lcommand)

This command is a utility used in a system comprised of handler and tester to output the contents of a file that contains the category and sort data of matrix measurement and remeasurement totaled in RETEST MODE 2 (selected by /STHAND) according to the format of controller command "L".

52. 1 Syntax

```
lcommand [-f file_name]  
lcommand [directory_name]
```

file-name

Specify the data file name.

directory_name

Specify the directory where the saved category and sort data file exists.

(If omitted, the current directory is assumed.)

Limitation

-f file-name and directory-name cannot be specified simultaneously.

52. 2 Method of Operation

When "-f file-name" is not specified

```
0 : exit
1 : display main test data
2 : display retest1 data
3 : display retest2 data
command ? d
```

The data to be displayed is selected by the command specified for d.

```
0    EXIT
1    Data of matrix measurement
2    Data of 1st remeasurement
3    Data of 2nd remeasurement
```

When data is selected, the following menu appears.

```
LIST UP COUNTERS COMMAND MENU
 1 : STATION1 DUTS COMBINED
 2 : STATION2 DUTS COMBINED
CA : ALL DUTS COMBINED
C1 : STATION1 DUTS COMBINED
C2 : STATION2 DUTS COMBINED
S1 : STATION1 DUTS SEPARATED
S2 : STATION2 DUTS SEPARATED
OF : OUTPUT FORMAT (k)
command ? n
```

The contents of various counters are listed by the command specified for n.

There are following types of counters that are listed:

- Total/Pass/Fail counter
- Category counter
- Sort counter

The contents listed here are determined by the value of command 'n' as shown below.

```
1/C1    The integrated value of all DUTs in station 1 is displayed.
2/C2    The integrated value of all DUTs in station 2 is displayed.
CA       The integrated value of all DUTs is displayed.
S1       The contents of all DUTs in station 1 are independently displayed.
S2       The contents of all DUTs in station 2 are independently displayed.
OF       The display formats of counters in 1 to S2 above are changed.
```

When the value of 'n' = OF, the display formats of category and sort counters in 1 to S2 above are changed.

```
SELECT OUTPUT FORMAT, ALL (AL) EXTRACT (EX) OR EXTRACT2 (E2) ?
```

Following three display formats are available to choose from.

- | | |
|----|---|
| AL | The category and sort counters are always displayed. |
| EX | The category counter is divided into blocks each consisting of 50 data entries (1 to 50, 51 to 100, ..., 900 to 950, and 951 to 999). If all counter values in a block are 0 s, the data of that block are not displayed. The sort counter is divided into blocks each consisting of 8 data entries (1 to 8 and 9 to 16). If all counter values in a block are 0 s, the data of that block are not displayed. |
| E2 | The category and sort counters are not displayed if the counter values of data in one horizontal row all are 0 s. |

When "-f file-name" is specified

```
LIST UP COUNTERS COMMAND MENU
C  : DUTS COMBINED
S  : DUTS SEPARATED
OF : OUTPUT FORMAT (EXTRACT2)
command ? m
```

The contents of various counters are listed by the command specified for m.

The listed counters are the same as when "-f file_name" is not specified.

The contents listed here are determined by the value of command 'm' as shown below.

- | | |
|----|---|
| C | The integrated value of all DUTs in a specified file |
| S | The contents of all DUTs in a specified file are independently displayed. |
| OF | The display formats of counters in C and S above are changed. |

The function of OF is the same as when "-f file-name" is not specified.

52. 3 Displayed File Names

When "-f file-name" is not specified

The following files are read in for data display on the screen according to the selected menu commands 'd' and 'n'.

d	n	Files read in
1	1/C1/S1	mtestd1.dat
1	2/C2/S2	mtestd2.dat
1	CA	mtestd1.dat, mtestd2.dat
2	1/C1/S1	rtestd11.dat
2	2/C2/S2	rtestd21.dat
2	CA	rtestd11.dat, rtestd21.dat
3	1/C1/S1	rtestd12.dat
3	2/C2/S2	rtestd22.dat
3	CA	rtestd12.dat, rtestd22.dat



Tip

The files to be read are searched for in a specified directory.

If no directory is specified, the current directory is searched.

If no file to be read is found, an error is assumed.

However, if when CA is selected one of two files is found, the file is read and the data in only that file is displayed.

When "-f file-name" is specified

The specified file is read and its data is displayed on the screen.

52. 4 File Output

The contents of listed data can be saved to a file.

This can be accomplished by specifying a file name after the startup command of lcommand as shown below. The listed contents will be output to the specified file.

```
lcommand -f data_file > log_file  
lcommand -f directory_name > log_file
```


52. 5 Output Examples

The following shows an output example when STATION1 COMBINED in matrix measurement is selected.

```

LIST UP COUNTER    AT 1998/08/19 15:29:55 (SAVED: 1997/07/20 14:57)
  LOT NO.: 0        PBID: #0000    PRO NAME: GPRTCR02
  STATION: 1        DUT: 1-32
  COMMENT:
    TOTAL = 32      PASS = 0 ( 0.00%)      FAIL = 32 (100.00%)
CAT-001 CAT-002 CAT-003 CAT-004 CAT-005 CAT-006 CAT-007 CAT-008 CAT-009 CAT-010
    1      1      1      1      1      1      1      1      1      1
  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%

CAT-011 CAT-012 CAT-013 CAT-014 CAT-015 CAT-016 CAT-017 CAT-018 CAT-019 CAT-020
    1      1      1      1      1      1      1      1      1      1
  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%

CAT-021 CAT-022 CAT-023 CAT-024 CAT-025 CAT-026 CAT-027 CAT-028 CAT-029 CAT-030
    1      1      1      1      1      1      1      1      1      1
  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%

CAT-031 CAT-032 CAT-033 CAT-034 CAT-035 CAT-036 CAT-037 CAT-038 CAT-039 CAT-040
    1      1      1      1      1      1      1      1      1      1
  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%

CAT-041 CAT-042 CAT-043 CAT-044 CAT-045 CAT-046 CAT-047 CAT-048 CAT-049 CAT-050
    1      1      1      1      1      1      1      1      1      1
  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%

CAT-051 CAT-052 CAT-053 CAT-054 CAT-055 CAT-056 CAT-057 CAT-058 CAT-059 CAT-060
    1      1      1      1      1      1      1      1      1      1
  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%

CAT-061 CAT-062 CAT-063 CAT-064 CAT-065 CAT-066 CAT-067 CAT-068 CAT-069 CAT-070
    1      1      1      1      1      1      1      1      1      1
  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%

CAT-071 CAT-072 CAT-073 CAT-074 CAT-075 CAT-076 CAT-077 CAT-078 CAT-079 CAT-080
    1      1      1      1      1      1      1      1      1      1
  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%

CAT-081 CAT-082 CAT-083 CAT-084 CAT-085 CAT-086 CAT-087 CAT-088 CAT-089 CAT-090
    1      1      1      1      1      1      1      1      1      1
  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%

CAT-091 CAT-092 CAT-093 CAT-094 CAT-095 CAT-096 CAT-097 CAT-098 CAT-099 CAT-100
    1      1      1      1      1      1      1      1      1      1
  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%  3.125%

```

CAT-101	CAT-102	CAT-103	CAT-104	CAT-105	CAT-106	CAT-107	CAT-108	CAT-109	CAT-110
1	1	1	1	1	1	1	1	1	1
3.125%	3.125%	3.125%	3.125%	3.125%	3.125%	3.125%	3.125%	3.125%	3.125%
CAT-111	CAT-112	CAT-113	CAT-114	CAT-115	CAT-116	CAT-117	CAT-118	CAT-119	CAT-120
1	1	1	1	1	1	1	1	1	1
3.125%	3.125%	3.125%	3.125%	3.125%	3.125%	3.125%	3.125%	3.125%	3.125%
CAT-121	CAT-122	CAT-123	CAT-124	CAT-125	CAT-126	CAT-127	CAT-128	CAT-129	CAT-130
1	1	1	1	1	1	1	1	0	0
3.125%	3.125%	3.125%	3.125%	3.125%	3.125%	3.125%	3.125%	0.000%	0.000%
CAT-131	CAT-132	CAT-133	CAT-134	CAT-135	CAT-136	CAT-137	CAT-138	CAT-139	CAT-140
0	0	0	0	0	0	0	0	0	0
0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%
CAT-141	CAT-142	CAT-143	CAT-144	CAT-145	CAT-146	CAT-147	CAT-148	CAT-149	CAT-150
0	0	0	0	0	0	0	0	0	0
0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%
:	:	:	:	:	:	:	:	:	:
:	:	:	:	:	:	:	:	:	:
CAT-981	CAT-982	CAT-983	CAT-984	CAT-985	CAT-986	CAT-987	CAT-988	CAT-989	CAT-990
0	0	0	0	0	0	0	0	0	0
0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%
CAT-991	CAT-992	CAT-993	CAT-994	CAT-995	CAT-996	CAT-997	CAT-998	CAT-999	
0	0	0	0	0	0	0	0	0	
0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	
SORT-01	SORT-02	SORT-03	SORT-04	SORT-05	SORT-06	SORT-07	SORT-08		
1	2	3	4	5	6	7	4		
3.125%	6.250%	9.375%	12.500%	15.625%	18.750%	21.875%	12.500%		

53. Limiter Fail Search Utility (LFSEARCH)

If a limiter failure occurs during device measurement, all DUTs fail in the test even when the device has faults.

However, the test for the DUTs where limiter failures do not occur can be executed by searching for the DUTs with limiter failures and by removing the DUTs from the test targets using the REJECTION statement.

This utility searches for the DUTs where limiter failures occurred and identifies the DUTs connected to the same child.

```

PRO      1

      .
      .
      ON DUTIF      :DERR
      .
      .
      .

DERR:
LOGI[2]  LFAIL(2)=0
        ECOD=0
        CALL LFSEARCH(LFAIL,ECOD)
        IF ECOD=0 THEN ENTRY
      .
      .
      .
      EXIT
      ONRTN

END

```

Tip

- This utility does not carry out DUT rejection processing.
- When re-testing a device after the rejection processing, measure the device after executing the RESET TESTER statement and setting measurement conditions.
- This utility cannot be used with any systems other than the T5581, T5581H, T5581P, T5571P, T5311, T5336 and T5371.

53. 1 Syntax

This utility calls the subroutine after detecting a limiter failure with ON DUTIF and other commands.

```
CALL LFSEARCH(v,e)
```

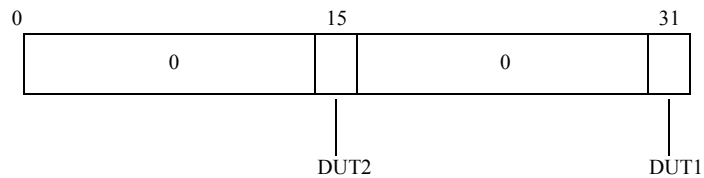
V

A variable for storing a failed DUT.

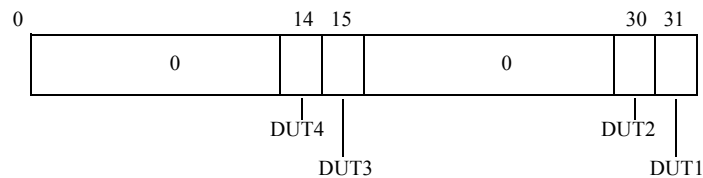
Specify using a logical variable.

A failed DUT format depends on device programs as shown below.

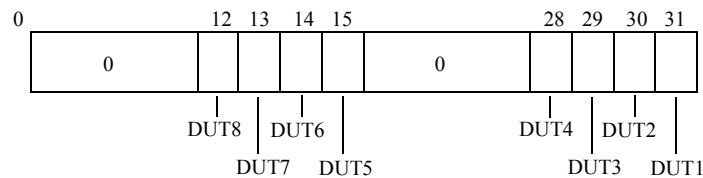
- When one DUT in one station, or two DUTs in two stations:



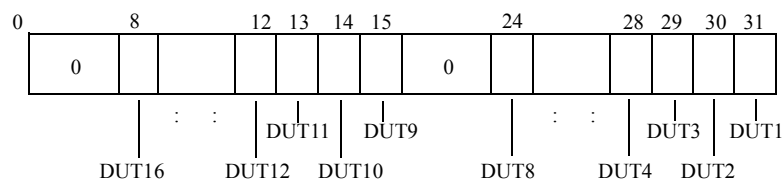
- When two DUTs in one station, or four DUTs in two stations:



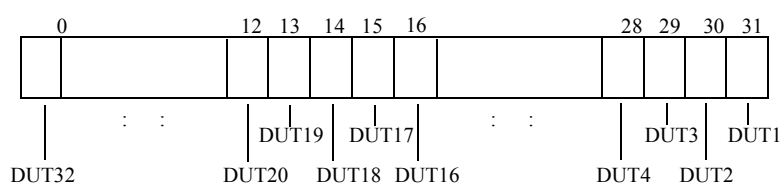
- When four DUTs in one station, or eight DUTs in two stations:



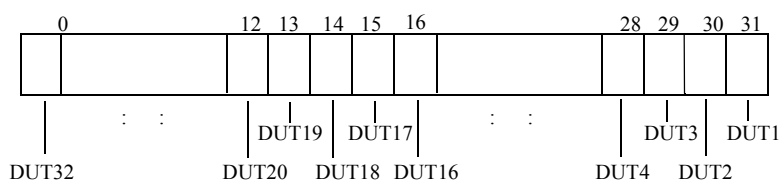
- When eight DUTs in one station, or 16 DUTs in two stations:



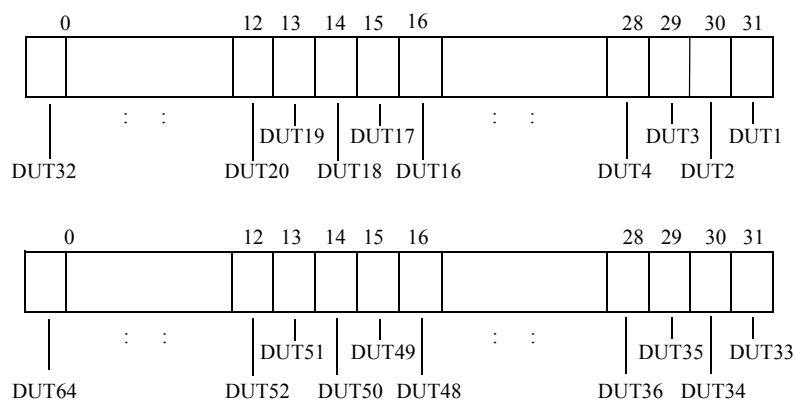
- When 16 DUTs in one station, or 32 DUTs in two stations:



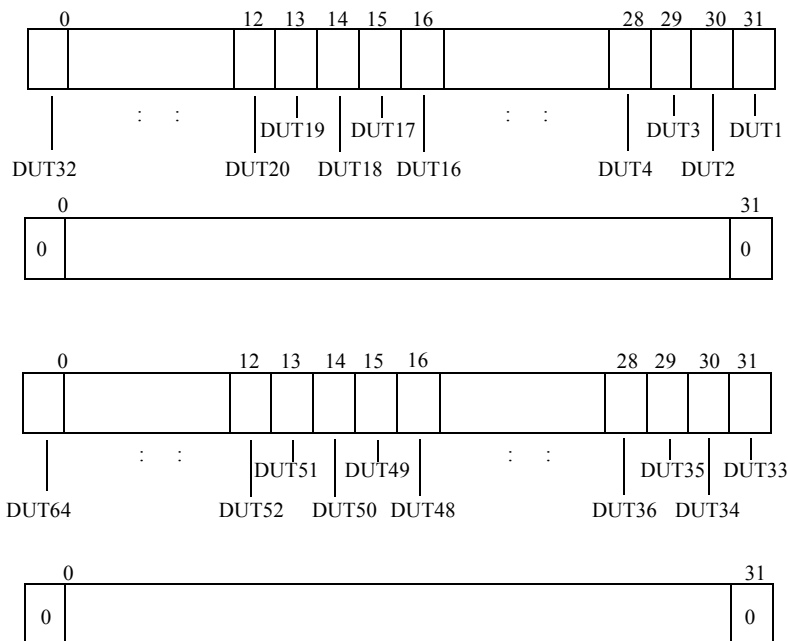
- When 32 DUTs in one station:



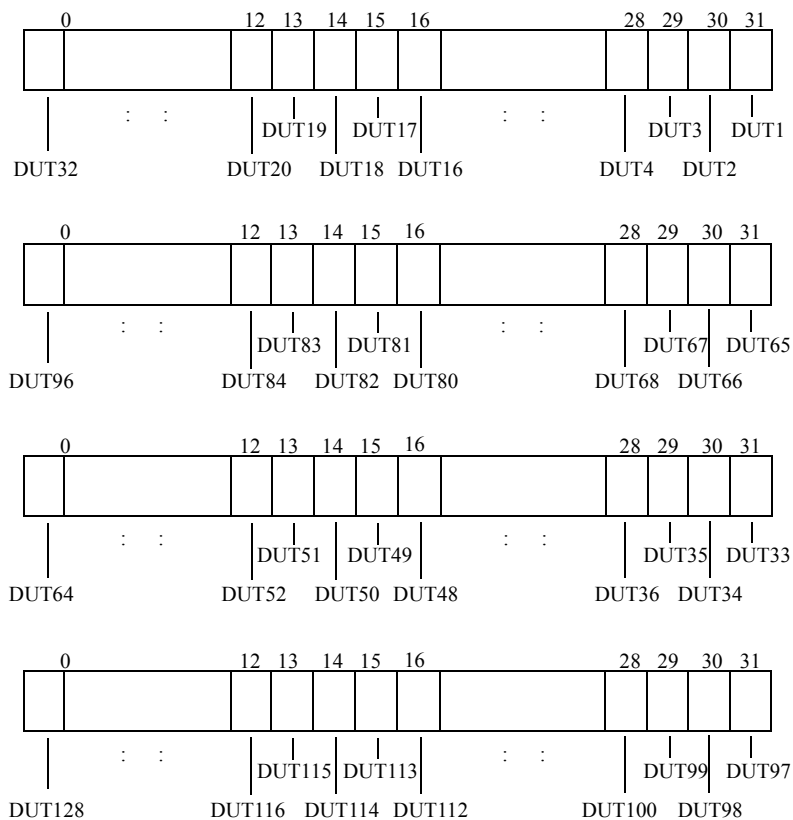
- When 64 DUTs in two stations:



- When 64 DUTs in one station:



- When 128 DUTs in two stations:



e

A variable for storing an error code

Error Codes	Description
0	Normal end
1	On other than the T5581, T5581H, T5581P, T5571P, T5311, T5336 and T5371

54. *OVERRIDE Program*

The utility program **OVERRIDE** temporarily disables a branch specified with the **ON** or **ONCALL** statement.

Normally, the test item failure is processed by branching with the **ON** or **ONCALL** statement, however, if any failures occur, the test may be terminated. If a test item failure occurs during debugging, the subsequent items cannot be debugged. At this time, to debug all test items, the user can use the utility program that disables the **ON** or **ONCALL** branch.

54. 1 Syntax

The following command must be entered during execution of the controller.

```
/OVERRIDE [mode] [event-name1 event-name n]↵
```

mode

Disables or releases the **ON** or **ONCALL** branch.

- **ON**
Disables the **ON** branch at the specified event-name. After this specification, the **ON** branch is not executed.
- **ONCALL**
Disables the **ONCALL** branch at the specified event-name. After this specification, **ONCALL** is not executed.
- **OFF**
Releases prohibition of the **ON** and **ONCALL** branches at the specified event-name. After this specification, the **ON** and **ONCALL** branches are executed.
- **LIST**
Displays the event-names disabling the **ON** and **ONCALL** branches.
- **Default**
Assumed **ON** is specified.

event-name

The following can be specified similar to the **ON** statement.

ERR (or **ERROR**), **STOP**, **TEST**, **FCT**, **DCT**, **OVERFLOW**, **UNDERFLOW**, **ZERODIVIDE**, **BTEST**, **FCTF**, **DCTF**, **KEY**, **TIME**, **DUTIF**, **BFCT**, **BDCT**, **WCNT**, **TRIP**, **SRQ0**, **SRQ1**, **GPSRQ**, **ALL**

When **ALL** is specified, all events are disabled.

When both mode and event-name are omitted, operation is equivalent to the following:

```
/OVERRIDE ON FCTF DCTF↵
```

event-name is considered to have been specified with mode OFF.

The specified items are equivalent to the following:

```
/OVERRIDE OFF ALL↵
```

When ON or ONCALL is specified in mode and event-name is omitted, no event-name is specified.

54. 2 Usage Examples

When disabling the ON branch for FCTF and DCTF

```
/OVERRIDE.↓
```

When disabling the ONCALL branch for STOP

```
/OVERRIDE ONCALL STOP.↓
```

When checking that FCTF, DCTF and STOP are specified

```
/OVERRIDE LIST.↓  
OVERRIDE      ONCALL      STOP  
OVERRIDE      ON          FCTF  
OVERRIDE      ON          DCTF
```

When releasing all the specified event-names

```
/OVERRIDE OFF.↓
```

When checking that all the event-names are released

```
/OVERRIDE LIST.↓
```

When disabling the ON branch for all events

```
/OVERRIDE ON ALL.↓  
  
/OVERRIDE LIST.↓  
OVERRIDE      ON      ERR  
OVERRIDE      ON      STOP  
OVERRIDE      ON      TEST  
OVERRIDE      ON      FCT  
OVERRIDE      ON      DCT  
OVERRIDE      ON      OVERFLOW  
OVERRIDE      ON      UNDERFLOW  
OVERRIDE      ON      ZERODIVIDE  
OVERRIDE      ON      BTEST  
OVERRIDE      ON      FCTF  
OVERRIDE      ON      DCTF  
OVERRIDE      ON      KEY  
OVERRIDE      ON      TIME  
OVERRIDE      ON      DUTIF  
OVERRIDE      ON      BFCT  
OVERRIDE      ON      BDCT  
OVERRIDE      ON      WCNT  
OVERRIDE      ON      TRIP  
OVERRIDE      ON      SRQ0  
OVERRIDE      ON      SRQ1  
OVERRIDE      ON      GPSRQ
```

54. 3 Error Message

The error message, "ILLEGAL EVENT NAME:event-name" is displayed, if an invalid mode or event-name is specified.

54. 4 Notes of OVERRIDE

- The OVERRIDE setting is reset when the controller mode is reselected.
- OVERRIDE applies only to PRO (test plan program).
Neither SUB (subroutine program) nor LIB (library program) is applicable.

55. ENDPRO

The ENDPRO utility program is used to run a specified program after the completion of a test program.

55. 1 Setting ENDPRO

The following commands are input for the controller:

```
/ENDPRO ↵
```

The following message is displayed on the screen:

```
SPECIFY THE CALLING PROGRAM NAME AND CONDITION ON END PROCESS.

STATION NO. (1,2,A) = 1

### CURRENT STATUS ###

      CONTENTS                                [FILENAME]

1 : EVERY TIME CALL                        ... DISABLE

SPECIFY TERM NO.=
```

Input either 1, 2 or A for "STATION NO.." 1 and 2 refer to Station 1 and 2 respectively. A sets both stations to be used simultaneously.

Enter 1 for the "SPECIFY TERM NO.=" item.

Pressing the return key instead of the number 1 terminates ENDPRO.

Entering 1 displays the following message.

```
MODE (DISABLE) ... (E OR D) ?
```

DISABLE

Current status

E

ENABLE mode

D

DISABLE mode

Entering D selects "DISABLE" and displays "CURRENT STATUS", and places the cursor at the SPECIFY TERM NO.?" entry.

Entering E selects "ENABLE" and displays the lines shown below, which specifies a file name.

```
FILE NAME (____*1)...____*2
```

**1 The file name specified previously*

**2 Enter a file name (up to eight characters). Specifiers (LIB or SUB) cannot be used. If files with different specifiers have the same name, LIB is checked first.*

If you do not change the file name, press ↵ only.

Example

```
### CURRENT STATUS ###
```

```
CONTENTS
```

```
[FILENAME]
```

```
1 : EVERY TIME CALL
```

```
... ENABLE
```

```
[WMAP ]
```

```
SPECIFY TERM NO.=
```

55. 2 Quitting ENDPRO

Select "D" for the message shown below. This will quit ENDPRO.

`MODE (ENABLE ) ... (E OR D) ?`

55. 3 Programs Called by ENDPRO

A program that the ENDPRO calls must be a subroutine program.

```
SUBR XXX
:
:
RETURN
END
```

XXX

Name of a program called by ENDPRO

RETURN

Must use RETURN to quit a program.

Argument

Arguments cannot be used. If the DEFINE statement is used instead, you can see the same variable areas between the main program and sub program.

55. 4 Notes of ENDPRO

- When performing a two-station parallel test, a program is executed using station 1 and 2 in this sequence.
- Even when the same program is specified for both stations 1 and 2 in a two-station parallel test, the program is executed only once.
- ENDPRO settings are saved even after ENDPRO quits the controller mode.
- Make sure the controller has finished updating each counter and the control box when a test ends, before the specified program is executed.
- When using CTRL/C or RESET to stop ENDPRO, all previously entered settings are lost.
- Use numbers 1 or 4 of the U command to quit ENDPRO.

56. VS Control Utility Using PPS Parallel Operation

If there are not enough VSs used for device power supply pins, specifying PPS parallel operation in the socket program assigns the PPSs to the remaining power supply pins. However, due to parallel operation, small currents cannot be measured, or PPSs cannot be used for measurement, individually.

This utility allows you to select PPSs instead of the VS to which PPS parallel operation is applied, and also to use all PPS ranges for measurements. In addition, PPSs can be used for measurements, individually.

This utility function allows PPSs to be used instead of the VSs listed below.

On the T5581/T5581H

VS1, 2

On other than the T5581/T5581H

VS1 to 12

56. 1 SETVSMASK

◆ Function

This utility is used to connect or disconnect the PPS from the device without turning the PPS off. Reference-off does not occur during execution.

However, the PPS cannot be controlled to DUTs that have been rejected.

This function allows you to temporarily cancel PPS parallel operation so that you can measure a small current.

If the specified PPS is not used for the test plan program, an error occurs and the error information is stored in the argument. If this occurs, the device test is terminated without performing any processes. Check the error information to fix the problem.

Even though this utility is used to disconnect the PPS from the device, executing the statements listed below will still connect the PPS.

- MULTIPLE DUT statement
- SET REJECTION statement
- SET/RESET EXCLUSION2 statement
- SELECT VS PARALLEL statement
- SELECT VS PPS statement

Even though the statement listed below is used to disconnect the PPS from the device, using this utility will still connect the PPS.

- SELECT VS SINGLE statement

◆ Syntax

```
CALL SETVSMK(sw, pdat, err)
```

sw

When sw is set to 1, the specified PPS is disconnected from the device.

When sw is set to 0, the specified PPS is connected to the device.

Use an integer constant or variable for sw.

pdat

Specifies a PPS for the currently operating test station.

PPS numbers should be specified in the format shown below.

Use the 2-word logical type of a non-array variable, subscripted variable or non-subscripted array variable for pdat.

When a non-subscripted array variable is used, up to eight elements can be used and the other elements are ignored.

On the T5581/T5581H

Data with more than two arrays is ignored.

On the T5593
Data with more than ten arrays is ignored.

	0	-----	30	31	
(1)	32	-----	2	1	PPS32-1
(2)	64	-----	34	33	PPS64-33
(3)	96	-----	66	65	PPS96-65
(4)	128	-----	98	97	PPS128-97
(5)	160	-----	130	129	PPS160-129
(6)	192	-----	162	161	PPS191-161
(7)	224	-----	194	193	PPS224-193
(8)	256	-----	226	225	PPS256-225
(9)	288	-----	258	257	PPS288-257
(10)	320	-----	290	289	PPS320-289

err
Stores an error code listed below using an integer, logical or real type of variable.

- 0 Normal
- 1 The data type or attribute specified for the argument is incorrect.
- 2 The value specified for sw is out of range.
- 3 The specified PPS number is not used for the test plan program.

If an error occurs, the SETVSMASK function is disabled.
Check the error information to fix the problem.

◆ Specification Example

When excluding PPS 1-10 and 97

```

LOGI[2] PDAT(4)=#000003FF,#0,#0,#00000001
INTE[1] EDAT
CALL SETVSMASK(1, PDAT, EDAT)
IF EDAT<>0 THEN .....
```

56. 2 SELVSRNG

◆ Function

When this utility is executed, the settings described below can be used only once.

Depending on the measurement, the items listed below can be specified to the VS for which PPS parallel operation can be applied.

For current measurements, the following items can be specified.

- Specifications for all measurement ranges
- Specifications for current clamp values
- Low-pass filter specifications
- Compensation capacity specifications

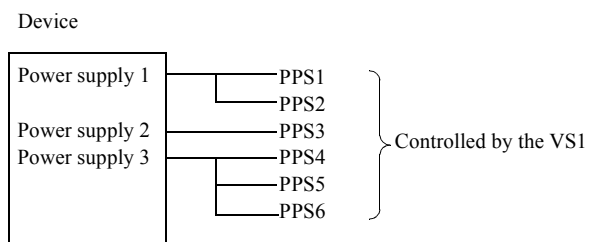
When measuring MVM

- Low-pass filter specifications

The WET specification is valid when the measurement range is changed.

Tip

To change the measurement range using this utility and the VS statement, use SETVSMASK to change the PPS connection so that one power supply pin is connected to one PPS. For example, in the following case, one side is connected in PPS1 and 2, and the other side is excluded. Either PPS4, 5, or 6 is connected, and the others excluded.



◆ Syntax

```
CALL SELVSRNG
```

◆ Specification Example

To change the VS1 or VS2 measurement range, write the program shown below.

```
SET WET                ← Specification of the WET
VS1=1V R(5V) M(400MA)
VS2=1V R(5V) M(400MA)
RON

CALL SELVSRNG
VS1=1V R(5V) M(40UA)   ← The measurement condition can be changed in a RON state.
CALL SELVSRNG
VS2=1V R(5V) M(40UA)   ← The measurement condition can be changed in a RON state.
```

56. 3 GETVSADC

◆ Function

The most recent values measured using the MEAS VS or START VS statements are read from the PPS and stored in variables.

However, the most recent values cannot be read from the disconnected PPS.

Variables are updated only when new measurement values are stored.

If an invalid value is specified, an error occurs and the error information is stored in the argument. If this occurs, the device test is terminated without performing any processes. Check the error information to fix the problem.

After the MEAS VS or START VS statement, execute the CALL GETVSADC to store measurement values in the variables. If a statement or utility listed below is executed between the MEAS VS or START VS statement and CALL GETVSADC, measurement values may not be correct.

- VS statement
- MEAS, START or JUDGE statement
- SELECT VS SINGLE/PARALLEL statement
- SHM2
- CALL SETVSMSK statement

◆ Syntax

```
CALL GETVSADC(stn, cnt, pdat, mdat, err)
```

stn

Specifies the number of the station to read.

Specify 0, 1, and 2 as an integer constant or variable

- | | |
|---|---|
| 0 | Measurement values are read from the currently operating test station.
When two test stations are tested in parallel, measurement values are read from Test Station 1. |
| 1 | Measurement values are read from Test Station 1 |
| 2 | Measurement values are read from Test Station 2 |

cnt

Stores the number of measurement values using an integer variable.

Specifies integer-type variables.

When cnt=0, neither a PPS number nor a measurement value is stored in pdat or mdat, respectively.

pdat

Stores the number of PPS numbers using an integer variable.
 Tested PPSs are stored by their numbers, in ascending order.
 The PPS numbers used for tests are stored in ascending order.

mdat

Specifies a name for a variable to store read measurement values.
 Tested PPSs are stored by their measurement results, in ascending order.
 Measurement values are transferred from the PPSs in ascending order of the PPS number, and then stored.

err

Stores an error code listed below using an integer, logical or real type of variable.

- | | |
|---|---|
| 0 | Normal |
| 1 | The data type or attribute specified for the argument is incorrect. |
| 2 | The test station number specified for the argument is out of range. |
| 3 | The VS used for tests is not located. |

If an error occurs, the GETVSADC function is disabled.
 Check the error information to fix the problem.

◆ Specification Example

To read measurement values from PPSs 1, 3, 5,..., or 31 specified as the nVS1 for the currently operating test station, write the program shown below.

```
INTE[1] CNT
INTE[1] PDAT(32)
REAL[3] MDAT(32)
INTE[1] EDAT

      MEAS VS1
      CALL GETVSADC(0, CNT, PDAT, MDAT, EDAT)
      IF EDAT<>0 THEN .....
```

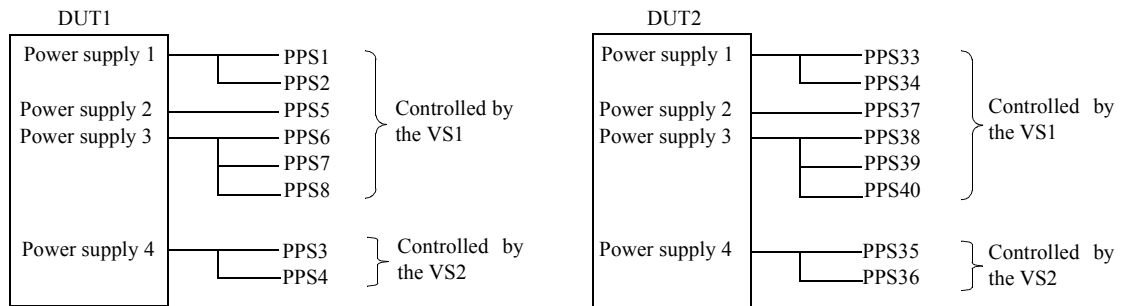
Measurement values are stored as shown below.

```
CNT      = 16
PDAT(1) = 1
PDAT(2) = 3
PDAT(3) = 5
      :
      :
PDAT(16) = 31
PDAT(17) = (The PPS number does not change because a value was not stored.)
```

:
PDAT(32) = (The PPS number does not change because a value was not stored.)
MDAT(1) = (Measured value of PPS1)
MDAT(2) = (Measured value of PPS3)
MDAT(3) = (Measured value of PPS5)
 :
MDAT(16) = (Measured value of PPS31)
MDAT(17) = (The measurement value does not change because a value was not stored.)
 :
MDAT(32) = (The measurement value does not change because a value was not stored.)

56. 4 Utility Usage Example

The example below shows how to measure a small current using the PPSs connected in parallel.



◆ A Socket Program Example is Shown Below

```

SOCKET S1

DEFINE MULTIPLE 2DUT 2STN
PPS=2
PARALLEL VS1=6
PARALLEL VS2=2
.....

DUT# = VS1          VS2 .....
DUT1 = PPS1,2,5-8   PPS3,4
DUT2 = PPS33,34,37-40 PPS35,36

END

```

◆ A Test Plan Program Example is Shown Below

```

PRO M1 S1                                ; Example for a wet state

      PCON=VS1,VS2

      VS1=1V R(10V) M(400MA)
      VS2=1V R(10V) M(400MA)
      MEAS VS1,VS2                        ; Parallel operation measurement

      SET WET

      PCON=0                             ; Restore after excluding from PPS using
                                          ; the SETVSMSK utility
                                          ; When updating the specification range
                                          ; during the interval leading up to
                                          ; performance, the maximum
                                          ; A setting to fill load capacity.

LOGI[2] VDAT(2)=#000000CA,#000000CA
INTE[1] EDAT

      CALL SETVSMSK(1,VDAT,EDAT)          ; Disconnect the PPSs listed below
      IF EDAT<>0 THEN .....              ;      DUT1: PPS2,7,8,4
                                          ;      DUT2: PPS34,39,40,36

      CALL SELVSRNG
      VS1=1V R(10V) M(40UA)              ; Changes the VS1 measurement range.
      CALL SELVSRNG
      VS2=1V R(10V) M(40UA)              ; Changes the VS2 measurement range.

INTE[1] CNT1,CNT2
INTE[1] PDAT1(32),PDAT2(32)
REAL[3] MDAT1(32),MDAT2(32)

      MEAS VS1,VS2                        ; Measures a small current.

      CALL GETVSADC(1,CNT1,PDAT1,MDAT1,EDAT) ; Reads measurement values ; from
                                          ; Test Station 1.
      IF EDAT<>0 THEN .....
      CALL GETVSADC(2,CNT2,PDAT2,MDAT2,EDAT) ; Reads measurement values ; from
                                          ; Test Station 2.
      IF EDAT<>0 THEN .....

      CALL SETVSMSK(0,VDAT,EDAT)          ; Connects the PPSs in parallel again.

      PCON=VS1,VS2

      RESET WET

END

```

After GETVSADC is executed, values are stored in the variables as shown below.

CNT1 = 8

PDAT1(1) = 1

PDAT1(2) = 3

PDAT1(3) = 5

PDAT1(4) = 6

PDAT1(5) = 33

PDAT1(6) = 35

PDAT1(7) = 37

PDAT1(8) = 38

PDAT1(9)=(The PPS number does not change because
a number was not stored.)

⋮

PDAT1(32)=(The PPS number does not change because
a number was not stored.)

MDAT1(1)=(Value measured by STN1 PPS1)

MDAT1(2)=(Value measured by STN1 PPS3)

MDAT1(3)=(Value measured by STN1 PPS5)

MDAT1(4)=(Value measured by STN1 PPS6)

MDAT1(5)=(Value measured by STN1 PPS33)

MDAT1(6)=(Value measured by STN1 PPS35)

MDAT1(7)=(Value measured by STN1 PPS37)

MDAT1(8)=(Value measured by STN1 PPS38)

MDAT1(9)=(The measurement value does not change
because a value was not stored.)

⋮

MDAT1(32)=(The measurement value does not change
because a value was not stored.)

CNT2 = 8

PDAT2(1) = 1

PDAT2(2) = 3

PDAT2(3) = 5

PDAT2(4) = 6

PDAT2(5) = 33

PDAT2(6) = 35

PDAT2(7) = 37

PDAT2(8) = 38

PDAT2(9)=(The PPS number does not change because
a number was not stored.)

⋮

PDAT2(32)=(The PPS number does not change because
a number was not stored.)

MDAT2(1)=(Value measured by STN2 PPS1)

MDAT2(2)=(Value measured by STN2 PPS3)

MDAT2(3)=(Value measured by STN2 PPS5)

MDAT2(4)=(Value measured by STN2 PPS6)

MDAT2(5)=(Value measured by STN2 PPS33)

MDAT2(6)=(Value measured by STN2 PPS35)

MDAT2(7)=(Value measured by STN2 PPS37)

MDAT2(8)=(Value measured by STN2 PPS38)

MDAT2(9)=(The measurement value does not change
because a value was not stored.)

⋮

MDAT2(32)=(The measurement value does not change
because a value was not stored.)

57. DQS Test Utility

The operation of the specified DQ pins on a Double Data Rate (DDR) SDRAM is synchronized with the associated DQS pin.

If this DQS pin has a jitter, the comparison result of this pin must be taken into consideration before making a PASS/FAIL judgment for the DQ pins. The comparison result for continuous cycles of the DQS pin must also be taken into consideration before making a PASS/FAIL judgment for this DQS pin.

This utility is used to reflect the comparison result of a DQS pin on the PASS/FAIL judgment of the associated DQ and DQS pins. In addition, the comparison result of the DQS pin for continuous cycles can be made to reflect the PASS/FAIL of this DQS pin by using this utility. Finally, the jitter distribution of a DQS pin can be measured.

The utility is compatible with the T5586 and T5593.

When the high-speed clock driver is used on the T5593, channels 55, 56, 119, 120, 183, 184, 247, and 248 cannot be used.

The odd-numbered phase STRB can be used on the T5586, also it can be used only for 2WAY on the T5593.

57. 1 DQS Test (DQSvsDQ Test) Utility

◆ Function

PASS/FAIL judgments for specified DQ pins can be made during each cycle by combining the DQS and DQ pin comparison results.

The combination used to determine the PASS/FAIL judgment for DQ pins can be specified as desired.

The test results (PASS/FAIL) specified by this utility are not reflected in the fail information (failure bits) in FM and the data logging of functional tests. The DQ and DQS pin comparison results is stored.

◆ Notes on the T5586

- 15 dummy cycles beginning with the last comparison cycle are required for a pattern program.
- DQ pin PASS/FAIL results can be judged each cycle from combinations of comparison results for multiple DQS pins.

◆ Format

```
CALL DQSSEL1 (ERR, PNNO, TYPE, TBL1, TBL2)
```

ERR

Stores the termination state (normal or abnormal completion) of the DQS test (DQSvsDQ test) utility.

Specify using an integer or logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For more information, refer to [57. 12 "ERR Description" on page 57-62](#).

PNNO

Specifies channel numbers to assign to DQ and DQS pins used for a DQSvsDQ test.

Specify using the PINLIST variable.

TYPE

Selects a channel number assigned to the DQS pin used for a PASS/FAIL judgment, in combination with the comparison results of the DQS and DQ pins that are specified by the PNNO.

Use an integer (or logical) constant, or a variable between 1 and 2.

Channel numbers available for assignment to DQS pins are listed in the table below.

Pin numbers refer to the tester channel numbers.

On the T5586

TYPE	Channel numbers assigned to DQS pins	PNNO
1	49	33 to 50
	113	97 to 114
	177	161 to 178
	241	225 to 242
2	50	33 to 50
	114	97 to 114
	178	161 to 178
	242	225 to 242

On the T5593

TYPE	Channel numbers assigned to DQS pins	PNNO
1	54	33 to 50
2	55	33 to 50

TBL1

Specifies the STRB test result for the STRB1 to 128.

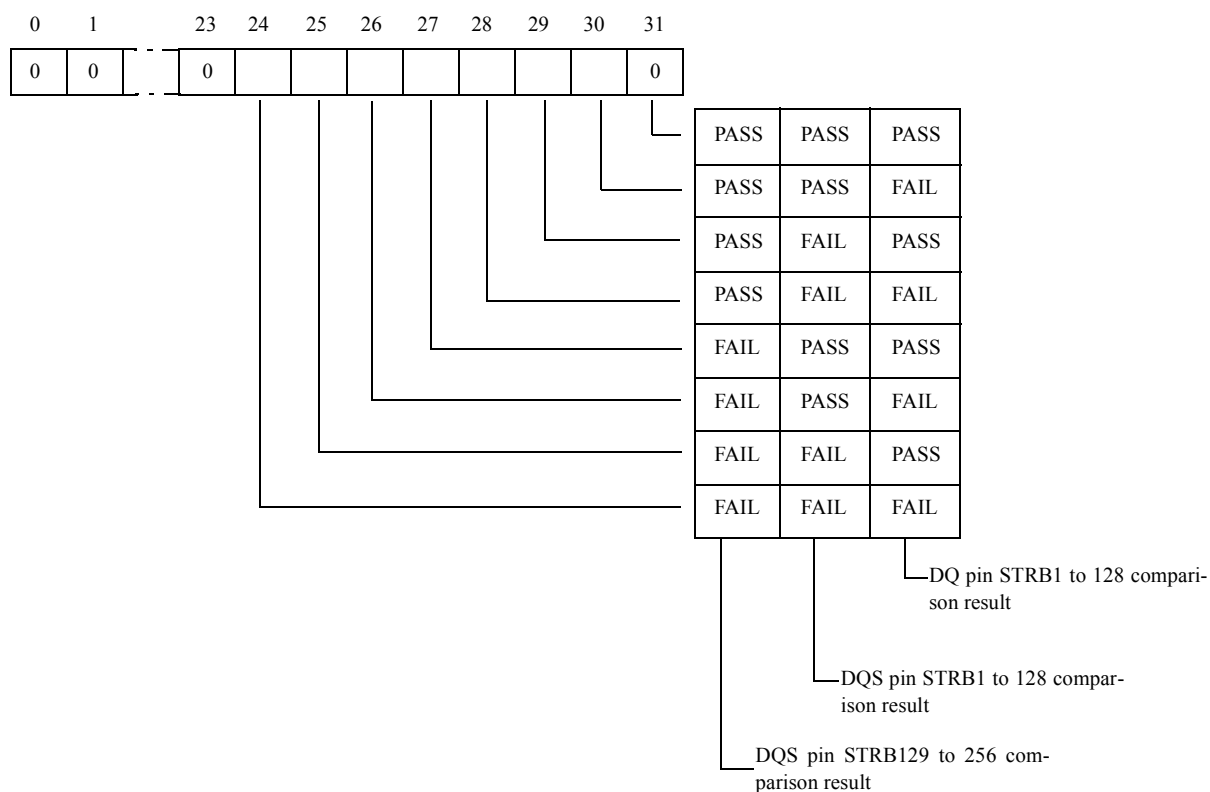
Specify using variables or logical constants.

Specify the STRB test result (PASS/FAIL) for the DQ pin STRB1 to 128 from the combination of the DQ pin STRB1 to 128, DQS pin STRB1 to 128, and DQS pin STRB129 to 256 comparison results.

Specify "0" for PASS, and "1" for FAIL.

The test result cannot be changed if each STRB comparison result is PASS.

TBL1 format is shown below.



For example, if the DQ pin STRB1 to 128 comparison result is FAIL, the DQS pin STRB1 to 128 comparison result is FAIL, and the DQS pin STRB129 to 256 comparison result is PASS, set Bit 28 to "0" so that the DQ pin STRB1 to 128 comparison result is set to PASS.

Tip

If only DQ pin STRB1 to 128 and DQS pin STRB1 to 128 test results are combined, set Bits 24 to 27 to the values of Bits 28 to 31, respectively.

If the DQ pin STRB1 to 128 and DQS pin STRB129 to 256 comparison test results are combined, set Bits 28 and 29 to the values of Bits 30 and 31, respectively and set the values of Bits 26 and 27 to Bits 24 and 25, respectively.

TBL2

Specifies the STRB test result for the STRB129 to 256.

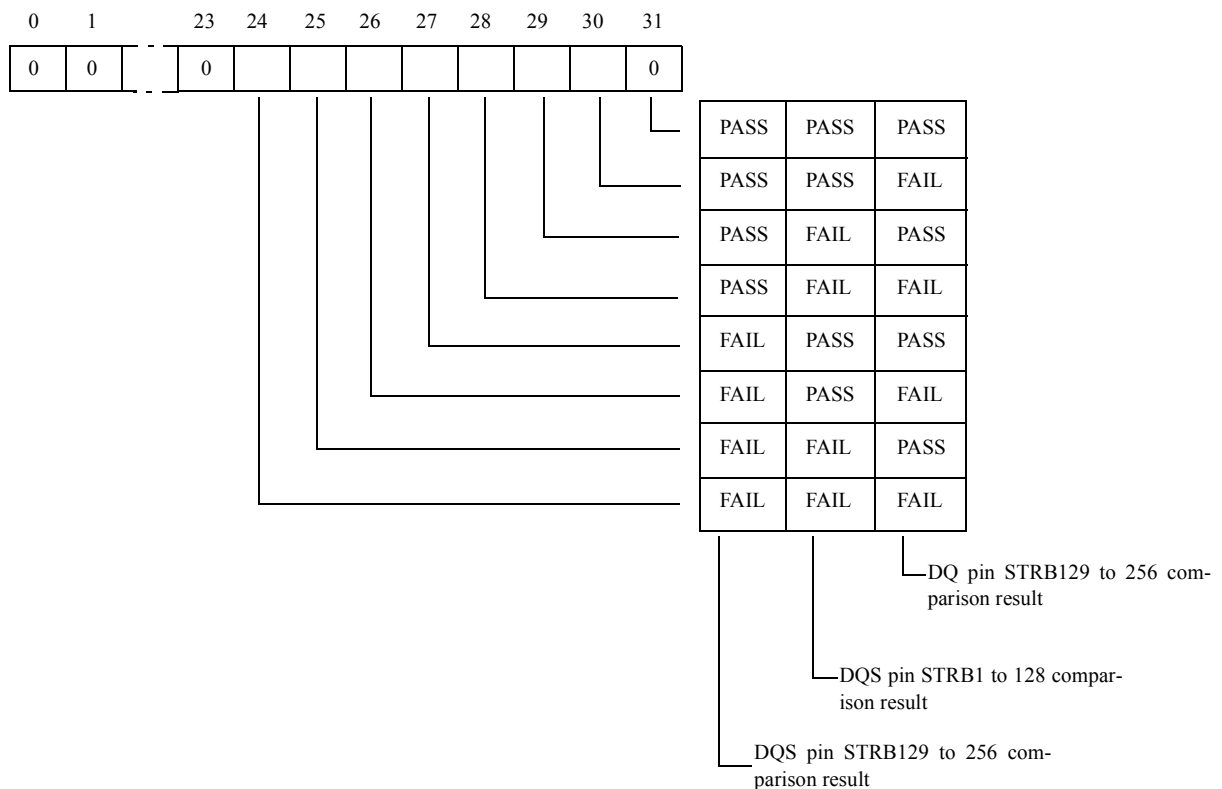
Specify using variables or logical constants.

Specify the STRB test result (PASS/FAIL) for the DQ pin STRB129 to 256 from the combination of the DQ pin STRB129 to 256, DQS pin STRB1 to 128, and DQS pin STRB129 to 256 comparison results.

Specify "0" for PASS, and "1" for FAIL.

The test result cannot be changed if each STRB comparison result is PASS.

The TBL2 format is shown below.



For example, if the DQ pin STRB1 to 128 comparison result is FAIL, the DQS pin STRB1 to 128 comparison result is FAIL, and the DQS pin STRB129 to 256 comparison result is PASS, set Bit 28 to "0" so that the DQ pin STRB129 to 256 comparison result is set to PASS.

**Tip**

If only DQ pin STRB129 to 256 and DQS pin STRB1 to 128 test results are combined, set Bits 24 to 27 to the values of Bits 28 to 31, respectively.

If the DQ pin STRB129 to 256 and DQS pin STRB129 to 256 comparison results are combined, set Bits 28 and 29 to the values of Bits 30 and 31, respectively and set the values of Bits 26 and 27 to Bits 24 and 25, respectively.

◆ Usage Examples

When DQS pin STRB129 to 256 strobes are not used

```
INTE[1]  ERR
INTE[1]  TYPE
LOGI[2]  TBL1,TBL2

PINLIST PNNO=P33-40

TYPE=1           ; Specify 49 channels on the T5586; 54 channels on the T5593.
TBL1=.00100010   ; Set Bits 24 thru 27 to the values of Bits 28 thru 31,
                  ; respectively.
TBL2=.01100110   ; Set Bits 24 thru 27 to the values of Bits 28 thru 31,
                  ; respectively.

CALL  DQSSEL1(ERR,PNNO,TYPE,TBL1,TBL2)

IF ERR<>0 THEN ...      ; Error processing.

MEAS MPAT SAMPLE
```

When DQS pin STRB1 to 128 strobes are not used:

```

INTE[1] ERR
INTE[1] TYPE
LOGI[2] TBL1,TBL2

PINLIST PNNO=P33-40

TYPE=1          ; Specify 49 channels on the T5586; 54 channels on the T5593.
TBL1=.01010000 ; Set Bits 28 and 29 to the values of Bits 30 and 31,
                ; respectively.
                ; Set Bits 24 and 25 to the values of Bits 26 and 27,
                ; respectively.
TBL2=.11111010 ; Set Bits 28 and 29 to the values of Bits 30 and 31,
                ; respectively.
                ; Set Bits 24 and 25 to the values of Bits 26 and 27,
                ; respectively.

CALL DQSSEL1(ERR,PNNO,TYPE,TBL1,TBL2)

IF ERR<>0 THEN ...      ; Error processing.

MEAS MPAT SAMPLE

```

When both DQS pin STRB1 to 128 and STRB129 to 256 strobes are used:

```

INTE[1] ERR
INTE[1] TYPE
LOGI[2] TBL1,TBL2

PINLIST PNNO=P33-40

TYPE=1          ; Specify 49 channels on the T5586; 54 channels on the T5593.
TBL1=.01110010
TBL2=.11100110

CALL DQSSEL1(ERR,PNNO,TYPE,TBL1,TBL2)

IF ERR<>0 THEN ...      ; Error processing.

MEAS MPAT SAMPLE

```

57. 2 DQS Test (CYCLEvsCYCLE Test) Utility

◆ Function

PASS/FAIL judgments for the specified DQS pins can be made by combining the comparison results from arbitrarily selected two or more strobes in continuous cycles.

The test results (PASS/FAIL) specified by this utility are not reflected in the fail information (failure bits) in FM and the data logging of functional tests. The DQ and DQS pin comparison results is stored.

◆ Notes on the T5586

- 15 dummy cycles beginning with the last comparison cycle are required for a pattern program.

◆ Syntax

```
CALL DQSSEL2 (ERR, PNNO, BCNT1, SSEL1, TBL1, BCNT2, SSEL2, TBL2)
```

ERR

Stores the termination state (normal or abnormal completion) of the DQS test (CYCLEvsCYCLE test) utility.

Specify using an integer or logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For more information, refer to [57. 12 "ERR Description" on page 57-62](#).

PNNO

Specifies channel numbers to assign to DQS pins used for a CYCLEvsCYCLE test.

Specifiable channels are as follows.

Specify using the PINLIST variable.

On the T5586

Channels 33 to 50, 97 to 114, 161 to 178, 225 to 242

On the T5593

Channels 33 to 56, 97 to 120, 161 to 184, 225 to 248

BCNT1

Specifies the number of comparison results selected using SSEL1.

Use an integer (or logical) constant, or a variable between 1 and 4.

BCNT2

Specifies the number of comparison results selected using SSEL2.

Use an integer (or logical) constant, or a variable between 1 and 4.

SSEL1

Selects the strobes and cycles to be used.

Use an integer (or logical) constant, or a variable between 0 and 7.

When using variables to make specifications, use array variables with up to four elements.

➔ Refer to [Figure 57-1](#), [Figure 57-2](#), and [Figure 57-3](#) for values to specify.

SSEL2

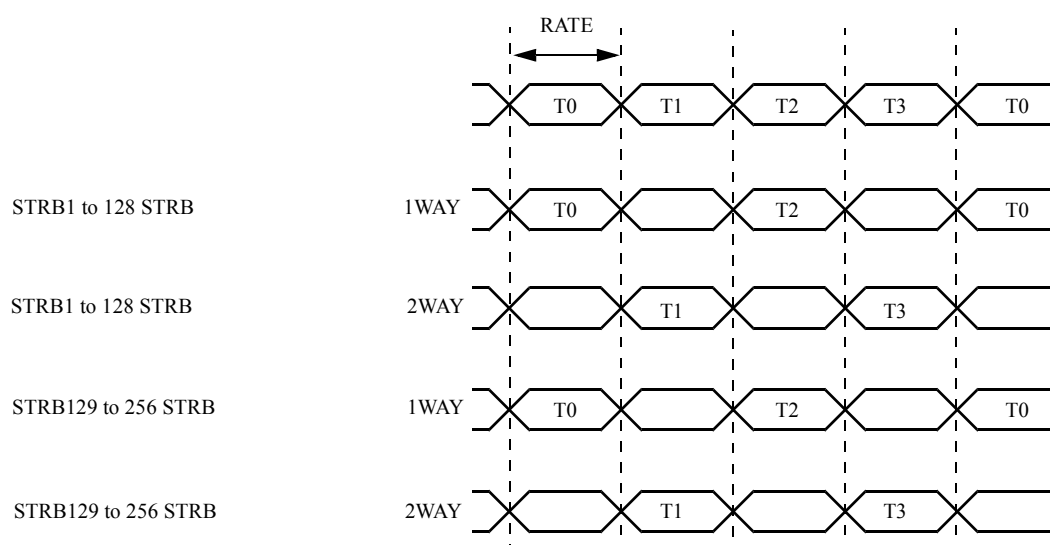
Selects the strobes and cycles to be used.

Use an integer (or logical) constant, or a variable between 0 and 7.

When using variables to make specifications, use array variables with up to four elements.

➔ Refer to [Figure 57-1](#), [Figure 57-2](#), and [Figure 57-3](#) for values to specify.

Figure 57-1 Relationship between SSEL1/SSEL2 and Strobes/Cycles (for 2WAY)



SSEL1 SSEL2	Strobes and Cycles to Be Used		
0	STRB1 to 128 STRB	During 1WAY	T0
1	STRB1 to 128 STRB	During 2WAY	T1
2	STRB129 to 256 STRB	During 1WAY	T0
3	STRB129 to 256 STRB	During 2WAY	T1
4	STRB1 to 128 STRB	During 1WAY	T2
5	STRB1 to 128 STRB	During 2WAY	T3
6	STRB129 to 256 STRB	During 1WAY	T2
7	STRB129 to 256 STRB	During 2WAY	T3

Figure 57-2 Relationship between SSEL1/SSEL2 in 1WAY and STRBs/cycles to be Used (on the T5586)

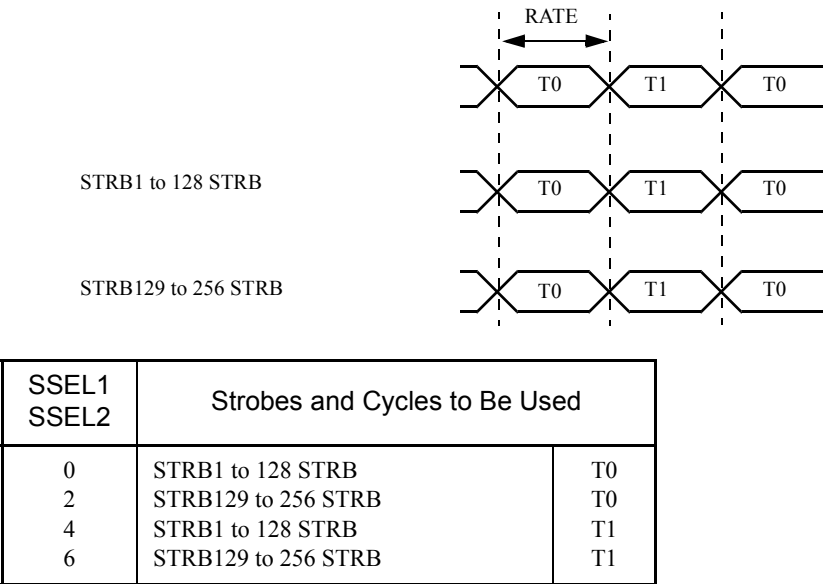
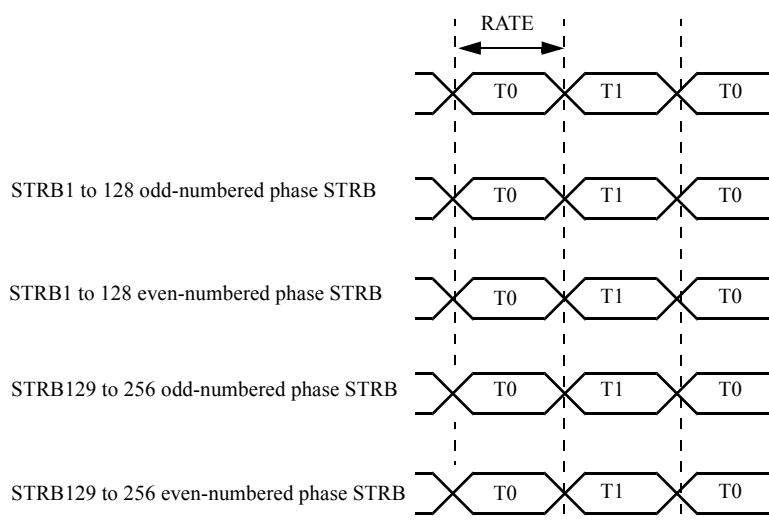


Figure 57-3 Relationship between SSEL1/SSEL2 and Strokes/Cycles (on the T5593)



SSEL1 SSEL2	Strokes and Cycles to Be Used	
0	STRB1 to 128 odd-numbered phase STRB	T0
1	STRB1 to 128 even-numbered phase STRB	T0
2	STRB129 to 256 odd-numbered phase STRB	T0
3	STRB129 to 256 even-numbered phase STRB	T0
4	STRB1 to 128 odd-numbered phase STRB	T1
5	STRB1 to 128 even-numbered phase STRB	T1
6	STRB129 to 256 odd-numbered phase STRB	T1
7	STRB129 to 256 even-numbered phase STRB	T1

TBL1

Specifies the STRB test result for the STRB1 to 128.

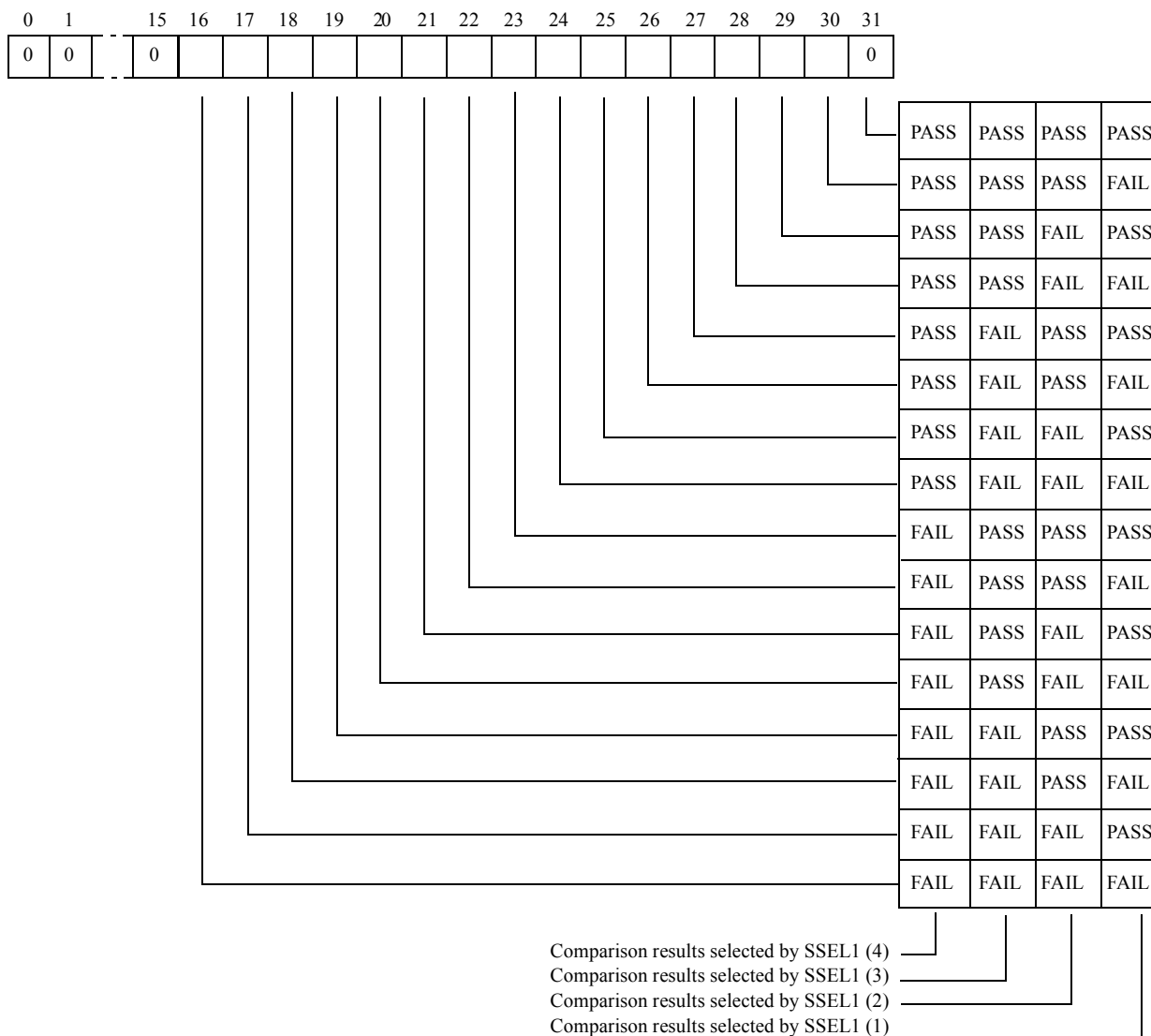
Specify using variables or logical constants.

Specify the STRB1-to-128 odd-numbered phase strobe test result (PASS/FAIL) from the combinations selected from SSEL1(1) thru SSEL1(4).

Specify "0" for PASS, and "1" for FAIL.

The test result cannot be changed if each STRB comparison result is PASS.

TBL1 format is shown below.



For example, if the comparison results selected from SSEL1(1) and SSEL1(2) and SSEL1(3) are FAIL, and if the comparison result selected from SSEL1(4) is PASS, set Bit 24 to "0" to set the STRB1-to-128 comparison result to PASS.

Tip

If specifying 1 using BCNT 1, specify bits 16 to 29 as "0."

If specifying 2 using BCNT 1, specify bits 16 to 27 as "0."

If specifying 3 using BCNT 1, specify bits 16 to 23 as "0."

TBL2

Specifies the STRB test result for the STRB129 to 256.

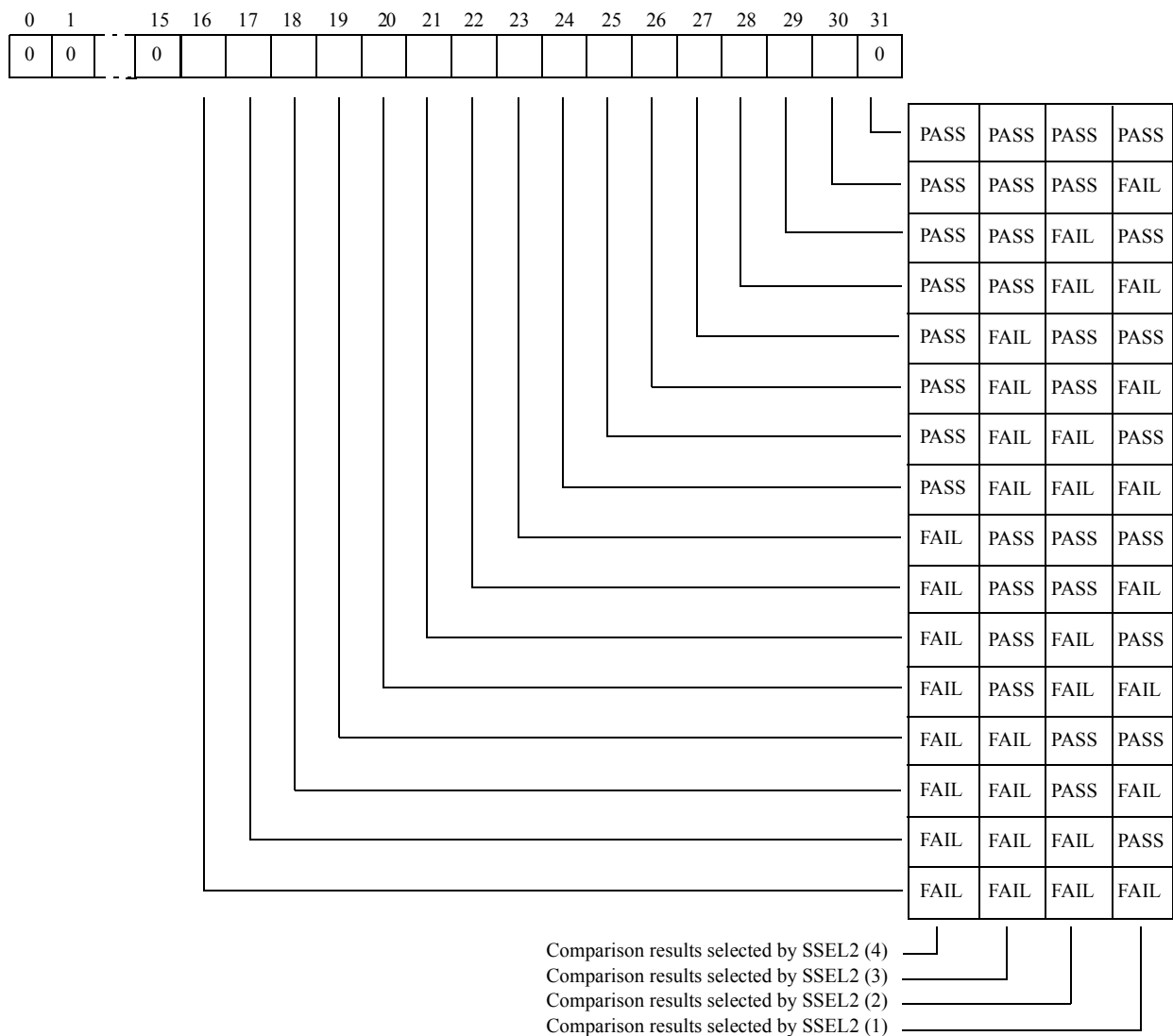
Specify using variables or logical constants.

Specify the test result (PASS/FAIL) from the combinations selected from SSEL2(1) thru SSEL2(4).

Specify "0" for PASS, and "1" for FAIL.

The test result cannot be changed if each STRB comparison result is PASS.

The TBL2 format is shown below.



For example, if the comparison results selected from SSEL2(1) and SSEL2(2) and SSEL2(3) are FAIL, and if the comparison result selected from SSEL2(4) is PASS, set Bit 24 to "0" to set the STRB129-to-256 comparison result to PASS.

Tip

If specifying 1 using BCNT2, specify bits 16 to 29 as "0."

If specifying 2 using BCNT2, specify bits 16 to 27 as "0."

If specifying 3 using BCNT2, specify bits 16 to 23 as "0."

! Important

When in bank multiplex mode, the STRB1 to 128 even-numbered phase comparison result is output as the STRB1 to 128 odd-numbered phase strobe test result, the STRB1 to 128 even-numbered phase strobe test result is always output as PASS.

The STRB129 to 256 even-numbered phase comparison result is output as the STRB129 to 256 odd-numbered phase strobe test result, the STRB129 to 256 even-numbered phase strobe test result is always output as PASS as well.

◆ Usage Examples

When a CYCLEvsCYCLE test is conducted for channel 49 (on the T5586) in 2WAY interleaved mode

```
INTE[1] ERR
INTE[1] BCNT1,BCNT2
INTE[1] SSEL1(2),SSEL2(2)
LOGI[1] TBL1,TBL2

PINLIST PNNO=P49

BCNT1=2
SSEL1(1)=0
SSEL1(2)=3
TBL1=#8888
BCNT2=2
SSEL2(1)=6
SSEL2(2)=5
TBL2=#8888

CALL DQSSEL2(ERR,PNNO,BCNT1,SSEL1,TBL1,BCNT2,SSEL2,TBL2)

IF ERR<>0 THEN ... ; Error processing.

MEAS MPAT SAMPLE
```

57. 3 Utility Used to Terminate a DQS Test

◆ Function

Terminates a DQS test (DQSvsDQ or CYCLEvsCYCLE test).

DQS tests (DQSvsDQ or CYCLEvsCYCLE test) are also terminated when:

- Starting the program
- Executing the RESET TESTER statement
- The current interleaved mode is changed using the SET ILMODE statement.

◆ Format

```
CALL DQSRST (ERR)
```

ERR

Stores the termination state (normal or abnormal completion) of the DQS test (DQSvsDQ test) utility.

Specify using an integer or logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For more information, refer to [57. 12 "ERR Description" on page 57-62](#).

◆ Usage Examples

When releasing the DQS test:

```
INTE[1] ERR  
  
CALL DQSRST (ERR)  
  
IF ERR<>0 THEN ...      ; Error processing.
```

57. 4 PASS/FAIL Counter Setting Utility

◆ Function

Sets the PASS/FAIL counter.

This utility counts the number of matches for the count conditions.

Count conditions are specified based on a combination of the STRB judgment results for STRB1 to 128 for STRB129 to 256.

Set the STRB 1 to 128 strobes and the STRB 129 to 256 strobes using the PIN statement when using this utility.

◆ Syntax

```
CALL STPFCNT (ERR, PNNO, MODE, STB)
```

ERR

Stores the termination state (normal or abnormal completion) of the PASS/FAIL counter setting utility.

Specify using an integer or logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For more information, refer to [57. 12 "ERR Description" on page 57-62](#).

PNNO

Specifies pin numbers used to enable or disable the PASS/FAIL counter.

Specifiable channels are as follows.

Specify using the PINLIST variable.

On the T5586

Channels 33 to 50, 97 to 114, 161 to 178, 225 to 242

On the T5593

Channels 33 to 56, 97 to 120, 161 to 184, 225 to 248

MODE

Sets whether or not the PASS/FAIL counter is enabled.

Use an integer (or logical) constant, or a variable between 0 and 1.

MODE	Count cycle
0	Disables the PASS/FAIL counter.
1	Enables the PASS/FAIL counter. (The CPE for the STRB 1 to 128 or the STRB 129 to 256 strobes causes the PASS/FAIL counter to be counted during the ENABLE cycles.)

STB

Specifies the conditions to count up the PASS/FAIL counter.

Use an integer (or logical) constant, or a variable between 0 and 3.

STB	Count Condition	
	STRB1 to 128 STRB	STRB129 to 256 STRB
0	PASS	PASS
1	FAIL	PASS
2	PASS	FAIL
3	FAIL	FAIL

Counter sizes are as follows:

On the T5586

The counter size is 12 bits (0 to 4095) in 1WAY interleaved mode, or 13 bits (0 to 8191) in 2WAY interleaved mode.

On the T5593

The counter size is 24 bits (0 to 16777215) in both 1WAY and 2WAY interleaved mode.

The counter is cleared every time a MEAS or START MPAT statement is executed.

It cannot be cleared by MEAS MPAT(1) or START MPAT(1).

The PASS/FAIL counter can also be disabled when:

- Starting the program
- Executing the RESET TESTER statement
- The current interleaved mode is changed using the SET ILMODE statement.

When this utility is used with the T5593, a limit value can be set for the counter value.

➔ For details about limit value settings, refer to [57. 10 "Limit Setting Utility for FAIL COUNTER and PASS/FAIL COUNTER \(for the T5593\)" on page 57-54.](#)

Also, the default value for the limit value in this utility is #FFFFFF.

 Tip

PASS/FAIL COUNTER and FAIL COUNTER cannot be mixed together.

When using the PASS/FAIL COUNTER setting utility (CALL STPFCNT) after using the FAIL COUNTER setting utility (CALL STFLCNT), set arguments MODE1 and 2 of the FAIL COUNTER setting utility to DISABLE.

◆ Usage Examples

When the PASS/FAIL counter is enabled for channel 49

```
INTE[1] ERR
INTE[1] MODE
INTE[1] STB

PINLIST PNNO=P49
PNNO=IN3,DNRZ,BCLK3,CCLK3,DRECLK2,DRE2,OUTL1,IOC,EXPDL1,CPEDLY1,@
      [STRB1,CPE1],EXPA1,[STRB129,CPE1],EXPB129,<C10,C11>

MODE=1
STB=1

CALL STPFCNT(ERR,PNNO,MODE,STB)

IF ERR<>0 THEN ...      ; Error processing.

SELECT MPAT INHIBIT
START MPAT
```

When the PASS/FAIL counter is disabled for channel 49

```
INTE[1] ERR
INTE[1] MODE
INTE[1] STB

PINLIST PNNO=P49

MODE=0
STB=0

CALL STPFCNT(ERR,PNNO,MODE,STB)

IF ERR<>0 THEN ...      ; Error processing.
```

57. 5 Utility Used to Read Data from the PASS/FAIL Counter

◆ Function

The data in the PASS/FAIL counter can be read.

◆ Syntax

CALL RDPFCNT (ERR, PNNO, CNT)

ERR

Stores the termination state (normal or abnormal completion) of the PASS/FAIL counter setting utility.

Specify using an integer or logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For more information, refer to [57. 12 "ERR Description" on page 57-62](#).

PNNO

Specifies channel numbers used to read data from the PASS/FAIL counter.

The following channels can be specified:

Specify using the PINLIST variable.

On the T5586

Channels 33 to 50, 97 to 114, 161 to 178, 225 to 242

On the T5593

Channels 33 to 56, 97 to 120, 161 to 184, 225 to 248

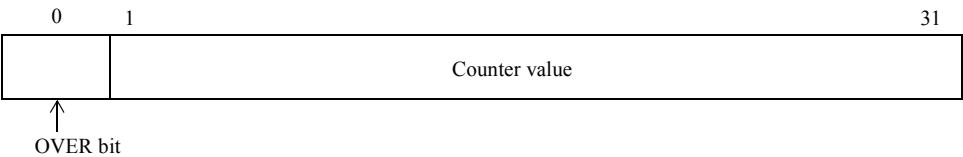
CNT

Specifies the variable with which the read counter values are stored.

Use a logical variable with 2 HW (32 bits).

Specify an array variable whose size is the number of channels × the number of DUTs in order to read the counter.

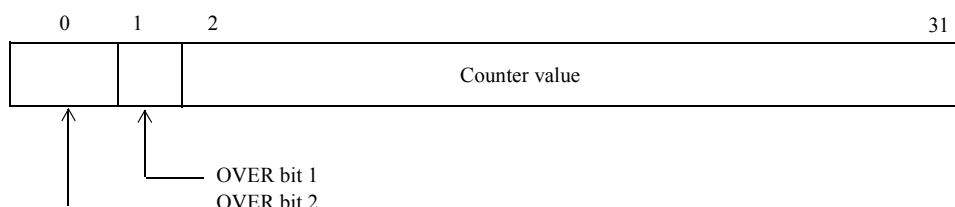
On the T5586



The OVER bit switches on when COUNTER's size exceeds its value.

COUNTER is 12 bits long.

On the T5593



OVER bit 1 switches on when the value of COUNTER matches LIMIT1.

OVER bit 2 switches on when the value of COUNTER matches LIMIT2.

COUNTER is 24 bits long.

The default values is #FFFFFF (24 bits) for both LIMIT1 and LIMIT2.

➔ For information about LIMIT1 and LIMIT2, refer to [57. 10 "Limit Setting Utility for FAIL COUNTER and PASS/FAIL COUNTER \(for the T5593\)" on page 57-54.](#)

The counter is read in ascending order starting from DUT 1.

If multiple channel numbers are specified, they are read in the specified order in PNNO for each DUT, then test the next DUT.

The target DUTs (DDUT) are those that are specified for a parallel test.

However, variables for other DUTs are not changed.

Tip

A error results when PASS/FAIL COUNTER is being read and the counter is set to FAIL COUNTER.

◆ Usage Examples

When reading the PASS/FAIL counter for channel 49

```

INTE[1] ERR
INTE[1] MODE
INTE[1] STB
LOGI[2] CNT(128)          ; 1PIN*128DUT

PINLIST PNNO=P49
PNNO=IN3,DNRZ,BCLK3,CCLK3,DRECLK2,DRE2,OUTL1,IOC,EXPDL1,CPEDLY1,@
      [STRB1,CPE1],EXPA1,[STRB129,CPE1],EXPB129,<C10,C11>

MODE=1
STB=1

CALL STPFCNT(ERR,PNNO,MODE,STB)

IF ERR<>0 THEN ...      ; Error processing.

SELECT MPAT INHIBIT
START MPAT

CALL RDPFCNT(ERR,PNNO,CNT) ; CNT(1)      : DUT1    P49
                          ; CNT(2)      : DUT2    P49
                          ;      :      :      :
                          ; CNT(127)   : DUT127 P49
                          ; CNT(128)   : DUT128 P49

IF ERR<>0 THEN ...      ; Error processing.

```

57. 6 DQS Test (DQSvsDQ Test) Utility (for the T5593)

The utility is compatible with the T5593.

◆ Function

PASS/FAIL judgments for specified DQ pins can be made during each cycle by combining the DQS and DQ pin comparison results.

The combination used to determine the PASS/FAIL judgment for DQ pins can be specified as desired.

The test results (PASS/FAIL) specified by this utility can be reflected in the fail information (failure bits) in FM and the data logging of functional tests.

◆ Syntax

```
CALL DQSSEL3 (ERR, PNNO, PNMD, PDQS, TBL1, TBL2, FSEL1, FSEL2)
```

ERR

Used to store the DQS test (DQSvsDQ test) utility end status (normal/abnormal).

Specify using an integer or logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For more information, refer to [57. 12 "ERR Description" on page 57-62](#).

PNNO

Specifies channel numbers to assign to DQ pins used for a DQSvsDQ test.

Specify using the PINLIST variable.

PNMD

Specifies channels assigned to DQ pins

Use an integer (or logical) constant, or a variable between 1 and 4.

PNMD	Number of channels assigned
1	up to 21 channels
2	up to 9 channels
3	up to 45 channels
4	up to 12 channels

PDQS

Specifies channel numbers to assign to DQS pins used for a DQSvsDQ test.

Specify using the PINLIST variable.

Up to 2 channels can be specified.

The DQS pin selected first is considered the DQS1 pin, and the one selected next is considered the DQS2 pin.

The same channel cannot be specified for the DQS1 and DQS2 pins.

A maximum of two pins from the same DQS pin group can be selected for a single test.

Each DQS pin group comprises three pins.

When using two DQS pins, one pin each from two respective groups cannot be used.

For instance, it is impossible to select channel 54 as the DQS1 pin from DQS pin group 54 to 56 while also selecting channel 118 as the DQS2 pin from pin group 118 to 120.

The channel assigned to DQ pin can not be specified from the pin group that includes the channel assigned to DQS pin.

For instance, it is impossible to specify channels 52 and 53 for the channel assigned to DQ pin from the pin group of channels 51 to 53 when specifying 51 for the channel assigned to DQS pin.

When two groups of DQ and DQS pins exist in one device, DQSSEL3 is specified twice as follows:

◆ Example :

```

PINLIST PNNO1=P33-41
PINLIST PDQS1=P51
PINLIST PNNO2=P42-50
PINLIST PDQS2=P54
PNMD=2
CALL DQSSEL3(ERR,PNNO1,PNMD,PDQS1,TBL1,TBL2,FSEL1,FSEL2)
IF ERR<>0 THEN ... ; Error processing.
CALL DQSSEL3(ERR,PNNO2,PNMD,PDQS2,TBL1,TBL2,FSEL1,FSEL2)
IF ERR<>0 THEN ... ; Error processing.
MEAS MPAT

```

PNMD	Channels assigned to DQS pins (reference A) ^{*1}	Channels assigned to DQS pins (reference B) ^{*2}	Channels assigned to DQ pins
1	54 to 56	----	33 to 53
	118 to 120	----	97 to 117
	182 to 184	----	161 to 181
	246 to 248	----	225 to 245
2	51 to 53	----	33 to 41
	54 to 56	----	42 to 50
	115 to 117	----	97 to 105
	118 to 120	----	106 to 114
	179 to 181	----	161 to 169
	182 to 184	----	170 to 178
	243 to 245	----	225 to 233
	246 to 248	----	234 to 242
3	----	51 to 53, 54 to 56, 115 to 117, 118 to 120	33 to 50, 97 to 114
	----	179 to 181, 182 to 184, 243 to 245, 246 to 248	161 to 178, 225 to 242
4	54 to 56	51 to 53	33 to 41
	118 to 120	115 to 117	97 to 105
	182 to 184	179 to 181	161 to 169
	246 to 248	243 to 245	225 to 233

**1 Channels used as DQS pins*

**2 Channels used as DQS or DQ pins*

TBL1

Specifies the STRB test result for the STRB1 to 128.

Specify using variables or logical constants.

Specify the DQ pin STRB1 to 128 strobe test result (PASS/FAIL) from the strobe comparison results for the combination of the DQ pin STRB1 to 128, the STRB1 to 128 for the DQS1 and DQS2 pins, and the STRB129 to 256 for the DQS1 and DQS2 pins.

Specify "0" for PASS, and "1" for FAIL.

The test result cannot be changed if each STRB comparison result is PASS.

For instance, if the DQ pin STRB1 to 128 comparison result is FAIL, the DQS1 pin STRB1 to 128 comparison result is FAIL, the DQS1 pin STRB129 to 256 comparison result is PASS, the DQS2 pin STRB1 to 128 comparison result is PASS, and the DQS2 pin STRB129 to 256 comparison result is PASS, set Bit 28 to "0" so that the DQ pin STRB1 to 128 comparison result is set to PASS.

Tip

- If only DQ pin STRB1 to 128 strobe comparison results and DQS1 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 3, 4 to 7, 8 to 11, 12 to 15, 16 to 19, 20 to 23, and 24 to 27 to the values of Bits 28 to 31, respectively.
- If only DQ pin STRB1 to 128 strobe comparison results and DQS1 pin STRB129 to 256 strobe comparison results are combined, set Bits 0 to 7, 8 to 15, and 16 to 23 to the values of Bits 24 to 31, respectively.
- If only DQ pin STRB1 to 128 strobe comparison results and DQS2 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 15 to the values of Bits 16 to 31, respectively.
- If only DQ pin STRB1 to 128 strobe comparison results and DQS2 pin STRB129 to 256 comparison results are combined, set Bits 16, 17 and 18, 19 and 20, 21 and 22, 23 and 24, 25 and 26, 27 and 28, and 29 to the values of Bits 30 to 31, respectively, and 0, 1 and 2, 3 and 4, 5 and 6, 7 and 8, 9 and 10, 11 and 12, and 13 to the values of Bits 14 and 15, respectively.
- If only DQ pin STRB1 to 128 strobe comparison results, DQS1 STRB1 to 128 strobe comparison results, and DQS1 pin STRB129 to 256 strobe comparison results are combined, set Bits 0 to 7, 8 to 15, and 16 to 23 to the values of Bits 24 to 31, respectively.
- If only DQ pin STRB1 to 128 strobe comparison results, DQS1 STRB1 to 128 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 15, 8 to 15, and 16 to 23 to the values of Bits 24 to 31, respectively.
- If only DQ pin STRB1 to 128 strobe comparison results, DQS1 STRB1 to 128 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 16 to 19, 20 to 23, and 24 to 27 to the values of Bits 28 to 31, and Bits 0 to 3, 4 to 7, and 8 to 11 to the values of Bits 12 to 15.
- If only DQ pin STRB1 to 128 strobe comparison results, DQS1 STRB129 to 256 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 15, 8 to 15, and 16 to 23 to the values of Bits 24 to 31, respectively.
- If only DQ pin STRB1 to 128 strobe comparison results, DQS1 STRB129 to 256 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 16 to 23 to the values of Bits 24 to 31, Bits 0 to 7 to the values of Bits 8 to 15.
- If only DQ pin STRB1 to 128 strobe comparison results, DQS2 pin STRB1 to 128 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 24 and 25, 26 and 27, and 28 and 29 to the values of Bits 30 and 31, Bits 16 and 17, 18 and 19, and 20 and 21 to the values of Bits 22 and 23, and Bits 8 and 9, 10 and 11, and 12 and 13 to the values of Bits 14 and 15, Bits 0 and 1, 2 and 3, and 4 and 5 to the values of Bits 6 and 7.
- If only DQ pin STRB1 to 128 strobe comparison results, DQS1 STRB1 to 128 strobe comparison results, and DQS1 pin STRB129 to 256 strobe comparison results, and DQS2 pin STRB1 to 128 comparison results are combined, set Bits 0 to 15 to the values of Bits 16 to 31.

- *If only DQ pin STRB1 to 128 strobe comparison results, DQS1 STRB1 to 128 strobe comparison results, and DQS1 pin STRB129 to 256 strobe comparison results, and DQS2 pin STRB129 to 256 comparison results are combined, set Bits 16 to 23 to the values of Bits 24 to 31, set Bits 0 to 7 to the values of Bits 8 to 15.*
- *If only DQ pin STRB1 to 128 strobe comparison results, DQS1 STRB1 to 128 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results, and DQS2 pin STRB129 to 256 comparison results are combined, set Bits 24 to 27 to the values of Bits 28 to 31, set Bits 16 to 19 to the values of Bits 20 to 23, set 8 to 11 to the values of Bits 12 to 15, set 0 to 3 to the values of Bits 4 to 7.*
- *If only DQ pin STRB1 to 128 comparison results, DQS1 pin STRB129 to 256 comparison results, DQS2 pin STRB1 to 128 comparison results, DQS2 pin STRB129 to 256 comparison results are combined, set Bits 28 and 29 to the values of Bits 30 and 31, set Bits 24 and 25 to the values of Bits 26 and 27, set Bits 20 and 21 to the values of Bits 22 and 23, set Bits 16 and 17 to the values of Bits 18 and 19, set Bits 12 and 13 to the values of Bits 14 and 15, set Bits 8 and 9 to the values of Bits 10 and 11, set Bits 4 and 5 to the values of Bits 6 and 7, set Bits 0 and 1 to the values of Bits 2 and 3.*

TBL2

Specifies the STRB test result for the STRB129 to 256.

Specify using variables or logical constants.

Specify the DQ pin STRB129 to 256 strobe test result (PASS/FAIL) from the strobe comparison results for the combination of the STRB129 to 256 for the DQ pin, the STRB1 to 128 for the DQS1 and DQS2 pins, and the STRB129 to 256 for the DQS1 and DQS2 pins.

Specify "0" for PASS, and "1" for FAIL.

The test result cannot be changed if each STRB comparison result is PASS.

For instance, if the DQ pin STRB129 to 256 comparison result is FAIL, the DQS1 pin STRB1 to 128 comparison result is FAIL, the DQS1 pin STRB129 to 256 comparison result is PASS, the DQS2 pin STRB1 to 256 comparison result is PASS, and the DQS2 pin STRB129 to 256 comparison result is PASS, set Bit 28 to "0" so that the DQ pin STRB129 to 256 comparison result is set to PASS.

Tip

- If only DQ pin STRB129 to 256 strobe comparison results and DQS1 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 3, 4 to 7, 8 to 11, 12 to 15, 16 to 19, 20 to 23, and 24 to 27 to the values of Bits 28 to 31, respectively.
- If only DQ pin STRB129 to 256 strobe comparison results and DQS1 pin STRB129 to 256 strobe comparison results are combined, set Bits 0 to 7, 8 to 15, and 16 to 23 to the values of Bits 24 to 31, respectively.
- If only DQ pin STRB129 to 256 strobe comparison results and DQS2 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 15 to the values of Bits 16 to 31, respectively.
- If only DQ pin STRB129 to 256 strobe comparison results and DQS2 pin STRB129 to 256 comparison results are combined, set Bits 16, 17 and 18, 19 and 20, 21 and 22, 23 and 24, 25 and 26, 27 and 28, and 29 to the values of Bits 30 to 31, respectively, and 0, 1 and 2, 3 and 4, 5 and 6, 7 and 8, 9 and 10, 11 and 12, and 13 to the values of Bits 14 and 15, respectively.
- If only DQ pin STRB129 to 256 strobe comparison results, DQS1 STRB1 to 128 strobe comparison results, and DQS1 pin STRB129 to 256 strobe comparison results are combined, set Bits 0 to 7, 8 to 15, and 16 to 23 to the values of Bits 24 to 31, respectively.
- If only DQ pin STRB129 to 256 strobe comparison results, DQS1 STRB1 to 128 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 15, 8 to 15, and 16 to 23 to the values of Bits 24 to 31, respectively.
- If only DQ pin STRB129 to 256 strobe comparison results, DQS1 STRB1 to 128 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 16 to 19, 20 to 23, and 24 to 27 to the values of Bits 28 to 31, and Bits 0 to 3, 4 to 7, and 8 to 11 to the values of Bits 12 to 15.
- If only DQ pin STRB129 to 256 strobe comparison results, DQS1 STRB129 to 256 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 15 to the values of Bits 16 to 31, respectively.
- If only DQ pin STRB129 to 256 strobe comparison results, DQS1 STRB129 to 256 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 16 to 23 to the values of Bits 24 to 31, Bits 0 to 7 to the values of Bits 8 to 15.
- If only DQ pin STRB129 to 256 strobe comparison results, DQS2 pin STRB1 to 128 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 24 and 25, 26 and 27, and 28 and 29 to the values of Bits 30 and 31, Bits 16 and 17, 18 and 19, and 20 and 21 to the values of Bits 22 and 23, and Bits 8 and 9, 10 and 11, and 12 and 13 to the values of Bits 14 and 15, Bits 0 and 1, 2 and 3, and 4 and 5 to the values of Bits 6 and 7.
- If only DQ pin STRB129 to 256 strobe comparison results, DQS1 STRB1 to 128 strobe comparison results, and DQS1 pin STRB129 to 256 strobe comparison results, and DQS2 pin STRB1 to 128 comparison results are combined, set Bits 0 to 15 to the values of Bits 16 to 31.

- If only DQ pin STRB129 to 256 strobe comparison results, DQS1 STRB1 to 128 strobe comparison results, and DQS1 pin STRB129 to 256 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 16 to 23 to the values of Bits 24 to 31, set Bits 0 to 7 to the values of Bits 8 to 15.
- If only DQ pin STRB129 to 256 strobe comparison results, DQS1 STRB1 to 128 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 24 to 27 to the values of Bits 28 to 31, Bits 16 to 19 to the values of Bits 20 to 23, Bits 8 to 11 to the values of Bits 12 to 15, Bits 0 to 3 to the values of Bits 4 to 7.
- If only DQ pin STRB129 to 256 strobe comparison results, and DQS1 pin 129 to 256 strobe comparison results, DQS2 STRB1 to 128 strobe comparison results, DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 28 and 29 to the values of Bits 30 and 31, Bits 24 and 25 to the values of Bits 26 and 27, Bits 20 and 21 to the values of Bits 22 and 23, Bits 16 and 17 to the values of Bits 18 and 19, Bits 12 and 13 to the values of Bits 14 and 15, Bits 8 and 9 to the values of Bits 10 and 11, Bits 4 and 5 to the values of Bits 6 and 7, Bits 0 and 1 to the values of Bits 2 and 3.

FSEL1

Select whether to capture either a DQ pin test result that does not use LOOKUP TABLE for the SC unit PIN-DFM or a DQ pin test result using LOOKUP TABLE.

Use an integer (or logical) constant, or a variable between 0 and 1.

FSEL1	Selection of test results
0	Captures DQ pin test results that do not use LOOKUP TABLE
1	Captures DQ pin test results that use LOOKUP TABLE

FSEL2

Select whether to capture either a DQ pin test result that does not use LOOKUP TABLE for the AFM or a DQ pin test result using LOOKUP TABLE.

Use an integer (or logical) constant, or a variable between 0 and 1.

FSEL2	Selection of test results
0	Captures DQ pin test results that do not use LOOKUP TABLE
1	Captures DQ pin test results that use LOOKUP TABLE

◆ Usage Examples

When setting DQS vs DQ test with channels 33 to 41 assigned to the DQ pins and channels 51 and 52 attached to the DQS pins

When using DQS1 pin STRB1 to 128 strobe, DQS1 pin STRB129 to 256 strobe, DQS2 pin STRB1 to 128 strobe, and DQS2 pin STRB129 to 256 strobe.

```
INTE[1]  ERR, PNMD, FSEL1, FSEL2
LOGI[2]  TBL1, TBL2

PINLIST  PNNO=P33-40
PINLIST  PDQS=P54-55

PNMD=1
TBL1=#12121212
TBL2=#21212120
FSEL1=0
FSEL2=0

CALL  DQSSEL3 (ERR, PNNO, PNMD, PDQS, TBL1, TBL2, FSEL1, FSEL2)

IF ERR<>0 THEN ...      ; Error processing.

MEAS MPAT SAMPLE
```

When using only DQS1 pin STRB1 to 128 strobe

```
INTE[1]  ERR, PNMD, FSEL1, FSEL2
LOGI[2]  TBL1, TBL2

PINLIST  PNNO=P33-40
PINLIST  PDQS=P54-55

PNMD=1
TBL1=#22222222      *1
TBL2=#22222222      *1
FSEL1=0
FSEL2=0

CALL  DQSSEL3 (ERR, PNNO, PNMD, PDQS, TBL1, TBL2, FSEL1, FSEL2)

IF ERR<>0 THEN ...      ; Error processing.

MEAS MPAT SAMPLE
```

**1 Set Bits 0 to 3, 4 to 7, 8 to 11, 12 to 15, 16 to 19, 20 to 23, and 24 to 27 to the values of Bits 28 to 31.*

57. 7 DQS Test (CYCLEvsCYCLE Test) Utility (for the T5593)

The utility is compatible with the T5593.

◆ Function

PASS/FAIL judgments for the specified DQS pins can be made by combining the comparison results from arbitrarily selected two or more strobes in continuous cycles.

The test results (PASS/FAIL) specified by this utility can be reflected in the fail information (failure bits) in FM and the data logging of functional tests.

◆ Syntax

```
CALL DQSSEL4 (ERR, PNNO, BCNT1, SSEL1, TBL1, BCNT2, SSEL2, TBL2, FSEL1, FSEL2)
```

ERR

Used to store the DQS test (CYCLEvsCYCLE test) utility end status (normal/abnormal).

Specify using an integer or logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For more information, refer to [57. 12 "ERR Description" on page 57-62](#).

PNNO

Specifies channel numbers to assign to DQS pins used for a CYCLEvsCYCLE test.

The following channels can be specified: 33 to 56, 97 to 120, 161 to 184 and 225 to 248.

Specify using the PINLIST variable.

BCNT1

Specifies the number of comparison results selected using SSEL1.

Use an integer (or logical) constant, or a variable between 1 and 4.

BCNT2

Specifies the number of comparison results selected using SSEL2.

Use an integer (or logical) constant, or a variable between 1 and 4.

SSEL1

Selects the strobes and cycles to be used.

Use an integer (or logical) constant, or a variable between 0 and 7.

When using variables to make specifications, use array variables with up to four elements.

➔ For information on values to specify, refer to [Figure 57-4](#) and [Figure 57-5](#).

SSEL2

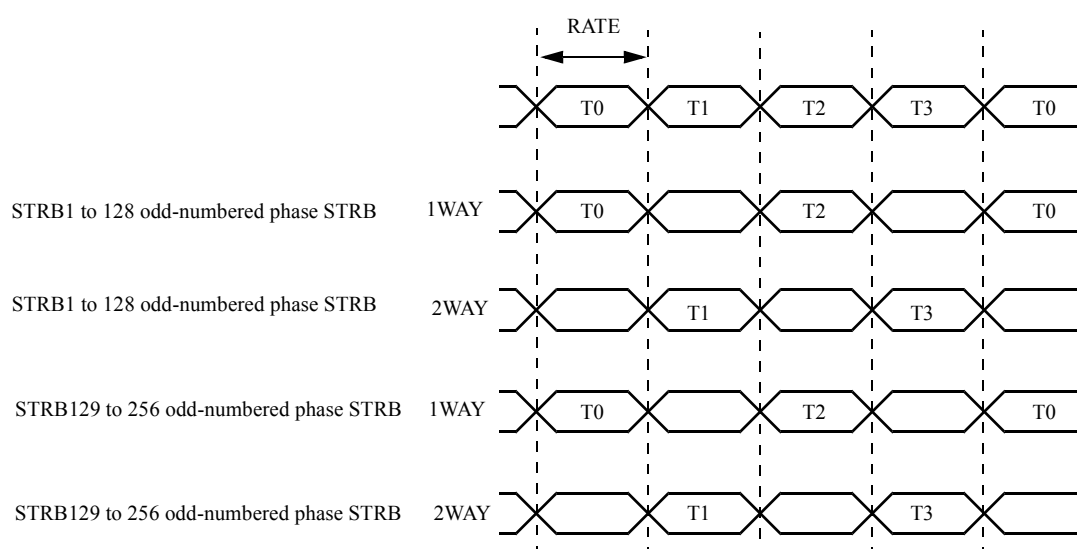
Selects the strobes and cycles to be used.

Use an integer (or logical) constant, or a variable between 0 and 7.

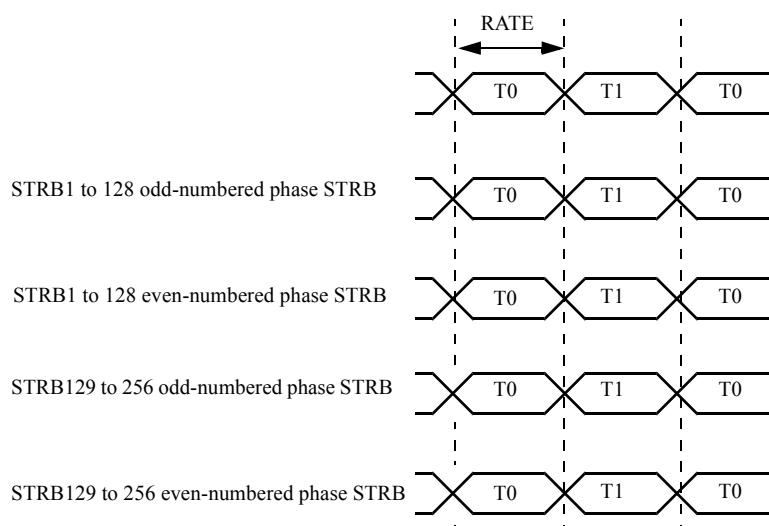
When using variables to make specifications, use array variables with up to four elements.

➔ For information on values to specify, refer to [Figure 57-4](#) and [Figure 57-5](#).

Figure 57-4 Relationship between SSEL1/SSEL2 and Strobes/Cycles (for 2WAY)



SSEL1 SSEL2	Strobes and Cycles to Be Used		
0	STRB1 to 128 odd-numbered phase STRB	During 1WAY	T0
1	STRB1 to 128 odd-numbered phase STRB	During 2WAY	T1
2	STRB129 to 256 odd-numbered phase STRB	During 1WAY	T0
3	STRB129 to 256 odd-numbered phase STRB	During 2WAY	T1
4	STRB1 to 128 odd-numbered phase STRB	During 1WAY	T2
5	STRB1 to 128 odd-numbered phase STRB	During 2WAY	T3
6	STRB129 to 256 odd-numbered phase STRB	During 1WAY	T2
7	STRB129 to 256 odd-numbered phase STRB	During 2WAY	T3

Figure 57-5 Relationship between SSEL1/SSEL2 and Strokes/Cycles (for 1WAY)

SSEL1 SSEL2	Strokes and Cycles to Be Used	
0	STRB1 to 128 odd-numbered phase STRB	T0
1	STRB1 to 128 even-numbered phase STRB	T0
2	STRB129 to 256 odd-numbered phase STRB	T0
3	STRB129 to 256 even-numbered phase STRB	T0
4	STRB1 to 128 odd-numbered phase STRB	T1
5	STRB1 to 128 even-numbered phase STRB	T1
6	STRB129 to 256 odd-numbered phase STRB	T1
7	STRB129 to 256 even-numbered phase STRB	T1

TBL1

Specifies the STRB test result for the STRB1 to 128.

Specify using variables or logical constants.

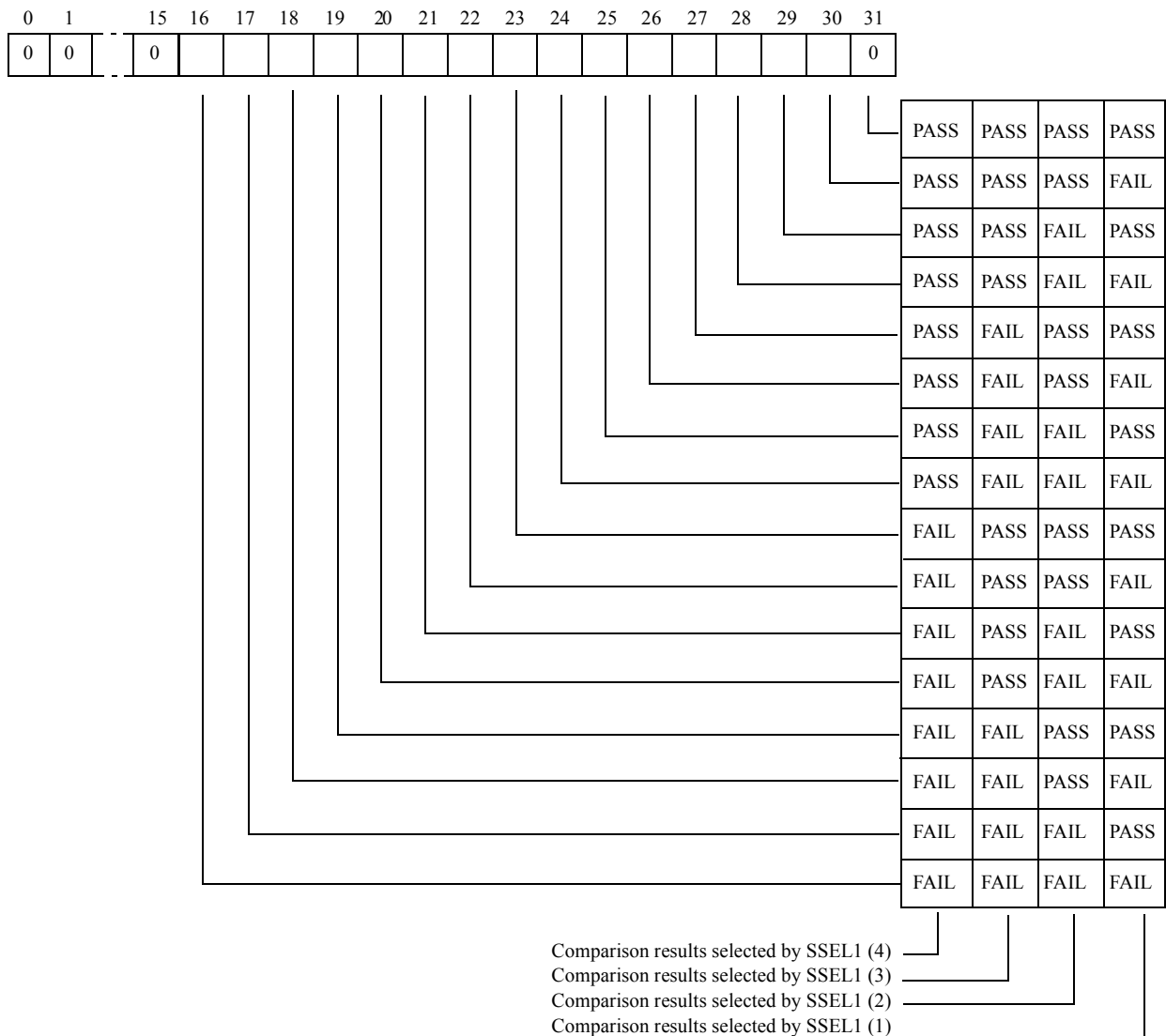
Specify the STRB1-to-128 odd-numbered phase strobe test result (PASS/FAIL) from the combinations selected from SSEL1(1) thru SSEL1(4).

Specify "0" for PASS, and "1" for FAIL.

The test result cannot be changed if each STRB comparison result is PASS.

57. 7 DQS Test (CYCLEvsCYCLE Test) Utility (for the T5593)

TBL1 format is shown below.



For example, if the comparison results selected from SSEL1(1) and SSEL1(2) and SSEL1(3) are FAIL, and if the comparison result selected from SSEL1(4) is PASS, set Bit 24 to "0" to set the STRB1 to 128 comparison result to PASS.

Tip

If specifying 1 using BCNT 1, specify bits 16 to 29 as "0."

If specifying 2 using BCNT 1, specify bits 16 to 27 as "0."

If specifying 3 using BCNT 1, specify bits 16 to 23 as "0."

TBL2

Specifies the STRB test result for the STRB129 to 256.

57. 7 DQS Test (CYCLEvsCYCLE Test) Utility (for the T5593)

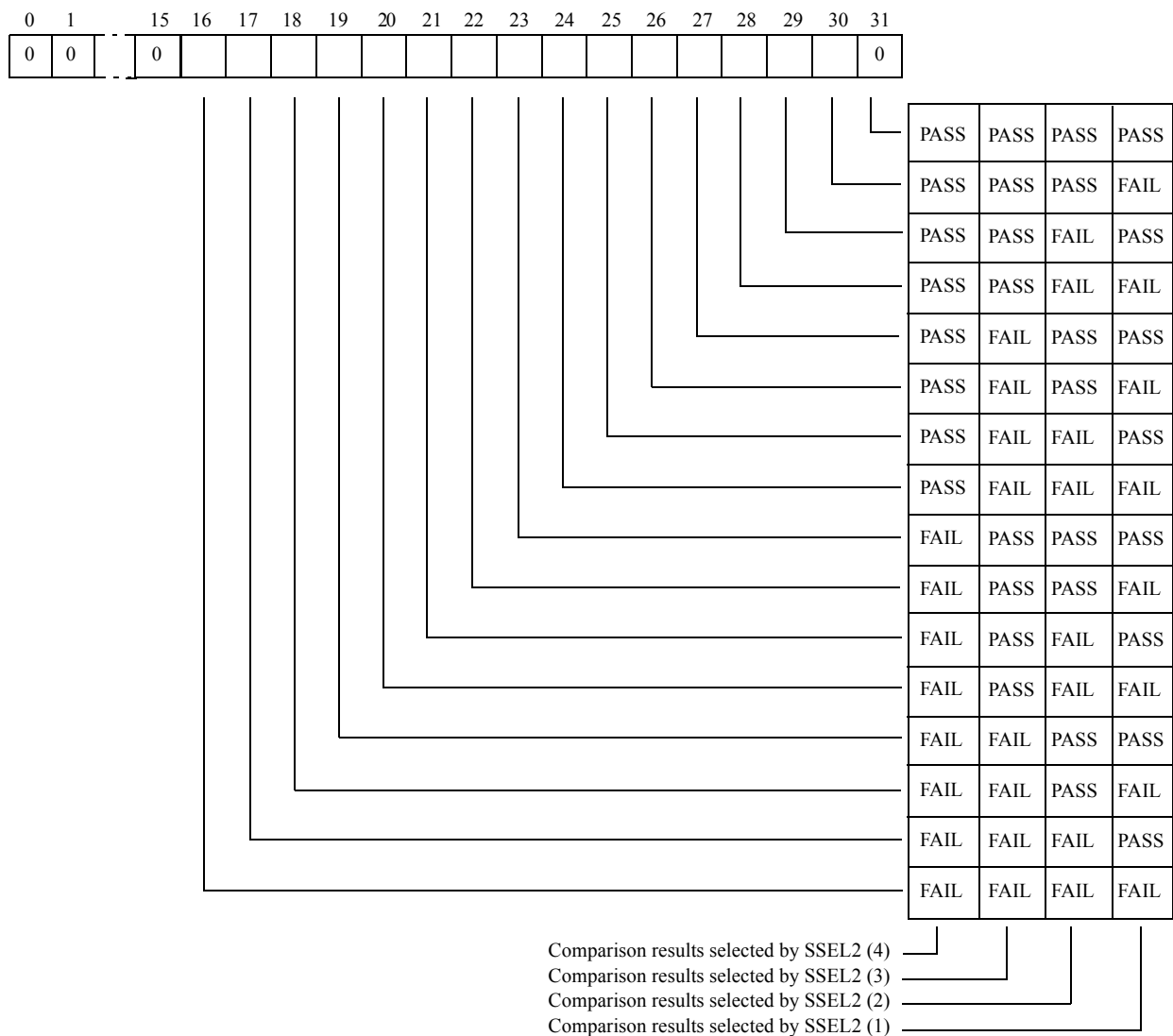
Specify using variables or logical constants.

Specify the test result (PASS/FAIL) from the combinations selected from SSEL2(1) thru SSEL2(4).

Specify "0" for PASS, and "1" for FAIL.

The test result cannot be changed if each STRB comparison result is PASS.

The TBL2 format is shown below.



For example, if the comparison results selected from SSEL2(1) and SSEL2(2) and SSEL2(3) are FAIL, and if the comparison result selected from SSEL2(4) is PASS, set Bit 24 to "0" to set the STRB129 to 256 comparison result to PASS.

Tip

If specifying 1 using BCNT2, specify bits 16 to 29 as "0."

If specifying 2 using BCNT2, specify bits 16 to 27 as "0."

If specifying 3 using BCNT2, specify bits 16 to 23 as "0."

FSEL1

Select whether to capture either a DQ pin test result that does not use LOOKUP TABLE for the SC unit PIN-DFM or a DQ pin test result using LOOKUP TABLE.

Use an integer (or logical) constant, or a variable between 0 and 1.

FSEL1	Selection of test results
0	Captures DQ pin test results that do not use LOOKUP TABLE
1	Captures DQ pin test results that use LOOKUP TABLE

FSEL2

Select whether to capture either a DQ pin test result that does not use LOOKUP TABLE for the AFM or a DQ pin test result using LOOKUP TABLE.

Use an integer (or logical) constant, or a variable between 0 and 1.

FSEL2	Selection of test results
0	Captures DQ pin test results that do not use LOOKUP TABLE
1	Captures DQ pin test results that use LOOKUP TABLE

! Important

When in bank multiplex mode, the STRB1 to 128 even-numbered phase comparison result is output as the STRB1 to 128 odd-numbered phase strobe test result, the STRB1 to 128 even-numbered phase strobe test result is always output as PASS.

The STRB129 to 256 even-numbered phase comparison result is output as the STRB129 to 256 odd-numbered phase strobe test result, the STRB129 to 256 even-numbered phase strobe test result is always output as PASS as well.

◆ Usage Examples

When a CYCLEvsCYCLE test is conducted for channel 49 in 2WAY interleaved mode

```

INTE[1] ERR
INTE[1] BCNT1,BCNT2,FSEL1,FSEL2
INTE[1] SSEL1(2),SSEL2(2)
LOGI[1] TBL1,TBL2

PINLIST PNNO=P49

BCNT1=2
SSEL1(1)=0
SSEL1(2)=3
TBL1=#8888
BCNT2=2
SSEL2(1)=6
SSEL2(2)=5
TBL2=#8888
FSEL1=0
FSEL2=0

CALL DQSSEL4(ERR,PNNO,BCNT1,SSEL1,TBL1,BCNT2,SSEL2,TBL2,FSEL1,FSEL2)

IF ERR<>0 THEN ... ; Error processing.

MEAS MPAT SAMPLE

```

57. 8 FAIL COUNTER Counter Setting Utility (for the T5593)

The utility is compatible with the T5593.

◆ Function

Sets FAIL COUNTER

This utility counts the number of FAILs.

When there are two (twelve-bit) FAIL COUNTERs, FAILs can be counted using two different conditions simultaneously.

Or, the two FAIL COUNTERs can be consolidated into a single, twenty-four-bit counter.

Set the STRB 1 to 128 strobes or the STRB 129 to 256 strobes using the PIN statement when using this utility.

Use this utility after executing the DQS testing (DQSvsDQ testing) utility (CALL DQSSEL3).

◆ Syntax

```
CALL STFLCNT (ERR, PNNO, MODE1, MODE2, CUPT, TBL1, TBL2)
```

ERR

Used to store the end status of the utility program for setting FAIL COUNTER (normal or abnormal end).

Specify using an integer or logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For more information, refer to [57. 12 "ERR Description" on page 57-62](#).

PNNO

Specifies pin numbers used to enable or disable FAIL COUNTER.

The following channels can be specified: 33 to 56, 97 to 120, 161 to 184 and 225 to 248.

Specify using the PINLIST variable.

MODE1

Selects DISABLE for FAIL COUNTER1 or the STRB for which to count FAILs.

Use an integer (or logical) constant, or a variable between 0 and 2.

MODE1	DISABLE and STRB selection for FAIL COUNTER1
0	Sets DISABLE for FAIL COUNTER1.
1	STRB1 to 128 STRB
2	STRB129 to 256 STRB

When used in 1WAY, odd-numbered phase STRB and even-numbered phase STRB are simultaneously selected.

MODE2

Selects DISABLE for FAIL COUNTER2 or the STRB for which to count FAILs.

Use an integer (or logical) constant, or a variable between 0 and 2.

MODE2	DISABLE and STRB selection for FAIL COUNTER2
0	Sets DISABLE for FAIL COUNTER2.
1	STRB1 to 128 STRB
2	STRB129 to 256 STRB

When used in 1WAY, odd-numbered phase STRB and even-numbered phase STRB are simultaneously selected.

When specifying "0" for MODE2, FAIL COUNTER1 changes from a twelve-bit to a twenty-four-bit counter.

CUPT

Selects whether to use a count condition table.

Use an integer (or logical) constant, or a variable between 0 and 1.

CUPT	Selects a count condition table
0	Does not use a count condition table
1	Uses a count condition table

TBL1

Specifies a count condition table when "1" is specified for CUPT.

The specification for TBL1 is reflected in the STRB count conditions selected in MODE1.

Specify using variables or logical constants.

Specify the count state as "1" from the combination of the DQ pin STRB and the DQS1 pin selected using MODE1, together with the DQS2 pin STRB1 to 128 STRB, DQS1 and DSQ2 pins STRB 129 to 256 STRB comparison result.

Specify the DQS1 and DQS2 pins using the DQS testing (DQSvsDQ testing) utility (CALL DQSSSEL3).

TBL1 cannot be omitted.

For example, to count when the DQ pin STRB comparison result selected by MODE1 is PASS, the DQS1 pin STRB1 to 128 comparison result is PASS, the DQS1 pin STRB129 to 256 comparison result is FAIL, the DQS2 pin STRB1 to 128 comparison result is PASS, and the DQS2 pin STRB129 to 256 comparison result is FAIL, specify "1" for Bit 11.

Tip

- If only the comparison results for STRB selected in DQ pin MODE1 and DQS1 pin STRB1 to 128 test results are combined, set Bits 0 to 3, 4 to 7, 8 to 11, 12 to 15, 16 to 19, 20 to 23, and 24 to 27 to the values of Bits 28 to 31, respectively.
- If only the comparison results for STRB selected in DQ pin MODE1 and DQS1 pin STRB129 to 256 strobe comparison results are combined, set Bits 0 to 7, 8 to 15, and 16 to 23 to the values of Bits 24 to 31, respectively.
- If only the comparison results for STRB selected in DQ pin MODE1 and DQS2 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 15 to the values of Bits 16 to 31, respectively.
- If only the comparison results for STRB selected in DQ pin MODE1 and DQS2 pin STRB129 to 256 test results are combined, set Bits 16, 17 and 18, 19 and 20, 21 and 22, 23 and 24, 25 and 26, 27 and 28, and 29 to the values of Bits 30 to 31, respectively, and 0, 1 and 2, 3 and 4, 5 and 6, 7 and 8, 9 and 10, 11 and 12, and 13 to the values of Bits 14 and 15, respectively.
- If only the comparison results for STRB selected in DQ pin MODE1 and DQS1 pin STRB1 to 128 strobe comparison results, and DQS1 pin STRB129 to 256 strobe comparison results are combined, set Bits 0 to 7, 8 to 15, and 16 to 23 to the values of Bits 24 to 31, respectively.
- If only the comparison results for STRB selected in DQ pin MODE1, DQS1 pin STRB1 to 128 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 15 to the values of Bits 16 to 31.
- If only the comparison results for STRB selected in DQ pin MODE1, DQS1 pin STRB1 to 128 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 16 to 19, 20 to 23, and 24 to 27 to the values of Bits 28 to 31, and the values of Bits 0 to 3, 4 to 7, and 8 to 11 to the values of Bits 12 to 15.
- If only the comparison results for STRB selected in DQ pin MODE1, DQS1 pin STRB129 to 256 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 15 to the values of Bits 16 to 31.
- If only the comparison results for STRB selected in DQ pin MODE1 and DQS1 pin STRB129 to 256 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 16 to 23 to the values of Bits 24 to 31, Bits 0 to 7 to the values of Bits 8 to 15.
- If only the comparison results for STRB selected in DQ pin MODE1, DQS2 pin STRB1 to 128 test results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 24 and 25, Bits 26 and 27, and Bits 28 and 29 to the values of Bits 30 and 31, Bits 16 and 17, Bits 18 and 19, and Bits 20 and 21 to the values of Bits 22 and 23, Bits 8 and 9, Bits 10 and 11, and Bits 12 and 13 to the values of Bits 14 and 15, and Bits 0 and 1, Bits 2 and 3, and Bits 4 and 5 to the values of Bits 6 and 7.

- *If only the comparison results for STRB selected in DQ pin MODE1, DQS1 pin STRB1 to 128 strobe comparison results, and DQS1 pin STRB129 to 256 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 15 to the values of Bits 16 to 31.*
- *If only the comparison results for STRB selected in DQ pin MODE1 and DQS1 pin STRB1 to 128 strobe comparison results, and DQS1 pin STRB129 to 256 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 16 to 23 to the values of Bits 24 to 31, Bits 0 to 7 to the values of Bits 8 to 15.*
- *If only the comparison results for STRB selected in DQ pin MODE1, DQS1 pin STRB1 to 128 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results, DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 24 to 27 to the values of Bits 28 to 31, Bits 16 to 19 to the values of Bits 20 to 23, Bits 8 to 11 to the values of Bits 12 to 15, Bits 0 to 3 to the values of 4 to 7.*
- *If only the comparison results for STRB selected in DQ pin MODE1 and DQS1 pin STRB129 to 256 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 28 and 29 to the values of Bits 30 and 31, Bits 24 and 25 to the values of Bits 26 and 27, Bits 20 and 21 to the values of Bits 22 and 23, Bits 16 and 17 to the values of Bits 18 and 19, Bits 12 and 13 to the values of Bits 14 and 15, Bits 8 and 9 to the values of Bits 10 and 11, Bits 4 and 5 to the values of Bits 6 and 7, Bits 0 and 1 to the values of Bits 2 and 3.*

TBL2

Specifies a count condition table when "1" is specified for CUPT.

The specification for TBL2 is reflected in the STRB count conditions selected in MODE2.

Specify using variables or logical constants.

Specify the count state as "1" from the combination of the DQ pin STRB and the DQS1 pin selected using MODE2, together with the DQS2 pin STRB1 to 128 STRB, DQS1 and DSQ2 pins STRB 129 to 256 STRB comparison results.

Specify the DQS1 and DQS2 pins using the DQS testing (DQSvsDQ testing) utility (CALL DQSSEL3).

TBL2 cannot be omitted.

For example, to count when the DQ pin STRB comparison result selected by MODE2 is PASS, the DQS1 pin STRB1 to 128 comparison result is PASS, the DQS1 pin STRB129 to 256 comparison result is FAIL, the DQS2 pin STRB1 to 128 comparison result is PASS, and the DQS2 pin STRB129 to 256 comparison result is FAIL, specify "1" for Bit 11.

Tip

- If only the comparison results for STRB selected in DQ pin MODE2 and DQS1 pin STRB1 to 128 test results are combined, set Bits 0 to 3, 4 to 7, 8 to 11, 12 to 15, 16 to 19, 20 to 23, and 24 to 27 to the values of Bits 28 to 31, respectively.
- If only the comparison results for STRB selected in DQ pin MODE2 and DQS1 pin STRB129 to 256 strobe comparison results are combined, set Bits 0 to 7, 8 to 15, and 16 to 23 to the values of Bits 24 to 31, respectively.
- If only the comparison results for STRB selected in DQ pin MODE2 and DQS2 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 15 to the values of Bits 16 to 31, respectively.
- If only the comparison results for STRB selected in DQ pin MODE2 and DQS2 pin STRB129 to 256 test results are combined, set Bits 16, 17 and 18, 19 and 20, 21 and 22, 23 and 24, 25 and 26, 27 and 28, and 29 to the values of Bits 30 to 31, respectively, and 0, 1 and 2, 3 and 4, 5 and 6, 7 and 8, 9 and 10, 11 and 12, and 13 to the values of Bits 14 and 15, respectively.
- If only the comparison results for STRB selected in DQ pin MODE2 and DQS1 pin STRB1 to 128 strobe comparison results, and DQS1 pin STRB129 to 256 strobe comparison results are combined, set Bits 0 to 7, 8 to 15, and 16 to 23 to the values of Bits 24 to 31, respectively.
- If only the comparison results for STRB selected in DQ pin MODE2, DQS1 pin STRB1 to 128 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 15 to the values of Bits 16 to 31.
- If only the comparison results for STRB selected in DQ pin MODE2, DQS1 pin STRB1 to 128 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 16 to 19, 20 to 23, and 24 to 27 to the values of Bits 28 to 31, and the values of Bits 0 to 3, 4 to 7, and 8 to 11 to the values of Bits 12 to 15.
- If only the comparison results for STRB selected in DQ pin MODE2, DQS1 pin STRB129 to 256 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 15 to the values of Bits 16 to 31.
- If only the comparison results for STRB selected in DQ pin MODE2 and DQS1 pin STRB129 to 256 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 16 to 23 to the values of Bits 24 to 31, Bits 0 to 7 to the values of Bits 8 to 15.
- If only the comparison results for STRB selected in DQ pin MODE2, DQS2 pin STRB1 to 128 test results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 24 and 25, Bits 26 and 27, and Bits 28 and 29 to the values of Bits 30 and 31, Bits 16 and 17, Bits 18 and 19, and Bits 20 and 21 to the values of Bits 22 and 23, Bits 8 and 9, Bits 10 and 11, and Bits 12 and 13 to the values of Bits 14 and 15, and Bits 0 and 1, Bits 2 and 3, and Bits 4 and 5 to the values of Bits 6 and 7.

57. 8 FAIL COUNTER Counter Setting Utility (for the T5593)

- If only the comparison results for STRB selected in DQ pin MODE2, DQS1 pin STRB1 to 128 strobe comparison results, and DQS1 pin STRB129 to 256 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results are combined, set Bits 0 to 15 to the values of Bits 16 to 31.
- If only the comparison results for STRB selected in DQ pin MODE2 and DQS1 pin STRB1 to 128 strobe comparison results, and DQS1 pin STRB129 to 256 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 16 to 23 to the values of Bits 24 to 31, Bits 0 to 7 to the values of Bits 8 to 15.
- If only the comparison results for STRB selected in DQ pin MODE2, DQS1 pin STRB1 to 128 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 24 to 27 to the values of Bits 28 to 31, Bits 16 to 19 to the values of Bits 20 to 23, Bits 8 to 11 to the values of Bits 12 to 15, Bits 0 to 3 to the values of Bits 4 to 7.
- If only the comparison results for STRB selected in DQ pin MODE2 and DQS1 pin STRB129 to 256 strobe comparison results, and DQS2 pin STRB1 to 128 strobe comparison results, and DQS2 pin STRB129 to 256 strobe comparison results are combined, set Bits 28 and 29 to the values of Bits 30 and 31, Bits 24 and 25 to the values of Bits 26 and 27, Bits 20 and 21 to the Bits 22 and 23, Bits 16 and 17 to the values of Bits 18 and 19, Bits 12 and 13 to the values of Bits 14 and 15, Bits 8 and 9 to the values of Bits 10 and 11, Bits 4 and 5 to the values of Bits 6 and 7, Bits 0 and 1 to the values of Bits 2 and 3.

- FAIL COUNTER is cleared every time a MEAS or START MPAT statement is executed. It cannot be cleared, however, by MEAS/START MPAT(1).
- The following timings can be performed to set FAIL COUNTER to DISABLE.
 - Starting the program
 - Executing the RESET TESTER statement
 - The current interleaved mode is changed using the SET ILMODE statement.
- A limit can be set for the counter value.
 - ➔ For details about limit value settings, refer to [57. 10 "Limit Setting Utility for FAIL COUNTER and PASS/FAIL COUNTER \(for the T5593\)" on page 57-54.](#)

Also, the default value for the limit value in this utility is #FFF when the counter is twelve bits, and #FFFFFF when it is twenty-four bits.

Tip

PASS/FAIL COUNTER and FAIL COUNTER cannot be mixed together.

When using the FAIL COUNTER setting utility (CALL STFLCNT) after the PASS/FAIL COUNTER setting utility (CALL STPFCNT), set the MODE argument of the PASS/FAIL COUNTER setting utility to DISABLE.

The same STRB can not be selected for the MODE1 and MODE2.

◆ Usage Examples

When setting FAIL COUNTER to ENABLE for channels 33 to 40

```

INTE[1] ERR,PNMD,FSEL1,FSEL2
INTE[1] MODE1,MODE2
INTE[1] CUPT
LOGI[2] TBL1,TBL2

PINLIST PNNO=P33-40
PINLIST PDQS=P54-55
PNNO=IN3,DNRZ,BCLK3,CCLK3,DRECLK2,DRE2,OUTL1,IOC,EXPDL1,CPEDLY1,@
[STRB1,CPE1],EXPA1,[STRB129,CPE1],EXPB129,<C10,C11>

PNMD=1
FSEL1=0
FSEL2=0
MODE1=1          ; STRB1 to 128 strobe selection (FAIL COUNTER1)
MODE2=2          ; STRB129 to 256 STRB selection (FAIL COUNTER2)
CUPT=1           ; Use count condition table
TBL1=#40         ; DQ pin STRB1 to 128 strobe comparison result is PASS,
                  ; DQS1 pin STRB1 to 128 strobe comparison result is FAIL,
                  ; DQS1 pin STRB129 to 256 strobe comparison result is FAIL,
                  ; DQS2 pin STRB1 to 128 strobe comparison result is PASS,
                  ; DQS2 pin STRB129 to 256 strobe comparison result is PASS,

TBL2=#40         ; DQ pin STRB129 to 256 strobe comparison result is PASS,
                  ; DQS1 pin STRB1 to 128 strobe comparison result is FAIL,
                  ; DQS1 pin STRB129 to 256 strobe comparison result is FAIL,
                  ; DQS2 pin STRB1 to 128 strobe comparison result is PASS,
                  ; DQS2 pin STRB129 to 256 strobe comparison result is PASS,

CALL DQSSEL3(ERR,PNNO,PNMD,PDQS,TBL1,TBL2,FSEL1,FSEL2)
IF ERR<>0 THEN ...      ; Error processing.

CALL STFLCNT(ERR,PNNO,MODE1,MODE2,CUPT,TBL1,TBL2)
IF ERR<>0 THEN ...      ; Error processing.

SELECT MPAT INHIBIT
START MPAT

```

57. 9 Utility Used to Read Data from FAIL COUNTER (for the T5593)

The utility is compatible with the T5593.

◆ Function

Reads FAIL COUNTER.

Reads the number of FAILs counted by the FAIL COUNTER setting utility (CALL STFLCNT).

Execute the FAIL COUNTER setting utility (CALL STFLCNT) in advance when using this utility.

◆ Syntax

```
CALL RDFLCNT (ERR, PNNO, BANK, CNT1, CNT2)
```

ERR

Used to store the end status of the FAIL COUNTER readout utility program (normal or abnormal end).

Specify using an integer or logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For more information, refer to [57. 12 "ERR Description" on page 57-62](#).

PNNO

Specifies channel numbers used to read data from FAIL COUNTER.

The following channels can be specified: 33 to 56, 97 to 120, 161 to 184 and 225 to 248.

Specify using the PINLIST variable.

BANK

Specifies banks used to read data from FAIL COUNTER.

Use an integer (or logical) constant, or a variable between 0 and 2.

BANK	Bank number
0	All banks
1	Bank1
2	Bank2

When all banks are specified in 1WAY, odd-numbered phase strobe and even-numbered phase strobe FAIL COUNTER are read out.

When bank1 is specified in 1WAY, odd-numbered phase strobe FAIL COUNTER is read out.

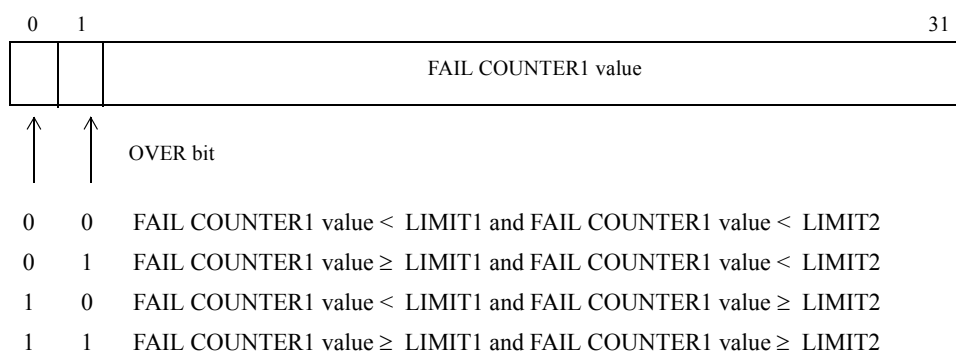
When bank2 is specified in 1WAY, even-numbered phase strobe FAIL COUNTER is read out.

CNT1

Specifies the variable with which the read FAIL COUNTER1 values are stored.

Use a logical variable with 2 HW (32 bits).

Specify an array variable whose size is the number of channels × the number of DUTs in order to read the counter.



When setting the LIMIT value, use the PASS/FAIL COUNTER and the FAIL COUNTER limit setting utility (CALL STLMT).

Reading takes place in order from DUT1.

If multiple channel numbers are specified, they are read in the specified order in PNNO for each DUT, then test the next DUT.

The target DUTs (DDUT) are those that are specified for a parallel test.

However, variables for other DUTs are not changed.

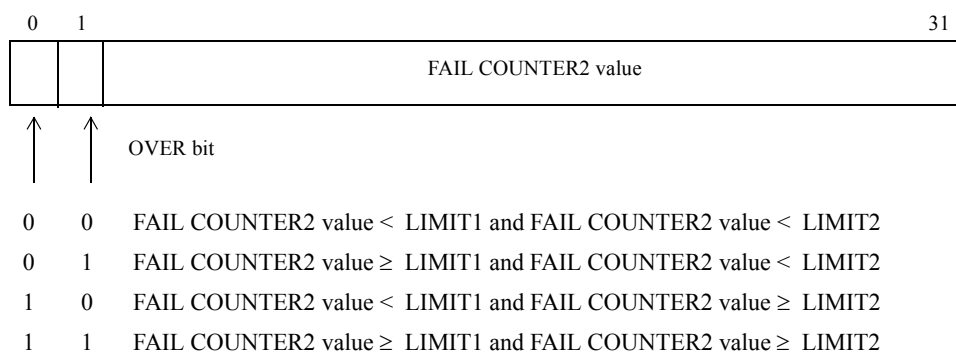
Also, contents of the variables cannot be changed when the MODE1 setting for the FAIL COUNTER setting utility (CALL STFLCNT) is DISABLE.

CNT2

Specifies the variable with which the read FAIL COUNTER2 values are stored.

Use a logical variable with 2 HW (32 bits).

Specify an array variable whose size is the number of channels × the number of DUTs in order to read the counter.



57. 9 Utility Used to Read Data from FAIL COUNTER (for the T5593)

When setting the LIMIT value, use the PASS/FAIL COUNTER and the FAIL COUNTER limit setting utility (CALL STLMT).

Reading takes place in order from DUT1.

If multiple channel numbers are specified, they are read in the specified order in PNNO for each DUT, then test the next DUT.

The target DUTs (DDUT) are those that are specified for a parallel test.

However, variables for other DUTs are not changed.

Also, contents of the variables cannot be changed when the MODE2 setting for the FAIL COUNTER setting utility (CALL STFLCNT) is DISABLE.

Tip

*A error results when FAIL COUNTER is being read and the counter is set to PASS/FAIL COUNTER.
When using a count condition table in the FAIL COUNTER setting utility, 0 is read out as the value for FAIL COUNTER of DQS pin.*

◆ Usage Examples

When reading FAIL COUNTER for channels 33 to 40

```

INTE[1]  ERR,PNMD,FSEL1,FSEL2
INTE[1]  MODE1,MODE2
INTE[1]  CUPT
LOGI[2]  TBL1,TBL2
INTE[1]  CNT=4
LOGI[2]  CNT1(1024),CNT2(1024)

PINLIST  PNNO=P33-40
PINLIST  PDQS=P54-55
PNNO=IN3,DNRZ,BCLK3,CCLK3,DRECLK2,DRE2,OUTL1,IOC,EXPDL1,CPEDLY1,@
        [STRB1,CPE1],EXPA1,[STRB129,CPE1],EXPB129,<C10,C11>

PNMD=1
FSEL1=0
FSEL2=0
MODE1=1          ; STRB1 to 128 strobe selection (FAIL COUNTER1)
MODE2=2          ; STRB129 to 256 STRB selection (FAIL COUNTER2)
CUPT=1           ; Use count condition table
TBL1=#40         ; DQ pin STRB1 to 128 strobe comparison result is PASS,
                  ; DQS1 pin STRB1 to 128 strobe comparison result is FAIL,
                  ; DQS1 pin STRB129 to 256 strobe comparison result is FAIL,
                  ; DQS2 pin STRB1 to 128 strobe comparison result is PASS,
                  ; DQS2 pin STRB129 to 256 strobe comparison result is PASS,

TBL2=#40         ; DQ pin STRB129 to 256 strobe comparison result is PASS,
                  ; DQS1 pin STRB1 to 128 strobe comparison result is FAIL,
                  ; DQS1 pin STRB129 to 256 strobe comparison result is FAIL,
                  ; DQS2 pin STRB1 to 128 strobe comparison result is PASS,
                  ; DQS2 pin STRB129 to 256 strobe comparison result is PASS,

CALL  DQSSEL3(ERR,PNNO,PNMD,PDQS,TBL1,TBL2.FSEL1.FSEL2)
IF ERR<>0 THEN ...      ; Error processing.

CALL  STFLCNT(ERR,PNNO,MODE1,MODE2,CUPT,TBL1,TBL2)
IF ERR<>0 THEN ...      ; Error processing.

SELECT MPAT INHIBIT
START MPAT

BANK=0           ; all banks

```

57. 9 Utility Used to Read Data from FAIL COUNTER (for the T5593)

```
                                ; STRB1 to 128 strobe STRB129 to 256 strobe
CALL RDFLCNT(ERR,PNN0,BANK,CNT1,CNT2) ; CNT1(1)          ,CNT2(1)      : DUT1   P33
                                ; CNT1(2)          ,CNT2(2)      : DUT1   P34
                                ;      :              :          :
                                ; CNT1(7)          ,CNT2(7)      : DUT1   P39
                                ; CNT1(8)          ,CNT2(8)      : DUT1   P40
                                ;      :              :          :
                                ; CNT1(1015)       ,CNT2(1015)   : DUT128 P33
                                ; CNT1(1016)       ,CNT2(1016)   : DUT128 P34
                                ;      :              :          :
                                ; CNT1(1023)       ,CNT2(1023)   : DUT128 P39
                                ; CNT1(1024)       ,CNT2(1024)   : DUT128 P40

IF ERR<>0 THEN ...      ; Error processing.
```

57.10 Limit Setting Utility for FAIL COUNTER and PASS/FAIL COUNTER (for the T5593)

The utility is compatible with the T5593.

◆ Function

Sets the limit value for FAIL COUNTER and PASS/FAIL COUNTER

Two different limit values can be set for one counter.

Counters 1 and 2 are available for each bank. Therefore, a limit value is judged for each bank.

Set the limit values for COUNTER2 (LIMIT1, 2) to 0 when specifying PASS/FAIL COUNTER limit values.

Set the limit values for COUNTER2 (LIMIT1, 2) to 0 when COUNTER2 is set to DISABLE using FAIL COUNTER.

◆ Syntax

```
CALL STLMT (ERR, PNNO, BANK, CT1LT1, CT1LT2, CT2LT1, CT2LT2)
```

ERR

Used to store the end status of the utility program for setting PASS/FAIL COUNTER and FAIL COUNTER limits (normal or abnormal end).

Specify using an integer or logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For more information, refer to [57. 12 "ERR Description" on page 57-62](#).

PNNO

Specifies channel numbers for which to set COUNTER limit conditions.

The following channels can be specified: 33 to 56, 97 to 120, 161 to 184 and 225 to 248.

Specify using the PINLIST variable.

BANK

Specifies banks for which to set two types of COUNTER limit values.

Use an integer (or logical) constant, or a variable between 0 and 2.

BANK	Bank number
0	All banks
1	Bank1
2	Bank2

57. 10 Limit Setting Utility for FAIL COUNTER and PASS/FAIL COUNTER (for the T5593)

When all banks are specified, the same limit value is specified for all of them.

When all banks are specified in 1WAY, set to odd/even-numbered phase strobes.

When bank1 is specified in 1WAY, set to odd-numbered phase strobe.

When bank2 is specified in 1WAY, set to even-numbered phase strobe.

CT1LT1

Specify LIMIT1 for COUNTER1.

Use an integer (or logical) constant, or a variable between 0 and 4095.

CT1LT2

Specify LIMIT2 for COUNTER1.

Use an integer (or logical) constant, or a variable between 0 and 4095.

CT2LT1

Specify LIMIT1 for COUNTER2.

Use an integer (or logical) constant, or a variable between 0 and 4095.

CT2LT2

Specify LIMIT2 for COUNTER2.

Use an integer (or logical) constant, or a variable between 0 and 4095.

When using PASS/FAIL COUNTER(COUNTER1), a twenty-four-bit counter is used.

When COUNTER2 is set to DISABLE by FAIL COUNTER, COUNTER1 uses a twenty-four-bit counter.

When a twenty-four-bit counter is used, the specifiable range for LIMIT1 (CT1LT1) and LIMIT2(CT1LT2) of COUNTER1 is 0 to 16777215 (#0 to #FFFFFF).

When this happens, specify "0" for COUNTER2 LIMIT1(CT2LT1) and LIMIT2(CT2LT2).

! Important

When the PASS/FAIL COUNTER setting utility (CALL STPFCNT) and the FAIL COUNTER setting utility (CALL STFLCNT) are set, the limits take on their default values.

➔ For default values, refer to each respective utility item.

◆ Usage Examples

When setting limits for FAIL COUNTER for channels 33 to 40

```

INTE[1] ERR
INTE[1] BANK
INTE[1] CT1LT1,CT1LT2,CT2LT1,CT2LT2
INTE[1] MODE1,MODE2
INTE[1] CUPT
LOGI[2] TBL1,TBL2

PINLIST PNNO=P33-40

MODE1=1          ; STRB1 to 128 strobe selection (FAIL COUNTER1)
MODE2=2          ; STRB129 to 256 STRB selection (FAIL COUNTER2)
CUPT=1           ; Use count condition table

TBL1=#40         ; DQ pin STRB1 to 128 strobe comparison result is PASS,
                  ; DQS1 pin STRB1 to 128 strobe comparison result is FAIL,
                  ; DQS1 pin STRB129 to 256 strobe comparison result is FAIL,
                  ; DQS2 pin STRB1 to 128 strobe comparison result is PASS,
                  ; DQS2 pin STRB129 to 256 strobe comparison result is PASS,
TBL2=#40         ; DQ pin STRB129 to 256 strobe comparison result is PASS,
                  ; DQS1 pin STRB1 to 128 strobe comparison result is FAIL,
                  ; DQS1 pin STRB129 to 256 strobe comparison result is FAIL,
                  ; DQS2 pin STRB1 to 128 strobe comparison result is PASS,
                  ; DQS2 pin STRB129 to 256 strobe comparison result is PASS,

CALL STFLCNT(ERR,PNNO,MODE1,MODE2,CUPT,TBL1,TBL2)
                  ;COUNTER1 LIMIT1=#FFF(default value)
                  ;COUNTER1 LIMIT2=#FFF(default value)
                  ;COUNTER2 LIMIT1=#FFF(default value)
                  ;COUNTER2 LIMIT2=#FFF(default value)

IF ERR<>0 THEN ... ; Error processing.

BANK=0           ; all banks
CT1LT1=10        ; COUNTER1 LIMIT1
CT1LT2=100       ; COUNTER1 LIMIT2
CT2LT1=10        ; COUNTER2 LIMIT1
CT2LT2=100       ; COUNTER2 LIMIT2

CALL STLMT(ERR,PNNO,BANK,CT1LT1,CT1LT2,CT2LT1,CT2LT2)
                  ; COUNTER1 LIMIT1=10
                  ; COUNTER1 LIMIT2=100
                  ; COUNTER2 LIMIT1=10
                  ; COUNTER2 LIMIT2=100

IF ERR<>0 THEN ... ; Error processing.
SELECT MPAT INHIBIT
START MPAT

```

57.11 Condition Information Reading Utility for FAIL COUNTER and PASS/FAIL COUNTER (for the T5593)

The utility is compatible with the T5593.

◆ Function

Reads the state of the OVER bit for each channel.

The state of the OVER bit is determined by the relationship between the values for COUNTER and LIMIT.

◆ Syntax

```
CALL RDCTCND (ERR, CTSEL, COND, BLK, BANK, CNT1, CNT2)
```

ERR

Used to store the end status of the utility program for reading PASS/FAIL COUNTER and FAIL COUNTER condition information (normal or abnormal end).

Specify using an integer or logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For more information, refer to [57. 12 "ERR Description" on page 57-62](#).

CTSEL

Specify a counter to read.

Use an integer (or logical) constant, or a variable between 1 and 3.

CTSEL	Counter selection
1	PASS/FAIL COUNTER
2	FAIL COUNTER
3	PASS/FAIL COUNTER and FAIL COUNTER.

COND:

Specifies a state for the OVER bit.

57. 11 Condition Information Reading Utility for FAIL COUNTER and PASS/FAIL COUNTER (for the T5593)

Use an integer (or logical) constant, or a variable between 1 and 4.

COND	Relationship between COUNTER and LIMIT values
1	COUNTER value < LIMIT1 and COUNTER value < LIMIT2
2	COUNTER value ≥ LIMIT1 and COUNTER value < LIMIT2
3	COUNTER value < LIMIT1 and COUNTER value ≥ LIMIT2
4	COUNTER value ≥ LIMIT1 and COUNTER value ≥ LIMIT2

When setting the value for LIMIT1 or LIMIT2, use the PASS/FAIL COUNTER and the FAIL COUNTER limit setting utility (CALL STLMT).

BLK

Specifies a block number corresponding to the channel number being read.

Use an integer (or logical) constant, or a variable with a value of 1, 3, 5, or 7.

Block number	Channel number
1	33 to 56
3	97 to 120
5	161 to 184
7	225 to 248

BANK

Specifies banks used to read information from the COUNTER condition information.

Use an integer (or logical) constant, or a variable between 0 and 2.

BANK	Bank number
0	All banks
1	Bank1
2	Bank2

When all banks are specified, the logical sum of bank1 and bank2 is read out.

When bank1 is specified in 1WAY, odd-numbered phase strobe condition information is read out.

When bank2 is specified in 1WAY, even-numbered phase strobe condition information is read out.

CNT1

Specifies the variable with which the read COUNTER1 values are stored.

Use a logical variable with 2 HW (32 bits).

Specify an array variable whose size is the number of channels × the number of DUTs in order to read the counter.

57. 11 Condition Information Reading Utility for FAIL COUNTER and PASS/FAIL COUNTER (for the T5593)

The read data is stored as thirty-two-bit logical-type data, and the number of channels is assigned to block numbers 1, 3, 5, and 7 for each thirty-two channels.

When there is a match with the conditions specified by COND, the bit for the corresponding channel becomes "0."

When there is no match with the conditions specified by COND, the bit for the corresponding channel becomes "1."

CNT2

Specifies the variable with which the read COUNTER2 values are stored.

Use a logical variable with 2 HW (32 bits).

Specify an array variable whose size is the number of channels × the number of DUTs in order to read the counter.

The read data is stored as thirty-two-bit logical-type data, and the number of channels is assigned to block numbers 1, 3, 5, and 7 for each thirty-two channels.

When there is a match with the conditions specified by COND, the bit for the corresponding channel becomes "0."

When there is no match with the conditions specified by COND, the bit for the corresponding channel becomes "1."

Data format for CNT1 and CNT2 (Socket conversion is not performed.)

0	1	2		21	22	23		31	
33	34	35		54	55	56			33 to 56 (Channels)
97	98	99		118	119	120			97 to 120 (Channels)
161	162	163		182	183	184			161 to 184 (Channels)
225	226	227		246	247	248			225 to 248 (Channels)

Reading takes place in order from DUT1.

The target DUTs (DDUT) are those that are specified for a parallel test.

However, variables for other DUTs are not changed.

For a twenty-four-bit counter, the result is stored in CNT1. When this happens, the contents of CNT remain unchanged.

!

Important

The bit that corresponds to the channel STRB is not specified for becomes "0".

The bit that corresponds to the channel other than the channels of read-out counter specified for CTSEL becomes "0".

When using a count conditions using an utility for FAIL COUNTER (CALL STFLCNT), FAIL COUTNER on the condition of "0" is read for the bit corresponds to the channel assigned to DQS pin.

57. 11 Condition Information Reading Utility for FAIL COUNTER and PASS/FAIL COUNTER (for the T5593)

The counter is cleared every time a MEAS or START MPAT statement is executed.
It cannot be cleared, however, by MEAS/START MPAT(1).

57. 11 Condition Information Reading Utility for FAIL COUNTER and PASS/FAIL COUNTER (for the T5593)

◆ Usage Examples

When reading FAIL COUNTER condition information for channels 33 to 40

```

INTE[1]  ERR,MODE1,MODE2,CUPT
LOGI[2]  TBL1,TBL2
INTE[1]  BANK,CT1LT1,CT1LT2,CT2LT1,CT2LT2
INTE[1]  CTSEL,COND,BLK
LOGI[2]  CNT1(128),CNT2(128)

PINLIST PNNO=P33-40
MODE1=1          ; STRB1 to 128 strobe selection (FAIL COUNTER1)
MODE2=2          ; STRB129 to 256 STRB selection (FAIL COUNTER2)
CUPT=1           ; Use count condition table
TBL1=#40         ; DQ pin STRB1 to 128 strobe comparison result is PASS,
                  ; DQS1 pin STRB1 to 128 strobe comparison result is FAIL,
                  ; DQS1 pin STRB129 to 256 strobe comparison result is FAIL,
                  ; DQS2 pin STRB1 to 128 strobe comparison result is PASS,
                  ; DQS2 pin STRB129 to 256 strobe comparison result is PASS,
TBL2=#40         ; DQ pin STRB129 to 256 strobe comparison result is PASS,
                  ; DQS1 pin STRB1 to 128 strobe comparison result is FAIL,
                  ; DQS1 pin STRB129 to 256 strobe comparison result is FAIL,
                  ; DQS2 pin STRB1 to 128 strobe comparison result is PASS,
                  ; DQS2 pin STRB129 to 256 strobe comparison result is PASS,
CALL STFLCNT(ERR,PNNO,MODE1,MODE2,CUPT,TBL1,TBL2)
IF ERR<>0 THEN ...      ; Error processing.

BANK=0           ; all banks
CT1LT1=10        ; COUNTER1 LIMIT1
CT1LT2=100       ; COUNTER1 LIMIT2
CT2LT1=10        ; COUNTER2 LIMIT1
CT2LT2=100       ; COUNTER2 LIMIT2

CALL STLMT(ERR,PNNO,BANK,CT1LT1,CT1LT2,CT2LT1,CT2LT2)
IF ERR<>0 THEN ...      ; Error processing.

SELECT MPAT INHIBIT
START MPAT

CTSEL=2
COND=1
BLK=1
CALL RDCTCND(ERR,CTSEL,COND,BLK,BANK,CNT1,CNT2)
                  ; CNT1(1)      ,CNT2(1)      : DUT1    P33-40
                  ; CNT1(2)      ,CNT2(2)      : DUT2    P33-40
                  ;      :          :          :
                  ; CNT1(127)   ,CNT2(127)   : DUT127 P33-40
                  ; CNT1(128)   ,CNT2(128)   : DUT128 P33-40
IF ERR<>0 THEN ...      ; Error processing.

```

57.12 ERR Description

ERR	Description
0	Normal termination
Other than 0	Abnormal end.
1	Cannot execute in this system.
2	Channel numbers are incorrectly specified (PNNO).
3	DQS pins are incorrectly selected (TYPE, PDQS).
4	Specifications for count condition table specifications (TB1, TBL2) or test results are incorrect.
5	The number of comparison results is incorrectly specified (BCNT1 or BCNT2).
6	The comparison result is incorrectly specified (SSEL1 or SSEL2).
7	The PASS/FAIL counter mode (enabled or disabled) is incorrect (MODE).
8	A condition under which the PASS/FAIL counter is counted is incorrectly specified (STB).
9	The variable used to store read counter values is incorrectly specified (CNT, CNT1, CNT2).
10	The system attempted to change data while the PG or TG was operating.
11	FAIL COUNTER is invalid or the STRB selections (MODE1, MODE2) are incorrect.
12	The specification (CUPT) for the count condition table is incorrect.
13	The high-speed clock driver channel cannot be used.
14	The number of channels assigned to DQS pins is specified (PNMD) incorrectly.
15	The test result capture selection (FSEL1, FSEL2) for PIN-DFM or the AFM is incorrect.
16	The bank setting (BANK) is incorrect.
17	The limit value setting (CT1LT1, CT1LT2, CT2LT1, CT2LT2) is incorrect.
18	The state of the COUNTER value is specified (COND) incorrectly with respect to the LIMIT value.
19	The specified block number (BLK) is incorrect.
20	The counter setting (CTSEL) is incorrect.
21	The selection mode for PASS/FAIL COUNTER enabled/disabled or FAIL COUNTER disabled or selection mode for STRB is incorrect.
22	PASS/FAIL COUNTER and FAIL COUNTER cannot be used with AUTO SCAN COUNTER at the same time.

58. Utility Used to Write Individual Data for Each DUT

The utility, which is used to write individual data for each DUT, allows the user to write manufacturing information and repair information to each memory devices (such as DRAM and FLASH) under test.

Normally, the same driver pattern data from the ALPG is used for all DUTs. However, this utility allows you to apply different driver patterns to each DUT using the AFM mask pattern generation function.

In addition to an I/O pin, a driver pin can be used with this utility, which allows you to allocate /WE to each DUT.

This utility consists of the following three programs.

- MFMSEND
Transfers driver pattern data from a data file to the AFM.
- IDSELFMT
Specifies a pin that uses the driver pattern transferred to the AFM.
- IDSELCLR
Clears the pin setting specified by IDSELFMT.

This utility can be used with the T5377, T5376 and T5375.

When using this utility, exclude any DUTs that are not assigned to the AFM from the test targets.

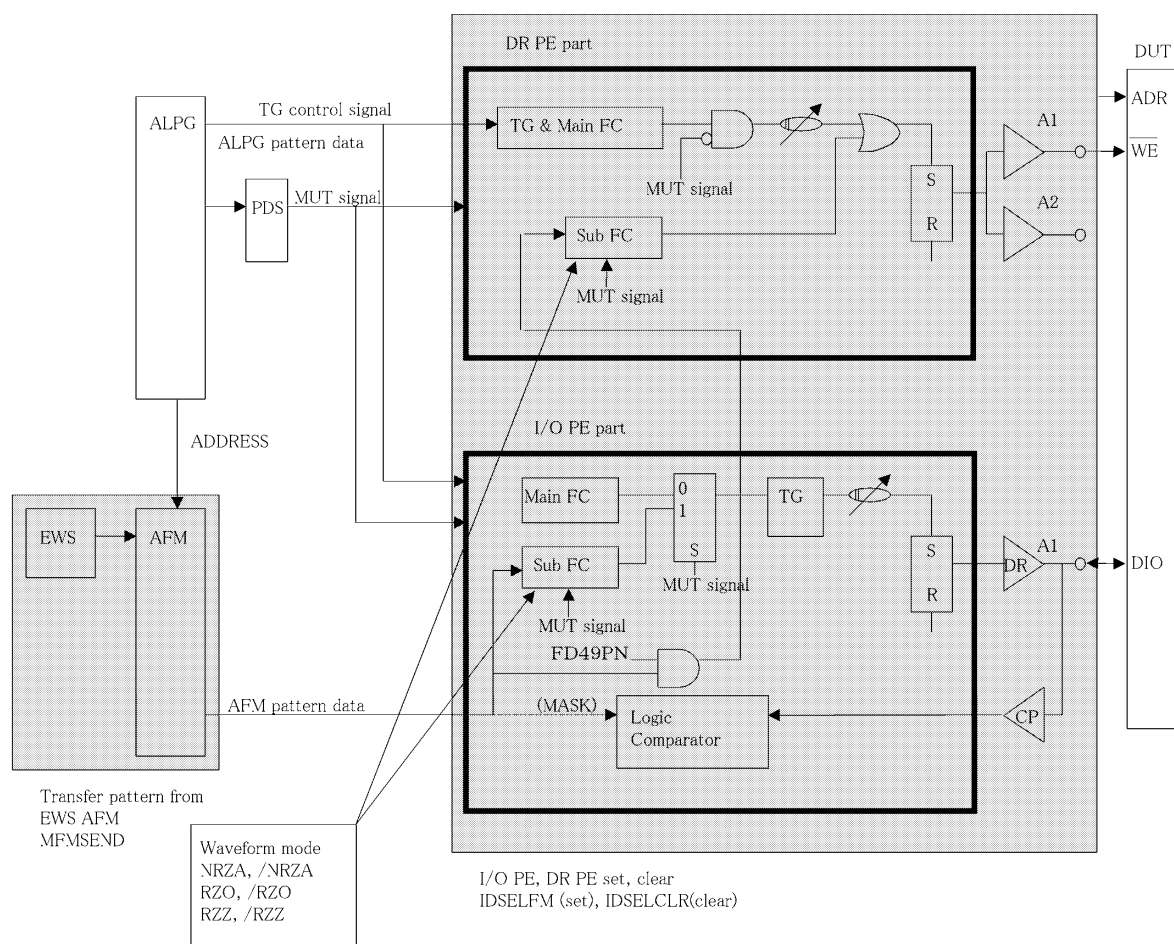
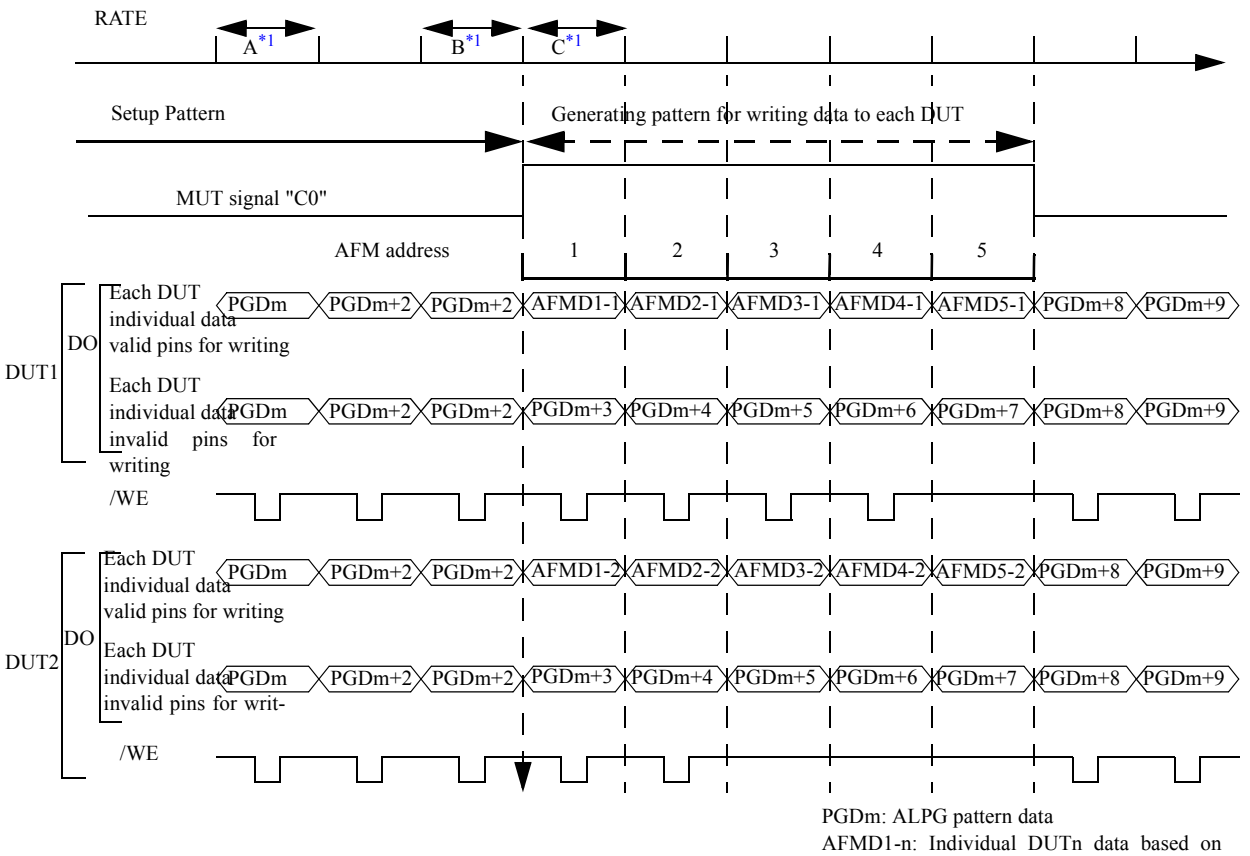
Figure 58-1 Block Diagram of the Function for Writing Individual Data to Each DUT

Figure 58-2 Timing Diagrams to be Used for Utility to Write Individual Data for Each DUT



*1 There are timing restrictions.
→ For details, refer to 6 of "Notes on Using IDSELUBM" on page 58-17.

58. 1 MFMSSEND

◆ Function

MFMSSEND is used to transfer driver pattern data stored in a data file (DAT file name) to AFM in 18-bit units.
The information included in one 18-bit unit appears in the table below.

Table 58-1 Information Included in One Unit (18 Bits) to be Transferred

Parallel test condition	AFM BITSIZE ^{*1}	Information included in one unit of data
PIN MODE=A With socket program Or, without socket program	1, 4, 8, 16, 18	Data for one DUT
	32, 36	Data for one DUT address. Data for pins 33 through 50 and 97 through 114 are contained in other data blocks.
PIN MODE=B With socket program	8, 9	Data for two DUTs set in the same Child
PIN MODE=C With socket program	4	Data for four DUTs set in the same Child

^{*1} MFMSSEND can only be executed for the AFM BITSIZE specified in the table.

Since MFMSSEND uses AFM mask patterns, it is necessary to set the MASK mode in AFM.

Any of the settings below can be specified.

- AFM ACTION=FMX>MASK
- AFM ACTION=FMX>MASK,PM>EXP
- AFM ACTION=FMX>MASK,PM>DBM

Data transferred to AFM can be confirmed using the READ AFM MASK statement or a fail memory data transfer utility (/FMXFER).

The data can be changed for each DUT using the AFM MASK statement.

Executing the RESET AFM ALL statement clears the transferred data.

◆ Syntax

```
CALL MFMSSEND(ERR,data file name)
```

ERR

Specifies a variable that stores the MFMSSEND termination status (normal termination or abnormal termination).

Specify using an integer or logical variable. value is 0, this indicates normal termination. When the ERR value is not 0, this indicates abnormal termination.

➔ For details, refer to ["ERR Description" on page 58-11](#).

data-file-name

Specifies a data file name.

- A STRING variable or character constant can be specified.
- Use up to 247 characters, including uppercase letters, lowercase, numbers, periods (.), and \$, to specify a file name.

- This data file is not compatible with the DIRLINK utility. The data in data files in the current directory or an arbitrary directory is transferred.

There are two methods using an arbitrary directory:

1. Add the directory name used to store a data file to the environment variable "ADI_ADPI" of the current customer name.
2. Specify a data file name, including the directory name. If the data file name begins with a slash (/), specify it using an absolute path in the file system of a host system. If the data file name does not begin with a slash (/), specify it using a relative path to the current customer directory from which the controller is usually started.

The following examples are provided to describe the two methods used when the data in "SamDat01" (which is stored in the /home/asx/user/ckdat directory) is transferred.

(Example 1)

Add the directory name to the current customer name.

```
%setenv ADI_ADPI ${ADI_ADPI}:/home/asx/user/ckdat
```

Specify only the file name in the CALL MFMSSEND statement in the test plan program.

```
PRO CK01
:
CALL MFMSSEND(ERR, "SamDat01")
:
END
```

(Example 2)

Specify a file name together with the directory name in the CALL MFMSSEND statement used in a test plan program.

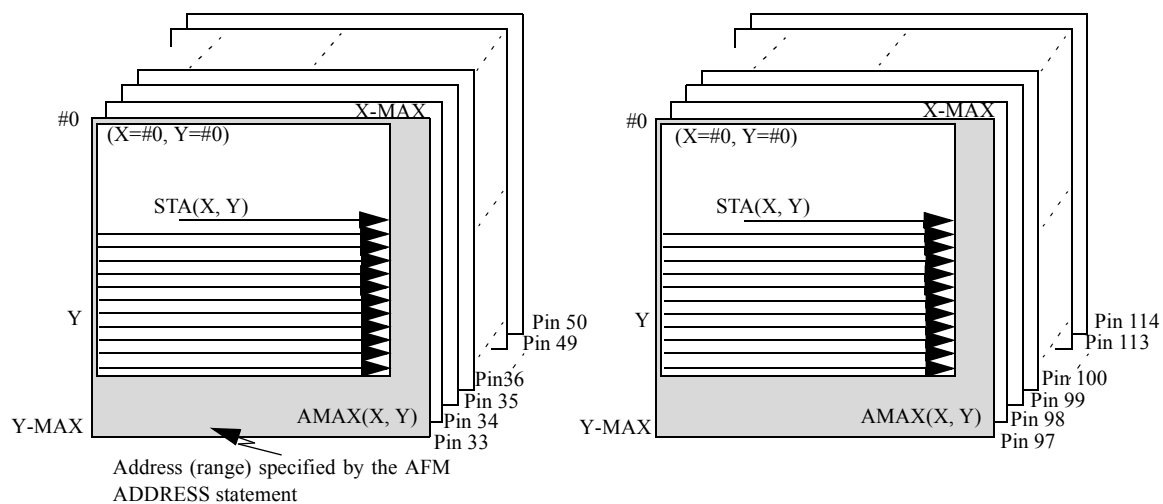
```
PRO CK01
:
CALL MFMSSEND(ERR, "/home/asx/user/ckdat/SamDat01")
:
END
```

◆ Data Transfer Method

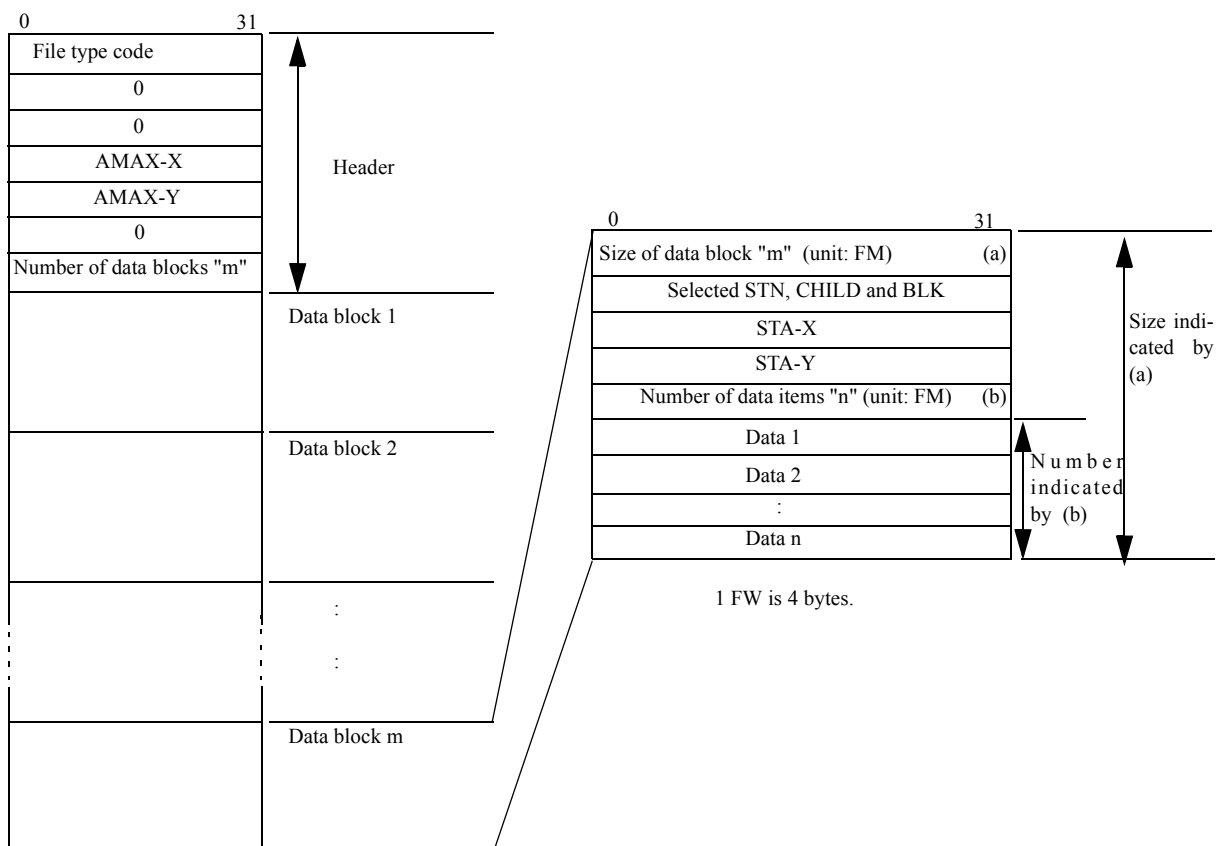
Concept of Data Transfer to AFM

Individual data for each DUT is transferred to AFM addresses ranging from the start address ([Figure 58-3](#) (X, Y) in Figure 58-3) to the end address (AMAX (X, Y) in Figure [Figure 58-3](#)) specified in the data file. A conceptual diagram of data transfer to AFM appears below.

➔ For detail about data file format, refer to ["Data file format" on page 58-6](#).

Figure 58-3 Conceptual Diagram of Data Transfer Method**Data file format**

The following data file format can be used.

**1. Details of data**

- File type code
#F7040000
- AMAX-X, AMAX-Y
Specifies the maximum X and Y addresses.
Specify values that do not exceed the X and Y addresses determined by the AFM ADDRESS statement
#0 ≤ AMAX-X ≤ #3FFFF
#0 ≤ AMAX-Y ≤ #3FFFF
- Number of data blocks
Specifies the number of data blocks in the file.
#1 ≤ Number of data blocks ≤ #1000
- Size of data block m (unit: FW)
#6 ≤ block m size ≤ #1FFFFFF8
- Selected STN, CHILD and BLK
Specifies the AFM where data is transferred and a DUT is assigned by selecting a Child, Station and pin block.
Specify 1 for the bit indicating transfer destination.
- STA-X, STA-Y
Specifies the X and Y addresses where the data transfer is started.
#0 ≤ STA-X ≤ AMAX-X
#0 ≤ STA-Y ≤ AMAX-Y
- The number of data
Specifies the number of data in data block m.
#1 ≤ Number of data ≤ #1FFFFFF3
- Data 1 to data "n"
Assign logical values of the driver pattern to the bits that correspond to the pin numbers shown in the figure below.

◆ Data for Pins 33 to 50

											1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	3	3	
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	
0															4	5	3	3	3	3	3	3	3	3	4	4	4	4	4	4	4	
															9	0	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8

◆ Data for Pins 97 to 114

											1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	3	3		
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1		
0															1	1	9	9	9	1	1	1	1	1	1	1	1	1	1	1	1		
															1	1	7	8	9	0	0	0	0	0	0	0	0	0	0	0	1	1	1
															3	4				0	1	2	3	4	5	6	7	8	9	0	1	2	

										1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	3	3	
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1
C	C	C	C	C	C	C	C	C	0	0	0	0	0	0	0	S	S	0	0	A	A	A	A	0	0	0	0	0	0	9	3
h	h	h	h	h	h	h	h	h								T	T			7	3	5	1						7	3	
i	i	i	i	i	i	i	i	i								N	N			7	3	5	1						7	3	
d	d	d	d	d	d	d	d	d								2	1			H	H	H	H						1	5	
A	B	C	D	E	F	G	H													7	3	5	1						4	0	
Child-1								Station number										Child-2						Pin block							

1. Station number specification

STN2	STN1	Parallel tests using two test stations	Parallel tests using one test station, or a single device test
0	0	Abnormal termination (ERR=11)	DUT at the Station being used for a test
0	1	DUT at Station 1	DUT at Station 1 ^{*1}
1	0	DUT at Station 2	DUT at Station 2 ^{*1}
1	1	DUTs at Station 1 and Station 2	Abnormal termination (ERR=11)

^{*1} If the number of the test station which is not being used for a test is specified, an abnormal termination (ERR = 11) occurs.

2. Specifying Child-1, Child-2, and pin block

• Child-1

Specify 1 for the bit corresponding to Child A to H (where the DUT targeted for data transfer is allocated).

Two or more pin blocks can be selected.

• Child-2

Specify 1 for the bit corresponding to Childs A1 to H1, A3 to H3, A5 to H5, and A7 to H7 (where the DUT targeted for data transfer is allocated).

A1 to H1	Indicates A1, B1, C1, D1, E1, F1, G1, and H1
A3 to H3	Indicates A3, B3, C3, D3, E3, F3, G3, and H3
A5 to H5	Indicates A5, B5, C5, D5, E5, F5, G5, and H5
A7 to H7	Indicates A7, B7, C7, D7, E7, F7, G7, and H7.

Two or more pin blocks can be selected.

- Pin blocks

Specify 1 for the bit corresponding to pins 33 to 50 and 97 to 114 (where the DUT targeted for data transfer is allocated). Two or more pin blocks can be selected.

- ◆ **When Creating Data for DUT11 and DUT12 Using the Following Socket Program**

```

SOCKET SOC1

DEFINE MULTIPLE 128DUT 1STN

BIT=9
PPS=1
PIN MODE=B
CHTYPE 11

DUT#      = ....PD1-12      PD26      P33-40      P49
DUT1      = ....PD1-12A     PD26A1    P33-40A1    P49A1
DUT2      = ....PD13-24A1   PD29A1    P41-48A1    P50A1
DUT3      = ....PD1-12A2    PD27A2    P97-104A1   P113A1
DUT4      = ....PD13-24A2   PD30A2    P105-112A1  P114A1
DUT5      = ....PD1-12A3    PD26A3    P33-40A3    P49A3
DUT6      = ....PD13-24A3   PD29A3    P41-48A3    P50A3
DUT7      = ....PD1-12A4    PD27A4    P97-104A3   P113A3
DUT8      = ....PD13-24A4   PD30A4    P105-112A3  P114A3
DUT9      = ....PD1-12A5    PD26A5    P33-40A5    P49A5
DUT10     = ....PD13-24A5   PD29A5    P41-48A5    P50A5
DUT11     = ....PD1-12A6    PD27A6    P97-104A5   P113A5
DUT12     = ....PD13-24A6   PD30A6    P105-112A5  P114A5
DUT13     = ....PD1-12A7    PD26A7    P33-40A7    P49A7
DUT14     = ....PD13-24A7   PD29A7    P41-48A7    P50A7
DUT15     = ....PD1-12A8    PD27A8    P97-104A7   P113A7
:
DUT16     = ....PD13-24A8   PD30A8    P105-112A7  P114A7
DUT128    = ....PD13-24H8   PD30H8    P105-112H7  P114H7
END

```


◆ ERR Description

ERR	Description
0	Normal termination
1	An internal error is detected during the write operation. (There is a mistake in the data file contents.)
2	Cannot execute in this system.
3	AFM assignment has not been completed. The AFM ACTION statement was not executed at the end of the AFM setting statements.
4	AFM double-speed mode cannot be specified for the AFM ADDRESS statement.
5	An illegal mode is specified in the AFM ACTION statement. ➔ For details, refer to "Concept of Data Transfer to AFM" on page 58-5 .
6	The AFM bit compression function cannot be used.
7	When PIN MODE = B or C is specified in the socket program, SINGLE mode cannot be specified in the AFM MULTIMODE statement.
8	An illegal character is used for the specified data file name.
9	The specified data file is not found.
10	The type of data file is incorrect.
11	AFM corresponding to the selected STN, CHILD and BLK cannot be found.
12	The data file address exceeds the value determined by the AFM ADDRESS statement.
13	The start address used for the write operation exceeds AMAX.
14	The number of data exceeds the number of addresses (AMAX) or the number of data is out of range. Or, the number of data is out of range.
15	A value other than 0 is stored in the data file area where 0 must be specified. ➔ For details, refer to "Data file format" on page 58-6 .
16	The specified number of data blocks is out of range.
17	The specified data block size is out of range.
18	The data file size is out of range.
19	Cannot be executed using the AFM BITSIZE specified. ➔ For details, refer to Table 58-1 of "Function" on page 58-3 .
20	The system attempted to change data while the PG or TG was operating.
21	DOUBLE cannot be executed using the SELECT AFM CONFIG statement when the AFM RATE is HS or MS.

◆ Notes on Using IDSELUBM

1. Data for a pin which has not been assigned to the AFM cannot be transferred.
2. The AFM bit compression function cannot be used.
3. Specify 0 for the start X and Y addresses in the AFM ADDRESS statement.
(Example) AFM ADDRESS = X0-7, Y0-9
4. Specify the data file size within the following range: $\#D \times 4 \text{ bytes} \leq \text{size} \leq \#1\text{FFFFFFF} \times 4 \text{ bytes}$.
To transfer data that exceeds this range, split the data file.
5. When MFMSSEND is executed, do not execute the following statements before the MEAS MPAT or START MPAT statement is executed:
 - RESET AFM ALL statement
 - RESET AFM MODE statement
 - AFM ACTION statement
6. While MFMSSEND is being executed, a CTRL/C command and the RESET switch of the control box is disabled.

58. 2 IDSELF

◆ Function

IDSELF is used to specify a device pin that is used for AFM driver patterns (containing individual data for each pin), not for a normal ALPG driver pattern. An I/O pin can be used for the AFM data. To clear this utility setting, use IDSELCLR.

◆ Syntax

```
CALL IDSELF(ERR, PIN, MUT, FD49PN)
```

ERR

Specifies a variable that stores the IDSELF termination status (normal termination or abnormal termination).

Specify using an integer or logical variable. If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For details, refer to ["ERR Description" on page 58-17](#).

PIN

Specifies a DUT I/O pin number that uses the AFM data. Specify using the PINLIST variable. Up to 40 pins can be specified for PIN.

When pin 51 is specified for PIN, the driver pattern data of pin 49 is used. In the same way, the following apply to pins 52, 115 and 116: When pin 52 is specified for PIN, the driver pattern data of pin 50 is used. When pin 115 is specified for PIN, the driver pattern data of pin 113 is used. When pin 116 is specified for PIN, the driver pattern data of pin 114 is used.

MUT

Specifies an MUT signal that is used to determine the cycle where the write operation of individual data for each DUT is enabled.

Use a character constant, STRING variable or character variable to specify MUT. Set a value between C0 to C11, or select M1 or M2.

FD49PN

Specifies a DUT driver pin number that uses the data of pin 49 that was transferred to AFM.

Using the PINLIST variable, specify only one pin.

If data sent to the AFM is not used for driver pins, specify 0 using an integer-type or logical-type constant, or an integer-type or logical-type variable.

DUT pin assignment in the socket program has the following restrictions:

- Pins (tester channels) that are assigned to the DUT pin numbers (FD49PN) must be different for each DUT.

- Assign pins corresponding to Pin 49 and to pin numbers specified using FD49PN in accordance with the following chart:

"FD49PN" in the table refers to DUT pin number specified by the FD49PN argument, and "Pin 49" refers to the tester channel number for Pin 49.

Table 58-2 Pin Assignment for Pin 49 and FD49PN

Driver output for FD49PN	PIN MODE	I/O bit		Comparator pin			
				33-40	41-48	97-104	105-112
INn	A	36	FD49PN	26A1			
			Pin 49	49A1			
		18	FD49PN	26A1		27A2	
			Pin 49	49A1		113A1	
	B.	9	FD49PN	26A1	29A1	27A2	30A2
			Pin 49	49A1	50A1	113A1	114A1
INn, INHn (FD49PN is HC driver pin)	A	36	FD49PN	8A1			
			Pin 49	49A1			
		18	FD49PN	8A1		24A2	
			Pin 49	49A1		113A1	
	B	9	FD49PN	8A1	16A1	24A2	30A2
			Pin 49	49A1	50A1	113A1	114A1
INn, INHn (FD49PN is the HV driver pin)	A	36	FD49PN	4A1			
			Pin 49	49A1			
		18	FD49PN	4A1		20A2	
			Pin 49	49A1		113A1	
	B	9	FD49PN	4A1	12A1	20A2	26A2
			Pin 49	49A1	50A1	113A1	114A1

The pin assignment used for Child A1 is also used for Childs A3, A5 and A7.

The pin assignment used for Child A2 is also used for Childs A4, A6 and A8.

The pin assignment used for Child A is also used for Childs B to H.

For the pin assignment used for a single device test, see the pin assignment used when PIN MODE = A, I/O bit = 36 is set in Table 58-2.

◆ Example of Socket Program Assigning AFM Pin 49 Data to a Driver Pin (PD26)

```

SOCKET SOC1

DEFINE MULTIPLE 128DUT 1STN

BIT=9
PPS=1
PIN MODE=B
CHTYPE 11

DUT#      = ..... , PD1-12      PD26      P33-40      P49
DUT1      = ..... , PD1-12A1    PD26A1    P33-40A1    P49A1
DUT2      = ..... , PD13-24A1   PD29A1    P41-48A1    P50A1
DUT3      = ..... , PD1-12A2    PD27A2    P97-104A1   P113A1
DUT4      = ..... , PD13-24A2   PD30A2    P105-112A1  P114A1
DUT5      = ..... , PD1-12A3    PD26A3    P33-40A3    P49A3
DUT6      = ..... , PD13-24A3   PD29A3    P41-48A3    P50A3
DUT7      = ..... , PD1-12A4    PD27A4    P97-104A3   P113A3
DUT8      = ..... , PD13-24A4   PD30A4    P105-112A3  P114A3
DUT9      = ..... , PD1-12A5    PD26A5    P33-40A5    P49A5
DUT10     = ..... , PD13-24A5   PD29A5    P41-48A5    P50A5
DUT11     = ..... , PD1-12A6    PD27A6    P97-104A5   P113A5
DUT12     = ..... , PD13-24A6   PD30A6    P105-112A5  P114A5
DUT13     = ..... , PD1-12A7    PD26A7    P33-40A7    P49A7
DUT14     = ..... , PD13-24A7   PD29A7    P41-48A7    P50A7
DUT15     = ..... , PD1-12A8    PD27A8    P97-104A7   P113A7
DUT16     = ..... , PD13-24A8   PD30A8    P105-112A7  P114A7
:
END

```

◆ Example of Socket Program Assigning AFM Pin 49 Data to an HC Driver Pin (PD8)

```

SOCKET SOC2

DEFINE MULTIPLE 128DUT 1STN

BIT=9
PPS=1
PIN MODE=B
CHTYPE 11

SEPARATE DR 2

DUT# =.....,PD1-7,9-15,17-23,25-29      PD8      P33-40      P49
DUT1 =.....,PD1-7,9-15,17-23,25-29A1    PD8A1    P33-40A1    P49A1
DUT2 =.....,PD1-7,9-15,17-23,25-29A1    PD16A1   P41-48A1    P50A1
DUT3 =.....,PD1-7,9-15,17-23,25-29A2    PD24A2   P97-104A1   P113A1
DUT4 =.....,PD1-7,9-15,17-23,25-29A2    PD30A2   P105-112A1  P114A1
DUT5 =.....,PD1-7,9-15,17-23,25-29A3    PD8A3    P33-40A3    P49A3
DUT6 =.....,PD1-7,9-15,17-23,25-29A3    PD16A3   P41-48A3    P50A3
DUT7 =.....,PD1-7,9-15,17-23,25-29A4    PD24A4   P97-104A3   P113A3
DUT8 =.....,PD1-7,9-15,17-23,25-29A4    PD30A4   P105-112A3  P114A3
DUT9 =.....,PD1-7,9-15,17-23,25-29A5    PD8A5    P33-40A5    P49A5
DUT10=.....,PD1-7,9-15,17-23,25-29A5    PD16A5   P41-48A5    P50A5
DUT11=.....,PD1-7,9-15,17-23,25-29A6    PD24A6   P97-104A5   P113A5
DUT12=.....,PD1-7,9-15,17-23,25-29A6    PD30A6   P105-112A5  P114A5
DUT13=.....,PD1-7,9-15,17-23,25-29A7    PD8A7    P33-40A7    P49A7
DUT14=.....,PD1-7,9-15,17-23,25-29A7    PD16A7   P41-48A7    P50A7
DUT15=.....,PD1-7,9-15,17-23,25-29A8    PD24A8   P97-104A7   P113A7
DUT16=.....,PD1-7,9-15,17-23,25-29A8    PD30A8   P105-112A7  P114A7
:
END

```

◆ ERR Description

ERR	Description
0	Normal termination
1	Cannot execute in this system.
2	An I/O pin (PIN) which uses AFM data is not specified correctly.
3	The MUT signal (MUT) setting is incorrect.
4	A driver pin that uses AFM data (FD49PN) is not specified correctly.
5	The SELECT APPLICATION MODE BBM _n , SELECT FLASH MODE statement, or the utility IDSELUBM is enabled.
6	The socket program specification of the pin assigned to pin 49 for each DUT is incorrect.
7	The socket program specification of the pin assigned to the pin specified by FD49PN for each DUT is incorrect.

◆ Notes on Using IDSELUBM

- When IDSELFM is executed, fail information cannot be transferred to DFM.
- Do not execute the AFM statements listed below between MFMSSEND or IDSELFM and the MEAS MPAT or START MPAT statement:
 - AFM MULTIMODE statement
 - AFM DUTSEL statement
 - AFM ADDRESS statement
 - AFM RATE statement
 - AFM BITSIZE statement
 - AFM BITSEL statement
 - AFM PGSEL statement
 - AFM ACTION statement
 - SELECT AFM CONFIG statement
 - SELECT COMPRESS statement
 - RESET AFM ALL statement
 - RESET AFM MODE statement
- Since the AFM MULTI MODE and AFM DUTSEL statements are executed in MODE 8 (SET DUT NO.) of the M command, do not set this mode to ENABLE.

4. IDSELFM cannot be executed when the following statements or utility program have been used. Clear the setting before executing IDSELFM.

Setting	Clear
SELECT FLASH MODE statement	SELECT FLASH DISABLE statement
SELECT APPLICATION MODE BBM statement	SELECT APPLICATION MODE BBM DISABLE statement

5. When IDSELFM is used, the waveform types that can be used for the pin used to write individual data for each DUT are NRZA, /NRZA, RZO, /RZO, RZZ, and /RZZ only.
6. There are restrictions listed below for the individual data write operation for each DUT.

For T5377, T5376 and T5375

	T5377	T5376 and T5375	Figure	Remarks
If no FD49PN pin is specified (I/O pin only)				
The RATE setting of the cycle where the setup pattern for the individual data write operation for each DUT is enabled.	RATE $\geq$ 7 ns	RATE $\geq$ 7 ns	Figure 58-2 (A)	
The RATE setting of the cycle which occurs immediately before the individual data write operation for each DUT starts.	RATE $\geq$ 7 ns 0 ns $\leq$ clock delay time $\leq$ RATE-27.34375 ps	RATE $\geq$ 38 ns 0 ns $\leq$ clock delay time $\leq$ RATE-38 ns	Figure 58-2 (B)	
The RATE setting of the cycle where the individual data write operation for each DUT is enabled.	RATE $\geq$ 7 ns 0 ns $\leq$ clock delay time $\leq$ RATE-27.34375 ps Pulse width $\geq$ 3.5 ns (clock resolution 27.34375 ps)	RATE $\geq$ 100 ns 38 ns $\leq$ clock delay time $\leq$ RATE-49 ns Pulse width $\geq$ 7 ns (clock resolution 3.5 ns)	Figure 58-2 (C)	
If an FD49PN pin is specified (the pattern is output to the driver pin)				
The RATE setting of the cycle where the setup pattern for the individual data write operation for each DUT is enabled.	RATE $\geq$ 7 ns	RATE $\geq$ 7 ns	Figure 58-2 (A)	
The RATE setting of the cycle which occurs immediately before the individual data write operation for each DUT starts.	RATE $\geq$ 38 ns 0 ns $\leq$ clock delay time $\leq$ RATE-38 ns	RATE $\geq$ 38 ns 0 ns $\leq$ clock delay time $\leq$ RATE-38 ns	Figure 58-2 (B)	The driver timing restrictions are applied because the FA49PN pin is specified.
The RATE setting of the cycle where the individual data write operation for each DUT is enabled.	RATE $\geq$ 100 ns 38 ns $\leq$ clock delay time $\leq$ RATE-49 ns I/O pin: Pulse width $\geq$ 3.5 ns (clock resolution 27.34375 ps) Pin specified for FD49PN: Pulse width $\geq$ 7 ns (clock resolution 3.5 ns)	RATE $\geq$ 100 ns 38 ns $\leq$ clock delay time $\leq$ RATE-49 ns Both the I/O pin and pin specified for FD49PN Pulse width $\geq$ 7 ns (clock resolution 3.5 ns)	Figure 58-2 (C)	The driver timing restrictions are applied because the FA49PN pin is specified.

T5376 and 5375 : Whether or not the FD49PN pin is specified, clocks used in cycles, where data is written for each DUT, are output -5 ns to -38 ns faster than the clocks used in other cycles.

T5377 : If the FD49PN pin is specified, the clocks for the FD49PN pin (converted by the socket program) in the individual data write operation are output -5 ns to -38 ns faster than clocks used in other cycles.

7. The setting for IDSELFM is cleared when one of the following occurs:

- Starting the program
- Executing the RESET TESTER statement
- The CTRL/C key combination is executed.
- IDSELCLR is executed.

The IDSELFM setting executed last takes effect when IDSELFM is executed repeatedly in the test program.

58. 3 IDSELCLR

◆ Function

IDSELCLR can be used to clear the IDSELFPM setting.

◆ Syntax

```
CALL IDSELCLR
```

When this utility is described in a test program on a tester other than the T5377, T5376 and T5375 test systems, this description is ignored.

◆ Notes on Using IDSELUBM

To allow fail information to be transferred to the FM execute one of the following statements:

- RESET AFM ALL statement
- RESET AFM MODE statement
- AFM ACTION statement

58. 4 An Example of the Writing Individual Data to Each DUT Function

The following example indicates a test program to generate an individual driver pattern for each DUT using the Writing Individual Data to Each DUT function from Channel P33-40 and PD26, a socket program, pattern program, and data file for the MFMSSEND utility.

An individual driver pattern used for the driver pin (PD26) uses data from Pin 49, 50, 113, and 114 which is transferred to the AFM by the MFMSSEND utility.

Socket program

The tester channel numbers that use the driver pattern are P33 to 40 and PD26.

The driver pattern used with the driver pin (PD26) uses the pin specification assigned to tester channel number P49.

The tester channel numbers PD26 and P49 are assigned to individual DUTs.

```

SOCKET SAMSOC

DEFINE MULTIPLE 128DUT 1STN

BIT=9
PPS=1
PIN MODE=B
CHTYPE11

DUT#   = VS1      PD1-12      PD26      P33-40      P49
DUT1   = PPS1     PD1-12A1    PD26A1    P33-40A1    P49A1
DUT2   = PPS2     PD13-24A1   PD29A1    P41-48A1    P50A1
DUT3   = PPS3     PD1-12A2    PD27A2    P97-104A1   P113A1
DUT4   = PPS4     PD13-24A2    PD30A2    P105-112A1  P114A1
DUT125= PPS125   PD1-12H7    PD26H7    P33-40H7    P49H7
:
DUT126= PPS126   PD13-24H7   PD29H7    P41-48H7    P50H7
DUT127= PPS127   PD1-12H8    PD27H8    P97-104H7   P113H7
DUT128= PPS128   PD13-24H8    PD30H8    P105-112H7  P114H7

END

```

58. 4 An Example of the Writing Individual Data to Each DUT Function

Pattern program

Specify C0 to the MUT signal in a cycle to enable the Writing Individual Data to Each DUT function.

```
MPAT SAMMPA

REGISTER
  XMAX=#FF
  YMAX=#FF
  IDX1=#8

START #0
  NOP XB<0 YB<0
ST0:JN11 ST0 W R X<XB Y<YB XB<XB+1 C0 TP<0
  NOP

  STPS

END
```

58. 4 An Example of the Writing Individual Data to Each DUT Function

A data file which is transferred by the MFMSSEND utility.

In the samdat01.dat (which is a data file transferred by the MFMSSEND utility), data for two DUTs is stored in one block.

#F704	0	Header
0		
0		
#FF		← AMAX-X
#FF		← AMAX-Y
0		
64		← The number of blocks
13		Block 1
#80004101		← child A1, 1STN, 33-50 pins
0		← STA-X
0		← STA-Y
8		← The number of data
#31112		← Data 1
#32122		← Data 2
:		← :
#34132		← Data 8
13		Block 2
#80004102		← Child A1, 1STN, Pin 97 to Pin 113
0		← STA-X
0		← STA-Y
8		← The number of data
#31314		← Data 1
#32324		← Data 2
:		← :
#34344		← Data 8
:		Block 3
:		Block 4
:		Block 62
13		Block 63
#1004801		← Child H7, 1STN, Pin 33 to Pin 55
0		← STA-X
0		← STA-Y
8		← The number of data
#38D8E		← Data 1
#39D9E		← Data 2
:		← :
#3BDBE		← Data 8
13		Block 64
#1004802		← Child H7, 1STN, Pin 97 to Pin 113
0		← STA-X
0		← STA-Y
8		← The number of data
#38F90		← Data 1
#39FA0		← Data 2
:		← :
#3BFC0		← Data 8

58. 4 An Example of the Writing Individual Data to Each DUT Function

A test program

Set the following conditions for the IDSELFM utility:

- Use the MFMSSEND utility to transfer the output data file samdat01.dat that will be a driver pattern to the AFM.
- Using the IDSELFM utility, sets a pin that uses the driver pattern transferred to the AFM. I/O pins P33 to P40, which generate driver patterns for individual DUTs.
- Specify the tester channel number PD26 to generate driver patterns for individual DUTs in the socket program.
- Specify the MUT signal C0 to enable the Writing Individual Data to Each DUT function that was specified in a pattern file.

```

PRO SAMPLE SAMSOC

    INTE[1] ERR1=0
    INTE[1] ERR2=0
    PINLIST PIN=P33-40
    STRING  MUT(2)="C0"
    PINLIST FD49PN=PD26

    RESET AFM ALL
    SELECT AFM CONFIG DOUBLE
    AFM MULTIMODE=ALL
    AFM RATE=LS
    AFM ADDRESS=X0-7,Y0-7
    AFM ACTION=FMX>MASK
    CALL MFMSSEND(ERR1,"samdat01.dat")
    IF ERR1<>0 THEN STOP / "ERROR:MFMSSEND"

    CALL IDSELFM(ERR2,PIN,MUT,FD49PN)
    IF ERR2<>0 THEN STOP / "ERROR:IDSELFM"

    START MPAT SAMMPA

    CALL IDSELCLR           ;The pin setting by the IDSELFM utility is cleared.

END

```

59. Utility Used to Output Individual Driver Levels for Each DUT

For preprocessing, it is sometimes required that individual driver values should be applied to specified DUTs only, by changing the driver output levels of I/O or driver pins.

However, if the IN or INH statement is used to change driver output levels for parallel tests, the driver output levels for all DUTs are changed.

Using this utility allows you to perform individual driver settings containing high-level and low-level voltage settings and the output setting (such that the driver is set to high-level fixed or low-level fixed, or used for pattern control) for a specified I/O or driver pin for each DUT.

This utility consists of the following two programs:

- **LEVELSET**
Makes individual driver settings containing high-level and low-level voltage settings and the output setting (such that the driver is set to high-level fixed or low-level fixed, or used for pattern control) for a specified I/O or driver pins for each DUT.
- **LEVELRST**
Clears the LEVELSET setting.

This utility can be used with the T5377, T5376 and T5375 test systems.

59. 1 LEVELSET

◆ Function

LEVELSET can perform the following settings for each DUT:

- Sets a high-level and low-level voltage values of driver output.
- Sets the driver level to high-level fixed or low-level fixed or specifies the driver for pattern control.

To cancel the LEVELSET setting, use LEVELRST.

◆ Syntax

```
CALL LEVELSET (ERR, STDUT, STPIN, UNT, VIH, VIL, DRLVL)
```

ERR

Specifies a variable that stores LEVELSET termination status (normal termination or abnormal termination).

Specify using an integer or logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For details, refer to ["ERR Description" on page 59-6](#).

STDUT

Specifies the DUT for which the driver output is specified.

Use a 2-word-type variable with no DIM or with a DIM, or a 2-word-type non-subscripted variable to specify STDUT.

The same manner used for the READ mDUT statement variable setting can be used to specify the DUT in this utility.

➔ For information about the data contents of the destination variable for the READ mDUT statement, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498\)](#).

STPIN

Specifies a pin used to control driver output.

Specify using the PINLIST variable.

UNT

When INH is connected to the pin that controls driver output, the value is 1; when IN is connected, it is 0.

Use an integer (or logical) constant, or a variable equal to 0 or 1.

VIH, VIL

Specifies a high-level voltage value (VIH) and low-level voltage value (VIL) for the driver.

Specify using variables or constants with voltage units.

The VIH and VIL setting ranges are as follows:

- When 1 is specified for UNIT, the same VIH and VIL range used for the INH statement is used.
- When 0 is specified in UNIT, the same VIH and VIL range for the IN statement is used.

➔ For information about the setting data ranges for IN statement and INH statement high-level voltage and low-level voltage, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498\)](#).

DRLVL

Sets the driver level setting to high-level fixed or low-level fixed, or specifies the driver for pattern control.

Use an integer (or logical) constant, or a variable between 1 and 2.

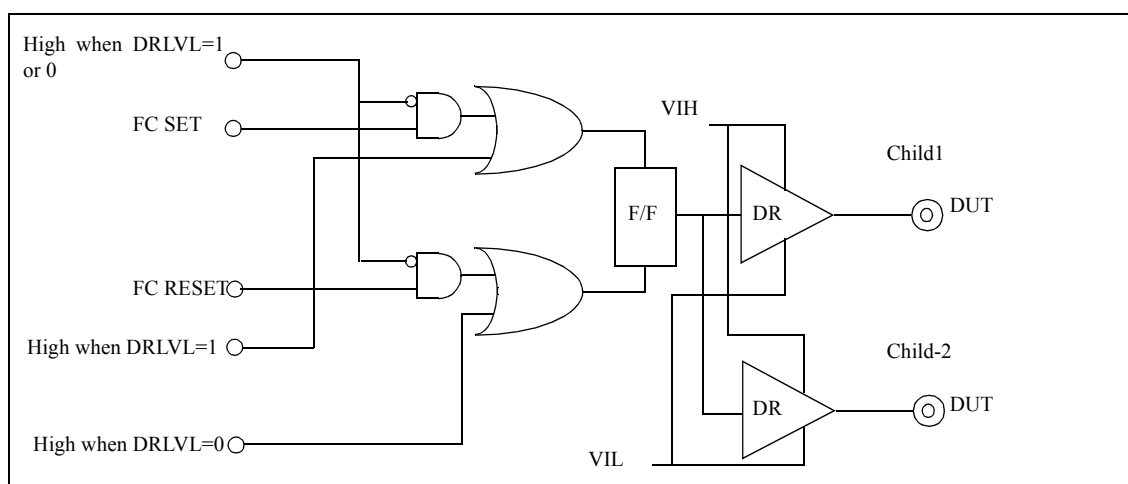
The following driver output settings can be selected.

DRLVL	Driver output setting
0	Low-level fixed
1	High-level fixed
2	Pattern control

When the LV driver pin is specified for STPIN, specify 1 for DRLVL.

◆ Notes on Using IDSELUBM

1. The driver pin circuit is paired (Child 1 and Child 2) as shown below.



When the driver output setting for Child 1 is changed by LEVELSET, the driver output setting for Child 2 is also changed.

This also applies to the pairs Child 3 and 4, Child 5 and 6, and Child 7 and 8.

◆ When the Following Socket Program is Used

```

SOCKET LEVSOC1
  DEFINE MULTIPLE 16DUT 1STN

  BIT=18
  PPS=1
  PIN MODE=B
  CHTYPE11

DUT# =VS1      PD1-30      P33-48
DUT1 =PPS1     PD1-30A1    P33-48A1
DUT2 =PPS2     PD1-30A2    P97-112A1

DUT3 =PPS3     PD1-30A3    P33-48A3
DUT4 =PPS4     PD1-30A4    P97-112A3

DUT5 =PPS5     PD1-30A5    P33-48A5
DUT6 =PPS6     PD1-30A6    P97-112A5

DUT7 =PPS7     PD1-30A7    P33-48A7
DUT8 =PPS8     PD1-30A8    P97-112A7

END

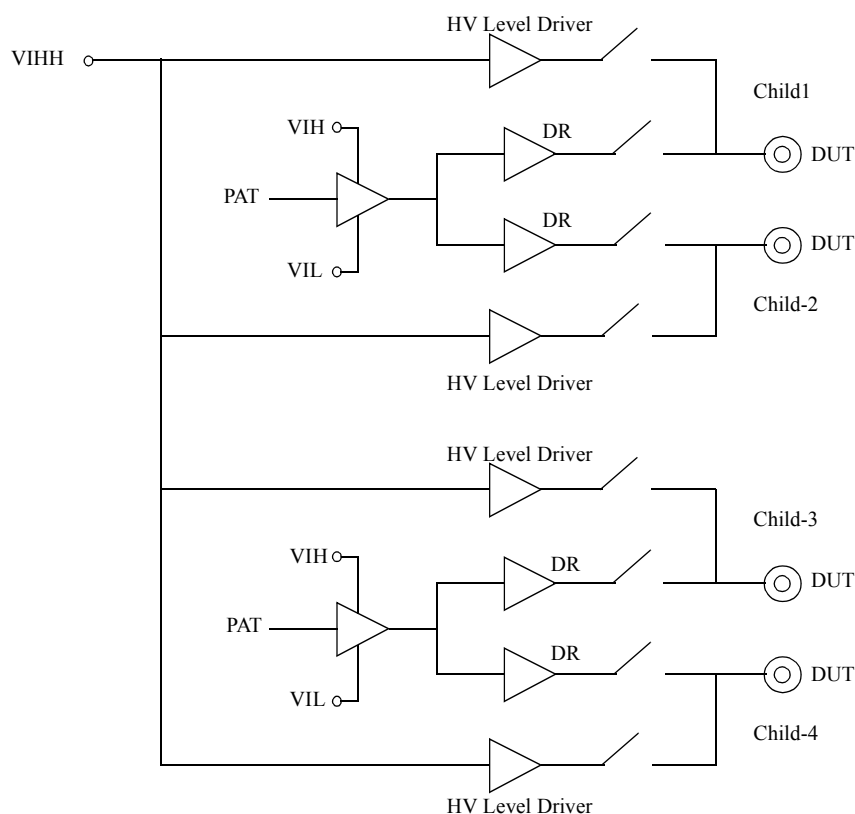
```

The PD1through 30 (driver pins) are connected to the DUTs as shown in the table below.

Driver Pin	Connected DUT
PD1-30A1(A2)	DUT1,DUT2
PD1-30A3(A4)	DUT3,DUT4
PD1-30A5(A6)	DUT5,DUT6
PD1-30A7(A8)	DUT7,DUT8

When LEVELSET sets a driver pin from PD1 to PD30 for DUT1, the same setting is made for DUT2. When the setting is made for DUT2 by LEVELSET, the same setting is made for DUT1. This works the same way for DUT3 and DUT4, DUT5 and DUT6, and DUT7 and DUT8.

2. When INHn is specified in a pin statement to connect a HV level driver to a HV level driver pin on the T5376, the HV level drivers used for Child 1, Child 2, Child 3, and Child 4 are connected to the same pin as shown below.



When LEVELSET is used to change the HV level driver output used for Child 1, the change affects the HV level drivers used for Child 2, Child 3, and Child 4. When the HV level driver output is changed for Child 5, the change also affects the HV level drivers used for Child 6, Child 7, and Child 8.

3. When the following statements or utilities are executed after CALL LEVELSET has been executed, the high-level (VIH) and low-level (VIL) settings of the driver set by LEVELSET are disabled.

The driver output setting (DRLVL) is enabled until LEVELRST is executed.

- RESET TESTER statement
- IN statement
- Pin statement
- Data scan (/DS)
- DATA SET (/unit name)
- Voltage measurement program (VMEAS)
- Two-dimensional shmoo plot (SHM2)

- Three-dimensional shmoo plot (SHM3)
 - MARGIN program (/MARGIN)
4. To execute CALL CALB, the CALL LEVELSET setting must have been cleared (using CALL LEVELRST).

◆ ERR Description

ERR	Description
0	Normal termination
1	Cannot execute in this system.
2	The DUT setting (STDUT) is incorrect.
3	The pin setting (STPIN) is incorrect.
4	The UNIT setting (that specifies the pin connection to IN or INH, is incorrect.
5	The high-level (VIH) or low-level (VIL) settings of the driver are incorrect.
6	The driver output setting (DRLVL) is incorrect.
7	An error occurred during data transfer.

59. 2 LEVELRST

◆ Function

LEVELRST is used to reset the driver output high-level and low-level voltage values set by LEVELSET to those specified by the IN or INH statement. The driver output setting is reset to pattern control.

◆ Syntax

CALL LEVELRST

When this utility is described in a test program on a tester other than the T5377, T5376 and T5375 test systems, this description is ignored.

59. 3 Usage Example of the Utility to Output Individual Driver Level for Each DUT

This section provides an example showing how to output individual driver levels for each DUT when 64-device parallel tests using a test station are executed under the following driver output conditions. 1 V is applied to PD26 of DUT1 through 32, 2 V is applied to PD26 of DUT33 through 48, and 3 V is applied to PD26 of DUT49 through 64.

The test plan program used in this example is as follows.

```

PRO LVL1 LVL SOC

LOGI[2] STDUT(4)
INTE[1] ERR
PINLIST STPIN=PD26
INTE[2] UNT=0
REAL[3] VIH=0V
REAL[3] VIL=0V
INTE[1] FIXHL=1

IN1=5V,0V
STPIN=IN1, FIXL

DIM STDUT=#0
STDUT(1)=#FFFFFFF ; DUT1-32
VIH=1V ; 1V is applied as the driver high-level voltage.

CALL LEVELSET(ERR,STDUT,STPIN,UNT,VIH,VIL,FIXHL)
IF ERR<>0 THEN STOP ; Error processing.

DIM STDUT=#0
STDUT(3)=#0000FFFF ; DUT33-48
VIH=2V ; 2V is applied as the driver high-level voltage.

CALL LEVELSET(ERR,STDUT,STPIN,UNT,VIH,VIL,FIXHL)
IF ERR<>0 THEN STOP ; Error processing.

DIM STDUT=#0
STDUT(3)=#FFFF0000 ; DUT3
VIH=3V ; 3V is applied as the driver high-level voltage.

CALL LEVELSET(ERR,STDUT,STPIN,UNT,VIH,VIL,FIXHL)
IF ERR<>0 THEN STOP ; Error processing.
:
CALL LEVELRST ;LEVELSET that resets the high-level and low-level
; voltage values of the pin driver to those specified
; by the IN statement

END

```

59. 3 Usage Example of the Utility to Output Individual Driver Level for Each DUT

The socket program used in this example is as shown below.

In this socket program, PD26 (for which individual driver level for each DUT is set by LEVELSET in the test plan program) is allocated to different drivers for each DUT.

```

SOCKET LVLSOC

DEFINE MULTIPLE 16DUT 1STN
  BIT=9
  PPS=1
  PIN MODE=B
  CHTYPE11

  DUT#   = VS1      PD1-12      PD26      P33-40      P49
  DUT1   = PPS1     PD1-12A1    PD26A1    P33-40A1    P49A1
  DUT2   = PPS2     PD13-24A1   PD29A1    P41-48A1    P50A1
  DUT3   = PPS3     PD1-12A2    PD27A2    P97-104A1   P113A1
  DUT4   = PPS4     PD13-24A2   PD30A2    P105-112A1  P114A1
  DUT5   = PPS5     PD1-12A3    PD26A3    P33-40A3    P49A3
  DUT6   = PPS6     PD13-24A3   PD29A3    P41-48A3    P50A3
  DUT7   = PPS7     PD1-12A4    PD27A4    P97-104A3   P113A3
  DUT8   = PPS8     PD13-24A4   PD30A4    P105-112A3  P114A3
  :
  DUT57  = PPS57    PD1-12H1    PD26H1    P33-40H1    P49H1
  DUT58  = PPS58    PD13-24H1   PD29H1    P41-48H1    P50H1
  DUT59  = PPS59    PD1-12H2    PD27H2    P97-104H1   P113H1
  DUT60  = PPS60    PD13-24H2   PD30H2    P105-112H1  P114H1
  DUT61  = PPS61    PD1-12H3    PD26H3    P33-40H3    P49H3
  DUT62  = PPS62    PD13-24H3   PD29H3    P41-48H3    P50H3
  DUT63  = PPS63    PD1-12H4    PD27H4    P97-104H3   P113H3
  DUT64  = PPS64    PD13-24H4   PD30H4    P105-112H3  P114H3

END

```

60. System Information Readout Utility Program (RDSYSTEM)

The system name and system configuration can be read using this utility program.

60. 1 Function

This utility program is used to read the following system information:

- System name
- FM (Fail Memory) board name
- Installed FM slot numbers (where FM boards are installed)
- PM (Pattern Memory) board name
- DBM (Data Buffer Memory) capacity

60. 2 Syntax

```
CALL RDSYSTEM(ERR, RDNO, RDDAT)
```

ERR

Used to store the end status of the system information readout utility program (normal or abnormal end).

Specify using an integer or logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For details, refer to [60. 5 "ERR Description" on page 60-7](#).

RDNO

Specifies system information numbers.

Specify using a constant or an integer or logical variable.

1 : System name

2 : FM board name

3 : Installed FM slot numbers (where FM boards are installed)

4 : PM board name

5 : DBM capacity

RDDAT

Used to store the system information.

If the system name is specified (RDNO=1), use a 3-word character-type variable; if any item other than the system name (RDNO<>1) is specified, use an integer-type or a logical-type variable.

60. 3 Details of the System Information

Reading the System Name

The system name is read as a character string.

Characterstring	Targetsystem
T5593	T5593
T5592	T5592
T5591	T5591
T5591R	T5591R
T5586	T5586
T5585	T5585
T5581	T5581
T5581H	T5581H
T5581P	T5581P
T5571P	T5571P
T5377	T5377
T5376	T5376
T5375	T5375
T5371	T5371
T5336	T5336
T5311	T5311

Reading an FM Board Name

The number that matches an installed FM board name is read.

Number	FM board name
0	Not installed
8	288 M board
11	576 M board
13	1 G board
15	2 G board
17	4 G board
18	1.44 G board
19	8 G board

Reading Installed FM Slot Numbers (Where FM Boards Are Installed)

Whether or not FM boards are installed is read in the format shown below:

Bit "1" indicates that an FM board is installed in the slot number; and Bit "0" indicates that an FM board is not installed in the slot number.

0	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	

The number is the FM slot number.

Reading an PM Board Name

The number that matches an installed PM board name is read.

Number	FM board name
0	Not installed
2	144 M board
3	288 M board
4	576 M board
5	1 G board
6	2 G board

Reading the DBM Capacity

The number that matches the capacity of an installed DBM is read.

Number	DBM capacity
0	Not installed
1	64 KW
2	256 KW
3	1 MW
4	16 MW
5	2MW

60. 4 Usage Examples

- ◆ The Example Below Shows How to Calculate the Total FM Capacity of the T5585

```

      :
INTE[1] ERR
CHAR[3] RDSYS
INTE[2] UNT=0
LOGI[2] FMBRD
INTE[2] UNT=0
      :
CALL RDSYSTEM(ERR,1,RDSYS)
IF ERR=0 THEN ENTRY
    IF RDSYS="T5585" THEN ENTRY
        CALL RDSYSTEM(ERR,2,FMCAP)
        CALL RDSYSTEM(ERR,3,FMBRD)
        BDCT=0
        DO LP01 J=0,15
            IF SR0(FMBRD,J).AND.#1=#1 THEN BDCT=BDCT+1
LP01:    CONT
        IF FMCAP=13 THEN ENTRY
            WRITE[1,2(10)]/,"FM VALUE=",BDCT*64*1024*1024*18
        EXIT
      :
    EXIT
  :
EXIT
  :

```

60. 5 ERR Description

ERR	Description
0	Normal termination
1	The specified argument for the system information number (RDNO) is incorrect.
2	The specified argument for storing the system information (RDDAT) is incorrect.

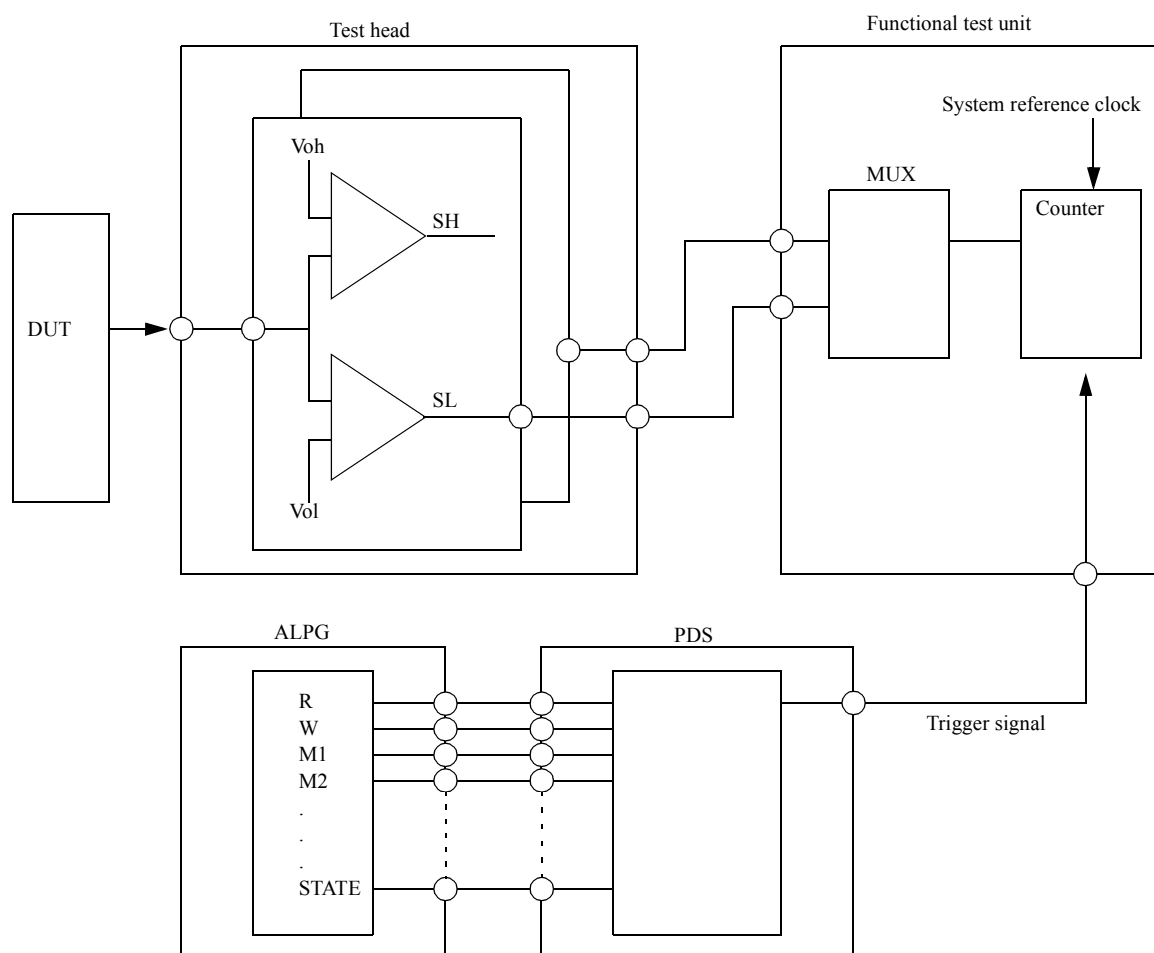
61. Pulse Count Measurement Utility (PULSECNT)

This utility is used to measure the period or the number of pulses of the specified pin.
This utility can be used with the T5377, T5376 and T5375.

61. 1 Principle of Operation

Figure 61-1 shows how signals are used during measurement.

Figure 61-1 PULSECNT Utility Principles of Operation



61. 1. 1 Measuring Period

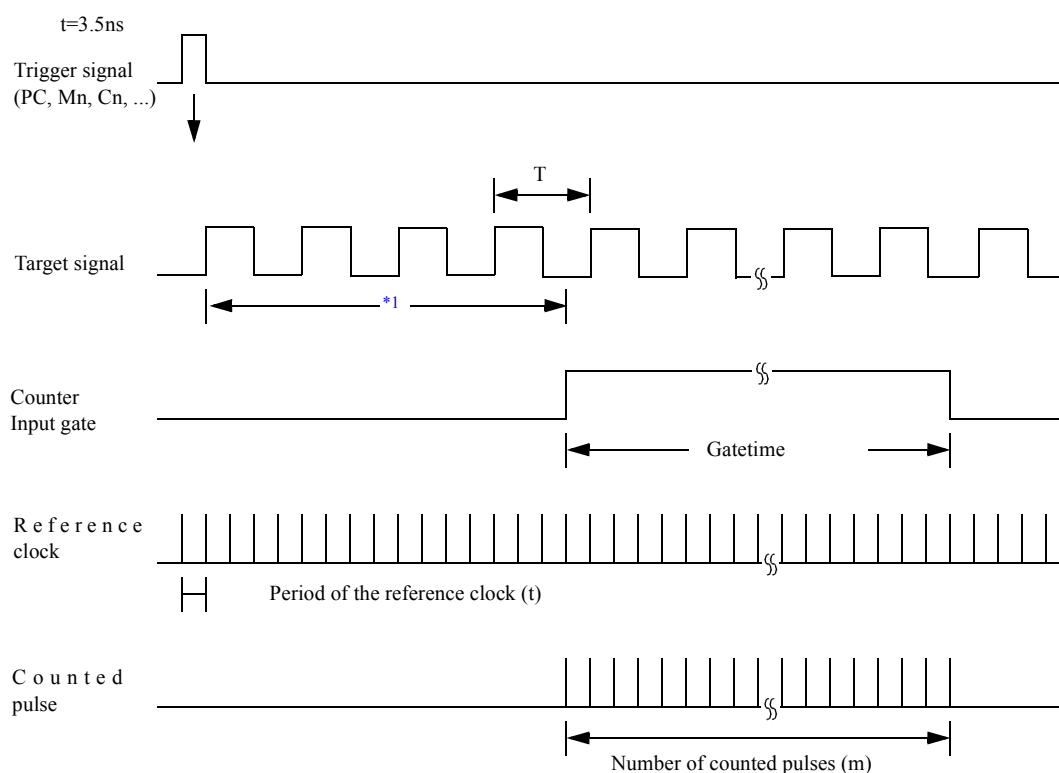
The period of the target signal is measured by counting the number of reference clock pulses while the input gate of the time counter, whose frequency is obtained by dividing the target signal by a dividing ratio of n , is opened after the trigger signal is generated.

Figure 61-2 shows that the period of the target signal is measured by counting the number of reference clock pulses while the input gate of the time counter is opened.

The period of the target signal is calculated using the formula shown below:

$$\text{Period of the target signal (T)} = \frac{\text{Counted pulses (m)} \times \text{Period of the reference clock (t)}}{\text{Dividing ratio (n)}}$$

Figure 61-2 Period Measurement Timing

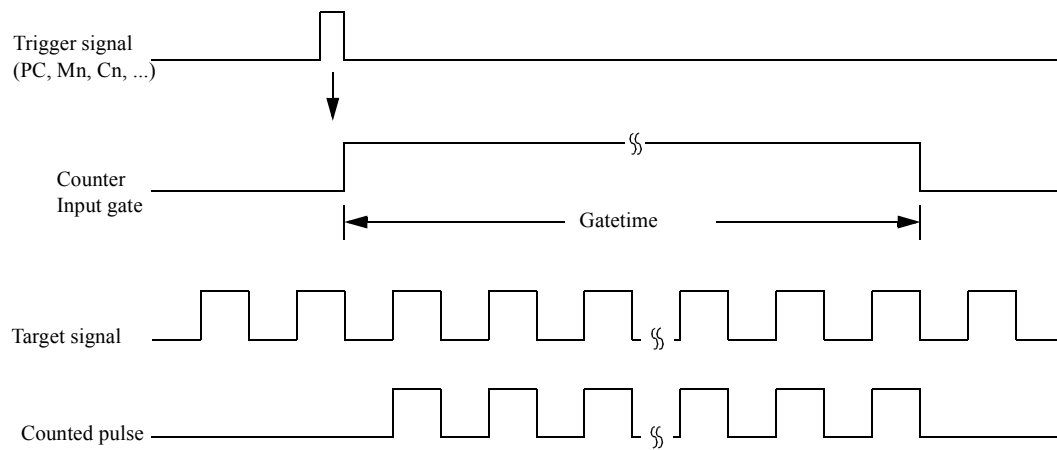


*1 The counter input gate is opened only 100 ns + four clock pulses has elapsed.

61. 1. 2 Measuring the Number of Pulses

The period of the target signal is measured by counting the number of target signal pulses while the input gate of the time counter is opened for a specified time after the trigger signal is generated.

Figure 61-3 shows how the number of target signal pulses is counted while the input gate of the time counter is opened.

Figure 61-3 Timing for Measuring Number of Pulses

61. 2 Syntax

```
CALL PULSECNT (ERR, MODE, RANG, PIN, TRIG, LIMIT, DATA)
```

ERR

Stores the end state.

Specify using 1W to 4W integer-type variables or a logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For details, refer to [61. 4 "ERR Description" on page 61-10](#).

Mode

Specifies the measurement mode.

Specify using 1W to 4W integer-type variables.

- 1 Period Measurement
The period is measured according to the value specified by RANG.
- 2 Measuring Pulse Count
The number of pulses entered during the time specified by LIMIT or the gate time specified by RANG.

RANG

Specifies the measurement range.

Specify using 1W to 4W integer-type variables.

- For period measurements (MODE=1):

RANG	Measurement range	Possible voltage setting	Time accuracy
1	3.5 ns to 229.376 μ s	3500.000 ps	Measured value $\pm$ 0.01% + resolution
2	3.5 ns to 114.688 μ s	1750.000 ps	
3	3.5 ns to 57.344 μ s	875.000 ps	
4	3.5 ns to 28.672 μ s	437.500 ps	
5	3.5 ns to 14.336 μ s	218.750 ps	
6	3.5 ns to 7.168 μ s	109.375 ps	
7	3.5 ns to 3.584 μ s	54.6875 ps	
8	3.5 ns to 1.792 μ s	27.34375 ps	
9	3.5 ns to 896 ns	13.671875 ps	
10	3.5 ns to 448 ns	6.8359375 ps	
11	3.5 ns to 224 ns	3.41796875 ps	
12	3.5 ns to 112 ns	1.708984375 ps	
13	3.5 ns to 56 ns	0.8544921875 ps	

- For pulse measurements (MODE = 2)

RANG	Gatetime	Gatetime accuracy
0	Time specified by LIMIT	±10%
1	0.0035 μs	±0.01% +1
2	0.007 μs	±0.01% +1
3	0.014 μs	±0.01% +1
4	0.028 μs	±0.01% +1
5	0.056 μs	±0.01% +1
6	0.112 μs	±0.01% +1
7	0.224 μs	±0.01% +1
8	0.448 μs	±0.01% +1
9	0.896 μs	±0.01% +1
10	1.792 μs	±0.01% +1
11	3.584 μs	±0.01% +1
12	7.168 μs	±0.01% +1
13	14.336 μs	±0.01% +1



Tip

Specify RANG=0 when a gate time equal to or larger than 14 μs is used.

In addition, the accuracy for the measurement gate time is within ±10% since the TP timer is used.

PIN

Specifies a pin to be measured.

Specify one comparator pin.

Specify using the PINLIST variable.

TRIG

Specifies the pattern cycle from which the measurement is started.

Specify using STRING- or CHAR-type 1W to 4W.

- When specifying the cycle on an MPAT program:

"R"	Cycle during which R is specified
"W"	Cycle during which W is specified
"M1"	Cycle during which M1 is specified
"M2"	Cycle during which M2 is specified
"C0"	Cycle during which C0 is specified
:	
"C11"	Cycle during which C11 is specified

- Signal specified by the T command used in the controller:

"STATE"	A signal specified by the T command used in the controller (PC, X-ADDRESS, Y-ADDRESS, PROGRAM COUNTER, MUT COMMAND, TIMING SET)
---------	---

LIMIT

Specifies a measurement timeout or measurement time.

Specify using real-type 3W.

Specify using a value from 0 s to 6.5535 s.

The measurement timeout time is specified when RANG≠0 and MODE=1 or MODE=2

The measurement time is specified if MODE=2 and if RANG=0.

Tip

Varies as follows according to MODE:

For MODE=1:

$$\begin{aligned}
 &\text{Software offset time} \\
 &\quad + \\
 &\quad 100 \text{ ns} \\
 &\quad + \\
 &\quad \text{Input signal period of the target device} \times 4 \\
 &\quad + \\
 &\quad \text{Pattern execution time until generation of the trigger signal} \\
 &\quad + \\
 &\quad \frac{\text{Synchronization} \times \text{division ratio (n) for target input signal}}{\text{Gatetime}} \times 1.2
 \end{aligned}$$

For MODE=2

- RANG=0

$$\begin{aligned}
 &\text{Software offset time} \\
 &\quad + \\
 &\quad \text{Gatetime}
 \end{aligned}$$

- RANG>1

$$\begin{aligned}
 &\text{Software offset time} \\
 &\quad + \\
 &\quad \text{Gate time} \\
 &\quad + \\
 &\quad \text{Pattern execution time until generation of the trigger signal}
 \end{aligned}$$

In addition, the software offset time is 3 ms, assuming that a RON is not performed when this utility program is called (because the software offset is greatly affected by the RON/ROF state prior to calling this utility program).

DATA

Specifies measurement values.

Specify an integer-type or logical-type 2W variable.

Specify using the array of the number of DUTs being measured. The maximum array is 512.

A value of 0 is entered when there is not a pulse.

Any of the following values is entered when the measurement value is overflowed:

INTEGER	-1
LOGICAL	#FFFFFFFF

The maximum measurement value is as follows:

MODE=1 $2^{32}-1$

MODE=2 $2^{16}-1$

The measurement value varies according to the mode used:

MODE=1 Integer with a resolution of 1 ps

MODE=2 Integer representing the count data

61. 3 Using the Utility Program

The utility program is used after a pattern program has been transferred.

The example below shows how to measure a period in a 32-device parallel test with the "C0" program.

◆ Main Program

```

PRO      EX1      SOC01

INTE[1]   RERR=0
INTE[1]   MODE=1           ;Specifies the measurement mode for a period
                           measurement.

INTE[1]   RANG=10          ;Specifies a range of 10.
STRING    TRIG(5)="C0"     ;Specifies "C0" for the measurement start trigger
REAL[3]   LIMT=1MS
INTE[2]   DATA(32)=0      ;Declares the array of 32 elements for the measurement
                           data variables.

        :
        :
PINLIST DD0=P33            ;Declares the pinlist for the measurement pins.
OUT1=1.5V,1.5V
DD0=OUT1

        :
        :
SEND MPAT MEX1             ;Transfers the pattern program.
REG MPAT PC=#0
CALL PULSECNT(RERR,MODE,RANG,DD0,TRIG,LIMT,DATA) ;Measures the period.
IF RERR<>0 THEN WRITE / "PULSE COUNT ERROR" /    ;An error is displayed if it
                                                occurs.

ELSE ENTRY
    WRITE "DUT1=" DATA(1) "PS" /
    WRITE "DUT2=" DATA(2) "PS" /
    WRITE "DUT3=" DATA(3) "PS" /
        :
        :
EXIT

```

◆ Pattern Program

```
MPAT    MEX1

START   #0
        NOP
        :           ; Write a setup pattern that outputs the oscillating
        :           ; clock from the device to P33.
        :
        NOP
        NOP         C0       ; Trigger that starts the counter measurement
        NOP
        STPS

END
```

This is an example of a device that starts oscillating by a setup pattern used in a pattern program.

61. 4 ERR Description

ERR	Description
0	Normal termination
1	Cannot execute in this system.
2	MODE is incorrectly specified.
3	RANGE is incorrectly specified.
4	PIN is incorrectly specified.
5	TRIG is incorrectly specified.
6	SELECT APPLICATION MODE BBM5/BBM6 is used.
7	LIMIT is incorrectly specified.
8	The pattern program is not found.
9	The PG is operating.
10	The T controller command setting is not available.

62. Utility to Use Universal Buffer Memory

This utility is used to write repair information on each DUT individually or write/read manufacturing information on each DUT to/from DRAM, FLASH, and other memory devices using Universal Buffer Memory (UBM).

When writing data to devices, driver patterns of each DUT are generated using UBM data.

When reading data from devices, fail data for each DUT is imported into the UBM.

This utility consists of the following programs:

IDSELUBM

Selects whether the UBM is used for driver pattern generation or fail data import.

IDSELCLR

Clears the setting of IDSELFM and IDSELUBM.

UBMWRT

Writes data to the UBM.

UBMREAD

Reads data from the UBM.

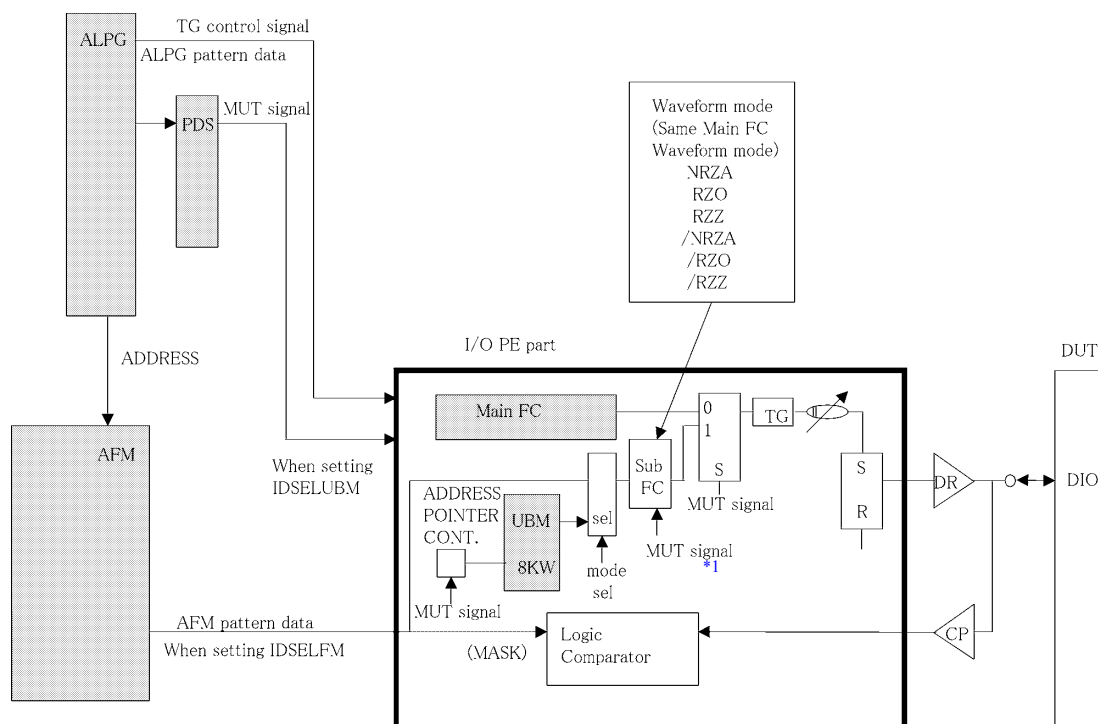
UBMCLR

Clears the UBM memory to zero.

This utility can be used with the T5377, T5376 and T5375 test systems.

62. 1 Configuration of the UBM

The functional block diagram of the UBM is as follows:

Figure 62-1 Functional Block Diagram of the UBM

*1 Switches whether to select IDSELF (refer to 58. "Utility Used to Write Individual Data for Each DUT" on page 58-1) or IDSELUBM.

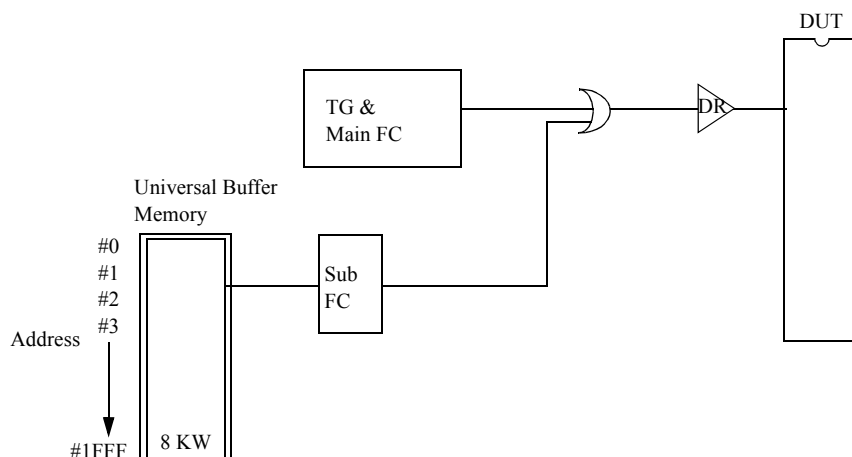
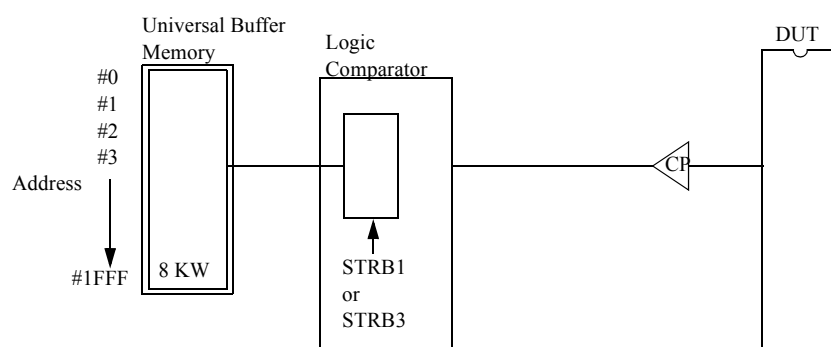
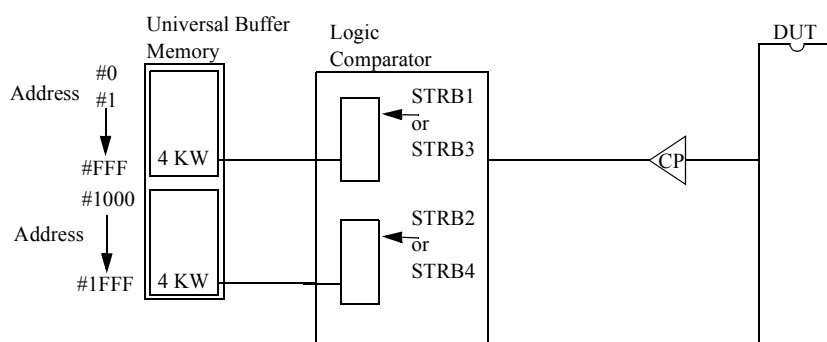
The UBM have a capacity of 8 KW for the I/O pins 33 to 52 and 97 to 116 respectively. The usage and the capacity partition of the UBM are shown in the table below.

Table 62-1 UBM Usage and Capacity Divisions

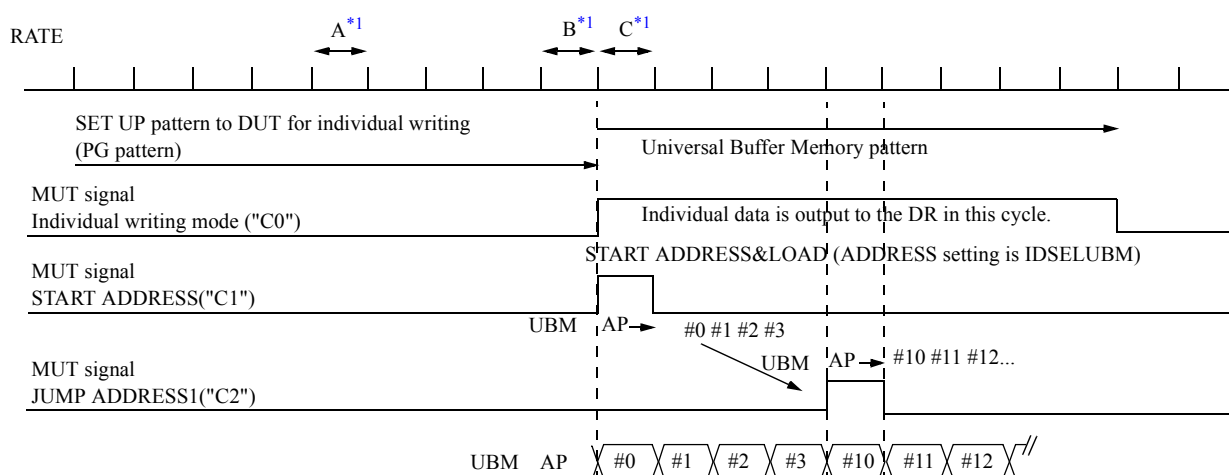
Usage of the UBM	STRB Selection	Capacity Partition of the UBM
Individual pattern generation	No selection	8 KW × 1 bit
Fail data import using single STRB	Odd-numbered phase STRB (select STRB1 or STRB3)	8 KW × 1 bit
Fail data import using double STRB	Odd-numbered phase STRB (select STRB1 or STRB3), odd-phased STRB (select STRB2 or STRB4)	4 KW × 2 bits

*1 1 W = 4 bytes

The schematic diagrams of the UBM are as shown below:

Figure 62-2 Individual Pattern Generation Conceptual Diagram**Figure 62-3 Fail Data Import Using Single STRB Conceptual Diagram****Figure 62-4 Fail Data Import Using double STRB Conceptual Diagram**

The timing diagrams when using the UBM are as shown below:

Figure 62-5 Timing Diagram when Using the UBM for Individual Pattern Generation

*1 There are timing restrictions.

➔ For details, refer to 3 of "Notes on Using IDSELUBM" on page 62-12.

```

MPAT  SAMMP1

NOP          ;Set up pattern
:
:
NOP C0C1    ;#0 (UBM AP) *1
NOP C0      ;#1
NOP C0      ;#2
NOP C0      ;#3
NOP C0 C2   ;#10
NOP C0      ;#11
:
:
STPS
END

```

*1 Write C1 in the first cycle in which C0 is written.

➔ For details, refer to 2 of "Notes on Using IDSELUBM" on page 62-12.

```

PRO    SAMPRO1    SAMPSOC

CALL  UBMWRT (ERR,DUTNO,BLKNO,STADR,CNT,DATA);Data Transfer Universal Buffer Memory

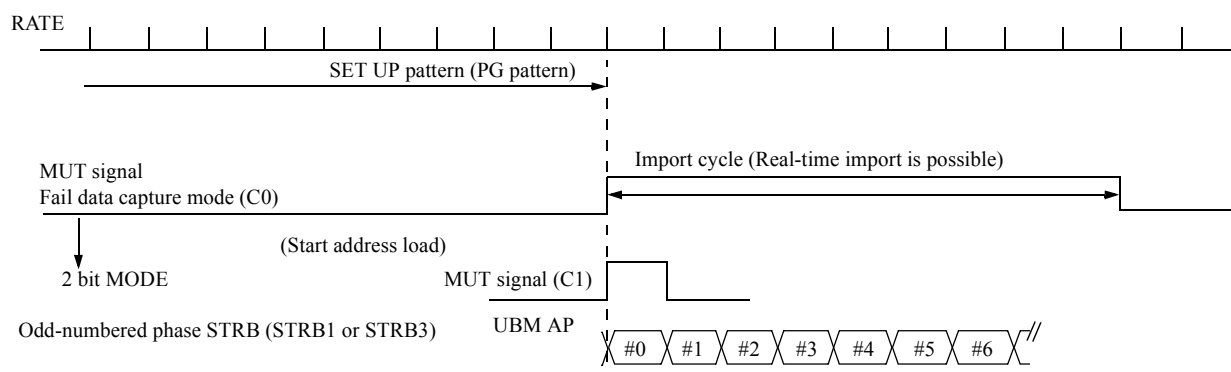
CALL  IDSELUBM (ARG1,ARG2,ARG3,ARG4*1,ARG5*1,ARG6,ARG7*1,ARG8*1,ARG9,ARG10)

      :
START MPAT
      :
      :
END

```

**1 ARG4 is MUT signal "C1", ARG5 is START ADDRESS #0, ARG7 is MUT signal "C2", and ARG8 is JUMP ADDRESS1 #10.*

Figure 62-6 Timing Diagram when Using the UBM for Fail Data Import with Single STRB



```

MPAT    SAMMP2

NOP      ;Set up pattern
      :
      :
NOP      ;UBM AP
NOP C0C1 XYINC R ;#0*1
NOP C0   XYINC R ;#1
NOP C0   XYINC R ;#2
      :
      :
STPS
END

```

**1 Write C1 in the first cycle in which C0 is written.*

➔ *For details, refer to 2 of "Notes on Using IDSELUBM" on page 62-12.*

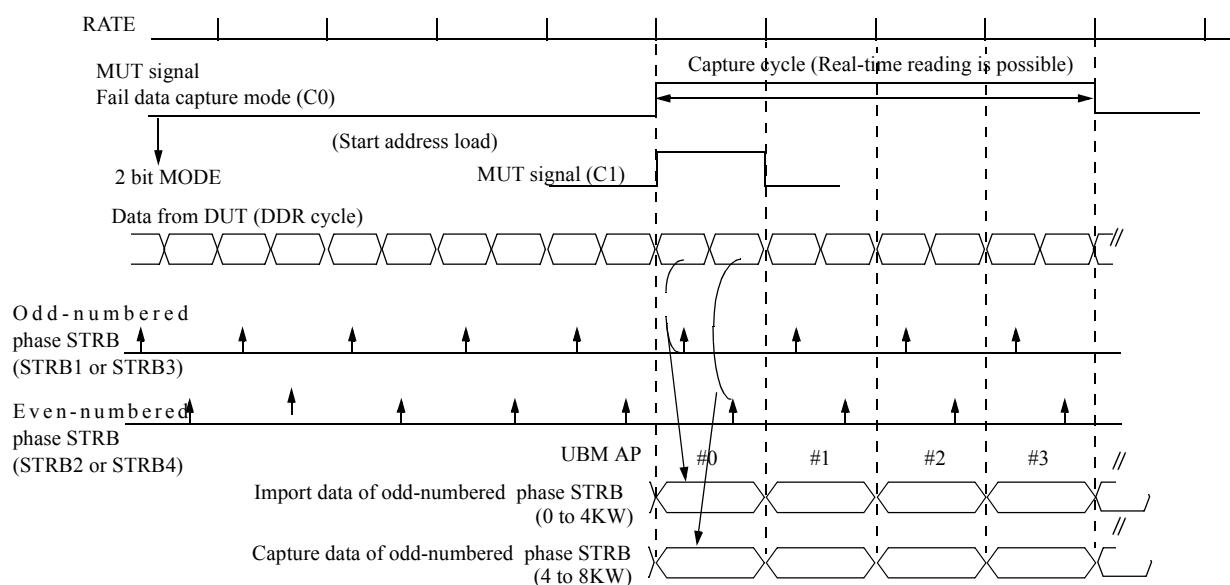
```
PRO    SAMPRO2    SAMP SOC

CALL  IDSELUBM(ARG1,ARG2,ARG3,ARG4*1,ARG5*1,ARG6,ARG7,ARG8,ARG9,ARG10)

      :
SELECT MPAT INHIBIT
START MPAT
      :
      :
END
```

**1 ARG4 is MUT signal "C1", and ARG5 is START ADDRESS #0.*

Figure 62-7 Timing Diagram when Using the UBM for Fail Data Import with Double STRB



MPAT SAMMP2

; UBM AP

NOP

NOP C0C1 XYINC R ;#0 *1

NOP C0 XYINC R ;#1

NOP C0 XYINC R ;#2

:

END

*1 Write C1 in the first cycle in which C0 is written.

➔ For details, refer to 2 of "Notes on Using IDSELUBM" on page 62-12.

PRO SAMPRO2

CALL IDSELUBM(ARG1,ARG2,ARG3,ARG4*1,ARG5*1,ARG6,ARG7,ARG8,ARG9,ARG10)

:

SELECT MPAT INHIBIT

START MPAT

:

:

END

**1 ARG4 is MUT signal "C1", and ARG5 is START ADDRESS #0.*

62. 2 IDSELUBM

◆ Function

Selects whether the UBM is used for driver pattern generation or fail data import.
The setting of IDSELUBM can be cleared by IDSELCLR.

◆ Syntax

```
CALL IDSELUBM (ARG1, ARG2, ARG3, ARG4, ARG5, ARG6, ARG7, ARG8, ARG9, ARG10)
```

ARG1

Specifies the variable (hereafter referred to as ERR) used to store the termination state (normal termination/abnormal termination) of IDSELUBM.

Specify an integer/logical variable of 1 HW.

ERR 0 indicates normal termination, and a number other than 0 indicates abnormal termination.

➔ For details, refer to ["ERR Description" on page 62-12](#).

ARG2

Specifies the UBM function (hereafter referred to as MODE).

Specify using an integer or logical variable or a constant.

1 and 2 can be specified.

If any other value is specified, an error occurs.

The values are described below:

- 1 The UBM is used for driver pattern generation.
- 2 The UBM is used for fail data import.

ARG3

Specifies the MUT signal (hereafter referred to as MUT) for the cycle in which the IDSELUBM function is enabled.

Specify a STRING variable, a CHAR variable, or a character constant of R, W, M1, M2, or C0 to C11.

ARG4

Specifies the MUT signal (hereafter referred to as STMUT) for the starting address (ARG5).

Specify a STRING variable, a CHAR variable, or a character constant of R, W, M1, M2, or C0 to C11.

ARG5

Specifies the UBM address (hereafter referred to as STADR) where individual pattern generation or fail data import starts.

Specify using an integer or logical variable or a constant.

Different setting methods are used for individual pattern generation (ARG2=1) and fail data import (ARG2=2).

- Individual pattern generation (ARG2=1):
#0 to #1FFF can be specified.
- Fail data import (ARG2=2):
The range of the value that can be specified for ARG5 varies depending on the value specified for ARG6.
When ARG6=1, #0 to #1FFF can be specified.
When ARG6=2, #0 to #FFF can be specified.

ARG6

Different setting methods are used for individual pattern generation (ARG2=1) and fail data import (ARG2=2).

- Individual pattern generation (ARG2=1):
Specifies the I/O pin (hereafter referred to as PIN) to use the written data.
Specify using the PINLIST variable.
P33 to 52, 97 to 116 can be specified.
If any pin outside the range is specified, an error occurs.
- Fail data import (ARG2=2):
Specifies the STRB mode (hereafter referred to as STBMD) for capture.
Specify using an integer or logical variable or a constant.
1 and 2 can be specified.
If any other value is specified, an error occurs.
The values are described below:
1: Single STRB
2: Double STRB

ARG7

Different setting methods are used for individual pattern generation (ARG2=1) and fail data import (ARG2=2).

- Individual pattern generation (ARG2=1):
Specifies the MUT signal (hereafter referred to as JP1MUT) for the cycle in which ARG8 is enabled.
Specify a STRING variable, a CHAR variable, a character constant of R, W, M1, M2, or C0 to C11, or a constant of 0.
When a constant of 0 is specified, the specification for ARG8 becomes invalid.
- Fail data import (ARG2=2):
Specification fixed at 0.
Specify 0 with a constant.
If any value other than 0 is specified, an error occurs.

ARG8

Different setting methods are used for individual pattern generation (ARG2=1) and fail data import (ARG2=2).

- Individual pattern generation (ARG2=1):
Specifies the UBM address (hereafter referred to as JP1ADR) to be accessed in the cycle indicated by the MUT signal specified by ARG7 (JP1MUT).
Specify using an integer or logical variable or a constant.
#0 to #1FFF can be specified.
When constant 0 is specified for ARG7, only constant 0 can be specified.
If any value other than the range is specified, an error occurs.
- For fail data read (ARG2=2):
Specification fixed at 0.
Specify 0 with a constant.
If any value other than 0 is specified, an error occurs.

ARG9

Different setting methods are used for individual pattern generation (ARG2=1) and reading of fail data (ARG2=2).

- Individual pattern generation (ARG2=1):
Specifies the MUT signal (hereafter referred to as JP2MUT) for the cycle in which ARG10 is enabled.
Specify a STRING variable, a CHAR variable, a character constant of R, W, M1, M2, or C0 to C11, or a constant of 0.
When a constant of 0 is specified, the specification for ARG10 becomes invalid.
- For fail data read (ARG2=2):
Specification fixed at 0.
Specify 0 with a constant.
If any value other than 0 is specified, an error occurs.

ARG10

Different setting methods are used for individual pattern generation (ARG2=1) and reading of fail data (ARG2=2).

- Individual pattern generation (ARG2=1):
Specifies the UBM address (hereafter referred to as JP2ADR) to be accessed in the cycle indicated by the MUT signal specified by ARG9 (JP2MUT).
Specify using an integer or logical variable or a constant.
#0 to #1FFF can be specified.
When constant 0 is specified for ARG9, only constant 0 can be specified.
If any value other than the range is specified, an error occurs.
- For fail data read (ARG2=2):
Specification fixed at 0.

Specify 0 with a constant.

If any value other than 0 is specified, an error occurs.

The usage and the setting method of ARG1 to 10 are shown in the table below.

Table 62-2 The Usage of the UBM and the Setting Method of ARG

Usage of the UBM	ARG1	ARG2	ARG3	ARG4	ARG5
Individual writing	ERR	Mode	MUT	STMUT	STADR
Fail data reading	ERR	Mode	MUT	STMUT	STADR

Usage of the UBM	ARG6	ARG7	ARG8	ARG9	ARG10
Individual writing	PIN	JP1MUT	JP1ADR	JP2MUT	JP2ADR
Fail data reading	STBMD	0	0	0	0

◆ ERR Description

ERR	Description
0	Normal termination
1	Cannot execute in this system.
2	The ARG2 setting is incorrect.
3	In ARG3, a character string other than R, W, M1, M2, or C0 to C11 is specified.
4	In ARG4, a character string other than R, W, M1, M2, or C0 to C11 is specified.
5	The ARG5 setting is incorrect.
6	The ARG6 setting is incorrect.
7	In ARG7, a character string other than R, W, M1, M2, or C0 to C11, or constant 0 is specified.
8	The ARG8 setting is incorrect.
9	In ARG9, a character string other than R, W, M1, M2, or C0 to C11, or constant 0 is specified.
10	The ARG10 setting is incorrect.
11	One of the following is valid: SELECT APPLICATION MODE BBMn statement, SELECT FLASH MODE statement, or utility IDSELFm.
12	The same MUT signal cannot be specified for ARG3, ARG4, ARG7, and ARG9.

◆ Notes on Using IDSELUBM

1. IDSELUBM cannot be executed when the following statements and utilities are set.

Clear the setting before executing IDSELUBM.

Setting	Clear
SELECT FLASH MODE statement	SELECT FLASH DISABLE statement
SELECT APPLICATION MODE BBMn statement	SELECT APPLICATION MODE BBM DISABLE statement
The utility used to write individual data on each DUT (IDSELFM)	The utility used to write individual data on each DUT (IDSELFM)

2. The following restrictions apply to the MUT signal:
 - a. The same MUT signal cannot be specified for ARG3, ARG4, ARG7, and ARG9.
 - b. The MUT signal used for the start address must be specified in the first cycle when the IDSELUBM function is enabled. In other words, the MUT signal specified for ARG4 must be written in the first line in which the MUT signal specified for ARG3 is written in the MPAT file.
3. The following restrictions apply to the timing:

	T5377	T5376 and T5375	Figure
The RATE setting of the cycle where the setup pattern for the individual data write operation for each DUT is enabled.	$\text{RATE} \geq 7 \text{ ns}$	$\text{RATE} \geq 7 \text{ ns}$	Figure 62-5 (A)
The RATE setting of the cycle which occurs immediately before the individual data write operation for each DUT starts.	$\text{RATE} \geq 7 \text{ ns}$ $0 \text{ ns} \leq \text{clock delay time} \leq \text{RATE}-27.34375 \text{ ps}$	$\text{RATE} \geq 38 \text{ ns}$ $0 \text{ ns} \leq \text{clock delay time} \leq \text{RATE}-38 \text{ ns}$	Figure 62-5 (B)
The RATE setting of the cycle where the individual data write operation for each DUT is enabled.	$\text{RATE} \geq 7 \text{ ns}$ $0 \text{ ns} \leq \text{clock delay time} \leq \text{RATE}-27.34375 \text{ ps}$ Pulse width $\geq 3.5 \text{ ns}$ (clock resolution 27.34375 ps)	$\text{RATE} \geq 100 \text{ ns}$ $38 \text{ ns} \leq \text{clock delay time} \leq \text{RATE}-49 \text{ ns}$ Pulse width $\geq 7 \text{ ns}$ (clock resolution 3.5 ns)	Figure 62-5 (C)

In the T5376 and T5375, clocks used in cycles, where the IDSEL and UBM functions are enabled, are output -5 ns to -38 ns faster than the clocks used in other cycles.

4. The cycle pattern number (CYPn) and driver pattern delay time (DRDLYn), which are specified in the PIN statement, are ignored for the cycle during which a driver pattern is being generated from UBM.

62. 3 IDSELCLR

◆ Function

Clears the setting of IDSELFM (refer to [58. "Utility Used to Write Individual Data for Each DUT" on page 58-1](#)) and IDSELUBM.

◆ Syntax

```
CALL IDSELCLR
```

UBMCLR is terminated without being executed when used on a system other than T5377, T5376 and T5375.

62. 4 UBMWRT

◆ Function

Writes data to the UBM.

◆ Syntax

```
CALL UBMWRT (ERR, DUTNO, BLKNO, STADR, CNT, DATA)
```

ERR

Specifies the variable used to store the termination state (normal termination/abnormal termination) of UBMWRT.

Specify an integer/logical variable of 1 HW.

ERR 0 indicates normal termination, and a number other than 0 indicates abnormal termination.

→ For details, refer to ["ERR Description" on page 62-23](#).

DUTNO

Specifies the DUT number.

Specify an integer/logical constant or variable of 1 HW or 2 HW.

1 to 512 can be specified.

If any other value is specified, an error occurs.

→ For details, refer to ["Details of Each Argument" on page 62-16](#).

BLKNO

Specifies the block number.

Specify an integer/logical constant or variable of 1 HW or 2 HW.

1 and 2 can be specified.

If any other value is specified, an error occurs.

→ For details, refer to ["Details of Each Argument" on page 62-16](#).

STADR

Specifies the starting address.

Specify an integer/logical constant or a variable for 1 HW or 2 HW.

#0 to #1FFF can be specified.

If any other value is specified, an error occurs.

→ For details, refer to ["Relationships between Arguments" on page 62-21](#).

CNT

Write count.

Specify an integer/logical constant or a variable for 1 HW or 2 HW.

1 to 8192 can be specified.

If any other value is specified, an error occurs.

➔ For details, refer to ["Relationships between Arguments" on page 62-21](#).

DATA
 Write data.
 Specify a logical variable or an array variable of 2 HW.

➔ For details, refer to ["Details of Each Argument" on page 62-16](#) and ["Relationships between Arguments" on page 62-21](#).

◆ Details of Each Argument

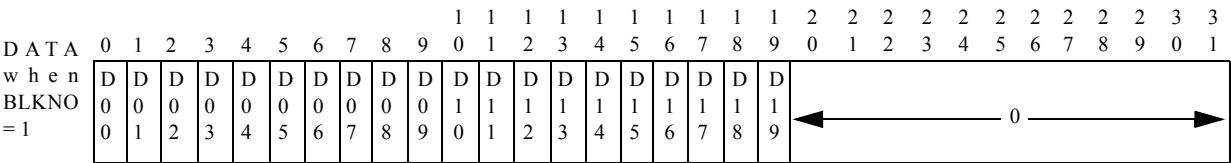
DUTNO
 DUTNO specifies the DUT number specified in the socket program.
 If an incorrect DUT number is specified, an error occurs.

BLKNO
 The value that can be specified as BLKNO is determined by the number of bits specified in the socket program.
 If 1 is specified, D0 to D19 of UBM data (=20 bits of I/O pins in the first half) can be specified.
 If 2 is specified, D20 to D39 of UBM data (= 20 bits of I/O pins in the latter half) can be specified.
 If the number of bits is 20 or less, only 1 can be specified. If 2 is specified in this situation, an error occurs.
 If the number of bits is larger than 20, 1 or 2 can be specified.
 (Example) When BIT = 40 (P33 to 52, P97 to 116) is specified in the socket program
 If 1 is specified as BLKNO, P33 to 52 can be specified.
 If 2 is specified as BLKNO, P97 to 116 can be specified.

DATA
 DATA format corresponds to the I/O pin specified in the socket program.
 The significant bit of DATA is the number of bits specified in the socket program.
 Other bits cannot be written to the UBM. When reading, 0 is specified.
 D0 to D39 are UBM data. They are not pin data.

The number of bits of I/O pins specified in the socket program corresponds to D0 to D39 as follows:

1. Data Format (When BIT = 1 to 20 in the socket program)



When BIT=1, the significant bit is D0.
 When BIT=2, the significant bits are D0 to D1.

4. Data Format (When BIT = 40 in the socket program)

[illegible]

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Significant bits are D0 to D39.

D0 to D39 are UBM data. They are not pin data.

(Example 1) When BIT = 10 (P33 to 40, 49, 51) is specified in the socket program

When BLKNO=1 is specified, D0 to 9 are written. Or, reading can be performed.

```
BLKNO=1
CALL UBMWRT (ERR,DUTNO,BLKNO,STADR,CNT,DATA)
CALL UBMREAD (ERR,DUTNO,BLKNO,STADR,CNT,DATA)
```

												1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	3	3						
	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1				
DATA	D	D	D	D	D	D	D	D	D	D																										
	0	0	0	0	0	0	0	0	0	0																										
	0	1	2	3	4	5	6	7	8	9																										
	P	P	P	P	P	P	P	P	P	P																										
	3	3	3	3	3	3	3	4	4	5																										
	3	4	5	6	7	8	9	0	9	1																										

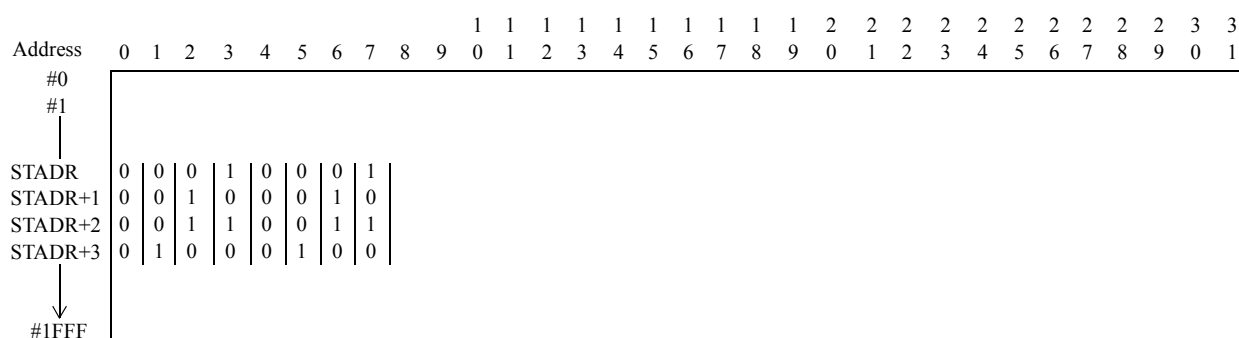
(Example 2) When BIT = 32 (P33 to 48, P97 to 112) is specified in the socket program

When BLKNO=1 is specified, D0 to 15 (P33 to 48) are written. Or, reading can be performed.

◆ An Example of Writing when CNT Number = Number of DATA Elements

```
INTE[1] CNT=4
LOGI[2] DATA(4) = #11000000,#22000000,#33000000,#44000000

CALL UBMWRT(ERR,DUTNO,BLKNO,STADR,CNT,DATA)
```

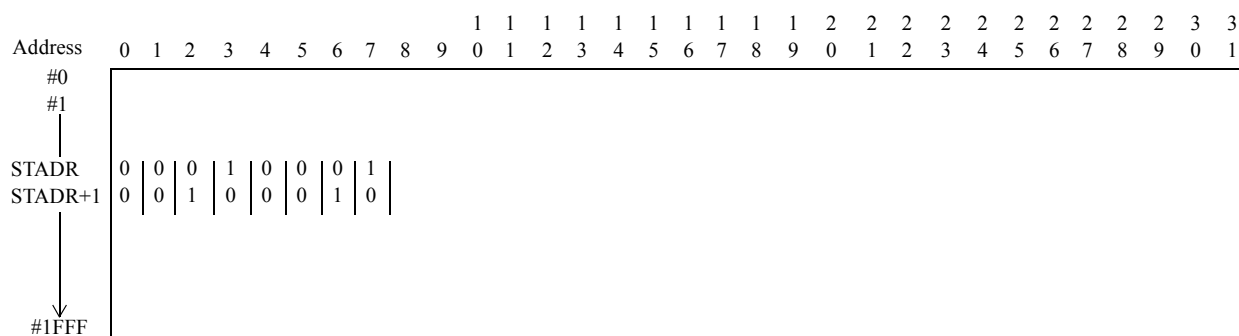


◆ An Example of Writing when CNT Number > Number of DATA Elements

```
INTE[1] CNT=4
LOGI[2] DATA(2) = #11000000,#22000000

CALL UBMWRT(ERR,DUTNO,BLKNO,STADR,CNT,DATA)
```

When BIT = 8 is specified in the socket program, data is stored in the UBM as follows:

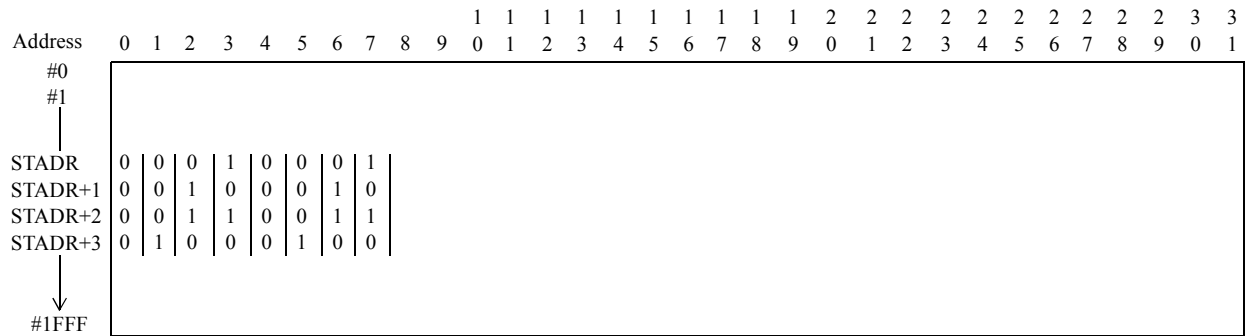


◆ An Example of Writing when CNT Number < Number of DATA Elements

```
INTE[1] CNT=4
LOGI[2] DATA(6) = #11000000,#22000000,#33000000,#44000000,#55000000,#66000000

CALL UBMWRT(ERR,DUTNO,BLKNO,STADR,CNT,DATA)
```

When BIT = 8 is specified in the socket program, data is stored in the UBM as follows:



◆ ERR Description

ERR	Description
0	Normal termination
1	Cannot execute in this system.
2	The specified DUT number (DUTNO) is incorrect.
3	The specified block number (BLKNO) is incorrect.
4	The specified starting address (STADR) is incorrect.
5	Write/read count (CNT) is incorrect.
6	Write/read data (DATA) is incorrect.
7	The specified DUT number is not included in the socket program currently being used.
8	The specified block number is not included in the socket program currently being used.

62. 5 UBMREAD

◆ Function

Reads data from the UBM.

◆ Syntax

```
CALL UBMREAD (ERR, DUTNO, BLKNO, STADR, CNT, DATA)
```

ERR

Specifies the variable used to store the termination state (normal termination/abnormal termination) of UBMREAD.

Specify an integer/logical variable of 1 HW.

ERR 0 indicates normal termination, and a number other than 0 indicates abnormal termination.

➔ For details, refer to ["ERR Description" on page 62-23](#).

DUTNO

Specifies the DUT number.

Specify an integer/logical constant or variable of 1 HW or 2 HW.

1 to 512 can be specified.

If any other value is specified, an error occurs.

➔ For details, refer to ["Details of Each Argument" on page 62-16](#).

BLKNO

Specifies the block number.

Specify an integer/logical constant or variable of 1 HW or 2 HW.

1 and 2 can be specified.

If any other value is specified, an error occurs.

➔ For details, refer to ["Details of Each Argument" on page 62-16](#).

STADR

Specifies the starting address.

Specify an integer/logical constant or a variable for 1 HW or 2 HW.

#0 to #1FFF can be specified.

If any other value is specified, an error occurs.

➔ For details, refer to ["Relationships between Arguments" on page 62-21](#).

CNT

Specifies the number of data items being read.

Specify an integer/logical constant or a variable for 1 HW or 2 HW.

1 to 8192 can be specified.

If any other value is specified, an error occurs.

→ For details, refer to ["Relationships between Arguments" on page 62-21](#).

DATA

Read data.

Specify a logical variable or an array variable of 2 HW.

→ For details, refer to ["Details of Each Argument" on page 62-16](#) and ["Relationships between Arguments" on page 62-21](#).

62. 6 UBMCLR

◆ Function

Clears the UBM.

Write 0 to all memory areas of the UBM.

◆ Syntax

CALL UBMCLR

UBMCLR is terminated without being executed when used on a system other than T5377, T5376 and T5375.

62. 7 Usage Examples

Sample programs of individual pattern generation and fail data import are shown below:

◆ Sample Program of Individual Pattern Generation:

- Test plan program

```

PRO SAM3    SKSAM1

INTE[1] ERR
INTE[1] MODE=2          ← Individual pattern generation selection
CHAR[1] MUT="C0"
CHAR[1] STMUT="C1"
LOGI[1] STADR=#0
CHAR[1] STMUT="C1"
LOGI[1] JP1ADR=#10
CHAR[1] STMUT="C1"
LOGI[1] JP2ADR=#20
INTE[1] DUTNO=1
INTE[1] BLKNO=1
LOGI[1] CNT=8
LOGI[2] DATA(8)=#0

PINLIST ADR = PD1-12
PINLIST WE  = PD25
PINLIST PIN = P33-40

IN1=3.0V,0.0V

ADR=IN1,NRZA,ACLK1,<X0-11>
PIN=IN1,NRZA,ACLK2,<D0-7>
WE=IN1,/RZO,BCLK1,CCLK1,<WT>
RATE=100NS
ACLK1=40NS
ACLK2=50NS
BCLK1=60NS
CCLK1=70NS

DUTNO=1
DATA(1)=#11800000
DATA(2)=#22000000
:
DATA(34)=#FF800000
CALL UBMWRT(ERR,DUTNO,BLKNO,STADR,CNT,DATA)
IF ERR<>0 THEN STOP / "UBMWRT ERROR" /
:
DUTNO=16
DATA(1)=#FF800000
DATA(2)=#EE000000
:
DATA(34)=#11800000
CALL UBMWRT(ERR,DUTNO,BLKNO,STADR,CNT,DATA)
IF ERR<>0 THEN STOP / "UBMWRT ERROR" /

CALL IDSELUBM(ERR,MODE,MUT,STMUT,STADR,PIN,JP1MUT,JP1ADR,JP2MUT,JP2ADR)
IF ERR<>0 THEN STOP / "IDSELUBM ERROR" /

REG MPAT PC=#0
SEND MPAT MUBM1
START MPAT
CALL IDSELCLR

END

```

Individual pattern setting for DUT 1

Individual pattern settings for DUT 2 to 15

Individual pattern setting for DUT 16

- Socket program

```
SOCKET SKSAM1
```

```
DEFINE MULTIPLE 16DUT 1STN
```

```
BIT=9
```

```
PPS=1
```

```
PIN MODE=B
```

```
CHTYPE11
```

DUT#	= VS1	PD1-12	PD25-27	PD65	PD31	P33-40	P49
DUT1	= PPS1	PD1-12A1	PD25-27A1	PD65A1	PD31A1	P33-40A1	P49A1
DUT2	= PPS2	PD13-24A1	PD28-30A1	PD66A1	PD32A1	P41-48A1	P50A1
DUT3	= PPS3	PD1-12A2	PD25-27A2	PD65A2	PD31A2	P97-104A1	P113A1
DUT4	= PPS4	PD13-24A2	PD28-30A2	PD66A2	PD32A2	P105-112A1	P114A1
DUT5	= PPS5	PD1-12A3	PD25-27A3	PD65A3	PD31A3	P33-40A3	P49A3
DUT6	= PPS6	PD13-24A3	PD28-30A3	PD66A3	PD32A3	P41-48A3	P50A3
DUT7	= PPS7	PD1-12A4	PD25-27A4	PD65A4	PD31A4	P97-104A3	P113A3
DUT8	= PPS8	PD13-24A4	PD28-30A4	PD66A4	PD32A4	P105-112A3	P114A3
DUT9	= PPS9	PD1-12B1	PD25-27B1	PD65B1	PD31B1	P33-40B1	P49B1
DUT10	= PPS10	PD13-24B1	PD28-30B1	PD66B1	PD32B1	P41-48B1	P50B1
DUT11	= PPS11	PD1-12B2	PD25-27B2	PD65B2	PD31B2	P97-104B1	P113B1
DUT12	= PPS12	PD13-24B2	PD28-30B2	PD66B2	PD32B2	P105-112B1	P114B1
DUT13	= PPS13	PD1-12B3	PD25-27B3	PD65B3	PD31B3	P33-40B3	P49B3
DUT14	= PPS14	PD13-24B3	PD28-30B3	PD66B3	PD32B3	P41-48B3	P50B3
DUT15	= PPS15	PD1-12B4	PD25-27B4	PD65B4	PD31B4	P97-104B3	P113B3
DUT16	= PPS16	PD13-24B4	PD28-30B4	PD66B4	PD32B4	P105-112B3	P114B3

```
END
```

- Pattern program

```

MPAT MUBM1

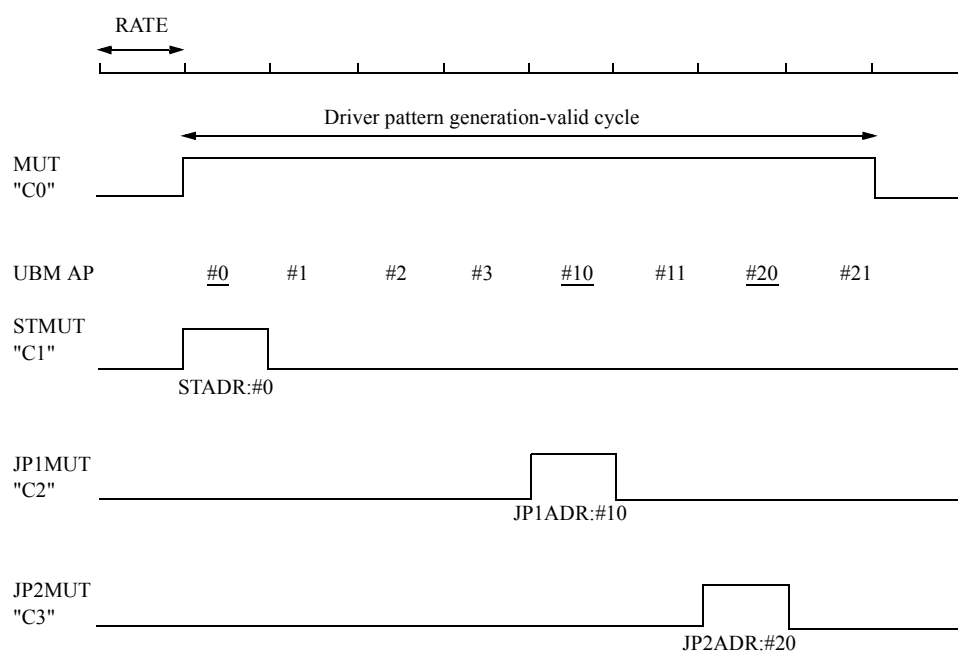
REGISTER

    START #0000
    NOP W
    NOP W C0 C3
    NOP W C0
    NOP W C0
    NOP W C0
    NOP W C0 C3
    NOP W C0
    NOP W C0 C3
    NOP W C0
    NOP
    STPS

    END
  
```

This pattern program operation is shown in the figure below.

Figure 62-8 Pattern Program Operation



◆ Sample Program of Fail Data Import (Single STRB):

- Test plan program

```

PRO SAM3    SKSAM1

INTE[1] ERR
INTE[1] MODE=2          ← Select fail data import.
CHAR[1] MUT="C0"
CHAR[1] STMUT="C1"
LOGI[1] STADR=#0
INTE[1] BITMD=1         ← Single STRB mode
INTE[1] DUTNO=1
INTE[1] BLKNO=1
LOGI[1] CNT=8
LOGI[2] DATA(8)=#0

PINLIST ADR = PD1-12
PINLIST WE  = PD25
PINLIST PIN = P33-40

IN1=3.0V,0.0V
OUT1=2.0V,1.0V

ADR=IN1,NRZA,ACLK1,<X0-7>
PIN=OUT1,STRB1,<D0-7>
WE=IN1,/RZO,BCLK1,CCLK1,<WT>
RATE=100NS
ACLK1=0NS
BCLK1=50NS
CCLK1=60NS
STRB1=70NS

CALL UBMCLR

CALL IDSELUBM(ERR,MODE,MUT,STMUT,STADR,BITMD,0,0,0,0)
IF ERR<>0 THEN STOP / "IDSELUBM ERROR" /

SELECT MPAT INHIBIT
REG MPAT PC=#0
SEND MPAT MUBM3
START MPAT

DO LPRD1 DUTNO=1,16
  CALL UBMREAD(ERR,DUTNO,BLKNO,STADR,CNT,DATA)
  IF ERR<>0 THEN STOP / "UBMREAD ERROR" /
  WRITE[1,1] "DUT" DUTNO
  DO LPRD2 I=1,CNT
    WRITE[1,1] #8 "DATA(" I ")=" DATA(I) /
LPRD2:
LPRD1:
  CALL IDSELCLR

END

```

- Socket program

The same SKSAM as in ["Sample Program of Individual Pattern Generation:"](#) on page 62-28.

- Pattern program

```

MPAT MUBM2

REGISTER

CPE1=R
SDEF ACLR  = XB<0 YB<0
SDEF XYINC = XB<XB+1 YB<YB+1^BX

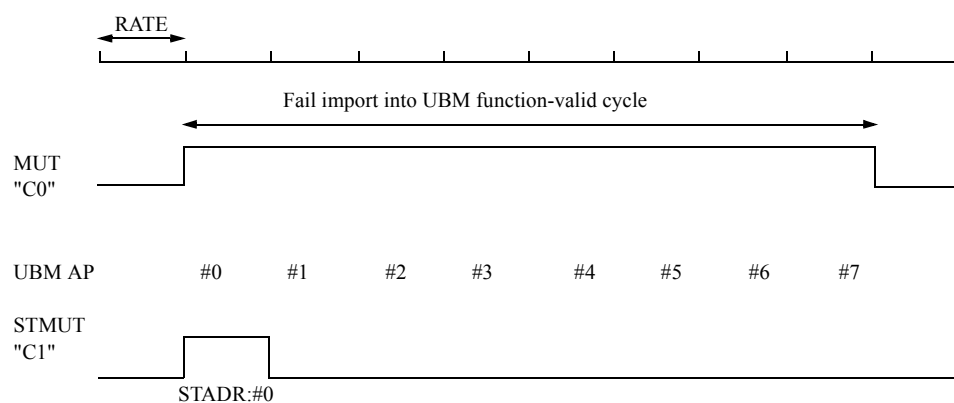
START #0000
NOP ACLR
:
NOP R XYINC C0 C1
NOP R XYINC C0
NOP R XYINC C0
NOP R XYINC C0
NOP R XYINC C0
NOP R XYINC C0
NOP R XYINC C0
NOP R XYINC C0
NOP R XYINC C0
STPS

END

```

This pattern program operation is shown in the figure below.

Figure 62-9 Pattern Program Operation



◆ Sample Program of Fail Data Import (Double STRB):

- Test plan program

```

PRO SAM3    SKSAM1

INTE[1] ERR
INTE[1] MODE=2          ← Select fail data import.
CHAR[1] MUT="C0"
CHAR[1] STMUT="C1"
LOGI[1] STADR=#0
INTE[1] BITMD=2         ← Double STRB mode
INTE[1] DUTNO=1
INTE[1] BLKNO=1
LOGI[1] CNT=8
LOGI[2] DATA(8)=#0

PINLIST ADR = PD1-12
PINLIST WE  = PD25
PINLIST PIN = P33-40

IN1=3.0V,0.0V
OUT1=2.0V,1.0V

ADR=IN1,NRZA,ACLK1,<X0-7>
PIN=OUT1,STRB1,STRB2,EXPAB,<D0-7,SD0-7>
WE=IN1,/RZO,BCLK1,CCLK1,<WT>
RATE=100NS
ACLK1=0NS
BCLK1=50NS
CCLK1=60NS
STRB1=70NS
STRB2=80NS

CALL UBMCLR

CALL IDSELUBM(ERR,MODE,MUT,STMUT,STADR,BITMD,0,0,0,0)
IF ERR<>0 THEN STOP / "IDSELUBM ERROR" /

SELECT MPAT INHIBIT
REG MPAT PC=#0
SEND MPAT MUBM3
START MPAT

DO LPRD1 DUTNO=1,16
  STADR=#0          ← STADR for odd-numbered phase STRB
  CALL UBMREAD(ERR,DUTNO,BLKNO,STADR,CNT,DATA)
  IF ERR<>0 THEN STOP / "UBMREAD ERROR" /

  WRITE[1,1] "DUT" DUTNO
  WRITE[1] #8 "STRB1"
  DO LPRD2 I=1,CNT
    WRITE[1,1] #8 "DATA("I")=" DATA(I) /
LPRD2:
LPRD1:

```

```

DO LPRD3 DUTNO=1,16
  STADR=#1000
  CALL UBMREAD(ERR,DUTNO,BLKNO,STADR,CNT,DATA)
  IF ERR<>0 THEN STOP / "UBMREAD ERROR" /

  WRITE[1,1] "DUT" DUTNO
  WRITE[1] #8 "STRB2"
  DO LPRD4 I=1,CNT
    WRITE[1,1] #8 "DATA("I")=" DATA(I) /
LPRD4:
LPRD3:
  CALL IDSELCLR
END

```

← STADR for even-numbered phase STRB

- Socket program

The same SKSAM as in ["Sample Program of Individual Pattern Generation:"](#) on page 62-28.

- Pattern program

```

MPAT MUBM3

REGISTER

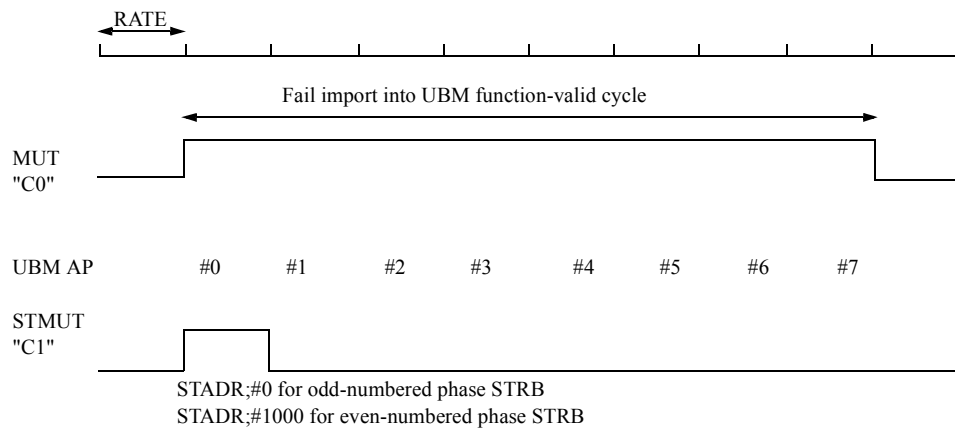
  CPE1=R
  CPE2=R
  SDEF ACLR = XB<0 YB<0
  SDEF XYINC = XB<XB+1 YB<YB+1^BX

  START #0000
  NOP ACLR
  :
  NOP R XYINC C0 C1
  NOP R XYINC C0
  NOP R XYINC C0
  NOP R XYINC C0
  NOP R XYINC C0
  NOP R XYINC C0
  NOP R XYINC C0
  NOP R XYINC C0
  STPS

END

```

This pattern program operation is shown in the figure below.

Figure 62-10 Pattern Program Operation

63. MEAS/START DC Control Utility for Branch Drivers

When a driver is assigned two or four DUTs in the socket program to configure a branch driver, executing the MEAS DC or START DC statement with this utility allows you to obtain measurement results for each DUT.

When a branch driver (see ^{*1}) is specified in the socket program, this utility can control the following statements:

- MEAS DC and START DC statements
- MULTIPLE DUT PARALLEL statement

The utility control is cleared and reset when one of the following operations occurs:

- Starting the program
 - Executing the RESET TESTER statement
 - Holding down the CTRL key and pressing the C key
- ➔ For information on testing when setting the branch driver, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498\)](#).

**1 A branch driver refers to a driver assigned to more than one DUT.*

63. 1 Syntax

1. Specifying how the DUTs are tested

```
CALL MSDCSEPA(dat)
```

dat

Selects a control state

Use an integer constant or variable for sw.

- 1 The DUTs assigned to the same branch driver are tested simultaneously.
- 2 The DUTs assigned to the same branch driver are tested individually.

2. Confirming the DUT control status

```
CALL RMSDCSEP(var)
```

var

Control status This is read to an integer variable.

Specifies integer-type variables.

- 0 Initial state without having executed CALL MSDCSEPA This is also read when the socket program is not used or when a socket program without a branch driver setting is used.
- 1 This is read when 2 is specified for the CALL MSDCSEPA command.
- 2 This is read when 2 is specified for the CALL MSDCSEPA command.

❗ Important

If an incorrect argument is specified, a runtime error occurs.

👉 Limitation

The MSDCSEPA is available when a socket program with a branch driver setting is used. When a socket program is not used, or when a socket program without a branch driver setting is used, MSDCSEPA is not used to control the target DUTs.

63. 2 Usage Examples

◆ A Socket Program Example is Shown Below

```
SOCKET SC128D9

DEFINE MULTIPLE 128DUT 2STN
    BIT = 9
    PPS = 2
    PIN MODE = B
    CHTYPE9
    SEPARATE PPS 2
    SEPARATE DR 2

DUT#    = VS1    VS2    PD1-22    PD23    PD157    P33-40    P49
DUT1    = PPS1    PPS97 PD1-22A1 PD23A1 PD157A1 P33-40A1 P49A1
DUT2    = PPS2    PPS97 PD1-22A1 PD24A1 PD157A1 P41-48A1 P50A1
DUT3    = PPS3    PPS98 PD65-86A1 PD87A1 PD159A1 P97-104A1 P113A1
DUT4    = PPS4    PPS98 PD65-86A1 PD88A1 PD159A1 P105-112A1 P114A1
.....
```

◆ A Test Plan Program Example is Shown Below

```
PRO A1 SC128D9

VSIM=1V R(10V) M(80UA)

MEAS DC (PD5)          *1
MEAS DC (P33)          *2

CALL MSDCSEPA(2)       *3
MEAS DC (PD5)          *4
MEAS DC (P33)          *5

END
```

- *1 The DUTs assigned to the branch driver are tested simultaneously.
PD5 is a driver pin, which is used to assign two DUTs to the same driver channel. The DUTs have the same test results.
- *2 The DUTs assigned to the branch driver are tested simultaneously.
P33 is an I/O pin, which is used to assign DUTs to different I/O channels. The DUTs assigned to the branch driver are simultaneously measured using the same DC measurement unit. As a result, each DUT has the same test result.
- *3 The individual test is set for the DUTs assigned to the branch driver.

**4 The DUTs assigned to the branch driver are tested individually.*

PD5 is a driver pin, which is used to assign two DUTs to the same driver channel. The DUTs are tested individually but have the same test results.

**5 The DUTs assigned to the branch driver are tested individually.*

P33 is an I/O pin, which is used to assign DUTs to different I/O channels. As a result, each DUT has a different test result.

64. Address Setting Utility for AFM Double-Speed Mode (FMADRSET)

Because this utility is used to assign addresses for AFM double-speed mode to the AFM, it is used in place of the AFM ADDRESS statement. While there are various combinations for device burst address progress, such as burst type, burst length, and start address, the double-speed mode specification for the AFM ADDRESS statement is not compatible with all of these combinations. (see [Table 64-1](#)). Using prescrambling to generate burst addresses, this utility can be used with all combinations. Therefore, when the AFM ADDRESS statement cannot be used to make a specification, defect analysis can be performed using this utility.

Table 64-1 Burst Address Progress and Benefits/Disadvantages of the AFM ADDRESS Statement

Burst length	Start address	Burst type	Burst address progress	Benefits/Disadvantages of the AFM ADDRESS Statement
2	0	Sequential	0, 1	Specifiable
		Interleave	0, 1	Specifiable
	1	Sequential	1, 0	Specifiable
		Interleave	1, 0	Specifiable
4	0	Sequential	0, 1, 2, 3	Specifiable
		Interleave	0, 1, 2, 3	Specifiable
	1	Sequential	1, 2, 3, 0	Not specifiable
		Interleave	1, 0, 3, 2	Specifiable
	2	Sequential	2, 3, 0, 1	Specifiable
		Interleave	2, 3, 0, 1	Specifiable
	3	Sequential	3, 0, 1, 2	Not specifiable
		Interleave	3, 2, 1, 0	Specifiable

Burst length	Start address	Burst type	Burst address progress	Benefits/Disadvantages of the AFM ADDRESS Statement
8	0	Sequential	0, 1, 2, 3, 4, 5, 6, 7	Specifiable
		Interleave	0, 1, 2, 3, 4, 5, 6, 7	Specifiable
	1	Sequential	1, 2, 3, 4, 5, 6, 7, 0	Not specifiable
		Interleave	1, 0, 3, 2, 5, 4, 7, 6	Specifiable
	2	Sequential	2, 3, 4, 5, 6, 7, 0, 1	Specifiable
		Interleave	2, 3, 0, 1, 6, 7, 4, 5	Specifiable
	3	Sequential	3, 4, 5, 6, 7, 0, 1, 2	Not specifiable
		Interleave	3, 2, 1, 0, 7, 6, 5, 4	Specifiable
	4	Sequential	4, 5, 6, 7, 0, 1, 2, 3	Specifiable
		Interleave	4, 5, 6, 7, 0, 1, 2, 3	Specifiable
	5	Sequential	5, 6, 7, 0, 1, 2, 3, 4	Not specifiable
		Interleave	5, 4, 7, 6, 1, 0, 3, 2	Specifiable
	6	Sequential	6, 7, 0, 1, 2, 3, 4, 5	Specifiable
		Interleave	6, 7, 4, 5, 2, 3, 0, 1	Specifiable
	7	Sequential	7, 0, 1, 2, 3, 4, 5, 6	Not specifiable
		Interleave	7, 6, 5, 4, 3, 2, 1, 0	Specifiable

Tip

- This utility can be used with the T5592/T5591/T5586/T5585/T5376/T5375/T5371.
- The AFM ADDRESS statement cannot be specified when the burst length is four or more, the start address is an odd number (1, 3, 5, 7), and the burst type is sequential.

64. 1 FMADRSET Syntax

◆ Specifying Addresses Directly

- ◆ When the Burst Length is 2:

```
CALL FMADRSET("<BA1, BA2>, AD1, AD2, ..., AD30", ECODE)
```

- ◆ When the Burst Length is 4:

```
CALL FMADRSET("<BA1, BA2, BA3, BA4>, AD1, AD2, ..., AD28", ECODE)
```

- ◆ When the Burst Length is 8:

```
CALL FMADRSET("<BA1, BA2, BA3, BA4, BA5, BA6>, AD1, AD2, ..., AD26", ECODE)
```

◆ When Specifying an Address as a Variable:

- ◆ The Same for Burst Lengths of 2, 4, and 8.

```
CALL FMADRSET(STRV, ECODE)
```

◆ Meaning of Each Argument

BA1 to 6

Specification of burst address. An address is arbitrarily assigned to each member from BA1 to 6 such that X and Y are both 0 to 15. Overlapping X and Y addresses cannot be selected. Also, addresses assigned to BA1 to 6 cannot be specified for AD1 to 30, discussed later.

- ◆ Example of Specification

```
CALL FMADRSET("<Y0, Y1, Y2, Y13, Y14, Y15>, X0, X1, X2, X3, X4, Y3, Y4", ECODE)
```

AD1 to 30

Specification of non-burst addresses. An address is arbitrarily assigned to each member from AD1 to 30 such that X and Y are both 0 to 15. Overlapping X and Y addresses cannot be selected. Also, addresses assigned to AD1 to 30 cannot be specified for BA1 to 6, discussed earlier.

Numbers of selectable address types

Thirty addresses for a burst length of 2; twenty-eight for a burst length of 4; and twenty-six for a burst length of 8.

STRV

Specifies string variables storing the addresses set for the AFM. The address specifications appearing in the quotes in [Specifying Addresses Directly](#) can be stored in, and specified using, the string variables.

- ◆ Example of Specification

```
STRING STRV(35) = "<Y0, Y1, Y2, Y13, Y14, Y15>, X0, X1, X2, X3, X4, Y3, Y4"
CALL FMADRSET(STRV, ECODE)
```


ECODE

Specifies a variable for storing an error code. Specify an integer/logical variable of 1 or 2 words.
The meaning for each error code is as follows:

Table 64-2 FMADRSET Error Code

Error Code	Meaning
0	Normal end
1	There is a mistake in the syntax.
2	The specified address number exceeds the maximum value.
3	There is a mistake in a burst address address specification.
4	The same address is selected more than once.
5	Cannot execute in this system.

64. 2 Relationship between Burst Addresses (BA1 to 6) and STRBn

The burst addresses (BA1 to 6) are related to the STRB to be used as described below.

- ♦ For CALL FMADRSET("<BA1, BA2>, AD1, AD2, ..., AD30", ECODE)
BA1 and BA2 are set for the least significant AFM block address bit.

On the T5592, T5591, T5586 and T5585

BA1 is used to select addresses for AFM blocks that capture fail addresses detected using the odd-numbered phase STRBs from 1 to 128.

BA2 is used to select addresses for AFM blocks that capture fail addresses detected using the odd-numbered phase STRBs from 129 to 256.

On the T5376, T5375 and T5371

BA1 is used to select addresses for AFM blocks that capture fail addresses detected using STRBs 1 and 3.

BA2 is used to select addresses for AFM blocks that capture fail addresses detected using STRBs 2 and 4.

- ♦ For CALL FMADRSET("<BA1, BA2, BA3, BA4>, AD1, AD2, ..., AD28", ECODE):
BA1, 2 and BA3, 4 are set for the two least significant AFM block address bits.

On the T5592, T5591, T5586 and T5585

BA1 and 2 are used to select addresses for AFM blocks that capture fail addresses detected using the odd-numbered phase STRBs from 1 to 128.

BA3 and 4 are used to select addresses for AFM blocks that capture fail addresses detected using the odd-numbered phase STRBs from 129 to 256.

On the T5376, T5375 and T5371

BA1 and 2 are used to select addresses for AFM blocks that capture fail addresses detected using STRBs 1 and 3.

BA3 and 4 are used to select addresses for AFM blocks that capture fail addresses detected using STRBs 2 and 4.

- ♦ For CALL FMADRSET("<BA1, BA2, BA3, BA4, BA5, BA6>, AD1, AD2, ..., AD26", ECODE):

BA1 to 3 and BA4 to 6 are set for the three least significant AFM block address bits.

On the T5592/T5591/T5586/T5585

BA1 to 3 are used to select addresses for AFM blocks that capture fail addresses detected using the odd-numbered phase STRBs from 1 to 128.

BA4 to 6 are used to select addresses for AFM blocks that capture fail addresses detected using the odd-numbered phase STRBs from 129 to 256.

On the T5376, T5375 and T5371

BA1 to 3 are used to select addresses for AFM blocks that capture fail addresses detected using STRBs 1 and 3.

BA4 to 6 are used to select addresses for AFM blocks that capture fail addresses detected using STRBs 2 and 4.

64. 3 Addresses for Defect Analysis

When the FMADRSET is used, the addresses specified using the FMADRSET utility serve as addresses for testing and for defect analysis.

Tip

- Testing addresses are addresses used for capturing fail data to the AFM during testing.
- Defect analysis addresses are used for reading fail data from the AFM after testing.
- When the FMADRSET utility is not used, the addresses specified in the defect analysis statements (the AFM memory, READ AFM, RESET AFM, and PRESET AFM statements) are used for defect analysis.

- ◆ For CALL FMADRSET("<BA1, BA2>, AD1, AD2, ..., AD30", ECODE)

When BA1 is the X address, the least significant bit of the defect analysis address serves as X0.

When BA1 is the Y address, the least significant bit of the defect analysis address serves as Y0.

◆ Example of Specification

```
CALL FMADRSET("<Y0, Y15>, X0, X1, X2, X3, X4, Y1, Y2, Y3, Y4", ECODE)
```

When the addresses mentioned above are specified, the testing and defect analysis addresses are as follows: (A0 to A31 are AFM address bits)

Test addresses

A31	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
0	0	Y4	Y3	Y2	Y1	X4	X3	X2	X1	X0	Y0 or Y15

Defect analysis addresses

A31	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
0	0	Y4	Y3	Y2	Y1	X4	X3	X2	X1	X0	Y0

- ◆ For CALL FMADRSET("<BA1, BA2, BA3, BA4>, AD1, AD2, ..., AD28", ECODE):

When BA1 is the X address, the two least significant bits of the defect analysis address serve as X0 and X1.

When BA1 is the Y address, the two least significant bits of the defect analysis address serves as Y0 and Y1.

◆ Example of Specification

```
CALL FMADRSET("<Y0, Y1, Y14, Y15>, X0, X1, X2, X3, X4, Y2, Y3, Y4", ECODE)
```

When the addresses mentioned above are specified, the testing and defect analysis addresses are as follows: (A0 to A31 are AFM address bits)

Test addresses

A31	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
0	0	Y4	Y3	Y2	X4	X3	X2	X1	X0	Y1 or Y15	Y0 or Y14

Defect analysis addresses

A31	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
0	0	Y4	Y3	Y2	X4	X3	X2	X1	X0	Y1	Y0

- ♦ For CALL FMADRSET("<BA1, BA2, BA3, BA4, BA5, BA6>, AD1, AD2, ..., AD26", ECODE):

When BA1 is the X address, the three least significant bits of the defect analysis address serves as X0, X1 and X2.

When BA1 is the Y address, the three least significant bits of the defect analysis address serves as Y0, Y1 and Y2.

♦ Example of Specification

```
CALL FMADRSET("<Y0, Y1, Y2, Y13, Y14, Y15>, X0, X1, X2, X3, X4, Y3, Y4", ECODE)
```

When the addresses mentioned above are specified, the testing and defect analysis addresses are as follows: (A0 to A31 are AFM address bits)

Test addresses

A31	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
0	0	Y4	Y3	X4	X3	X2	X1	X0	Y2 or Y15	Y1 or Y14	Y0 or Y13

Defect analysis addresses

A31	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
0	0	Y4	Y3	X4	X3	X2	X1	X0	Y2	Y1	Y0

64. 4 Notes on Using IDSELUBM

- This utility does not have a function equivalent to the DELAY or SCRAM specification for the AFM ADDRESS specification. To use these functions in combination, specify SCRAM or DELAYn using the AFM ADDRESS statement before executing this utility.

- ♦ **Example of Specifying SCRAM and DELAYn**

```
AFM ADDRESS = SCRAM, DELAY7  
CALL FMADRSET("<Y0, Y12>, X0, X1, X2, X3, X4, Y1, Y2, Y3, Y4", ECODE)
```

- Because this utility uses a prescrambler, it is necessary to create a prescrambling program to generate burst addresses, and send it to the prescrambler in advance.
- As with the AFM ADDRESS statement, the addresses specified using this utility are set for the AFM when the AFM ACTION, MEAS MPAT, or START MPAT statement is executed.

64. 5 Program Example

Program examples for each burst length (for the T5592/T5591) appear below.

◆ When the Burst Length is 2:

◆ Test Plan Program

```
PRO  SAMP1
      :
      INTEGER[1] ECODE = 0
      :
      SET ILMODE 4WAY
      RESET PRESCRAM
      SELECT PRESCRAM MODE NORMAL
      SELECT PRESCRAM ENABLE
      SELECT PRESCRAM ADDRESS FP, FM
      SEND PRESCRAM SAMPSCR

      RESET AFM ALL
      CALL FMADRSET("<Y0,Y10>,X0,X1,X2,X3,X4,X5,X6,X7,X8,Y1,Y2,Y3,Y4,Y5,Y6,Y7,Y8,Y9",ECODE)
      IF ECODE<>0 THEN STOP / "FMADRSET ERROR!!" /
      AFM ACTION = FT>FMX

      REG MPAT PC = #0
      MEAS MPAT MSAMP1
      :
END
```

◆ Pattern Program

```

MPAT      MSAMP1

SDEF INIT = XC<0 YC<0 Z<0 TP<TPH, TP2<TPH2 D4<D4B D3<D3B XB<0 YB<0
SDEF COUT = X<XC Y<YC                      ; Current Register
SDEF XINC = XC<XC+1                          ; Row Address (+1) Increment
SDEF YDI4 = YC<YC+D4                         ; Column Address (+D4B) Increment

ILMODE 4WAY

REGISTER
    D4B = #2
    :
    START #0

    NOP      : INIT
              :
              :
              :
    NOP      : R COUT Z<Z+1 XINC YDI4 ; Z<Z+1 Binary Counter for Burst Address
              : R COUT Z<Z+1 XINC YDI4
              : R COUT Z<Z+1 XINC YDI4
              : R COUT Z<Z+1 XINC YDI4
    NOP      : R COUT Z<Z+1 XINC YDI4
              : R COUT Z<Z+1 XINC YDI4
              : R COUT Z<Z+1 XINC YDI4
              : R COUT Z<Z+1 XINC YDI4
              :
    STPS

END

```


◆ Scramble Program (For Generating Sequential and Interleaved Addresses)

```
SCRAM      SAMPSCR

RDX 10

AMAX = #1FF, #7FF

EXPRESSION
    YA0  = Y0
    YA10 = .NOT.Y0

ADDRESS
    Y0   = YA0
    Y1   = YA1
    Y2   = YA2
    Y3   = YA3
    Y4   = YA4
    Y5   = YA5
    Y6   = YA6
    Y7   = YA7
    Y8   = YA8
    Y9   = YA9
    Y10  = YA10

END
```

◆ When the Burst Length is 4:

◆ Test Plan Program

```
PRO SAMP2
:
INTEGER[1] ECODE = 0
:
SET ILMODE 4WAY
RESET PRESCRAM
SELECT PRESCRAM MODE NORMAL
SELECT PRESCRAM ENABLE
SELECT PRESCRAM ADDRESS FP, FM
IF FK = #1 THEN SEND PRESCRAM INTSCR2      ; Interleave Address
                SEND PRESCRAM SEQSCR2      ; Sequential Address

RESET AFM ALL
CALL FMADRSET("<Y0,Y1,Y10,Y11>,X0,X1,X2,X3,X4,X5,X6,X7,X8,Y2,Y3,Y4,Y5,Y6,Y7,Y8,Y9",ECODE)
IF ECODE<>0 THEN STOP / "FMADRSET ERROR!!" /
AFM ACTION = FT>FMX

REG MPAT PC = #0
MEAS MPAT MSAMP2
:
END
```

◆ Pattern Program

```

MPAT      MSAMP2

SDEF INIT = XC<0 YC<0 Z<0 TP<TPH, TP2<TPH2 D4<D4B D3<D3B XB<0 YB<0
SDEF COUT = X<XC Y<YC                ; Current Register
SDEF XINC = XC<XC+1                  ; Row Address (+1) Increment
SDEF YDI4 = YC<YC+D4                 ; Column Address (+D4B) Increment

ILMODE 4WAY

REGISTER
    D4B = #4
    :
    START #0

    NOP      : INIT
              :
              :
              :

    NOP      : COUT Z<Z+1          ; Z<Z+1 Binary Counter for Burst Address
              : COUT Z<Z+1 XINC YDI4
              : COUT Z<Z+1
              : COUT Z<Z+1 XINC YDI4
    NOP      : COUT Z<Z+1
              : COUT Z<Z+1 XINC YDI4
              : COUT Z<Z+1
              : COUT Z<Z+1 XINC YDI4

    :
    STPS

END

```

◆ Scramble Program (For Generating Sequential Addresses)

```
SCRAM    SEQSCR2

RDX 10

AMAX = #1FF, #FFF

EXPRESSION
    YA0  = Y0
    YA1  = Y1.XOR.Z0
    YA10 = .NOT.Y0
    YA11 = Y1.XOR.Z0.XOR.Y0

ADDRESS
    Y0   = YA0
    Y1   = YA1
    Y2   = YA2
    Y3   = YA3
    Y4   = YA4
    Y5   = YA5
    Y6   = YA6
    Y7   = YA7
    Y8   = YA8
    Y9   = YA9
    Y10  = YA10
    Y11  = YA11

END
```

◆ Scramble Program (For Generating Interleaved Addresses)

```
SCRAM      INTSCR2

RDX 10

AMAX = #1FF, #FFF

EXPRESSION
    YA0  = Y0
    YA1  = Y1.XOR.Z0
    YA10 = .NOT.Y0
    YA11 = Y1.XOR.Z0

ADDRESS
    Y0   = YA0
    Y1   = YA1
    Y2   = YA2
    Y3   = YA3
    Y4   = YA4
    Y5   = YA5
    Y6   = YA6
    Y7   = YA7
    Y8   = YA8
    Y9   = YA9
    Y10  = YA10
    Y11  = YA11

END
```

◆ When the Burst Length is 8:

◆ Test Plan Program

```
PRO SAMP3
:
  INTEGER[1] ECODE = 0
:
  SET ILMODE 4WAY
  RESET PRESCRAM
  SELECT PRESCRAM MODE NORMAL
  SELECT PRESCRAM ENABLE
  SELECT PRESCRAM ADDRESS FP, FM
  IF FK = #1 THEN    SEND PRESCRAM INTSCR3    ; Interleave Address
                    SEND PRESCRAM SEQSCR3    ; Sequential Address

  RESET AFM ALL
  CALL FMADRSET("<Y0,Y1,Y2,Y10,Y11,Y12>,X0,X1,X2,X3,X4,X5,X6,X7,X8,Y3,Y4,Y5,Y6,Y7,Y8,Y9",ECODE)
  IF ECODE<>0 THEN STOP / "FMADRSET ERROR!!" /
  AFM ACTION = FT>FMX

  REG MPAT PC = #0
  MEAS MPAT MSAMP3
:
END
```

◆ Pattern Program

```

MPAT      MSAMP3

SDEF INIT = XC<0 YC<0 Z<0 TP<TPH, TP2<TPH2 D4<D4B D3<D3B XB<0 YB<0
SDEF COUT = X<XC Y<YC                ; Current Register
SDEF XINC = XC<XC+1                  ; Row Address (+1) Increment
SDEF YDI4 = YC<YC+D4                 ; Column Address (+D4B) Increment

ILMODE 4WAY

REGISTER
    D4B = #8
    :
    START #0

    NOP      : INIT
              :
              :
              :
    NOP      : COUT Z<Z+1          ; Z<Z+1 Binary Counter for Burst Address
              : COUT Z<Z+1
              : COUT Z<Z+1
              : COUT Z<Z+1 XINC YDI4
    NOP      : COUT Z<Z+1
              : COUT Z<Z+1
              : COUT Z<Z+1
              : COUT Z<Z+1 XINC YDI4
    :
    STPS

END

```

◆ Scramble Program (For Generating Sequential Addresses)

```
SCRAM      SEQSCR3

RDX 10

AMAX = #1FF, #1FFF

EXPRESSION
    YA0  = Y0
    YA1  = Y1.XOR.Z0
    YA2  = Y2.XOR.Z1.XOR.(Y1.AND.Z0
)
    YA10 = .NOT.Y0
    YA11 = Y1.XOR.Z0.XOR.Y0
    YA12 = Y2.XOR.Z1.(Y1.AND.Z0).OR.(Y1.OR.Z0).AND.Y0)

ADDRESS
    Y0   = YA0
    Y1   = YA1
    Y2   = YA2
    Y3   = YA3
    Y4   = YA4
    Y5   = YA5
    Y6   = YA6
    Y7   = YA7
    Y8   = YA8
    Y9   = YA9
    Y10  = YA10
    Y11  = YA11
    Y12  = YA12

END
```


◆ Scramble Program (For Generating Interleaved Addresses)

```
SCRAM      INTSCR3

RDX 10

AMAX = #1FF, #1FFF

EXPRESSION
    YA0  = Y0
    YA1  = Y1.XOR.Z0
    YA2  = Y2.XOR.Z1
    YA10 = .NOT.Y0
    YA11 = Y1.XOR.Z0
    YA12 = Y2.XOR.Z1

ADDRESS
    Y0   = YA0
    Y1   = YA1
    Y2   = YA2
    Y3   = YA3
    Y4   = YA4
    Y5   = YA5
    Y6   = YA6
    Y7   = YA7
    Y8   = YA8
    Y9   = YA9
    Y10  = YA10
    Y11  = YA11
    Y12  = YA12

END
```

65. DFM Simultaneous Capture Utility

This chapter explains the utility that performs DFM simultaneous capture.

65. 1 Overview of the DFM Simultaneous Capture Utility

Analysis of fail information (pass/fail data) uses the DFM (data fail memory).

There are two types of DFM, used respectively on PIN-DFM boards and AFM (address fail memory) boards.

When interleave mode is 1WAY, 1024 cycles of data can be stored in memory; when it is 2WAY, 2048 cycles can be stored.

Fail data stored to the DFM can be selected as either fails generated by comparisons with expected values for each I/O channel, or fails resulting from comparisons between DQ and DQS in DQS testing (fails conforming to the look-up table).

➔ For more information, refer to [57. "DQS Test Utility" on page 57-1](#).

In a single test, all DUTs can be analyzed for a PIN-DFM board. For an AFM board, however, DUTs are analyzed for DFM in accordance with AFM assignments.

AFM assignment conditions

- AFM MULTIMODE statement
- AFM DUTSEL statement (when the AFM MULTIMODE statement is SINGLE)
- AFM PGSEL statement
- AFM double-speed on/off for AFM ADDRESS statement
- AFM ACTION statement (necessary that FT>FMX)

This utility performs settings for assigning multiple DUTs to two types of DFMs, and stores to the two types of DFM pass/fail data up to pattern n for successive pattern programs specified using the DFM MODE statement (DFM MODE = ALL UP TO PCNTn). It then reads data from j cycles before pattern n up to pattern n for each DUT. It can also display that data.

n

Multiples of 1024 are used when the interleave mode is 1WAY.

1024, 2048, 3072, 4096, ... maximum 536869888(#1FFFC00).

Multiples of 2048 are used when the interleave mode is 2WAY.

2048, 4096, 6144, 8192, ... maximum 536868864(#1FFF800).

j

1024 is used when the interleave mode is 1WAY.

2048 is used when the interleave mode is 2WAY.

Single-measurement operation (no socket program, no socket program parallel test definition) is also possible.

This utility consists of the following three programs.

PDFMSET

Setting and clearing functions of the DFM simultaneous capture utility
Specify before testing.

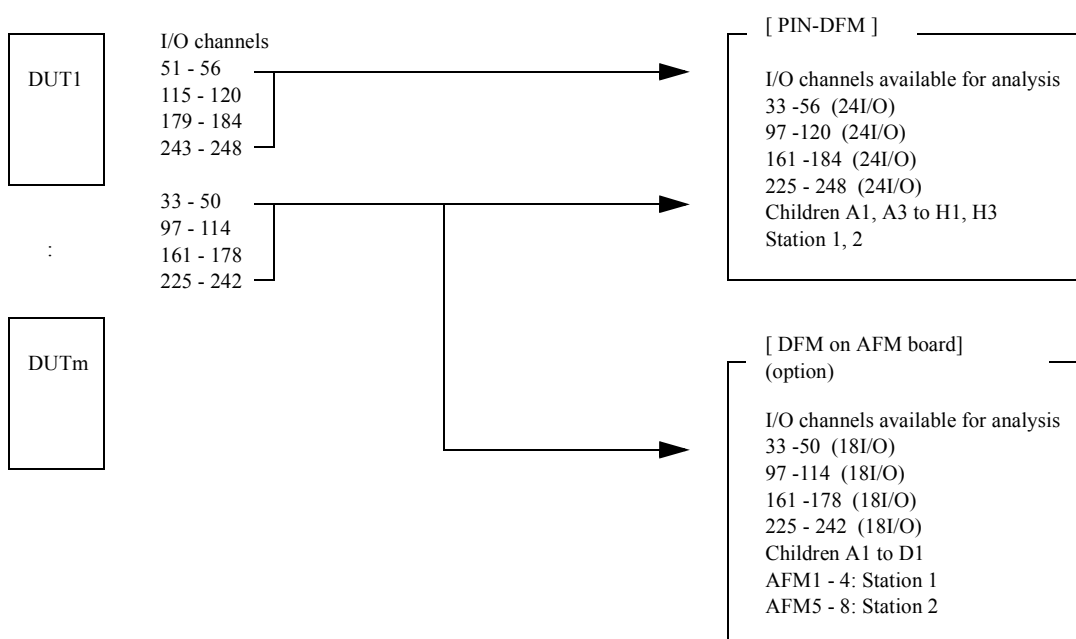
PDFMREAD

Reads fail information captured to the DFM for each I/O channel block of each DUT

PDFMDISP

Displays fail information captured to the DFM for each DUT.

Figure 65-1 Flow of Fail Information



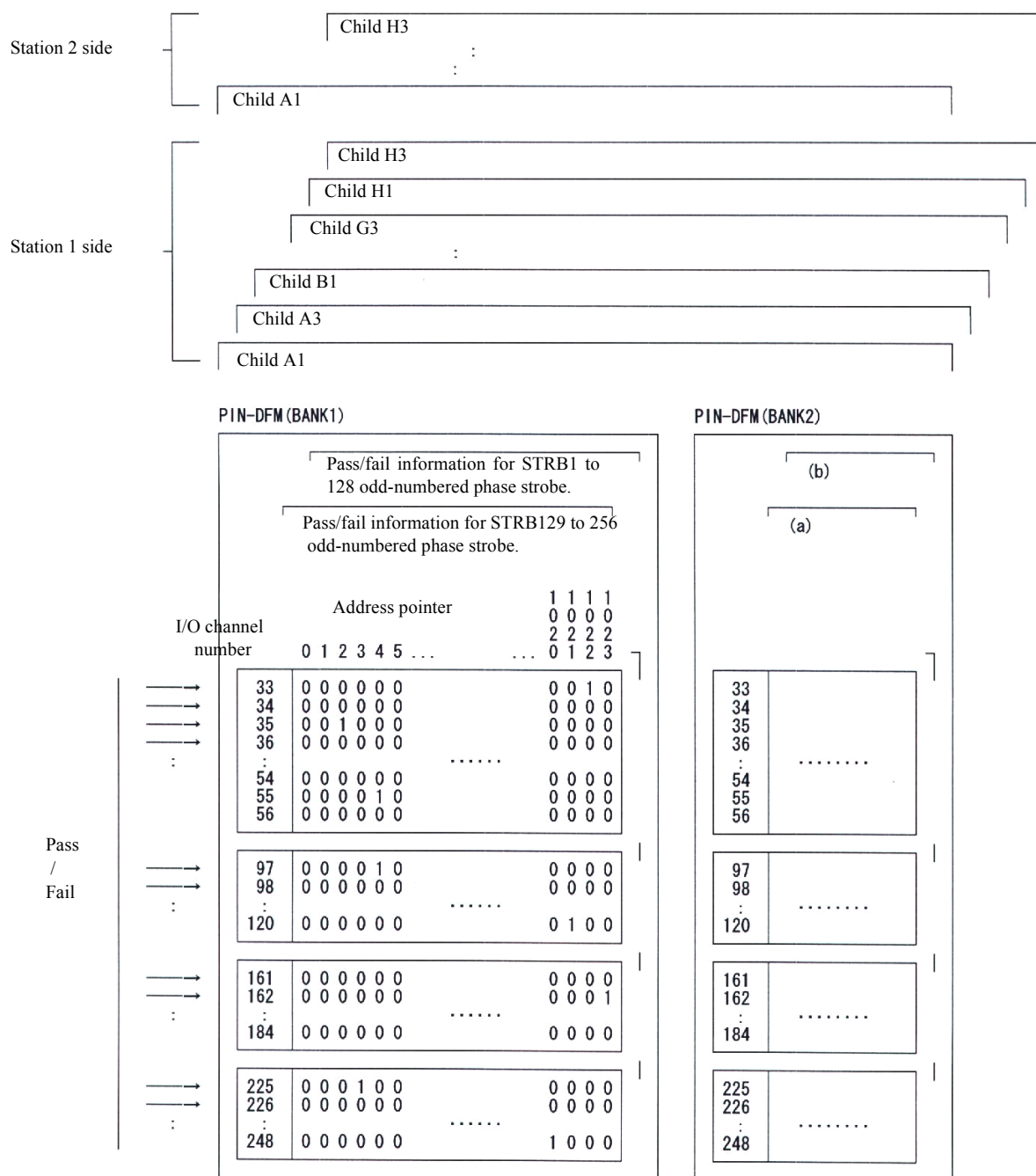
For DFM on an AFM board:

DUTs available for analysis follow the AFM assignment.

- AFM MULTIMODE statement
- AFM DUTSEL statement (when AFM MULTIMODE is SINGLE)

➔ For more information, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498\)](#).

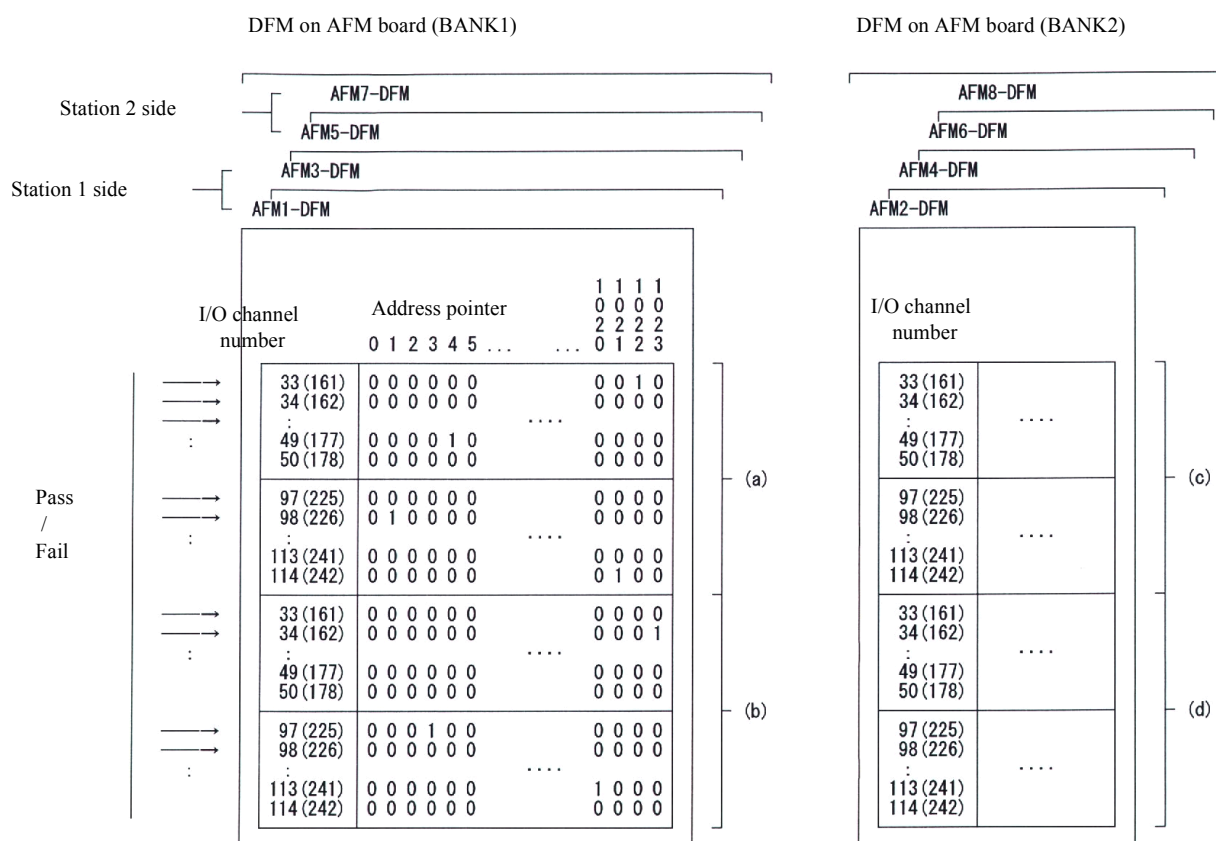
Figure 65-2 Image of PIN-DFM



Pass/fail information for (a), (b)

AFM PGSEL	Pass/fail information for (a), (b) target STRB
MAIN	(a) : STRB1 to 128 odd-numbered phase STRB (b) : STRB129 to 256 odd-numbered phase STRB
MAIN, SUB	(a) : STRB1 to 128 even-numbered phase STRB (b) : STRB129 to 256 even-numbered phase STRB

Figure 65-3 Image of DFM on an AFM Board:



Pass/fail information for (a) to (d)

Inter-leave mode	AFM PGSEL	AFM double-speed mode	Pass/fail information for (a) to (d) target STRB	Supplement
1WAY	MAIN	Absent	(a) to (d) : STRB1 to 256 odd-numbered phase strobe	--
		Present	(a), (c) : STRB1 to 128 odd-numbered phase STRB (b), (d) : STRB129 to 256 odd-numbered phase STRB	(a) and (b) are on the same I/O channel. (c) and (d) are on the same I/O channel.
	MAIN, SUB	Absent	(a), (b) : STRB1 to 256 odd-numbered phase strobe (c), (d) : STRB1 to 256 even-numbered phase strobes	(a) and (c) are on the same I/O channel. (b) and (d) are on the same I/O channel.
		Present	(a) : STRB1 to 128 odd-numbered phase STRB (b) : STRB129 to 256 odd-numbered phase STRB (c) : STRB1 to 256 even-numbered phase STRB (d) : STRB129 to 256 even-numbered phase STRB	(a) to (d) are on the same I/O channel.
2WAY	MAIN	Absent	(a) to (d) : STRB1 to 256 odd-numbered phase strobe	(a) and (c) are on the same I/O channel. (b) and (d) are on the same I/O channel.
		Present	(a), (c) : STRB1 to 128 odd-numbered phase STRB (b), (d) : STRB129 to 256 odd-numbered phase STRB	(a) to (d) are on the same I/O channel.

65. 2 PDFMSET

◆ Function

Setting and clearing functions of the DFM simultaneous capture utility
Specify in advance, before testing.

◆ Syntax

```
CALL PDFMSET(err, grpno)
```

err:

The end state of CALL PDFMSET (normal/abnormal end) is stored.

Specify using integer and logical variables (two word format).

If err is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For more information, refer to ["Description of Err" on page 65-19](#).

grpno

Turns the DFM parallel test function on/off (released)

Specify using a logical and integer variable (two-word format) or logical and integer constant equal to 0 to 1.

grpno	Meaning
0	Off
1	On

65. 3 PDFMREAD

◆ Function

Reads fail information captured to the DFM for each I/O channel block of each DUT
 DFM RWSEL must be specified when the AFM PGSEL statement data are MAIN and SUB.
 The correspondence between the AFM PGSEL and DFM RWSEL statements is as follows:

AFM PGSEL	DFM RWSEL
MAIN	Off
MAIN, SUB	On MAIN : Specify DFM RWSEL = MAIN in advance. SUB : Specify DFM RWSEL = SUB in advance.

◆ Syntax

```
CALL PDFMREAD(err, untsl, dutno, pnblk, rdcnt, rddat)
```

err:

The end state of CALL PDFMREAD (normal/abnormal end) is stored.
 Specify using integer and logical variables (two word format).
 If err is 0, it means a normal end; if ERR is not 0, it means an abnormal end.
 ➔ For more information, refer to ["Description of Err" on page 65-19](#).

untsl

Specifies the memory to read.
 Specify using a logical and integer variable (two-word format) or logical and integer constant equal to 1 to 2.

untsl	Target memory
1	PIN-DFM
2	DFM on AFM board

dutno

Specifies the DUT number to read.
 Specify using a logical and integer variable (two-word format) or logical and integer constant.

When untsl = 1

Specifiable DUT numbers are 1 or the DUT numbers for the parallel test definition range for the socket program.

When untsl = 2

Specifiable DUT numbers are the same as for AFM, and conform to the AFM assignments.
Use the READ AFM DUT SELECT statement for information on assigned DUT numbers.

pnblk

Specifies the I/O channel block to read.

Specify using a logical and integer variable (two-word format) or logical and integer constant equal to 1 to 2.

pnblk	I/O channel block
1	33 to 56
2	97 to 120
3	161 to 184
4	225 to 248

rdcnt

Stores the number of valid data stored.

Specify using integer and logical variables (two word format).

rdcnt data is invalid when err≠0.

Used when the number of patterns is less than the maximum number of data

The maximum number of valid data that can be stored to the DFM is as follows:

Interleave mode	AFM double-speed mode	Maximum number of valid data
1WAY	Absent	1024
	Present	2048
2WAY	Absent	2048
	Present	4096

rddat

Stores pass/fail information in sequence, from the data stored in the DFM lead address.

Specify logical variables (two word format) specified for all DIM specifications.

rddat data is invalid when err≠0 or rdcnt=0.

Data exceeding the number of valid data for rdcnt data is invalid.

The number of buffers necessary for reading data varies depending on operation conditions (interleave move and AFM double-speed mode). (see rdcnt).

When AFM double-speed mode is not specified, data for all STRBs are read.

When AFM double-speed mode is specified, there are two pass/fail data per cycle.

The odd-numbered read variables are data for STRB1 to 128, while the even-numbered ones are data for STRB129 to 256.

◆ Data Format

The numbers in the boxes are I/O channel numbers.

For each piece of data, 0=pass, and 1=fail (a blank field or unused I/O channel is treated as a 0.)

0	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
.....	5 6	5 5	5 4	5 3	5 2	5 1	5 0	4 9	4 8	4 7	4 6	4 5	4 4	4 3	4 2	4 1	4 0	3 9	3 8	3 7	3 6	3 5	3 4	3 3	
.....	1 2 0	1 1 9	1 1 8	1 1 7	1 1 6	1 1 5	1 1 4	1 1 3	1 1 2	1 1 1	1 1 0	1 0 9	1 0 8	1 0 7	1 0 6	1 0 5	1 0 4	1 0 3	1 0 2	1 0 1	1 0 0	9 9	9 8	9 7	
.....	1 8 4	1 8 3	1 8 2	1 8 1	1 8 0	1 7 9	1 7 8	1 7 7	1 7 6	1 7 5	1 7 4	1 7 3	1 7 2	1 7 1	1 7 0	1 6 9	1 6 8	1 6 7	1 6 6	1 6 5	1 6 4	1 6 3	1 6 2	1 6 1	
.....	2 4 8	2 4 7	2 4 6	2 4 5	2 4 4	2 4 3	2 4 2	2 4 1	2 4 0	2 3 9	2 3 8	2 3 7	2 3 6	2 3 5	2 3 4	2 3 3	2 3 2	2 3 1	2 3 0	2 2 9	2 2 8	2 2 7	2 2 6	2 2 5	

When the bits specified by the socket program are 9 and 12 (the positions of I/O channels 49, 51, 53 and 55 are different)

.....																5 5	5 3	5 1	4 9	4 0	3 9	3 8	3 7	3 6	3 5	3 4	3 3
-------	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--------	--------	--------	--------	--------	--------	--------	--------	--------	--------	--------	--------

65. 4 PDFMDISP

◆ Function

Displays fail information captured to the DFM for each DUT.

To execute PDFMDISP, turn on the DFM parallel test function using CALL PDFMSET, set the test to a post-execution pause state.

To pause the program, execute the PAUSE statement or set MANUAL and HOLD for the test mode. Also, pass/fail information can be output to a file by selecting output device for the S command.

◆ Start Up

Enter the utility start command from VKT.

```
/PDFMDISP <cr>
```

The following menu is output.

```
*****
*
*          PARALLEL DFM DISPLAY UTILITY          *
*
*****

  _DISPLAY
  . SELECT DFM _____ : PIN-DFM
  . SELECT DUT _____ : 1
  . DFM SKIP PATTERN COUNT : 0
  . OUTPUT DATA _____ : FAIL
  . PAGE PAUSE _____ : YES
  . SELECT DATA _____ : MAIN *1

[WORK]

[MEMO]
PF1:EXIT
```

**1 Displayed when AFM PGSEL=MAIN,SUB.*

Select the item to execute using the [↓] and [↑] keys, and press the Return key.

.DISPLAY

Data output is started according to the conditions specified using the following menu.

- SELECT DFM
- SELECT DUT
- DFM SKIP PATTERN COUNT
- OUTPUT DATA
- PAGE PAUSE
- SELECT DATA

. SELECT DFM

Select the DFM memory.

[WORK]

```
SELECT DFM(0:PIN-DFM , 1:DFM) = __ 0.↓
```

Select 0 for PIN-DFM and 1 for DFM on an AFM board.

The default value is PIN-DFM.

. SELECT DUT

Selects the DUT.

[WORK]

```
SELECT DUT(1-ddd) = __ 1.↓
```

Input a DUT number in the 1 to ddd range.

The largest DUT number currently being measured is displayed in ddd.

The default value is 1.

. DFM SKIP PATTERN COUNT

Enters the number of patterns to skip reading for the final valid data captured to the DFM.

[WORK]

```
DFM SKIP PATTERN COUNT(0-xxxx) = __ 0.↓
```

Input a number in the 0 to xxxx range for the number of patterns to skip reading.

1023 is displayed on xxxx for 1WAY, and 2047 is displayed for 2WAY.

The default value is 0.

. OUTPUT DATA

Selects the pass/fail data to display.

```
[WORK]
      OUTPUT DATA(0:FAIL , 1:ALL) = __ 0.↓
```

Input 0 to display only data for failed cycles, and 1 to display passed cycles, also.
The default value is FAIL.

. PAGE PAUSE

Selects whether to pause at units of twenty-four lines of data.

```
[WORK]
      PAGE PAUSE(0:YES , 1:NO) = __ 0.↓
```

Enter 0 to pause, and 1 not to pause.
The default value is YES.

. SELECT DATA

Selects the DFM data to display.

```
[WORK]
      SELECT DATA(0:MAIN , 1:SUB , 2:MAIN,SUB) = __ 0.↓
```

Enter 0 to display MAIN data, 1 to display SUB data, and 2 to display both MAIN and SUB data.
The default value is MAIN.

◆ Output Format

Depending on the number of bits, there are three available data formats for selecting PIN-DFM as the target memory and for selecting DFM on an AFM board:

Number of bits	PIN-DFM	DFM on AFM board
1 2 4 8 9 12 16 18 24	24-bit display	18-bit display
32 36 48	48-bit display	36-bit display
64 72 96	96-bit display	72-bit display

The display formats for 48 bits, PIN-DFM, and all data appear below.

For each I/O channel, "." signifies a pass, and "#" signifies a fail.

DFM COUNT counts the number of DFM patterns, and assigns a value of 1 to the cycle of the earliest pattern last captured to the DFM.

When there is no AFM double-speed mode, one cycle is displayed per line in the AFM PGSEL and AFM ADDRESS statements.

```
*****
*      PARALLEL DFM DISPLAY UTILITY      *
*****

DUT NO.0001

                                I/O CHANNEL NO.
                                11111111 11111111 11111
DFM                            21111111 11100000 00000999 55555554 44444444 43333333
COUNT      Y      X 09876543 21098765 43210987 65432109 87654321 09876543
-----
00001  #00000 #00000 .....
00002  #00000 #00100 .....
00003  #00000 #00101 .....
00004  #00000 #00102 .....
00005  #00000 #00103 .....
00006  #00000 #00104 .....
00007  #00000 #00105 .....
00008  #00000 #00106 .....
00009  #00000 #00107 .....
00010  #00000 #00108 .....
00011  #00000 #00109 .....
00012  #00000 #0010A .....
00013  #00000 #0010B .....
00014  #00000 #0010C .....
00015  #00000 #0010D .....
00016  #00000 #0010E .....
00017  #00000 #0010F .....
00018  #00000 #00110 .....#
00019  #00000 #00111 .....
00020  #00000 #00112 .....
00021  #00000 #00113 .....
00022  #00000 #00114 .....
00023  #00000 #00115 .....
00024  #00000 #00116 .....
[MEMO] PF1:LAST MENU PF2:NEXT PAGE *1
```

**1 This line is displayed when PAGE PAUSE=YES.*

To return to the last menu, press the PF1 key. To display the next data, press the PF2 key.

When data is displayed up to the very last item, the following message is displayed:

```
      :      :      :      :      :      :      :      :
01023  #00800 #00F15 .....
01024  #00800 #00F16 .....
      DATA END!!
[MEMO] PF1:LAST MENU
```

Press the PF1 key to return to the last menu.

Display format for when the AFM PGSEL statement data is MAIN, and AFM double-speed mode is in the AFM ADDRESS statement

Data for one cycle is displayed in two lines.

		I/O CHANNEL NO.							
		11111111 11111111 11111							
DFM		21111111 11100000 00000999 55555554 44444444 43333333							
COUNT	Y	X	09876543	21098765	43210987	65432109	87654321	09876543	

00001	#00000	#00000							*1
	#00001	#00100							*2
00002	#00002	#00101							*1
	#00003	#00102							*2
:	:	:	:	:	:	:	:	:	

*1 Strobe data for odd-numbered phase STRB1 to 128.

*2 Strobe data for odd-numbered phase STRB129 to 256.

Display format for when the AFM PGSEL statement data are MAIN and SUB, and there is no AFM double-speed mode in the AFM ADDRESS statement

When only MAIN is displayed, data for one cycle is displayed in one line.

		I/O CHANNEL NO.							
		11111111 11111111 11111							
DFM	M	21111111 11100000 00000999 55555554 44444444 43333333							
COUNT	S	Y	X	09876543	21098765	43210987	65432109	87654321	09876543

00001	M	#00000	#00000						*1
00002	M	#00000	#00100						*1
00003	M	#00000	#00101						*1
00004	M	#00000	#00102						*1
:	:	:	:	:	:	:	:	:	

*1 MAIN data

Display format for when the AFM PGSEL statement data are MAIN and SUB, and there is no AFM double-speed mode in the AFM ADDRESS statement

When only SUB is displayed, data for one cycle is displayed in one line.

				I/O CHANNEL NO.						
M				11111111	11111111	11111				
DFM	/				21111111	11100000	00000999	55555554	44444444	43333333
COUNT	S	Y	X	09876543	21098765	43210987	65432109	87654321	09876543	

00001	S	#00000	#00000							*1
00002	S	#00000	#00200							*1
00003	S	#00000	#00201							*1
00004	S	#00000	#00202							*1
:	:	:	:	:	:	:	:	:	:	

*1 SUB data

Display format for when the AFM PGSEL statement data are MAIN and SUB, and there is no AFM double-speed mode in the AFM ADDRESS statement

When both MAIN and SUB are displayed, data for one cycle is displayed in two lines.

				I/O CHANNEL NO.						
M				11111111	11111111	11111				
DFM	/				21111111	11100000	00000999	55555554	44444444	43333333
COUNT	S	Y	X	09876543	21098765	43210987	65432109	87654321	09876543	

00001	M	#00000	#00000							*1
	S	#00000	#00000							*2
00002	M	#00000	#00100							*1
	S	#00000	#00200							*2
00003	M	#00000	#00101							*1
	S	#00000	#00201							*2
00004	M	#00000	#00102							*1
	S	#00000	#00202							*2
:	:	:	:	:	:	:	:	:	:	

*1 MAIN data

*2 SUB data

Display format for when the AFM PGSEL statement data are MAIN and SUB, and AFM double-speed mode is in the AFM ADDRESS statement

When only MAIN is displayed, data for one cycle is displayed in two lines.

				I/O CHANNEL NO.							
M				11111111 11111111 11111							
DFM	/			21111111	11100000	00000999	55555554	44444444	43333333		
COUNT	S	Y	X	09876543	21098765	43210987	65432109	87654321	09876543		

00001	M	#00000	#00000								*1
	M	#00000	#10000								*2
00002	M	#00001	#10001								*1
	M	#00002	#10001								*2
00003	M	#00003	#10002								*1
	M	#00004	#10002								*2
00004	M	#00005	#10003								*1
	M	#00006	#10003								*2
:	:	:	:	:	:	:	:	:	:	:	

*1 Data for MAIN (odd-numbered phase strobes from STRB1 to 128)

*2 Data for MAIN (odd-numbered phase strobes from STRB129 to 256)

Display format for when the AFM PGSEL statement data are MAIN and SUB, and AFM double-speed mode is in the AFM ADDRESS statement

When only SUB is displayed, data for one cycle is displayed in two lines.

				I/O CHANNEL NO.											
M				11111111 11111111 11111											
DFM	/			21111111		11100000		00000999		55555554		44444444		43333333	
COUNT	S	Y	X	09876543	21098765	43210987	65432109	87654321	09876543						

00001	S	#00000	#00000												*1
	S	#00800	#20000												*2
00002	S	#00801	#20001												*1
	S	#00802	#20001												*2
00003	S	#00803	#20002												*1
	S	#00804	#20002												*2
00004	S	#00805	#20003												*1
	S	#00806	#20003												*2
:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	

*1 Data for SUB (even-numbered phase strobes from STRB1 to 128)

*2 Data for SUB (even-numbered phase strobes from STRB129 to 256)

Display format for when the AFM PGSEL statement data are MAIN and SUB, and AFM double-speed mode is in the AFM ADDRESS statement

When both MAIN and SUB are displayed, data for one cycle is displayed in four lines.

				I/O CHANNEL NO.						
M				11111111	11111111	11111				
DFM	/			21111111	11100000	00000999	55555554	44444444	43333333	
COUNT	S	Y	X	09876543	21098765	43210987	65432109	87654321	09876543	

00001	M	#00000	#00000							*1
	M	#00000	#10000							*2
	S	#00800	#00000							*3
	S	#00800	#20000							*4
00002	M	#00001	#10001							*1
	M	#00002	#10001							*2
	S	#00801	#20001							*3
	S	#00802	#20001							*4
:	:	:	:	:	:	:	:	:	:	

*1 Data for MAIN (odd-numbered phase strobes from STRB1 to 128)

*2 Data for MAIN (odd-numbered phase strobes from STRB129 to 256)

*3 Data for SUB (even-numbered phase strobes from STRB1 to 128)

*4 Data for SUB (even-numbered phase strobes from STRB129 to 256)

◆ Description of Err

err	Description
0	Normal termination
1	Not an executable system.
2	The specification for the DFM MODE statement is not ALL UP TO PCNTn. Or, the value of n is incorrect.
3	There is no FT>FMX specification in the AFM ACTION statement.
4	The specification (grpno) setting/clearing the DFM simultaneous capture function is incorrect.
5	The settings for the DFM simultaneous capture utility are not valid.
6	The unit specification (untsl) is incorrect.
7	The specified DUT number (dut) is incorrect.
8	The specification for the I/O channel block (pnblk) is incorrect.
9	The specification (rdcnt) for reading the number of valid data is incorrect.
10	The specification (rddat) for reading the data is incorrect.
11	The AFM ADDRESS statement is the AFM double-speed mode (two-pass) specification.
12	The PG is operating.

◆ Usage Examples

When failed cycles and data are displayed using PIN-DFM under the following conditions:
 BIT=24, parallel test of sixty-four devices (thirty-two doctors devices/station), 2WAY
 interleave mode, and AFM double-speed mode.

Omits the details of the pattern program.

The number of patterns is 32768 (2048 patter x 16)

```

:
LOGI[2] ERR, RDDUT(2), RDDAT(4096)
INTE[2] K, J, SPCNT, DUTNO, RDCNT
:
SET ILMODE 2WAY
:
AFM ADDRESS=<DOUBLE>,X...           ; AFM double-speed mode
AFM ACTION=FT>FMX                   ; AFM confirmation
:
SEND MPAT PTN01
SELECT MPAT INHIBIT
:
CALL PDFMSET(ERR, 1)                 ; DFM parallel test function setting
IF ERR<>0 THEN ENTRY
:
EXIT
:
DO LOOP1 K=1,16                      ; Loop for the number of patterns
  SPCNT=2048*K                       ; 2WAY=2048
  :
  DFM MODE=ALL UP TO PCNT(SPCNT) ; settings for DFM operation mode and
                                cycles to be analyzed

  REG MPAT PC=#100
  MEAS MPAT

  DO LOOP2 DUTNO=1,64                ; DUT loop
    CALL PDFMREAD(ERR, 1, DUTNO, 1, RDCNT, RDDAT) ; read data
    IF ERR<>0 OR RDCNT=0 THEN ENTRY
    :
    EXIT
  DO LOOP3 J=1,RDCNT
    IF RDDAT(J)<>#0 THEN ENTRY          ; Output fail
                                      information

    WRITE[1,2(2)] /,"DUT NO.",DUTNO
    WRITE[1,2(8)] "  COUNT=",2048*(K-1)*2+J
    WRITE[1,5(8)] "  DATA=#",RDDAT(J)
  EXIT
LOOP3:  CONT
LOOP2:  CONT
LOOP1:  CONT
:

```

◆ Precautions on Use

1. System compatibility

Cannot be executed on systems other than the T5593.

2. Pattern Generator Operation Mode

Execute the pattern after setting a fail inhibit state (SELECT MPAT INHIBIT)

3. Incorrect use

When data is read without performing a test, or when CALL PDFMSET is executed (changed) after a test, read data is invalidated.

4. Using DFM on an AFM board

An AFM board is required.

If no AFM is installed, no error (err=7) occurs when the DUT pass/fail information assigned to children A1, B1, C1, and D1 is read, and the pass data is read.

Specification of CALL PDFMSET(err,1) is required.

5. Operation mode for the DFM MODE statement

If the following specification is not made, an error (err=2) results, as proper data analysis becomes impossible.

DFM MODE = ALL UP TO PCNTn

When interleave mode is 1WAY, the value of n is a multiple of 1 (1, 2, 3, 4, ... ,536870911).

When interleave mode is 2WAY, the value of n is a multiple of 2 (2, 4, 6, 8, ... , 536870910).

6. The number of pattern program patterns used

When pass/fail information for all patterns cannot be captured to the DFM in one attempt, the data for the last and second-to-last pattern tests captured to the DFM may overlap.

When executing with an arbitrary pattern cycle, recognize and analyze overlapping data.

It is also possible to avoid data overlap by matching the number of pattern program patterns used to the interleave mode.

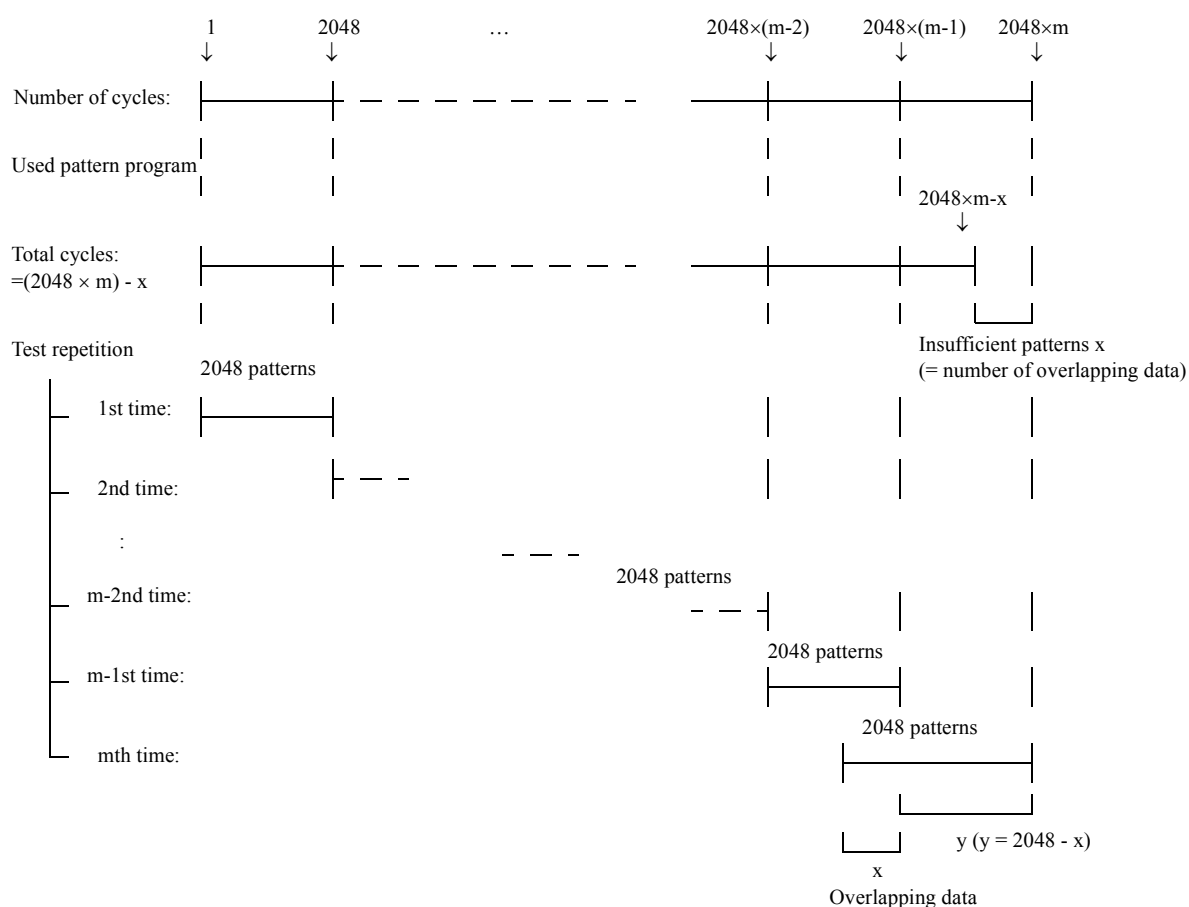
- A multiple of 1024 applies when the interleave mode is 1WAY.
- A multiple of 2048 applies when the interleave mode is 2WAY.

Also, the number of pattern program patterns can be confirmed using the READ DFM COUNTER statement.

- ◆ **When the Number of Patterns is not a Multiple of 2048, Interleave Mode is 2WAY, No AFM Double-Speed Mode, and Information for All Patterns cannot be Captured to the DFM in One Attempt**

In this case, the number of valid data for CALL PDFMREAD is 2WAY, and a total of 2048 results (even for the final test, the number of valid data is 2048).

DFM MODE=ALL UP TO PCNTn (n=2048, 4096, ..., 2048×(m-2), 2048×(m-1), 2048×m)



In this case, preparing x dummy cycles will eliminate overlapping data (i.e., 0 instances).

7. Initial state of this function

The initial state is CALL PDFMSET (err, 0).

The conditions of the initial state are as follows; CALL PDFMREAD data cannot be read in the initial state.

- Controller startup
- Starting the program
- RESET TESTER statement

- The current interleaved mode is changed using the SET ILMODE statement.
 - RESET AFM ALL statement
 - RESET AFM MODE statement
 - AFM DUTSEL statement
 - When M command MODE 8 (change of target DUT number) is ENABLE
 - CTRL/C
 - C-BOX reset button
 - Utility /FBMAP(START MPAT=YES), /FMXFER execution
8. When this function is specified as valid using CALL PDFMSET, restricted statements and utilities are as follows:
- Restricted statements
The READ DFM statement causes a runtime error.
 - Restricted utilities
Function data log
The data log cannot be executed. However, when M command MODE 7 (all DUT pass/fail information output) is valid, all device pass/fail information is output.
Also, when M command MODE 8 (change of target DUT number) is valid, the DUT to view is selected in the MEASMPAT statement, and the function data log function becomes valid, whereupon the CALL PDFMSET function is released.

66. Motherboard ID Reading Utility

This utility program is used to read ID numbers set by the ID switches on the motherboard.
The utility is compatible with the T5593.

66. 1 Motherboard ID Reading Utility

◆ Syntax

```
CALL RDMBID (ERR, SW, V)
```

ERR

Used to store the end status of the Motherboard ID readout utility program (normal or abnormal end).

Specify using an integer or logical variable.

Specify using a 1HW- or 2HW-type variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For details, refer to ["ERR Description" on page 66-3](#).

SW

Select a motherboard to read.

Specify using an integer (or logical) constant, or a variable between 1 to 2.

If using a variable, specify using a 1HW- or 2HW-type variable.

When SW=1, MB0 to 15 is read for Motherboard A.

When SW=2, MB32 to 47 is read for Motherboard B.

V

A variable that stores read ID numbers

Preformatted variables without, with, or for all integer-/logical-type DIM specifications can be specified.

Specify using a 1HW- or 2HW-type variable.

Tip

When the numbers of variables and motherboard IDs (number of stations) do not match, the smaller value becomes the ID number.

◆ Data Format

Motherboard ID numbers are read in units of stations.

When SW=1, MB0 to 15 is read.

When SW=2, MB32 to 47 is read.

Figure 66-1 SW=1 Data Format

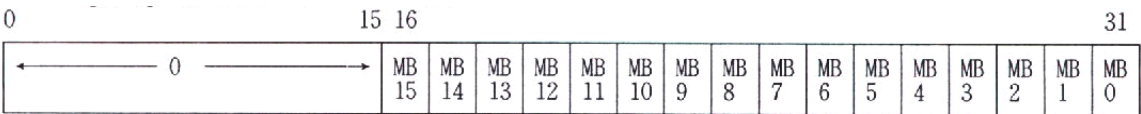
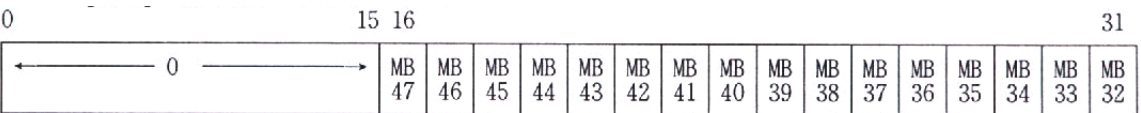


Figure 66-2 SW=2 Data Format



◆ Usage Examples

When reading MB0 to 15

```
PRO PRO1 SOC1

LOGI[2] V(2)=0
INTE[1] ERR=0
INTE[1] SW=1
CALL RDMBID(ERR,SW,V)

END
```

When reading MB32 to 47

```
PRO PRO1 SOC1

LOGI[2] V(2)=0
INTE[1] ERR=0
INTE[1] SW=2
CALL RDMBID(ERR,SW,V)

END
```

 Tip

When there is one socket program station, or when a socket program is not used, the ID number for the station being executed is read to V(1). V(2) does not change.

When there are 2 socket program stations, the first station ID number is read to V(1) The second station ID number is read to V(2).

◆ ERR Description

ERR	Description
0	Normal end
1	Cannot be executed in this system.
2	The selection (SW) on the motherboard is incorrect.
3	The read data (V) is incorrect.

67. Autoscan Mode Utility (AUTOSCAN)

This section describes the autoscan mode utility (AUTOSCAN).

67. 1 Autoscan Mode Setting Utility

◆ Function

The AUTOSCAN MODE utility is used to set a mode for adding or subtracting the designated resolution for the designated clock and STRB setting for each cycle.

Addition or subtraction is performed in the control bit cycle specified by ASCNT in the REGISTER statement described in the pattern program or in the REP MPAT statement.

A reset is performed in the the control-bit cycle specified by ASINI, and the value returns to that specified by the clock and STRB statement.

This utility can be used on the T5593.

◆ Syntax

```
CALL AUTOSCAN(ERR,"ARG1, resolution, unit name, unit name, unit name,...,
COUNTER=unit name, unit name...")
```

ERR

Stores the end state.

Specify using 1W to 2W integer-type variables or a logical variable.

If ERR is 0, it means a normal end; if ERR is not 0, it means an abnormal end.

➔ For more information, refer to [67. 3 "ERR Description" on page 67-12](#).

ARG1

Sets the mode.

0

Resets AUTOSCAN and disables AUTOSCAN COUNTER.

1

Sets AUTOSCAN

The AUTOSCAN COUNTER of the units specified after "COUNTER=" is enabled. The AUTOSCAN COUNTER of the units unspecified after "COUNTER=" is disabled.

➔ For more information on the AUTOSCAN COUNTER reading utility, refer to [67. 2 "AUTOSCAN COUNTER Reading Utility" on page 67-7](#).

Specify the following argument when specifying 1 in ARG1.

Possible voltage setting

Specify the value to be added or subtracted during each cycle for the STRB setting value and specified clock.

The specified resolution is enabled for all timing sets.

Specify using real numbers with time units within the following ranges.

$(-937.5\text{ps} + \text{minimum resolution}) \leq \text{resolution} \leq -7.32421875\text{ps}$

$7.32421875\text{ps} \leq \text{resolution} \leq (937.5\text{ps} - \text{minimum resolution})$

Minimum resolution: 7.32421875ps

Unit name

BCLK_n, CCLK_n, DREL_n, DRET_n, STRB_n

n specifies a phase number. Specify using an integer-type constant.

Unit	n
BCLK CCLK	$1 \leq n \leq 128$
DREL DRET	$1 \leq n \leq 256$
STRB	$1 \leq n \leq 256$ (only odd-numbered phases can be specified)

COUNTER=unit name

Specifies the unit in which AUTOSCAN COUNTER is enabled.

The unit name specifies the STRB_n.

n specifies a phase number. Specify using an integer-type constant.

Unit	n
STRB	$1 \leq n \leq 256$

This argument can be omitted.

If omitted, AUTOSCAN COUNTER of all units are disabled.

◆ Example of Specification

```
CALL AUTOSCAN(ERR,"1,7.32421875PS,BCLK1")
CALL AUTOSCAN(ERR,"1,7.32421875PS,BCLK1,CCLK1")
CALL AUTOSCAN(ERR,"1,7.32421875PS,BCLK1,CCLK1,COUNTER=STRB1")
```

Important

- Clocks and STRB settings, as well as ranges for results of resolution operations, are as follows:

For 1WAY

$0\text{ns} \leq \text{Operation result}^{*1} \leq (\text{clock and STRB setting value} + 3.75\text{ ns} - \text{minimum resolution})$
and $(4 \times \text{RATE} - \text{minimum resolution})$

For 2WAY

$0\text{ns} \leq \text{Operation result}^{*1} \leq (\text{clock and STRB setting value} + 3.75\text{ ns} - \text{minimum resolution})$
and $(8 \times \text{RATE} - \text{minimum resolution})$

**1 The number of control bit cycles specified by the clock and STRB setting + resolution $\times$ ASCNT*

➔ Refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498 \)](#) regarding other restrictions.

- When the MEAS MPAT or START MPAT statement is executed, the clock and STRB values to be scanned are equal to the values specified in the clock statement and STRB statement plus or minus the specified resolution.
- When AUTOSCAN is described consecutively, the resolution last specified in AUTOSCAN and the unit name take effect.
- Set the pin conditions before performing AUTOSCAN settings.
- When STRBn is specified for a unit name, STRB for STRBn+1 is scanned at the same time.
- If AUTOSCAN COUNTER is enabled, the ASCNT register value is stored in the RHZA and RHZB registers every time the MEAS MPAT or START MPAT statement is executed. The ASCNT register value cannot be stored by MEAS MPAT(1) or START MPAT(1).
- PASS/FAIL COUNTER, FAIL COUNTER and AUTOSCAN COUNTER cannot be used at the same time.
DISABLE must be set to PASS/FAIL COUNTER or FAIL COUNTER if enabling AUTOSCAN COUNTER after using the PASS/FAIL COUNTER or FAIL COUNTER setting utility of the DQS test utility.

Limitation

- AUTOSCAN cannot be used at the same time as SELECT DBM DELAY INHIBIT.
- The control bits for which ASINI and ASCNT are specified cannot be specified in the same cycle.

◆ How to Use

The following example shows addition taking place in cycles described as C1, and a reset taking place in a the cycle described as C0.

(Example 1) When Using Timing Set 1

◆ Main Program

```

PRO B
  (omitted)
RATE=3.75NS,3.75NS
STRB1=0NS,0NS
CALL AUTOSCAN(ERR,"1,7.32421875PS,STRB1")
IF ERR<>0 THEN ... ; Error processing
MEAS MPAT MPA1
  (omitted)
END

```

◆ Pattern Program

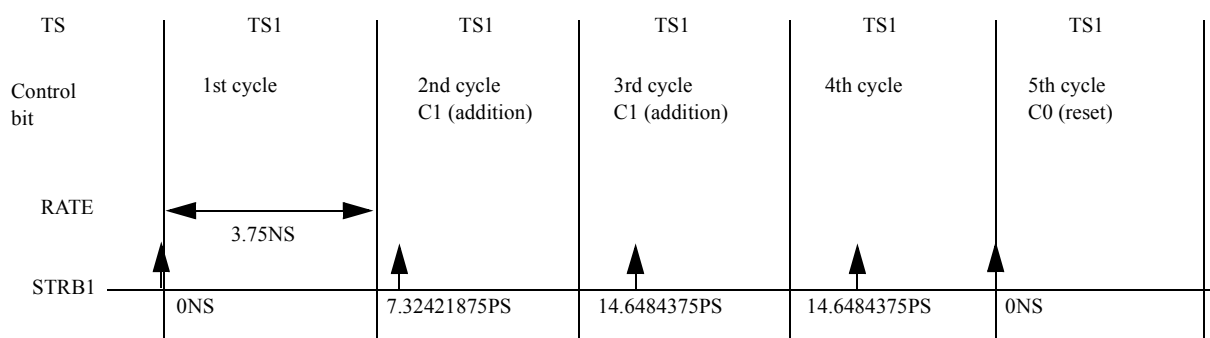
```

MPAT MPA1

REGISTER
  ASINI = C0
  ASCNT = C1
  (omitted)
START #0
  NOP          TS1 ; (1st cycle)
  NOP C1       TS1 ; (2nd cycle)
  NOP C1       TS1 ; (3rd cycle)
  NOP          TS1 ; (4th cycle)
  NOP C0       TS1 ; (5th cycle)
STPS
END

```

Based on the settings shown above, the value STRB1 takes on the following values for the MEAS MPAT statement.



(Example 2) When Using Timing Sets 1 and 2

◆ **Main Program**

```

PRO C
  (omitted)
RATE=3.75NS,3.75NS
STRB1=0NS,1.875NS
CALL AUTOSCAN(ERR,"1,7.32421875PS,STRB1")
IF ERR<>0 THEN ... ; Error processing
MEAS MPAT MPA2
  (omitted)
END

```

◆ **Pattern Program**

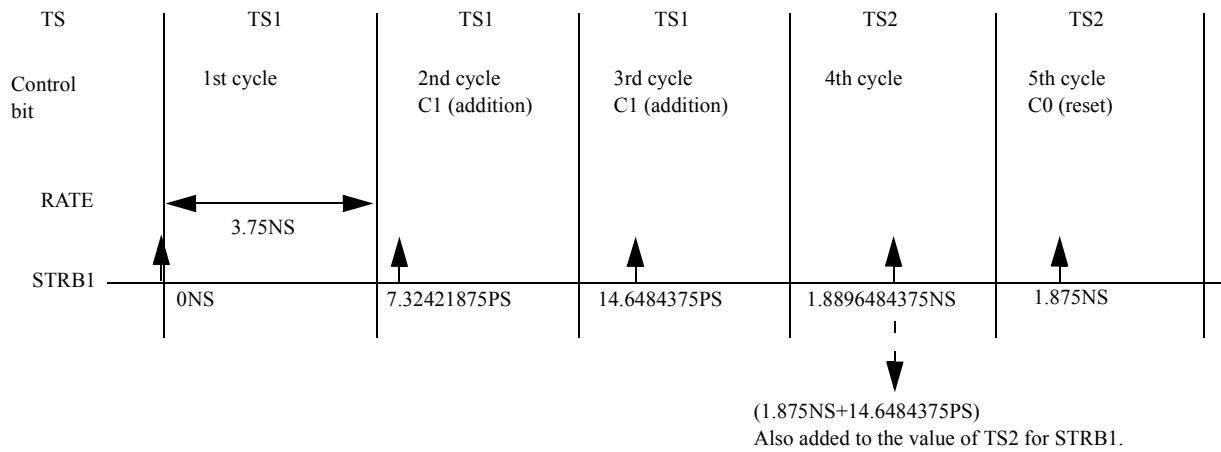
```

MPAT MPA2

REGISTER
  ASINI  = C0
  ASCNT  = C1
  (omitted)
START #0
  NOP          TS1      ; (1st cycle)
  NOP C1       TS1      ; (2nd cycle)
  NOP C1       TS1      ; (3rd cycle)
  NOP          TS2      ; (4th cycle)
  NOP C0       TS2      ; (5th cycle)
STPS
END

```


Based on the settings shown above, the value STRB1 takes on the following values for the MEAS MPAT statement.



67. 2 AUTOSCAN COUNTER Reading Utility

◆ Function

This utility reads the AUTOSCAN COUNTER data registered in the control bit cycles which are specified by ASCNT, if AUTOSCAN COUNTER is enabled by using the AUTOSCAN utility.

The AUTOSCAN COUNTER stops when FIRST FAIL is detected.

The AUTOSCAN COUNTER value no longer increases, even in the control bit cycles which are specified by ASCNT after detecting FIRST FAIL.

If the 2WAY mode is used, the utility reads out the AUTOSCAN COUNTER data which is registered when odd-numbered strobes are generated.

To use this utility, specify the SELECT MPAT INHIBIT statement.

The utility is compatible with the T5593.

◆ Syntax

```
CALL RDASCNT (ERR, PNNO, BANK, CNT1, CNT2)
```

ERR

Stores the end state.

Specify by using a 1W through 2W integer-type variable or a logical variable.

If ERR is 0, the scan was completed correctly, if ERR is not 0, the scan was completed incorrectly.

PNNO

Specifies channel numbers whose AUTOSCAN COUNTER data is read.

The specifiable channels are 33 to 56, 97 to 120, 161 to 184, and 225 to 248.

Specify by using the PINLIST variable.

BANK

Specifies the banks from which AUTOSCAN COUNTER data is read.

Use an integer (or logical) constant, or a variable equal to 0 through 2.

BANK	Bank number
0	All banks
1	Bank 1
2	Bank 2

When all banks are specified in the 1WAY mode, the utility read out the AUTOSCAN COUNTER data, which is registered when the odd-numbered strobes and even-numbered strobes are generated.

When bank 1 is specified in the 1WAY mode, the utility reads out the AUTOSCAN COUNTER data which is registered when the odd-numbered strobes are generated.

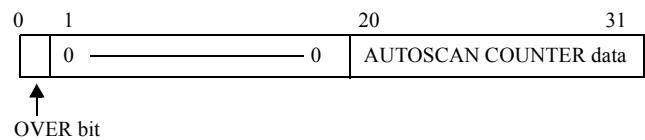
When bank 2 is specified in the 1WAY mode, the utility reads out the AUTOSCAN COUNTER data which is registered when the even-numbered strobes are generated.

CNT1

Specifies the variable which stores the read AUTOSCAN COUNTER data of STRB1 through STRB128.

Use a logical variable with 2 HW (32 bits).

Specify an array variable whose size is the number of channels multiplied by the number of DUTs for which the counter data is read.



COUNTER is 12 bits (0 through 4095) long.

When the COUNTER value is 4095 or more, 1 is set to the OVER bit.

DUTs are read out in order from DUT1.

If multiple channel numbers are specified, the counter data for each DUT is read out in the order of the specified channel numbers to PNNO, and then the next DUT is tested.

The target DUTs are those that are specified in the program as the parallel test target DUTs (DDUT).

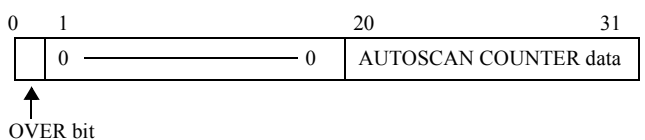
However, variables of DUTs which are not currently target DUTs in the parallel test (other than CDUT) and channels of the unit in which AUTOSCAN COUNTER is disabled are unchanged.

CNT2

Specifies the variable which stores the read AUTOSCAN COUNTER data of STRB129 through STRB256.

Use a logical variable with 2 HW (32 bits).

Specify an array variable whose size is the number of channels multiplied by the number of DUTs for which the counter data is read.



COUNTER is 12 bits (0 through 4095) long.

When the COUNTER value is 4095 or more, 1 is set to the OVER bit.

DUTs are read out in order from DUT1.

If multiple channel numbers are specified, the counter data for each DUT is read out in the order of the specified channel numbers to PNNO, and then the next DUT is tested.

The target DUTs are those that are specified in the program as the parallel test target DUTs (DDUT).

However, variables of DUTs which are not currently target DUTs in the parallel test (other than CDUT) and channels of the unit in which AUTOSCAN COUNTER is disabled are unchanged.

AUTOSCAN COUNTER is reset every time a MEAS MPAT or START MPAT statement is executed.

It cannot be cleared, however, by MEAS MPAT(1) or START MPAT(1).

AUTOSCAN COUNTER is also disabled when one of the following events occurs:

- A program is started
- The RESET TESTER statement is executed
- The current interleaved mode setting is changed by using the SET ILMODE statement.

Tip

In 2WAY, a FIRST FAIL is detected for each bank.

(Example) If a FAIL is detected in bank2 first:

In bank1, a FIRST FAIL is detected in (2) and the counter value is 2.

In bank2, a FIRST FAIL is detected in (1) and the counter value is 1.

◆ Pattern Program

```
MPAT MPA4

REGISTER
  CPE1=R
  ASCNT=C1
  (omitted)
START #0
  NOP: R C1 ; PASS (Bank1)
      : R C1 ; FAIL (Bank2) ... (1)
  NOP: R C1 ; FAIL (Bank1) ... (2)
      : R C1 ; FAIL (Bank2)

STPS
END
```

◆ Main Program

```
PRO E
  (omitted)
CALL RDASCNT(ERR, PNNO, BANK, CNT1, ...)
                                ; CNT1=3 when BANK=0
                                ; CNT1=2 when BANK=1
                                ; CNT1=1 when BANK=2
IF ERR<>0 THEN ... ; Error processing
  (omitted)
END
```

◆ How to Use

(Example) When reading AUTOSCAN COUNTER for channels 33 through 40

```

PRO D      SOC1
  (omitted)
INTE[1] ERR,BANK
LOGI[2] CNT1(1024)=0
LOGI[2] CNT2(1024)=0
PINLIST DQP=P33-40

RATE      = 4NS
BCLK1     = 1NS
CCLK1     = 3NS
STRB1     = 9NS
STRB129=11NS

IN1 = 3V,0V
OUT1= 2.5V,0.5V
VT1 = 0.5V

DQP=IN1,DNRZ,BCLK1,CCLK1,IOC,OUTL1,VT1,DRECLK1,DRE1,@
      [STRB1,CPE1],EXPA1,[STRB129,CPE1],EXPB129,<D0-7,SD0-7>

CALL AUTOSCAN(ERR,"1,0.5NS,STRB1,STRB129,COUNTER=STRB1,STRB129")
IF ERR<>0 THEN ... ; Error processing

SELECT MPAT INHIBIT
MEAS MPAT MPA3

BANK=0 ; All banks

CALL RDASCNT(ERR,DQP,BANK,CNT1,CNT2) ; STRB1 to 128      STRB129 to 256
; CNT1(1) ,CNT2(1) : DUT1 P33
; CNT1(2) ,CNT2(2) : DUT1 P34
; (omitted) (omitted) (omitted)
; CNT1(7) ,CNT2(7) : DUT1 P39
; CNT1(8) ,CNT2(8) : DUT1 P40
; (omitted) (omitted) (omitted)
; CNT1(1015) ,CNT2(1015) : DUT128 P33
; CNT1(1016) ,CNT2(1016) : DUT128 P34
; (omitted) (omitted) (omitted)
; CNT1(1023) ,CNT2(1023) : DUT128 P39
; CNT1(1024) ,CNT2(1024) : DUT128 P40

IF ERR<>0 THEN ... ; Error processing
(omitted)
END

```

67. 3 ERR Description

ERR	Description
0	Normal termination
1	Cannot execute on this system.
2	The ARG1 setting is incorrect.
3	Out of resolution range or units are wrong
4	The unit name is incorrect.
5	Cannot be specified at the same time as SELECT DBM DELAY INHIBIT
6	TG or PG is operating.
7	Cannot be used simultaneously with the real-time Hi-z mode.
8	Channel numbers are incorrectly specified (PNNO).
9	The bank setting (BANK) is incorrect.
10	The variable used to store read counter values is incorrectly specified (CNT1, CNT2).
11	Cannot be used simultaneously with the PASS/FAIL COUNTER setting utility (CALL STPFCNT) of the DQS test utility or the FAIL COUNTER setting utility (CALL STFLCNT).
12	The high-speed clock driver channel cannot be used.

68. *PMONIT (RON/ROF MONITOR)*

This section describes PMONIT (RON/ROF MONITOR).

68. 1 PMONIT Overview

PMONIT displays the reference-on/off log data of the PPS (Programmable Power Supply).

The PMONIT utility can save up to 200 reference-on/off log data items, while checking reference-on/off occurrences, and displays the data.

This utility can be set by two ways; starting from a test plan program or starting from a TCLI.

For more information, refer to the description for each setting method.

Limitation

This utility program cannot be executed in the T5581, T5581H, T5581P, T5571P, T5591, T5591R, T5336, and T5511 test systems.

68. 2 PMONIT Syntax

68. 2. 1 Starting up from the Program

```
CALL LIB PMONIT(str, flg)
```

str

Character string or character string variable

"ENABLE"	→	Starts checking RON/ROF
"READ"	→	Reads the change from RON into ROF after CALL PMONIT("ENABLE", flg).
"DISABLE"	→	Stops checking RON/ROF

flg

Integer-type variable

When str="ENABLE"

flg=0	→	Current state is RON
flg=1	→	Current state is ROF

When str="READ"

flg=0	→	ROF is not generated between ENABLE through READ.
flg=1	→	ROF is generated between ENABLE through READ.

When str="DISABLE"

flg	→	No change
-----	---	-----------

! Important

- A runtime error occurs if the argument is omitted.

68. 2. 2 Starting up from the TCLi

```
/PMONIT global switch
```

Global switch

/E	Starts checking RON/ROF and displays the current state.
/D	Ends checking RON/ROF.
/R	Displays the ROF data obtained from the start of RON/ROF check until now.
/H	Displays the file name, line number, and RON/ROF state on the display.
No specification	Displays the ROF data obtained from the start of RON/ROF check until now.
(same as the /R specification)	

- Up to 32 alphanumeric characters can be displayed for the file name when the global switch /H is used.

In addition, the library name is displayed when a hybrid controller is used.

- When executing the global switch /E, the current state is displayed as follows.

CURRENT STATUS : RON or ROFF

VS state	Display
ON	RON
OFF	ROFF

- When executing the global switch /R or specifying no global switch, the following messages are displayed.

ROF state	Display
Occurred	The ROF state has been occurred.
Not occurred	The ROF state has NOT been occurred.

- If specifying more than one global switch, the global switch selected first is enabled.

/PMONIT/E/A ← /E is selected.

- If the global switch is not /E, /D, /R, /H, or no specification, an error message is displayed.

/PMONIT/B

ILLEGAL PMONIT SPECIFIER: B

Tip

- The check ends in the following cases.
 - If the command to end checking RON/ROF was executed.
 - If a program end statement, such as an END statement and STOP statement, was executed.
 - If the program was terminated using "control - C".

4. *If a runtime error occurred.*

- *If the command which starts RON/ROF-checking is executed a second time while checking RON/ROF, RON/ROF checking is performed twice.*

If the command which stops RON/ROF-checking is executed without executing the command which starts checking RON/ROF, the command which ends RON/ROF-checking is ignored.

- *If the RON/ROF data items reach 200, subsequent items overwrites the old data and only a maximum of 200 data items are displayed.*
 - *Available range for checked data*
 1. *Available until the test plan program is executed.*
 2. *Available until the command which starts checking RON/ROF is executed.*
-

68. 3 Usage Examples

68. 3. 1 Starting up from the Program

```
INTE[2] FLG=0
IF FK=1 THEN ENTRY
    CALL LIB PMONIT("ENABLE",FLG)           ; Starts checking RON/ROF
    IF FLG=0 THEN WRITE "Current state : RON" /
    ELSE WRITE "Current state : ROF" /
EXIT
:
:
:
IF FK=1 THEN ENTRY
    CALL LIB PMONIT("READ",FLG)             ; Reads whether or not ROF occurred
    IF FLG=1 THEN ENTRY
        WRITE "ROF HAS OCCURRED" /
        GOSUB VSRON                        ; Turns VS on
    EXIT
EXIT
:
:
IF FK=1 THEN CALL LIB PMONIT("DISABLE",FLG) ; Ends checking RON/ROF
```

68. 3. 2 Starting up from the TCLi

```

BREAK! 01B  PRO SAMPLE1    8
%/PMONIT/E                      ; Starts checking RON/ROF
  CURRENT STATUS : ROFF        ; Outputs the current RON/ROF state
%/800$B                          ; Sets the stop position
%$P

BREAK! 02B  PRO SAMPLE1    800
%/PMONIT/R                      ; Calls for the data output of RON/ROF change

  Displays the stored program name (library name), line number, and RON/ROF state.

  PRO SAMPLE1                  :    10          : RON
  PRO SAMPLE1                  :   245          : ROF
  SUB SBSAMP1                   :    19          : RON
  AT_set_rejection              :                : ROF
  PRO SAMPLE1                  :   345          : RON
  PRO SAMPLE1                  :   500          : ROF

%/PMONIT                        ; Calls for the data output of RON/ROF change

  The ROF state has been occurred.      ; ROF occurred
  or
  The ROF state has NOT been occurred.   ; ROF not occurred

%/PMONIT/D                      ; Sets the DISABLE to RON/ROF monitor

```

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